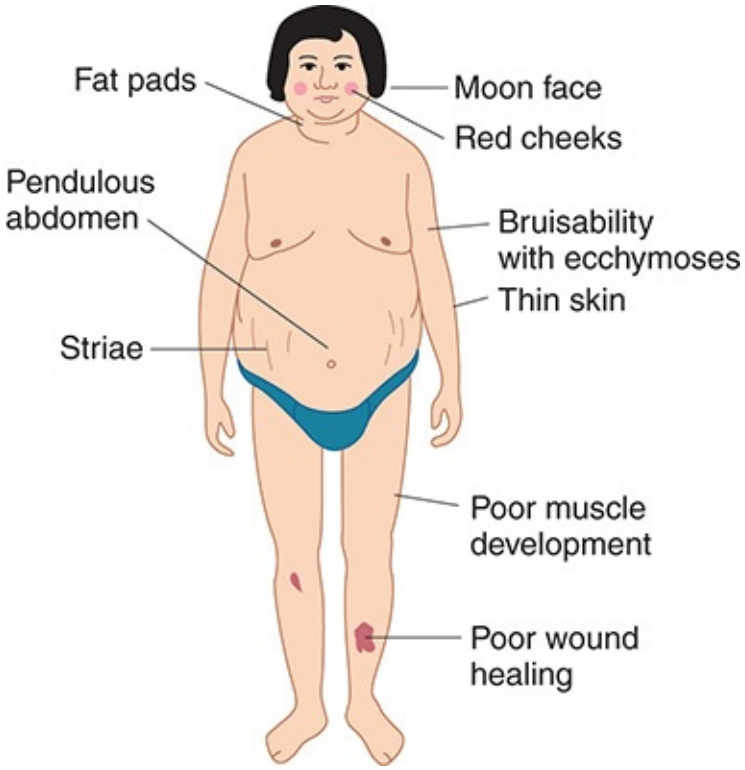


FIGURE 19–13 Typical findings in Cushing syndrome. (Reproduced with permission from Forsham PH, Di Raimondo VC: *Traumatic Medicine and Surgery for the Attorney*. Butterworth; 1960.)

Patients with Cushing syndrome are protein-depleted as a result of excess protein catabolism. The skin and subcutaneous tissues are therefore thin and the muscles are poorly developed. Wounds heal poorly, and minor injuries cause bruises and ecchymoses. The hair is thin and scraggly. Many patients with the disease have some increase in facial hair and acne, but this is caused by the increased secretion of adrenal androgens and often accompanies the increase in glucocorticoid secretion.

Body fat is redistributed in a characteristic way. The extremities are thin, but fat collects in the abdominal wall, face, and upper back, where it produces a "buffalo hump." As the thin skin of the abdomen is stretched by the increased subcutaneous fat depots, the subdermal tissues rupture to form prominent reddish purple **striae.** These scars are seen normally whenever a rapid stretching of skin occurs, but in normal individuals the striae are usually inconspicuous and lack the intense purplish color.

Many of the amino acids liberated from catabolized proteins are converted into glucose in the liver and the resultant hyperglycemia and decreased peripheral utilization of glucose may be sufficient to precipitate insulin-resistant

diabetes mellitus, especially in patients genetically predisposed to diabetes. Hyperlipidemia and ketosis are associated with the diabetes, but acidosis is usually not severe.

The glucocorticoids are present in such large amounts in Cushing syndrome that they may exert a significant mineralocorticoid action. Deoxycorticosterone secretion is also elevated in cases due to ACTH hypersecretion. The salt and water retention plus the facial obesity cause the characteristic plethoric, rounded "moon-faced" appearance, and there may be significant K^+ depletion and weakness. About 85% of patients with Cushing syndrome are hypertensive. The hypertension may be due to increased deoxycorticosterone secretion, increased angiotensinogen secretion, or a direct glucocorticoid effect on blood vessels (see Chapter 32).

Glucocorticoid excess leads to bone dissolution by decreasing bone formation and increasing bone resorption. This leads to **osteoporosis,** a loss of bone mass that leads eventually to collapse of vertebral bodies and other fractures. The mechanisms by which glucocorticoids produce their effects on bone are discussed in Chapter 21.

Glucocorticoids in excess accelerate the basic electroencephalographic rhythms and produce mental aberrations ranging from increased appetite, insomnia, and euphoria to frank toxic psychoses. As noted above, glucocorticoid deficiency is also associated with mental symptoms, but the symptoms produced by glucocorticoid excess are more severe.

ANTI-INFLAMMATORY & ANTI-ALLERGIC EFFECTS OF GLUCOCORTICOIDS

Glucocorticoids inhibit the inflammatory response to tissue injury. The glucocorticoids also suppress manifestations of allergic disease that are due to the release of histamine from mast cells and basophils. Both of these effects require high levels of circulating glucocorticoids and cannot be produced by administering steroids without producing the other manifestations of glucocorticoid excess. Furthermore, large doses of exogenous glucocorticoids inhibit ACTH secretion to the point that severe adrenal insufficiency can be a dangerous problem when therapy is stopped. However, local administration of glucocorticoids, for example, by injection into an inflamed joint or near an irritated nerve, produces a high local concentration of the steroid, often without enough systemic absorption to cause serious side effects.

The actions of glucocorticoids in patients with bacterial infections are dramatic but dangerous. For example, in pneumococcal pneumonia or active tuberculosis, the febrile reaction, the toxicity, and the lung symptoms disappear, but unless antibiotics are given at the same time, the bacteria spread throughout the body. It is important to remember that the symptoms are the warning that disease is present; when these symptoms are masked by treatment with glucocorticoids, there may be serious and even fatal delays in diagnosis and the institution of treatment with antimicrobial drugs.

The role of NF-κB in the anti-inflammatory and anti-allergic effects of glucocorticoids has been mentioned above and is discussed in Chapter 3. An additional action that combats local inflammation is inhibition of phospholipase A_2. This reduces the release of arachidonic acid from tissue phospholipids and consequently reduces the formation of leukotrienes, thromboxanes, prostaglandins, and prostacyclin (see Chapter 32).

OTHER EFFECTS

Large doses of glucocorticoids inhibit growth, decrease growth hormone secretion (see Chapter 18), induce PNMT, and decrease thyroid-stimulating hormone secretion. During fetal life, glucocorticoids accelerate the maturation of surfactant in the lungs (see Chapter 34).

REGULATION OF GLUCOCORTICOID SECRETION

ROLE OF ACTH

Both basal secretion of glucocorticoids and the increased secretion provoked by stress depend on ACTH from the anterior pituitary (Chapter 18). Angiotensin II also stimulates the adrenal cortex, but its effect is mainly on aldosterone secretion. Large doses of a number of other naturally occurring substances, including vasopressin, serotonin, and vasoactive intestinal polypeptide (VIP), are capable of stimulating the adrenal directly, but there is no evidence that these agents play any role in the physiologic regulation of glucocorticoid secretion.

CHEMISTRY & METABOLISM OF ACTH

ACTH is a single-chain polypeptide containing 39 amino acids. Its origin from proopiomelanocortin (POMC) in the pituitary is discussed in Chapter 18. The first 23 amino acids in the chain generally constitute the active "core" of the molecule. Amino acids 24–39 constitute a "tail" that stabilizes the molecule and varies slightly in composition from species to species. The ACTHs that have been isolated are generally active in all species but antigenic in heterologous species.

ACTH is inactivated in blood in vitro more slowly than in vivo; its half-life in the circulation in humans is about 10 min. A large part of an injected dose of ACTH is found in the kidneys, but neither nephrectomy nor evisceration appreciably enhances its in vivo activity, and the site of its inactivation is not known.

EFFECT OF ACTH ON THE ADRENAL

After hypophysectomy, glucocorticoid synthesis and output decline within 1 h to very low levels, although some hormone is still secreted. Within a short time after an injection of ACTH (in dogs, less than 2 min), glucocorticoid output is increased. With low doses of ACTH, the relationship between the log of the dose and the increase in glucocorticoid secretion is linear. However, the maximal rate at which glucocorticoids can be secreted is rapidly reached, and this "ceiling on output" also exists in humans. The effects of ACTH on adrenal morphology and the mechanism by which it increases steroid secretion have been discussed above.

ADRENAL RESPONSIVENESS

ACTH not only produces prompt increases in glucocorticoid secretion but also increases the sensitivity of the adrenal to subsequent doses of ACTH. Conversely, single doses of ACTH do not increase glucocorticoid secretion in chronically hypophysectomized animals and patients with hypopituitarism, and repeated injections or prolonged infusions of ACTH are necessary to restore normal adrenal responses to ACTH. Decreased responsiveness is also produced by doses of glucocorticoids that inhibit ACTH secretion. The decreased adrenal responsiveness to ACTH is detectable within 24 h after hypophysectomy and increases progressively with time **(Figure 19–14)**. It is marked when the adrenal is atrophic but develops before visible changes occur in adrenal size or

morphology.

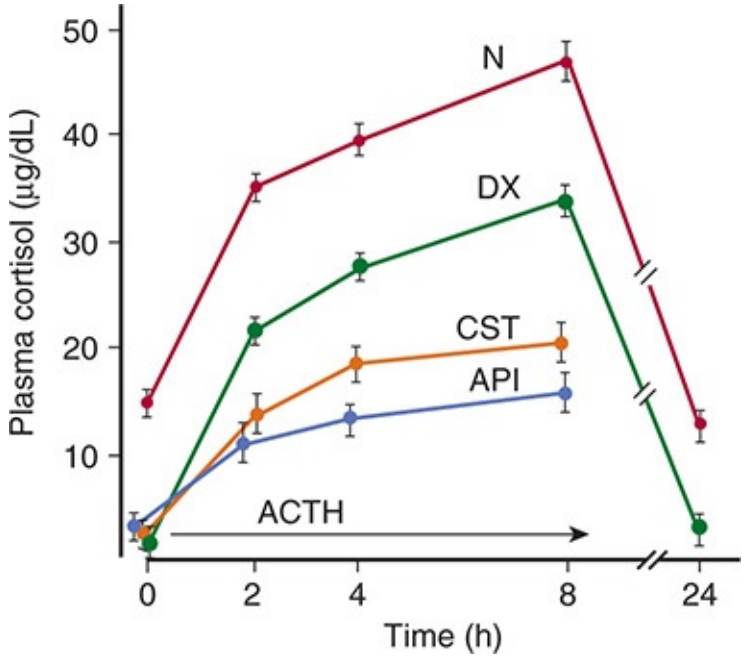


FIGURE 19–14 Loss of ACTH responsiveness when ACTH secretion is decreased in humans. The 1- to 24-amino-acid sequence of ACTH was infused intravenously (IV) in a dose of 250 µg over 8 h. CST, long-term corticosteroid therapy; DX, dexamethasone 0.75 mg every 8 h for 3 days; API, anterior pituitary insufficiency; N, normal subjects. (Reproduced with permission from Kolanowski J, et al: Adrenocortical response upon repeated stimulation with corticotropin in patients lacking endogenous corticotropin secretion. Acta Endocrinol [Kbh] 1977;85:595.)

CIRCADIAN RHYTHM

ACTH is secreted in irregular bursts throughout the day and plasma cortisol tends to rise and fall in response to these bursts **(Figure 19–15)**. In humans, the bursts are most frequent in the early morning, and about 75% of the daily production of cortisol occurs between 4:00 AM and 10:00 AM. The bursts are least frequent in the evening. This **diurnal (circadian) rhythm** in ACTH secretion is present in patients with adrenal insufficiency receiving constant doses of glucocorticoids. It is not due to the stress of getting up in the morning, traumatic as that may be, because the increased ACTH secretion occurs before waking up. If the “day” is lengthened experimentally to more than 24 h, that is, if the individual is isolated and the day’s activities are spread over more than 24

h, the adrenal cycle also lengthens, but the increase in ACTH secretion still occurs during the period of sleep. The biologic clock responsible for the diurnal ACTH rhythm is located in the suprachiasmatic nuclei of the hypothalamus (see Chapter 17).

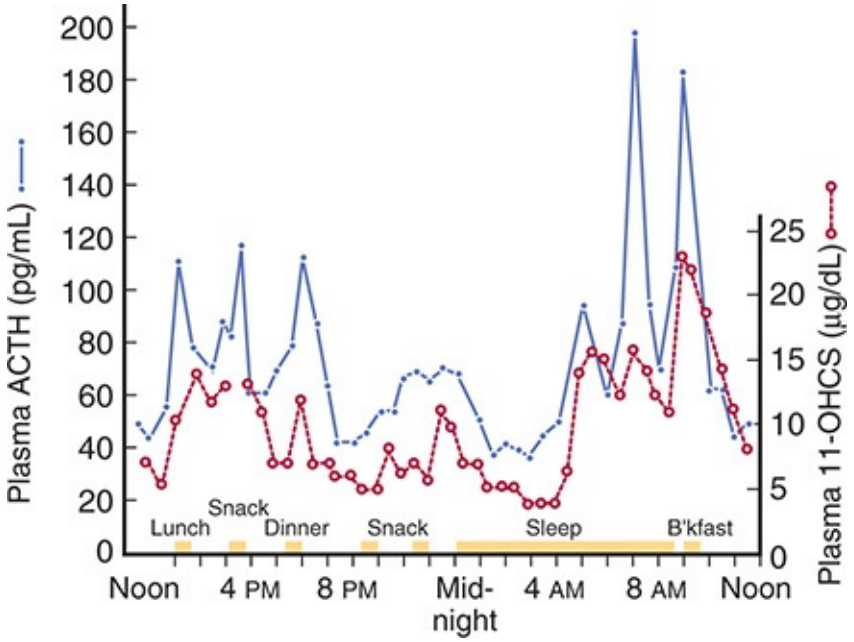


FIGURE 19–15 Fluctuations in plasma ACTH and glucocorticoids throughout the day in a normal girl (age 16). The ACTH was measured by immunoassay and the glucocorticoids as 11-oxysteroids (11-OHCS). Note the greater ACTH and glucocorticoid rises in the morning, before awakening. ACTH, adrenocorticotropic hormone. (Reproduced, with permission, from Krieger DT, et al: Characterization of the normal temporal pattern of plasma corticosteroid levels. J Clin Endocrinol Metab 1971; Feb; 32(2):266–284.)

THE RESPONSE TO STRESS

The morning plasma ACTH concentration in a healthy resting human is about 25 pg/mL (5.5 pmol/L). ACTH and cortisol values in various abnormal conditions are summarized in **Figure 19–16**. During severe stress, the amount of ACTH secreted exceeds the amount necessary to produce maximal glucocorticoid output. However, prolonged exposure to ACTH in conditions such as the ectopic ACTH syndrome increases the adrenal maximum.

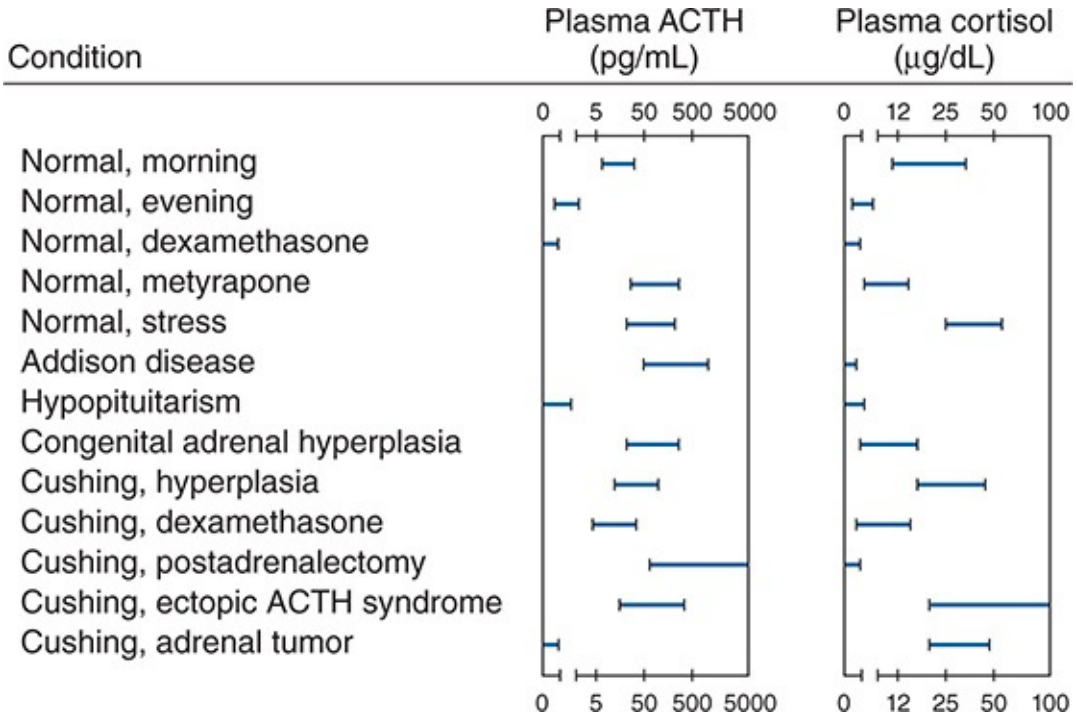


FIGURE 19–16 Plasma concentrations of ACTH and cortisol in various clinical states. ACTH, adrenocorticotropic hormone. (Reproduced with permission from Williams RH [editor]: *Textbook of Endocrinology*, 5th ed. Saunders, 1974.)

Increases in ACTH secretion to meet emergency situations are mediated almost exclusively through the hypothalamus via release of CRH. This polypeptide is produced by neurons in the paraventricular nuclei. It is secreted in the median eminence and transported in the portal-hypophysial vessels to the anterior pituitary, where it stimulates ACTH secretion (see Chapter 18). If the median eminence is destroyed, increased secretion in response to many different stresses is blocked. Afferent nerve pathways from many parts of the brain converge on the paraventricular nuclei. Fibers from the amygdaloid nuclei mediate responses to emotional stresses, and fear, anxiety, and apprehension cause marked increases in ACTH secretion. Input from the suprachiasmatic nuclei provides the drive for the diurnal rhythm. Impulses ascending to the hypothalamus via the nociceptive pathways and the reticular formation trigger increased ACTH secretion in response to injury (Figure 19–16). The baroreceptors exert an inhibitory input via the nucleus of the tractus solitarius.

GLUCOCORTICOID FEEDBACK

Free glucocorticoids inhibit ACTH secretion, and the degree of pituitary inhibition is proportional to the circulating glucocorticoid level. The inhibitory effect is exerted at both the pituitary and the hypothalamic levels. The inhibition is due primarily to an action on DNA, and maximal inhibition takes several hours to develop, although more rapid "fast feedback" also occurs. The ACTH-inhibiting activity of the various steroids parallels their glucocorticoid potency. A drop in resting corticoid levels stimulates ACTH secretion, and in chronic adrenal insufficiency the rate of ACTH synthesis and secretion is markedly increased.

Thus, the rate of ACTH secretion is determined by two opposing forces: the sum of the neural and possibly other stimuli converging through the hypothalamus to increase ACTH secretion, and the magnitude of the braking action of glucocorticoids on ACTH secretion, which is proportional to their level in the circulating blood **(Figure 19–17)**.

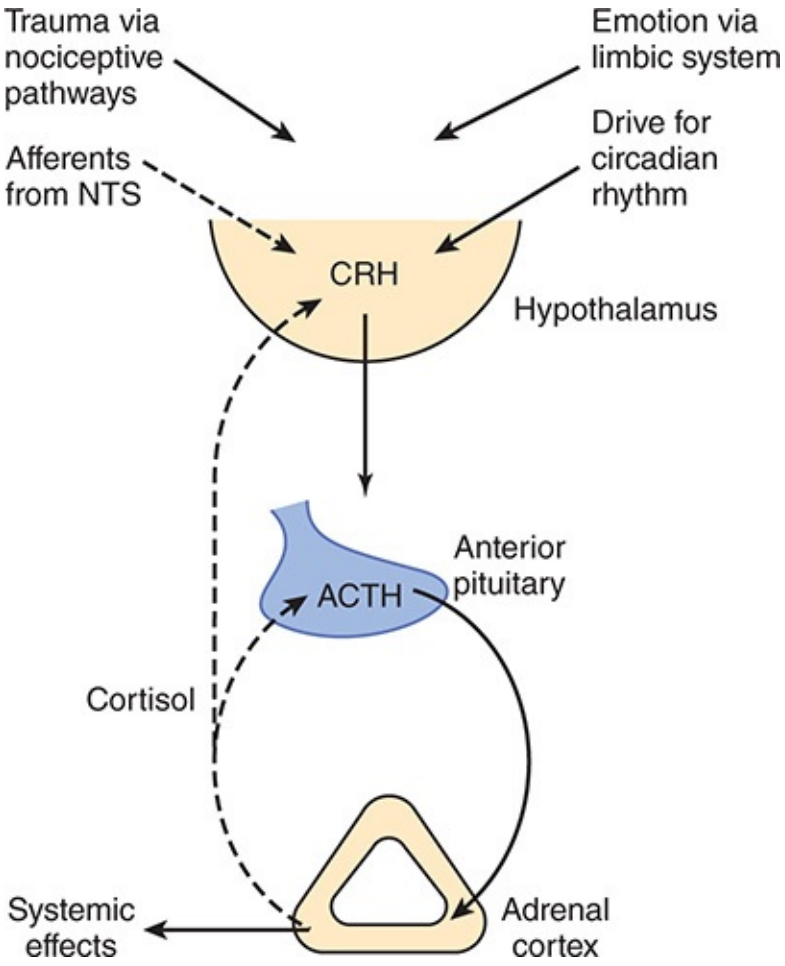


FIGURE 19–17 Feedback control of the secretion of cortisol and other

glucocorticoids via the hypothalamic–pituitary–adrenal axis. The dashed arrows indicate inhibitory effects and the solid arrows indicate stimulating effects. ACTH, adrenocorticotropic hormone; CRH, corticotropin-releasing hormone; NTS, nucleus tractus solitarius.

The dangers involved when prolonged treatment with anti-inflammatory doses of glucocorticoids is stopped deserve emphasis. Not only is the adrenal atrophic and unresponsive after such treatment, but even if its responsiveness is restored by injecting ACTH, the pituitary may be unable to secrete normal amounts of ACTH for as long as a month. The cause of the deficiency is presumably diminished ACTH synthesis. Thereafter, ACTH secretion slowly increases to supranormal levels. These in turn stimulate the adrenal, and glucocorticoid output rises, with feedback inhibition gradually reducing the elevated ACTH levels to normal **(Figure 19–18)**. The complications of sudden cessation of steroid therapy can usually be avoided by slowly decreasing the steroid dose over a long period of time.

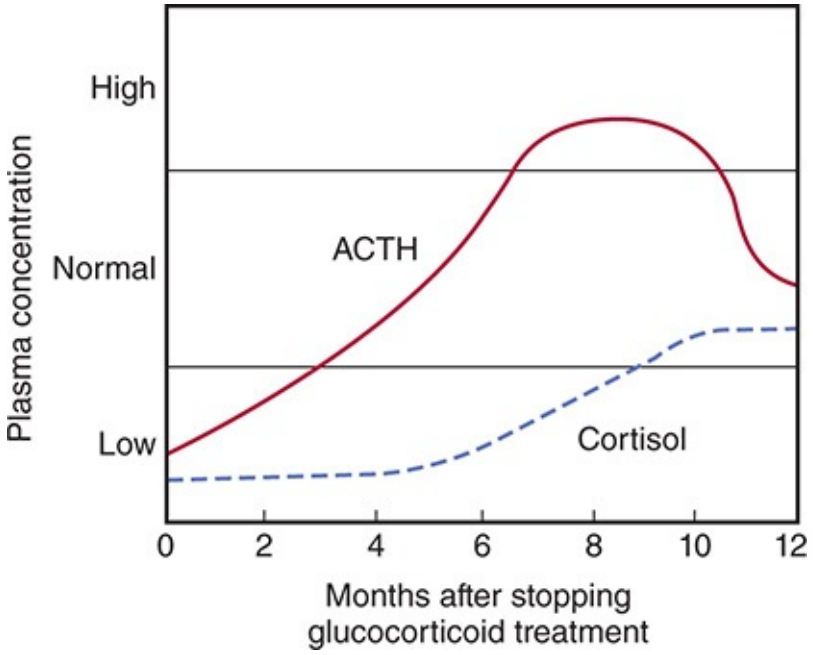


FIGURE 19–18 Pattern of plasma ACTH and cortisol values in patients recovering from prior long-term daily treatment with large doses of glucocorticoids. ACTH, adrenocorticotropic hormone. (Used with permission of R Ney.)

EFFECTS OF MINERALOCORTICOIDS

ACTIONS

Aldosterone and other steroids with mineralocorticoid activity increase the reabsorption of Na^+ from the urine, sweat, saliva, and the contents of the colon. Thus, mineralocorticoids cause retention of Na^+ in the ECF. This expands ECF volume. In the kidneys, they act primarily on the **principal cells (P cells)** of the collecting ducts (see Chapter 37). Under the influence of aldosterone, increased amounts of Na^+ are in effect exchanged for K^+ and H^+ in the renal tubules, producing a K^+ diuresis **(Figure 19–19)** and an increase in urine acidity.

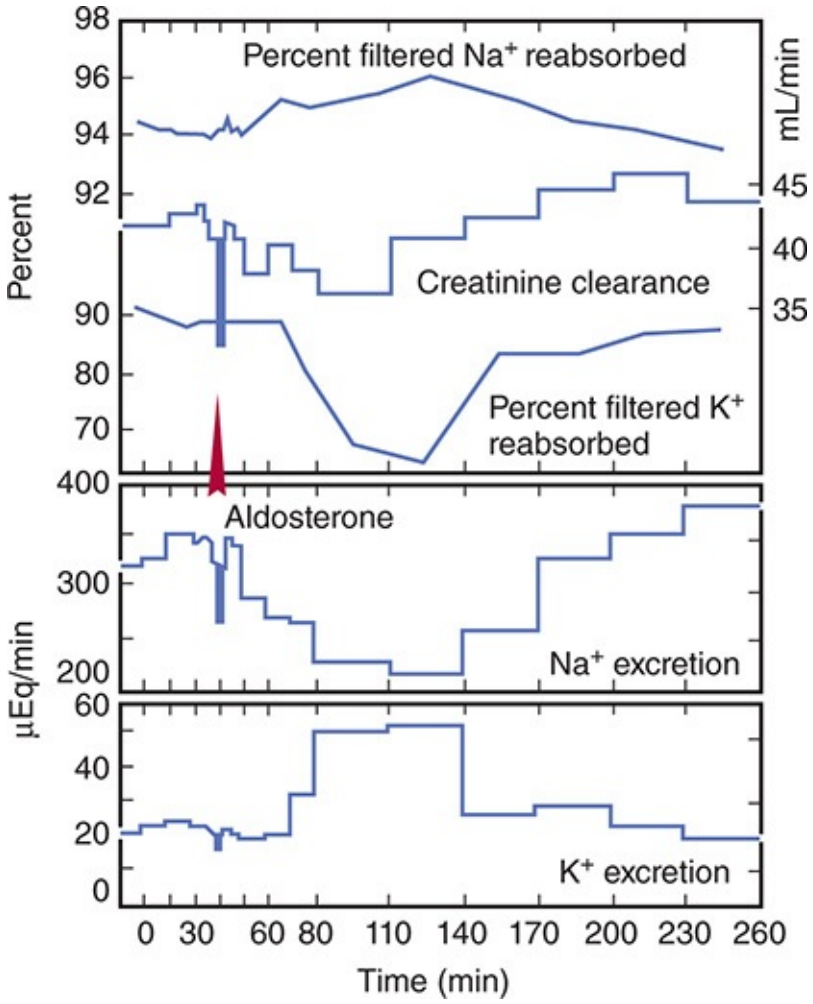


FIGURE 19–19 Effect of aldosterone (5 μg as a single dose injected into the aorta) on electrolyte excretion in an adrenalectomized dog. The scale for creatinine clearance is on the right.

MECHANISM OF ACTION

Like many other steroids, aldosterone binds to a cytoplasmic receptor, and the receptor–hormone complex moves to the nucleus where it alters the transcription of mRNAs. This in turn increases the production of proteins that alter cell function. The aldosterone-stimulated proteins have two effects—a rapid effect, to increase the activity of epithelial sodium channels (ENaCs) by increasing the insertion of these channels into the cell membrane from a cytoplasmic pool; and a slower effect to increase the synthesis of ENaCs. Among the genes activated by aldosterone is the gene for **serum- and glucocorticoid-regulated kinase (sgk),** a serine-threonine protein kinase. The gene for sgk is an early response gene, and sgk increases ENaC activity. Aldosterone also increases the mRNAs for the three subunits that make up ENaCs. The fact that sgk is activated by glucocorticoids as well as aldosterone is not a problem because glucocorticoids are inactivated at mineralocorticoid receptor sites. However, aldosterone activates the genes for other proteins in addition to sgk and ENaCs and inhibits others. Therefore, the exact mechanism by which aldosterone-induced proteins increase Na^+ reabsorption is still unsettled.

Evidence is accumulating that aldosterone also binds to the cell membrane and by a rapid, nongenomic action increases the activity of membrane Na^+–K^+ exchangers. This produces an increase in intracellular Na^+, and the second messenger involved is probably IP_3. In any case, the principal effect of aldosterone on Na^+ transport takes 10–30 min to develop and peaks even later (Figure 19-19), indicating that it depends on the synthesis of new proteins by a genomic mechanism.

RELATION OF MINERALOCORTICOID TO GLUCOCORTICOID RECEPTORS

It is intriguing that in vitro, the mineralocorticoid receptor (MR) has an appreciably higher affinity for glucocorticoids than the glucocorticoid receptor does, and glucocorticoids are present in large amounts in vivo. This raises the question of why glucocorticoids do not bind to the mineralocorticoid receptors in the kidneys and other locations and produce mineralocorticoid effects. At least in part, the answer is that the kidneys and other mineralocorticoid-sensitive tissues also contain the enzyme **11β-hydroxysteroid dehydrogenase type 2.** This enzyme leaves aldosterone untouched, but it converts cortisol to cortisone

(Figure 19–11) and corticosterone to its 11-oxy derivative. These 11-oxy derivatives do not bind to the receptor **(Clinical Box 19–3)**.

CLINICAL BOX 19–3

Apparent Mineralocorticoid Excess

If 11β-hydroxysteroid dehydrogenase type 2 is inhibited or absent, cortisol has marked mineralocorticoid effects. The resulting syndrome is called **apparent mineralocorticoid excess (AME).** Patients with this condition have the clinical picture of hyperaldosteronism because cortisol is acting on their mineralocorticoid receptors, and their plasma aldosterone level as well as their plasma renin activity is low. The condition can be due to congenital absence of the enzyme.

THERAPEUTIC HIGHLIGHTS

Prolonged ingestion of licorice can also cause an increase in blood pressure. Outside of the United States, licorice contains glycyrrhetinic acid, which inhibits 11β-hydroxysteroid dehydrogenase type 2. Individuals who eat large amounts of licorice have an increase in MR-activated sodium absorption via the epithelial sodium channel ENaC in the renal collecting duct, and blood pressure can rise.

OTHER STEROIDS THAT AFFECT NA^+ EXCRETION

Aldosterone is the principal mineralocorticoid secreted by the adrenals, although corticosterone is secreted in sufficient amounts to exert a minor mineralocorticoid effect (Tables 19–1 and 19–2). Deoxycorticosterone, which is secreted in appreciable amounts only in abnormal situations, has about 3% of the activity of aldosterone. Large amounts of progesterone and some other steroids cause natriuresis, but there is little evidence that they play any normal role in the control of Na^+ excretion.

EFFECT OF ADRENALECTOMY

In adrenal insufficiency, Na^+ is lost in the urine; K^+ is retained, and the plasma K^+ rises. When adrenal insufficiency develops rapidly, the amount of Na^+ lost from the ECF exceeds the amount excreted in the urine, indicating that Na^+ also must be entering cells. When the posterior pituitary is intact, salt loss exceeds water loss, and the plasma Na^+ falls **(Table 19–5)**. However, the plasma volume is also reduced, resulting in hypotension, circulatory insufficiency and, eventually, fatal shock. These changes can be prevented to a degree by increasing dietary NaCl intake. Rats survive indefinitely on extra salt alone, but in dogs and most humans, the amount of supplementary salt needed is so large that it is almost impossible to prevent eventual collapse and death unless mineralocorticoid treatment is also instituted **(see Clinical Box 19–4)**.

TABLE 19–5 Typical plasma electrolyte levels in normal humans and patients with adrenocortical diseases.

	Plasma Electolytes (mEq/L)			
State	Na^+	K^+	Cl^-	HCO_3^-
Normal	142	4.5	105	25
Adrenal insufficiency	120	6.7	85	25
Primary hyperaldosteronism	145	2.4	96	41

REGULATION OF ALDOSTERONE SECRETION

STIMULI

The principal conditions that increase aldosterone secretion are summarized in **Table 19–6**. Some of them also increase glucocorticoid secretion; others selectively affect the output of aldosterone. The primary regulatory factors involved are ACTH from the pituitary, renin from the kidney via angiotensin II, and a direct stimulatory effect on the adrenal cortex of a rise in plasma K^+ concentration.

TABLE 19–6 Conditions that increase aldosterone secretion.

Glucocorticoid secretion also increased

Surgery
Anxiety
Physical trauma
Hemorrhage

Glucocorticoid secretion unaffected

High potassium intake
Low sodium intake
Constriction of inferior vena cava in thorax
Standing
Secondary hyperaldosteronism (in some cases of heart failure, cirrhosis, and nephrosis)

CLINICAL BOX 19–4

Secondary Effects of Excess Mineralocorticoids

A prominent feature of prolonged mineralocorticoid excess (Table 20–5) is K^+ depletion due to prolonged K^+ diuresis. H^+ is also lost in the urine. Na^+ is retained initially, but the plasma Na^+ is elevated only slightly if at all, because water is retained with the osmotically active sodium ions. Consequently, ECF volume is expanded and the blood pressure rises. When the ECF expansion passes a certain point, Na^+ excretion is usually increased in spite of the continued action of mineralocorticoids on the renal tubules. This **escape phenomenon (Figure 19–20)** is probably due to increased secretion of atrial natriuretic peptide (ANP) (see Chapter 38). Because of increased excretion of Na^+ when the ECF volume is expanded, mineralocorticoids do not produce edema in normal individuals and patients with hyperaldosteronism. However, escape may not occur in certain disease states, and in these situations, continued expansion of ECF volume leads to edema (see Chapters 37 and 38).

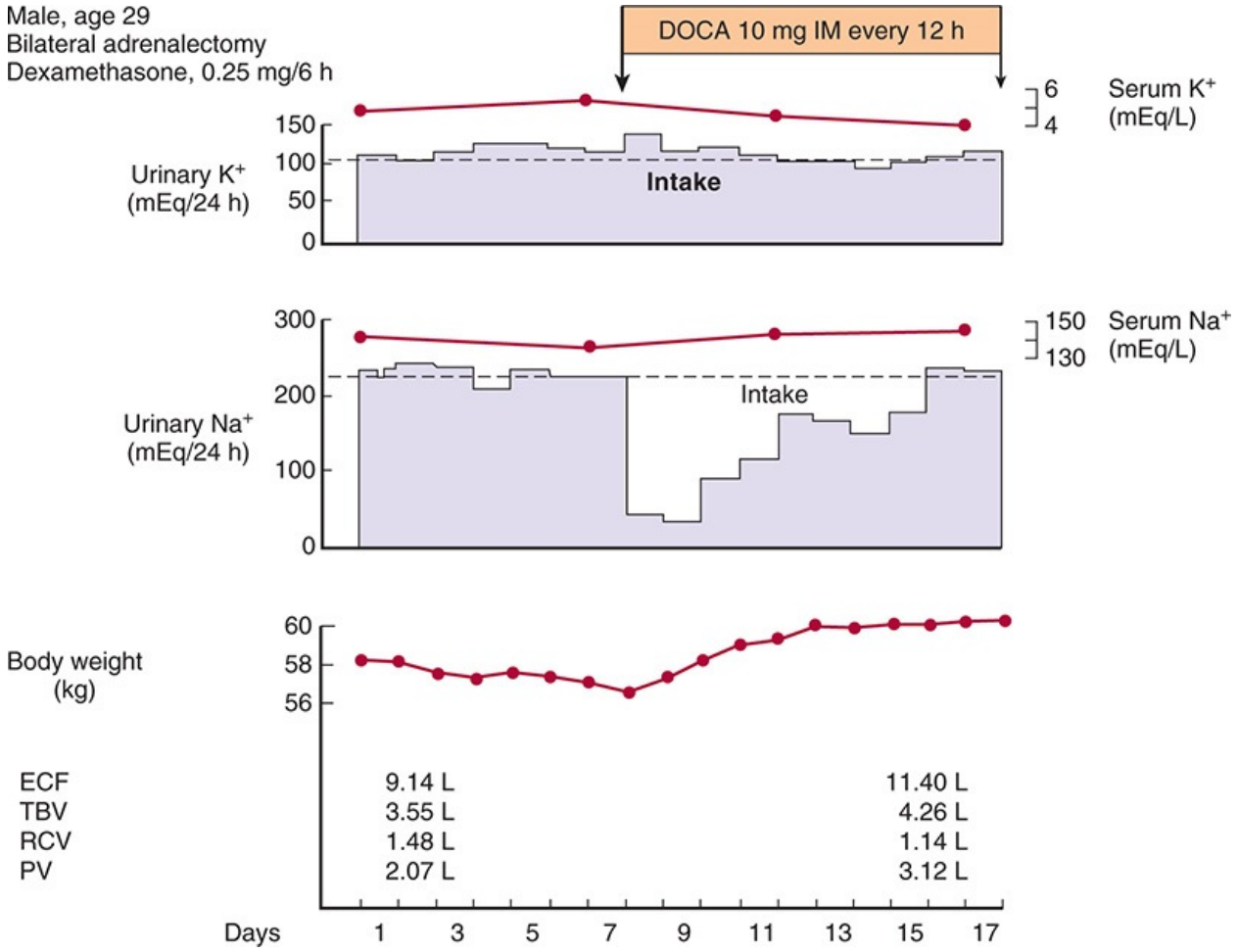


FIGURE 19–20 “Escape” from the sodium-retaining effect of desoxycorticosterone acetate (DOCA) in an adrenalectomized patient. ECF, extracellular fluid volume; PV, plasma volume; RCV, red cell volume; TBV, total blood volume. (Used with permission of EG Biglieri.)

EFFECT OF ACTH

When first administered, ACTH stimulates the output of aldosterone as well as that of glucocorticoids and sex hormones. Although the amount of ACTH required to increase aldosterone output is somewhat greater than the amount that stimulates maximal glucocorticoid secretion **(Figure 19–21)**, it is well within the range of endogenous ACTH secretion. The effect is transient, and even if ACTH secretion remains elevated, aldosterone output declines in 1 or 2 days. On the other hand, the output of the mineralocorticoid deoxycorticosterone remains elevated. The decline in aldosterone output is partly due to decreased renin

secretion secondary to hypervolemia, but it is possible that some other factor also decreases the conversion of corticosterone to aldosterone. After hypophysectomy, the basal rate of aldosterone secretion is normal. The increase normally produced by surgical and other stresses is absent, but the increase produced by dietary salt restriction is unaffected for some time. Later on, atrophy of the zona glomerulosa complicates the picture in long-standing hypopituitarism, and this may lead to salt loss and hypoaldosteronism.

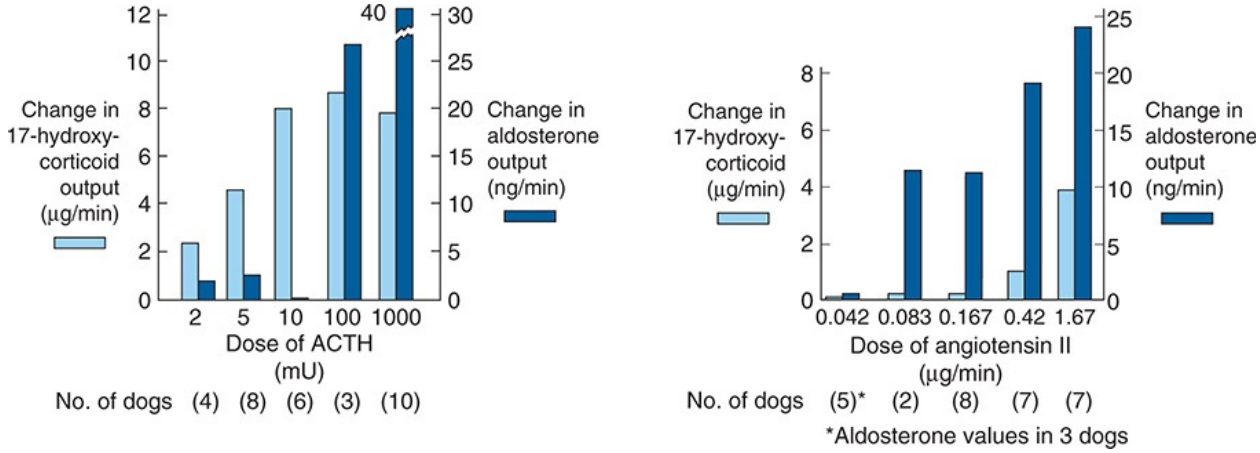


FIGURE 19–21 Changes in adrenal venous output of steroids produced by ACTH in nephrectomized hypophysectomized dogs. ACTH, adrenocorticotropic hormone.

Normally, glucocorticoid treatment does not suppress aldosterone secretion. However, an interesting recently described syndrome is **glucocorticoid-remediable aldosteronism (GRA).** This is an autosomal dominant disorder in which the increase in aldosterone secretion produced by ACTH is no longer transient. The hypersecretion of aldosterone and the accompanying hypertension are remedied when ACTH secretion is suppressed by administering glucocorticoids. The genes encoding aldosterone synthase and 11β-hydroxylase are 95% identical and are close together on chromosome 8. In individuals with GRA, there is unequal crossing over so that the 5′regulatory region of the 11β-hydroxylase gene is fused to the coding region of the aldosterone synthase gene. The product of this hybrid gene is an ACTH-sensitive aldosterone synthase.

EFFECTS OF ANGIOTENSIN II & RENIN

The octapeptide angiotensin II is formed in the body from angiotensin I, which is

liberated by the action of renin on circulating angiotensinogen (see Chapter 38). Injections of angiotensin II stimulate adrenocortical secretion and, in small doses, affect primarily the secretion of aldosterone. The sites of action of angiotensin II are both early and late in the steroid biosynthetic pathway. The early action is on the conversion of cholesterol to pregnenolone, and the late action is on the conversion of corticosterone to aldosterone (Figure 19–8). Angiotensin II does not increase the secretion of deoxycorticosterone, which is controlled by ACTH.

Renin is secreted from the juxtaglomerular cells that surround the renal afferent arterioles as they enter the glomeruli (see Chapter 38). Aldosterone secretion is regulated via the renin–angiotensin system in a feedback manner **(Figure 19–22)**. A drop in ECF volume or intra-arterial vascular volume leads to a reflex increase in renal nerve discharge and decreases renal arterial pressure. Both changes increase renin secretion, and the angiotensin II formed by the action of renin increases the rate of secretion of aldosterone. The aldosterone causes Na^+ and, secondarily, water retention, expanding ECF volume, and shutting off the stimulus that initiated increased renin secretion.

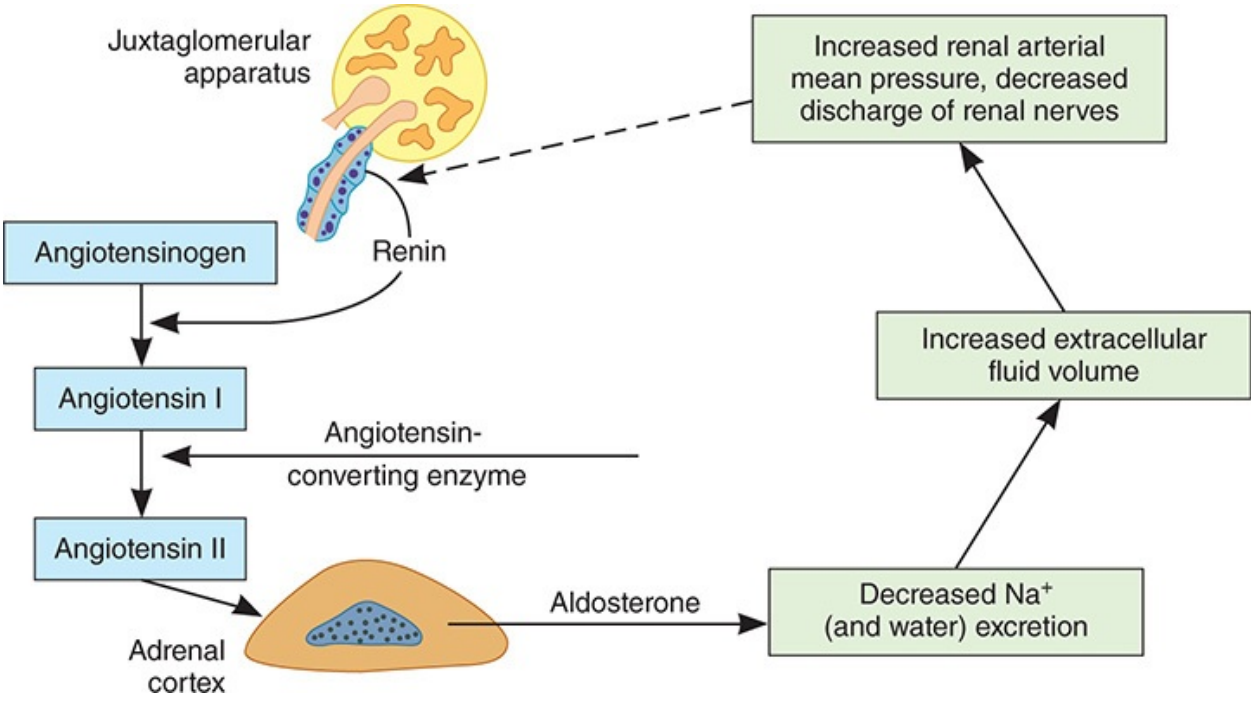


FIGURE 19–22 Feedback mechanism regulating aldosterone secretion. The dashed arrow indicates inhibition.

Hemorrhage stimulates ACTH and renin secretion. Like hemorrhage,

standing and constriction of the thoracic inferior vena cava decrease intrarenal arterial pressure. Dietary sodium restriction also increases aldosterone secretion via the renin–angiotensin system **(Figure 19–23)**. Such restriction reduces ECF volume, but aldosterone and renin secretion are increased before any consistent decrease in blood pressure takes place. Consequently, the initial increase in renin secretion produced by dietary sodium restriction is probably due to a reflex increase in the activity of the renal nerves. The increase in circulating angiotensin II produced by salt depletion upregulates the angiotensin II receptors in the adrenal cortex and hence increases the response to angiotensin II, whereas it down-regulates the angiotensin II receptors in the blood vessels.

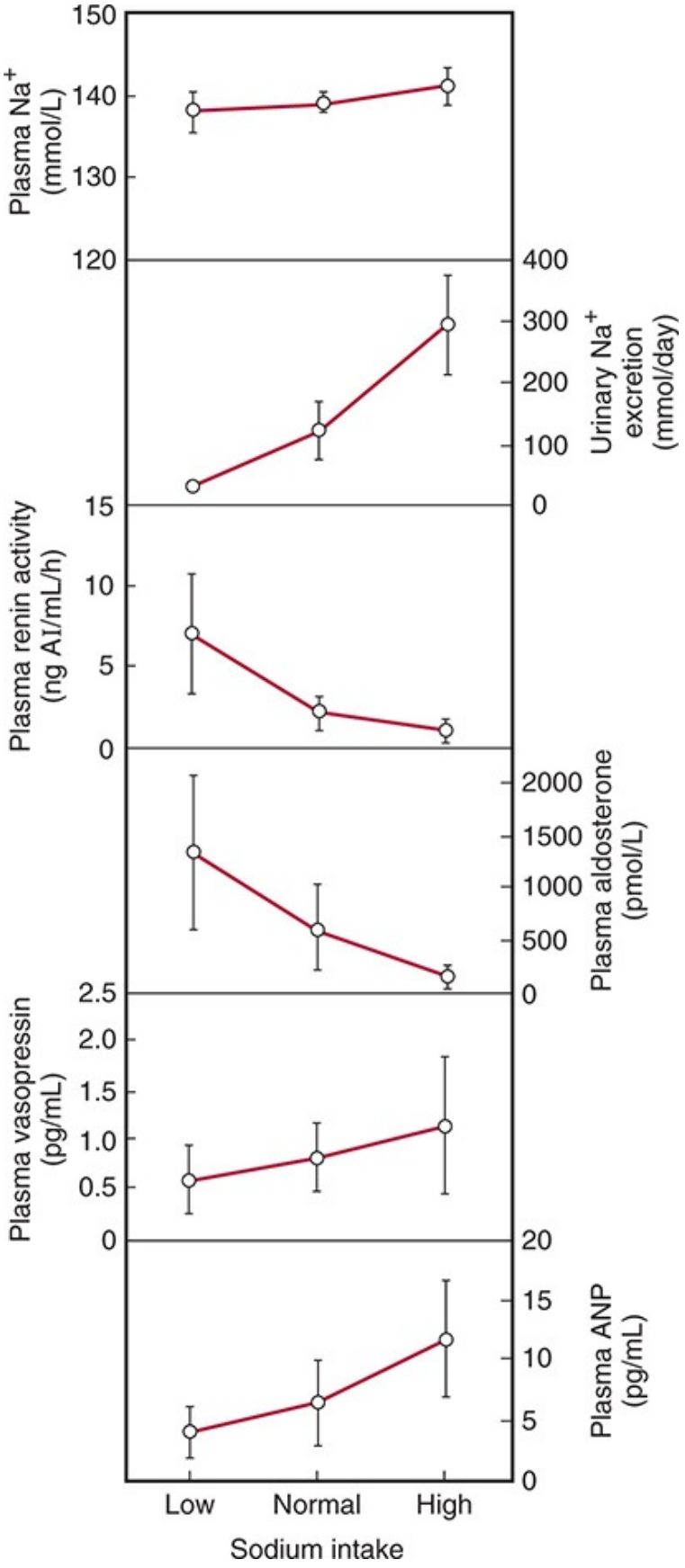

Plasma Na^+ (mmol/L)
150
140
130
120
Urinary Na^+ excretion (mmol/day)
400
300
200
100
0
Plasma renin activity (ng AI/mL/h)
15
10
5
0
Plasma aldosterone (pmol/L)
2000
1500
1000
500
0
Plasma vasopressin (pg/mL)
2.5
2.0
1.5
1.0
0.5
0
Plasma ANP (pg/mL)
20
15
10
5
0
Low
Normal
High
Sodium intake

FIGURE 19–23 Effect of low-, normal-, and high-sodium diets on sodium metabolism and plasma renin activity, aldosterone, vasopressin, and ANP in normal humans. (Data from Sagnella GA, et al: Plasma atrial natriuretic peptide: Its relationship to changes in sodium in-take, plasma renin activity, and aldosterone in man. Clin Sci 1987; 72:25.)

ELECTROLYTES & OTHER FACTORS

An acute decline in plasma Na^+ of about 20 mEq/L stimulates aldosterone secretion, but changes of this magnitude are rare. However, the plasma K^+ level need increase only 1 mEq/L to stimulate aldosterone secretion, and transient increases of this magnitude may occur after a meal, particularly if it is rich in K^+. Like angiotensin II, K^+ stimulates the conversion of cholesterol to pregnenolone and the conversion of deoxycorticosterone to aldosterone. It appears to act by depolarizing the cell, which opens voltage-gated Ca^{2+} channels, increasing intracellular Ca^{2+}. The sensitivity of the zona glomerulosa to angiotensin II and consequently to a low-sodium diet is decreased by a low-potassium diet.

In normal individuals, plasma aldosterone concentrations increase during the portion of the day that the individual is carrying on activities in the upright position. This increase is due to a decrease in the rate of removal of aldosterone from the circulation by the liver and an increase in aldosterone secretion due to a postural increase in renin secretion. Individuals who are confined to bed show a circadian rhythm of aldosterone and renin secretion, with the highest values in the early morning before awakening. Atrial natriuretic peptide (ANP) inhibits renin secretion and decreases the responsiveness of the zona glomerulosa to angiotensin II (see Chapter 38). The mechanisms by which ACTH, angiotensin II, and K^+ stimulate aldosterone secretion are summarized in **Table 19–7**.

TABLE 19–7 Second messengers involved in the regulation of aldosterone secretion.

Secretagogue	Intracellular Mediator
ACTH	Cyclic AMP, protein kinase A
Angiotensin II	Diacylglycerol, protein kinase C
K^+	Ca^{2+} via voltage-gated Ca^{2+} channels

ROLE OF MINERALOCORTICOIDS IN THE REGULATION OF SALT BALANCE

Variations in aldosterone secretion is only one of many factors affecting Na^+ excretion. Other major factors include the glomerular filtration rate, ANP, the presence or absence of osmotic diuresis, and changes in tubular reabsorption of Na^+ independent of aldosterone. It takes some time for aldosterone to act. When one rises from the supine to the standing position, aldosterone secretion increases and Na^+ is retained from the urine. However, the decrease in Na^+ excretion develops too rapidly to be explained solely by increased aldosterone secretion. The primary function of the aldosterone-secreting mechanism is the defense of intravascular volume, but it is only one of the homeostatic mechanisms involved in this regulation.

SUMMARY OF THE EFFECTS OF ADRENOCORTICAL HYPER- & HYPOFUNCTION IN HUMANS

Recapitulating the manifestations of excess and deficiency of the adrenocortical hormones in humans is a convenient way to summarize the multiple and complex actions of these steroids. A characteristic clinical syndrome is associated with excess secretion of each of the types of hormones.

Excess androgen secretion causes masculinization **(adrenogenital syndrome)** and precocious pseudopuberty or female pseudohermaphroditism.

Excess glucocorticoid secretion produces a moon-faced, plethoric appearance, with trunk obesity, purple abdominal striae, hypertension, osteoporosis, protein depletion, mental abnormalities and, frequently, diabetes mellitus **(Cushing syndrome).** Excess mineralocorticoid secretion leads to K^+ depletion and Na^+ retention, usually without edema but with weakness, hypertension, tetany, polyuria, and hypokalemic alkalosis **(hyperaldosteronism).** This condition may be due to primary adrenal disease **(primary hyperaldosteronism; Conn syndrome)** such as an adenoma of the zona glomerulosa, unilateral or bilateral adrenal hyperplasia, adrenal carcinoma, or by GRA. In patients with primary hyperaldosteronism, renin secretion is depressed. **Secondary hyperaldosteronism** with high plasma renin activity is caused by cirrhosis, heart failure, and nephrosis. Increased renin secretion is also found in individuals with

the salt-losing form of the adrenogenital syndrome (see above), because their ECF volume is low. In patients with elevated renin secretion due to renal artery constriction, aldosterone secretion is increased; in those in whom renin secretion is not elevated, aldosterone secretion is normal. The relationship of aldosterone to hypertension is discussed in Chapter 32.

Primary adrenal insufficiency due to disease processes that destroy the adrenal cortex is called **Addison disease.** The condition used to be a relatively common complication of tuberculosis, but now it is usually due to autoimmune inflammation of the adrenal. Patients lose weight, are tired, and become chronically hypotensive. They have small hearts, probably because the hypotension decreases the work of the heart. Eventually, severe hypotension and shock **(addisonian crisis)** develop. This is due not only to mineralocorticoid deficiency but to glucocorticoid deficiency as well. Fasting causes fatal hypoglycemia, and any stress causes collapse. Water is retained, and there is always the danger of water intoxication. Circulating ACTH levels are elevated. The diffuse tanning of the skin and the spotty pigmentation characteristic of chronic glucocorticoid deficiency are due, at least in part, to the melanocyte-stimulating hormone (MSH) activity of the ACTH in the blood. Pigmentation of skin creases on the hands and the gums are common. Minor menstrual abnormalities occur in women, but the deficiency of adrenal sex hormones usually has little effect in the presence of normal testes or ovaries.

Secondary adrenal insufficiency is caused by pituitary diseases that decrease ACTH secretion, and **tertiary adrenal insufficiency** is caused by hypothalamic disorders disrupting CRH secretion. Both are usually milder than primary adrenal insufficiency because electrolyte metabolism is affected to a lesser degree. In addition, there is no pigmentation because in both of these conditions, plasma ACTH is low, not high.

Cases of isolated aldosterone deficiency have also been reported in patients with renal disease and a low circulating renin level **(hyporeninemic hypoaldosteronism).** In addition, **pseudohypoaldosteronism** is produced when there is resistance to the action of aldosterone. Patients with these syndromes have marked hyperkalemia, salt wasting, and hypotension, and they may develop metabolic acidosis.

CHAPTER SUMMARY

- The adrenal gland consists of the adrenal medulla that secretes dopamine and the catecholamines epinephrine and norepinephrine, and the adrenal cortex

that secretes steroid hormones.

- Norepinephrine and epinephrine act on two classes of receptors, α- and β-adrenergic receptors, and exert metabolic effects that include glycogenolysis in liver and skeletal muscle, mobilization of FFA, increased plasma lactate, and stimulation of the metabolic rate.
- The hormones of the adrenal cortex are derivatives of cholesterol and include the mineralocorticoid aldosterone, the glucocorticoids cortisol and corticosterone, and the androgens dehydroepiandrosterone (DHEA) and androstenedione.
- Androgens are the hormones that exert masculinizing effects, and they promote protein anabolism and growth. The adrenal androgen androstenedione is converted to testosterone and to estrogens (aromatized) in fat and other peripheral tissues. This is an important source of estrogens in men and postmenopausal women.
- The mineralocorticoid aldosterone has effects on Na^+ and K^+ excretion and glucocorticoids affect glucose and protein metabolism.
- Glucocorticoid secretion is dependent on ACTH from the anterior pituitary and is increased by stress. Angiotensin II increases the secretion of aldosterone.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. Which of the following is produced only by *large amounts* of glucocorticoids?

   A. Normal responsiveness of fat depots to norepinephrine
   B. Maintenance of normal vascular reactivity
   C. Increased excretion of a water load
   D. Inhibition of the inflammatory response
   E. Inhibition of ACTH secretion

2. Which of the following are *incorrectly* paired?

   A. Gluconeogenesis : Cortisol
   B. Free fatty acid mobilization : Dehydroepiandrosterone
   C. Muscle glycogenolysis : Epinephrine
   D. Kaliuresis : Aldosterone

E. Hepatic glycogenesis : Insulin

3. Which of the following hormones has the shortest plasma half-life?

A. Corticosterone

B. Renin

C. Dehydroepiandrosterone

D. Aldosterone

E. Norepinephrine

4. Mole for mole, which of the following has the greatest effect on Na^+ excretion?

A. Progesterone

B. Cortisol

C. Vasopressin

D. Aldosterone

E. Dehydroepiandrosterone

5. Mole for mole, which of the following has the greatest effect on plasma osmolality?

A. Progesterone

B. Cortisol

C. Vasopressin

D. Aldosterone

E. Dehydroepiandrosterone

6. The secretion of which of the following would be *least* affected by a decrease in extracellular fluid volume?

A. CRH

B. Arginine vasopressin

C. Dehydroepiandrosterone

D. Estrogens

E. Aldosterone

7. A young man presents with a blood pressure of 175/110 mm Hg. He is found to have a high circulating aldosterone but a low circulating cortisol. Glucocorticoid treatment lowers his circulating aldosterone and lowers his blood pressure to 140/85 mm Hg. He probably has an abnormal

A. 17α-hydroxylase.

B. 21β-hydroxylase.

C. 3β-hydroxysteroid dehydrogenase.

D. aldosterone synthase.

E. cholesterol desmolase.

8. A 32-year-old woman presents with a blood pressure of 155/96 mm Hg. In response to questioning, she admits that she loves licorice and eats some at least three times a week. She probably has a low level of

A. type 2 11β-hydroxysteroid dehydrogenase activity.

B. ACTH.

C. 11β-hydroxylase activity.

D. glucuronyl transferase.

E. norepinephrine.

9. In its action in cells, aldosterone

A. increases transport of ENaCs from the cytoplasm to the cell membrane.

B. does not act on the cell membrane.

C. binds to a receptor excluded from the nucleus.

D. may activate a heat shock protein.

E. also binds to glucocorticoid receptors.

CHAPTER 20

The Thyroid Gland

OBJECTIVES

After studying this chapter, you should be able to:

- Describe the development, anatomy, and histology of the thyroid gland, and how these relate to its function, as well as the changes in histology that accompany the activation of secretion with reabsorption of thyroglobulin from the colloid.
- Define the chemical nature of the thyroid hormones and how they are synthesized. Understand the critical role of iodine in the thyroid gland and how its transport is controlled, as well as how it is incorporated into the tyrosine residues of thyroglobulin at the interface between thyrocytes and colloid via the process of organification.
- Describe the relative roles of binding to albumin, transthyretin, and thyroxine-binding globulin in the transport of thyroid hormones, the availability of free hormones, and peripheral metabolism.
- Identify the role of the hypothalamus and pituitary in regulating thyroid function and the feedback loops that establish the level of secretion of thyroid hormones.
- Define the effects of the thyroid hormones in homeostasis, metabolism, and growth.

- Understand the basis of conditions where thyroid function is abnormal and how they can be treated.

INTRODUCTION

The thyroid gland is one of the larger endocrine glands of the body. The gland has two primary functions. The first is to secrete the thyroid hormones, which maintain the level of metabolism in the tissues that is optimal for their normal function. Thyroid hormones stimulate O_2 consumption by most of the cells in the body, help regulate lipid and carbohydrate metabolism, and thereby influence body mass and mentation. Consequences of thyroid gland dysfunction depend on the life stage at which they occur. The thyroid is not essential for life, but its absence or hypofunction during fetal and neonatal life results in severe mental retardation and dwarfism. In adults, hypothyroidism is accompanied by mental and physical slowing and poor resistance to cold. Conversely, excess thyroid secretion leads to body wasting, nervousness, tachycardia, tremor, and excess heat production. Thyroid function is controlled by the thyroid-stimulating hormone (TSH, thyrotropin) that is secreted by the anterior lobe of the pituitary. The secretion of this hormone is in turn increased by thyrotropin-releasing hormone (TRH) from the hypothalamus and is also subject to negative feedback control by high circulating levels of thyroid hormones acting on the anterior pituitary and the hypothalamus.

The second function of the thyroid gland is to secrete calcitonin, a hormone that regulates circulating levels of calcium. This function of the thyroid gland is discussed in Chapter 21 in the broader context of whole body calcium homeostasis.

DEVELOPMENT, STRUCTURE, AND CELL TYPES OF THE THYROID GLAND

GROSS ANATOMY

The thyroid is a butterfly-shaped gland that straddles the trachea in the front of the neck. It develops from an evagination of the floor of the pharynx, and a **thyroglossal duct** marking the path of the thyroid from the tongue to the neck

sometimes persists in the adult. The two lobes of the human thyroid are connected by a bridge of tissue, the **thyroid isthmus,** and there is sometimes a **pyramidal lobe** arising from the isthmus in front of the larynx **(Figure 20–1)**. The gland is well vascularized, and the thyroid has one of the highest rates of blood flow per gram of tissue of any organ in the body.

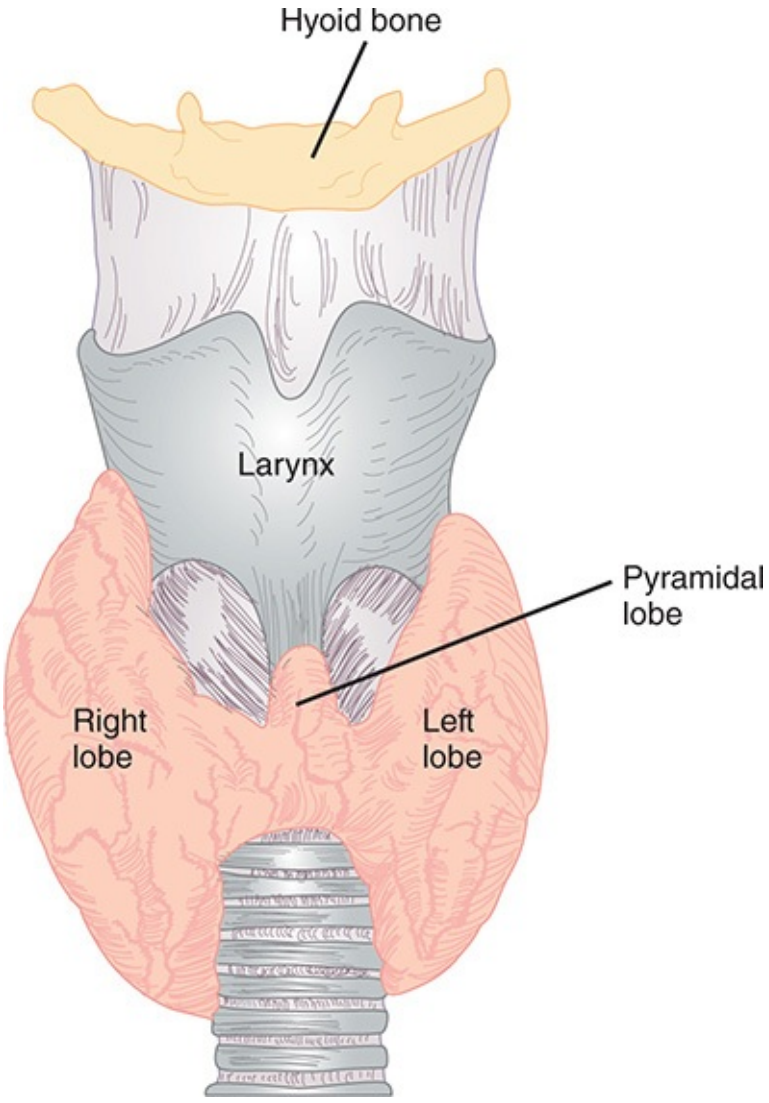


FIGURE 20–1 The human thyroid.

HISTOLOGY

The portion of the thyroid concerned with the production of thyroid hormone consists of multiple **follicles.** Each spherical follicle is surrounded by a single layer of polarized epithelial cells and filled with proteinaceous material called

colloid. Colloid consists predominantly of the glycoprotein, thyroglobulin. When the gland is inactive, the colloid is abundant, the follicles are large, and the cells lining them are flat. When the gland is active, the follicles are small, the cells are cuboid or columnar, and areas where the colloid is being actively reabsorbed into the thyrocytes are visible on histology as "reabsorption lacunae" **(Figure 20–2)**.

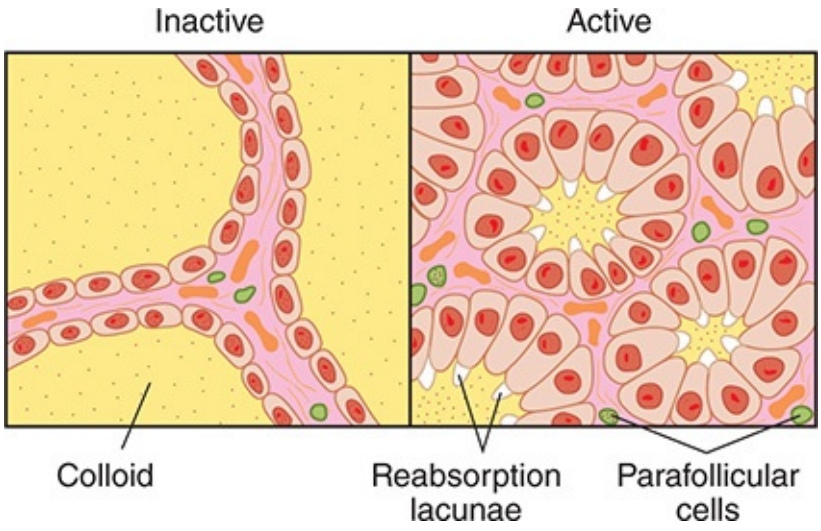


FIGURE 20–2 Thyroid histology. The appearance of the gland when it is inactive (left) and actively secreting (right) is shown. Note the small, punched-out "reabsorption lacunae" in the colloid next to the cells in the active gland.

Microvilli project into the colloid from the apexes of the thyrocytes and canaliculi extend into them. The endoplasmic reticulum is prominent, a feature common to most glandular cells, and secretory granules containing thyroglobulin are seen **(Figure 20–3)**. The individual thyroid cells rest on a basal lamina that separates them from the adjacent capillaries. The capillaries are fenestrated, like those of other endocrine glands (see Chapter 31).

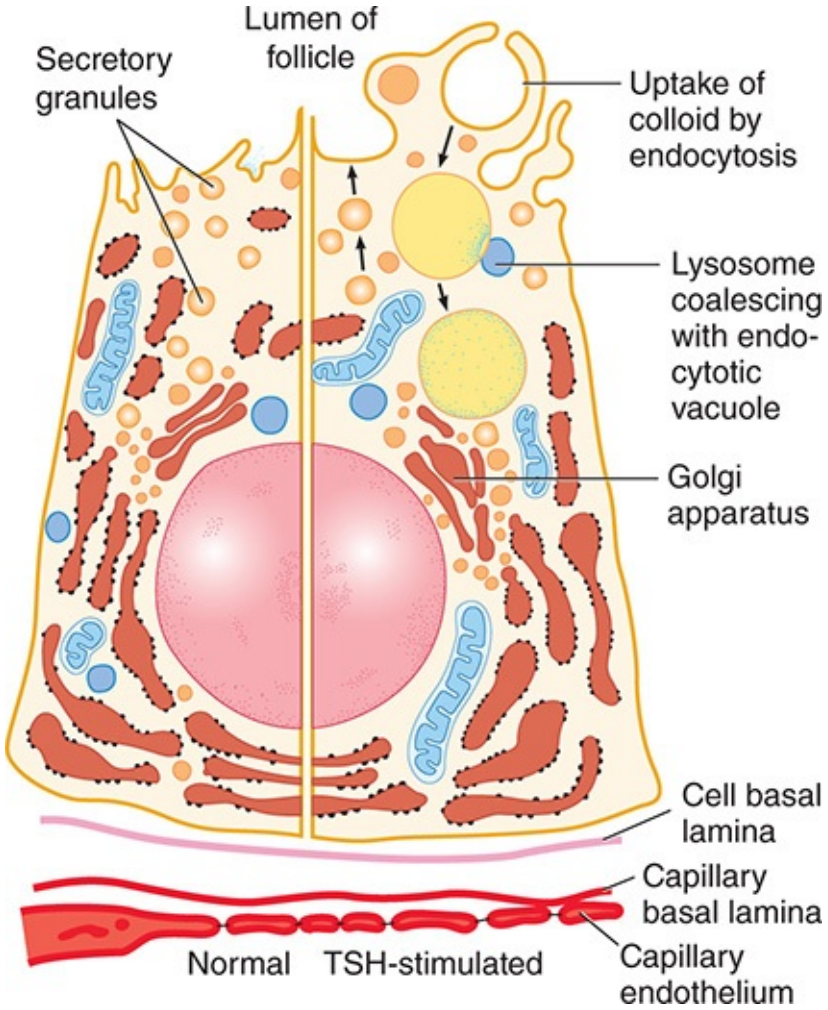


FIGURE 20–3 Thyrocyte. Left: Normal pattern. **Right:** After TSH stimulation. The arrows on the right show the secretion of thyroglobulin into the colloid. On the right, endocytosis of the colloid and merging of a colloid-containing vacuole with a lysosome are also shown. The cell rests on a capillary with gaps (fenestrations) in the endothelial wall.

FORMATION & SECRETION OF THYROID HORMONES

CHEMISTRY

The primary hormone secreted by the thyroid is **thyroxine (T_4),** along with much lesser amounts of **triiodothyronine (T_3).** T_3 has much greater biologic activity than T_4 and is also specifically generated at its site of action in

peripheral tissues by deiodination of T_4 (see below). Both hormones are iodine-containing amino acids **(Figure 20–4)**. Small amounts of reverse triiodothyronine (3,3′,5′-triiodothyronine, RT_3) and other compounds are also found in thyroid venous blood. Whether RT_3 is biologically active remains unclear.

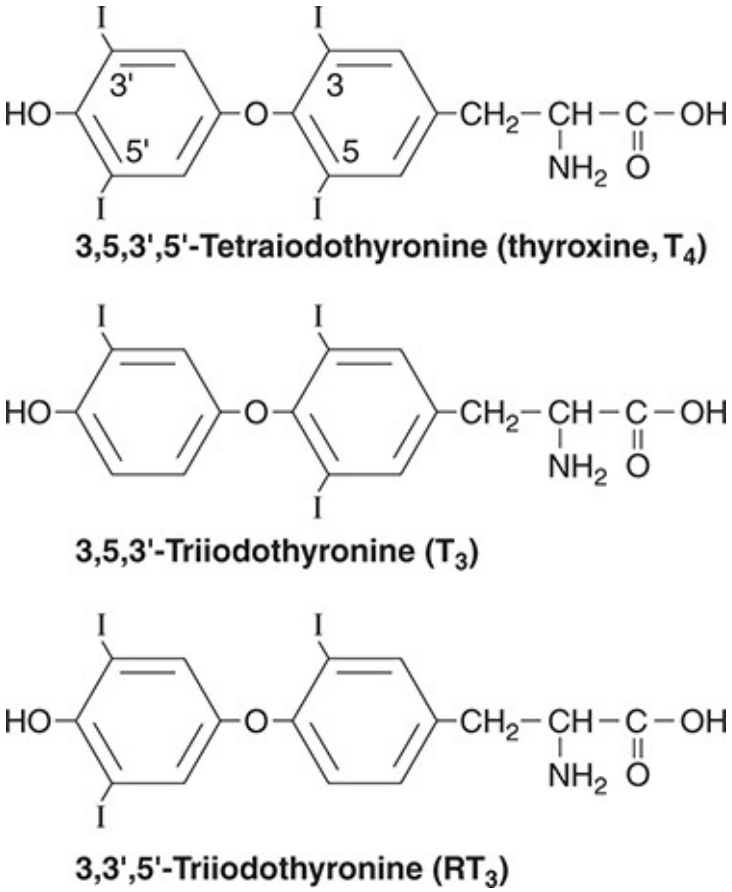


FIGURE 20–4 Thyroid hormones. The numbers in the rings in the T_4 formula indicate the numbers of positions in the molecule. RT_3, reverse triiodothyronine.

IODINE HOMEOSTASIS

Iodine is an essential raw material for thyroid hormone synthesis. Dietary iodide is absorbed by the intestine and enters the circulation; its subsequent fate is summarized in **Figure 20–5**. The minimum daily iodine intake that will maintain normal thyroid function is 150 µg in adults. In most developed countries, supplementation of table salt means that the average dietary intake is

approximately 500 µg/day. The principal organs that take up circulating I^- are the thyroid, which uses it to make thyroid hormones, and the kidneys, which excrete it in the urine. About 120 µg/day enter the thyroid at normal rates of thyroid hormone synthesis and secretion. The thyroid secretes 80 µg/day in the form of T_3 and T_4, while 40 µg/day diffuses back into the extracellular fluid (ECF). Circulating T_3 and T_4 are metabolized in the liver and other tissues, with the release of a further 60 µg of I^- per day into the ECF. Some thyroid hormone derivatives are excreted in the bile, and some of the iodine in them is reabsorbed (enterohepatic circulation), but there is a net loss of I^- in the stool of approximately 20 µg/day. The total amount of I^- entering the ECF is thus 500 + 40 + 60, or 600 µg/day; 20% of this I^- enters the thyroid, whereas 80% is excreted in the urine.

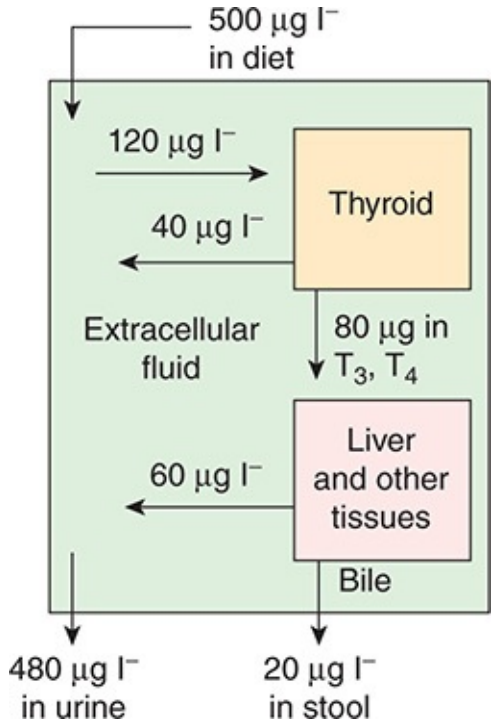


FIGURE 20–5 Iodide metabolism. The figure shows the movement of iodide among various body compartments on a daily basis.

IODIDE TRANSPORT ACROSS THYROCYTES

The basolateral membranes of thyrocytes facing the capillaries contain a **symporter** that transports two Na^+ ions and one I^- ion into the cell with each

cycle, against the electrochemical gradient for I^-. This Na^+/I^- symporter **(NIS)** is capable of producing intracellular I^- concentrations that are 20–40 times as great as the concentration in plasma **(Figure 20–6)**. The process involved is secondary active transport (see Chapter 2), with the energy provided by active transport of Na^+ out of thyroid cells by Na, K ATPase. NIS is regulated both by transcriptional means and by active trafficking into and out of the thyrocyte basolateral membrane; in particular, TSH (see below) induces both NIS expression and the retention of NIS in the basolateral membrane, where it can mediate sustained iodide uptake.

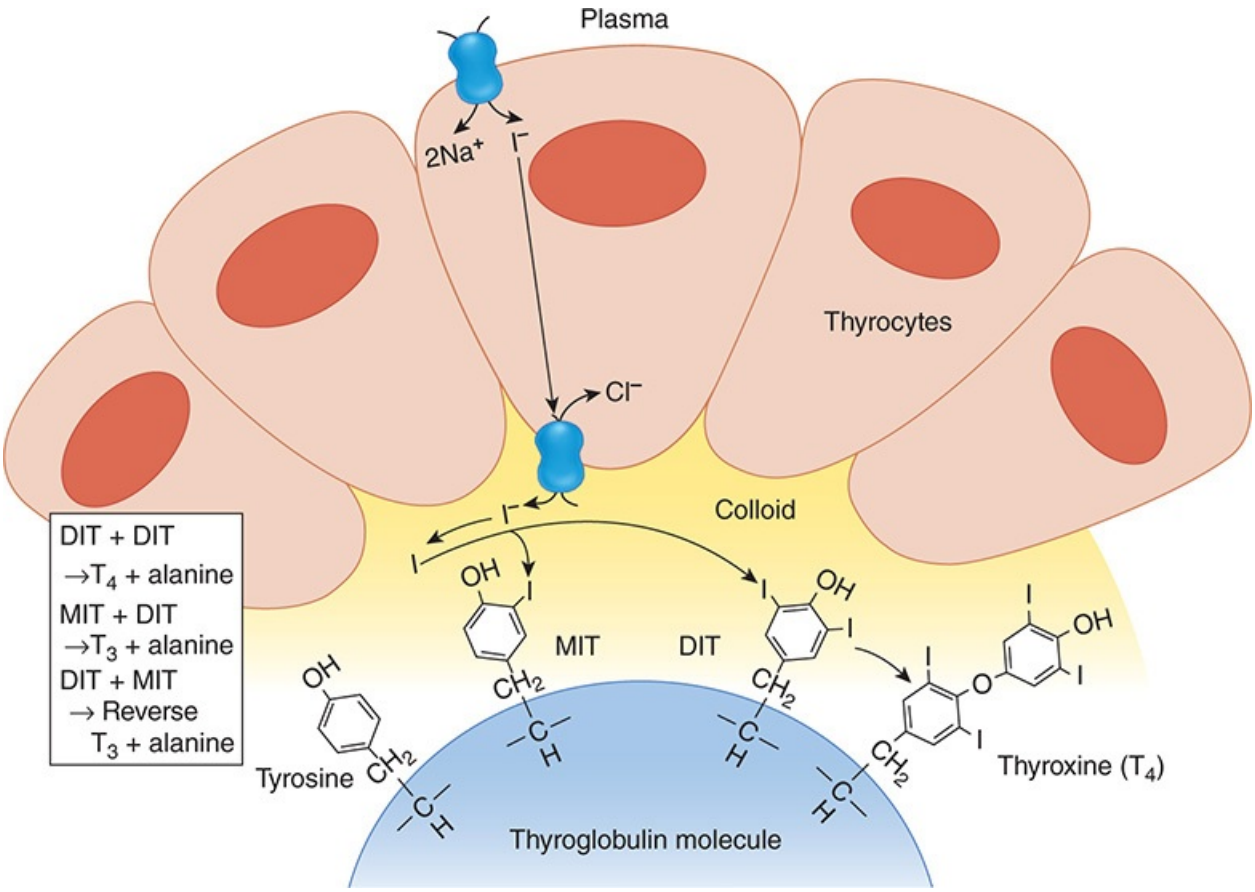


FIGURE 20–6 Outline of thyroid hormone biosynthesis. Iodide (I^-) is transported vectorially from the plasma across the epithelial cells of the thyroid gland by specific transporters. The iodide is converted to iodine, which reacts with tyrosine residues exposed on the surface of thyroglobulin molecules resident in the colloid. Iodination of tyrosine takes place at the apical border of the thyroid cells while the tyrosine moieties remain bound to thyroglobulin via peptide linkages.

Iodide must also exit the thyrocyte across the apical membrane to access the colloid, where the initial steps of thyroid hormone synthesis occur. This transport step is believed to be mediated, at least in part, by a Cl^-/I^- exchanger known as **pendrin** (Figure 20–6). This protein was first identified as the product of the gene responsible for the Pendred syndrome, which causes thyroid dysfunction and deafness. Pendrin (SLC26A4) is one member of the larger family of SLC26 anion exchangers.

The relationship of thyroid function to iodide is unique. Iodide is essential for normal thyroid function, but iodide deficiency and iodide excess both inhibit thyroid function.

The salivary glands, the gastric mucosa, the placenta, the ciliary body of the eye, the choroid plexus, the mammary glands, and certain cancers derived from these tissues also express NIS and can transport iodide against a concentration gradient, but the transporter in these tissues is not affected by TSH. The physiologic significance of all these extrathyroidal iodide-concentrating mechanisms is obscure, but they may provide pathways for radioablation of NIS-expressing cancer cells using iodide radioisotopes. This approach is also useful for the ablation of thyroid cancers.

THYROID HORMONE SYNTHESIS & SECRETION

At the interface between the thyrocyte and the colloid, iodide undergoes a process referred to as organification. First, it is oxidized to iodine, and then incorporated into the carbon 3 position of tyrosine residues that are part of the thyroglobulin molecule in the colloid (Figure 20–6). **Thyroglobulin** is a glycoprotein made up of two subunits. It contains 123 tyrosine residues, but only 4–8 of these are normally incorporated into thyroid hormones. Thyroglobulin is synthesized in the thyrocytes and secreted into the colloid by exocytosis of granules. The oxidation and reaction of iodide with the secreted thyroglobulin is mediated by **thyroid peroxidase,** a membrane-bound enzyme found in the thyrocyte apical membrane. The thyroid hormones so produced remain part of the thyroglobulin molecule until needed. As such, colloid represents a reservoir of thyroid hormones, and humans can ingest a diet completely devoid of iodide for up to 2 months before a decline in circulating thyroid hormone levels is seen. When there is a need for thyroid hormone secretion, colloid is internalized by the thyrocytes by endocytosis, and directed toward lysosomal degradation. Thus, the

peptide bonds of thyroglobulin are hydrolyzed, and free T_4 and T_3 are discharged into cytosol and thence to the capillaries (see below). Thyrocytes thus have four functions: They collect and transport iodine, they synthesize thyroglobulin and secrete it into the colloid, they fix iodine to the thyroglobulin to generate thyroid hormones, and they remove the thyroid hormones from thyroglobulin and secrete them into the circulation.

Thyroid hormone synthesis is a multistep process. Thyroid peroxidase generates reactive iodine species that can attack thyroglobulin. The first product is monoiodotyrosine (MIT). MIT is next iodinated on the carbon 5 position to form diiodotyrosine (DIT). Two DIT molecules then undergo an oxidative condensation to form T_4 with the elimination of the alanine side chain from the molecule that forms the outer ring. There are two theories of how this **coupling reaction** occurs. One holds that the coupling occurs with both DIT molecules attached to thyroglobulin (intramolecular coupling). The other holds that the DIT that forms the outer ring is first detached from thyroglobulin (intermolecular coupling). In either case, thyroid peroxidase is involved in coupling as well as iodination. T_3 is formed by condensation of MIT with DIT. RT_3 is likely formed by condensation of DIT with MIT. In the normal human thyroid, the average distribution of iodinated compounds is 3% MIT, 33% DIT, 35% T_4, and 7% T_3. Only traces of RT_3 and other components are present.

The human thyroid secretes about 80 µg (103 nmol) of T_4, 4 µg (7 nmol) of T_3, and 2 µg (3.5 nmol) of RT_3 per day **(Figure 20–7)**. MIT and DIT are not secreted. These iodinated tyrosines can be deiodinated by a microsomal **iodotyrosine deiodinase.** This represents a mechanism to recover iodine and bound tyrosines and recycle them for additional rounds of hormone synthesis. The iodine liberated by deiodination of MIT and DIT is reutilized in the gland and normally provides about twice as much iodide for hormone synthesis as NIS does. In patients with congenital absence of the iodotyrosine deiodinase, MIT and DIT appear in the urine and there are symptoms of iodine deficiency (see below). Iodinated thyronines are resistant to the activity of iodotyrosine deiodinase, thus allowing T_4 and T_3 to pass into the circulation.

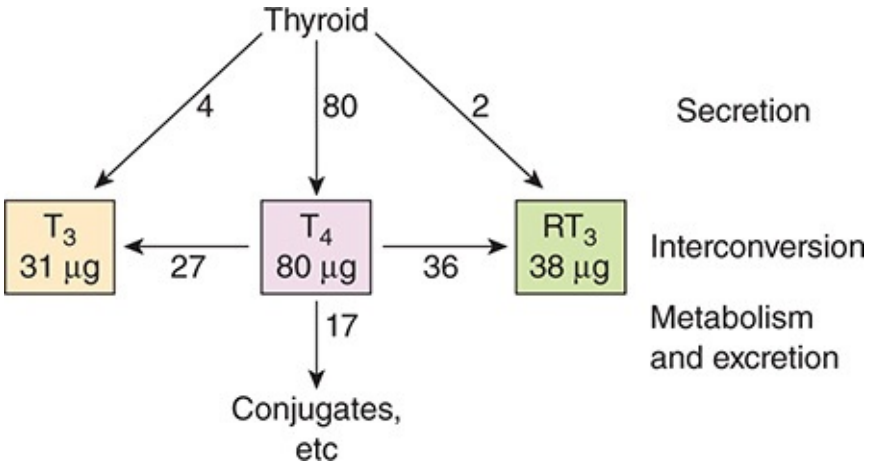


FIGURE 20–7 Secretion and interconversion of thyroid hormones in normal adult humans. Figures are in micrograms per day. Note that most of the T_3 and RT_3 are formed from T_4 deiodination in the tissues and only small amounts are secreted by the thyroid. T_4 is also conjugated for subsequent excretion from the body.

TRANSPORT & METABOLISM OF THYROID HORMONES

PROTEIN BINDING

The normal total **plasma T_4** level in adults is approximately 8 µg/dL (103 nmol/L), and the **plasma T_3** level is approximately 0.15 µg/dL (2.3 nmol/L). T_4 and T_3 are relatively lipophilic; thus, their free forms in plasma are in equilibrium with a much larger pool of protein-bound thyroid hormones in plasma and in tissues. Free thyroid hormones are added to the circulating pool by the thyroid. It is the free thyroid hormones in plasma that are physiologically active and that feed back to inhibit pituitary secretion of TSH **(Figure 20–8)**. The function of protein-binding appears to be the maintenance of a large pool of hormone that can readily be mobilized as needed. In addition, at least for T_3, hormone-binding prevents excess uptake by the first cells encountered and promotes uniform tissue distribution. Total T_4 and T_3 can both be measured by radioimmunoassay. There are also direct assays that specifically measure only the free forms of the hormones. The latter are the more clinically relevant measures given that these are the active forms, and also due to both acquired and

congenital variations in the concentrations of binding proteins between individuals.

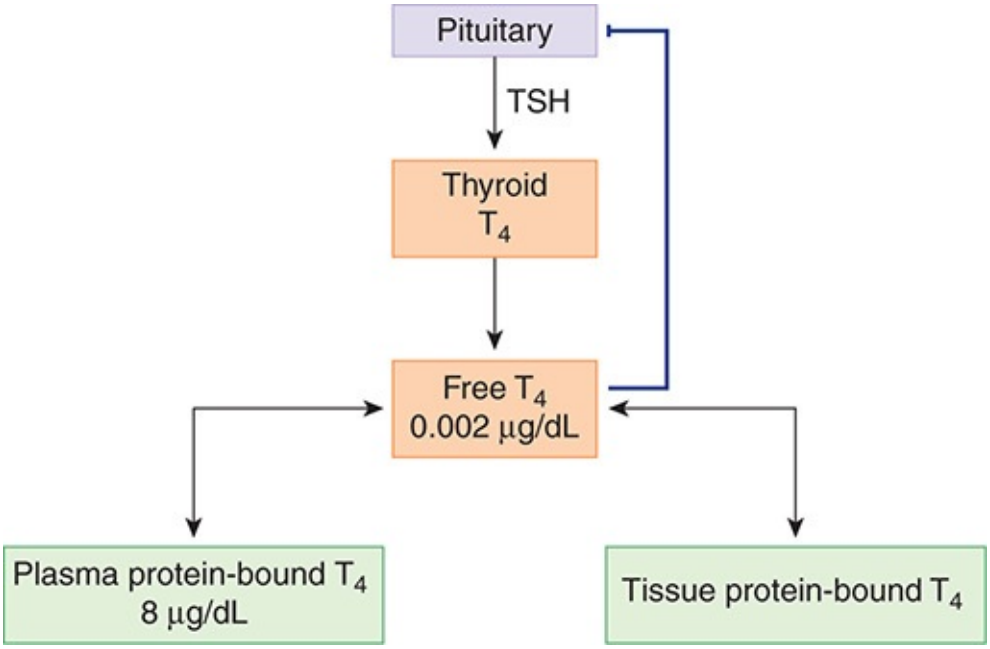


FIGURE 20–8 Regulation of thyroid hormone synthesis. T_4 is secreted by the thyroid in response to TSH. Free T_4 secreted by the thyroid into the circulation is in equilibrium with T_4 bound to both plasma and tissue proteins. Free T_4 also feeds back to inhibit TSH secretion by the pituitary. Not shown, T_3 is also secreted (in small amounts) by the thyroid and produced (in larger amounts) from T_4 in the periphery by deiodination. T_3 also feeds back to reduce secretion of TSH from the pituitary.

The plasma proteins that bind thyroid hormones are **albumin, transthyretin** (formerly called **thyroxine-binding prealbumin),** and **thyroxine-binding globulin (TBG).** Of the three proteins, albumin has the largest **capacity** to bind T_4 (ie, it can bind the most T_4 before becoming saturated) and TBG has the smallest capacity. However, the **affinities** of the proteins for T_4 (ie, the avidity with which they bind T_4 under physiologic conditions) are such that most of the circulating T_4 is bound to TBG **(Table 20–1)**. Smaller amounts of T_4 are bound to transthyretin and albumin.

TABLE 20–1 Binding of thyroid hormones to plasma proteins in normal adult humans.

Protein	Plasma Concentration (mg/dL)	Amounts of Circulating Hormone Bound (%)	
		T_4	T_3
Thyroxine-binding globulin (TBG)	2	67	46
Transthyretin (thyroxine-binding prealbumin, TBPA)	15	20	1
Albumin	3500	13	53

Normally, 99.98% of the T_4 in plasma is bound; the free T_4 level is only about 2 ng/dL. There is very little T_4 in the urine. Its biologic half-life is long (about 6–7 days), and its volume of distribution is less than that of ECF (10 L, or about 15% of body weight). All of these properties are characteristic of a substance that is strongly bound to protein.

T_3 is not bound to quite as great an extent; of the 0.15 µg/dL normally found in plasma, 0.2% (0.3 ng/dL) is free. The remaining 99.8% is protein-bound, 46% to TBG and most of the remainder to albumin, with very little binding to transthyretin (Table 20–1). The lesser binding of T_3 correlates with the facts that T_3 has a shorter half-life than T_4 and that its action on the tissues is much more rapid. RT_3 also binds to TBG.

FLUCTUATIONS IN BINDING

When a sudden, sustained increase in the concentration of thyroid-binding proteins in the plasma takes place, the concentration of free thyroid hormones falls. This change is temporary, however, because the decrease in the concentration of free thyroid hormones in the circulation stimulates TSH secretion, which in turn causes an increase in the production of free thyroid hormones. A new equilibrium is eventually reached at which the total quantity of thyroid hormones in the blood is elevated but the concentration of free hormones, the rate of their metabolism, and the rate of TSH secretion are normal. Corresponding changes in the opposite direction occur when the concentration of thyroid-binding protein is reduced. Consequently, patients with

elevated or decreased concentrations of binding proteins, particularly TBG, are typically neither hyperthyroid nor hypothyroid; that is, they are **euthyroid.**

TBG levels are elevated in estrogen-treated patients and during pregnancy, as well as after treatment with various drugs **(Table 20–2)**. They are depressed by glucocorticoids, androgens, the weak androgen danazol, and the cancer chemotherapeutic agent L-asparaginase. A number of other drugs, including salicylates, the anticonvulsant phenytoin, and the cancer chemotherapeutic agents mitotane (o, p′-DDD) and 5-fluorouracil, inhibit binding of T_4 and T_3 to TBG and consequently produce changes similar to those produced by a decrease in TBG concentration. Changes in total plasma T_4 and T_3 can also be produced by changes in plasma concentrations of albumin and prealbumin.

TABLE 20–2 Effect of variations in the concentrations of thyroid hormone–binding proteins in the plasma on various parameters of thyroid function after equilibrium has been reached.

Condition	Concentrations of Binding Proteins	Total Plasma T_4, T_3, RT_3	Free Plasma T_4, T_3, RT_3	Plasma TSH	Clinical State
Hyperthyroidism	Normal	High	High	Low	Hyperthyroid
Hypothyroidism	Normal	Low	Low	High	Hypothyroid
Estrogens, methadone, heroin, antipsychotic drugs, clofibrate	High	High	Normal	Normal	Euthyroid
Glucocorticoids, androgens, danazol, asparaginase	Low	Low	Normal	Normal	Euthyroid

METABOLISM OF THYROID HORMONES

T_4 and T_3 are deiodinated in the liver, the kidneys, and many other tissues. These deiodination reactions serve not only to catabolize the hormones, but also to provide a local supply specifically of T_3, which is believed to be the primary mediator of the physiologic effects of thyroid secretion. One-third of the circulating T_4 is normally converted to T_3 in adult humans, and 45% is converted to RT_3. As shown in Figure 20–7, only about 13% of the circulating T_3 is secreted by the thyroid while 87% is formed by deiodination of T_4; similarly, only 5% of the circulating RT_3 is secreted by the thyroid and 95% is formed by deiodination of T_4. It should be noted as well that marked differences in the ratio of T_3 to T_4 occur in various tissues. Two tissues that have very high T_3/T_4 ratios are the pituitary and the cerebral cortex, due to the expression of specific deiodinases, as discussed below. In the brain, in particular, high levels of

deiodinase activity ensure an ample supply of active T_3.

Three different deiodinases act on thyroid hormones: D_1, D_2, and D_3. All are unique in that they contain the rare amino acid selenocysteine, with selenium in place of sulfur, which is essential for their enzymatic activity. D_1 is present in high concentrations in the liver, kidneys, thyroid, and pituitary. It appears primarily to be responsible for maintaining the formation of T_3 from T_4 in the periphery. D_2 is present in the brain, pituitary, and brown fat. It also contributes to the formation of T_3. In the brain, it is located in astroglia and provides a supply of T_3 to neurons. D_3 is also present in the brain and in reproductive tissues. It acts only on the 5 position of T_4 and T_3 and is probably the main source of RT_3 in the blood and tissues.

Some of the T_4 and T_3 are further converted to deiodotyrosines by deiodinases. T_4 and T_3 are also conjugated in the liver to form sulfates and glucuronides. These conjugates enter the bile and pass into the intestine. The thyroid conjugates are hydrolyzed, and some are thereafter reabsorbed (enterohepatic circulation), but others are excreted in the stool. In addition, some T_4 and T_3 pass directly from the circulation to the intestinal lumen. The iodide lost by these routes amounts to about 4% of the total daily iodide loss.

FLUCTUATIONS IN DEIODINATION

Much more RT_3 and much less T_3 are formed during fetal life, and the ratio shifts to that of adults about 6 weeks after birth. Various drugs inhibit deiodinases, producing a fall in plasma T_3 levels and a reciprocal rise in RT_3. Selenium deficiency has the same effect. A wide variety of nonthyroidal illnesses also suppress deiodinases. These include burns, trauma, advanced cancer, cirrhosis, chronic kidney disease, myocardial infarction, and febrile states. The low-T_3 state produced by these conditions disappears with recovery.

Diet also has a clear-cut effect on conversion of T_4 to T_3. In fasted individuals, plasma T_3 is reduced by 10–20% within 24 h and by about 50% in 3–7 days, with a corresponding rise in RT_3 **(Figure 20–9)**. Free and bound T_4 levels remain essentially normal. During more prolonged starvation, RT_3 returns to normal but T_3 remains depressed. At the same time, the basal metabolic rate (BMR) falls and urinary nitrogen excretion, an index of protein breakdown, is

decreased. Thus, the decline in T_3 conserves calories and protein. Conversely, overfeeding increases T_3 and reduces RT_3.

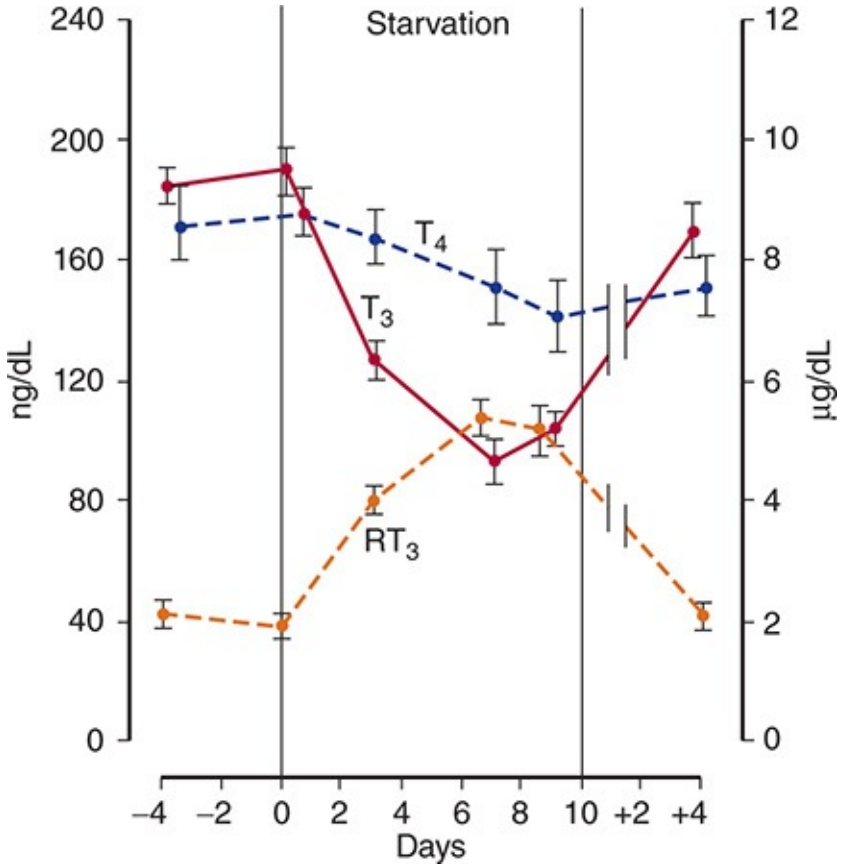


FIGURE 20–9 Effect of starvation on plasma levels of T_4, T_3, and RT_3 in humans. The scale for T_3 and RT_3 is on the left and the scale for T_4 is on the right. The most pronounced effect is a reduction in T_3 levels with a reciprocal rise in RT_3. The changes, which conserve calories by reducing tissue metabolism, are reversed promptly by refeeding. Similar changes occur in wasting diseases. (Reproduced with permission from Burger AG: New aspects of the peripheral action of thyroid hormones. Triangle 1983;22:175. Copyright © 1983 Sandoz Ltd., Basel, Switzerland.)

REGULATION OF THYROID SECRETION

Thyroid function is regulated primarily by variations in the circulating level of pituitary TSH (Figure 20–8). TSH secretion is increased by the hypothalamic hormone TRH (see Chapter 17) and inhibited in a negative feedback manner by

circulating free T_4 and T_3. The effect of T_4 is enhanced by production of T_3 in the cytoplasm of the pituitary cells by the D_2 they contain. TSH secretion is also inhibited by stress, and in experimental animals it is increased by cold and decreased by warmth.

CHEMISTRY & METABOLISM OF TSH

Human TSH is a glycoprotein made up of α and β subunits that become noncovalently linked in the pituitary thyrotropes. TSH-α is identical to the α subunit of luteinizing hormone, follicle-stimulating hormone, and human chorionic gonadotropin (hCG) (see Chapters 18 and 22). The functional specificity of TSH is therefore conferred by the β subunit.

The biologic half-life of human TSH is about 60 min. TSH is degraded for the most part in the kidneys and to a lesser extent in the liver. Secretion is pulsatile, and mean output starts to rise at about 9:00 PM, peaks at midnight, and then declines thereafter. The normal secretion rate is about 110 µg/day. The average plasma level is about 2 µg/mL.

Because the α subunit in hCG is the same as that in TSH, large amounts of hCG can activate TSH receptors nonspecifically. In some patients with benign or malignant tumors of placental origin, plasma hCG levels can rise so high that they produce mild hyperthyroidism.

EFFECTS OF TSH ON THE THYROID

When the pituitary is removed, thyroid function is depressed and the gland atrophies; when TSH is administered, thyroid function is stimulated. TSH acts via its receptor, a typical G-protein-coupled, seven-transmembrane receptor that activates adenylyl cyclase through G_s. It also activates phospholipase C (PLC). Within a few minutes after the injection of TSH, there are increases in iodide binding, synthesis of T_3, T_4, and iodotyrosines, secretion of thyroglobulin into the colloid, and endocytosis of colloid. Iodide trapping is increased in a few hours; blood flow increases; and, with long-term TSH treatment, the thyrocytes hypertrophy and the weight of the gland increases.

Whenever TSH stimulation is prolonged, the thyroid becomes detectably enlarged. Enlargement of the thyroid is called a **goiter.**

OTHER FACTORS AFFECTING THYROID GROWTH

In addition to TSH receptors, thyrocytes express receptors for insulin-like growth factor I (IGF-I) and epidermal growth factor (EGF). IGF-I and EGF promote growth, whereas interferon-γ and tumor necrosis factor-α inhibit growth. The effect of the cytokines implies that thyroid function might be inhibited in the setting of chronic inflammation, which could contribute to cachexia, or weight loss.

CONTROL MECHANISMS

The mechanisms regulating thyroid secretion are summarized in Figure 20–8. The negative feedback effect of thyroid hormones on TSH secretion is exerted in part at the hypothalamic level, but it is also due in large part to an action on the pituitary, since T_4 and T_3 block the increase in TSH secretion produced by TRH. Infusion of either T_4 or T_3 reduces circulating TSH, which declines measurably within 1 h. The day-to-day maintenance of thyroid secretion depends on the feedback interplay of thyroid hormones with TSH and TRH (Figure 20–8). The adjustments that appear to be mediated via TRH include the increased secretion of thyroid hormones produced by cold and, presumably, the decrease produced by heat. It is worth noting that although cold produces clear-cut increases in circulating TSH in human infants, the rise produced by cold in adults is negligible. Consequently, in adults, increased heat production due to increased thyroid hormone secretion **(thyroid hormone thermogenesis)** plays little if any role in the response to cold. Stress has an inhibitory effect on TRH secretion. Glucocorticoids also inhibit TSH secretion.

EFFECTS OF THYROID HORMONES

Some of the widespread effects of thyroid hormones in the body are secondary to stimulation of O_2 consumption **(calorigenic action),** although the hormones also affect growth and development in mammals, help regulate lipid metabolism, and increase the absorption of carbohydrates from the intestine **(Table 20–3)**. They also increase the dissociation of oxygen from hemoglobin by increasing red cell 2,3-diphosphoglycerate (DPG) (see Chapter 35).

TABLE 20–3 Physiologic effects of thyroid hormones.

Target Tissue	Effect	Mechanism
Heart	Chronotropic and inotropic	Increased number of β-adrenergic receptors Enhanced responses to circulating catecholamines Increased proportion of α-myosin heavy chain (with higher ATPase activity)
Adipose tissue	Catabolic	Stimulated lipolysis
Muscle	Catabolic	Increased protein breakdown
Bone	Developmental	Promote normal growth and skeletal development
Nervous system	Developmental	Promote normal brain development
Gut	Metabolic	Increased rate of carbohydrate absorption
Lipoprotein	Metabolic	Formation of LDL receptors
Other	Calorigenic	Stimulated oxygen consumption by metabolically active tissues (exceptions: testes, uterus, lymph nodes, spleen, anterior pituitary) Increased metabolic rate

Modified with permission from McPhee SJ, Lingarra VR, Ganong WF (editors): *Pathophysiology of Disease*, 6th ed. New York, NY: McGraw-Hill; 2010.

MECHANISM OF ACTION

Thyroid hormones enter cells and T_3 binds to thyroid hormone receptors **(TRs),** members of the superfamily of hormone-sensitive nuclear transcription factors. T_4 can also bind, but not as avidly. The hormone–receptor complex then

migrates to the nucleus and binds to DNA via zinc fingers and increases (or in some cases, decreases) the expression of a variety of different genes that code for proteins regulating cell function (see Chapters 1 and 16).

There are two human TR genes: an α receptor gene on chromosome 17 and a β receptor gene on chromosome 3. By alternative splicing, each forms at least two different mRNAs and therefore two different receptor proteins. TRβ2 is found only in the brain, but TRα1, TRα2, and TRβ1 are widely distributed. TRα2 differs from the other three in that it does not bind thyroid hormones due to a unique C-terminus, and may actually antagonize the activity of the other TRs. TRs bind to DNA as monomers, homodimers, and heterodimers with other nuclear receptors, particularly the retinoid X receptor **(RXR).** The TR/RXR heterodimer does not bind to 9-*cis* retinoic acid, the usual ligand for RXR, but TR binding to DNA is greatly enhanced in response to thyroid hormones when the receptor is in the form of this heterodimer. There are also coactivator and corepressor proteins that affect the actions of TRs. Presumably, this complexity underlies the ability of thyroid hormones to produce many different effects in the body.

In most of its actions, T_3 acts more rapidly and is three to five times more potent than T_4 **(Figure 20–10)**. This is because T_3 is less tightly bound to plasma proteins than is T_4, but binds more avidly to thyroid hormone receptors. As previously noted, RT_3 is inert.

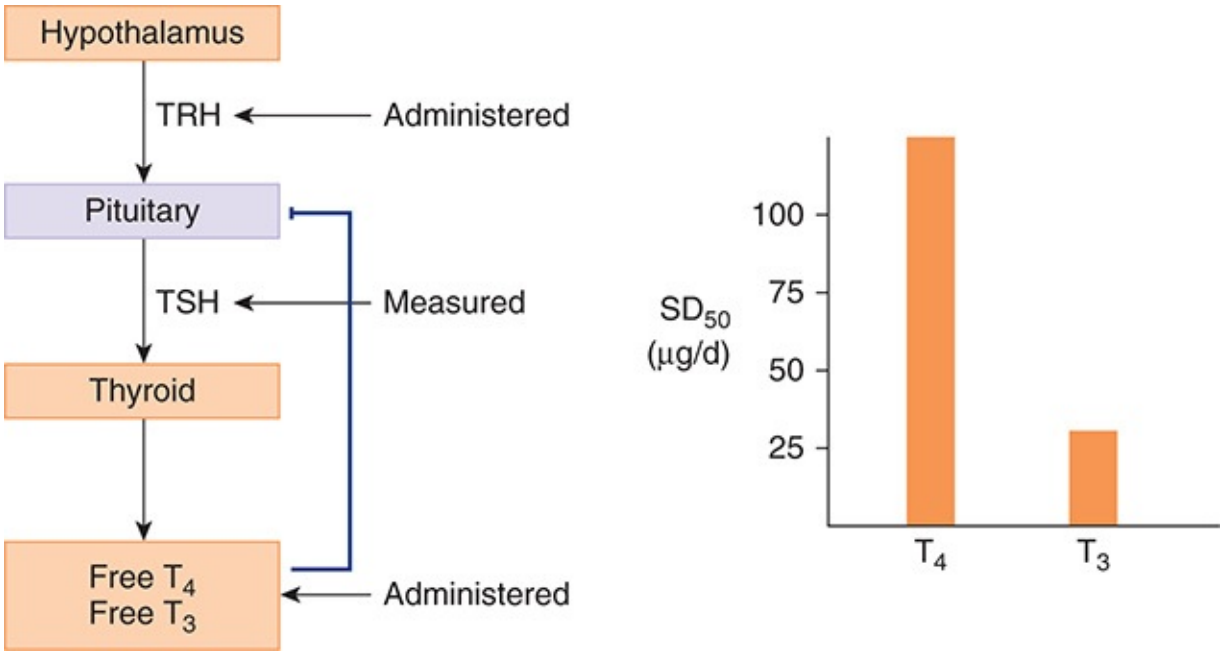


FIGURE 20–10 Experiment demonstrating the relative potency of T_4 and T_3. Healthy male volunteers were given TRH to stimulate TSH production in the absence or presence of various doses of T_3 or T_4. The left hand portion of the figure shows the relationships between the hormones that were administered or measured. On the right, the calculated doses of T_3 and T_4 needed to reduce the TSH response induced by TRH by 50% (SD_{50}) are shown. Note the substantially greater potency of T_3. (Data from Sawin CT, Hershman JM, Chopra IJ: The comparative effect of T_4 and T_3 on the TSH response to TRH in young adult men. J Clin Endo Metab 1977; 44:273–278.)

CALORIGENIC ACTION

T_4 and T_3 increase O_2 consumption by almost all metabolically active tissues. The exceptions are the adult brain, testes, uterus, lymph nodes, spleen, and anterior pituitary. T_4 actually depresses the O_2 consumption of the anterior pituitary, presumably because it inhibits TSH secretion. The increase in metabolic rate produced by a single dose of T_4 becomes measurable after a latent period of several hours and lasts 6 days or more.

Some of the calorigenic effect of thyroid hormones is due to metabolism of the fatty acids they mobilize. In addition, thyroid hormones increase the activity of the membrane-bound Na, K ATPase in many tissues.

EFFECTS SECONDARY TO CALORIGENESIS

When the metabolic rate is increased by T_4 and T_3 in adults, nitrogen excretion is increased; if food intake is not increased, endogenous protein and fat stores are catabolized and weight is lost. In hypothyroid children, small doses of thyroid hormones cause a positive nitrogen balance because they stimulate growth, but large doses cause protein catabolism similar to that produced in the adult. The potassium liberated during protein catabolism appears in the urine, and there is also an increase in urinary hexosamine and uric acid excretion.

When the metabolic rate is increased, the need for all vitamins is increased and vitamin deficiency syndromes may be precipitated. Thyroid hormones are necessary for hepatic conversion of carotene to vitamin A, and the accumulation of carotene in the bloodstream **(carotenemia)** in hypothyroidism is responsible

for the yellowish tint of the skin. Carotenemia can be distinguished from jaundice because in the former condition the sclera are not yellow.

The skin normally contains a variety of proteins combined with polysaccharides, hyaluronic acid, and chondroitin sulfuric acid. In hypothyroidism, these complexes accumulate, promoting water retention and the characteristic puffiness of the skin (myxedema). When thyroid hormones are administered, the proteins are metabolized, and diuresis continues until the myxedema is cleared.

Milk secretion is decreased in hypothyroidism and stimulated by thyroid hormones, a fact sometimes put to practical use in the dairy industry. Thyroid hormones do not stimulate the metabolism of the uterus but are essential for normal menstrual cycles and fertility.

EFFECTS ON THE CARDIOVASCULAR SYSTEM

Large doses of thyroid hormones cause enough extra heat production to lead to a slight rise in body temperature (Chapter 17), which in turn activates heat-dissipating mechanisms. Peripheral resistance decreases because of cutaneous vasodilation, and this increases levels of renal Na^+ and water absorption, expanding blood volume. Cardiac output is increased by the direct action of thyroid hormones, as well as that of catecholamines, on the heart, so that pulse pressure and cardiac rate are increased and circulation time is shortened.

T_3 is not formed from T_4 in cardiac myocytes, but circulating T_3 enters myocytes, combines with its receptors, and enters the nucleus, where the complex promotes the expression of some genes and inhibits the expression of others. Those that are enhanced include the genes for α-myosin heavy chain, sarcoplasmic reticulum Ca^{2+} ATPase, β-adrenergic receptors, G-proteins, Na, K ATPase, and certain K^+ channels. Those that are inhibited include the genes for β-myosin heavy chain, phospholamban, two types of adenylyl cyclase, TRs, and NCX, the Na^+–Ca^{2+} exchanger. The net result is increased heart rate and force of contraction.

The two myosin heavy chain (MHC) isoforms, α-MHC and β-MHC, produced by the heart are encoded by two highly homologous genes located on the short arm of chromosome 17. Each myosin molecule consists of two heavy chains and two pairs of light chains (see Chapter 5). The myosin containing β-MHC has less ATPase activity than the myosin containing α-MHC. α-MHC predominates in the atria in adults, and its level is increased by treatment with

thyroid hormone. This increases the speed of cardiac contraction. Conversely, expression of the α-MHC gene is depressed and that of the β-MHC gene is enhanced in hypothyroidism.

EFFECTS ON THE NERVOUS SYSTEM

In hypothyroidism, mentation is slow and the cerebrospinal fluid (CSF) protein level is elevated. Thyroid hormones reverse these changes, and large doses cause rapid mentation, irritability, and restlessness. Overall, cerebral blood flow and glucose and O_2 consumption by the brain are normal in adult hypothyroidism and hyperthyroidism. However, thyroid hormones enter the brain in adults and are found in gray matter in numerous different locations. In addition, astrocytes in the brain convert T_4 to T_3, and there is a sharp increase in brain D_2 activity after thyroidectomy that is reversed within 4 h by a single intravenous dose of T_3. Some of the effects of thyroid hormones on the brain are probably secondary to increased responsiveness to catecholamines, with consequent increased activation of the reticular activating system (see Chapter 14). In addition, thyroid hormones have marked effects on brain development. The parts of the central nervous system (CNS) most affected are the cerebral cortex and the basal ganglia. In addition, the cochlea is also affected. Consequently, thyroid hormone deficiency during development causes mental retardation, motor rigidity, and deaf–mutism. Deficiencies in thyroid hormone synthesis secondary to a failure of thyrocytes to transport iodide presumably also contribute to deafness in Pendred syndrome, discussed above.

Thyroid hormones also exert effects on reflexes. The reaction time of stretch reflexes (see Chapter 12) is shortened in hyperthyroidism and prolonged in hypothyroidism. Measurement of the reaction time of the ankle jerk (Achilles reflex) once attracted attention as a clinical test for evaluating thyroid function, but this reaction time is also affected by other diseases and thus is not a specific assessment of thyroid activity.

RELATION TO CATECHOLAMINES

The actions of thyroid hormones and the catecholamines norepinephrine and epinephrine are intimately interrelated. Epinephrine increases the metabolic rate, stimulates the nervous system, and produces cardiovascular effects similar to those of thyroid hormones, although the duration of these actions is brief.

Norepinephrine has generally similar actions. The toxicity of the catecholamines is markedly increased in rats treated with T_4. Although plasma catecholamine levels are normal in hyperthyroidism, the cardiovascular effects, tremulousness, and sweating that are seen in the setting of excess thyroid hormones can be reduced or abolished by sympathectomy. They can also be reduced by drugs such as propranolol that block β-adrenergic receptors. Indeed, propranolol and other β-blockers are used extensively in the treatment of thyrotoxicosis and in the treatment of the severe exacerbations of hyperthyroidism called **thyroid storms.** However, even though β-blockers can weakly inhibit extrathyroidal conversion of T_4 to T_3, and consequently may produce a small fall in plasma T_3, they have little effect on the other actions of thyroid hormones. Presumably, the functional synergism observed between catecholamines and thyroid hormones, particularly in pathologic settings, arises from their overlapping biologic functions as well as the ability of thyroid hormones to increase expression of catecholamine receptors and the signaling effectors to which they are linked.

EFFECTS ON SKELETAL MUSCLE

Muscle weakness occurs in most patients with hyperthyroidism **(thyrotoxic myopathy),** and when the hyperthyroidism is severe and prolonged, the myopathy may be severe. The muscle weakness may be due in part to increased protein catabolism. Thyroid hormones affect the expression of the MHC genes in skeletal as well as cardiac muscle (see Chapter 5). However, the effects produced are complex and their relation to the myopathy is not established. Hypothyroidism is also associated with muscle weakness, cramps, and stiffness.

EFFECTS ON CARBOHYDRATE METABOLISM

Thyroid hormones increase the rate of absorption of carbohydrates from the gastrointestinal tract, an action that is probably independent of their calorigenic action. In hyperthyroidism, therefore, the plasma glucose level rises rapidly after a carbohydrate meal, sometimes exceeding the renal threshold. However, it falls again at a rapid rate.

EFFECTS ON CHOLESTEROL METABOLISM

Thyroid hormones lower circulating cholesterol levels. The plasma cholesterol

level drops before the metabolic rate rises, which indicates that this action is independent of the stimulation of O_2 consumption. The decrease in plasma cholesterol concentration is due to increased formation of low-density lipoprotein (LDL) receptors in the liver, resulting in increased hepatic removal of cholesterol from the circulation. Despite considerable effort, however, it has not been possible to produce a clinically useful thyroid hormone analog that lowers plasma cholesterol without increasing metabolism.

EFFECTS ON GROWTH

Thyroid hormones are essential for normal growth and skeletal maturation (see Chapter 21). In hypothyroid children, bone growth is slowed and epiphysial closure delayed. In the absence of thyroid hormones, growth hormone secretion is also depressed. This further impairs growth and development, since thyroid hormones normally potentiate the effect of growth hormone on tissues.

DISORDERS OF THYROID FUNCTION

Based on the foregoing, some of the causes and effects of either under- or overactive thyroid function (hypo- and hyperthyroid conditions, respectively) should be predictable **(Clinical Boxes 20–1** and **20–2)**. Altered regulation of thyroid-responsive physiological systems can also be seen in the setting of thyroid hormone resistance, most commonly attributed to mutations in TRβ **(Clinical Box 20–3)**.

The amount of thyroid hormone required to maintain normal function in thyroidectomized individuals is defined as the amount necessary to return plasma TSH to normal. Indeed, measurement of TSH is regarded as one of the best tests of thyroid function. The amount of T_4 that normalizes plasma TSH in individuals lacking a functional thyroid gland (**athyreotic)** averages 112 µg of T_4 by mouth per day in adults. About 80% of this dose is absorbed from the gastrointestinal tract. It produces a slightly greater than normal level of free T_4 but a normal level of free T_3, indicating that circulating T_3 may be the principal feedback regulator of TSH secretion in humans.

CLINICAL BOX 20–1

Reduced Thyroid Function

The syndrome of adult **hypothyroidism** is generally called **myxedema,** although this term is also used to refer specifically to the skin changes in the syndrome. Hypothyroidism may be the end result of a number of diseases of the thyroid gland, or it may be secondary to pituitary or hypothalamic failure. In the latter two conditions, the thyroid remains able to respond to TSH. Thyroid function may be reduced by a number of conditions **(Table 20–4)**. For example, when the dietary iodine intake falls below 50 µg/day, thyroid hormone synthesis is inadequate and secretion declines. As a result of increased TSH secretion, the thyroid hypertrophies, producing an **iodine deficiency goiter** that may become very large. Such "endemic goiters" have been substantially reduced by the practice of adding iodide to table salt. Drugs may also inhibit thyroid function. Most do so either by interfering with the iodide-trapping mechanism or by blocking the organic binding of iodine. In either case, TSH secretion is stimulated by the decline in circulating thyroid hormones, and a goiter is produced. Paradoxically, another substance that inhibits thyroid function under certain conditions is iodide itself. In normal individuals, large doses of iodide act directly on the thyroid to produce a mild and transient inhibition of organic binding of iodide and hence of hormone synthesis. This inhibition is known as the **Wolff–Chaikoff effect.**

In completely athyreotic adults, the BMR falls to about 40%. The hair is coarse and sparse, the skin is dry and yellowish (carotenemia), and cold is poorly tolerated. Mentation is slow, memory is poor, and in some patients there are severe mental symptoms ("myxedema madness"). Plasma cholesterol is elevated. Children who are hypothyroid from birth or before are called **cretins.** They are dwarfed and mentally retarded. Worldwide, congenital hypothyroidism is one of the most common causes of preventable mental retardation. The main causes are included in Table 20–4. They include not only maternal iodine deficiency and various congenital abnormalities of the fetal hypothalamo–pituitary–thyroid axis, but also maternal antithyroid antibodies that cross the placenta and damage the fetal thyroid. T_4 also crosses the placenta, and unless the mother is hypothyroid, growth and development are normal until birth. If treatment is started at birth, the prognosis for normal growth and development is good, and mental retardation can generally be avoided; for this reason, screening tests for congenital hypothyroidism are becoming routine. When the mother is hypothyroid as well, as in the case of iodine deficiency, the mental deficiency is more severe and less responsive to treatment after birth. It has been estimated that 20

million people in the world now have various degrees of brain damage caused by iodine deficiency in utero.

Uptake of tracer doses of radioactive iodine can be used to assess thyroid function (contrast this with the use of large doses to ablate thyroid tissue in cases of hyperthyroidism (Clinical Box 20–2)).

THERAPEUTIC HIGHLIGHTS

The treatment of hypothyroidism depends on the underlying mechanisms. Iodide deficiency can be addressed by adding it to the diet, as is done routinely in developed countries. In congenital hypothyroidism, levothyroxine—a synthetic form of the thyroid hormone T_4—can be given. It is important that this take place as soon as possible after birth, with levels regularly monitored, to minimize long-term adverse effects.

TABLE 20–4 Causes of congenital hypothyroidism.

Maternal iodine deficiency
Fetal thyroid dysgenesis
Inborn errors of thyroid hormone synthesis
Maternal antithyroid antibodies that cross the placenta
Fetal hypopituitary hypothyroidism

CLINICAL BOX 20–2

Hyperthyroidism

The symptoms of an overactive thyroid gland follow logically from the actions of thyroid hormone discussed in this chapter. Thus, hyperthyroidism is characterized by nervousness; weight loss; hyperphagia; heat intolerance; increased pulse pressure; a fine tremor of the outstretched fingers; warm, soft skin; sweating; and a BMR from +10 to as high as +100. It has various causes (Table 20–5); however, the most common cause is **Graves disease (Graves hyperthyroidism),** which accounts for 60–80% of the cases. This is an autoimmune disease, more common in women, in which antibodies to the TSH receptor stimulate the receptor. This produces marked T_4 and T_3

secretion and enlargement of the thyroid gland (goiter). However, due to the feedback effects of T_4 and T_3, plasma TSH is low, not high. Another hallmark of Graves disease is the occurrence of swelling of tissues in the orbits, producing protrusion of the eyeballs **(exophthalmos).** This occurs in 50% of patients and often precedes the development of obvious hyperthyroidism. Other antithyroid antibodies are present in Graves disease, including antibodies to thyroglobulin and thyroid peroxidase. In Hashimoto thyroiditis, autoimmune antibodies and infiltrating cytotoxic T cells ultimately destroy the thyroid, but during the early stage the inflammation of the gland causes excess thyroid hormone secretion and thyrotoxicosis similar to that seen in Graves disease.

THERAPEUTIC HIGHLIGHTS

Some of the symptoms of hyperthyroidism can be controlled by the **thioureylenes.** These are a group of compounds related to thiourea, which inhibit the iodination of monoiodotyrosine and block the coupling reaction. The two used clinically are propylthiouracil and methimazole. Iodination of tyrosine is inhibited because propylthiouracil and methimazole compete with tyrosine residues for iodine and become iodinated. In addition, propylthiouracil but not methimazole inhibits D_2 deiodinase, reducing the conversion of T_4 to T_3 in many extrathyroidal tissues. In severe cases, hyperthyroidism can also be treated by the infusion of radioactive iodine, which accumulates in the gland and then partially destroys it. Surgery is also considered if the thyroid becomes so large that it affects swallowing and/or breathing.

TABLE 20–5 Causes of hyperthyroidism.

Thyroid overactivity
Graves disease
Solitary toxic adenoma
Toxic multinodular goiter
Early stages of Hashimoto thyroiditis[a]
TSH-secreting pituitary tumor
Mutations causing constitutive activation of TSH receptor
Other rare causes

Extrathyroidal

Administration of T_3 or T_4 (factitious or iatrogenic hyperthyroidism)

Ectopic thyroid tissue

[a]Note that ultimately the thyroid will be destroyed in Hashimoto disease, resulting in hypothyroidism. Many patients only present after they become hypothyroid, and do not recall a transient phase of hyperthyroidism.

CLINICAL BOX 20–3

Thyroid Hormone Resistance

Some mutations in the gene that codes for TRβ are associated with resistance to the effects of T_3 and T_4. Most commonly, there is resistance to thyroid hormones in the peripheral tissues and the anterior pituitary gland. Patients with this abnormality are usually not clinically hypothyroid, because they maintain plasma levels of T_3 and T_4 that are high enough to overcome the resistance, and hTRα is unaffected. However, plasma TSH is inappropriately high relative to the high circulating T_3 and T_4 levels and is difficult to suppress with exogenous thyroid hormone. Some patients have thyroid hormone resistance only in the pituitary. They have hypermetabolism and elevated plasma T_3 and T_4 levels with normal, nonsuppressible levels of TSH. A few patients apparently have peripheral resistance with normal pituitary sensitivity. They have hypometabolism despite normal plasma levels of T_3, T_4, and TSH. An interesting finding is that **attention deficit hyperactivity disorder,** a condition frequently diagnosed in children who are overactive and impulsive, is much more common in individuals with thyroid hormone resistance than in the general population. This suggests that hTRβ may play a special role in brain development.

THERAPEUTIC HIGHLIGHTS

Most patients remain euthyroid in this condition, even in the face of a goiter. It is important to consider thyroid hormone resistance in the differential diagnosis of Graves disease to avoid the inappropriate use of antithyroid medications or

even thyroid ablation. Isolated peripheral resistance to thyroid hormones can be treated by supplying large doses of synthetic T_4 exogenously. These are sufficient to overcome the resistance and increase the metabolic rate.

CHAPTER SUMMARY

- The thyroid gland straddles the trachea in the front of the neck and consists of multiple acini (follicles). Each follicle is surrounded by a single layer of epithelial cells and is filled with colloid, consisting predominantly of thyroglobulin. When the gland is active, areas where the colloid is being reabsorbed are visible (reabsorption lacunae).
- The gland transports and fixes iodide to amino acids present in thyroglobulin to generate the thyroid hormones thyroxine (T_4) and triiodothyronine (T_3). Iodide is transported into the follicle epithelial cells via the Na^+/I^- symporter (NIS) and from there to the colloid via the chloride/iodide exchanger, pendrin. Thyroid peroxidase converts iodide to iodine then incorporates this into the carbon 3 position of tyrosine residues in thyroglobulin in the process of organification.
- Thyroid hormones circulate in the plasma predominantly in protein-bound forms. Only the free hormones are biologically active, and both feed back to reduce secretion of TSH.
- Synthesis and secretion of thyroid hormones is stimulated by thyroid-stimulating hormone (TSH) from the pituitary, which in turn is released in response to thyrotropin-releasing hormone (TRH) from the hypothalamus. These releasing factors are controlled by changes in whole body status (eg, exposure to cold or stress).
- Thyroid hormones exert their effects by entering cells and binding to thyroid receptors. The liganded forms of thyroid receptors are nuclear transcription factors that alter gene expression.
- Thyroid hormones stimulate metabolic rate, calorigenesis, cardiac function, and normal mentation, and interact synergistically with catecholamines. Thyroid hormones also play critical roles in development, particularly of the nervous system, and growth.
- Disease results with both underactivity and overactivity of the thyroid gland. Hypothyroidism is accompanied by mental and physical slowing in adults,

and by mental retardation and dwarfism if it occurs in neonatal life. Overactivity of the thyroid gland, which most commonly is caused by autoantibodies that trigger secretion (Graves disease), results in body wasting, nervousness, and tachycardia.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. A 40-year-old woman comes to her primary care clinician complaining of nervousness and an unexplained weight loss of 20 pounds over the past 3 months despite her impression that she is eating all the time. On physical examination, her eyes are found to be protruding, her skin is moist and warm, and her fingers have a slight tremor. Compared to a healthy individual, a biopsy of her thyroid gland would most likely reveal which of the following:

   A. Decreased numbers of reabsorption lacunae

   B. Decreased evidence of endocytosis

   C. A decrease in the cross-sectional area occupied by colloid

   D. Increased levels of NIS in the basolateral membrane of thyrocytes

   E. Decreased evidence of lysosomal activity

2. Which of the following is *not* essential for normal biosynthesis of thyroid hormones?

   A. Iodine

   B. Ferritin

   C. Thyroglobulin

   D. Protein synthesis

   E. TSH

3. The enzyme primarily responsible for the conversion of T_4 to T_3 in the periphery is

   A. D_1 thyroid deiodinase.

   B. D_2 thyroid deiodinase.

   C. D_3 thyroid deiodinase.

   D. thyroid peroxidase.

   E. None of the above.

4. Increasing intracellular I^- due to the action of NIS is an example of

A. endocytosis.

B. passive diffusion.

C. Na^+ and K^+ cotransport.

D. primary active transport.

E. secondary active transport.

5. The metabolic rate is *least* affected by an increase in the plasma level of

A. TSH.

B. TRH.

C. TBG.

D. Free T_4.

E. Free T_3.

6. In which of the following conditions is it *most* likely that the TSH response to TRH will be reduced?

A. Hypothyroidism due to tissue resistance to thyroid hormone

B. Hypothyroidism due to disease destroying the thyroid gland

C. Hyperthyroidism due to circulating antithyroid antibodies with TSH activity

D. Hyperthyroidism due to diffuse hyperplasia of thyrotropes of the anterior pituitary

E. Iodine deficiency

7. Hypothyroidism due to disease of the thyroid gland is associated with increased plasma levels of

A. cholesterol.

B. albumin.

C. RT_3.

D. iodide.

E. TBG.

8. A young woman has puffy skin and a hoarse voice. Her plasma TSH concentration is low but increases markedly when she is given TRH. She probably has

A. hyperthyroidism due to a thyroid tumor.

B. hypothyroidism due to a primary abnormality in the thyroid gland.

C. hypothyroidism due to a primary abnormality in the pituitary gland.

D. hypothyroidism due to a primary abnormality in the hypothalamus.
E. hyperthyroidism due to a primary abnormality in the hypothalamus.

9. Which of the following would be *least* affected by injections of TSH?
A. Thyroidal uptake of iodine
B. Synthesis of thyroglobulin
C. Cyclic adenosine monophosphate (cAMP) in thyroid cells
D. Cyclic guanosine monophosphate (cGMP) in thyroid cells
E. Size of the thyroid

10. Thyroid hormone receptors bind to DNA in which of the following forms?
A. A heterodimer with the prolactin receptor
B. A heterodimer with the growth hormone receptor
C. A heterodimer with the retinoid X receptor
D. A heterodimer with the insulin receptor
E. A heterodimer with the progesterone receptor

CHAPTER 21

Hormonal Control of Calcium & Phosphate Metabolism & the Physiology of Bone

OBJECTIVES

After studying this chapter, you should be able to:

- Understand the importance of maintaining homeostasis of body calcium and phosphate concentrations.
- Describe the body pools of calcium, their rates of turnover, and the organs that play central roles in regulating movement of calcium between stores.
- Delineate the mechanisms of calcium and phosphate absorption and excretion.
- Identify the major hormones—vitamin D, parathyroid hormone, and calcitonin—and other factors that regulate calcium and phosphate homeostasis, their sites of synthesis, targets of their action, and consequences of dysfunction.
- Define the basic anatomy of bone and understand how linear bone growth is arrested after puberty.
- Delineate the cell types that regulate bone formation and resorption and their mechanism of action, and discuss diseases that result from abnormalities in bone homeostasis.

INTRODUCTION

Calcium is an essential intracellular signaling molecule and also plays a variety of extracellular functions, thus the control of body calcium concentrations is vitally important. The components of the system that maintains calcium homeostasis include cell types that sense changes in extracellular calcium and release calcium-regulating hormones, and the targets of these hormones, including the kidneys, bones, and intestine, that respond with changes in calcium mobilization, excretion, or uptake. Three hormones are primarily concerned with the regulation of calcium homeostasis. **1,25-Dihydroxycholecalciferol** is a steroid hormone formed from vitamin D by successive hydroxylations in the liver and kidneys. Its primary action is to increase calcium absorption from the intestine. **Parathyroid hormone (PTH)** is secreted by the parathyroid glands. Its main action is to mobilize calcium from bone and increase urinary phosphate excretion. **Calcitonin,** a calcium-lowering hormone that in mammals is secreted primarily by cells in the thyroid gland, inhibits bone resorption. Although the role of calcitonin seems to be relatively minor, all three hormones probably operate in concert to maintain the constancy of the calcium level in the body fluids. Phosphate homeostasis is likewise critical to normal body function, particularly given its inclusion in adenosine triphosphate (ATP), its role as a biologic buffer, and its role as a modifier of proteins, thereby altering their functions. Many of the systems that regulate calcium homeostasis also contribute to that of phosphate, albeit sometimes in a reciprocal manner, and thus will also be discussed in this chapter.

CALCIUM & PHOSPHORUS BALANCE IN THE BODY

CALCIUM

The body of a young adult human contains about 1100 g (27.5 moles) of calcium. Ninety-nine percent of the calcium is in the skeleton. Plasma calcium, normally at a concentration of around 10 mg/dL (5 mEq/L, 2.5 mmol/L), is partly bound to protein and partly diffusible **(Table 21–1)**. The distribution of calcium inside the cells and the role of Ca^{2+} as a second messenger molecule are discussed in Chapter 2.

TABLE 21–1 Distribution (mg/dL) of calcium in normal human plasma.

Total diffusible		**5.36**
Ionized (Ca^{2+})	4.72	
Complexed to HCO_3^-, citrate, etc	0.64	
Total nondiffusible (protein-bound)		**4.64**
Bound to albumin	3.68	
Bound to globulin	0.96	
Total plasma calcium		**10.00**

It is the free, ionized calcium (Ca^{2+}) in the body fluids that is a vital second messenger and is necessary for blood coagulation, muscle contraction, and nerve function. A decrease in extracellular Ca^{2+} exerts a net excitatory effect on nerve and muscle cells in vivo (see Chapters 4 and 5). The result is **hypocalcemic tetany,** which is characterized by extensive spasms of skeletal muscle, involving especially the muscles of the extremities and the larynx. Laryngospasm can become so severe that the airway is obstructed and fatal asphyxia is produced. Ca^{2+} also plays an important role in blood clotting (see Chapter 31), but in vivo, fatal tetany would occur before the clotting reaction would be compromised.

Because the extent of Ca^{2+} binding by plasma proteins is proportional to the plasma protein level, it is important to know the plasma protein level when evaluating the total plasma calcium. Other electrolytes and pH also affect the free Ca^{2+} level. Thus, for example, symptoms of tetany can occur at higher total calcium levels if the patient hyperventilates, thereby increasing plasma pH. Plasma proteins are more ionized when the pH is high, providing more protein anions to bind with and sequester free Ca^{2+}.

The calcium in bone is of two types: a readily exchangeable reservoir and a much larger pool of stable calcium that is only slowly exchangeable. Two independent but interacting homeostatic systems affect the calcium in bone. One is the system that regulates plasma Ca^{2+}, providing for the movement of about 500 mmol of Ca^{2+} per day into and out of the readily exchangeable pool in the bone **(Figure 21–1)**. The other system involves bone remodeling by the constant interplay of bone resorption and deposition (see following text). However, the Ca^{2+} interchange between plasma and this stable pool of bone calcium is only about 7.5 mmol/d.

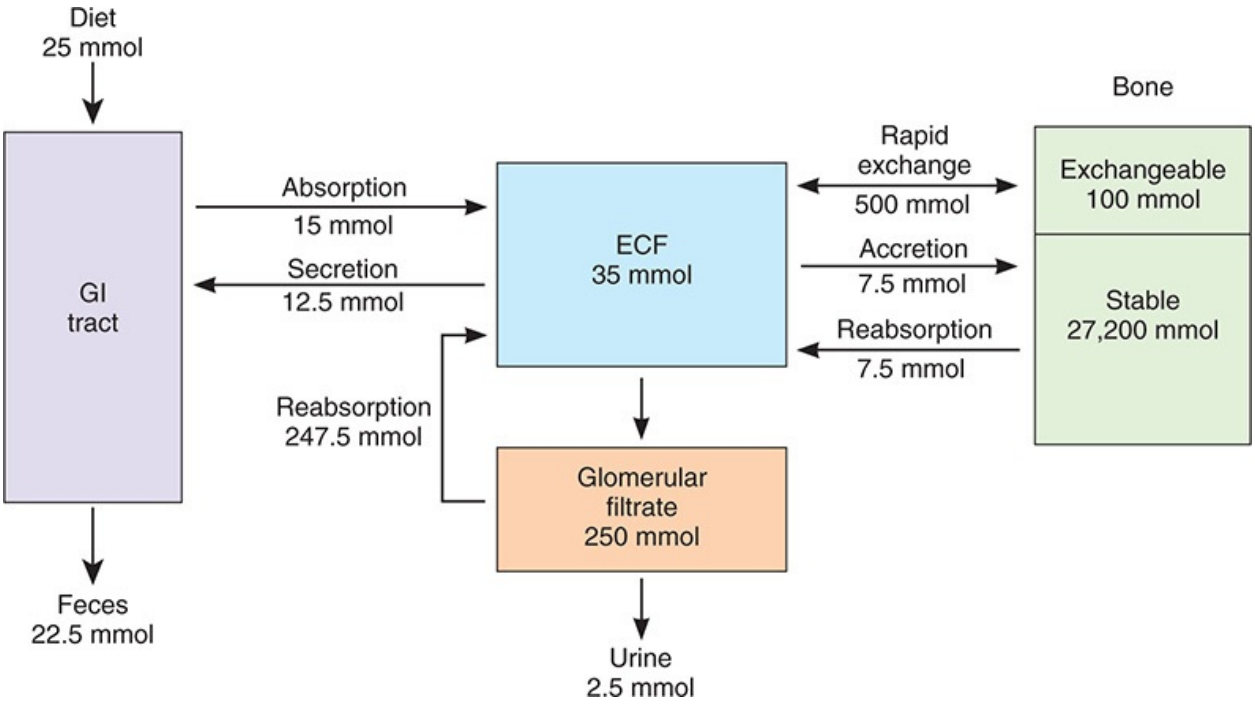


FIGURE 21–1 Calcium metabolism in an adult human. A typical daily dietary intake of 25 mmol Ca^{2+} (1000 mg) moves through many body compartments. Note that the majority of body calcium is in bones, in a pool that is only slowly exchangeable with the extracellular fluid (ECF).

Ca^{2+} is taken up into the body across the brush border of intestinal epithelial cells via channels known as transient receptor potential vanilloid type 6 (TRPV6) and binds to an intracellular protein known as calbindin-D_{9k}. Calbindin-D_{9k} sequesters the absorbed calcium so that it does not disturb epithelial signaling processes that involve calcium. The absorbed Ca^{2+} is thereby delivered to the basolateral membrane of the epithelial cell, from where it can be transported into the bloodstream by either a Na^+/Ca^{2+} exchanger (NCX1) or the Ca^{2+}-Mg^{2+} ATPase, $PMCA_{1b}$ **(Figure 21-2)**. Nevertheless, it should be noted that recent studies indicate that some intestinal Ca^{2+} uptake persists even in the absence of TRPV6 and calbindin-D_{9k}, suggesting that additional pathways are likely also involved in this critical process. The overall transport process is regulated by 1,25-dihydroxycholecalciferol (see below). As Ca^{2+} uptake rises, moreover, 1,25-dihydroxycholecalciferol levels fall in response to increased plasma Ca^{2+}.

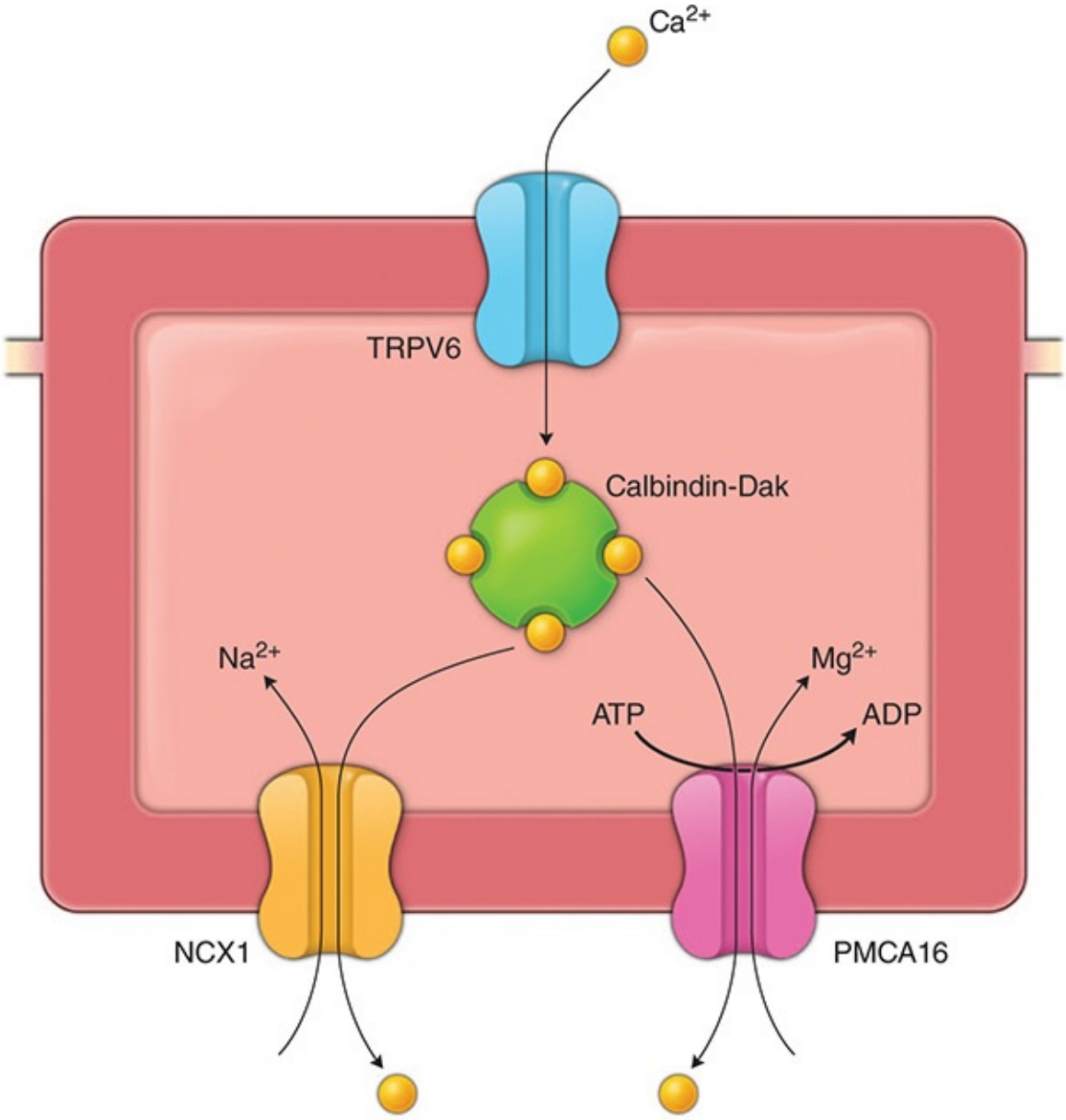


FIGURE 21–2 Intestinal absorption of calcium. Calcium is taken up across the enterocyte apical membrane via TRPV6 calcium channels. It is sequestered in the cytosol by calbindin-D_{9k} and trafficked to the basolateral membrane, from which it is exported to the bloodstream by either NCX1 (taking advantage of the low intracellular sodium concentration to drive calcium efflux) or $PMCA_{1b}$, with expenditure of cellular energy. The expression levels of TRPV6, calbindin-D_{9k}, and $PMCA_{1b}$ are all upregulated by 1,25-dihydroxycholecalciferol.

Plasma Ca^{2+} is filtered in the kidneys, but 98–99% of the filtered Ca^{2+} is reabsorbed. About 60% of the reabsorption occurs in the proximal tubules and

the remainder in the ascending limb of the loop of Henle and the distal tubule. Distal tubular reabsorption depends on the TRPV5 channel, whose expression is regulated by PTH.

PHOSPHORUS

Phosphate is found in ATP, cyclic adenosine monophosphate (cAMP), 2,3-diphosphoglycerate, many proteins, and other vital compounds in the body. Phosphorylation and dephosphorylation of proteins are involved in the regulation of cell function (see Chapter 2). Therefore, it is not surprising that, like calcium, phosphate metabolism is closely regulated. Total body phosphorus is 500–800 g (16.1–25.8 moles), 85–90% of which is in the skeleton. Total plasma phosphorus is about 12 mg/dL, with two-thirds of this total in organic compounds and the remainder as inorganic phosphorus (P_i) (mostly in PO_4^{3-}, HPO_4^{2-}, and $H_2PO_4^-$). The amount of phosphorus normally entering bone is about 3 mg (97 µmol)/kg/d, with an equal amount leaving via resorption.

P_i in the plasma is filtered in the glomeruli, and 85–90% of the filtered P_i is reabsorbed. Active transport in the proximal tubule accounts for most of the reabsorption and involves two related sodium-dependent P_i cotransporters, NaPi-IIa and NaPi-IIc. NaPi-IIa is powerfully inhibited by PTH, which causes its internalization and degradation and thus a reduction in renal P_i reabsorption (see below).

P_i is absorbed in the duodenum and small intestine. Uptake occurs by a transporter related to those in the kidney, NaPi-IIb, that takes advantage of the low intracellular Na^+ concentration established by the Na, K ATPase on the basolateral membrane of intestinal epithelial cells to load P_i against its concentration gradient. However, the pathway by which P_i exits into the bloodstream is not known. Many stimuli that increase Ca^{2+} absorption, including 1,25-dihydroxycholecalciferol, also increase P_i absorption via increased NaPi-IIb expression and/or its insertion into the enterocyte apical membrane.

VITAMIN D & THE HYDROXYCHOLECALCIFEROLS

CHEMISTRY

The active transport of Ca^{2+} and PO_4^{3-} from the intestine is increased by a metabolite of **vitamin D.** The term "vitamin D" is used to refer to a group of closely related sterols produced by the action of ultraviolet light on certain provitamins **(Figure 21–3)**. Vitamin D_3, which is also called cholecalciferol, is produced in the skin of mammals from 7-dehydrocholesterol by the action of sunlight. The reaction involves the rapid formation of previtamin D_3, which is then converted more slowly to vitamin D_3. Vitamin D_3 and its hydroxylated derivatives are transported in the plasma bound to vitamin D-binding protein (DBP). Vitamin D_3 is also ingested in the diet.

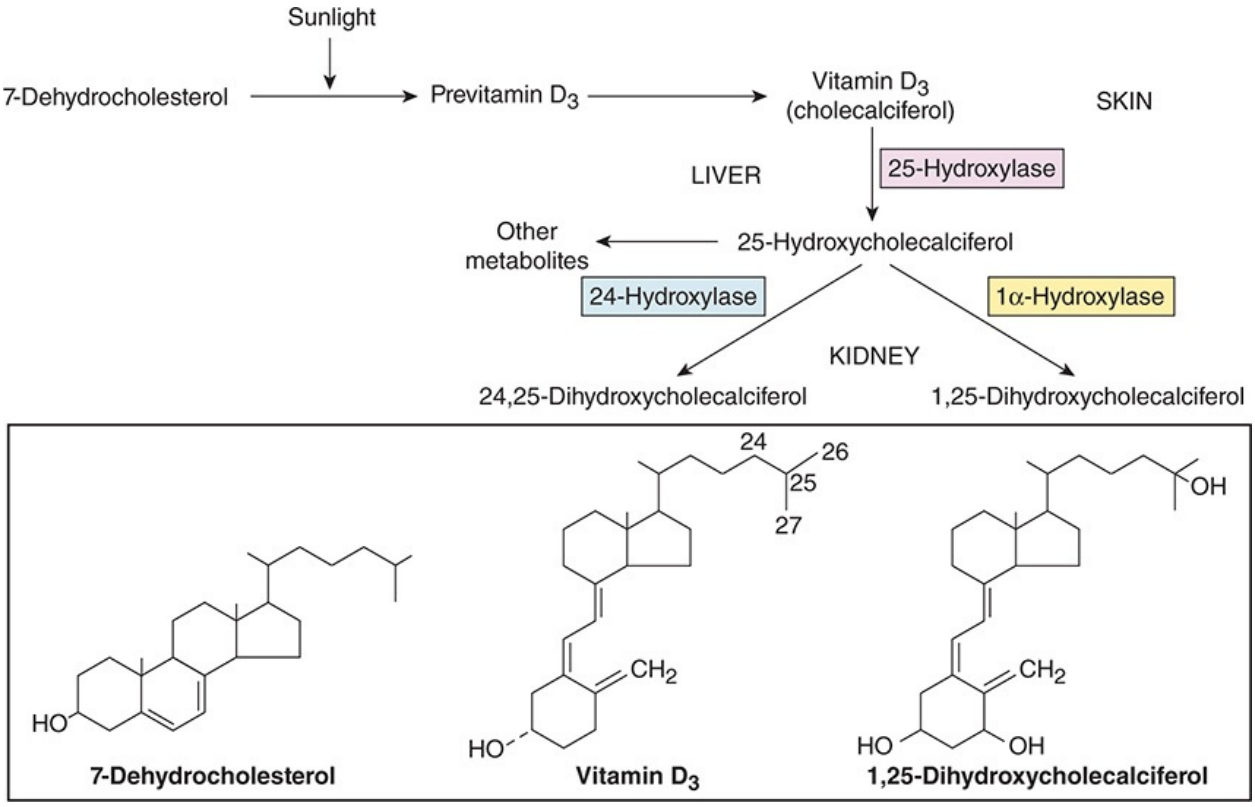


FIGURE 21–3 Formation and hydroxylation of vitamin D_3. 25-Hydroxylation takes place in the liver, and the other hydroxylations occur primarily in the kidneys. The structures of 7-dehydrocholesterol, vitamin D_3, and 1,25-dihydroxycholecalciferol are also shown in the boxed area.

Vitamin D_3 is metabolized by enzymes that are members of the cytochrome P450 (CYP) superfamily (see Chapters 1 and 28). In the liver, vitamin D_3 is converted to **25-hydroxycholecalciferol** (calcidiol, 25-(OH)D_3). The 25-

hydroxycholecalciferol is further converted in the cells of the proximal tubules of the kidneys to the more active metabolite **1,25-dihydroxycholecalciferol,** which is also called calcitriol or 1,25-$(OH)_2D_3$. 1,25-Dihydroxycholecalciferol is also made in the placenta, in keratinocytes in the skin, and in macrophages. The normal plasma level of 25-hydroxycholecalciferol is about 30 ng/mL, and that of 1,25-dihydroxycholecalciferol is about 0.03 ng/mL (approximately 100 pmol/L). The less active metabolite 24,25-dihydroxycholecalciferol is also formed in the kidneys (Figure 21–3).

MECHANISM OF ACTION

1,25-Dihydroxycholecalciferol stimulates the expression of a number of gene products involved in Ca^{2+} transport and handling via its receptor (VDR), which acts as a transcriptional regulator in its ligand-bound form. One target is the family of **calbindin-D** proteins. These are members of the troponin C superfamily of Ca^{2+}-binding proteins that also includes calmodulin (see Chapter 2). Calbindin-Ds are found in human intestine, brain, and kidneys. In the intestinal epithelium and many other tissues, two calbindins are induced: calbindin-D_{9K} and calbindin-D_{28K}, with molecular weights of 9000 and 28,000, respectively. 1,25-Dihydroxycholecalciferol also increases the number of Ca^{2+}–ATPase and TRPV6 molecules in the intestinal cells, and thus, the overall capacity for absorption of dietary calcium.

In addition to increasing Ca^{2+} absorption from the intestine, 1,25-dihydroxycholecalciferol facilitates Ca^{2+} reabsorption in the kidneys via increased TRPV5 expression in the proximal tubules, increases the synthetic activity of osteoblasts, and is necessary for normal calcification of matrix **(Clinical Box 21–1)**. The stimulation of osteoblasts brings about a secondary increase in the activity of osteoclasts (see below).

REGULATION OF SYNTHESIS

The formation of 25-hydroxycholecalciferol does not appear to be stringently regulated. However, the formation of 1,25- dihydroxycholecalciferol in the kidneys, which is catalyzed by the renal 1α-hydroxylase, is regulated in a feedback manner by plasma Ca^{2+} and PO_4^{3+} **(Figure 21–4)**. When the plasma Ca^{2+} level is high, little 1,25-dihydroxycholecalciferol is produced, and the

kidneys produce the relatively inactive metabolite 24,25-dihydroxycholecalciferol instead. This effect of Ca^{2+} on production of 1,25-dihydroxycholecalciferol thereby reduces the signal for Ca^{2+} absorption from the intestine (see previous text). Conversely, when the plasma Ca^{2+} level is low, PTH secretion is increased, and expression of 1α-hydroxylase is stimulated by PTH. The production of 1,25-dihydroxycholecalciferol is also increased by low plasma PO_4^{3-} levels and inhibited by high plasma PO_4^{3-} levels, by a direct inhibitory effect of PO_4^{3-} on the 1α-hydroxylase. Additional control of 1,25-dihydroxycholecalciferol formation results from its direct negative feedback effect on 1α-hydroxylase activity, a positive feedback action on the formation of 24,25-dihydroxycholecalciferol, and a direct action on the parathyroid gland to inhibit PTH expression.

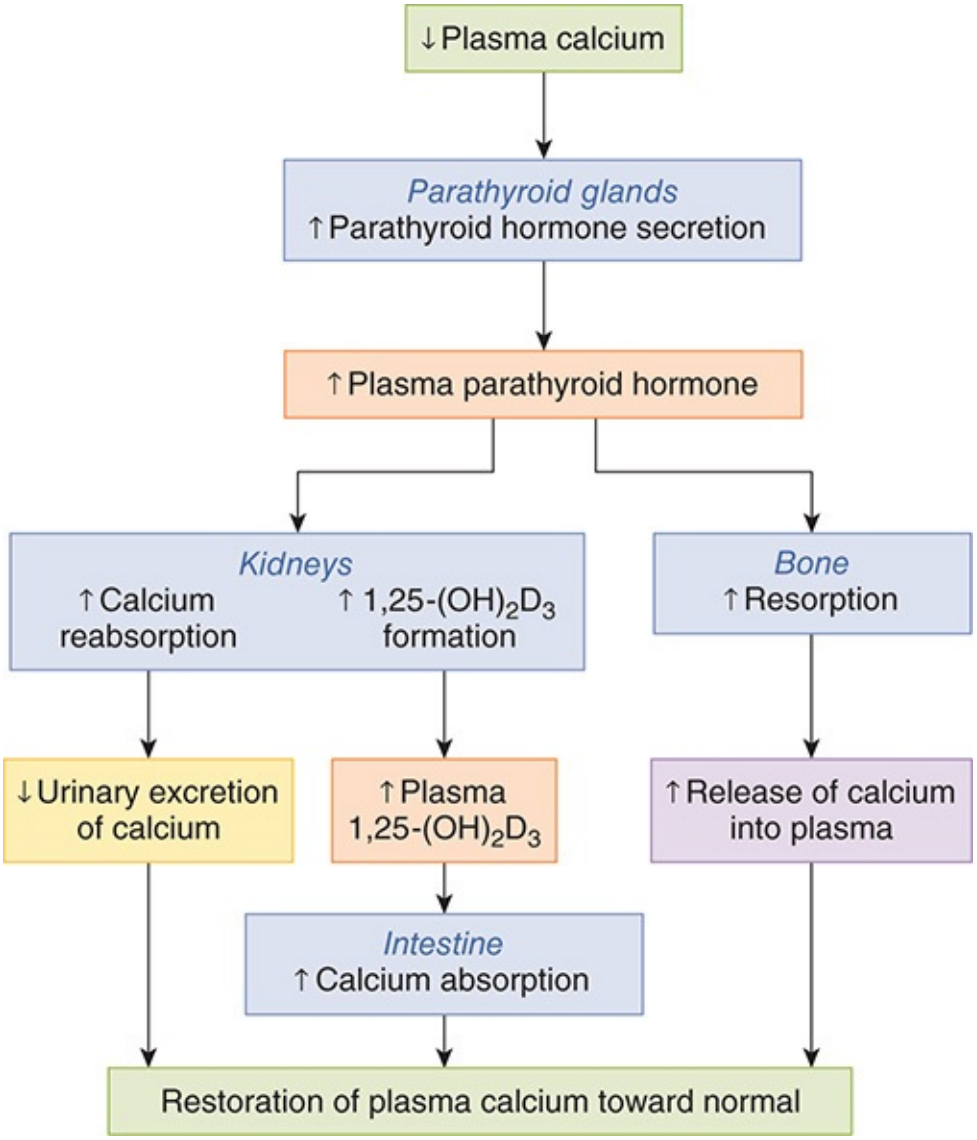


FIGURE 21–4 Effects of PTH and 1,25-dihydroxycholecalciferol on whole body calcium homeostasis. A reduction in plasma calcium stimulates parathyroid hormone secretion. PTH in turn causes calcium conservation and production of 1,25-dihydroxycholecalciferol in the kidneys, the latter of which increases calcium uptake in the intestine. PTH also releases calcium from the readily exchangeable pool in the bone. All of these actions act to restore normal plasma calcium. (Reproduced with permission from Widmaier EP, Raff H, Strang KT: *Vander's Human Physiology*, 10th ed. New York, NY: McGraw-Hill; 2006.)

CLINICAL BOX 21–1

Rickets & Osteomalacia

Vitamin D deficiency causes defective calcification of bone matrix and the disease called **rickets** in children and **osteomalacia** in adults. Even though 1,25-dihydroxycholecalciferol is necessary for normal mineralization of bone matrix, the main defect in this condition is failure to deliver adequate amounts of Ca^{2+} and PO_4^{3-} to the sites of mineralization. The full-blown condition in children is characterized by weakness and bowing of weight-bearing bones, dental defects, and hypocalcemia. In adults, the condition is less obvious. It used to be most commonly due to inadequate exposure to the sun in smoggy cities, but now it is more commonly due to inadequate intake of the provitamins on which the sun acts in the skin. These cases respond to administration of vitamin D. The condition can also be caused by inactivating mutations of the gene for renal 1α-hydroxylase, or in severe kidney or liver diseases, in which case there is no response to vitamin D but a normal response to 1,25- dihydroxycholecalciferol **(type I vitamin D–resistant rickets).** In rare instances, it can be due to inactivating mutations of the gene for the VDR **(type II vitamin D–resistant rickets),** in which case there is a deficient response to both vitamin D and 1,25-dihydroxycholecalciferol.

THERAPEUTIC HIGHLIGHTS

Treatment of these conditions depends on the underlying biochemical basis, as indicated above. Routine supplementation of milk with vitamin D has greatly reduced the occurrence of rickets in Western countries, but the condition remains among the most common childhood diseases in developing countries. Orthopedic surgery may be necessary in severely affected children.

An “anti-aging” protein called α-Klotho has also recently been discovered to play important roles in calcium and phosphate homeostasis, in part by reciprocal effects on 1,25- dihydroxycholecalciferol levels. Mice deficient in α-Klotho displayed accelerated aging, decreased bone mineral density, calcifications, and hypercalcemia and hyperphosphatemia. α-Klotho stabilizes the membrane localization of proteins important in calcium and phosphate (re)absorption, such as TRPV5 and Na, K ATPase. Likewise, it enhances the activity of another factor, fibroblast growth factor 23 (FGF23), at its receptor. FGF23 thereby decreases renal NaPi-IIa and NaPi-IIc expression and inhibits the production of 1α-hydroxylase, reducing levels of 1,25-dihydroxycholecalciferol.

THE PARATHYROID GLANDS

ANATOMY

Humans usually have four parathyroid glands: two embedded in the superior poles of the posterior thyroid and two in its inferior poles **(Figure 21–5)**. Each parathyroid gland is a richly vascularized disk, about 3 × 6 × 2 mm, containing two distinct types of cells **(Figure 21–6)**. The abundant **chief cells,** which contain a prominent Golgi apparatus plus endoplasmic reticulum and secretory granules, synthesize and secrete **PTH.** The less abundant and larger **oxyphil cells** contain oxyphil granules and large numbers of mitochondria in their cytoplasm. In humans, few oxyphil cells are seen before puberty, and thereafter they increase in number with age. Their function is unknown. Consequences of loss of the parathyroid glands are discussed in **Clinical Box 21–2**.

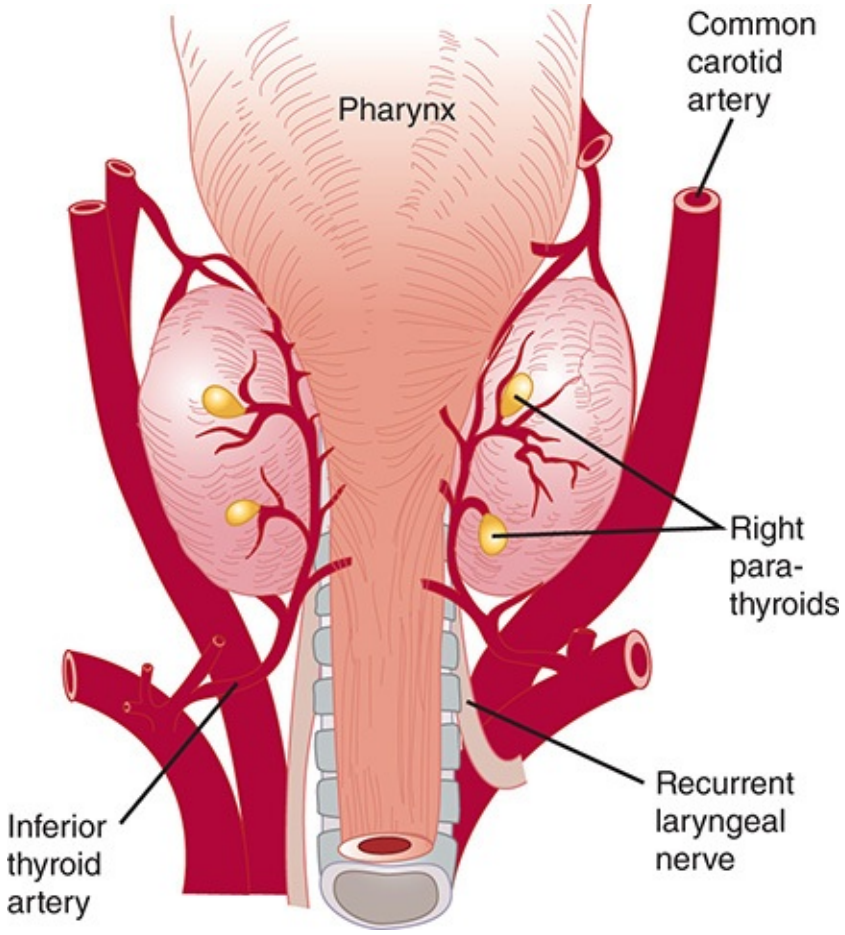


FIGURE 21–5 The human parathyroid glands, viewed from behind. The glands are small structures adherent to the posterior surface of the thyroid gland.

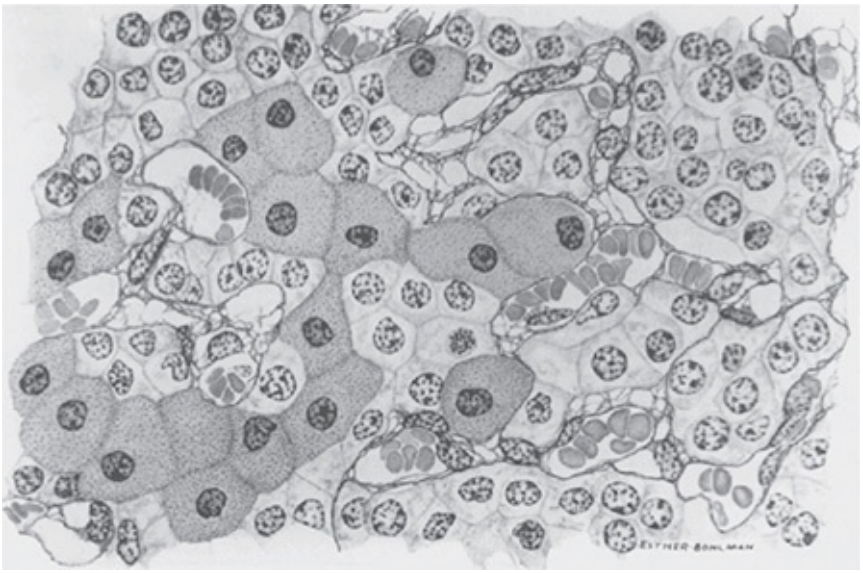


FIGURE 21–6 Section of human parathyroid. (Reduced 50% from ×960.) Small cells are chief cells; large stippled cells (especially prominent in the lower left of picture) are oxyphil cells. (Reproduced with permission from Fawcett DW: *Bloom and Fawcett, A Textbook of Histology*, 11th ed. Saunders, 1986.)

CLINICAL BOX 21–2

Effects of Parathyroidectomy

Occasionally, inadvertent parathyroidectomy occurs in humans during thyroid surgery. This can have serious consequences as PTH is essential for life. After parathyroidectomy, there is a steady decline in the plasma Ca^{2+} level. Signs of neuromuscular hyperexcitability appear, followed by full-blown hypocalcemic tetany (see text). Plasma phosphate levels usually rise as the plasma Ca^{2+} level falls. Symptoms usually develop 2–3 days postoperatively but may not appear for several weeks or more. The signs of tetany in humans include **Chvostek sign,** a quick contraction of the ipsilateral facial muscles elicited by tapping over the facial nerve at the angle of the jaw, and **Trousseau sign,** a spasm of the muscles of the upper extremity that causes flexion of the wrist and thumb with extension of the fingers. In individuals with mild tetany in whom spasm is not yet evident, Trousseau sign can sometimes be produced by occluding the circulation for a few minutes with a blood pressure cuff.

THERAPEUTIC HIGHLIGHTS

Treatment centers around replacing the PTH that would normally be produced by the missing glands. Injections of PTH can be given to correct the chemical abnormalities, and the symptoms then disappear. Injections of Ca^{2+} salts can also give temporary relief.

SYNTHESIS & METABOLISM OF PTH

Human PTH is a linear polypeptide with a molecular weight of 9500 that contains 84 amino acid residues **(Figure 21–7)**. It is synthesized as part of a larger molecule containing 115 amino acid residues **(preproPTH).** On entry of preproPTH into the endoplasmic reticulum, a leader sequence is removed from the amino terminal to form the 90-amino-acid polypeptide **proPTH.** Six additional amino acid residues are removed from the amino terminal of proPTH in the Golgi apparatus, and the 84-amino-acid polypeptide PTH is packaged in secretory granules and released as the main secretory product of the chief cells.

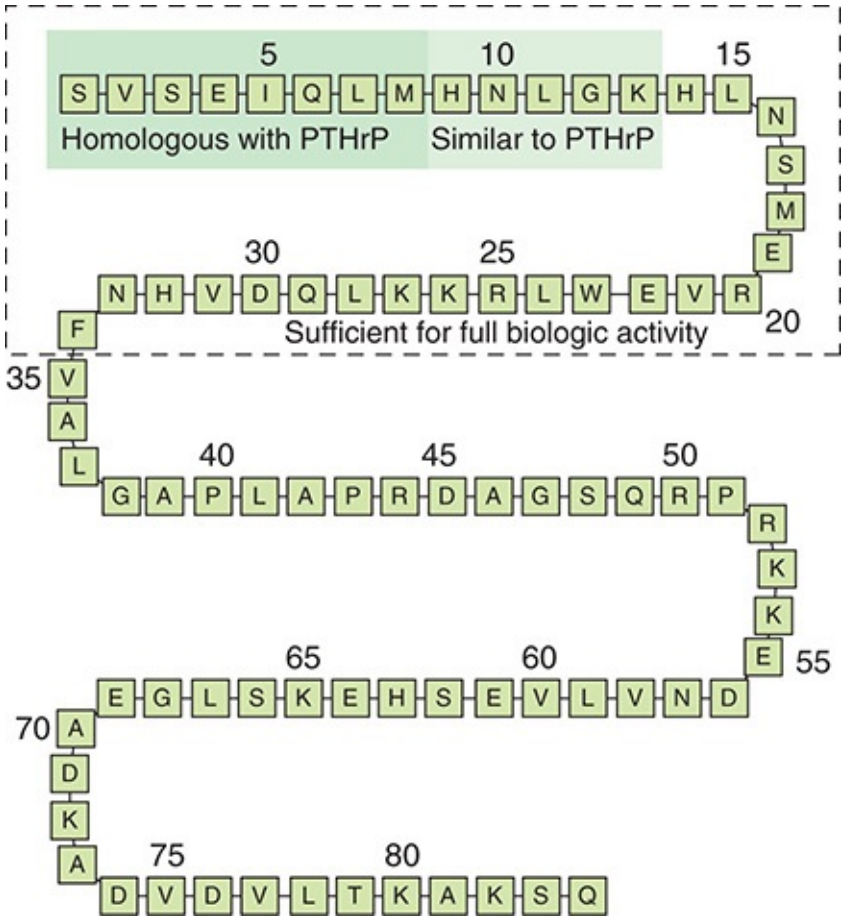


FIGURE 21–7 Parathyroid hormone. The solid boxes show residues that are identical to, or similar to, those at the N-terminus of parathyroid hormone–related peptide (PTHrP). The dashed box indicates that the first 34 amino acids of PTH are sufficient for full biologic activity. Interestingly, this is also the case for PTHrP, although after the first 13 amino acids, its sequence is divergent from that of PTH.

The normal plasma level of intact PTH is 10–55 pg/mL. The half-life of PTH is approximately 10 min, and the secreted polypeptide is rapidly cleaved by Kupffer cells in the liver into inactive fragments. PTH and these fragments are then cleared by the kidneys. Currently used immunoassays for PTH are designed only to measure mature PTH (ie, 84 amino acids) and not its fragments in order to obtain an accurate measure of "active" PTH.

ACTIONS

PTH acts directly on bone to increase bone resorption and mobilize Ca^{2+}. In

addition to increasing plasma Ca^{2+}, PTH increases phosphate excretion in the urine and thereby depresses plasma phosphate levels. This **phosphaturic action** is due to a decrease in reabsorption of phosphate via effects on NaPi-IIa in the proximal tubules, as discussed previously. PTH also increases reabsorption of Ca^{2+} in the distal tubules, although Ca^{2+} excretion in the urine is often increased in hyperparathyroidism because the increase in the load of filtered calcium overwhelms the effect on reabsorption **(Clinical Box 21–3)**. PTH also increases the formation of 1,25- dihydroxycholecalciferol, and this increases Ca^{2+} absorption from the intestine. On a longer time scale, PTH stimulates both osteoblasts and osteoclasts.

MECHANISM OF ACTION

It now appears that there are at least three different PTH receptors. One also binds parathyroid hormone–related protein (PTHrP; see below) and is known as the hPTH/PTHrP receptor. A second receptor, PTH2 (hPTH2-R), does not bind PTHrP and is found in the brain, placenta, and pancreas. In addition, there is evidence for a third receptor, CPTH, which reacts with the carboxyl terminal rather than the amino terminal of PTH. The first two receptors are coupled to G_s, and via this heterotrimeric G-protein they activate adenylyl cyclase, increasing intracellular cAMP. The hPTH/PTHrP receptor also activates PLC via G_q, increasing intracellular Ca^{2+} concentrations and activating protein kinase C **(Figure 21–8)**. However, the way these second messengers affect Ca^{2+} in bone is unsettled.

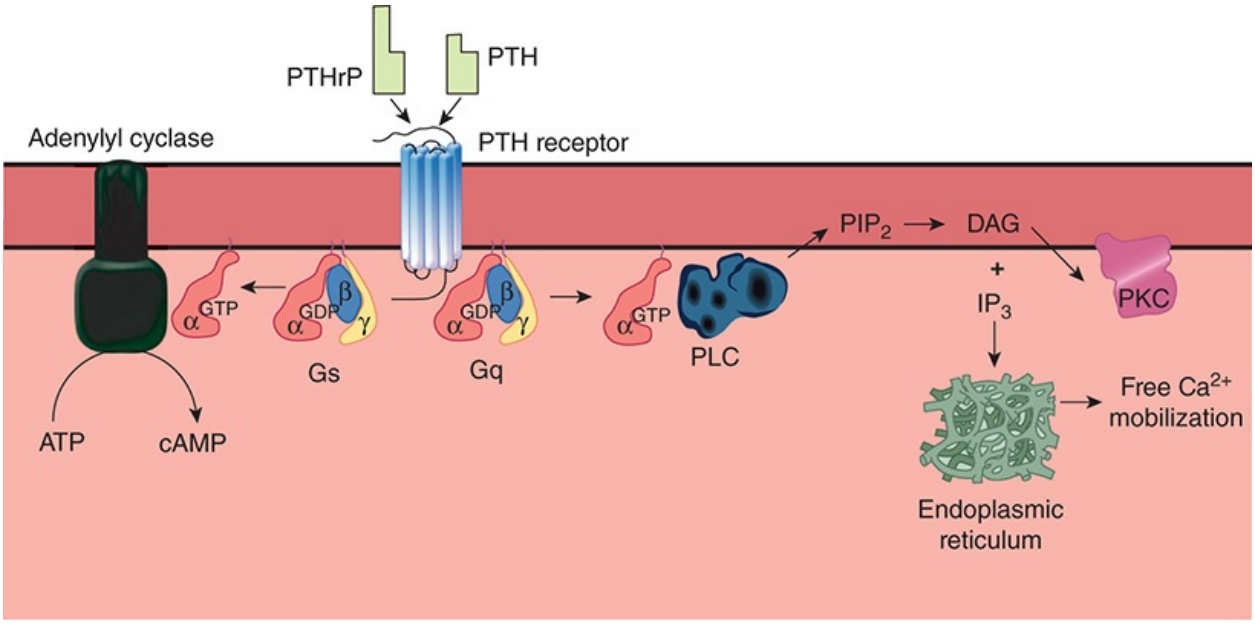


FIGURE 21–8 Signal transduction pathways activated by PTH or PTHrP binding to the hPTH/PTHrP receptor. Intracellular cAMP is increased via G_s and adenylyl cyclase (AC). Diacylglycerol (DAG) and inositol 1,4,5-trisphosphate (IP_3) are increased via G_q and phospholipase C (PLC). DAG and IP_3 activate protein kinase C (PKC) and mobilization of calcium from the endoplasmic reticulum.

CLINICAL BOX 21–3

Diseases of Parathyroid Excess

Hyperparathyroidism due to hypersecretion from a functioning parathyroid tumor in humans is characterized by hypercalcemia and hypophosphatemia. Humans with PTH-secreting adenomas are usually asymptomatic, with the condition detected when plasma Ca^{2+} is measured in conjunction with a routine physical examination. However, there may be minor changes in personality, and calcium-containing kidney stones occasionally form. In conditions such as chronic kidney disease and rickets, in which the plasma Ca^{2+} level is chronically low, the associated stimulation of the parathyroid glands causes compensatory parathyroid hypertrophy and secondary hyperparathyroidism. The plasma Ca^{2+} level is low in chronic kidney disease primarily because the diseased kidneys lose the ability to form 1,25-dihydroxycholecalciferol. Finally, mutations in the Ca^{2+}-sensing receptor

gene *CASR* cause predictable long-term changes in plasma Ca^{2+}. Individuals heterozygous for inactivating mutations have familial benign hypocalciuric hypercalcemia, a condition in which there is a chronic moderate elevation in plasma Ca^{2+} because the feedback inhibition of PTH secretion by Ca^{2+} is reduced. Plasma PTH levels are normal or even elevated. However, children who are homozygous for inactivating mutations develop neonatal severe primary hyperparathyroidism. Conversely, individuals with gain-of-function mutations in the *CASR* gene develop familial hypercalciuric hypocalcemia due to increased sensitivity of the parathyroid glands to plasma Ca^{2+}.

THERAPEUTIC HIGHLIGHTS

Subtotal parathyroidectomy is sometimes necessary in patients in whom parathyroid adenoma or hyperplasia with associated hypercalcemia and resulting symptoms develops. However, because parathyroid disease is often benign or only slowly progressing, surgery remains controversial in most patients and is typically reserved for those who have experienced life-threatening complications of hypercalcemia.

In the disease called **pseudohypoparathyroidism,** the signs and symptoms of hypoparathyroidism develop but the circulating level of PTH is normal or even elevated. Because tissues fail to respond to the hormone, this is a receptor disease. There are two forms. In the more common form, a congenital 50% reduction of the activity of G_s occurs and PTH fails to produce a normal increase in cAMP concentration. In a different, less common form, the cAMP response is normal but the phosphaturic action of the hormone is defective.

REGULATION OF SECRETION

Circulating Ca^{2+} acts directly on the parathyroid glands in a negative feedback manner to regulate the secretion of PTH. The key to this regulation is a cell membrane Ca^{2+} sensing receptor, CaSR. Activation of this G-protein-coupled receptor leads to phosphoinositide turnover in many tissues. In the parathyroid, its activation inhibits PTH secretion. In this way, when the plasma Ca^{2+} level is high, PTH secretion is inhibited and Ca^{2+} is deposited in the bones. When it is low, secretion is increased and Ca^{2+} is mobilized from the bones.

1,25-Dihydroxycholecalciferol acts directly on the parathyroid glands to decrease preproPTH mRNA. Increased plasma P_i stimulates PTH secretion by lowering plasma levels of free Ca^{2+} and inhibiting the formation of 1,25-dihydroxycholecalciferol. Magnesium is required to maintain normal parathyroid secretory responses. Impaired PTH release along with diminished target organ responses to PTH account for the hypocalcemia that occasionally occurs in magnesium deficiency (Clinical Boxes 21–2 and 21–3).

PTHrP

PTHrP, another protein with PTH activity, is produced by many different tissues in the body. PTHrP and PTH have marked homology at their amino terminal ends and they both bind to the hPTH/PTHrP receptor, yet their physiologic effects are very different. How is this possible when they bind to the same receptor? For one thing, PTHrP is primarily a paracrine factor, acting close to where it is produced. It may be that circulating PTH cannot reach at least some of these sites. Second, subtle conformational differences may be produced by binding of PTH versus PTHrP to their receptor, despite their structural similarities. Another possibility is action of one or the other hormone on additional, more selective receptors.

PTHrP has a marked effect on the growth and development of cartilage in utero. Mice in which both alleles of the PTHrP gene are knocked out have severe skeletal deformities and die soon after birth. In normal animals, on the other hand, PTHrP-stimulated cartilage cells proliferate and their terminal differentiation is inhibited. PTHrP is also expressed in the brain, where evidence indicates that it inhibits excitotoxic damage to developing neurons. In addition, there is evidence that it is involved in Ca^{2+} transport in the placenta. PTHrP is also found in keratinocytes in the skin, in smooth muscle, and in the teeth, where it is present in the enamel epithelium that caps each tooth. In the absence of PTHrP, teeth cannot erupt.

HYPERCALCEMIA OF MALIGNANCY

Hypercalcemia is a common metabolic complication of cancer. About 20% of hypercalcemic patients have bone metastases that produce hypercalcemia secondary to bone erosion **(local osteolytic hypercalcemia).** Evidence suggests that this erosion is produced by prostaglandins such as prostaglandin E_2 arising

from the tumor. Hypercalcemia in the remaining 80% of the patients is due to elevated circulating levels of PTHrP **(humoral hypercalcemia of malignancy).** The tumors responsible for this hypersecretion include cancers of the breast, kidney, ovary, and skin.

CALCITONIN

ORIGIN

In dogs, perfusion of the thyroparathyroid region with solutions containing high concentrations of Ca^{2+} leads to a fall in peripheral plasma Ca^{2+}, and after damage to this region, Ca^{2+} infusions cause a greater increase in plasma Ca^{2+} than they do in control animals. These and other observations led to the discovery that a Ca^{2+}-lowering as well as a Ca^{2+}-elevating peptide hormone was secreted by structures in the neck. This Ca^{2+}-lowering hormone has been named **calcitonin.** In mammals, calcitonin is produced by the **parafollicular cells** of the thyroid gland, which are also known as the clear or C cells.

SECRETION & METABOLISM

Calcitonin secretion is increased when the thyroid gland is exposed to a plasma calcium level of approximately 9.5 mg/dL. Above this level, plasma calcitonin is directly proportional to plasma calcium. β-Adrenergic agonists, dopamine, and estrogens also stimulate calcitonin secretion. Gastrin, cholecystokinin (CCK), glucagon, and secretin have also been reported to stimulate calcitonin secretion, with gastrin being the most potent stimulus (see Chapter 25). Thus, the plasma calcitonin level is elevated in Zollinger–Ellison syndrome and in pernicious anemia (see Chapter 25). However, the dose of gastrin needed to stimulate calcitonin secretion is supraphysiologic and not seen after eating in normal individuals, so dietary calcium in the intestine probably does not induce secretion of a calcium-lowering hormone prior to the calcium being absorbed. In any event, the actions of calcitonin are short-lived because it has a half-life of less than 10 min in humans.

ACTIONS

Receptors for calcitonin are found in bones and the kidneys. Calcitonin lowers

circulating calcium and phosphate levels. It exerts its calcium-lowering effect by inhibiting bone resorption. This action is direct, and calcitonin also inhibits the activity of osteoclasts in vitro. It also increases Ca^{2+} excretion in the urine.

The exact physiologic role of calcitonin is uncertain. The calcitonin content of the human thyroid is low, and after thyroidectomy, bone density and plasma Ca^{2+} level are normal as long as the parathyroid glands are intact. In addition, after thyroidectomy, there are only transient abnormalities of Ca^{2+} homeostasis when a Ca^{2+} load is injected. This may be explained in part by secretion of calcitonin from tissues other than the thyroid. However, there is general agreement that the hormone has little long-term effect on the plasma Ca^{2+} level in adult animals and humans. Further, unlike PTH and 1,25-dihydroxycholecalciferol, calcitonin does not appear to be involved in phosphate homeostasis. Moreover, patients with medullary carcinoma of the thyroid have a very high circulating calcitonin level but no symptoms directly attributable to the hormone, and their bones are essentially normal. No syndrome due to calcitonin deficiency has been described. More hormone is secreted in young individuals, and it may play a role in skeletal development. In addition, it may protect the bones of the mother from excess calcium loss during pregnancy. Bone formation in the infant and lactation are major drains on Ca^{2+} stores, and plasma concentrations of 1,25-dihydroxycholecalciferol are elevated in pregnancy. They would cause bone loss in the mother if bone resorption were not simultaneously inhibited by an increase in the plasma calcitonin level.

SUMMARY OF CALCIUM HOMEOSTATIC MECHANISMS

The actions of the three principal hormones that regulate the plasma concentration of Ca^{2+} can now be summarized. PTH increases plasma Ca^{2+} by mobilizing this ion from bone. It increases Ca^{2+} reabsorption in the kidney, but this may be offset by the increase in filtered Ca^{2+}. It also increases the formation of 1,25-dihydroxycholecalciferol. 1,25-Dihydroxycholecalciferol increases Ca^{2+} absorption from the intestine and increases Ca^{2+} reabsorption in the kidneys. Calcitonin inhibits bone resorption and increases the amount of Ca^{2+} in the urine.

EFFECTS OF OTHER HORMONES & HUMORAL

AGENTS ON CALCIUM METABOLISM

Calcium metabolism is affected by various hormones in addition to 1,25-dihydroxycholecalciferol, PTH, and calcitonin. **Glucocorticoids** lower plasma Ca^{2+} levels by inhibiting osteoclast formation and activity acutely, but over long periods they cause osteoporosis by decreasing bone formation and increasing bone resorption. They decrease bone formation by inhibiting protein synthesis in osteoblasts. They also decrease the absorption of Ca^{2+} and PO_4^{3-} from the intestine and increase the renal excretion of these ions. The decrease in plasma Ca^{2+} concentration also increases the secretion of PTH, and bone resorption is facilitated. **Growth hormone** increases Ca^{2+} excretion in the urine, but it also increases intestinal absorption of Ca^{2+}, and this effect may be greater than the effect on excretion, with a resultant positive calcium balance. Insulin-like growth factor I (IGF-I) generated by the action of growth hormone stimulates protein synthesis in bone. As noted previously, **thyroid hormones** may cause hypercalcemia, hypercalciuria, and, in some instances, osteoporosis. **Estrogens** prevent osteoporosis by inhibiting the stimulatory effects of certain cytokines on osteoclasts. **Insulin** increases bone formation, and there is significant bone loss in untreated diabetes.

BONE PHYSIOLOGY

Bone is a special form of connective tissue with a collagen framework impregnated with Ca^{2+} and PO_4^{3-} salts, particularly **hydroxyapatites,** which have the general formula $Ca_{10}(PO_4)_6(OH)_2$. Bone is also involved in overall Ca^{2+} and PO_4^{3-} homeostasis. It protects vital organs, and the rigidity it provides permits locomotion and the support of loads against gravity. Old bone is constantly being resorbed and new bone formed, permitting remodeling that allows it to respond to the stresses and strains that are put upon it. It is a living tissue that is well vascularized and has a total blood flow of 200–400 mL/min in adult humans.

STRUCTURE

There are two types of bone: **compact** or **cortical bone,** which makes up the outer layer of most bones **(Figure 21–9)** and accounts for 80% of the bone in the

body; and **trabecular** or **spongy bone** inside the cortical bone, which makes up the remaining 20% of bone in the body. In compact bone, the surface-to-volume ratio is low, and bone cells lie in lacunae. They receive nutrients by way of canaliculi that ramify throughout the compact bone (Figure 21–9). Trabecular bone is made up of spicules or plates, with a high surface to volume ratio and many cells sitting on the surface of the plates. Nutrients diffuse from bone extracellular fluid (ECF) into the trabeculae, but in compact bone, nutrients are provided via **haversian canals** (Figure 21–9), which contain blood vessels. Around each haversian canal, collagen is arranged in concentric layers, forming cylinders called **osteons** or **haversian systems.**

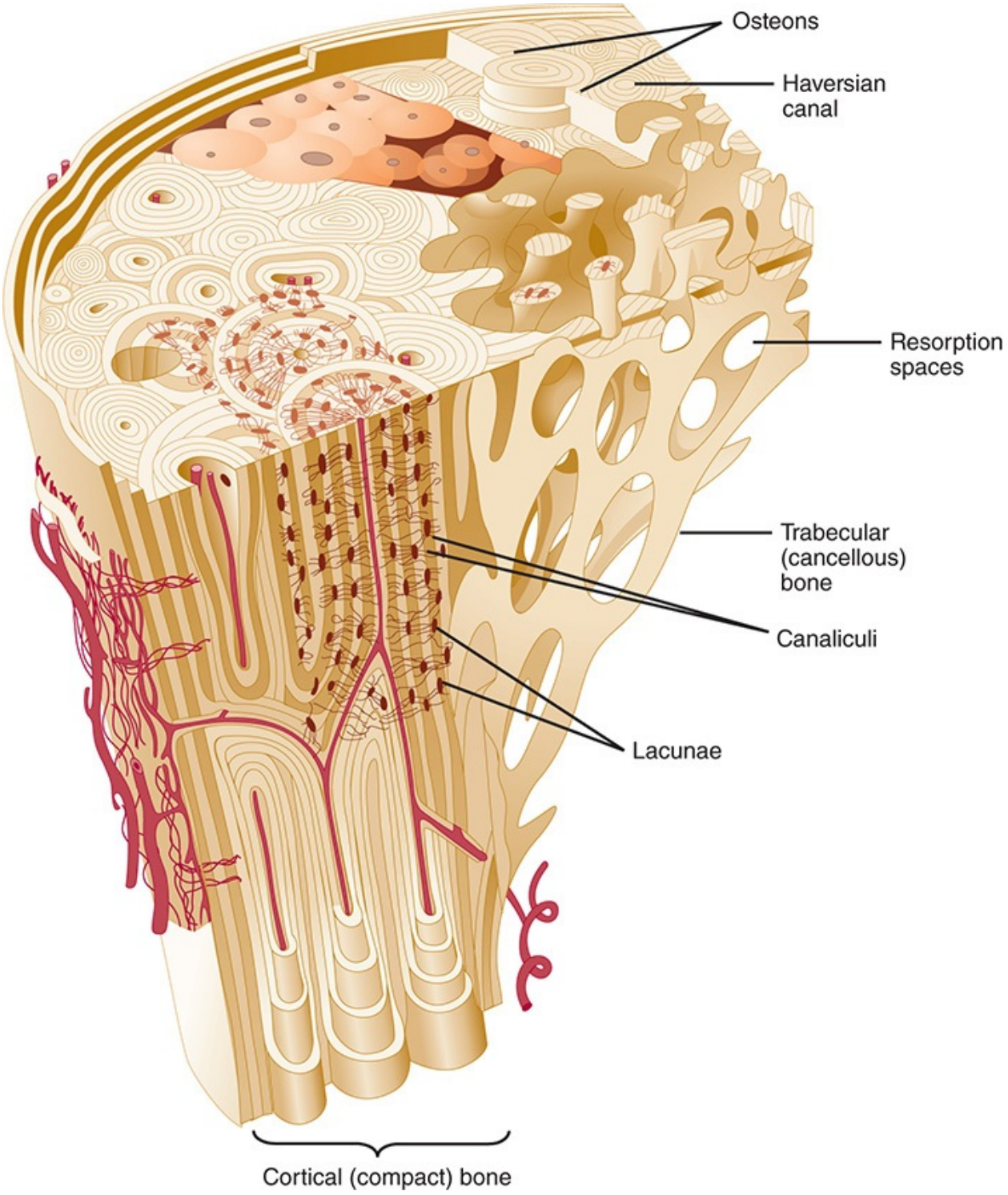


FIGURE 21–9 Structure of compact and trabecular bone. The compact bone is shown in horizontal section (top) and vertical section (bottom). (Reproduced with permission from Williams PL et al (editors): *Gray's Anatomy*, 37th ed. Churchill Livingstone; 1989.)

The protein in bone matrix is over 90% type I collagen, which is also the major structural protein in tendons and skin. This collagen, which weight for weight is as strong as steel, is made up of a triple helix of three polypeptides bound tightly together. Two of these are identical α_1 polypeptides encoded by one gene, and one is an α_2 polypeptide encoded by a different gene. Collagens make up a family of structurally related proteins that maintain the integrity of many different organs.

BONE GROWTH

During fetal development, most bones are modeled in cartilage and then transformed into bone by ossification **(enchondral bone formation).** The exceptions are the clavicles, the mandibles, and certain bones of the skull, in which mesenchymal cells form bone directly **(intramembranous bone formation).**

During growth, specialized areas at the ends of each long bone **(epiphyses)** are separated from the shaft of the bone by a plate of actively proliferating cartilage, the **epiphysial plate** **(Figure 21–10)**. The bone increases in length as this plate lays down new bone on the end of the shaft. The width of the epiphysial plate is proportional to the rate of growth. The width is affected by a number of hormones, but most markedly by growth hormone and IGF-I (see Chapter 18).

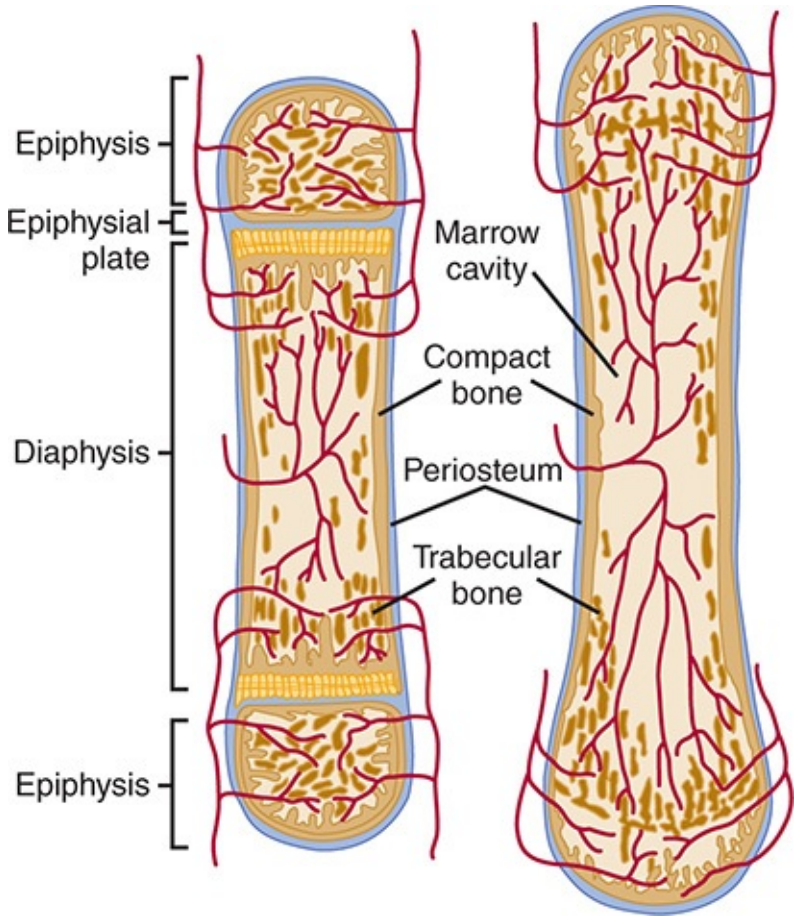


FIGURE 21–10 Structure of a typical long bone before (left) and after (right) epiphysial closure. Note the rearrangement of cells and growth of the bone as the epiphysial plate closes (see text for details).

Linear bone growth can occur as long as the epiphyses are separated from the shaft of the bone, but such growth ceases after the epiphyses unite with the shaft **(epiphysial closure)**. The cartilage cells stop proliferating, become hypertrophic, and secrete vascular endothelial growth factor (VEGF), leading to vascularization and ossification. The epiphyses of the various bones close in an orderly temporal sequence, the last epiphyses closing after puberty. The normal age at which each of the epiphyses closes is known, and the "bone age" of a young individual can be determined by radiographing the skeleton and noting which epiphyses are open and which are closed.

The **periosteum** is a dense fibrous, vascular, and innervated membrane that covers the surface of bones. This layer consists of an outer layer of collagenous tissue and an inner layer of fine elastic fibers containing cells that contribute to bone growth. The periosteum covers all surfaces of the bone except for those

capped with cartilage (eg, at the joints) and serves as a site of attachment of ligaments and tendons. As one ages, the periosteum becomes thinner and loses some of its vasculature. This renders bones more susceptible to injury and disease.

BONE FORMATION & RESORPTION

The cells responsible for bone formation are **osteoblasts** and the cells responsible for bone resorption are **osteoclasts.**

Osteoblasts are modified fibroblasts. Their initial development from the mesenchyme is the same as that of fibroblasts. Later, ossification-specific transcription factors, such as runt-related transcription factor 2 (Runx2; also known as core binding factor subunit alpha-1), contribute to their differentiation. The importance of this transcription factor in bone development is underscored in knockout mice deficient for the *RUNX2* gene. These mice develop to term with their skeletons made exclusively of cartilage; no ossification occurs. Normal osteoblasts, on the other hand, are able to lay down type 1 collagen and form new bone.

Osteoclasts are members of the monocyte family. Stromal cells in the bone marrow, osteoblasts, and T lymphocytes all express receptor activator for nuclear factor kappa beta ligand (RANKL) on their surface. When these cells come in contact with appropriate monocytes expressing RANK (ie, the RANKL receptor) two distinct signaling pathways are initiated: (1) There is a RANKL/RANK interaction between the cell pairs and (2) mononuclear phagocyte colony stimulating factor (M-CSF) is secreted by the nonmonocytic cells and it binds to its corresponding receptor on the monocytes (colony stimulating factor 1 receptor, CSF1R). The combination of these two signaling events leads to differentiation of the monocytes into osteoclasts. The precursor cells also secrete **osteoprotegerin (OPG),** which limits differentiation of the monocytes by competing with RANK for binding of RANKL.

Osteoclasts erode and absorb previously formed bone. They become attached to bone via integrins in a membrane extension called the **sealing zone.** This creates an isolated area (bone-resorbing compartment) between the bone and a portion of the osteoclast. Proton pumps (ie, H^+-dependent ATPases) then move from endosomes into the cell membrane apposed to the isolated area, and they acidify the area to approximately pH 4.0. Similar proton pumps are found in the endosomes and lysosomes of all eukaryotic cells, but in only a few other instances do they move into the cell membrane. Thus, the sealed-off space

formed by the osteoclast resembles a large lysosome. The acidic pH dissolves hydroxyapatite, and acid proteases secreted by the cell break down collagen, forming a shallow depression in the bone **(Figure 21–11)**. The products of bone digestion are then endocytosed and move across the osteoclast by transcytosis (see Chapter 2), with release into the interstitial fluid. The collagen breakdown products have pyridinoline structures, and pyridinolines can be measured in the urine as an index of the rate of bone resorption.

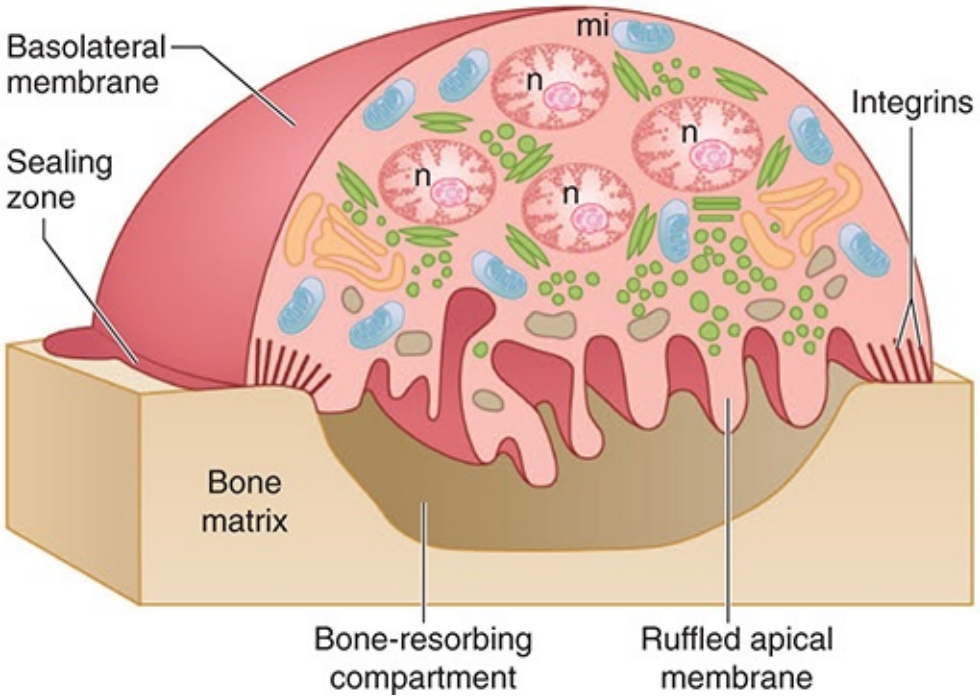


FIGURE 21–11 Osteoclast resorbing bone. The edges of the cell are tightly sealed to bone, permitting secretion of acid from the ruffled apical membrane and consequent erosion of the bone underneath the cell. Note the multiple nuclei (n) and mitochondria (mi). (Used with permission of R. Baron.)

Throughout life, bone is being constantly resorbed and new bone is being formed. The calcium in bone turns over at a rate of 100% per year in infants and 18% per year in adults. Bone remodeling is mainly a local process carried out in small areas by populations of cells called bone-remodeling units. First, osteoclasts resorb bone, and then osteoblasts lay down new bone in the same general area. This cycle takes about 100 days. The shape of bones may also change as bone is resorbed in one location but added in another. Osteoclasts tunnel into cortical bone followed by osteoblasts, whereas trabecular bone remodeling occurs on the surface of the trabeculae. About 5% of the bone mass

is being remodeled by about 2 million bone-remodeling units in the human skeleton at any one time. The renewal rate for bone is about 4% per year for compact bone and 20% per year for trabecular bone. The remodeling is related in part to the stresses and strains imposed on the skeleton by gravity.

At the cellular level, there is some regulation of osteoclast formation by osteoblasts via the RANKL–RANK and the M-CSF–CSF1R mechanism; however, specific feedback mechanisms of osteoclasts on osteoblasts are not well defined. In a broader sense, the bone-remodeling process is primarily under endocrine control. Not surprisingly, hormonal control of bone metabolism can be quite complex, and this can be illustrated by examining effects of the weight-associated hormone leptin on bone metabolism. When administered intracerebroventricularly, leptin decreases bone formation. This is thought to occur via release of various substances from the hypothalamus that can reduce osteoblast function. However, circulating leptin can increase bone mass through osteoblast and pre-osteoblast cell signaling pathways. More generally, PTH accelerates bone resorption, and estrogens slow bone resorption by inhibiting the production of bone-eroding cytokines.

BONE DISEASE

The diseases produced by selective abnormalities of the cells and processes discussed above illustrate the interplay of factors that maintain normal bone function.

In **osteopetrosis,** a rare and often severe disease, the osteoclasts are defective and are unable to resorb bone in their usual manner so the osteoblasts operate unopposed. The result is a steady increase in bone density, neurologic defects due to narrowing and distortion of foramina through which nerves normally pass, and hematologic abnormalities due to crowding out of the marrow cavities. Mice lacking the protein encoded by the immediate-early gene c-*fos* develop osteopetrosis; osteopetrosis also occurs in mice lacking the PU.1 transcription factor. This suggests that all these factors are involved in normal osteoclast development and function.

On the other hand, **osteoporosis** is caused by a relative excess of osteoclastic function. Loss of bone matrix in this condition **(Figure 21–12)** is marked, and the incidence of fractures is increased. Fractures are particularly common in the distal forearm (Colles fracture), vertebral body, and hip. All of these areas have a high content of trabecular bone, and because trabecular bone is more active metabolically, it is lost more rapidly. Fractures of the vertebrae with compression

cause kyphosis, with the production of a typical “widow’s hump” that is common in elderly women with osteoporosis. Fractures of the hip in elderly individuals are associated with a mortality rate of 12–20%, and half of those who survive require prolonged expensive care.

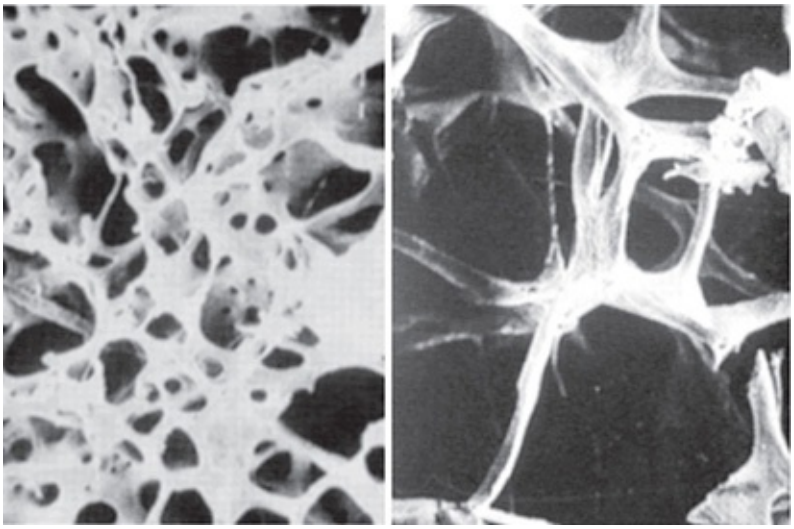


FIGURE 21–12 Normal trabecular bone (left) compared with trabecular bone from a patient with osteoporosis (right). The loss of mass in osteoporosis leaves bones more susceptible to breakage.

Osteoporosis has multiple causes, but by far the most common form is **involutional osteoporosis.** Humans normally gain bone during growth early in life. After a plateau, they begin to lose bone as they grow older **(Figure 21–13)**. When this loss is accelerated or exaggerated, it leads to osteoporosis **(Clinical Box 21–4)**. Increased intake of calcium, particularly from natural sources such as milk, and moderate exercise may help prevent or slow the progress of osteoporosis, although their effects are not very large. Bisphosphonates such as etidronate, which inhibit osteoclastic activity, increase the mineral content of bone and decrease the rate of new vertebral fractures when administered in a cyclical manner. Fluoride stimulates osteoblasts, making bone more dense, but it has proven to be of little value in the treatment of the disease.

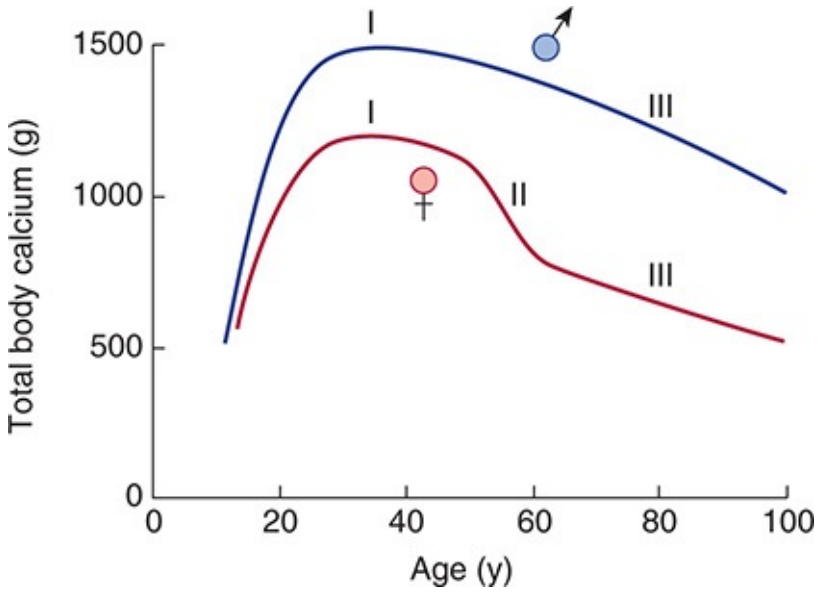


FIGURE 21–13 Total body calcium, an index of bone mass, at various ages in men and women. Note the rapid increase to young adult levels (phase I) followed by the steady loss of bone with advancing age in both sexes (phase III) and the superimposed rapid loss in women after menopause (phase II). (Reproduced with permission from Evans TG, Williams TF (editors): *Oxford Textbook of Geriatric Medicine*. Oxford University Press; 1992.)

CHAPTER SUMMARY

- Calcium and inorganic phosphate ions fulfill physiologic roles in cell signaling, nerve function, muscle contraction, and blood coagulation, among others. The majority of the calcium in the body is stored in the bones but it is the free, ionized calcium in the cells and extracellular fluids that is physiologically relevant. Phosphate is likewise predominantly stored in the bones.
- Calcium and phosphate are absorbed from the diet in a regulated fashion, enter the extracellular fluid, and move into and out of the bones. Circulating calcium and phosphate are filtered by the kidneys, and typically largely reabsorbed unless renal wasting is called for when circulating levels of the minerals are elevated.
- Circulating levels of calcium and phosphate ions are controlled by cells that sense the levels of these electrolytes in the blood and are thereby stimulated to release hormones, and these hormones mobilize the minerals from the bones, increase intestinal absorption, and/or modulate renal wasting.

- The major hormones regulating calcium and phosphate homeostasis are 1,25-dihydroxycholecalciferol (a derivative of vitamin D that is produced cooperatively by the skin, liver and kidneys) and parathyroid hormone, secreted by the parathyroid glands. Calcitonin is also capable of regulating levels of these ions, but its full physiologic contribution is unclear.
- 1,25-Dihydroxycholecalciferol acts to elevate plasma calcium and phosphate by predominantly transcriptional mechanisms, whereas parathyroid hormone elevates calcium but decreases phosphate by increasing the latter's renal excretion. Calcitonin lowers both calcium and phosphate levels. Deficiencies of 1,25-dihydroxycholecalciferol, or mutations in its receptor, also lead to decreases in circulating calcium, defective calcification of the bones, and bone weakness. Disease states also result from either deficiencies or overproduction of parathyroid hormone, with reciprocal effects on calcium and phosphate.
- Bone is a highly structured mass with outer cortical and inner trabecular layers. Regulated bone growth through puberty occurs through epiphysial plates. These plates are located near the end of the bone shaft and fuse with the shaft of the bone to cease linear bone growth.
- Bone is constantly remodeled by osteoclasts, which erode and absorb bone, and osteoblasts, which lay down new bone. An imbalance between the activities of these two cell types can lead to a steady increase in bone density when osteoblasts function unopposed (osteopetrosis) or a loss of bone mass when there is a relative excess of osteoclast activity (osteoporosis).

CLINICAL BOX 21–4

Osteoporosis

Adult women have less bone mass than adult men, and after menopause they initially lose it more rapidly than men of comparable age. Consequently, they are more prone to development of serious osteoporosis. The cause of bone loss after menopause is primarily estrogen deficiency, and estrogen treatment arrests the progress of the disease. Estrogens inhibit secretion of cytokines such as interleukin-1 (IL-1), IL-6, and tumor necrosis factor (TNF-α), cytokines that otherwise foster the development of osteoclasts. Estrogen also stimulates production of transforming growth factor (TGF-β), and this

cytokine increases apoptosis of osteoclasts.

Bone loss can also occur in both men and women as a result of inactivity. In patients who are immobilized for any reason, and during space flight, bone resorption exceeds bone formation and **disuse osteoporosis** develops. The plasma calcium level is not markedly elevated, but plasma concentrations of PTH and 1,25-dihydroxycholecalciferol fall and large amounts of calcium are lost in the urine.

THERAPEUTIC HIGHLIGHTS

Hormone therapy has traditionally been used to offset osteoporosis. **Estrogen replacement therapy** begun shortly after menopause can help maintain bone density. However, it appears that even small doses of estrogens may increase the incidence of uterine and breast cancer, and in carefully controlled studies, estrogens do not protect against cardiovascular disease. Therefore, treatment of a postmenopausal woman with estrogens is no longer used as a primary option. **Raloxifene** is a selective estrogen receptor modulator that can mimic the beneficial effects of estrogen on bone density in postmenopausal women without some of the risks associated with estrogen. However, this too carries risk of side effects (eg, blood clots). Other hormone treatments include the use of **calcitonin** and the PTH analogue **teriparatide.** An alternative to hormone treatments is the **bisphosphonates.** These drugs can inhibit bone breakdown, preserve bone mass, and even increase bone density in the spine and hip to reduce the risk of fractures. Unfortunately, these drugs also can cause mild to serious side effects and require monitoring for patient suitability. In addition to hormones and medications listed above, **physical therapy** to increase appropriate mechanical load and improve balance and muscle strength can significantly improve quality of life.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. A patient with parathyroid deficiency developing 10 days after inadvertent damage to the parathyroid glands during thyroid surgery would probably have

   A. low plasma phosphate and Ca^{2+} levels and tetany.

   B. low plasma phosphate and Ca^{2+} levels and tetanus.

C. a low plasma Ca^{2+} level, increased muscular excitability, and spasm of the muscles of the upper extremity (Trousseau sign).

D. high plasma phosphate and Ca^{2+} levels and bone demineralization.

E. increased muscular excitability, a high plasma Ca^{2+} level, and bone demineralization.

2. In an experiment, a rat is infused with a small volume of a calcium chloride solution, or sodium chloride as a control. Compared to the control condition, which of the following would result from the calcium load?

A. Bone demineralization

B. Increased formation of 1,25-dihydroxycholecalciferol

C. Decreased secretion of calcitonin

D. Decreased blood coagulability

E. Increased formation of 24,25-dihydroxycholecalciferol

3. In the same rat, which of the following would be increased?

A. Intestinal absorption of calcium

B. Expression of TRPV5 in the renal proximal tubule

C. NaPi-IIb abundance in the enterocyte apical membrane

D. Urinary excretion of calcium

E. Expression of calbindins

4. Which of the following is *not* involved in regulating plasma Ca^{2+} levels?

A. Kidneys

B. Skin

C. Liver

D. Lungs

E. Intestine

5. 1,25-Dihydroxycholecalciferol affects intestinal Ca^{2+} absorption through a mechanism that

A. includes alterations in the activity of genes.

B. activates adenylyl cyclase.

C. decreases cell turnover.

D. changes gastric acid secretion.

E. involves degradation of apical calcium channels.

6. Which of the following would you expect to find in a patient whose diet has

been low in calcium for 2 months?

A. Increased formation of 24,25-dihydroxycholecalciferol

B. Decreased expression of TRPV6 in intestinal epithelial cells

C. Increased parathyroid hormone secretion

D. A high plasma calcitonin concentration

E. Increased plasma phosphate

7. The skeleton of a normal male college student would be expected to display which of the following features, relative to that of his 7-year-old brother?

A. Merging of cortical bone and trabecular bone.

B. Differentiation of osteoclasts and osteoblasts.

C. An extended amount of proliferating cartilage that contributes to bone elongation.

D. A meeting of the lacunae with the trabecular bone.

E. Ephyses that are united with the bone shaft.

8. A mouse is engineered to lack a transcription factor necessary for the normal development of osteoclasts. Compared to normal littermate mice, which of the following would be reduced in the knock-out animals?

A. Phosphate deposition in trabecular bone

B. Hydroxyapatite levels in bone

C. Osteoblast proliferation

D. Secretion of acid proteases

E. Bone collagen

CHAPTER 22

Reproductive Development & Function of the Female Reproductive System

OBJECTIVES

After studying this chapter, you should be able to:

- Outline the role of chromosomes, hormones, and related factors in sex determination and development.
- Summarize the hormonal changes that occur at puberty in females.
- Describe oogenesis within the ovarian follicle. Explain the roles of FSH, LH and inhibin in oogenesis and maturation of follicles.
- Describe ovulation and the role of the corpus luteum in the physiologic changes that occur in the female reproductive organs during the menstrual cycle.
- Outline the hormonal changes and their physiologic effects during perimenopause and menopause.
- Describe the hormonal regulation of estrogen and progesterone biosynthesis, transport, metabolism, targets, and actions.
- Describe the roles of the pituitary and the hypothalamus in the regulation of ovarian function, and the role of feedback loops in this process.
- Describe the process of fertilization, implantation and the development of

the placenta

- Discuss the hormonal changes that accompany pregnancy and parturition.
- Outline the processes involved in lactation including neuroendocrine regulation of milk secretion and ejection.

INTRODUCTION

Modern genetics and experimental embryology make it clear that, in most species of mammals, the multiple differences between the male and the female depend primarily on a single chromosome (the Y chromosome) and a single pair of endocrine structures, namely the testes in the male and the ovaries in the female. The differentiation of the primitive gonads into testes or ovaries in utero is genetically determined in humans, but the formation of male genitalia depends on the presence of a functional, secreting testis; in the absence of testicular tissue, development is female. Evidence indicates that male sexual behavior and, in some species, the male pattern of gonadotropin secretion are due to the action of male hormones on the brain in early development. After birth, the gonads remain quiescent until adolescence, when they are activated by gonadotropins from the anterior pituitary. Hormones secreted by the gonads at this time cause the appearance of features typical of the adult male or female and the onset of the sexual cycle in the female. In human females, ovarian function regresses after a number of years and sexual cycles cease (the menopause). In males, gonadal function slowly declines with advancing age, but the ability to produce viable gametes persists.

In both sexes, the gonads have a dual function: the production of germ cells **(gametogenesis)** and the secretion of **sex hormones.** The **androgens** are steroid sex hormones that are masculinizing in their action; the **estrogens** are those that are feminizing. Both types of hormones are normally secreted in both sexes. The ovaries secrete large amounts of estrogens and small amounts of androgens, a pattern that is reversed in males. Androgens are secreted from the adrenal cortex in both sexes, and some of the androgens are converted to estrogens in fat and other extragonadal and extra-adrenal tissues. The ovaries also secrete **progesterone,** a steroid that has special functions in preparing the uterus for pregnancy.

Particularly during pregnancy, the ovaries secrete the polypeptide hormone **relaxin,** which loosens the ligaments of the pubic symphysis and softens the cervix, facilitating delivery of the fetus. In both sexes, the gonads secrete other

polypeptides, including **inhibin B,** a polypeptide that inhibits follicle-stimulating hormone (FSH) secretion.

The secretory and gametogenic functions of the gonads are both dependent on the secretion of the anterior pituitary gonadotropins, FSH, and luteinizing hormone (LH). The sex hormones and inhibin B feed back to inhibit gonadotropin secretion. In males, gonadotropin secretion is noncyclic; but in postpubertal females an orderly, sequential secretion of gonadotropins is necessary for the occurrence of menstruation, pregnancy, and lactation.

SEX DIFFERENTIATION & DEVELOPMENT

CHROMOSOMAL SEX

The Sex Chromosomes

Sex is determined genetically by two chromosomes, called the **sex chromosomes,** to distinguish them from the **somatic chromosomes (autosomes).** In humans and many other mammals, the sex chromosomes are called X and Y. The Y chromosome is necessary and sufficient for the production of testes, and the testis-determining gene product is called SRY (for sex- determining region of the Y chromosome). SRY is a DNA-binding regulatory protein. It bends the DNA and acts as a transcription factor that initiates transcription of a cascade of genes necessary for testicular differentiation, including the gene for **müllerian-inhibiting substance (MIS;** see below). The gene for SRY is located near the tip of the short arm of the human Y chromosome. Diploid male cells contain an X and a Y chromosome (XY pattern), whereas female cells contain two X chromosomes (XX pattern). As a consequence of meiosis during gametogenesis, each normal ovum contains a single X chromosome, but half of the normal sperm contain an X chromosome and half contain a Y chromosome **(Figure 22–1)**. When a sperm containing a Y chromosome fertilizes an ovum, an XY pattern results and the zygote develops into a **genetic male.** When fertilization occurs with an X-containing sperm, an XX pattern and a **genetic female** results. Cell division and the chemical nature of chromosomes are discussed in Chapter 1.

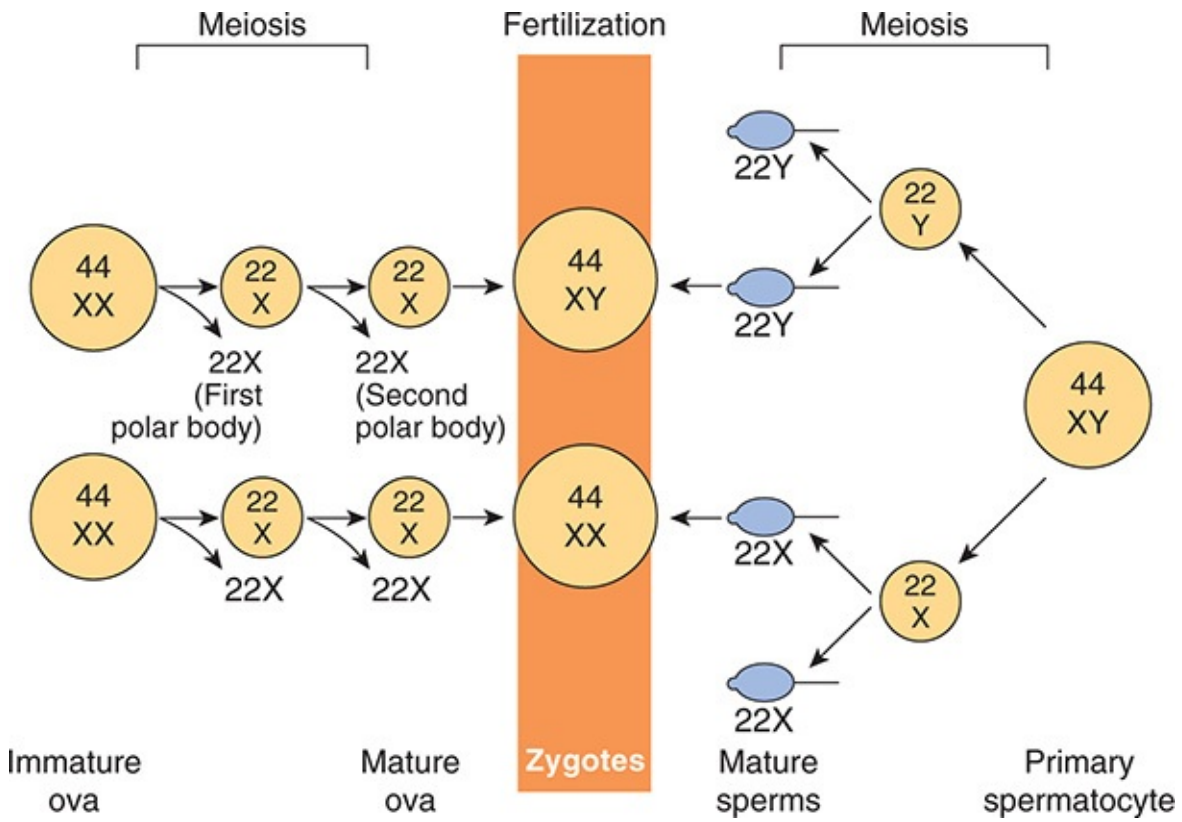


FIGURE 22–1 Basis of genetic sex determination. In the two-stage meiotic division in the female, only one cell survives as the mature ovum. In the male, the meiotic division results in the formation of four sperms, two containing the X and two the Y chromosome. Fertilization thus produces a male zygote with 22 pairs of autosomes plus an X and a Y or a female zygote with 22 pairs of autosomes and two X chromosomes. Note that for clarity, this figure and Figure 22–6 and 22–7 differ from the current international nomenclature for karyotypes, which lists the total number of chromosomes followed by the sex chromosome pattern. Thus, XO is 45, X; XY is 46, XY; XXY is 47, XXY, and so on.

Human Chromosomes

Human chromosomes can be studied in detail. Human cells are grown in tissue culture; treated with the drug colchicine, which arrests mitosis at the metaphase; exposed to a hypotonic solution that makes the chromosomes swell and disperse; and then “squashed” onto slides. Staining techniques make it possible to identify the individual chromosomes **(Figure 22–2)**. There are 46 chromosomes: in males, 22 pairs of autosomes plus an X chromosome and a Y chromosome; in females, 22 pairs of autosomes plus two X chromosomes. The individual

chromosomes are usually arranged in an arbitrary pattern **(karyotype).** The individual autosome pairs are identified by the numbers 1–22 on the basis of their morphologic characteristics.



FIGURE 22–2 Karyotype of chromosomes from a normal male. The chromosomes have been stained with Giemsa stain, which produces a characteristic banding pattern. (Reproduced with permission from Lingappa VJ, Farey K: *Physiological Medicine*. New York, NY: McGraw-Hill; 2000.)

Sex Chromatin

Soon after cell division has started during embryonic development, one of the two X chromosomes of the somatic cells in normal females becomes functionally inactive. In abnormal individuals with more than two X chromosomes, only one remains active. The process that is normally responsible for inactivation is initiated in an X-inactivation center in the chromosome, probably via the transactivating factor CTCF (for CCCTC-binding factor), which is also induced during gene imprinting. However, the details of the inactivation

process are still incompletely understood. The choice of which X chromosome remains active is random, so normally one X chromosome remains active in approximately half of the cells and the other X chromosome is active in the other half. The selection persists through subsequent divisions of these cells, and consequently some of the somatic cells in adult females contain an active X chromosome of paternal origin and some contain an active X chromosome of maternal origin.

In normal cells, the inactive X chromosome condenses and can be seen in various types of cells, usually near the nuclear membrane, as the **Barr body,** also called sex chromatin **(Figure 22–3)**. Thus, there is a Barr body for each X chromosome in excess of one in the cell. The inactive X chromosome is also visible as a small "drumstick" of chromatin projecting from the nuclei of 1–15% of the polymorphonuclear leukocytes in females but not in males (Figure 22–3).

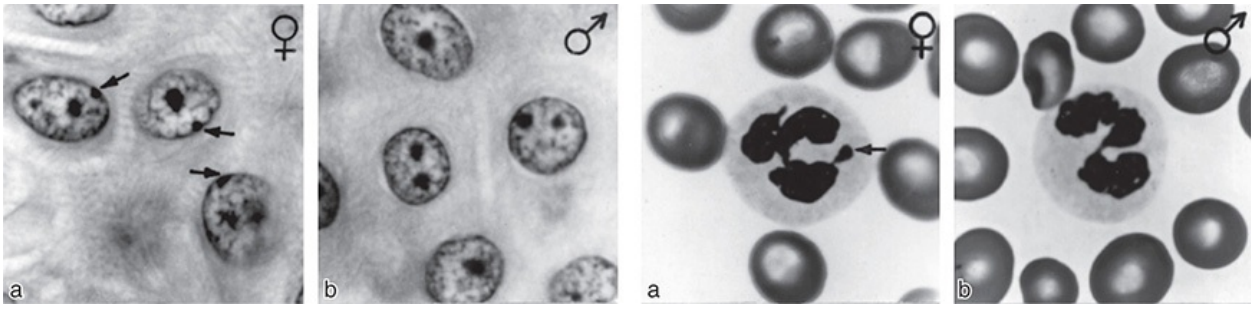


FIGURE 22–3 Left: Barr body (arrows) in the epidermal spinous cell layer. Right: Nuclear appendage ("drumstick") identified by arrow in white blood cells. (Reproduced with permission from Grumbach MM, Barr ML: Cytologic tests of chromosomal sex in relation to sex anomalies in man. Recent Prog Horm Res 1958;14:255–324.)

EMBRYOLOGY OF THE HUMAN REPRODUCTIVE SYSTEM

Development of the Gonads

On each side of the embryo, a primitive gonad arises from the genital ridge, a condensation of tissue near the adrenal gland. The gonad develops a **cortex** and a **medulla.** Until the sixth week of development, these structures are identical in both sexes. In genetic males, the medulla develops during the seventh and eighth weeks into a testis, and the cortex regresses. Leydig and Sertoli cells appear, and

testosterone and MIS are secreted. In genetic females, the cortex develops into an ovary and the medulla regresses. The embryonic ovary does not secrete hormones. Hormonal treatment of the mother has no effect on gonadal (as opposed to ductal and genital) differentiation in humans, although it does in some experimental animals.

Embryology of the Genitalia

The embryology of the gonads is summarized in **Figures 22–4** and **22–5**. In the seventh week of gestation, the embryo has both male and female primordial genital ducts (Figure 22–4). In a normal female fetus, the müllerian duct system then develops into uterine tubes (oviducts) and a uterus. In the normal male fetus, the wolffian duct system on each side develops into the epididymis and vas deferens. The external genitalia are similarly bipotential until the eighth week (Figure 22–5). Thereafter, the urogenital slit disappears and male genitalia form, or, alternatively, it remains open and female genitalia form.

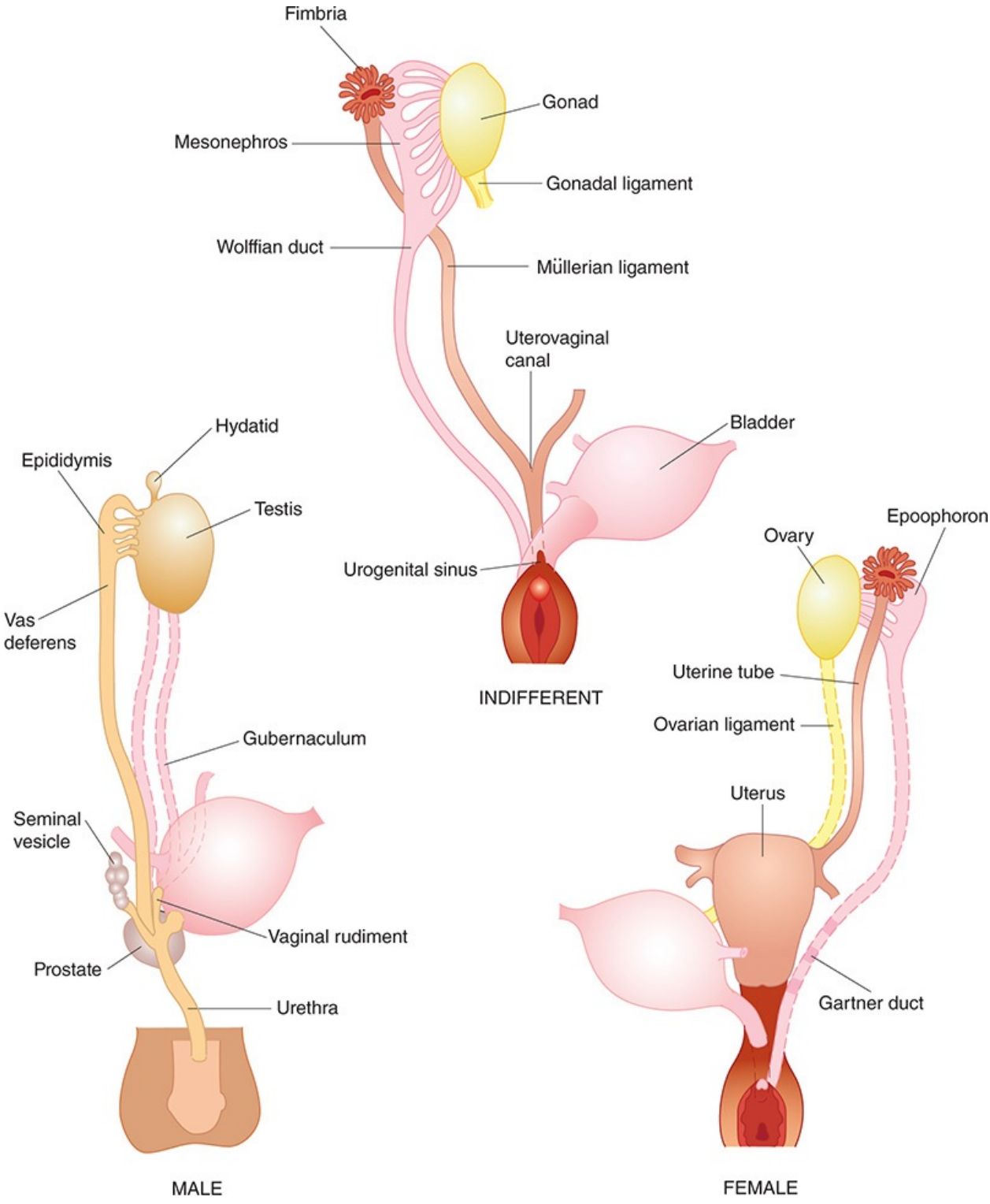


FIGURE 22–4 Embryonic differentiation of male and female internal genitalia (genital ducts) from wolffian (male) and müllerian (female) primordia. (Reproduced with permission from Wilson JD, Foster DW [editors]: *Williams Textbook of Endocrinology*, 7th ed. Saunders; 1985.)

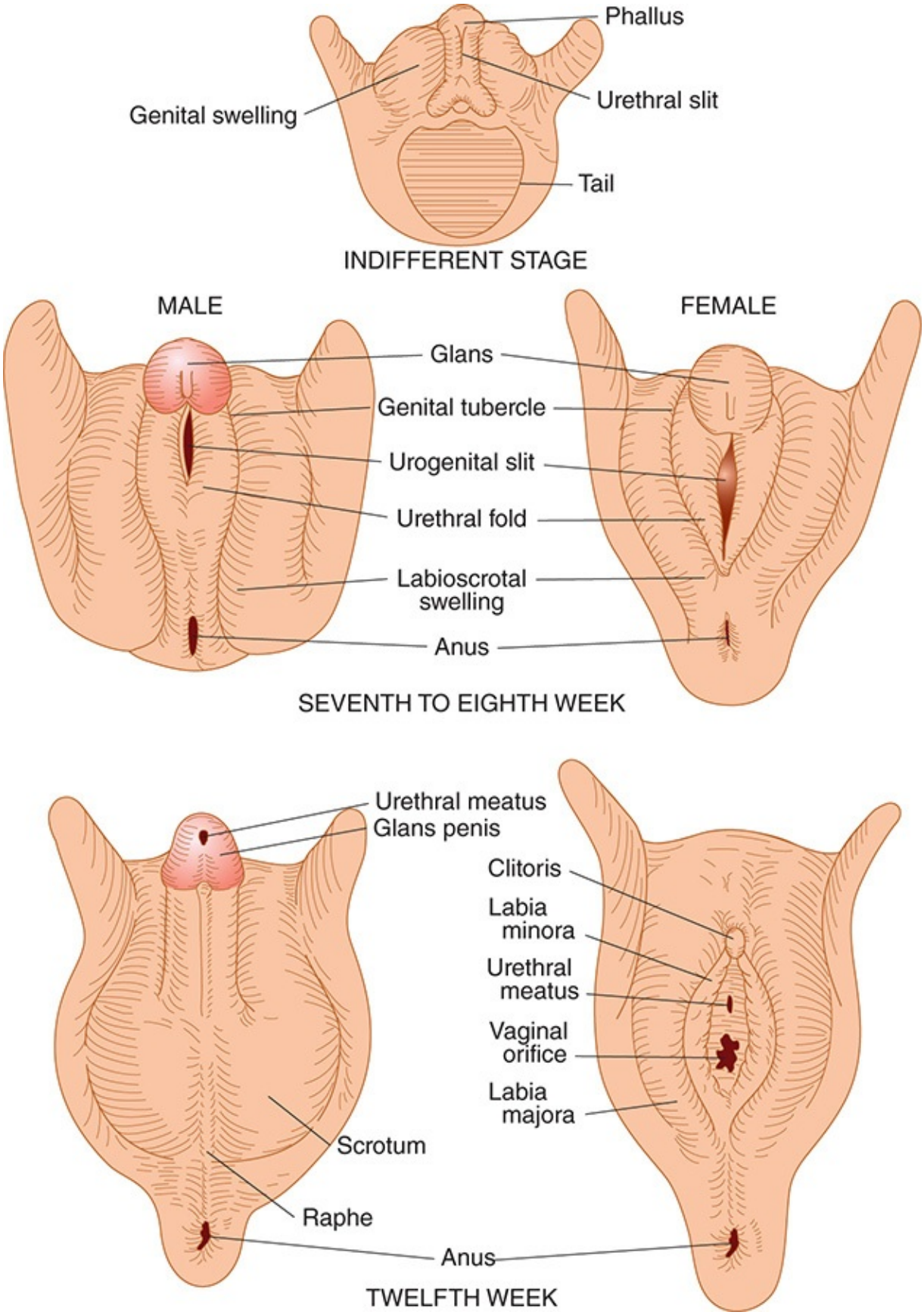

Phallus
Genital swelling
Urethral slit
Tail
INDIFFERENT STAGE
MALE
FEMALE
Glans
Genital tubercle
Urogenital slit
Urethral fold
Labioscrotal swelling
Anus
SEVENTH TO EIGHTH WEEK
Urethral meatus
Glans penis
Clitoris
Labia minora
Urethral meatus
Vaginal orifice
Labia majora
Scrotum
Raphe
Anus
TWELFTH WEEK

FIGURE 22–5 Differentiation of male and female external genitalia from indifferent primordial structures in the embryo.

When the embryo has functional testes, male internal and external genitalia develop. The Leydig cells of the fetal testis secrete testosterone, and the Sertoli cells secrete MIS (also called müllerian regression factor, or MRF). MIS is a 536-amino-acid homodimer that is a member of the transforming growth factor β (TGF-β) super-family of growth factors, which includes inhibins and activins.

In their effects on the internal as opposed to the external genitalia, MIS and testosterone act unilaterally. MIS causes regression of the müllerian ducts by apoptosis on the side on which it is secreted, and testosterone fosters the development of the vas deferens and related structures from the wolffian ducts. The testosterone metabolite dihydrotestosterone induces the formation of male external genitalia and male secondary sex characteristics **(Figure 22–6)**.

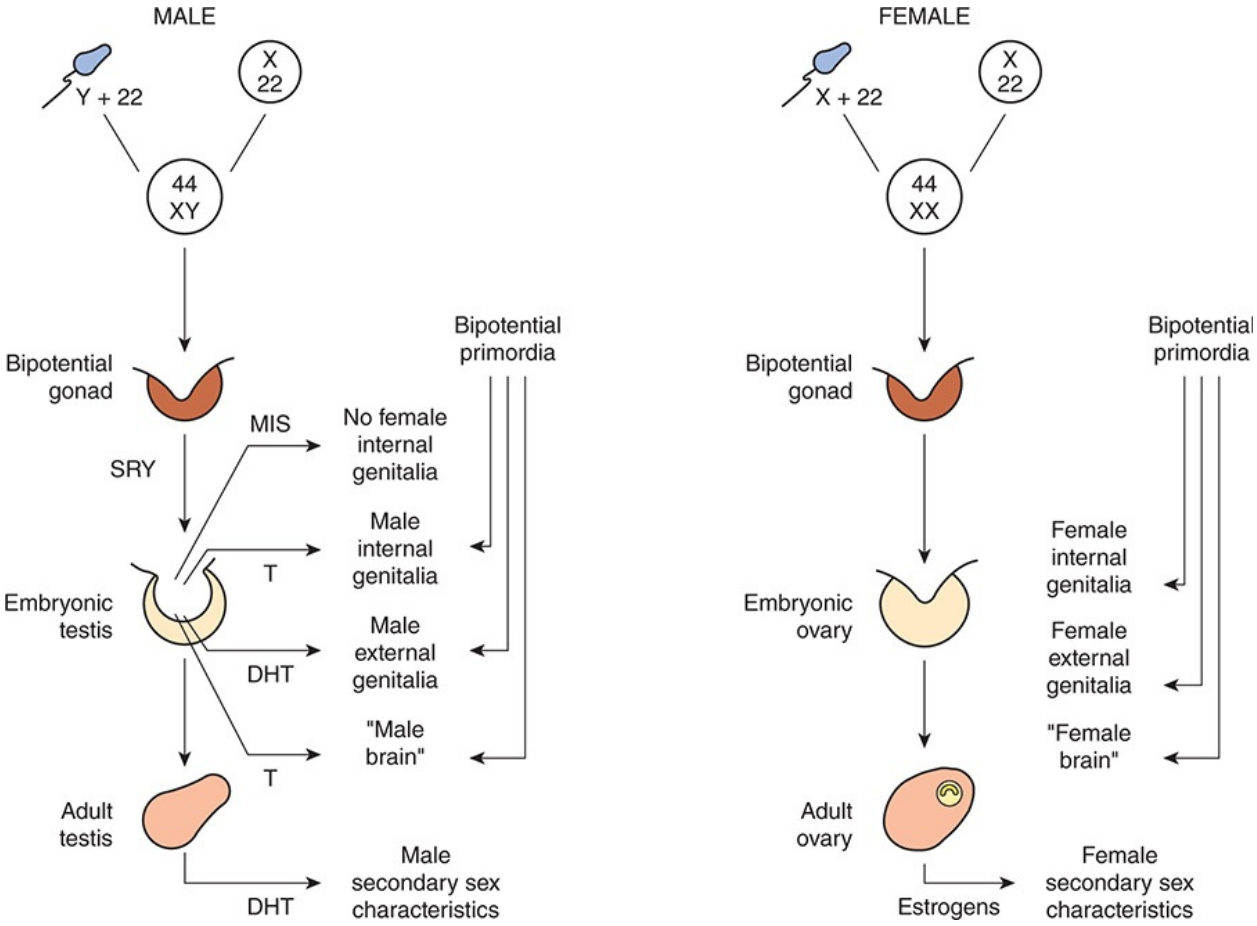


FIGURE 22–6 Diagrammatic summary of normal sex determination,

differentiation, and development in humans. DHT, dihydrotestosterone, MIS, müllerian-inhibiting substance; T, testosterone.

MIS continues to be secreted by the Sertoli cells, and it reaches mean values of 48 ng/mL in plasma in 1- to 2-year-old boys. Thereafter, it declines to low levels by the time of puberty and persists at low but detectable levels throughout life. In girls, MIS is produced by granulosa cells in small follicles in the ovaries, but plasma levels are very low or undetectable until puberty. Thereafter, plasma MIS is about the same as in adult men, that is, about 2 ng/mL. The functions of MIS after early embryonic life are unsettled, but it is probably involved in germ cell maturation in both sexes and in control of testicular descent in boys.

Development of the Brain

At least in some species, the development of the brain as well as the external genitalia is affected by androgens early in life. In rats, a brief exposure to androgens during the first few days of life causes the male pattern of sexual behavior and the male pattern of hypothalamic control of gonadotropin secretion to develop after puberty. In the absence of androgens, female patterns develop (see Chapter 17). In monkeys, similar effects on sexual behavior are produced by exposure to androgens in utero, but the pattern of gonadotropin secretion remains cyclical. Early exposure of female human fetuses to androgens also appears to cause subtle but significant masculinizing effects on behavior. However, women with adrenogenital syndrome due to congenital adrenocortical enzyme deficiency (see Chapter 20) develop normal menstrual cycles when treated with cortisol. Thus, the human, like the monkey, appears to retain the cyclical pattern of gonadotropin secretion despite exposure to androgens in utero.

ABERRANT SEXUAL DIFFERENTIATION

Chromosomal Abnormalities

From the preceding discussion, it might be expected that abnormalities of sexual development could be caused by genetic or hormonal abnormalities as well as by other nonspecific teratogenic influences, and this is indeed the case. The major classes of abnormalities are listed in **Table 22–1**.

TABLE 22–1 Classification of the major disorders of sex differentiation in humans.[a]

Chromosomal disorders
Gonadal dysgenesis (XO and variants)
"Superfemales" (XXX)
Seminiferous tubule dysgenesis (XXY and variants)
True hermaphroditism
Developmental disorders
Female pseudohermaphroditism
Congenital virilizing adrenal hyperplasia of fetus
Maternal androgen excess
Virilizing ovarian tumor
Iatrogenic: Treatment with androgens or certain synthetic progestational drugs
Male pseudohermaphroditism
Androgen resistance
Defective testicular development
Congenital 17α-hydroxylase deficiency
Congenital adrenal hyperplasia due to blockade of pregnenolone formation
Various nonhormonal anomalies

[a]Many of these syndromes can have great variation in degree and, consequently, in manifestations.

Nondisjunction of sex chromosomes during the first division in meiosis results in distinct defects **(see Clinical Box 22–1; Figure 22–7)**. Meiosis is a two-stage process, and although nondisjunction usually occurs during the first meiotic division, it can occur in the second, producing more complex chromosomal abnormalities. In addition, nondisjunction or simple loss of a sex chromosome can occur during the early mitotic divisions after fertilization. The consequence of faulty mitoses in the early zygote is **mosaicism,** in which two or

more populations of cells have different chromosome complements. **True hermaphroditism,** the condition in which the individual has both ovaries and testes, is probably due to XX/XY mosaicism and related mosaic patterns, although other genetic aberrations are possible.

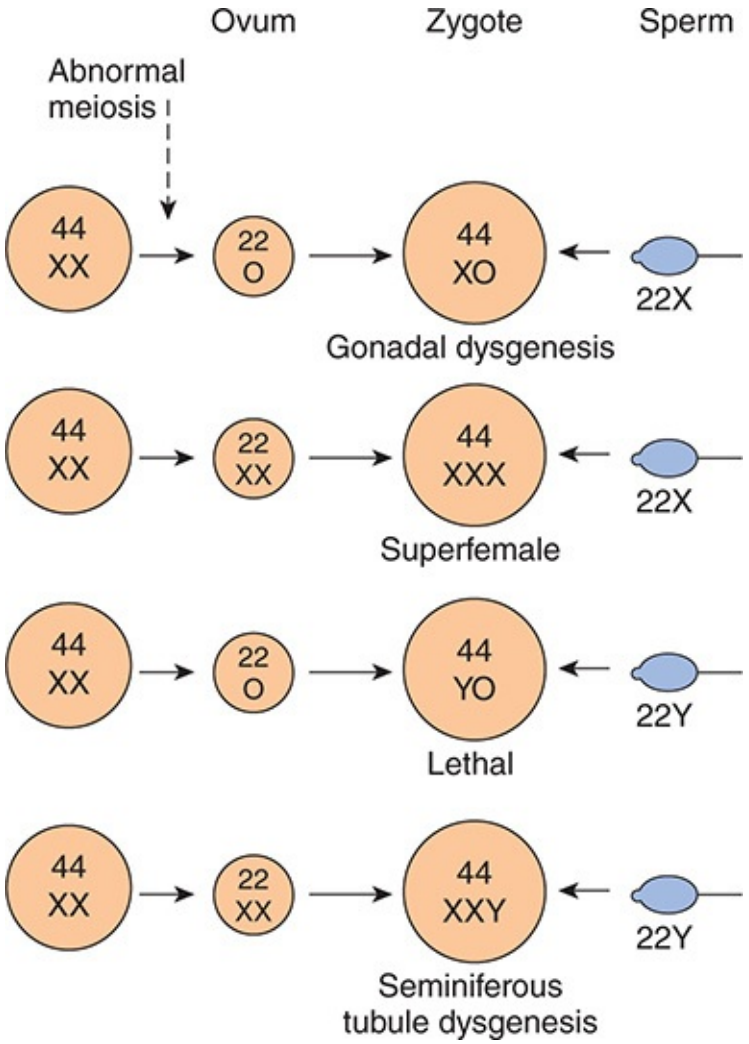


FIGURE 22–7 Summary of four possible defects produced by maternal nondisjunction of the sex chromosomes at the time of meiosis. The YO combination is believed to be lethal, and the fetus dies in utero.

Chromosomal abnormalities also include transposition of parts of chromosomes to other chromosomes. Rarely, genetic males are found to have the XX karyotype because the short arm of their father's Y chromosome was transposed to their father's X chromosome during meiosis and they received that X chromosome along with their mother's. Similarly, deletion of the small portion of the Y chromosome containing SRY produces females with the XY karyotype.

Hormonal Abnormalities

Development of the male external genitalia occurs normally in genetic males in response to androgen secreted by the embryonic testes, but male genital development may also occur in genetic females exposed to androgens from some other source during the 8th to the 13th weeks of gestation. The syndrome that results is **female pseudohermaphroditism.** A pseudohermaphrodite is an individual with the genetic constitution and gonads of one sex and the genitalia of the other. After the 13th week, the genitalia are fully formed, but exposure to androgens can cause hypertrophy of the clitoris. Female pseudohermaphroditism may be due to congenital virilizing adrenal hyperplasia (see Chapter 19), or it may be caused by androgens administered to the mother. Conversely, one cause of the development of female external genitalia in genetic males **(male pseudohermaphroditism)** is defective testicular development. Because the testes also secrete MIS, genetic males with defective testes have female internal genitalia.

Another cause of male pseudohermaphroditism is **androgen resistance,** in which, as a result of various congenital abnormalities, male hormones cannot exert their full effects on the tissues. One form of androgen resistance is a **5α-reductase deficiency,** in which the enzyme responsible for the formation of dihydrotestosterone, the active form of testosterone, is decreased **(Figure 22–8)**. The consequences of this deficiency are discussed in Chapter 23. Other forms of androgen resistance are due to various mutations in the androgen receptor gene, and the resulting defects in receptor function range from minor to severe. Mild defects cause infertility with or without gynecomastia. When the loss of receptor function is complete, the **testicular feminizing syndrome,** now known as **complete androgen resistance syndrome,** results. In this condition, MIS is present and testosterone is secreted at normal or even elevated rates. The external genitalia are female, but the vagina ends blindly because there are no female internal genitalia. At puberty, enlarged breasts develop in persons with this syndrome. The syndrome usually remains undiagnosed until women seek medical advice because of lack of menstruation.

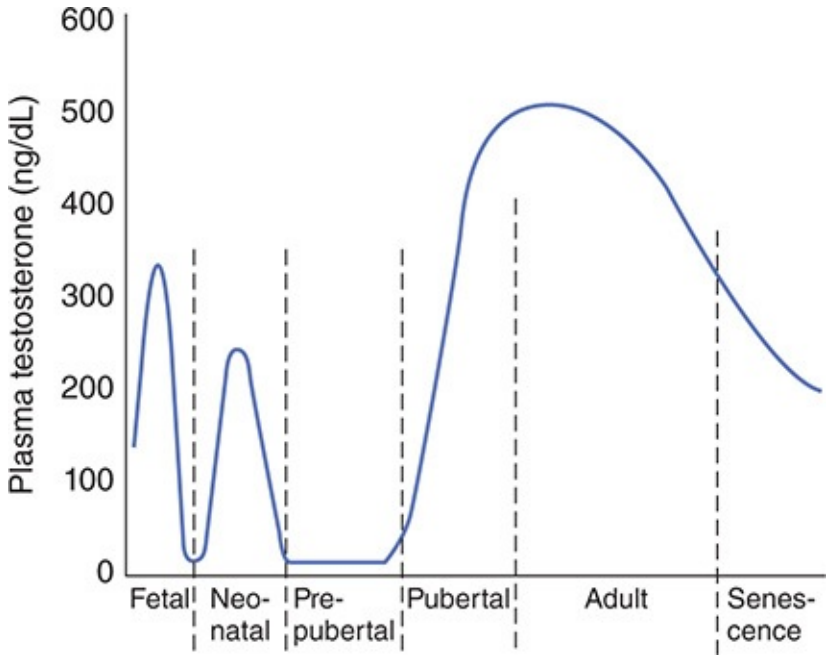


FIGURE 22–8 Plasma testosterone levels at various ages in human males.

CLINICAL BOX 22–1

Chromosomal Abnormalities

An established defect in gametogenesis is **nondisjunction,** a phenomenon in which a pair of chromosomes fail to separate, so that both go to one of the daughter cells during meiosis. Four of the abnormal zygotes that can form as a result of nondisjunction of one of the X chromosomes during oogenesis are shown in Figure 25–7. In individuals with the XO chromosomal pattern, the gonads are rudimentary or absent, so that female external genitalia develop, stature is short, other congenital abnormalities are often present, and no sexual maturation occurs at puberty. This syndrome is called **gonadal dysgenesis** or, alternatively, **ovarian agenesis** or **Turner syndrome.** Individuals with the XXY pattern, the most common sex chromosome disorder, have the genitalia of a normal male. Testosterone secretion at puberty is often great enough for the development of male characteristics; however, the seminiferous tubules are abnormal, and the incidence of mental retardation is higher than normal. This syndrome is known as **seminiferous tubule dysgenesis** or **Klinefelter syndrome.** The XXX (“superfemale”) pattern is second in frequency only to the XXY pattern and may be even more common in the general population, since it does not seem to be associated

with any characteristic abnormalities. The YO combination is probably lethal.

Nondisjunction of chromosome 21 produces **trisomy 21,** the chromosomal abnormality associated with **Down syndrome.** The additional chromosome 21 is normal, so Down syndrome is a pure case of gene excess causing abnormalities.

Many other chromosomal abnormalities occur as well as numerous diseases due to defects in single genes. These conditions are generally diagnosed in utero by analysis of fetal cells in a sample of amniotic fluid collected by inserting a needle through the abdominal wall **(amniocentesis)** or, earlier in pregnancy, by examining fetal cells obtained by a needle biopsy of chorionic villi **(chorionic villus sampling)**.

THERAPEUTIC HIGHLIGHTS

Many of the syndromes mentioned have effects on multiple organ systems, and patients must be carefully monitored with a multidisciplinary approach to avert the consequences of cardiovascular defects, infections secondary to urinary tract and renal malformations, and the psychological impact of reproductive implications. Girls with Turner syndrome and evidence of gonadal failure are also treated with low-dose estrogen to evoke puberty, followed by gradual replacement of mature estrogen levels to permit feminization. Conversely, patients with Klinefelter syndrome are often supplemented with androgens to improve virilization and libido.

It is worth noting that genetic males with congenital blockage of the formation of pregnenolone are pseudohermaphrodites because testicular as well as adrenal androgens are normally formed from pregnenolone. Male pseudohermaphroditism also occurs when there is a congenital deficiency of 17α-hydroxylase (see Chapter 19).

PUBERTY

As noted above, a burst of testosterone secretion occurs in male fetuses before birth (see Chapter 23). In the neonatal period there is another burst, with unknown function, but thereafter the Leydig cells become quiescent. There follows in all mammals a period in which the gonads of both sexes are quiescent until they are activated by gonadotropins from the pituitary to bring about the

final maturation of the reproductive system. This period of final maturation is known as **adolescence.** It is often also called **puberty,** although puberty, strictly defined, is the period when the endocrine and gametogenic functions of the gonads have first developed to the point where reproduction is possible. In girls, the first event is **thelarche,** the development of breasts, followed by **pubarche,** the development of axillary and pubic hair, and then by **menarche,** the first menstrual period. Initial menstrual periods are generally anovulatory, and regular ovulation appears about a year later. In contrast to the situation in adulthood, removal of the gonads during the period from soon after birth to puberty causes only a small increase in gonadotropin secretion, so gonadotropin secretion is not being held in check by the gonadal hormones. In children between the ages of 7 and 10, a slow increase in estrogen and androgen secretion precedes the more rapid rise in the early teens **(Figure 22–9)**.

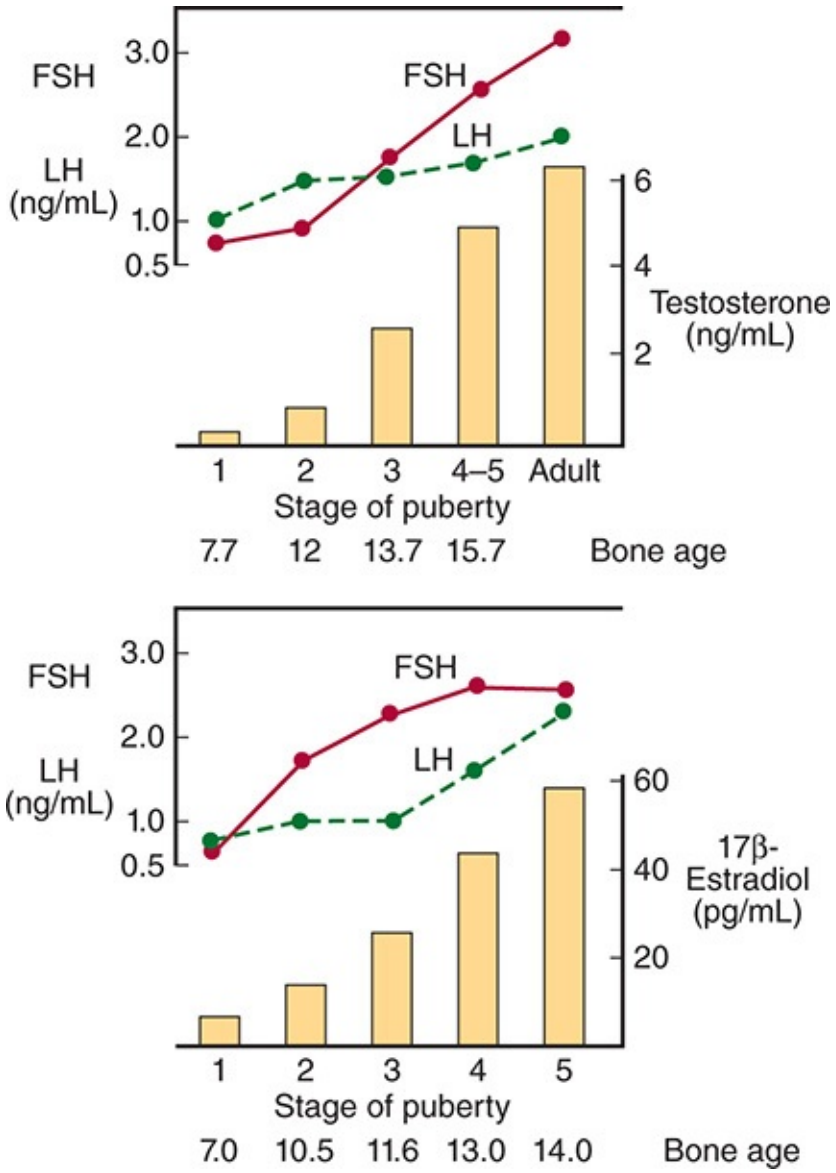


FIGURE 22–9 Changes in plasma hormone concentrations during puberty in boys (top) and girls (bottom). Stage 1 of puberty is preadolescence in both sexes. In boys, stage 2 is characterized by beginning enlargement of the testes, stage 3 by penile enlargement, stage 4 by growth of the glans penis, and stage 5 by adult genitalia. In girls, stage 2 is characterized by breast buds, stage 3 by elevation and enlargement of the breasts, stage 4 by projection of the areolas, and stage 5 by adult breasts. FSH, follicle-stimulating hormone; LH, luteinizing hormone. (Modified and reproduced with permission from Berenberg SR [ed]: *Puberty: Biologic and Psychosocial Components*. HE Stenfoert Kroese BV; 1975.)

The age at the time of puberty is variable. In Europe and the United States, it

has been declining at the rate of 1–3 months per decade for more than 175 years. In the United States in recent years, puberty generally occurs between the ages of 8 and 13 in girls and 9 and 14 in boys.

Another event that occurs in humans at the time of puberty is an increase in the secretion of adrenal androgens (see Figure 19–12). The onset of this increase is called **adrenarche.** It occurs at age 8–10 years in girls and age 10–12 years in boys. Dehydroepiandrosterone (DHEA) values peak at about age 25 in women and slightly later than that in men. They then decline slowly to low values in old age. The rise appears to be due to an increase in the activity of 17α-hydroxylase.

Control of the Onset of Puberty

The gonads of children can be stimulated by gonadotropins; their pituitaries contain gonadotropins and their hypothalami contain gonadotropin-releasing hormone (GnRH) (see Chapter 17). However, their gonadotropins are not secreted. In immature monkeys, normal menstrual cycles can be brought on by pulsatile injection of GnRH, and they persist as long as the pulsatile injection is continued. Thus, it seems clear that pulsatile secretion of GnRH brings on puberty. During the period from birth to puberty, a neural mechanism is operating to prevent the normal pulsatile release of GnRH. The nature of the mechanism inhibiting the GnRH pulse generator is unknown. However, one or more genes produce products that stimulate secretion of GnRH, and inhibition of these genes before puberty is an interesting possibility **(Clinical Box 22–2)**.

PRECOCIOUS & DELAYED PUBERTY

Sexual Precocity

The major causes of precocious sexual development in humans are listed in **Table 22–2**. Early development of secondary sexual characteristics without gametogenesis is caused by abnormal exposure of immature males to androgen or females to estrogen. This syndrome should be called **precocious pseudopuberty** to distinguish it from **true precocious puberty** due to an early but otherwise normal pubertal pattern of gonadotropin secretion from the pituitary.

TABLE 22–2 Classification of the causes of precocious sexual development in humans.

True precocious puberty
Constitutional
Cerebral: Disorders involving posterior hypothalamus
Tumors
Infections
Developmental abnormalities
Gonadotropin-independent precocity
Precocious pseudopuberty (no spermatogenesis or ovarian development)
Adrenal
Congenital virilizing adrenal hyperplasia
Androgen-secreting tumors (in males)
Estrogen-secreting tumors (in females)
Gonadal
Leydig cell tumors of testis
Granulosa cell tumors of ovary
Miscellaneous

CLINICAL BOX 22–2

Leptin

It has been argued for some time that a critical body weight must normally be reached for puberty to occur. Thus, for example, young women who engage in strenuous athletics lose weight and stop menstruating, as do girls with anorexia nervosa. If these girls start to eat and gain weight, they menstruate again, that is, they "go back through puberty." It now appears that leptin, the satiety-producing hormone secreted by fat cells, may be the link between body weight and puberty. Obese ob/ob mice that cannot make leptin are infertile, and their fertility is restored by injections of leptin. Leptin treatment also induces precocious puberty in immature female mice. However, the way that leptin fits into the overall control of puberty remains to be determined.

Constitutional precocious puberty, that is, precocious puberty in which no cause can be determined, is more common in girls than in boys. In both sexes, tumors or infections involving the hypothalamus cause precocious puberty.

Indeed, in one large series of cases, precocious puberty was the most common endocrine symptom of hypothalamic disease. In experimental animals, precocious puberty can be produced by hypothalamic lesions. Apparently the lesions interrupt a pathway that normally holds pulsatile GnRH secretion in check. Pineal tumors are sometimes associated with precocious puberty, but evidence indicates that these tumors are associated with precocity only when there is secondary damage to the hypothalamus.

Precocious gametogenesis and steroidogenesis can occur without the pubertal pattern of gonadotropin secretion (gonadotropin-independent precocity). At least in some cases of this condition, the sensitivity of LH receptors to gonadotropins is increased because of an activating mutation in the G-protein that couples the receptors to adenylyl cyclase.

CLINICAL BOX 22–3

Hyperprolactinemia

Up to 70% of the patients with chromophobe adenomas of the anterior pituitary have elevated plasma prolactin levels. In some instances, the elevation may be due to damage to the pituitary stalk, but in most cases, the tumor cells are actually secreting the hormone. The hyperprolactinemia may cause galactorrhea, but in many individuals no demonstrable endocrine abnormalities are present. Conversely, most women with galactorrhea have normal prolactin levels; definite elevations are found in less than one-third of patients with this condition.

Another interesting observation is that 15–20% of women with secondary amenorrhea have elevated prolactin levels, and when prolactin secretion is reduced, normal menstrual cycles and fertility return. Prolactin may produce amenorrhea by blocking the action of gonadotropins on the ovaries. The hypogonadism produced by prolactinomas is associated with osteoporosis due to estrogen deficiency.

As noted previously, hyperprolactinemia in men is associated with erectile dysfunction and hypogonadism that disappear when prolactin secretion is reduced.

THERAPEUTIC HIGHLIGHTS

Prescription drug use is a common cause of hyperprolactinemia. Prolactin

secretion in the pituitary is suppressed by the brain chemical dopamine. Use of drugs that block the effects of dopamine can cause the pituitary to secrete prolactin. Examples of some prescription drugs that can cause hyperprolactinemia include haloperidol and phenothiazines; most antipsychotic medications; and cisapride, which is used to treat nausea and gastroesophageal reflux in cancer patients. If possible, the drug suspected as causing hyperprolactinemia should be withdrawn, or the dose titrated. Regardless of the etiology, treatment should endeavor to restore normal prolactin levels to avoid suppressive effects on the ovaries and preserve bone density. Dopamine agonists also provide benefit in many cases, including prolactinoma, and can be used in patients for whom an inciting pharmaceutical agent cannot be withdrawn.

Delayed or Absent Puberty

The normal variation in the age at which adolescent changes occur is so wide that puberty cannot be considered to be pathologically delayed until the menarche has failed to occur by the age of 17 or testicular development by the age of 20. Failure of maturation due to panhypopituitarism is associated with dwarfing and evidence of other endocrine abnormalities. Patients with the XO chromosomal pattern and gonadal dysgenesis are also dwarfed. In some individuals, puberty is delayed even though the gonads are present and other endocrine functions are normal. In males, this clinical picture is called **eunuchoidism.** In females, it is called **primary amenorrhea** **(Clinical Box 22–3)**.

THE FEMALE REPRODUCTIVE SYSTEM

THE MENSTRUAL CYCLE

The reproductive system of women **(Figure 22–10)**, unlike that of men, shows regular cyclic changes that teleologically may be regarded as periodic preparations for fertilization and pregnancy. In humans and other primates, the cycle is a **menstrual** cycle, and its most conspicuous feature is the periodic vaginal bleeding that occurs with the shedding of the uterine mucosa **(menstruation).** The length of the cycle is notoriously variable in women, but an average figure is 28 days from the start of one menstrual period to the start of the

next. By common usage, the days of the cycle are identified by number, starting with the first day of menstruation.

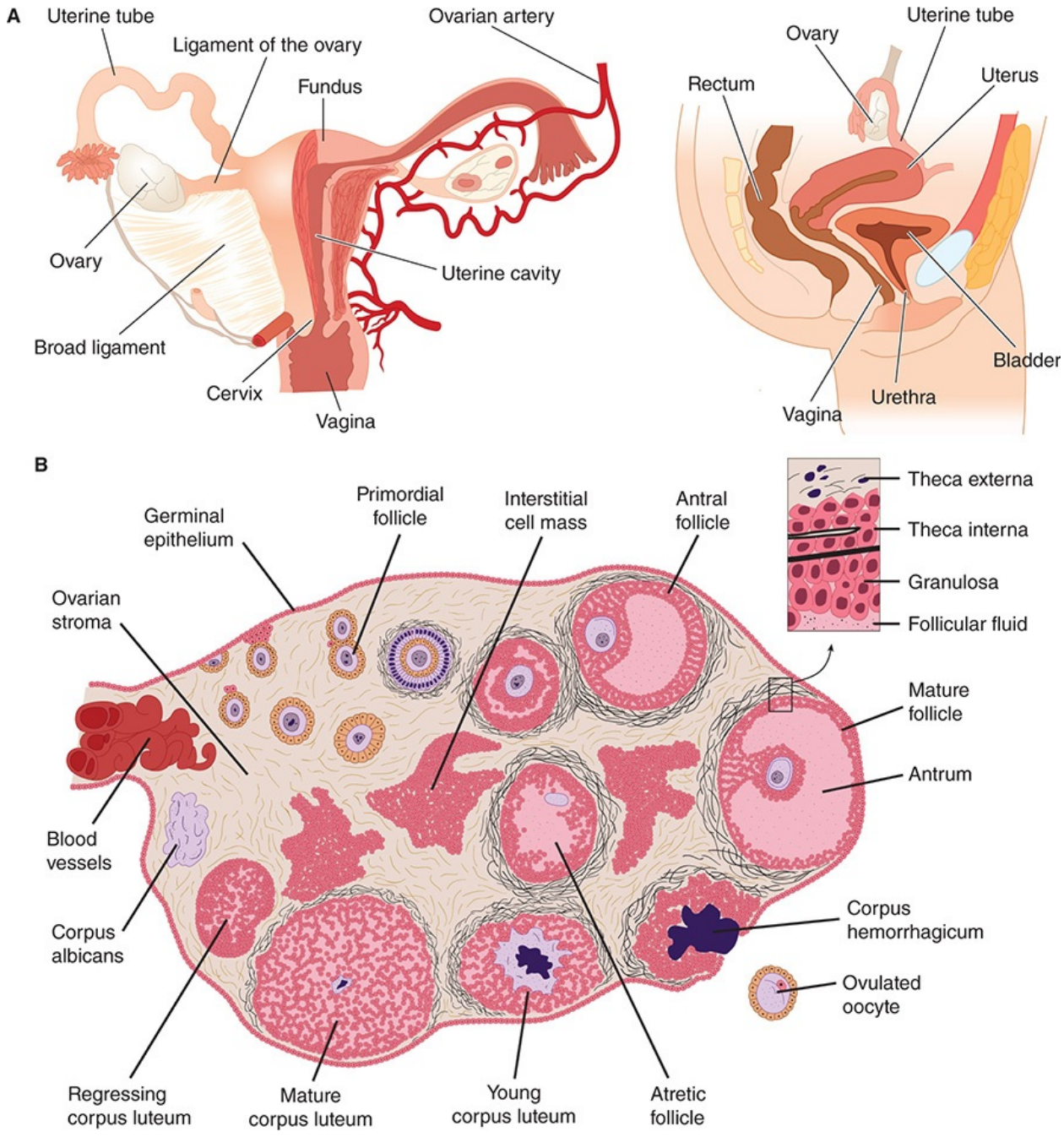


FIGURE 22–10 Functional anatomy of the female reproductive tract. The female reproductive organs include the ovaries, the uterus and the fallopian tubes, and the breast/mammary glands. The sequential development of a follicle, the formation of a corpus luteum and follicular atresia are shown.

Ovarian Cycle

From the time of birth, there are many **primordial follicles** under the ovarian capsule. Each contains an immature ovum (Figure 22–10). At the start of each cycle, several of these follicles enlarge, and a cavity forms around the ovum **(antrum formation).** This cavity is filled with follicular fluid. In humans, usually one of the follicles in one ovary starts to grow rapidly on about the sixth day and becomes the **dominant follicle,** while the others regress, forming **atretic follicles.** The atretic process involves apoptosis. It is uncertain how one follicle is selected to be the dominant follicle in this **follicular phase** of the menstrual cycle, but it seems to be related to the ability of the follicle to secrete the estrogen inside it that is needed for final maturation. When women are given human pituitary gonadotropin preparations by injection, many follicles develop simultaneously.

The structure of a maturing ovarian **(graafian)** follicle is shown in Figure 22–10. The primary source of circulating estrogen is the granulosa cells of the ovaries; however, the cells of the **theca interna** of the follicle are necessary for the production of estrogen as they secrete androgens that are aromatized to estrogen by the granulosa cells.

At about the 14th day of the cycle, the distended follicle ruptures, and the ovum is extruded into the abdominal cavity. This is the process of **ovulation.** The ovum is picked up by the fimbriated ends of the uterine tubes (oviducts). It is transported to the uterus and, unless fertilization occurs, out through the vagina.

The follicle that ruptures at the time of ovulation promptly fills with blood, forming what is sometimes called a **corpus hemorrhagicum.** Minor bleeding from the follicle into the abdominal cavity may cause peritoneal irritation and fleeting lower abdominal pain (“mittelschmerz”). The granulosa and theca cells of the follicle lining promptly begin to proliferate, and the clotted blood is rapidly replaced with yellowish, lipid-rich **luteal cells,** forming the **corpus luteum.** This initiates the **luteal phase** of the menstrual cycle, during which the luteal cells secrete estrogen and progesterone. Growth of the corpus luteum depends on its developing an adequate blood supply, and there is evidence that vascular endothelial growth factor (VEGF) (see Chapter 31) is essential for this process.

If pregnancy occurs, the corpus luteum persists and usually there are no more periods until after delivery. If pregnancy does not occur, the corpus luteum begins to degenerate about 4 days before the next menses (24th day of the cycle)

and is eventually replaced by scar tissue, forming a **corpus albicans.**

The ovarian cycle in other mammals is similar, except that in many species more than one follicle ovulates and multiple births are the rule. Corpora lutea form in some submammalian species but not in others.

In humans, no new ova are formed after birth. During fetal development, the ovaries contain over 7 million primordial follicles. However, many undergo atresia (involution) before birth and others are lost after birth. At the time of birth, there are 2 million ova, but 50% of these are atretic. The million that are normal undergo the first part of the first meiotic division at about this time and enter a stage of arrest in prophase in which those that survive persist until adulthood. Atresia continues during development, and the number of ova in both of the ovaries at the time of puberty is less than 300,000 (Figure 22–14). Only one of these ova per cycle (or about 500 in the course of a normal reproductive life) normally reaches maturity; the remainder degenerate. Just before ovulation, the first meiotic division is completed. One of the daughter cells, the **secondary oocyte,** receives most of the cytoplasm, while the other, the **first polar body,** fragments and disappears. The secondary oocyte immediately begins the second meiotic division, but this division stops at metaphase and is completed only when a sperm penetrates the oocyte. At that time, the **second polar body** is cast off and the fertilized ovum proceeds to form a new individual. The arrest in metaphase is due, at least in some species, to formation in the ovum of the protein **pp39mos**, which is encoded by the **cmos** protooncogene. When fertilization occurs, the pp39mos is destroyed within 30 min by **calpain,** a calcium-dependent cysteine protease.

Uterine Cycle

At the end of menstruation, all but the deep layers of the endometrium have sloughed. A new endometrium then regrows under the influence of estrogens from the developing follicle. The endometrium increases rapidly in thickness from the 5th to the 14th days of the menstrual cycle. As the thickness increases, the uterine glands are drawn out so that they lengthen **(Figure 22–11)**, but they do not become convoluted or secrete to any degree. These endometrial changes are called proliferative, and this part of the menstrual cycle is sometimes called the **proliferative phase.** It is also called the preovulatory or follicular phase of the cycle. After ovulation, the endometrium becomes more highly vascularized and slightly edematous under the influence of estrogen and progesterone from the corpus luteum. The glands become coiled and tortuous and they begin to

secrete a clear fluid. Consequently, this phase of the cycle is called the **secretory** or **luteal phase.** Late in the luteal phase, the endometrium, like the anterior pituitary, produces prolactin, but the function of this endometrial prolactin is unknown.

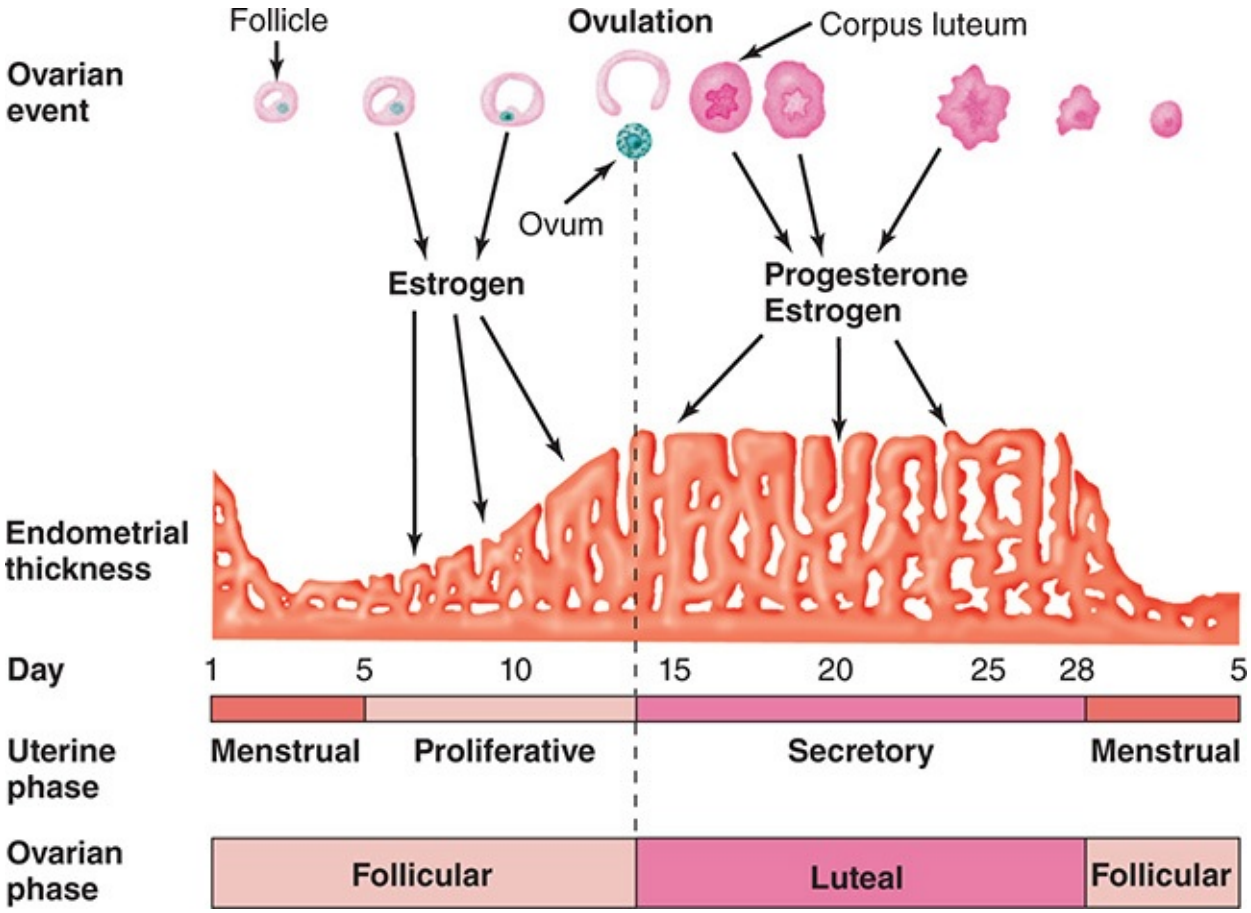


FIGURE 22–11 Relationship between ovarian and uterine changes during the menstrual cycle. (Reproduced with permission from Windmaier EP, Raff H, Strang KT: *Vander's Human Physiology: The Mechanisms of Body Function*, 11th ed. New York, NY: McGraw-Hill; 2008.)

The endometrium is supplied by two types of arteries. The superficial two-thirds of the endometrium that is shed during menstruation, the **stratum functionale,** is supplied by long, coiled **spiral arteries,** whereas the deep layer that is not shed, the **stratum basale,** is supplied by short, straight **basilar arteries.**

When the corpus luteum regresses, hormonal support for the endometrium is withdrawn. The endometrium becomes thinner, which adds to the coiling of the spiral arteries. Foci of necrosis appear in the endometrium, and these coalesce.

In addition, spasm and degeneration of the walls of the spiral arteries take place, leading to spotty hemorrhages that become confluent and produce the menstrual flow.

The vasospasm is probably produced by locally released prostaglandins. Large quantities of prostaglandins are present in the secretory endometrium and in menstrual blood, and infusions of prostagladin $F_{2\alpha}$($PGF_{2\alpha}$) produce endometrial necrosis and bleeding.

From the point of view of endometrial function, the proliferative phase of the menstrual cycle represents restoration of the epithelium from the preceding menstruation, and the secretory phase represents preparation of the uterus for implantation of the fertilized ovum. The length of the secretory phase is remarkably constant at about 14 days, and the variations seen in the length of the menstrual cycle are due for the most part to variations in the length of the proliferative phase. When fertilization fails to occur during the secretory phase, the endometrium is shed and a new cycle starts.

Normal Menstruation

Menstrual blood is predominantly arterial, with only 25% of the blood being of venous origin. It contains tissue debris, prostaglandins, and relatively large amounts of fibrinolysin from endometrial tissue. The fibrinolysin lyses clots, so that menstrual blood does not normally contain clots unless the flow is excessive.

The usual duration of the menstrual flow is 3–5 days, but flows as short as 1 day and as long as 8 days can occur in normal women. The amount of blood lost may range normally from slight spotting to 80 mL; the average amount lost is 30 mL. Loss of more than 80 mL is abnormal. Obviously, the amount of flow can be affected by various factors, including the thickness of the endometrium, medication, and diseases that affect the clotting mechanism.

Anovulatory Cycles

In some instances, ovulation fails to occur during the menstrual cycle. Such anovulatory cycles are common for the first 12–18 months after menarche and again before the onset of the menopause. When ovulation does not occur, no corpus luteum is formed and the effects of progesterone on the endometrium are absent. Estrogens continue to cause growth, however, and the proliferative endometrium becomes thick enough to break down and begins to slough. The

time it takes for bleeding to occur is variable, but it usually occurs in less than 28 days from the last menstrual period. The flow is also variable and ranges from scanty to relatively profuse.

Cyclical Changes in the Uterine Cervix

Although it is continuous with the body of the uterus, the cervix of the uterus is different in a number of ways. The mucosa of the uterine cervix does not undergo cyclical desquamation, but there are regular changes in the cervical mucus. Estrogen makes the mucus thinner and more alkaline, changes that promote the survival and transport of sperm. Progesterone makes it thick, tenacious, and cellular. The mucus is thinnest at the time of ovulation, and its elasticity, or **spinnbarkeit,** increases so that by midcycle, a drop can be stretched into a long, thin thread that may be 8–12 cm or more in length. In addition, it dries in an arborizing, fern-like pattern **(Figure 22–12)** when a thin layer is spread on a slide. After ovulation and during pregnancy, it becomes thick and fails to form the fern pattern.

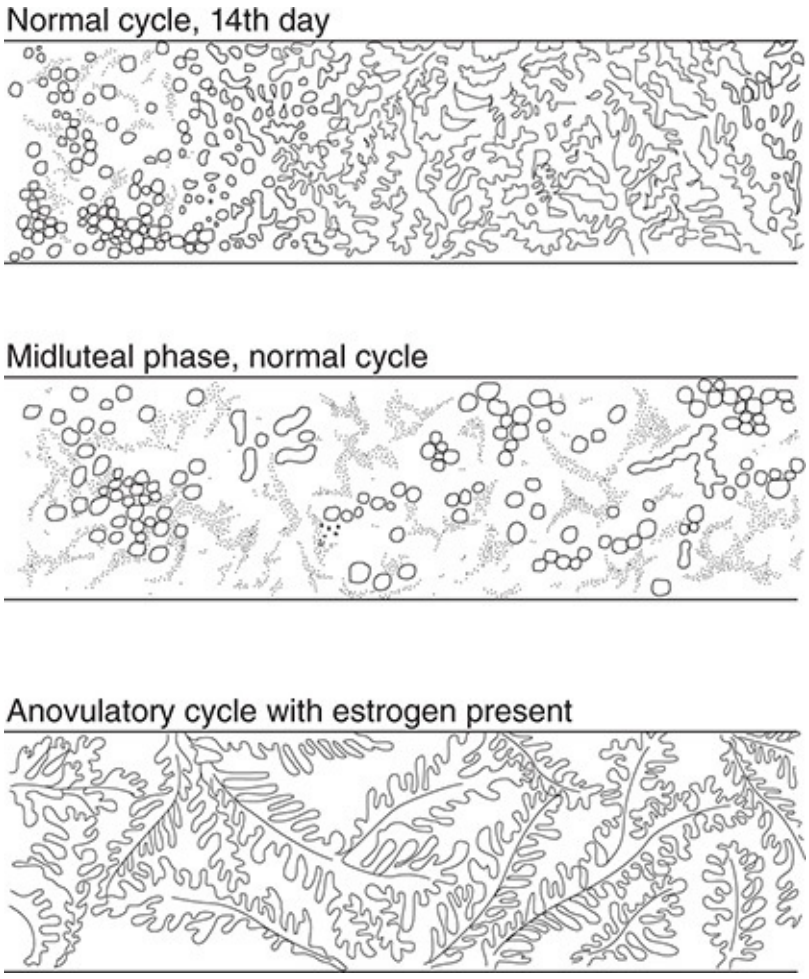


FIGURE 22–12 Patterns formed when cervical mucus is smeared on a slide, permitted to dry, and examined under the microscope. Progesterone makes the mucus thick and cellular. In the smear from a patient who did not ovulate **(bottom),** no progesterone is present to inhibit the estrogen-induced fern pattern.

Vaginal Cycle

Under the influence of estrogens, the vaginal epithelium becomes cornified, and cornified epithelial cells can be identified in the vaginal smear. Under the influence of progesterone, a thick mucus is secreted, and the epithelium proliferates and becomes infiltrated with leukocytes. The cyclical changes in the vaginal smear in rats are relatively marked. The changes in humans and other species are similar but not so clear-cut.

Cyclical Changes in the Breasts

Although lactation normally does not occur until the end of pregnancy, cyclical changes take place in the breasts during the menstrual cycle. Estrogens cause proliferation of mammary ducts, whereas progesterone causes growth of lobules and alveoli. The breast swelling, tenderness, and pain experienced by many women during the 10 days preceding menstruation are probably due to distension of the ducts, hyperemia, and edema of the interstitial tissue of the breast. All these changes regress, along with the symptoms, during menstruation.

Changes During Intercourse

During sexual excitement in women, fluid is secreted onto the vaginal walls, probably because of release of vasoactive intestinal polypeptide (VIP) from vaginal nerves. A lubricating mucus is also secreted by the vestibular glands. The upper part of the vagina is sensitive to stretch, while tactile stimulation from the labia minora and clitoris adds to the sexual excitement. These stimuli are reinforced by tactile stimuli from the breasts and, as in men, by visual, auditory, and olfactory stimuli, which may build to the crescendo known as orgasm. During orgasm, autonomically mediated rhythmic contractions occur in the vaginal walls. Impulses also travel via the pudendal nerves and produce rhythmic contraction of the bulbocavernosus and ischiocavernosus muscles. The vaginal contractions may aid sperm transport but are not essential for it, since fertilization of the ovum is not dependent on female orgasm.

Indicators of Ovulation

Knowing when during the menstrual cycle ovulation occurs is important in increasing fertility or, conversely, in family planning. A convenient and reasonably reliable indicator of the time of ovulation is a change—usually a rise—in the basal body temperature **(Figure 22–13)**. The rise starts 1–2 days after ovulation. Women interested in obtaining an accurate temperature chart should use a digital thermometer and take their temperatures (oral or rectal) in the morning before getting out of bed. The cause of the temperature change at the time of ovulation is probably the increase in progesterone secretion, since progesterone is thermogenic.

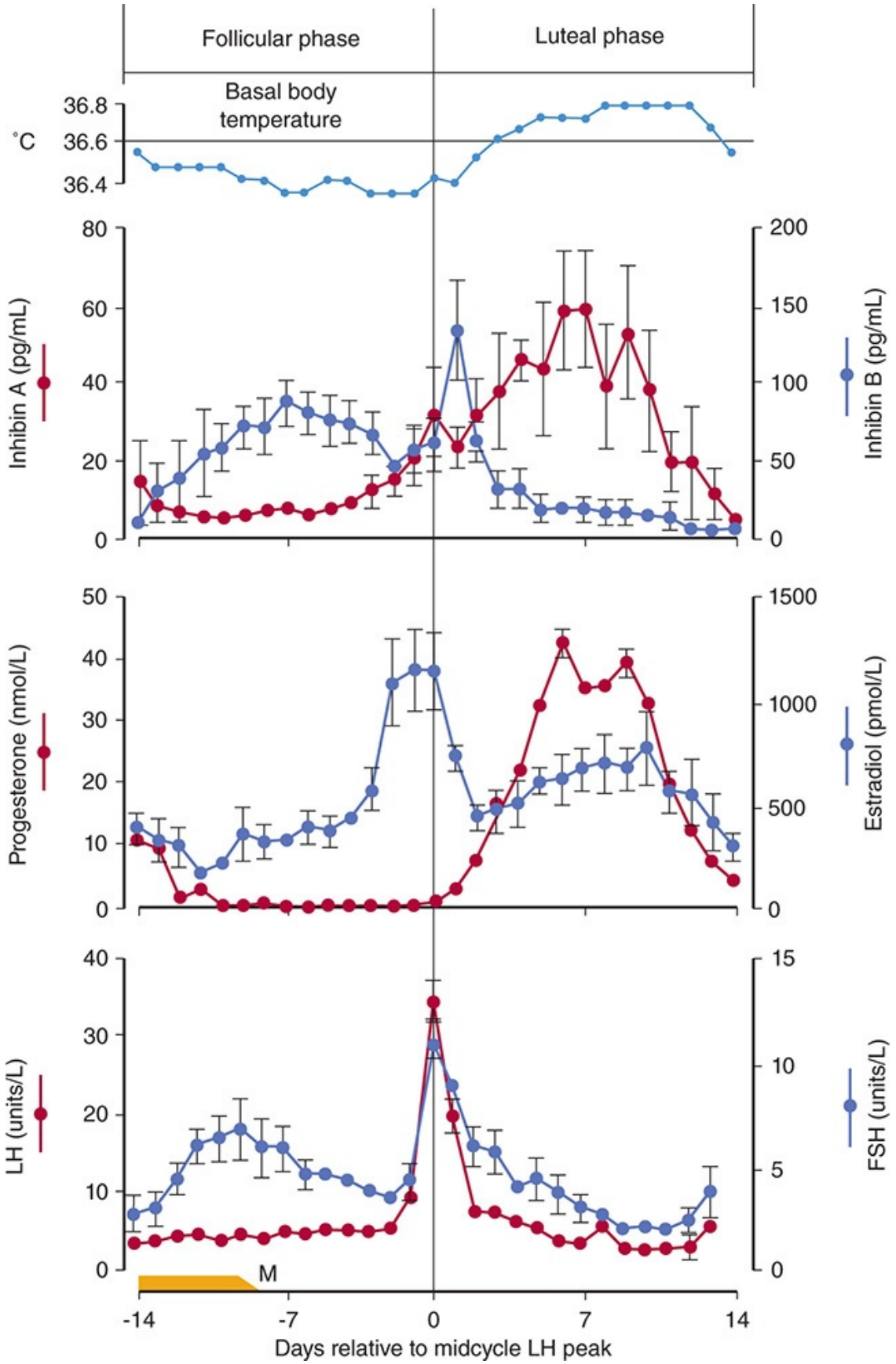

Follicular phase
Luteal phase
Basal body temperature
°C
36.8
36.6
36.4
Inhibin A (pg/mL)
Inhibin B (pg/mL)
80
60
40
20
0
200
150
100
50
0
Progesterone (nmol/L)
Estradiol (pmol/L)
50
40
30
20
10
0
1500
1000
500
0
LH (units/L)
FSH (units/L)
40
30
20
10
0
15
10
5
0
M
-14
-7
0
7
14
Days relative to midcycle LH peak

FIGURE 22–13 Basal body temperature and plasma hormone concentrations (mean ± standard error) during the normal human menstrual cycle. Values are aligned with respect to the day of the midcycle LH peak. FSH, follicle-stimulating hormone; LH, luteinizing hormone; M, menses.

A surge in LH secretion triggers ovulation, and ovulation normally occurs about 9 h after the peak of the LH surge at midcycle (Figure 22–13). The ovum lives for approximately 72 h after it is extruded from the follicle, but it is fertilizable for a much shorter time than this. In a study of the relation of isolated intercourse to pregnancy, 36% of women had a detected pregnancy following intercourse on the day of ovulation, but with intercourse on days after ovulation, the percentage was zero. Isolated intercourse on the first and second day before ovulation also led to pregnancy in about 36% of women. A few pregnancies resulted from isolated intercourse on day 3, 4, or 5 before ovulation, although the percentage was much lower, for example, 8% on day 5 before ovulation. Thus, some sperms can survive in the female genital tract and fertilize the ovum for up to 120 h before ovulation, but the most fertile period is clearly the 48 h before ovulation. However, for those interested in the "rhythm method" of contraception, it should be noted that there are rare but documented cases in the literature of pregnancy resulting from isolated coitus on every day of the cycle.

The Estrous Cycle

Mammals other than primates do not menstruate, and their sexual cycle is called an **estrous cycle.** It is named for the conspicuous period of "heat" **(estrus)** at the time of ovulation, normally the only time during which the sexual interest of the female is aroused. In spontaneously ovulating species with estrous cycles, such as the rat, no episodic vaginal bleeding occurs but the underlying endocrine events are essentially the same as those in the menstrual cycle. In other species, ovulation is induced by copulation (reflex ovulation).

MENOPAUSE

The human ovaries become unresponsive to gonadotropins with advancing age, and their function declines, so that sexual cycles disappear **(menopause).** This unresponsiveness is associated with and probably caused by a decline in the number of primordial follicles, which becomes precipitous at the time of

menopause **(Figure 22–14)**. The ovaries no longer secrete progesterone and 17β-estradiol in appreciable quantities, and estrogen is formed only in small amounts by aromatization of androstenedione in peripheral tissues (see Chapter 20). The uterus and the vagina gradually become atrophic. As the negative feedback effect of estrogens and progesterone is reduced, secretion of FSH is increased, and plasma FSH increases to high levels, LH levels are moderately high. Old female mice and rats have long periods of diestrus and increased levels of gonadotropin secretion. In women, a period called perimenopause precedes menopause and can last up to 10 years. During perimenopause FSH levels will increase before an increase in LH is observed due to a decrease in estrogen, progesterone, and inhibins and the menses become irregular. This usually occurs between the ages of 45 and 55. **The average age at onset of the menopause is 51 years.**

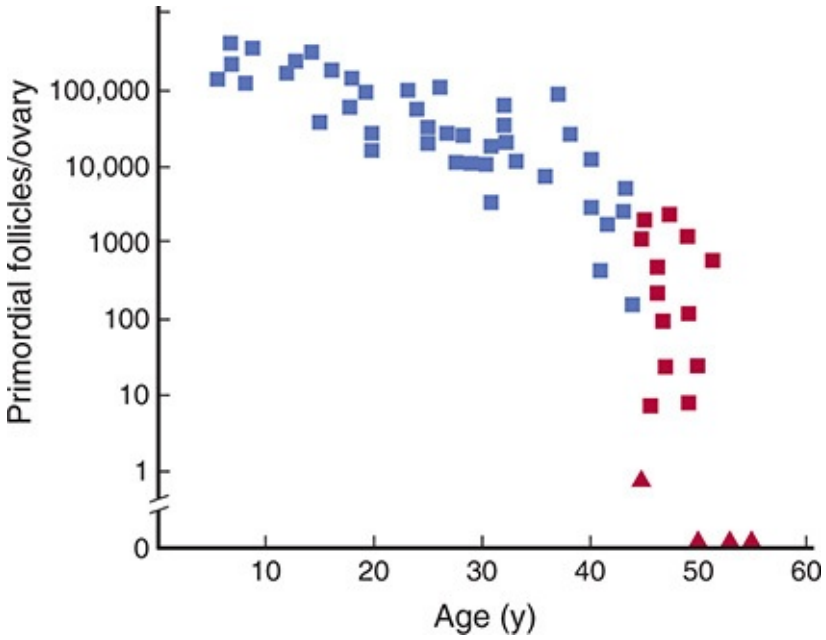


FIGURE 22–14 Number of primordial follicles per ovary in women at various ages. Blue squares, premenopausal women (regular menses); red squares, perimenopausal women (irregular menses for at least 1 year); red triangles, postmenopausal women (no menses for at least 1 year). Note that the vertical scale is a log scale and that the values are from one rather than two ovaries. (Redrawn by PM Wise and reproduced with permission from Richardson SJ, Senikas V, Nelson JF: Follicular depletion during the menopausal transition: Evidence for accelerated loss and ultimate exhaustion. J Clin Endocrinol Metab 1987;65:1231.)

The loss of ovarian function causes many symptoms such as sensations of warmth spreading from the trunk to the face (hot flushes; also called hot flashes) and night sweats. In addition, the onset of menopause increases the risk of many diseases such as osteoporosis, ischemic heart disease, and renal disease.

Hot flushes are said to occur in 75% of menopausal women and may continue intermittently for as long as 40 years. They also occur when early menopause is produced by bilateral ovariectomy, and they are prevented by estrogen treatment. In addition, they occur after castration in men. Their cause is unknown. However, they coincide with surges of LH secretion. LH is secreted in episodic bursts at intervals of 30–60 min or more **(circhoral secretion),** and in the absence of gonadal hormones these bursts are large. Each hot flush begins with the start of a burst. However, LH itself is not responsible for the symptoms, because they can continue after removal of the pituitary. Instead, it appears that some estrogen-sensitive event in the hypothalamus initiates both the release of LH and the episode of flushing.

Although the function of the testes tends to decline slowly with advancing age, the evidence is unclear whether there is a "male menopause" **(andropause)** similar to that occurring in women.

OVARIAN HORMONES

Chemistry, Biosynthesis, & Metabolism of Estrogens

The naturally occurring estrogens are **17β-estradiol, estrone,** and **estriol (Figure 22–15)**. They are C18 steroids that do not have an angular methyl group attached to the 10 position or a Δ4-3-keto configuration in the A ring. They are secreted primarily by the granulosa cells of the ovarian follicles, the corpus luteum, and the placenta. Their biosynthesis depends on the enzyme **aromatase** (CYP19), which converts testosterone to estradiol and androstenedione to estrone (Figure 22–15). The latter reaction also occurs in fat, liver, muscle, and the brain.

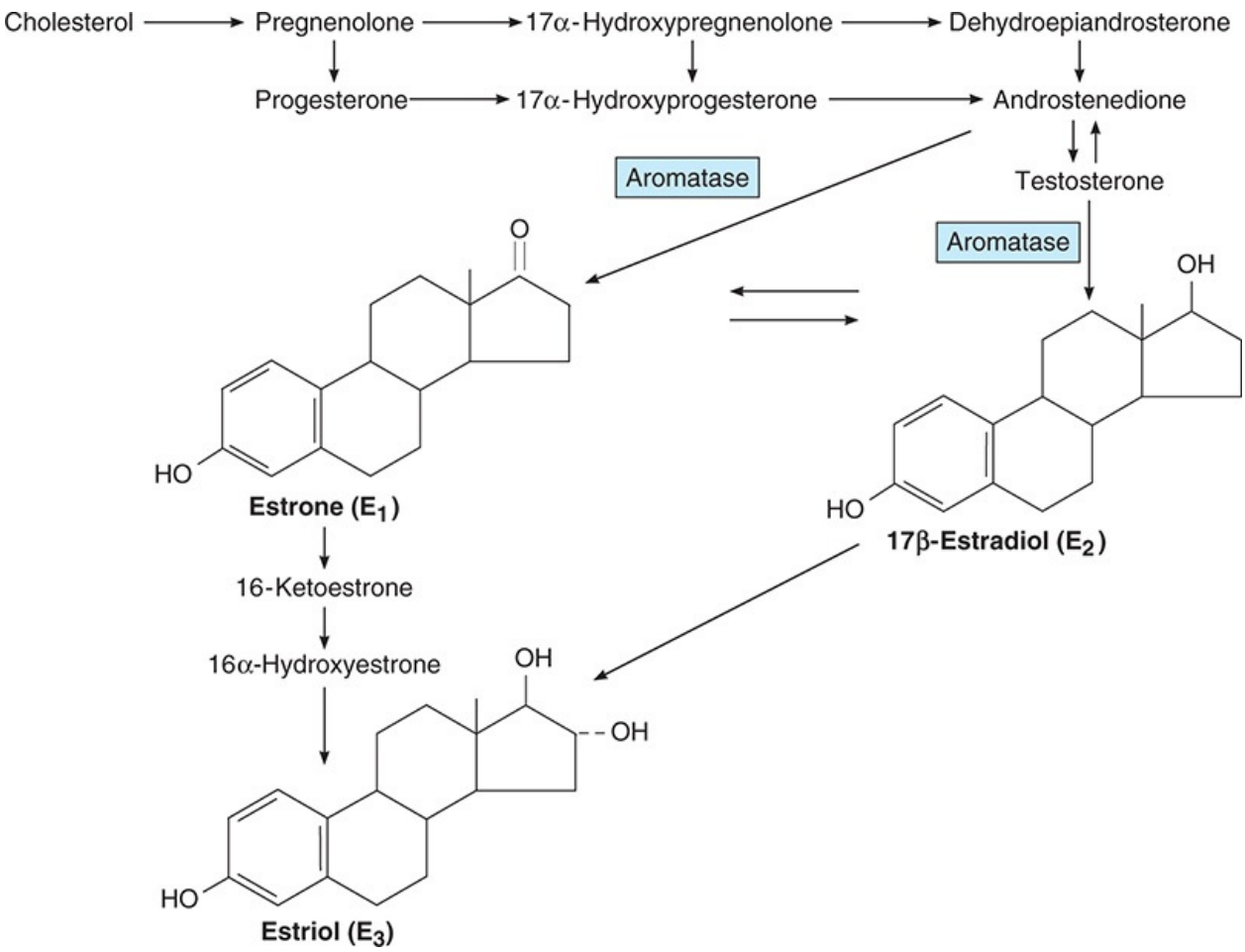


FIGURE 22–15 Biosynthesis and metabolism of estrogens. The formulas of the precursor steroids are shown in Figure 19–7.

Theca interna cells have many LH receptors, and LH acts via cAMP to increase conversion of cholesterol to androstenedione. The theca interna cells supply androstenedione to the granulosa cells. The granulosa cells make estradiol when provided with androgens **(Figure 22–16)**, and it appears that the estradiol they form in primates is secreted into the follicular fluid. Granulosa cells have many FSH receptors, and FSH facilitates their secretion of estradiol by acting via cAMP to increase their aromatase activity. Mature granulosa cells also acquire LH receptors, and LH also stimulates estradiol production.

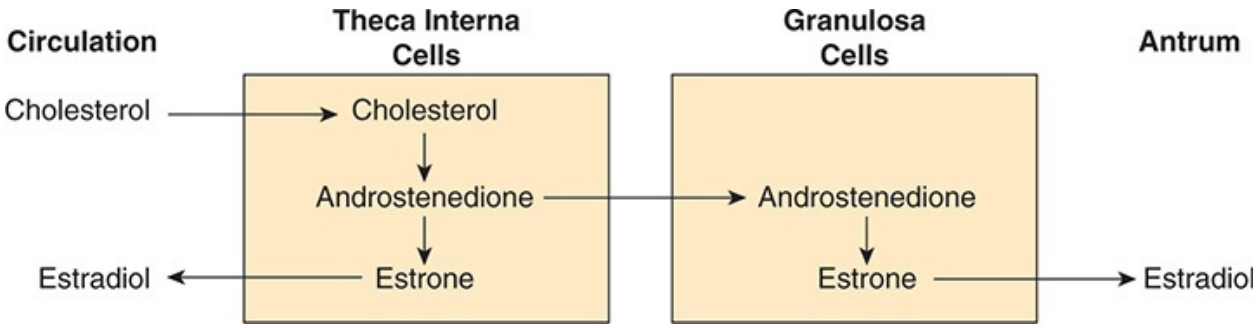


FIGURE 22–16 Interactions between theca and granulosa cells in estradiol synthesis and secretion.

Two percent of the circulating estradiol is free, and the remainder is bound to protein: 60% to albumin and 38% to the same gonadal steroid-binding globulin (GBG) that binds testosterone.

In the liver, estradiol, estrone, and estriol are converted to glucuronide and sulfate conjugates. All these compounds, along with other metabolites, are excreted in the urine. Appreciable amounts are secreted in the bile and reabsorbed into the bloodstream (enterohepatic circulation).

Secretion

The concentration of estradiol in the plasma during the menstrual cycle is shown in Figure 22–14. Almost all of this estrogen comes from the ovary, and two peaks of secretion occur: one just before ovulation and one during the midluteal phase. The estradiol secretion rate is 36 µg/day (133 nmol/day) in the early follicular phase, 380 µg/day just before ovulation, and 250 µg/day during the midluteal phase **(Table 22–3)**. After menopause, estrogen secretion declines to low levels.

TABLE 22–3 Twenty-four-hour production rates of sex steroids in women at different stages of the menstrual cycle.

Sex Steroids	Early Follicular	Preovulatory	Midluteal
Progesterone (mg)	1.0	4.0	25.0
17-Hydroxyprogesterone (mg)	0.5	4.0	4.0
Dehydroepiandrosterone (mg)	7.0	7.0	7.0
Androstenedione (mg)	2.6	4.7	3.4
Testosterone (µg)	144.0	171.0	126.0
Estrone (µg)	50.0	350.0	250.0
Estradiol (µg)	36.0	380.0	250.0

Data from Baird DT, Fraser IS: Blood production and ovarian secretion rates of estradiol-17 beta and estrone in women throughout the menstrual cycle. J Clin Endocrinal Metab 1974 Jun; 38(6):1009-1017.

As noted previously, the estradiol production rate in men is about 50 µg/day (184 nmol/day).

Effects on the Female Genitalia

Estrogens facilitate the growth of the ovarian follicles and increase the motility of the uterine tubes. Their role in the cyclic changes in the endometrium, cervix, and vagina has been discussed previously. They increase uterine blood flow and have important effects on the smooth muscle of the uterus. In immature and castrated females, the uterus is small and the myometrium atrophic and inactive. Estrogens increase the amount of uterine muscle and its content of contractile proteins. Under the influence of estrogens, the muscle becomes more active and excitable, and action potentials in the individual fibers become more frequent. The “estrogen-dominated” uterus is also more sensitive to oxytocin.

Chronic treatment with estrogens causes the endometrium to hypertrophy. When estrogen therapy is discontinued, sloughing takes place with **withdrawal bleeding.** Some “breakthrough” bleeding may occur during treatment when estrogens are given for long periods. Prolonged exposure to estrogen alone (unopposed by progesterone) has been indicated as a risk factor in the development of endometrial cancer.

Effects on Endocrine Organs

Estrogens decrease FSH secretion. Under some circumstances, they inhibit LH secretion (negative feedback); in other circumstances, they increase LH secretion (positive feedback). Women are sometimes given large doses of estrogens for 4–6 days to prevent conception after coitus during the fertile period (postcoital or "morning-after" contraception). However, in this instance, pregnancy is probably prevented by interference with implantation of the ovum rather than changes in gonadotropin secretion.

Estrogens cause increased secretion of angiotensinogen and thyroid-binding globulin. They exert an important protein anabolic effect in chickens and cattle, possibly by stimulating the secretion of androgens from the adrenal, and estrogen treatment has been used commercially to increase the weight of domestic animals. They cause epiphysial closure in humans (see Chapter 21).

Effects on the Central Nervous System

The estrogens are responsible for estrous behavior in animals, and they increase libido in humans. They apparently exert this action by a direct effect on certain neurons in the hypothalamus. Estrogens also increase the proliferation of dendrites on neurons and the number of synaptic knobs in rats.

Effects on the Breasts

Estrogens produce duct growth in the breasts and are largely responsible for breast enlargement at puberty in girls; they have been called the growth hormones of the breast. They are responsible for the pigmentation of the areolas, although pigmentation usually becomes more intense during the first pregnancy than it does at puberty. The role of estrogens in the overall control of breast growth and lactation is discussed below.

Female Secondary Sex Characteristics

The body changes that develop in girls at puberty—in addition to enlargement of breasts, uterus, and vagina—are due in part to estrogens, which are the "feminizing hormones," and in part simply to the absence of testicular androgens. Women have narrow shoulders and broad hips, thighs that converge, and arms that diverge (wide **carrying angle**). This body configuration, plus the

female distribution of fat in the breasts and buttocks, is seen also in castrate males. In women, the larynx retains its prepubertal proportions and the voice stays high-pitched. Women have less body hair and more scalp hair, and the pubic hair generally has a characteristic flat-topped pattern (female escutcheon). However, growth of pubic and axillary hair in both sexes is due primarily to androgens rather than estrogens.

Other Actions

Normal women retain salt and water and gain weight just before menstruation. Estrogens cause some degree of salt and water retention. However, aldosterone secretion is slightly elevated in the luteal phase, and this also contributes to the premenstrual fluid retention.

Estrogens are said to make sebaceous gland secretions more fluid and thus to counter the effect of testosterone and inhibit formation of **comedones** ("black-heads") and acne. The liver palms, spider angiomas, and slight breast enlargement seen in advanced liver disease are due to increased circulating estrogens. The increase appears to be due to decreased hepatic metabolism of androstenedione, making more of this androgen available for conversion to estrogens.

Estrogens have a significant plasma cholesterol-lowering action, and they rapidly produce vasodilation by increasing the local production of NO. Their action on bone is discussed in Chapter 21.

Mechanism of Action

There are two principal types of nuclear estrogen receptors: estrogen receptor α (ERα) encoded by a gene on chromosome 6; and estrogen receptor β (ERβ), encoded by a gene on chromosome 14. Both are members of the nuclear receptor super-family (see Chapter 2). After binding estrogen, they form homodimers and bind to DNA, altering its transcription. Some tissues contain one type or the other, but overlap also occurs, with some tissues containing both ERα and ERβ. ERα is found primarily in the uterus, kidneys, liver, and heart, whereas ERβ is found primarily in the ovaries, prostate, lungs, gastrointestinal tract, hemopoietic system, and central nervous system (CNS). ERα and ERβ can also form heterodimers. Male and female mice in which the gene for ERα has been knocked out are sterile, develop osteoporosis, and continue to grow because their epiphyses do not close. ERβ female knockouts are infertile, but ERβ male

knockouts are fertile even though they have hyperplastic prostates and loss of fat. Both receptors exist in isoforms and, like thyroid receptors, can bind to various activating and stimulating factors. In some situations, ERβ can inhibit ERα transcription. Thus, their actions are complex, multiple, and varied.

Most of the effects of estrogens are genomic, that is, due to actions on the nucleus, but some are so rapid that it is difficult to believe they are mediated via production of mRNAs. These include effects on neuronal discharge in the brain and, possibly, feedback effects on gonadotropin secretion. Evidence is accumulating that these effects are mediated by cell membrane receptors that appear to be structurally related to the nuclear receptors and produce their effects by intracellular mitogen-activated protein kinase pathways. Similar rapid effects of progesterone, testosterone, glucocorticoids, aldosterone, and 1,25-dihydroxycholecalciferol may also be produced by membrane receptors (see Chapter 16).

Synthetic & Environmental Estrogens

The ethinyl derivative of estradiol is a potent estrogen and, unlike the naturally occurring estrogens, is relatively active when given by mouth because it is resistant to hepatic metabolism. The activity of the naturally occurring hormones is low when they are administered by mouth because the portal venous drainage of the intestine carries them to the liver, where they are inactivated before they can reach the general circulation. Some nonsteroidal substances and a few compounds found in plants also have estrogenic activity. The plant estrogens are rarely a problem in human nutrition, but they may cause undesirable effects in farm animals. **Dioxins,** which are found in the environment and are produced by a variety of industrial processes, can activate estrogen response elements on genes. However, they have been reported to have antiestrogenic as well as estrogenic effects, and their role, if any, in the production of human disease remains a matter of disagreement and debate.

Because natural estrogens have undesirable as well as desirable effects (eg, they preserve bone in osteoporosis but can cause uterine and breast cancer), there has been an active search for “tailor-made” estrogens that have selective effects in humans. Two compounds, **tamoxifen** and **raloxifene,** show promise in this regard. Neither combats the symptoms of menopause, but both have the bone-preserving effects of estradiol. In addition, tamoxifen does not stimulate the breast, and raloxifene does not stimulate the breast or uterus. The way the effects of these selective estrogen receptor modulators **(SERMs)** are brought

about is related to the complexity of the estrogen receptors and hence to differences in the way that the receptor–ligand complexes they form bind to DNA.

Chemistry, Biosynthesis, & Metabolism of Progesterone

Progesterone is a C_{21} steroid **(Figure 22–17)** secreted by the corpus luteum, the placenta, and (in small amounts) the follicle. It is an important intermediate in steroid biosynthesis in all tissues that secrete steroid hormones, and small amounts apparently enter the circulation from the testes and adrenal cortex. About 2% of the circulating progesterone is free whereas 80% is bound to albumin and 18% is bound to corticosteroid-binding globulin. Progesterone has a short half-life and is converted in the liver to pregnanediol, which is conjugated to glucuronic acid and excreted in the urine.

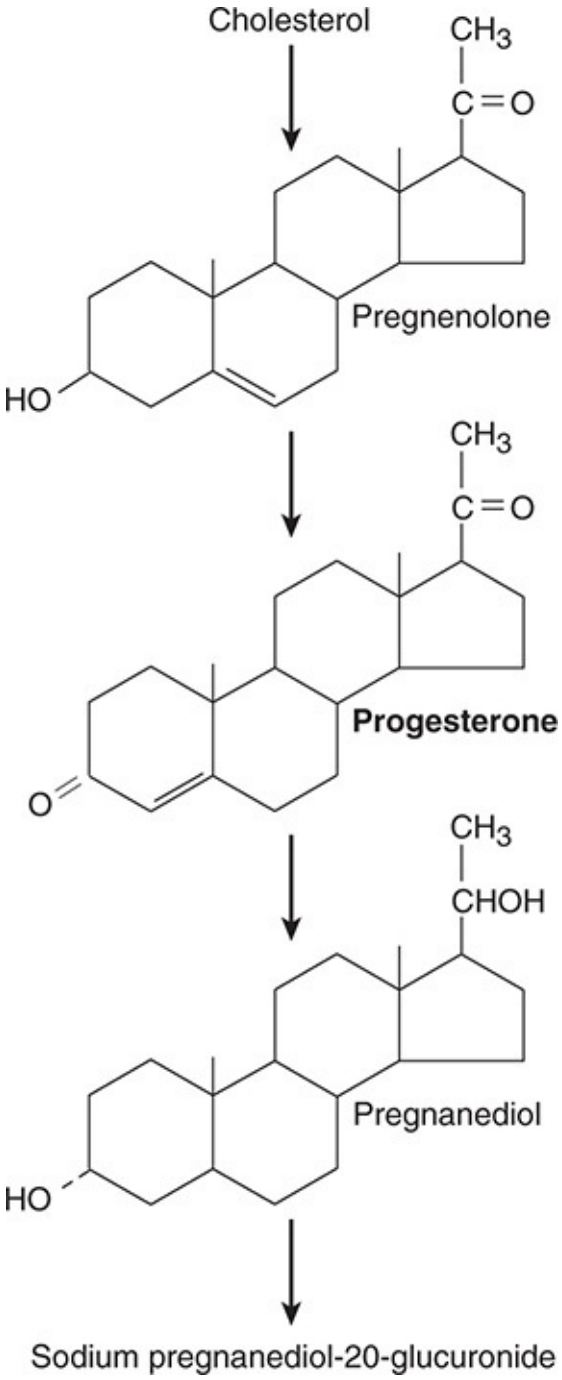


FIGURE 22–17 Biosynthesis of progesterone and major pathway for its metabolism. Other metabolites are also formed.

Secretion

In men, the plasma progesterone level is approximately 0.3 ng/mL (1 nmol/L). In women, the level is approximately 0.9 ng/mL (3 nmol/L) during the follicular

phase of the menstrual cycle (Figure 22–14). The difference is due to secretion of small amounts of progesterone by cells in the ovarian follicles; theca cells provide pregnenolone to the granulosa cells, which convert it to progesterone. Late in the follicular phase, progesterone secretion begins to increase. During the luteal phase, the corpus luteum produces large quantities of progesterone (Table 22–3) and plasma progesterone is markedly increased to a peak value of approximately 18 ng/mL (60 nmol/L).

The stimulating effect of LH on progesterone secretion by the corpus luteum is due to activation of adenylyl cyclase and involves a subsequent step that is dependent on protein synthesis.

Actions

The principal target organs of progesterone are the uterus, the breasts, and the brain. Progesterone is responsible for the progestational changes in the endometrium and the cyclical changes in the cervix and vagina described above. It has an antiestrogenic effect on the myometrial cells, decreasing their excitability, their sensitivity to oxytocin, and their spontaneous electrical activity while increasing their membrane potential. It also decreases the number of estrogen receptors in the endometrium and increases the rate of conversion of 17β-estradiol to less active estrogens.

In the breast, progesterone stimulates the development of lobules and alveoli. It induces differentiation of estrogen-prepared ductal tissue and supports the secretory function of the breast during lactation.

The feedback effects of progesterone are complex and are exerted at both the hypothalamic and pituitary levels. Large doses of progesterone inhibit LH secretion and potentiate the inhibitory effect of estrogens, preventing ovulation.

Progesterone is thermogenic and is probably responsible for the rise in basal body temperature at the time of ovulation. It stimulates respiration, and the alveolar PCO_2 (see Chapter 34) in women during the luteal phase of the menstrual cycle is lower than that in men. In pregnancy, the PCO_2 falls as progesterone secretion rises. However, the physiologic significance of this respiratory response is unknown.

Large doses of progesterone produce natriuresis, probably by blocking the action of aldosterone on the kidney. The hormone does not have a significant anabolic effect.

Mechanism of Action

The effects of progesterone, like those of other steroids, are brought about by an action on DNA to initiate synthesis of new mRNA. The progesterone receptor is bound to a heat shock protein in the absence of the steroid, and progesterone binding releases the heat shock protein, exposing the DNA-binding domain of the receptor. The synthetic steroid **mifepristone (RU 486)** binds to the receptor but does not release the heat shock protein, and it blocks the binding of progesterone. Because the maintenance of early pregnancy depends on the stimulatory effect of progesterone on endometrial growth and its inhibition of uterine contractility, mifepristone combined with a prostaglandin can be used to produce elective abortions.

There are two isoforms of the progesterone receptor—PR_A and PR_B—that are produced by differential processing from a single gene. PR_A is a truncated form, but it is likely that both isoforms mediate unique subsets of progesterone action.

Substances that mimic the action of progesterone are sometimes called **progestational agents, gestagens,** or **progestins.** They are used along with synthetic estrogens as oral contraceptive agents.

Relaxin

Relaxin is a polypeptide hormone that is produced in the corpus luteum, uterus, placenta, and mammary glands in women and in the prostate gland in men. During pregnancy, it relaxes the pubic symphysis and other pelvic joints and softens and dilates the uterine cervix. Thus, it facilitates delivery. It also inhibits uterine contractions and may play a role in the development of the mammary glands. In nonpregnant women, relaxin is found in the corpus luteum and the endometrium during the secretory but not the proliferative phase of the menstrual cycle. Its function in nonpregnant women is unknown. In men, it is found in semen, where it may help maintain sperm motility and aid in sperm penetration of the ovum.

In most species there is only one relaxin gene, but in humans there are two genes on chromosome 9 that code for two structurally different polypeptides that both have relaxin activity. However, only one of these genes is active in the ovary and the prostate.

CONTROL OF OVARIAN FUNCTION

FSH from the pituitary is responsible for the early maturation of the ovarian follicles, and FSH and LH together are responsible for their final maturation. A burst of LH secretion (Figure 22–14) is responsible for ovulation and the initial formation of the corpus luteum. A smaller midcycle burst of FSH secretion also occurs, the significance of which is uncertain. LH stimulates the secretion of estrogen and progesterone from the corpus luteum.

Hypothalamic Components

The hypothalamus occupies a key position in the control of gonadotropin secretion. Hypothalamic control is exerted by GnRH secreted into the portal hypophysial vessels. GnRH stimulates the secretion of FSH as well as LH.

GnRH is normally secreted in episodic bursts, and these bursts produce the circhoral peaks of LH secretion. They are essential for normal secretion of gonadotropins. If GnRH is administered by constant infusion, the GnRH receptors in the anterior pituitary downregulate and LH secretion declines to zero. However, if GnRH is administered episodically at a rate of one pulse per hour, LH secretion is stimulated. This is true even when endogenous GnRH secretion has been prevented by a lesion of the ventral hypothalamus.

It is now clear not only that episodic secretion of GnRH is a general phenomenon but also that fluctuations in the frequency and amplitude of the GnRH bursts are important in generating the other hormonal changes that are responsible for the menstrual cycle. Frequency is increased by estrogens and decreased by progesterone and testosterone. The frequency increases late in the follicular phase of the cycle, culminating in the LH surge. During the secretory phase, the frequency decreases as a result of the action of progesterone **(Figure 22–18)**, but when estrogen and progesterone secretion decrease at the end of the cycle, the frequency once again increases.

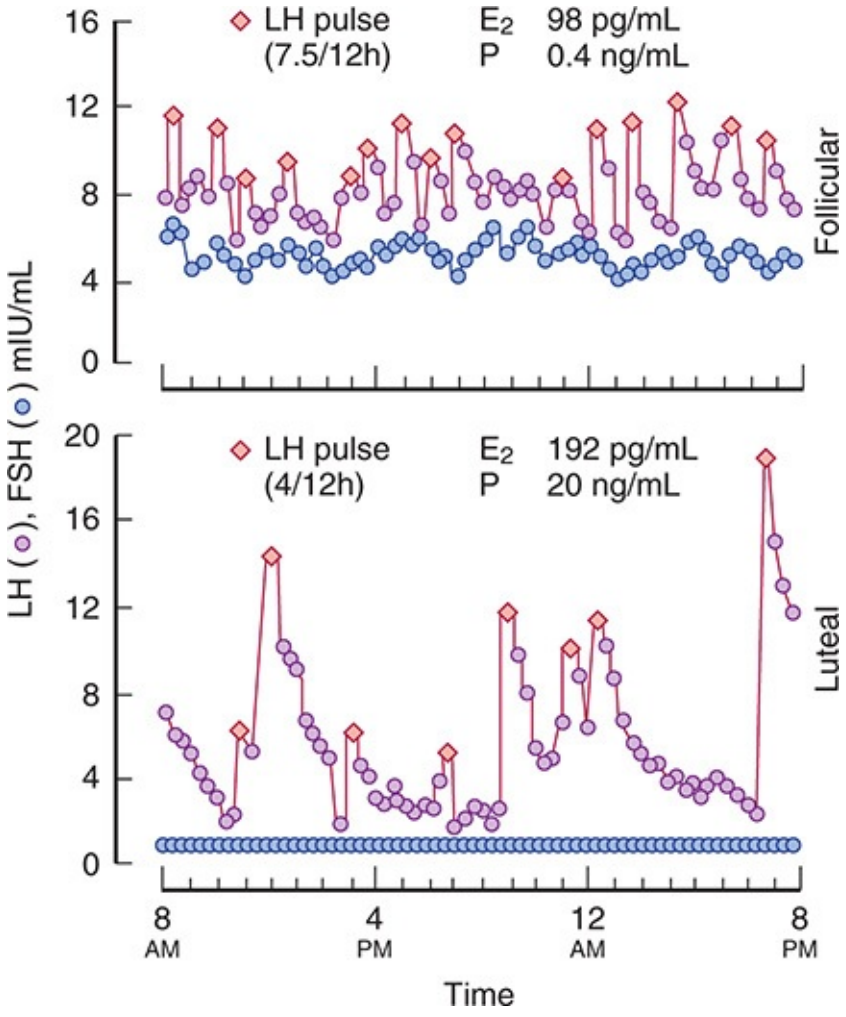


FIGURE 22–18 Episodic secretion of LH (s) and FSH (d) during the follicular stage (top) and the luteal stage (bottom) of the menstrual cycle. The numbers above each graph indicate the numbers of LH pulses per 12 hours and the plasma estradiol (E_2) and progesterone (P) concentrations at these two times of the cycle. (Reproduced with permission from Marshall JC, Kelch RO: Gonadotropin-releasing hormone: Role of pulsatile secretion in the regulation of reproduction. N Engl J Med 1986; Dec 4; 315(23):1459–1468.)

At the time of the midcycle LH surge, the sensitivity of the gonadotropes to GnRH is greatly increased because of their exposure to GnRH pulses at a specific frequency. This self-priming effect of GnRH is important in producing a maximum LH response.

The nature and the exact location of the GnRH pulse generator in the hypothalamus are still unsettled. However, it is known in a general way that norepinephrine and possibly epinephrine in the hypothalamus increase GnRH

pulse frequencies. Conversely, opioid peptides such as the enkephalins and β-endorphin reduce the frequency of GnRH pulses.

The downregulation of pituitary receptors and the consequent decrease in LH secretion produced by constantly elevated levels of GnRH has led to the use of long-acting GnRH analogs to inhibit LH secretion in precocious puberty and in cancer of the prostate.

Feedback Effects

Changes in plasma LH, FSH, sex steroids, and inhibin during the menstrual cycle are shown in Figure 22–13, and their feedback relations are diagrammed in **Figure 22–19**. During the early part of the follicular phase, inhibin B is low and FSH is modestly elevated, fostering follicular growth. LH secretion is held in check by the negative feedback effect of the rising plasma estrogen level. At 36–48 h before ovulation, the estrogen feedback effect becomes positive, and this initiates the burst of LH secretion (LH surge) that produces ovulation. Ovulation occurs about 9 h after the LH peak. FSH secretion also peaks, despite a small rise in inhibin, probably because of the strong stimulation of gonadotropes by GnRH. During the luteal phase, the secretion of LH and FSH is low because of the elevated levels of estrogen, progesterone, and inhibin.

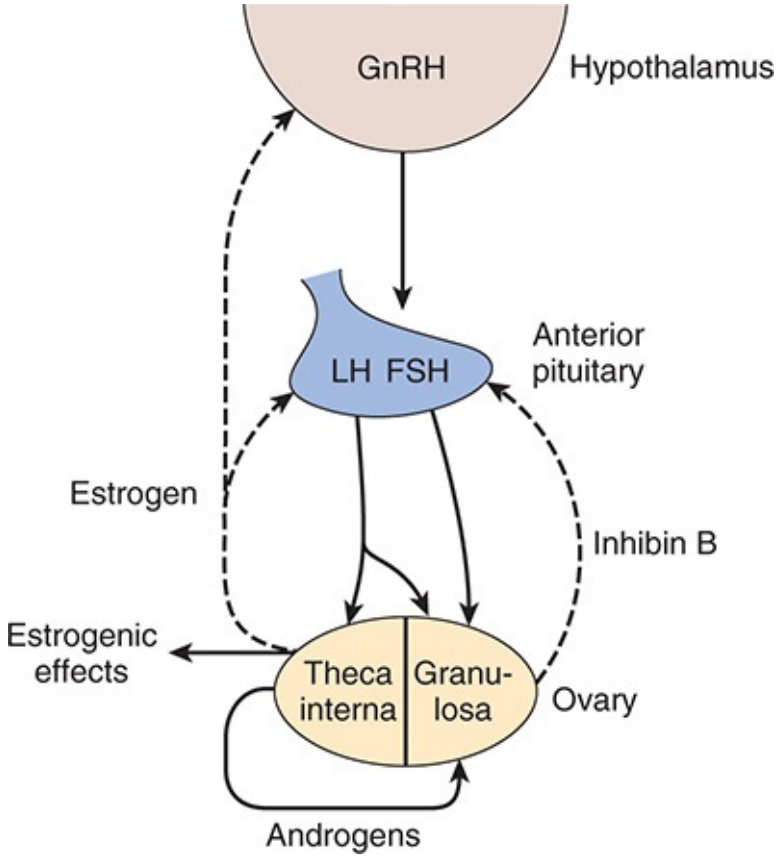


FIGURE 22–19 Feedback regulation of ovarian function. The cells of the theca interna provide androgens to the granulosa cells, and theca cells also produce the circulating estrogens that inhibit the secretion of GnRH, LH, and FSH. Inhibin from the granulosa cells inhibits FSH secretion. LH regulates the thecal cells, whereas the granulosa cells are regulated by both LH and FSH. The dashed arrows indicate inhibitory effects and the solid arrows stimulatory effects. FSH, follicle-stimulating hormone; GnRH, gonadotropin-releasing hormone; LH, luteinizing hormone.

It should be emphasized that a moderate, constant level of circulating estrogen exerts a negative feedback effect on LH secretion, whereas during the cycle, an elevated estrogen level exerts a positive feedback effect and stimulates LH secretion. In monkeys, it has been demonstrated that estrogens must also be elevated for a minimum time to produce positive feedback. When circulating estrogen was increased about 300% for 24 h, only negative feedback was seen; but when it was increased about 300% for 36 h or more, a brief decline in secretion was followed by a burst of LH secretion that resembled the midcycle surge. When circulating levels of progesterone were high, the positive feedback effect of estrogen was inhibited. In primates, there is evidence that both the

negative and the positive feedback effects of estrogen are exerted in the hypothalamus and recent studies have identified a role for the peptide **kisspeptin**, which is expressed in neurons in the hypothalamus (in humans, kisspeptin positive neurons are concentrated in the infundibular nucleus). Estrogen and progesterone modulate kisspeptin activity though receptors expressed on kisspeptin neurons.

Control of the Cycle

In an important sense, regression of the corpus luteum **(luteolysis)** starting 3–4 days before menses is the key to the menstrual cycle. $PGF_{2\alpha}$ appears to be a physiologic luteolysin, but this prostaglandin is only active when endothelial cells producing ET-1 (see Chapter 32) are present. Therefore, it appears that at least in some species luteolysis is produced by the combined action of $PGF_{2\alpha}$ and ET-1. In some domestic animals, oxytocin secreted by the corpus luteum appears to exert a local luteolytic effect, possibly by causing the release of prostaglandins. Once luteolysis begins, the estrogen and progesterone levels fall and the secretion of FSH and LH increases. A new crop of follicles develops, and then a single dominant follicle matures as a result of the action of FSH and LH. Near midcycle, estrogen secretion from the follicle rises. This rise augments the responsiveness of the pituitary to GnRH and triggers a burst of LH secretion. The resulting ovulation is followed by formation of a corpus luteum. Estrogen secretion drops, but progesterone and estrogen levels then rise together, along with inhibin B. The elevated levels inhibit FSH and LH secretion for a while, but luteolysis again occurs and a new cycle starts.

Reflex Ovulation

Female cats, rabbits, mink, and some other animals have long periods of estrus, during which they ovulate only after copulation. Such **reflex ovulation** is brought about by afferent impulses from the genitalia and the eyes, ears, and nose that converge on the ventral hypothalamus and provoke an ovulation-inducing release of LH from the pituitary. In species such as rats, monkeys, and humans, ovulation is a spontaneous periodic phenomenon, but neural mechanisms are also involved. Ovulation can be delayed 24 h in rats by administering pentobarbital or various other neurally active drugs 12 h before the expected time of follicle rupture.

Contraception

Methods commonly used to prevent conception are listed in Table 22–4, along with their failure rates. Once conception has occurred, abortion can be produced by progesterone antagonists such as mifepristone.

TABLE 22–4 Relative effectiveness of frequently used contraceptive methods.

Method	Failures per 100 Woman-Years
Vasectomy	0.02
Tubal ligation and similar procedures	0.13
Oral contraceptives	
>50 ug estrogen and progestin	0.32
<50 ug estrogen and progestin	0.27
Progestin only	1.2
IUD	
Copper 7	1.5
Loop D	1.3
Diaphragm	1.9
Condom	3.6
Withdrawal	6.7
Spermicide	11.9
Rhythm	15.5

Data from Vessey M, Lawless M, Yeates D: Efficacy of different contraceptive methods. Lancet 1982;1:841. Reproduced with permission.

Implantation of foreign bodies in the uterus causes changes in the duration of the sexual cycle in a number of mammalian species. In humans, such foreign bodies do not alter the menstrual cycle, but they act as effective contraceptive devices. Intrauterine implantation of pieces of metal or plastic **(intrauterine devices**, **IUDs)** has been used in programs aimed at controlling population growth. Although the mechanism of action of IUDs is still unsettled, they seem in general to prevent sperm from fertilizing ova. Those containing copper appear to exert a spermatocidal effect. IUDs that slowly release progesterone or

synthetic progestins have the additional effect of thickening cervical mucus so that entry of sperm into the uterus is impeded. IUDs can cause intrauterine infections, but these usually occur in the first month after insertion and in women exposed to sexually transmitted diseases.

Women undergoing long-term treatment with relatively large doses of estrogen do not ovulate, probably because they have depressed FSH levels and multiple irregular bursts of LH secretion rather than a single midcycle peak. Women treated with similar doses of estrogen plus a progestational agent do not ovulate because the secretion of both gonadotropins is suppressed. In addition, the progestin makes the cervical mucus thick and unfavorable to sperm migration, and it may also interfere with implantation. For contraception, an orally active estrogen such as ethinyl estradiol is often combined with a synthetic progestin such as norethindrone. The pills are administered for 21 days, then withdrawn for 5–7 days to permit menstrual flow, and started again. Like ethinyl estradiol, norethindrone has an ethinyl group on position 17 of the steroid nucleus, so it is resistant to hepatic metabolism and consequently is effective by mouth. In addition to being a progestin, it is partly metabolized to ethinyl estradiol, and for this reason it also has estrogenic activity. Small as well as large doses of estrogen are effective (Table 22–4).

Implants made up primarily of progestins such as levonorgestrel are now seeing increased use in some parts of the world. These are inserted under the skin and can prevent pregnancy for up to 5 years. They often produce amenorrhea, but otherwise they appear to be effective and well tolerated.

ABNORMALITIES OF OVARIAN FUNCTION

Menstrual Abnormalities

Some women who are infertile have **anovulatory cycles;** they fail to ovulate but have menstrual periods at fairly regular intervals. As noted above, anovulatory cycles are the rule for the first 1–2 years after menarche and again before the menopause. **Amenorrhea** is the absence of menstrual periods. If menstrual bleeding has never occurred, the condition is called **primary amenorrhea.** Some women with primary amenorrhea have small breasts and other signs of failure to mature sexually. Cessation of cycles in a woman with previously normal periods is called **secondary amenorrhea.** The most common cause of secondary amenorrhea is pregnancy, and the old clinical maxim that “secondary amenorrhea should be considered to be due to pregnancy until proved otherwise”

has considerable merit. Other causes of amenorrhea include emotional stimuli and changes in the environment, hypothalamic diseases, pituitary disorders, primary ovarian disorders, and various systemic diseases. Evidence suggests that in some women with hypothalamic amenorrhea, the frequency of GnRH pulses is slowed as a result of excess opioid activity in the hypothalamus. In encouraging preliminary studies, the frequency of GnRH pulses was increased by administration of the orally active opioid blocker naltrexone.

The terms **hypomenorrhea** and **menorrhagia** refer to scanty and abnormally profuse flow, respectively, during regular periods. **Metrorrhagia** is bleeding from the uterus between periods, and **oligomenorrhea** is reduced frequency of periods. **Dysmenorrhea** is painful menstruation. The severe menstrual cramps that are common in young women quite often disappear after the first pregnancy. Most of the symptoms of dysmenorrhea are due to accumulation of prostaglandins in the uterus, and symptomatic relief has been obtained by treatment with inhibitors of prostaglandin synthesis.

Some women develop symptoms such as irritability, bloating, edema, decreased ability to concentrate, depression, headache, and constipation during the last 7–10 days of their menstrual cycles. These symptoms of the **premenstrual syndrome (PMS)** have been attributed to salt and water retention. However, it seems unlikely that this or any of the other hormonal alterations that occur in the late luteal phase are responsible because the time course and severity of the symptoms are not modified if the luteal phase is terminated early and menstruation produced by administration of mifepristone. The antidepressant fluoxetine, which is a serotonin reuptake inhibitor, and the benzodiazepine alprazolam produce symptomatic relief, and so do GnRH-releasing agonists in doses that suppress the pituitary–ovarian axis. How these diverse clinical observations fit together to produce a picture of the pathophysiology of PMS is still unknown **(Clinical Box 22–4)**.

PREGNANCY

Fertilization & Implantation

In humans, **fertilization** of the ovum by the sperm (see Chapter 23) usually occurs in the ampulla of the uterine tube. Fertilization involves (1) chemoattraction of the sperm to the ovum by substances produced by the ovum; (2) adherence to the **zona pellucida,** the membranous structure surrounding the ovum; (3) penetration of the zona pellucida and the acrosome reaction; and (4)

adherence of the sperm head to the cell membrane of the ovum, with breakdown of the area of fusion and release of the sperm nucleus into the cytoplasm of the ovum **(Figure 22–20)**. Millions of sperm are deposited in the vagina during intercourse. Eventually, 50–100 sperm reach the ovum, and many of them contact the zona pellucida. Sperm bind to a receptor in the zona, and this is followed by the **acrosomal reaction,** that is, the breakdown of the acrosome, the lysosome-like organelle on the head of the sperm (Figure 23–4). Various enzymes are released, including the trypsin-like protease **acrosin.** Acrosin facilitates but is not required for the penetration of the sperm through the zona pellucida. When one sperm reaches the membrane of the ovum, fusion to the ovum membrane is mediated by **fertilin,** a protein on the surface of the sperm head that resembles the viral fusion proteins that permit some viruses to attack cells. The fusion provides the signal that initiates development. In addition, the fusion sets off a reduction in the membrane potential of the ovum that prevents polyspermy, the fertilization of the ovum by more than one sperm. This transient potential change is followed by a structural change in the zona pellucida that provides protection against polyspermy on a more long-term basis.

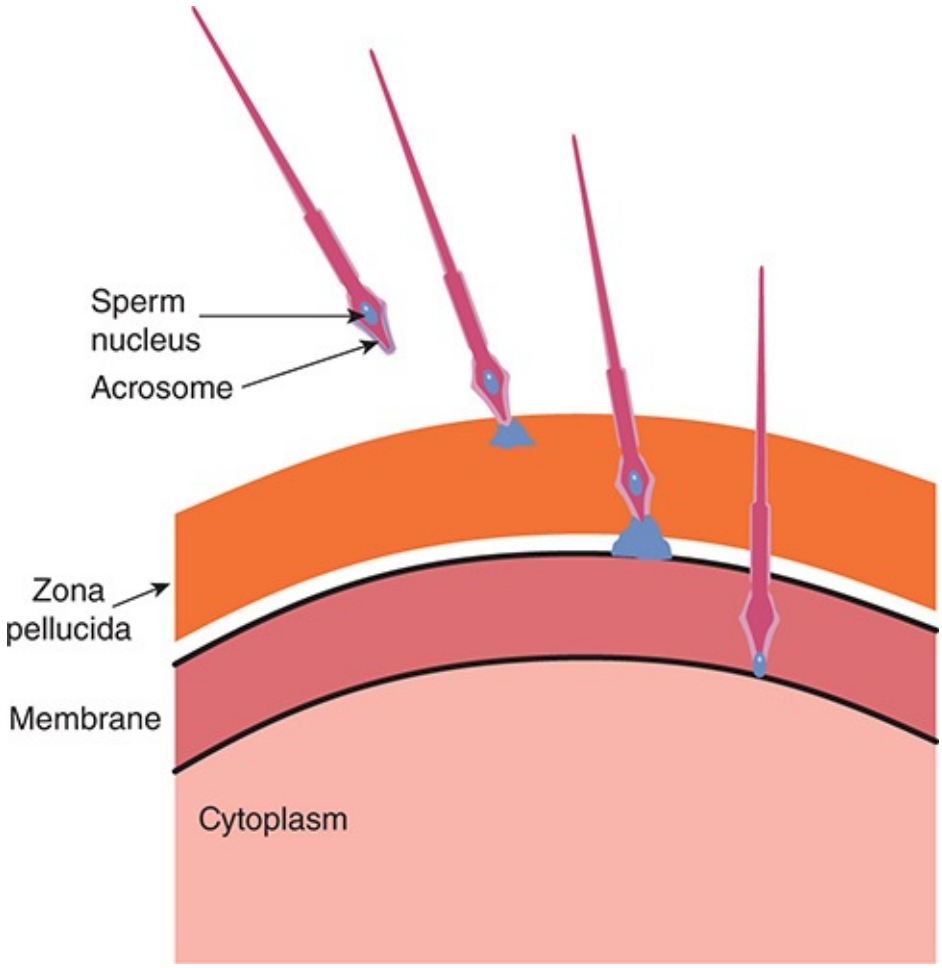


FIGURE 22–20 Sequential events in fertilization in mammals. Sperm are attracted to the ovum, bind to the zona pellucida, release acrosomal enzymes, penetrate the zona pellucida, and fuse with the membrane of the ovum, releasing the sperm nucleus into its cytoplasm. Current evidence indicates that the side, rather than the tip, of the sperm head fuses with the egg cell membrane. (Modified with permission from Vacquier VD: Evolution of gamete recognition proteins. Science 1998; Sep 25; 281(5385):1995–1998.)

CLINICAL BOX 22–4

Genetic Defects Causing Reproductive Abnormalities

A number of single-gene mutations cause reproductive abnormalities when they occur in women. Examples include (1) Kallmann syndrome, which causes hypogonadotropic hypogonadism; (2) GnRH resistance, FSH resistance, and LH resistance, which are due to defects in the GnRH, FSH, or

LH receptors, respectively; and (3) aromatase deficiency, which prevents the formation of estrogens. These are all caused by loss-of-function mutations. An interesting gain-of-function mutation causes the **McCune–Albright syndrome,** in which Gsα becomes constitutively active in certain cells but not others (mosaicism) because a somatic mutation after initial cell division has occurred in the embryo. It is associated with multiple endocrine abnormalities, including precocious puberty and amenorrhea with galactorrhea.

The developing embryo, now called a **blastocyst,** moves down the tube into the uterus. This journey takes about 3 days, during which the blastocyst reaches the 8- or 16-cell stage. Once in contact with the endometrium, the blastocyst becomes surrounded by an outer layer of **syncytiotrophoblast,** a multi-nucleate mass with no discernible cell boundaries, and an inner layer of **cytotrophoblast** made up of individual cells. The syncytiotrophoblast erodes the endometrium, and the blastocyst burrows into it **(implantation).** The implantation site is usually on the dorsal wall of the uterus. A placenta then develops, and the trophoblast remains associated with it.

Failure to Reject the "Fetal Graft"

It should be noted that the fetus and the mother are two genetically distinct individuals, and the fetus is in effect a transplant of foreign tissue in the mother. However, the transplant is tolerated, and the rejection reaction that is characteristically produced when other foreign tissues are transplanted (see Chapter 3) fails to occur. The way the "fetal graft" is protected is unknown. However, one explanation may be that the placental trophoblast, which separates maternal and fetal tissues, does not express the polymorphic class I and class II major histocompatibility complex (MHC) genes and instead expresses *HLA-G*, a nonpolymorphic gene. Therefore, antibodies against the fetal proteins do not develop. In addition, there is Fas ligand on the surface of the placenta, and this binds to T cells, causing them to undergo apoptosis (see Chapter 3).

Infertility

The vexing clinical problem of infertility often requires extensive investigation before a cause is found. In 30% of cases, the problem is in the man; in 45%, the

problem is in the woman; in 20%, both partners have a problem; and in 5%, no cause can be found. **In vitro fertilization,** that is, removing mature ova, fertilizing them with sperm, and implanting one or more of them in the uterus at the four-cell stage is of some value in these cases. It has a 5–10% chance of producing a live birth.

Endocrine Changes

In all mammals, the corpus luteum in the ovary at the time of fertilization fails to regress and instead enlarges in response to stimulation by gonadotropic hormones secreted by the placenta. The placental gonadotropin in humans is called **human chorionic gonadotropin (hCG).** The enlarged **corpus luteum of pregnancy** secretes estrogens, progesterone, and relaxin. Progesterone and relaxin help maintain pregnancy by inhibiting myometrial contractions; progesterone prevents prostaglandin production by the uterus, which stops contractions from occurring. In humans, the placenta produces sufficient estrogen and progesterone from maternal and fetal precursors to take over the function of the corpus luteum after the sixth week of pregnancy. Ovariectomy before the sixth week thus leads to abortion, but ovariectomy thereafter has no effect on the pregnancy. The function of the corpus luteum begins to decline after 8 weeks of pregnancy, but it persists throughout pregnancy. hCG secretion decreases after an initial marked rise, but estrogen and progesterone secretion increase until just before parturition **(Table 22–5)**.

TABLE 22–5 Hormone levels in human maternal blood during normal pregnancy.

Hormone	Approximate Peak Value	Time of Peak Secretion
hCG	5 mg/mL	First trimester
Relaxin	1 ng/mL	First trimester
hCS	15 mg/mL	Term
Estradiol	16 ng/mL	Term
Estriol	14 ng/mL	Term
Progesterone	190 ng/mL	Term
Prolactin	200 ng/mL	Term

Human Chorionic Gonadotropin

hCG is a glycoprotein that contains galactose and hexosamine. It is produced by the syncytiotrophoblast. Like the pituitary glycoprotein hormones, it is made up of α and β subunits. hCG-α is identical to the α subunit of LH, FSH, and TSH. The molecular weight of hCG-α is 18,000, and that of hCG-β is 28,000. hCG is primarily luteinizing and luteotropic and has little FSH activity. It can be measured by radioimmunoassay and detected in the blood as early as 6 days after conception. Its presence in the urine in early pregnancy is the basis of the various laboratory tests for pregnancy, and it can sometimes be detected in the urine as early as 14 days after conception. It appears to act on the same receptor as LH. hCG is not absolutely specific for pregnancy. Small amounts are secreted by a variety of gastrointestinal and other tumors in both sexes, and hCG has been measured in individuals with suspected tumors as a "tumor marker." It also appears that the fetal liver and kidney normally produce small amounts of hCG.

Human Chorionic Somatomammotropin

The syncytiotrophoblast also secretes large amounts of a protein hormone that is lactogenic and has a small amount of growth-stimulating activity. This hormone has been called **chorionic growth hormone-prolactin (CGP)** and **human placental lactogen (hPL),** but it is now generally called **human chorionic somatomammotropin (hCS).** The structure of hCS is very similar to that of human growth hormone (see Chapter 18), and it appears that these two hormones and prolactin evolved from a common progenitor hormone. Large quantities of hCS are found in maternal blood, but very little reaches the fetus. Secretion of growth hormone from the maternal pituitary is not increased during pregnancy and may actually be decreased by hCS. However, hCS has most of the actions of growth hormone and apparently functions as a "maternal growth hormone of pregnancy" to bring about the nitrogen, potassium, and calcium retention, lipolysis, and decreased glucose utilization seen in this state. These latter two actions divert glucose to the fetus. The amount of hCS secreted is proportional to the size of the placenta, which normally weighs about one-sixth as much as the fetus. Low hCS levels are a sign of placental insufficiency.

Other Placental Hormones

In addition to hCG, hCS, progesterone, and estrogens, the placenta secretes other

hormones. Human placental fragments probably produce proopiomelanocortin (POMC). In culture, they release corticotropin-releasing hormone (CRH), β-endorphin, α-melanocyte-stimulating hormone (MSH), and dynorphin A, all of which appear to be identical to their hypothalamic counterparts. They also secrete GnRH and inhibin, and since GnRH stimulates and inhibin inhibits hCG secretion, locally produced GnRH and inhibin may act in a paracrine fashion to regulate hCG secretion. The trophoblast cells and amnion cells also secrete leptin, and moderate amounts of this satiety hormone enter the maternal circulation. Some also enters the amniotic fluid. Its function in pregnancy is unknown. The placenta also secretes prolactin in a number of forms.

Finally, the placenta secretes the α subunits of hCG, and the plasma concentration of free α subunits rises throughout pregnancy. These α subunits acquire a carbohydrate composition that makes them unable to combine with β subunits, and their prominence suggests that they have a function of their own. It is interesting in this regard that the secretion of the prolactin produced by the endometrium also appears to increase throughout pregnancy, and it may be that the circulating α subunits stimulate endometrial prolactin secretion. The cytotrophoblast of the human chorion contains prorenin (see Chapter 38). A large amount of prorenin is also present in amniotic fluid, but its function in this location is unknown.

Fetoplacental Unit

The fetus and the placenta interact in the formation of steroid hormones. The placenta synthesizes pregnenolone and progesterone from cholesterol. Some of the progesterone enters the fetal circulation and provides the substrate for the formation of cortisol and corticosterone in the fetal adrenal glands **(Figure 22–21)**. Some of the pregnenolone enters the fetus and, along with pregnenolone synthesized in the fetal liver, is the substrate for the formation of dehydroepiandrosterone sulfate (DHEAS) and 16-hydroxydehydroepiandrosterone sulfate (16-OHDHEAS) in the fetal adrenal. Some 16- hydroxylation also occurs in the fetal liver. DHEAS and 16-OHDHEAS are transported back to the placenta, where DHEAS forms estradiol and 16-OHDHEAS forms estriol. The principal estrogen formed is estriol, and since fetal 16- OHDHEAS is the principal substrate for the estrogens, the urinary estriol excretion of the mother can be monitored as an index of the state of the fetus.

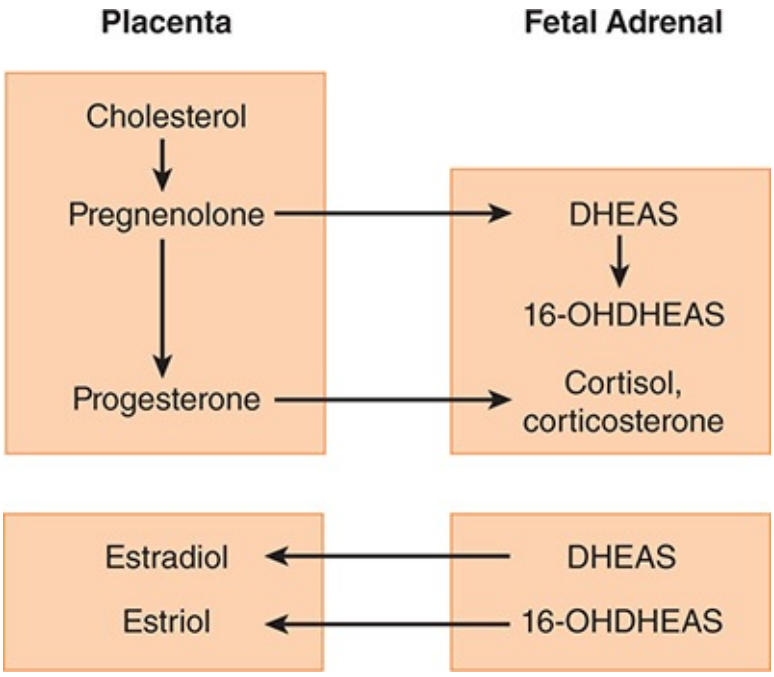


FIGURE 22–21 Interactions between the placenta and the fetal adrenal cortex in the production of steroids.

Parturition

The duration of pregnancy in humans averages 270 days from fertilization (284 days from the first day of the menstrual period preceding conception). Irregular uterine contractions increase in frequency in the last month of pregnancy.

The difference between the body of the uterus and the cervix becomes evident at the time of delivery. The cervix, which is firm in the nonpregnant state and throughout pregnancy until near the time of delivery, softens and dilates, while the body of the uterus contracts and expels the fetus.

There is still considerable uncertainty about the mechanisms responsible for the onset of labor. One factor is the increase in circulating estrogens produced by increased circulating DHEAS. This makes the uterus more excitable, increases the number of gap junctions between myometrial cells, and causes production of more prostaglandins, which in turn cause uterine contractions. In humans, CRH secretion by the fetal hypothalamus increases and is supplemented by increased placental production of CRH. This increases circulating adrenocorticotropic hormone (ACTH) in the fetus, and the resulting increase in cortisol hastens the maturation of the respiratory system. Thus, in a sense, the fetus picks the time to be born by increasing CRH secretion.

The number of oxytocin receptors in the myometrium and the decidua (the endometrium of pregnancy) increases more than 100-fold during pregnancy and reaches a peak during early labor. Estrogens increase the number of oxytocin receptors, and uterine distension late in pregnancy may also increase their formation. In early labor, the oxytocin concentration in maternal plasma is not elevated from the prelabor value of about 25 pg/mL. It is possible that the marked increase in oxytocin receptors causes the uterus to respond to normal plasma oxytocin concentrations. However, at least in rats, the amount of oxytocin mRNA in the uterus increases, reaching a peak at term; this suggests that locally produced oxytocin also participates in the process.

Premature onset of labor is a problem because premature infants have a high mortality rate and often require intensive, expensive care. Intramuscular 17α-hydroxyprogesterone causes a significant decrease in the incidence of premature labor. The mechanism by which it exerts its effect is uncertain, but it may be that the steroid provides a stable level of circulating progesterone. Progesterone relaxes uterine smooth muscle, inhibits the action of oxytocin on the muscle, and reduces the formation of gap junctions between the muscle fibers. All these actions would be expected to inhibit the onset of labor.

Once labor is started, the uterine contractions dilate the cervix, and this dilation in turn sets up signals in afferent nerves that increase oxytocin secretion **(Figure 22–22)**. The plasma oxytocin level rises and more oxytocin becomes available to act on the uterus. Thus, a positive feedback loop is established that aids delivery and terminates on expulsion of the products of conception. Oxytocin increases uterine contractions in two ways: (1) It acts directly on uterine smooth muscle cells to make them contract and (2) it stimulates the formation of prostaglandins in the decidua. The prostaglandins enhance the oxytocin-induced contractions.

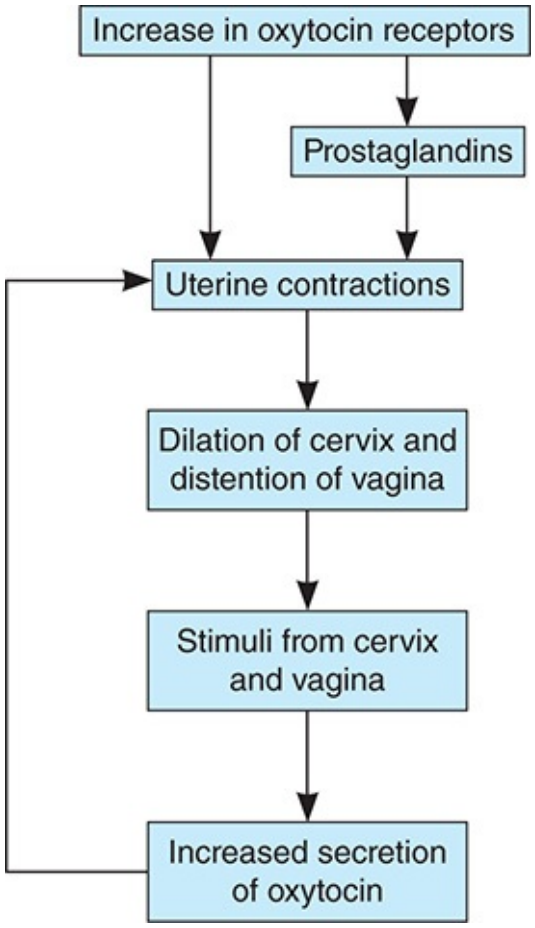


FIGURE 22–22 Role of oxytocin in parturition.

During labor, spinal reflexes and voluntary contractions of the abdominal muscles ("bearing down") also aid in delivery. However, delivery can occur without bearing down and without a reflex increase in secretion of oxytocin from the posterior pituitary gland, since paraplegic women can go into labor and deliver.

LACTATION

Development of the Breasts

Many hormones are necessary for full mammary development. In general, estrogens are primarily responsible for proliferation of the mammary ducts and progesterone for the development of the lobules. In rats, some prolactin is also needed for development of the glands at puberty, but it is not known if prolactin is necessary in humans. During pregnancy, prolactin levels increase steadily until

term, and levels of estrogens and progesterone are elevated as well, producing full lobuloalveolar development.

Secretion & Ejection of Milk

The composition of human and cows' milk is shown in **Table 22–6**. In estrogen- and progesterone-primed rodents, injections of prolactin cause the formation of milk droplets and their secretion into the ducts. Oxytocin causes contraction of the myoepithelial cells lining the duct walls, with consequent ejection of the milk through the nipple.

TABLE 22–6 Composition of colostrum and milk.[a]

Component	Human Colostrum	Human Milk	Cows' Milk
Water (g)	...	88	88
Lactose (g)	5.3	6.8	5.0
Protein (g)	2.7	1.2	3.3
Casein: lactalbumin ratio	...	1:2	3:1
Fat (g)	2.9	3.8	3.7
Linoleic acid	...	8.3% of fat	1.6% of fat
Sodium (mg)	92	15	58
Potassium (mg)	55	55	138
Chloride (mg)	117	43	103
Calcium (mg)	31	33	125
Magnesium (mg)	4	4	12
Phosphorus (mg)	14	15	100
Iron (mg)	0.09[b]	0.15[b]	0.10[b]
Vitamin A (µg)	89	53	34
Vitamin D (µg)	...	0.03[b]	0.06[b]
Thiamine (µg)	15	16	42
Riboflavin (µg)	30	43	157
Nicotinic acid (µg)	75	172	85
Ascorbic acid (mg)	4.4[b]	4.3[b]	1.6[b]

[a]Weights per deciliter.

[b]Poor source.

Reproduced with permission from Findlay ALR: Lactation. Res Reprod (Nov) 1974;6(6).

Initiation of Lactation after Delivery

The breasts enlarge during pregnancy in response to high circulating levels of estrogens, progesterone, prolactin, and possibly hCG. Some milk is secreted into the ducts as early as the fifth month, but the amounts are small compared with the surge of milk secretion that follows delivery. In most animals, milk is secreted within an hour after delivery, but in women it takes 1–3 days for the milk to "come in."

After expulsion of the placenta at parturition, the levels of circulating estrogens and progesterone abruptly decline. The drop in circulating estrogen initiates lactation. Prolactin and estrogen synergize in producing breast growth, but estrogen antagonizes the milk-producing effect of prolactin on the breast. Indeed, in women who do not wish to breastfeed their babies, estrogens may be administered to stop lactation.

Suckling not only evokes reflex oxytocin release and milk ejection, it also maintains and augments the secretion of milk because of the stimulation of prolactin secretion it produces.

Effect of Lactation on Menstrual Cycles

Women who do not breastfeed their infants usually have their first menstrual period 6 weeks after delivery. However, women who breastfeed regularly have amenorrhea for 25–30 weeks. Breastfeeding stimulates prolactin secretion, and evidence suggests that prolactin inhibits GnRH secretion, inhibits the action of GnRH on the pituitary, and antagonizes the action of gonadotropins on the ovaries. Ovulation is inhibited, and the ovaries are inactive, so estrogen and progesterone output falls to low levels. Consequently, only 5–10% of women become pregnant again during the suckling period, and breastfeeding has long been known to be an important, if only partly effective, method of birth control. Furthermore, almost 50% of the cycles in the first 6 months after resumption of menses are anovulatory **(Clinical Box 22–5)**.

Gynecomastia

Breast development in the male is called **gynecomastia.** It may be unilateral but is more commonly bilateral. It is common in newborns because of transplacental passage of maternal estrogens (occurring in about 75%). It also occurs in a mild, transient form in 70% of normal boys at the time of puberty and in many men

over the age of 50. It occurs in androgen resistance. It is a complication of estrogen therapy and is seen in patients with estrogen-secreting tumors. It is found in a wide variety of seemingly unrelated conditions, including eunuchoidism, hyperthyroidism, and cirrhosis of the liver. Digitalis can produce it, apparently because cardiac glycosides are weakly estrogenic. It can also be caused by many other drugs. It has been seen in malnourished prisoners of war, but only after they were liberated and eating an adequate diet. A feature common to many and perhaps all cases of gynecomastia is an increase in the plasma estrogen:androgen ratio due to either increased circulating estrogens or decreased circulating androgens.

CLINICAL BOX 22–5

Chiari–Frommel Syndrome

An interesting, although rare, condition is persistence of lactation **(galactorrhea)** and amenorrhea in women who do not breastfeed after delivery. This condition, called the **Chiari–Frommel syndrome,** may be associated with some genital atrophy and is due to persistent prolactin secretion without the secretion of the FSH and LH necessary to produce maturation of new follicles and ovulation. A similar pattern of galactorrhea and amenorrhea with high circulating prolactin levels is seen in nonpregnant women with chromophobe pituitary tumors and in women in whom the pituitary stalk has been sectioned during treatment of cancer.

HORMONES & CANCER

About 35% of carcinomas of the breast in women of childbearing age are **estrogen-dependent;** their continued growth depends on the presence of estrogens in the circulation. The tumors are not cured by decreasing estrogen secretion, but symptoms are dramatically relieved, and the tumor regresses for months or years before recurring (see Chapter 16). Women with estrogen-dependent tumors often have a remission when their ovaries are removed. Inhibition of the action of estrogens with **tamoxifen** also produces remissions, and inhibition of estrogen formation with drugs that inhibit **aromatase** (Figure 22–15) is even more effective.

CHAPTER SUMMARY

- Differences between males and females depend primarily on a single chromosome (the Y chromosome) and a single pair of endocrine structures (the gonads); testes in the male and ovaries in the female.
- The gonads have a dual function: the production of germ cells (gametogenesis) and the secretion of sex hormones. The testes secrete large amounts of androgens, principally testosterone, but they also secrete small amounts of estrogens. The ovaries secrete large amounts of estrogens and small amounts of androgens.
- The reproductive system of women has regular cyclical changes that can be thought of as periodic preparations for fertilization and pregnancy. In humans and other primates, the cycle is a **menstrual** cycle, and features the periodic vaginal bleeding that occurs with the shedding of the uterine mucosa **(menstruation).**
- Ovaries also secrete progesterone, a steroid that has special functions in preparing the uterus for pregnancy. During pregnancy, the ovaries secrete relaxin, which facilitates the delivery of the fetus. In both sexes, the gonads secrete other polypeptides, including inhibin B, a polypeptide that inhibits FSH secretion.
- In women, a period called perimenopause precedes menopause, and can last up to 10 years; during this time the menstrual cycles become irregular and the level of inhibins decrease.
- Once in menopause, the ovaries no longer secrete progesterone and 17β-estradiol and estrogen is formed only in small amounts by aromatization of androstenedione in peripheral tissues.
- The naturally occurring estrogens are **17β-estradiol, estrone,** and **estriol.** They are secreted primarily by the granulosa cells of the ovarian follicles, the corpus luteum, and the placenta. Their biosynthesis depends on the enzyme **aromatase** (CYP19), which converts testosterone to estradiol and androstenedione to estrone. The latter reaction also occurs in fat, liver, muscle, and the brain.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. If a young woman has high plasma levels of T_3, cortisol, and renin activity but her blood pressure is only slightly elevated and she has no symptoms or signs of thyrotoxicosis or Cushing syndrome, the most likely explanation is that
   A. she has been treated with TSH and ACTH.
   B. she has been treated with T_3 and cortisol.
   C. she is in the third trimester of pregnancy.
   D. she has an adrenocortical tumor.
   E. she has been subjected to chronic stress.
2. In humans, fertilization usually occurs in the
   A. vagina.
   B. cervix.
   C. uterine cavity.
   D. uterine tubes.
   E. abdominal cavity.
3. Which of the following is *not* a steroid?
   A. 17α-hydroxyprogesterone
   B. Estrone
   C. Relaxin
   D. Pregnenolone
   E. Etiocholanolone
4. Which of the following probably triggers the onset of labor?
   A. ACTH in the fetus
   B. ACTH in the mother
   C. Prostaglandins
   D. Oxytocin
   E. Placental renin

CHAPTER 23

Function of the Male Reproductive System

OBJECTIVES

After studying this chapter, you should be able to:

- Name the key hormones secreted by Leydig cells and Sertoli cells of the testes.
- Outline the steps involved in spermatogenesis.
- Outline the mechanisms that produce erection and ejaculation.
- Know the general structure of testosterone and describe its biosynthesis, transport, metabolism, and actions.
- Describe the processes involved in regulation of testosterone secretion.

INTRODUCTION

The role for a functional, secreting testis in the formation of male genitalia, the action of male hormones on the brain in early development, and development of the male reproductive system through adolescence and into adulthood were discussed in the previous chapter. As observed in the female, male gonads have a dual function: the production of germ cells **(gametogenesis)** and the secretion of **sex hormones.** The **androgens** are the steroid sex hormones that are masculinizing in their action. The testes secrete large amounts of androgens,

principally **testosterone,** but they also secrete small amounts of estrogens. Unlike that observed in females, male gonadotropin secretion is noncyclical, and once mature, male gonadal function slowly declines with advancing age, but the ability to produce viable gametes persists. In this chapter, the discussion will be focused on the structure and physiology of the mature male reproductive system.

THE MALE REPRODUCTIVE SYSTEM

STRUCTURE

The testes are made up of loops of convoluted **seminiferous tubules,** in the walls of which the spermatozoa are formed from the primitive germ cells **(spermatogenesis).** Both ends of each loop drain into a network of ducts in the head of the **epididymis.** From there, spermatozoa pass through the tail of the epididymis into the **vas deferens.** They enter through the **ejaculatory ducts** into the urethra in the body of the **prostate** at the time of ejaculation **(Figure 23–1)**. Between the tubules in the testes are nests of cells containing lipid granules, the **interstitial cells of Leydig** (Leydig cells; **Figures 23–2** and **23–3**), which secrete testosterone into the bloodstream. The spermatic arteries to the testes are tortuous, and blood in them runs parallel but in the opposite direction to blood in the pampiniform plexus of spermatic veins. This anatomic arrangement may permit countercurrent exchange of heat and testosterone. The principles of countercurrent exchange are considered in detail in relation to the kidney in Chapter 37.

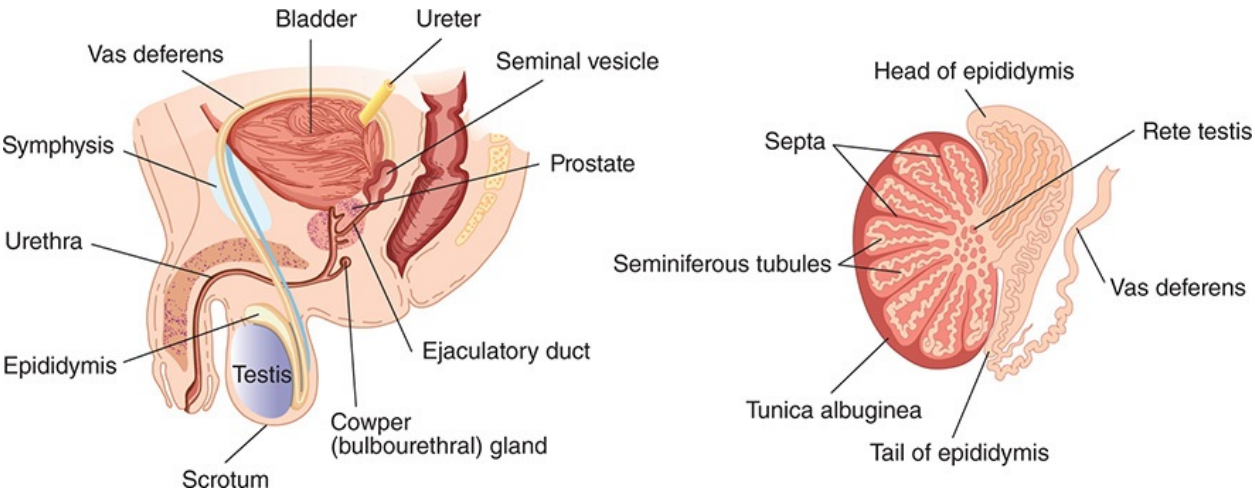


FIGURE 23–1 Male reproductive system. Anatomic features of the male reproductive system in relation to the bladder are shown at left. A cross section

of the duct system within the testis is shown at right.

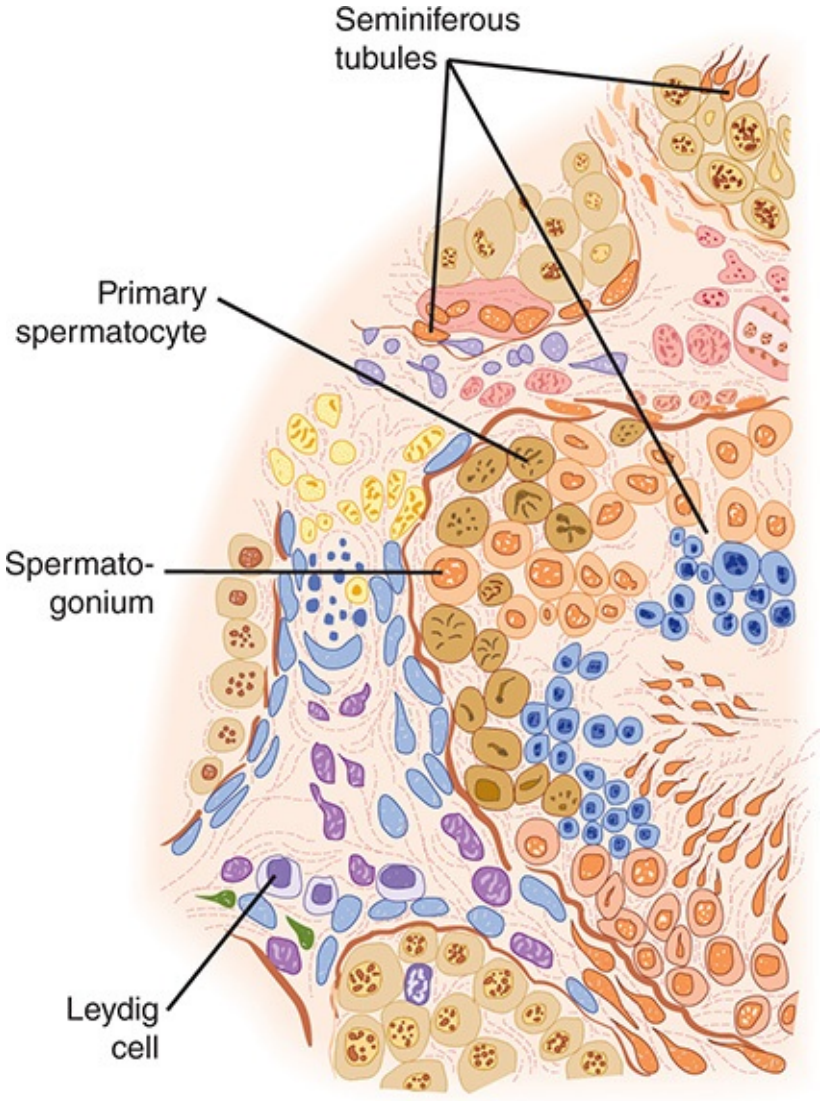


FIGURE 23–2 Section of the human testis. Cross section shows the diversity of cellular anatomy within the testis.

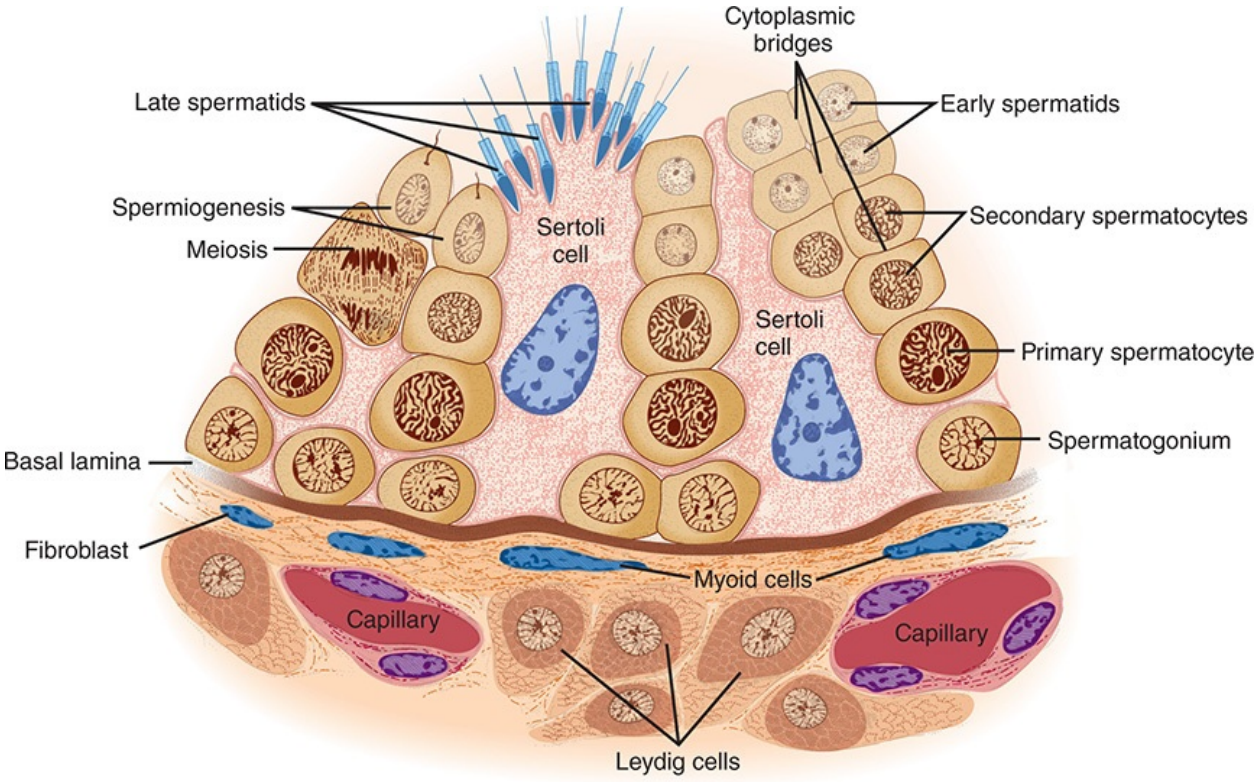


FIGURE 23–3 Seminiferous epithelium. Note that maturing germ cells remain connected by cytoplasmic bridges through the early spermatid stage and that these cells are closely invested by Sertoli cell cytoplasm as they move from the basal lamina to the lumen. (Reproduced with permission from Junqueira LC, Carneiro J: *Basic Histology: Text & Atlas*, 10th ed. New York, NY: McGraw-Hill; 2003.)

GAMETOGENESIS & EJACULATION

Blood-Testis Barrier

The walls of the seminiferous tubules are lined by primitive germ cells and **Sertoli cells,** large, complex glycogen-containing cells that stretch from the basal lamina of the tubule to the lumen (Figure 23–3). Germ cells must stay in contact with Sertoli cells to survive; this contact is maintained by cytoplasmic bridges. Tight junctions between adjacent Sertoli cells near the basal lamina form a **blood-testis barrier** that prevents many large molecules from passing from the interstitial tissue and the part of the tubule near the basal lamina (basal compartment) to the region near the tubular lumen (adluminal compartment) and

the lumen. However, steroids penetrate this barrier with ease, and evidence suggests that some proteins also pass from the Sertoli cells to the Leydig cells, and vice versa, to function in a paracrine manner. In addition, maturing germ cells must pass through the barrier as they move to the lumen. This appears to occur without disruption of the barrier by coordinated breakdown of the tight junctions above the germ cells and formation of new tight junctions below them.

The fluid in the lumen of the seminiferous tubules is quite different from plasma; it contains very little protein and glucose but is rich in androgens, estrogens, K^+, inositol, and glutamic and aspartic acids. Maintenance of its composition depends on the blood-testis barrier. The barrier also protects the germ cells from bloodborne noxious agents, prevents antigenic products of germ cell division and maturation from entering the circulation and generating an autoimmune response, and may help establish an osmotic gradient that facilitates movement of fluid into the tubular lumen.

Spermatogenesis

Spermatogonia, the primitive germ cells next to the basal lamina of the seminiferous tubules, mature into **primary spermatocytes** (Figure 23–3). This process begins during adolescence. The primary spermatocytes undergo meiotic division, reducing the number of chromosomes. In this two-stage process, they divide into **secondary spermatocytes** and then into **spermatids,** which contain the haploid number of 23 chromosomes. The spermatids mature into **spermatozoa (sperm).** As a single spermatogonium divides and matures, its descendants remain tied together by cytoplasmic bridges until the late spermatid stage. This arrangement helps ensure synchrony of the differentiation of each clone of germ cells. The estimated number of spermatids formed from a single spermatogonium is 512. The formation of a mature sperm from a primitive germ cell by spermatogenesis in humans spans approximately 74 days.

Each sperm is an intricate motile cell, rich in DNA, with a head that is made up mostly of chromosomal material **(Figure 23–4)**. Covering the head like a cap is the **acrosome,** a lysosome-like organelle rich in enzymes involved in sperm penetration of the ovum and other events associated with fertilization. The motile tail of the sperm is wrapped in its proximal portion by a sheath holding numerous mitochondria. The membranes of late spermatids and spermatozoa contain a special small form of angiotensin-converting enzyme (ACE) called **germinal angiotensin-converting enzyme (gACE).** Germinal ACE is transcribed from the same gene as the somatic ACE (sACE); however, gACE

displays tissue-specific expression based on alternative transcription initiation sites and alternate splicing patterns. The full function of gACE has yet to be elucidated, although gACE-specific knockout mouse models are sterile.

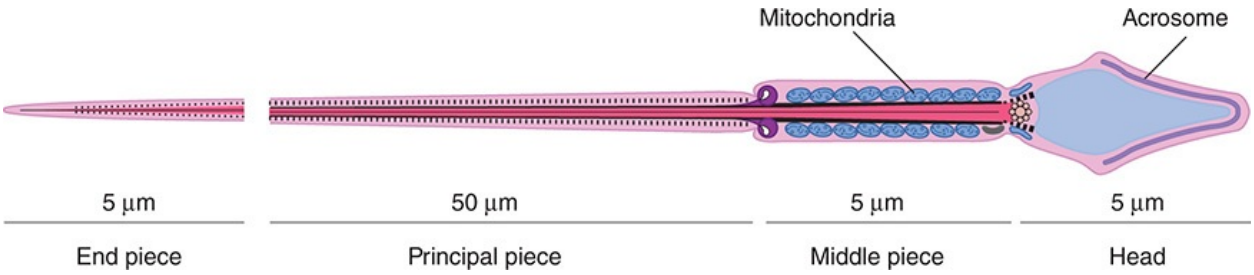


FIGURE 23–4 Human spermatozoon, profile view. Note the acrosome, an organelle that covers half the sperm head inside the plasma membrane of the sperm. (Reproduced with permission from Junqueira LC, Carneiro J: *Basic Histology: Text & Atlas*, 11th ed. New York, NY: McGraw-Hill; 2005.)

Spermatids mature into spermatozoa in deep folds of the cytoplasm of the Sertoli cells (Figure 23–3). Mature spermatozoa are released from the Sertoli cells and become free in the lumens of the tubules. The Sertoli cells secrete **androgen-binding protein (ABP), inhibin,** and **Mullerian inhibiting substance (MIS**). They do not synthesize androgens, but they contain **aromatase (CYP19),** the enzyme responsible for conversion of androgens to estrogens, and they can produce estrogens. ABP probably functions to maintain a high, stable supply of androgen in the tubular fluid. Inhibin inhibits follicle-stimulating hormone (FSH) secretion.

FSH and androgens maintain the gametogenic function of the testis. After hypophysectomy, injection of luteinizing hormone (LH) produces a high local concentration of androgen in the testes, and this maintains spermatogenesis. The stages from spermatogonia to spermatids appear to be androgen-independent. However, the maturation from spermatids to spermatozoa depends on androgen acting on the Sertoli cells in which the developing spermatozoa are embedded. FSH acts on the Sertoli cells to facilitate the last stages of spermatid maturation. In addition, it promotes the production of ABP.

An interesting observation is that the estrogen content of the fluid in the rete testis (Figure 23–1) is high, and the walls of the rete testis contain numerous α estrogen receptors (ERα). In this region, fluid is reabsorbed and the spermatozoa are concentrated. If this does not occur, the sperm entering the epididymis are diluted in a large volume of fluid, resulting in reduced fertility.

Further Development of Spermatozoa

Spermatozoa leaving the testes are not fully mobile. They continue their maturation and acquire motility during their passage through the epididymis. Motility is obviously important in vivo, but fertilization occurs in vitro if an immotile spermatozoon from the head of the epididymis is microinjected directly into an ovum. The ability to move forward **(progressive motility),** which is acquired in the epididymis, involves activation of a unique set of cation channel proteins from the **CatSper** family, which are localized to the principal piece of the sperm tail. CatSpers form an alkaline-sensitive Ca^{2+} channel that becomes more active as the sperm go from the acidic vagina (pH ~5) to the cervical mucus (pH ~8). Sperm from knockout mice that do not express CatSper1-4 have altered motility and are infertile, emphasizing the important role of these channel proteins. In addition, spermatozoa express olfactory receptors, and ovaries produce odorant-like molecules. Recent evidence indicates that these molecules and their receptors interact, fostering movement of the spermatozoa toward the ovary (chemotaxis).

Ejaculation of the spermatozoon involves contractions of the vas deferens mediated in part by P2X purinoreceptors, ligand-gated cation channels that open in response to ATP (see Chapter 7), and fertility is reduced in mice in which these receptors are knocked out.

Once ejaculated into the female, the spermatozoa move up the uterus to the isthmus of the uterine tubes, where they slow down and undergo **capacitation.** This further maturation process involves two components: increasing the motility of the spermatozoa and facilitating their preparation for the acrosome reaction. However, the role of capacitation appears to be facilitatory rather than obligatory, because fertilization is readily produced in vitro. From the isthmuses, the capacitated spermatozoa move rapidly to the tubal ampullas, where fertilization takes place.

Effect of Temperature

Spermatogenesis requires a temperature considerably lower than that of the interior of the body. The testes are normally maintained at a temperature of about 32°C. They are kept cool by air circulating around the scrotum and probably by heat exchange in a countercurrent manner between the spermatic arteries and veins. When the testes are retained in the abdomen or when, in experimental animals, they are held close to the body, degeneration of the tubular walls and

sterility result. Situations that increase heat around the testes in humans (e.g., hot baths [43–45°C for 30 min/d] and insulated athletic supporters) can reduce sperm counts, in some cases by 90%. However, the reductions produced in this manner are not consistent enough to make the procedures reliable forms of male contraception. In addition, evidence suggests a seasonal effect in men, with sperm counts being greater in the winter regardless of the temperature to which the scrotum is exposed.

Semen

The fluid that is ejaculated at the time of orgasm, the **semen,** contains sperm and the secretions of the seminal vesicles, prostate, Cowper glands, and, probably, the urethral glands **(Table 23–1)**. An average volume per ejaculate is 2.5–3.5 mL after several days of abstinence from sexual activity. The volume of semen and the sperm count decrease rapidly with repeated ejaculation. Even though it takes only one sperm to fertilize the ovum, each milliliter of semen normally contains about 100 million sperm. Reduction in sperm production is associated with infertility: 50% of men with counts of 20–40 million/mL and essentially all of those with counts under 20 million/mL are sterile. The presence of many morphologically abnormal or immotile spermatozoa also correlates with infertility. The **prostaglandins** in semen, which come from the seminal vesicles, are at high concentrations, but their function in semen is unknown. The causes of male infertility, as well as the underlying mechanisms of sperm in fertilization, are used as clues in developing male contraception **(Clinical Box 23–1)**.

TABLE 23–1 Composition of human semen.

Color: White, opalescent		
Specific gravity: 1.028		
pH: 7.35–7.50		
Sperm count: Average about 100 million/mL, with < 20% abnormal forms		
Other components: Fructose (1.5–6.5 mg/mL) Phosphorylcholine Ergothioneine Ascorbic acid Flavins Prostaglandins	}	From seminal vesicles (contributes 60% of total volume)
Spermine Citric acid Cholesterol, phospholipids Fibrinolysin, fibrinogenase Zinc Acid phosphatase	}	From prostate (contributes 20% of total volume)
Phosphate Bicarbonate	}	Buffers
Hyaluronidase		From the acrosome of the sperm cells after it has reached the oocyte

CLINICAL BOX 23–1

Male Contraception

Several methods independent of physical intervention (such as hormonal control of sperm development, targeting of cation channel proteins important in fertilization [e.g., CatSpers], and the use of natural compounds that limit sperm function) have been explored as male contraceptives. However, considering the number of sperm and their ability to regenerate, methods that

sufficiently reduce sperm production or limit function with an absence of side effects have been difficult to obtain. In lieu of pharmacological control, the most common male contraception continues to be bilateral ligation of the vas deferens **(vasectomy),** a relatively safe and convenient contraceptive procedure. Interestingly, antibodies against spermatozoa develop in ~50% of the men who have undergone vasectomy; in monkeys, the presence of such antibodies is associated with a higher incidence of infertility after restoration of the patency of the vas. However, the anti-sperm antibodies do not appear to have any other adverse effects. Alternatives to ligation are vas occlusion methods, such as silicone plugs, that aim to block the vas deferens while leaving the tube intact, making reversal easier if desired. Not surprisingly, such methods are not as effective as traditional vasectomy.

THERAPEUTIC HIGHLIGHTS

Vasectomy reversal: Whereas it was once quite difficult to restore the patency of the vas in those wishing to restore fertility, the success rate for such operations has improved steadily. Successful reversal of vasectomy usually results in meaningful sperm counts within months, although delays of a year or more are not unusual. The ultimate success, as measured by pregnancy, is recorded in ~50% of reversals within 2 years.

Human sperm move at a speed of about 3 mm/min through the female genital tract. Sperm reach the uterine tubes 30–60 min after copulation. Contractions of the female organs may facilitate the transport of the sperm to the uterine tubes.

Erection

Erection is initiated by dilation of the arterioles of the penis. As the erectile tissue of the penis fills with blood, the veins are compressed, blocking outflow and adding to the turgor of the organ. The integrating centers in the lumbar segments of the spinal cord are activated by impulses in afferents from the genitalia and descending tracts that mediate erection in response to erotic psychological stimuli. The efferent parasympathetic fibers are in the pelvic splanchnic nerves **(nervi erigentes).** The fibers presumably release acetylcholine and the vasodilator vasoactive intestinal polypeptide (VIP) as cotransmitters (see Chapter 7).

Nonadrenergic noncholinergic fibers are also present in the nervi erigentes, and these contain large amounts of **nitric oxide synthase (NOS),** the enzyme that catalyzes the formation of nitric oxide (NO; see Chapter 32). NO activates soluble guanylyl cyclase, resulting in increased production of cyclic guanosine monophosphate (cGMP), and cGMP is a potent vasodilator. Injection of NOS inhibitors prevents the erection normally produced by stimulation of the pelvic nerve in experimental animals. Thus, it seems clear that NO plays a prominent role in the production of erection. The drugs sildenafil (Viagra), tadalafil (Cialis), and vardenafil (Levitra) all inhibit the breakdown of cGMP by phosphodiesterases and have gained worldwide fame for the treatment of erectile dysfunction. The multiple phosphodiesterases (PDEs) in the body have been divided into 11 isoenzyme families, and these drugs are all most active against PDE-5, the type of phosphodiesterase found in the corpora cavernosa (Clinical Box 5–7). It is worth noting, however, that these drugs can also produce significant inhibition of PDE-6 (and others, if taken at high doses). PDE-6 is found in the retina, and one of the side effects of these drugs is a transient loss of the ability to discriminate between blue and green (see Chapter 9).

Normally, erection is terminated by sympathetic vasoconstrictor impulses to the penile arterioles.

Ejaculation

Ejaculation is a two-part spinal reflex that involves **emission,** the movement of the semen into the urethra; and **ejaculation** proper, the propulsion of the semen out of the urethra at the time of orgasm. The afferent pathways are mostly fibers from touch receptors in the glans penis that reach the spinal cord through the internal pudendal nerves. Emission is a sympathetic response, integrated in the upper lumbar segments of the spinal cord and effected by contraction of the smooth muscle of the vasa deferentia and seminal vesicles in response to stimuli in the hypogastric nerves. The semen is propelled out of the urethra by contraction of the bulbocavernosus muscle, a skeletal muscle. The spinal reflex centers for this part of the reflex are in the upper sacral and lowest lumbar segments of the spinal cord, and the motor pathways traverse the first to third sacral roots and the internal pudendal nerves.

Prostate-Specific Antigen

The prostate produces and secretes into the semen and the bloodstream a 30-kDa

serine protease generally called **prostate-specific antigen (PSA).** The gene for PSA has two androgen response elements. PSA hydrolyzes the sperm motility inhibitor semenogelin in semen, and it has several substrates in plasma, but its precise function in the circulation is unknown. An elevated plasma PSA occurs in prostate cancer and has been widely used as a screening test for this disease. However, PSA is also elevated in benign prostatic hyperplasia and prostatitis, and the effectiveness of PSA screening as a sole tool in diagnosis of prostate cancer has been called into question.

ENDOCRINE FUNCTION OF THE TESTES

Chemistry & Biosynthesis of Testosterone

Testosterone, the principal hormone of the testes, is a C_{19} steroid with a hydroxyl group in the 17 position **(Figure 23–5)**. It is synthesized from cholesterol in the Leydig cells and is also formed from androstenedione secreted by the adrenal cortex. The biosynthetic pathways in all endocrine organs that form steroid hormones are similar, the organs differing only in the enzyme systems they contain. In the Leydig cells, the 11- and 21-hydroxylases found in the adrenal cortex (see Figure 20–7) are absent, but 17α-hydroxylase is present. Pregnenolone is therefore hydroxylated in the 17 position and then subjected to side chain cleavage to form dehydroepiandrosterone. Androstenedione is also formed via progesterone and 17-hydroxyprogesterone, but this pathway is less prominent in humans. Dehydroepiandrosterone and androstenedione are then converted to testosterone.

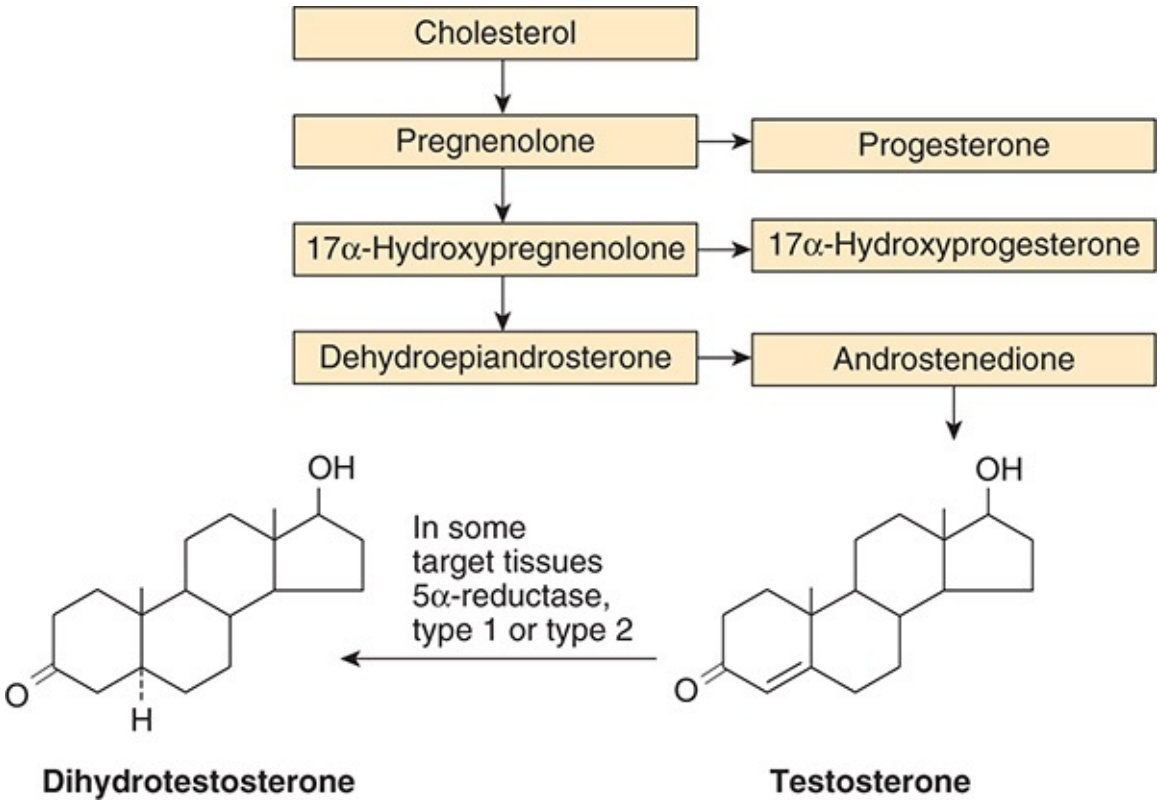


FIGURE 23–5 Biosynthesis of testosterone. The formulas of the precursor steroids are shown in Figure 20–7. Although the main secretory product of the Leydig cells is testosterone, some of the precursors also enter the circulation.

The secretion of testosterone is under the control of luteinizing hormone (LH), and the mechanism by which LH stimulates Leydig cells involves increased formation of cyclic adenosine monophosphate (cAMP) via the G-protein- coupled LH receptor and G_s. Cyclic AMP increases the formation of cholesterol from cholesterol esters and the conversion of cholesterol to pregnenolone via the activation of protein kinase A.

Secretion

The testosterone secretion rate is 4–9 mg/d (13.9–31.33 µmol/d) in normal adult males. Small amounts of testosterone are also secreted in females, with the major source being the ovary, but possibly from the adrenal as well.

Transport & Metabolism

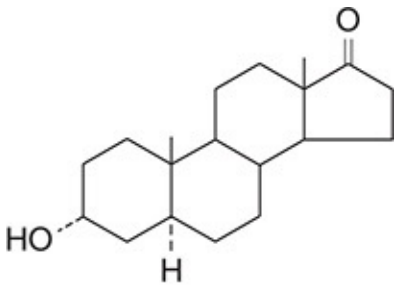
Ninety-eight percent of the testosterone in plasma is bound to protein: 65% is bound to a β-globulin called **gonadal steroid–binding globulin (GBG)** or **sex steroid–binding globulin,** and 33% to albumin **(Table 23–2)**. GBG also binds estradiol. The plasma testosterone level (free and bound) is 300–1000 ng/dL (10.4–34.7 nmol/L) in adult men (Figure 22–8), compared with 30–70 ng/dL (1.04–2.43 nmol/L) in adult women. It declines somewhat with age in men.

TABLE 23–2 Distribution of gonadal steroids and cortisol in plasma.

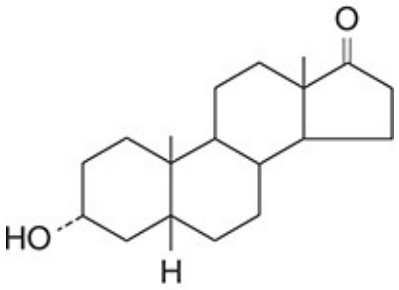
		% Bound to		
Steroid	**% Free**	**CBG**	**GBG**	**Albumin**
Testosterone	2	0	65	33
Androstenedione	7	0	8	85
Estradiol	2	0	38	60
Progesterone	2	18	0	80
Cortisol	4	90	0	6

CBG, corticosteroid-binding globulin; GBG, gonadal steroid–binding globulin. (Used with permission of S Munroe.)

A small amount of circulating testosterone is converted to estradiol, but most of the testosterone is converted to 17-ketosteroids, principally androsterone and its isomer etiocholanolone **(Figure 23–6)**, and excreted in the urine. About two-thirds of the urinary 17-ketosteroids are of adrenal origin, and one-third are of testicular origin. Although most of the 17-ketosteroids are weak androgens (they have 20% or less the potency of testosterone), it is worth emphasizing that not all 17-ketosteroids are androgens and not all androgens are 17-ketosteroids. Etiocholanolone, for example, has no androgenic activity, and testosterone itself is not a 17-ketosteroid.



Androsterone



Etiocholanolone

FIGURE 23–6 Testosterone metabolites. Androsterone and etiocholanolone are two 17-ketosteroid metabolites of testosterone that are secreted in the urine.

Actions

In addition to their actions during development, testosterone and other androgens exert an inhibitory feedback effect on pituitary LH secretion; develop and maintain the male secondary sex characteristics; exert an important protein-anabolic, growth-promoting effect; and, along with FSH, maintain spermatogenesis.

Secondary Sex Characteristics

The widespread changes in hair distribution, body configuration, and genital size that develop in boys at puberty—the male **secondary sex characteristics**—are summarized in Table 23–3. The prostate and seminal vesicles enlarge, and the seminal vesicles begin to secrete fructose. This sugar appears to function as the main nutritional supply for the spermatozoa. The psychological effects of testosterone are difficult to define in humans, but in experimental animals, androgens provoke boisterous and aggressive play. The effects of androgens and

estrogens on sexual behavior are considered in detail in Chapter 22. Although body hair is increased by androgens, scalp hair is decreased **(Figure 23–7)**. Hereditary baldness often fails to develop unless dihydrotestosterone (DHT) is present.

TABLE 23–3 Changes at puberty in boys (male secondary sex characteristics).

External genitalia: Penis increases in length and width. Scrotum becomes pigmented and rugose.
Internal genitalia: Seminal vesicles enlarge and secrete and begin to form fructose. Prostate and bulbourethral glands enlarge and secrete.
Voice: Larynx enlarges, vocal cords increase in length and thickness, and voice becomes deeper.
Hair growth: Beard appears. Hairline on scalp recedes anterolaterally. Pubic hair grows with male (triangle with apex up) pattern. Hair appears in axillas, on chest, and around anus; general body hair increases.
Mental: More aggressive, active attitude. Interest in opposite sex develops.
Body conformation: Shoulders broaden; muscles enlarge.
Skin: Sebaceous gland secretion thickens and increases (predisposing to acne).



FIGURE 23–7 Hairline in children and adults. The hairline of the woman is like that of the child, whereas that of the man is indented in the lateral frontal region.

Anabolic Effects

Androgens increase the synthesis and decrease the breakdown of protein, leading to an increase in the rate of growth. It used to be argued that they cause the epiphyses to fuse to the long bones, thus eventually stopping growth, but it now appears that epiphysial closure is due to estrogens (see Chapter 21). Secondary to their anabolic effects, androgens cause moderate Na^+, K^+, H_2O, Ca^{2+}, SO_4^{2-}, and PO_4^{3-} retention, and they also increase the size of the kidneys. Doses of exogenous testosterone that exert significant anabolic effects are also masculinizing and increase libido, which limits the usefulness of the hormone as an anabolic agent in patients with wasting diseases. Attempts to develop synthetic steroids in which the anabolic action is separated from the androgenic action have not been successful.

Mechanism of Action

Like other steroids, testosterone binds to an intracellular receptor, for example, androgen receptor (or NR3C4), and the receptor/steroid complex then binds to DNA in the nucleus, facilitating transcription of various genes. In addition, testosterone is converted to **DHT** by 5α-reductase in some target cells (Figure 23–5 and **Figure 23–8**), and DHT binds to the same intracellular receptor as testosterone. DHT also circulates, with a plasma level that is about 10% of the testosterone level. Testosterone–receptor complexes are less stable than DHT–receptor complexes in target cells, and they conform less well to the DNA-binding state. Thus, DHT formation is a way of amplifying the action of testosterone in target tissues. Humans have two 5α-reductases that are encoded by different genes. Type 1 5α-reductase is present in skin throughout the body and is the dominant enzyme in the scalp. Type 2 5α-reductase is present in genital skin, the prostate, and other genital tissues.

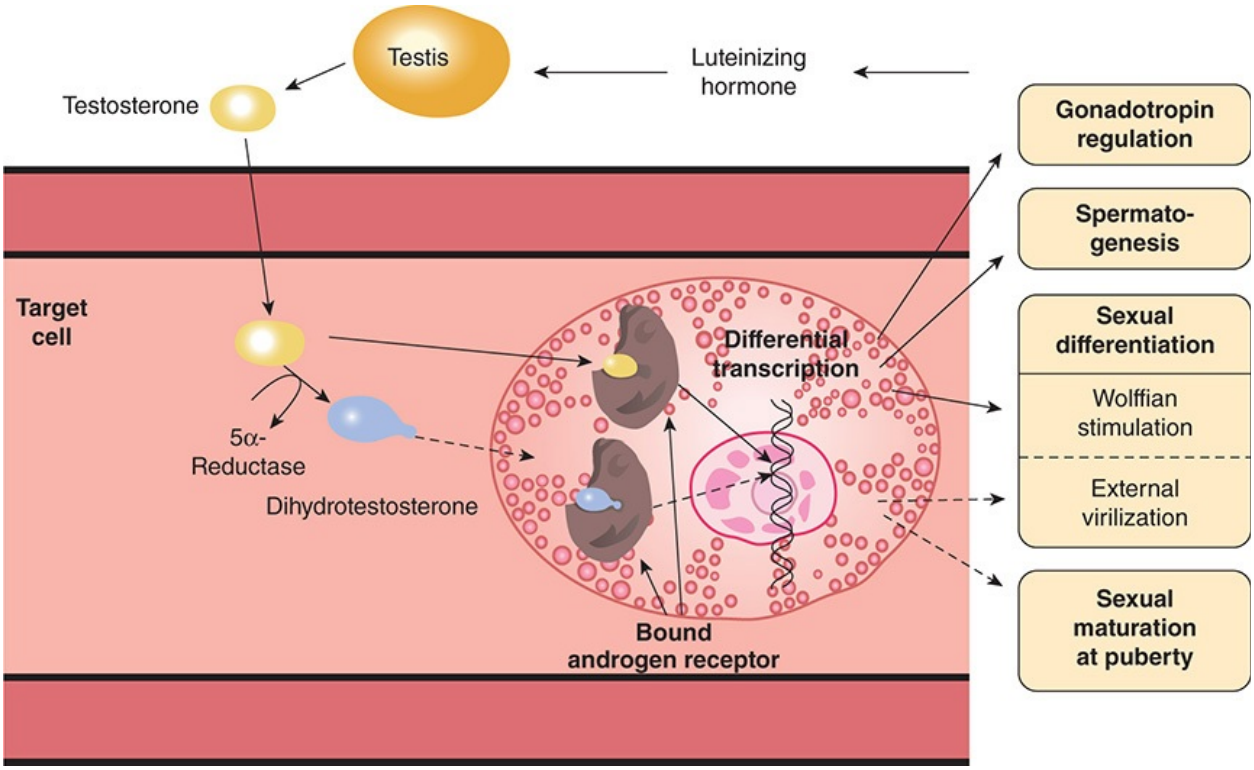


FIGURE 23–8 Schematic diagram of the actions of testosterone (solid arrows) and dihydrotestosterone (dashed arrows). Note that they both bind to the same receptor, but DHT binds more effectively. (Reproduced with permission of Wilson JD, Griffin JE, Russell DW: Steroid 5 alpha-reductase 2 deficiency. Endocr Rev 1993 Oct; 14(5):577–593.)

Testosterone–receptor complexes are responsible for the maturation of Wolffian duct structures and consequently for the formation of male internal genitalia during development, but DHT–receptor complexes are needed to form male external genitalia (Figure 23–8). DHT–receptor complexes are also primarily responsible for enlargement of the prostate and probably of the penis at the time of puberty, as well as for the facial hair, the acne, and the temporal recession of the hairline. On the other hand, the increase in muscle mass and the development of male sex drive and libido depend primarily on testosterone rather than DHT **(see Clinical Box 23–2)**.

Testicular Production of Estrogens

Over 80% of the estradiol and 95% of the estrone in the plasma of adult men are formed by extragonadal and extra-adrenal aromatization of circulating

testosterone and androstenedione. The remainder comes from the testes. Some of the estradiol in testicular venous blood comes from the Leydig cells, but some is also produced by aromatization of androgens in Sertoli cells. In men, the plasma estradiol level is 20–50 pg/mL (73–184 pmol/L) and the total production rate is approximately 50 µg/d (184 nmol/d). In contrast to the situation in women, estrogen production moderately increases with advancing age in men.

CLINICAL BOX 23–2

Congenital 5α-Reductase Deficiency

Congenital 5α-reductase deficiency, in which the gene for type 2 5α-reductase is mutated, is common in certain parts of the Dominican Republic. It produces an interesting form of male pseudohermaphroditism. Individuals with this syndrome are born with male internal genitalia including testes, but they have female external genitalia and are usually raised as girls. However, when they reach puberty, LH secretion and circulating testosterone levels are increased. Consequently, they develop male body contours and male libido. At this point, they usually change their gender identities and "become boys." The clitoris enlarges ("penis-at-12 syndrome") to the point that some of the individuals can have intercourse with women. This enlargement probably occurs because, with the high LH, enough testosterone is produced to overcome the need for DHT amplification in the genitalia.

THERAPEUTIC HIGHLIGHTS

5α-Reductase–inhibiting drugs are being used clinically to treat benign prostatic hyperplasia, and **finasteride,** the most extensively used drug, has its greatest effect on type 2 5α-reductase.

CONTROL OF TESTICULAR FUNCTION

FSH is tropic for Sertoli cells, and FSH and androgens maintain the gametogenic function of the testes. FSH also stimulates the secretion of ABP and inhibin. Inhibin feeds back to inhibit FSH secretion. LH is tropic for Leydig cells and stimulates the secretion of testosterone, which in turn feeds back to inhibit LH secretion. Hypothalamic lesions in animals and hypothalamic disease in humans

lead to atrophy of the testes and loss of their function.

Inhibins

Testosterone reduces plasma LH, but, except in large doses, it has no effect on plasma FSH. Plasma FSH is elevated in patients who have atrophy of the seminiferous tubules but normal levels of testosterone and LH secretion. These observations led to the search for **inhibin,** a factor of testicular origin that inhibits FSH secretion. There are two inhibins in extracts of testes in men and in antral fluid from ovarian follicles in women. They are formed from three polypeptide subunits: a glycosylated α subunit with a molecular weight of 18,000, and two nonglycosylated β subunits, β_A and β_B, each with a molecular weight of 14,000. The subunits are formed from precursor proteins **(Figure 23–9)**. The α subunit combines with β_A to form a heterodimer and with β_B to form another heterodimer, with the subunits linked by disulfide bonds. Both $\alpha\beta_A$ (inhibin A) and $\alpha\beta_B$ (inhibin B) inhibit FSH secretion by a direct action on the pituitary, although it appears that it is inhibin B that is the FSH-regulating inhibin in adult men and women. Inhibins are produced by Sertoli cells in males and granulosa cells in females.

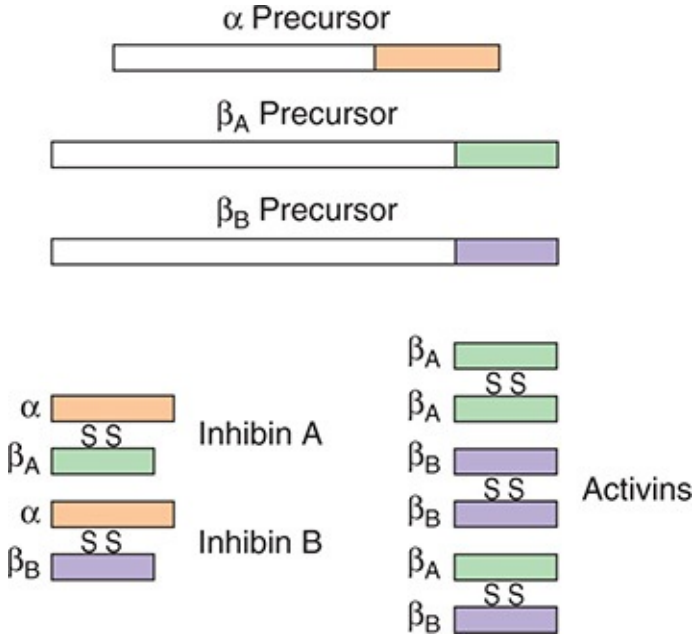


FIGURE 23–9 Inhibin and activin production. From three precursor proteins (α, β_A, and β_B), five separate inhibins and activins are produced. SS, disulfide bonds.

The heterodimer $\beta_A\beta_B$ and the homodimers $\beta_A\beta_A$ and $\beta_B\beta_B$ are also formed. They stimulate rather than inhibit FSH secretion and consequently are called **activins.** Their function in reproduction is unsettled. However, the inhibins and activins are members of the transforming growth factor beta (TGF-β) superfamily of dimeric growth factors that also includes MIS. **Activin receptors** have been identified and belong to the serine/threonine kinase receptor family. Inhibins and activins are found not only in the gonads but also in the brain and many other tissues. In the bone marrow, activins are involved in the development of white blood cells. In embryonic life, activins are involved in the formation of mesoderm. All mice with a targeted deletion of the α-inhibin subunit gene initially exhibit normal growth but then develop gonadal stromal tumors, so the gene is a tumor suppressor gene.

In plasma, α_2-macroglobulin binds activins and inhibins. In tissues, activins bind to a family of four glycoproteins called **follistatins.** Binding of the activins inactivates their biological activity, but the relation of follistatins to inhibin and their physiological function remain unsettled.

Steroid Feedback

A current “working hypothesis” of the way the functions of the testes are regulated by steroids is shown in **Figure 23–10**. Castration is followed by a rise in the pituitary content and secretion of FSH and LH, and hypothalamic lesions prevent this rise. Testosterone inhibits LH secretion by acting directly on the anterior pituitary and by inhibiting the secretion of gonadotropin-releasing hormone (GnRH) from the hypothalamus. Inhibin acts directly on the anterior pituitary to inhibit FSH secretion.

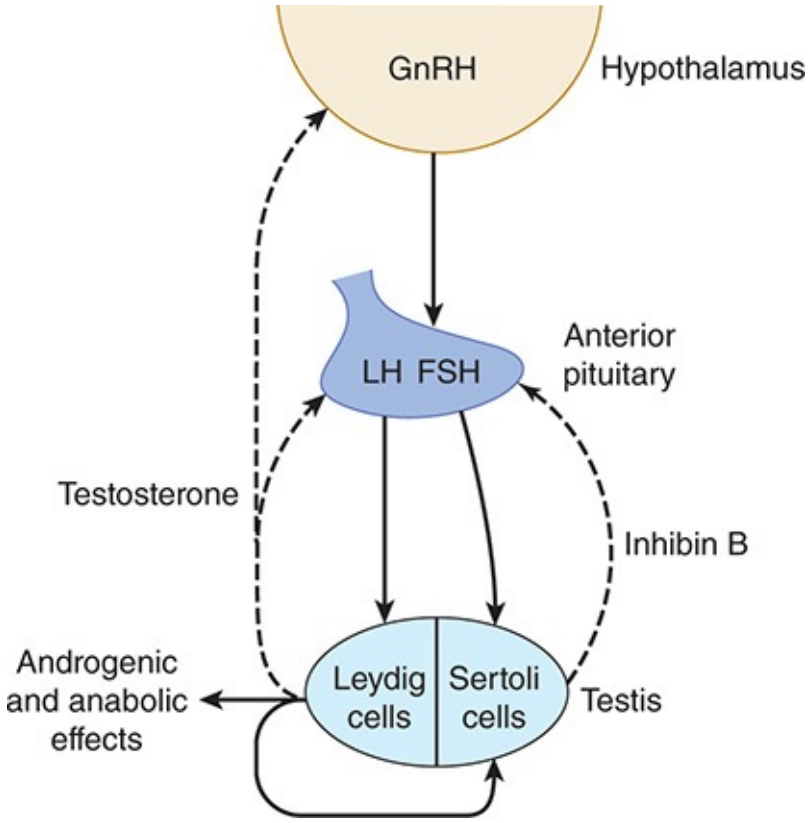


FIGURE 23–10 The hypothalamus, pituitary, and testes interact via signaling molecules. Postulated signaling pathways between the hypothalamus, anterior pituitary, and testis are shown. Solid arrows indicate excitatory effects, whereas dashed arrows indicate inhibitory effects.

In response to LH, some of the testosterone secreted from the Leydig cells bathes the seminiferous epithelium and provides the high local concentration of androgen to the Sertoli cells that is necessary for normal spermatogenesis. Systemically administered testosterone does not raise the androgen level in the testes to as great a degree, and it inhibits LH secretion. Consequently, the net effect of systemically administered testosterone is generally a decrease in sperm count. Testosterone therapy has been suggested as a means of male contraception. However, the dose of testosterone needed to suppress spermatogenesis causes sodium and water retention. The possible use of inhibins as male contraceptives is being explored.

ABNORMALITIES OF TESTICULAR FUNCTION

Cryptorchidism

The testes develop in the abdominal cavity and normally migrate to the scrotum during fetal development. **Testicular descent** to the inguinal region depends on MIS, and descent from the inguinal region to the scrotum depends on other factors. Descent is incomplete on one or, less commonly, both sides in 10% of newborn males, with the testes remaining in the abdominal cavity or inguinal canal. Gonadotropic hormone treatment speeds descent in some cases, or the defect can be corrected surgically. Spontaneous descent of the testes is the rule, and the proportion of boys with undescended testes **(cryptorchidism)** falls to 2% at age 1 year and 0.3% after puberty. However, early treatment is now recommended despite these figures because the incidence of malignant tumors is higher in undescended than in scrotal testes and because after puberty the higher temperature in the abdomen eventually causes irreversible damage to the spermatogenic epithelium.

Male Hypogonadism

The clinical picture of male hypogonadism depends on whether testicular deficiency develops before or after puberty. In adults, if it is due to testicular disease, circulating gonadotropin levels are elevated **(hypergonadotropic hypogonadism);** if it is secondary to disorders of the pituitary or the hypothalamus (eg, Kallmann syndrome), circulating gonadotropin levels are decreased **(hypogonadotropic hypogonadism).** If the endocrine function of the testes is lost in adulthood, the secondary sex characteristics regress slowly because it takes very little androgen to maintain them once they are established. The growth of the larynx during adolescence is permanent, and the voice remains deep. Men castrated in adulthood suffer some loss of libido, although the ability to copulate persists for some time. They occasionally have hot flushes and are generally more irritable, passive, and depressed than men with intact testes. When the Leydig cell deficiency dates from childhood, the clinical picture is that of **eunuchoidism.** Eunuchoid individuals over the age of 20 are characteristically tall, although not as tall as hyperpituitary giants, because their epiphyses remain open and some growth continues past the normal age of puberty. They have narrow shoulders and small muscles, a body configuration resembling that of the adult female. The genitalia are small and the voice high-pitched. Pubic hair and axillary hair are present because of adrenocortical androgen secretion. However, the hair is sparse, and the pubic hair has the female “triangle with the base up” distribution rather than the “triangle with the base down” pattern (male escutcheon) seen in normal males.

Androgen-Secreting Tumors

"Hyperfunction" of the testes in the absence of tumor formation is not a recognized entity. Androgen-secreting Leydig cell tumors are rare and cause detectable endocrine symptoms only in prepubertal boys, in whom precocious pseudopuberty develops (Table 22–2).

Hormones & Cancer

Some carcinomas of the prostate are **androgen-dependent** and regress temporarily after the removal of the testes or treatment with GnRH agonists in doses that are sufficient to produce downregulation of the GnRH receptors on gonadotropes and decrease LH secretion.

CHAPTER SUMMARY

- The gonads have a dual function: the production of germ cells (gametogenesis) and the secretion of sex hormones. The testes secrete large amounts of androgens, principally testosterone, but they also secrete small amounts of estrogens.
- Spermatogonia develop into mature spermatozoa in the seminiferous tubules via a process called spermatogenesis. This is a multistep process that includes maturation of spermatogonia into primary spermatocytes, which undergo meiotic division, resulting in haploid secondary spermatocytes. Several further divisions result in spermatids. Each cell division from a spermatogonium to a spermatid is incomplete, with cells remaining connected via cytoplasmic bridges. Spermatids eventually mature into motile spermatozoa to complete spermatogenesis; this last part of maturation is called spermiogenesis.
- Testosterone is the principal hormone of the testis. It is synthesized from cholesterol in Leydig cells. The secretion of testosterone from Leydig cells is under control of luteinizing hormone at a rate of 4–9 mg/d in adult males. Most testosterone is bound to albumin or to gonadal steroid–binding globulin in the plasma. Testosterone plays an important role in the development and maintenance of male secondary sex characteristics, as well as other defined functions.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. Full development and function of the seminiferous tubules require
   A. somatostatin.
   B. LH.
   C. oxytocin.
   D. FSH.
   E. androgens and FSH.
2. In human males, testosterone is produced mainly by the
   A. Leydig cells.
   B. Sertoli cells.
   C. seminiferous tubules.
   D. epididymis.
   E. vas deferens.
3. NOS contributes to erection by:
   A. raising cAMP levels that relax smooth muscles and increase blood flow.
   B. blocking phosphodiesterases to increase cGMP levels that release smooth muscle and increase blood flow.
   C. activating soluble guanylyl cyclases to increase cGMP levels that relax smooth muscle and increase blood flow.
   D. raising intracellular Ca^{2+} concentrations that relax smooth muscles and increase blood flow.
4. Testosterone is produced
   A. in the testes after reduction of dihydrotestosterone.
   B. in Leydig cells from cholesterol and pregnenolone precursors.
   C. by LH in Leydig cells.
   D. as a precursor for several membrane lipids.

CHAPTER 24

Endocrine Functions of the Pancreas & Regulation of Carbohydrate Metabolism

OBJECTIVES

After studying this chapter, you should be able to:

- List the hormones that affect the plasma glucose concentration and briefly describe the action of each.
- Describe the structure of the pancreatic islets and name the hormones secreted by each of the cell types in the islets.
- Describe the structure of insulin and outline the steps involved in its biosynthesis and release into the bloodstream.
- List the consequences of insulin deficiency and explain how each of these abnormalities is produced.
- Describe insulin receptors, the way they mediate the effects of insulin, and the way they are regulated.
- Describe the types of glucose transporters found in the body and the function of each.
- List the major factors that affect the secretion of insulin.
- Describe the structure of glucagon and other physiologically active peptides produced from its precursor.

- List the physiologically significant effects of glucagon and the factors that regulate glucagon secretion.
- Describe the physiologic effects of somatostatin in the pancreas.
- Outline the mechanisms by which thyroid hormones, adrenal glucocorticoids, catecholamines, and growth hormone affect carbohydrate metabolism.
- Understand the major differences between type 1 and type 2 diabetes.

INTRODUCTION

The pancreas is a glandular organ that contributes to the physiological regulation of the endocrine and digestive system. It is located in the abdominal cavity, below the liver and behind the stomach (Figure 25-11, Chapter 25). The **islets of Langerhans** are the endocrine cells within the pancreas that secrete four polypeptides with regulatory activity. Two of these, **insulin** and **glucagon,** are hormones and have important functions in the regulation of the intermediary metabolism of carbohydrates, proteins, and fats. The third polypeptide, **somatostatin,** plays a role in the regulation of islet cell secretion, and the fourth, **pancreatic polypeptide,** is probably concerned primarily with the regulation of ion transport in the intestine. Glucagon, somatostatin, and possibly pancreatic polypeptide are also secreted by cells in the mucosa of the gastrointestinal tract (Chapter 25).

Insulin is anabolic, increasing the storage of glucose, fatty acids, and amino acids. Glucagon is catabolic, mobilizing glucose, fatty acids, and the amino acids from stores into the bloodstream. The two hormones are thus reciprocal in their overall action and are reciprocally secreted in most circumstances. Insulin excess causes hypoglycemia, which leads to convulsions and coma. Insulin deficiency, either absolute or relative, causes **diabetes mellitus** (chronic elevated blood glucose), a complex and debilitating disease that if untreated is eventually fatal. Glucagon deficiency can cause hypoglycemia, and glucagon excess makes diabetes worse. Excess pancreatic production of somatostatin causes hyperglycemia and other manifestations of diabetes.

A variety of other hormones also have important roles in the regulation of carbohydrate metabolism.

ISLET CELL STRUCTURE

The islets of Langerhans **(Figure 24–1)** are ovoid, 76- × 175-µm collections of cells. The islets are scattered throughout the pancreas, although they are more plentiful in the tail than in the body and head. β-Islets make up about 2% of the volume of the gland, whereas the exocrine portion of the pancreas (see Chapter 25) makes up 80%, and ducts and blood vessels make up the remainder. Humans have 1–2 million islets. Each has a copious blood supply; blood from the islets, like that from the gastrointestinal tract (but unlike that from any other endocrine organs) drains into the hepatic portal vein.

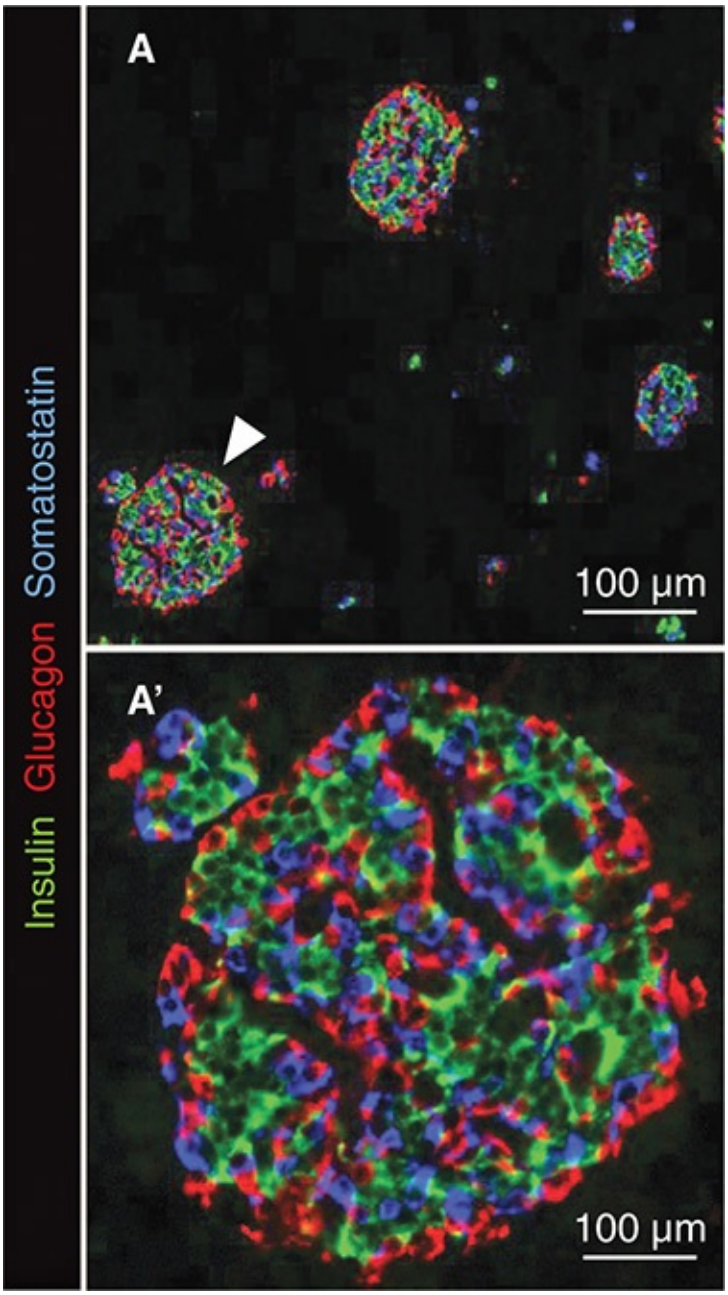


FIGURE 24–1 Islet of Langerhans in the human pancreas. Green fluorescent labeling are insulin containing B cells within the islets, red fluorescent labeling are glucagon containing A cells and blue fluorescent labeling are somatostatin containing D cells. Surrounding dark area is pancreatic acinar tissue. (Used with permission from N. Hart).

The cells in the islets can be divided into types on the basis of their staining properties and morphology. Humans have at least four distinct cell types: A, B, D, and F cells. A, B, and D cells are also called α, β, and δ cells. However, this leads to confusion in view of the use of Greek letters to refer to other structures in the body, particularly adrenergic receptors (see Chapter 7). The A cells secrete glucagon, the B cells secrete insulin, the D cells secrete somatostatin, and the F cells secrete pancreatic polypeptide. The B cells, which are the most common and account for 60–75% of the cells in the islets, are generally located in the center of each islet. They tend to be surrounded by the A cells, which make up 20% of the total, and the less common D and F cells. The islets in the tail, the body, and the anterior and superior part of the head of the human pancreas have many A cells and few if any F cells in the outer rim, whereas in rats and probably in humans, the islets in the posterior part of the head of the pancreas have a relatively large number of F cells and few A cells. The A-cell-rich (glucagon-rich) islets arise embryologically from the dorsal pancreatic bud, and the F-cell-rich (pancreatic polypeptide-rich) islets arise from the ventral pancreatic bud. These buds arise separately from the duodenum.

The B cell granules are packets of insulin in the cell cytoplasm. The shape of the packets varies from species to species; in humans, some are round whereas others are rectangular **(Figure 24–2)**. In the B cells, the insulin molecule forms polymers and also complexes with zinc. The differences in the shape of the packets are probably due to differences in the size of polymers or zinc aggregates of insulin. The A granules, which contain glucagon, are relatively uniform from species to species **(Figure 24–3)**. The D cells also contain large numbers of relatively homogeneous granules.

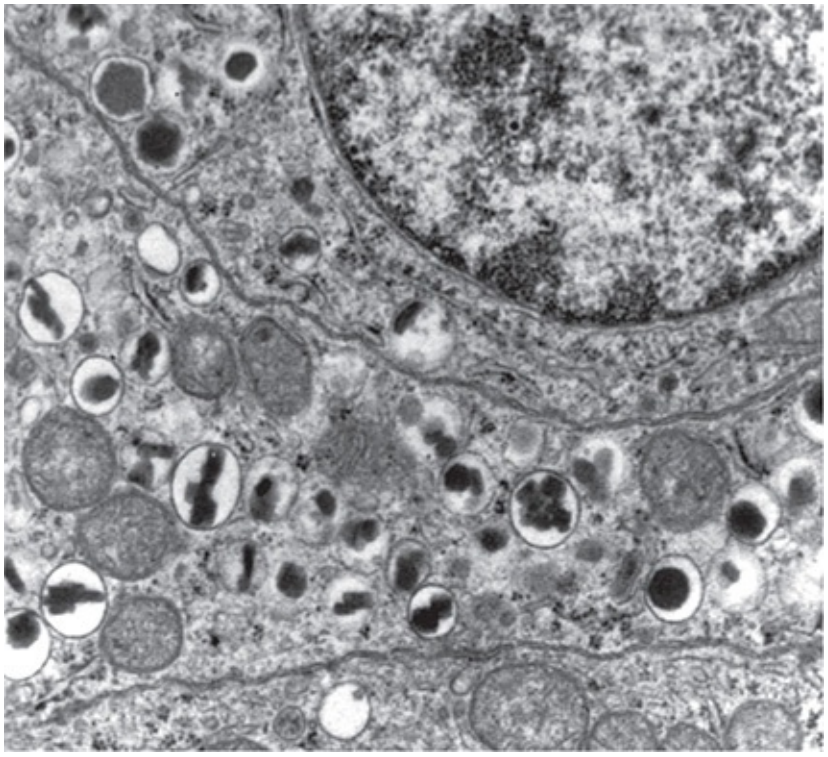


FIGURE 24–2 Electronmicrograph of two adjoining B cells in a human pancreatic islet. The B granules are the crystals in the membrane-lined vesicles. They vary in shape from rhombic to round (× 26,000). (Reproduced with permission from Fawcett DW: *Bloom and Fawcett A Textbook of Histology*, 11th ed. St. Louis, MO: Saunders; 1986.)

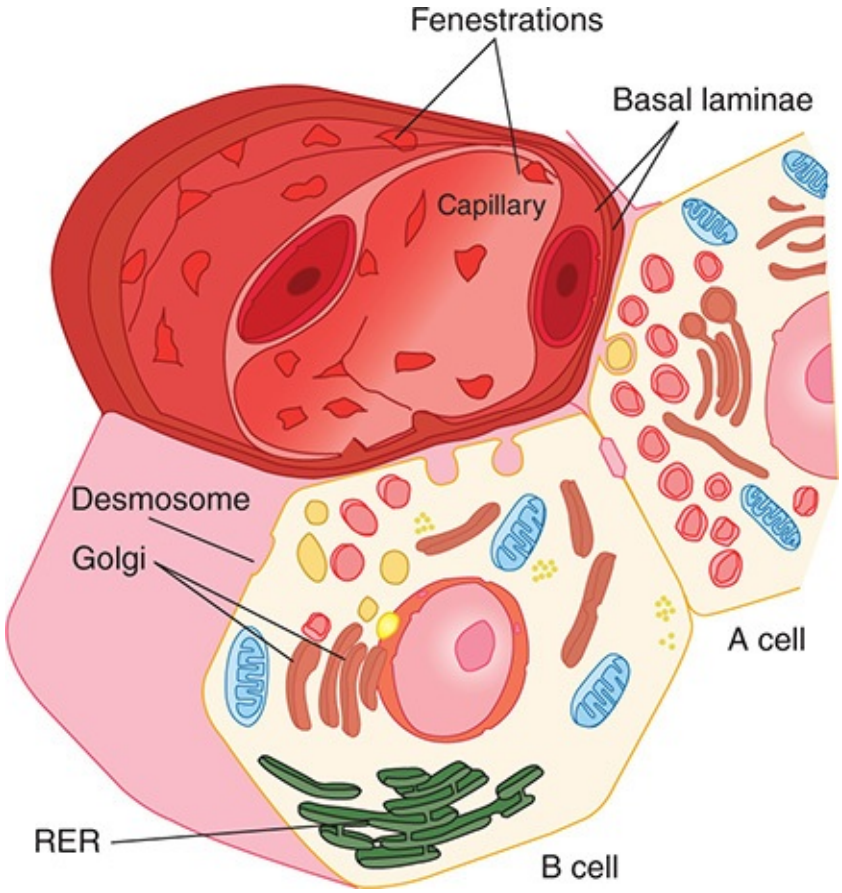


FIGURE 24–3 A and B cells, showing their relation to a blood vessel. Insulin from the B cell and glucagon from the A cell are secreted by exocytosis and cross the basal lamina of the cell and the basal lamina of the capillary before entering the lumen of the fenestrated capillary. RER, rough endoplasmic reticulum. (Reproduced with permission from Junqueira IC, Carneiro J: *Basic Histology: Text and Atlas*, 10th ed. New York, NY: McGraw-Hill; 2003.)

STRUCTURE, BIOSYNTHESIS, & SECRETION OF INSULIN

STRUCTURE & SPECIES SPECIFICITY

Insulin is a polypeptide containing two chains of amino acids linked by disulfide bridges. Minor differences occur in the amino acid composition of the molecule from species to species. The differences are generally not sufficient to affect the biologic activity of a particular insulin in heterologous species but are sufficient to make the insulin antigenic. If insulin of one species is injected for a prolonged

period into another species, the anti-insulin antibodies formed inhibit the injected insulin. Almost all humans who have received commercial bovine insulin for more than 2 months have antibodies against bovine insulin, but the titer is usually low. Porcine insulin differs from human insulin by only one amino acid residue and has low antigenicity. Human insulin produced in bacteria by recombinant DNA technology is now widely used to avoid antibody formation.

BIOSYNTHESIS & SECRETION

Insulin is synthesized in the rough endoplasmic reticulum of the B cells (Figure 24–3). It is then transported to the Golgi apparatus, where it is packaged into membrane-bound granules. These granules move to the plasma membrane by a process involving microtubules, and their contents are expelled by exocytosis (see Chapters 2 and 16). The insulin then crosses the basal lamina of the B cell and a neighboring capillary and the fenestrated endothelium of the capillary to reach the bloodstream. The fenestrations are discussed in detail in Chapter 31.

Like other polypeptide hormones and related proteins that enter the endoplasmic reticulum, insulin is synthesized as part of a larger preprohormone (see Chapter 1). The gene for insulin is located on the short arm of chromosome 11 in humans. It has two introns and three exons. **Preproinsulin** originates from the endoplasmic reticulum. The remainder of the molecule is then folded, and the disulfide bonds are formed to make **proinsulin.** The peptide segment connecting the A and B chains, the **connecting peptide (C peptide),** facilitates the folding and then is detached in the granules before secretion. Two proteases are involved in processing the proinsulin. Normally, 90–97% of the product released from the B cells is insulin along with equimolar amounts of C peptide. The rest is mostly proinsulin. C peptide can be measured by radioimmunoassay, and its level in blood provides an index of B cell function in patients receiving exogenous insulin.

FATE OF SECRETED INSULIN

INSULIN & INSULIN-LIKE ACTIVITY IN BLOOD

Plasma contains a number of substances with insulin-like activity in addition to insulin. The activity that is not suppressed by anti-insulin antibodies has been called **nonsuppressible insulin-like activity (NSILA).** Most, if not all, of this

activity persists after pancreatectomy and is due to the insulin-like growth factors **IGF-I** and **IGF-II** (see Chapter 18). These IGFs are polypeptides. Small amounts are free in the plasma (low-molecular-weight fraction), but large amounts are bound to proteins (high-molecular-weight fraction).

One may well ask why pancreatectomy causes diabetes mellitus when NSILA persists in the plasma. However, the insulin-like activities of IGF-I and IGF-II are weak compared to that of insulin and likely subserve other specific functions.

METABOLISM

The half-life of insulin in the circulation in humans is about 5 min. Insulin binds to insulin receptors, and some is internalized. It is destroyed by proteases in the endosomes formed by the endocytotic process.

EFFECTS OF INSULIN

The physiologic effects of insulin are far-reaching and complex. They are conveniently divided into rapid, intermediate, and delayed actions **(Table 24–1)**. The best known is the hypoglycemic effect, but there are additional effects on amino acid and electrolyte transport, many enzymes, and growth. The net effect of the hormone is storage of carbohydrate, protein, and fat. Therefore, insulin is appropriately called the "hormone of abundance."

TABLE 24–1 Principal actions of insulin.

Rapid (seconds)
Increased transport of glucose, amino acids, and K^+ into insulin-sensitive cells
Intermediate (minutes)
Stimulation of protein synthesis
Inhibition of protein degradation
Activation of glycolytic enzymes and glycogen synthase
Inhibition of phosphorylase and gluconeogenic enzymes
Delayed (hours)
Increase in mRNAs for lipogenic and other enzymes

Courtesy of ID Goldfine.

The actions of insulin on adipose tissue; skeletal, cardiac, and smooth muscle; and the liver are summarized in **Table 24–2**.

TABLE 24–2 Effects of insulin on various tissues.

Adipose tissue
Increased glucose entry
Increased fatty acid synthesis
Increased glycerol phosphate synthesis
Increased triglyceride deposition
Activation of lipoprotein lipase
Inhibition of hormone-sensitive lipase
Increased K^+ uptake
Muscle
Increased glucose entry
Increased glycogen synthesis
Increased amino acid uptake
Increased protein synthesis in ribosomes
Decreased protein catabolism
Decreased release of gluconeogenic amino acids
Increased ketone uptake
Increased K^+ uptake
Liver
Decreased ketogenesis
Increased protein synthesis
Increased lipid synthesis
Decreased glucose output due to decreased gluconeogenesis, increased glycogen synthesis, and increased glycolysis
General
Increased cell growth

GLUCOSE TRANSPORTERS

Glucose enters cells by **facilitated diffusion** (see Chapter 1) or, in the intestine and kidneys, by secondary active transport with Na^+. In muscle, adipose, and

some other tissues, insulin stimulates glucose entry into cells by increasing the number of glucose transporters (GLUTs) in the cell membranes.

The GLUTs that are responsible for facilitated diffusion of glucose across cell membranes are a family of closely related proteins that span the cell membrane 12 times and have their amino and carboxyl terminals inside the cell. They differ from and have no homology with the **Sodium**-dependent **glucose** cotransporters (SGLT-1 and SGLT-2), which are responsible for the secondary active transport of glucose in the intestine (see Chapter 26) and renal tubules (see Chapter 38), although the SGLTs also have 12 transmembrane domains.

Seven different GLUTs, named GLUT 1–7 in order of discovery, have been characterized **(Table 24–3)**. They contain 492–524 amino acid residues and their affinity for glucose varies. Each transporter appears to have evolved for special tasks. GLUT-4 is the transporter in muscle and adipose tissue that is stimulated by insulin. A pool of GLUT-4 molecules is maintained within vesicles in the cytoplasm of insulin-sensitive cells. When the insulin receptors of these cells are activated, the vesicles move rapidly to the cell membrane and fuse with it, inserting the transporters into the cell membrane **(Figure 24–4)**. When insulin action ceases, the transporter-containing patches of membrane are endocytosed and the vesicles are ready for the next exposure to insulin. Activation of the insulin receptor brings about the movement of the vesicles to the cell membrane by activating phosphatidylinositol 3-kinase (Figure 24–4). Most of the other GLUT transporters that are not insulin-sensitive appear to be constitutively expressed in the cell membrane.

TABLE 24–3 Glucose transporters in mammals.

	Function	K_m (mM)[a]	Major Sites of Expression
Secondary active transport (Na^+-glucose cotransport)			
SGLT 1	Absorption of glucose	0.1–1.0	Small intestine, renal tubules
SGLT 2	Absorption of glucose	1.6	Renal tubules
Facilitated diffusion			
GLUT 1	Basal glucose uptake	1–2	Placenta, blood-brain barrier, brain, red cells, kidneys, colon, many other organs
GLUT 2	B-cell glucose sensor; transport out of intestinal and renal epithelial cells	12–20	B cells of islets, liver, epithelial cells of small intestine, kidneys
GLUT 3	Basal glucose uptake	<1	Brain, placenta, kidneys, many other organs
GLUT 4	Insulin-stimulated glucose uptake	5	Skeletal and cardiac muscle, adipose tissue, other tissues
GLUT 5	Fructose transport	1–2	Jejunum, sperm
GLUT 6	Unknown	—	Brain, spleen, and leukocytes
GLUT 7	Glucose 6-phosphate transporter in endoplasmic reticulum	—	Liver

[a]The K_m is the glucose concentration at which transport is half-maximal.

Data from Stephens JM, Pilch PF: The metabolic regulation and vesicular transport of GLUT 4, the major insulin-responsive glucose transporter. Endocr Rev 1995;16:529.

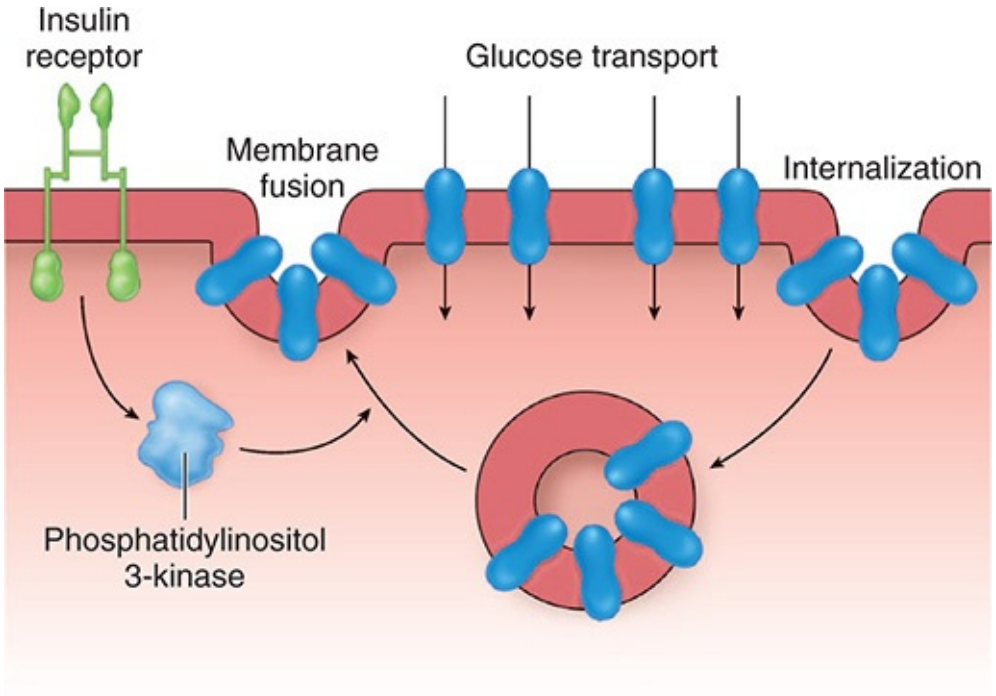


FIGURE 24–4 Cycling of GLUT-4 transporters through endosomes in insulin-sensitive tissues. Activation of the insulin receptor causes activation of phosphatidylinositol 3-kinase, which speeds translocation of the GLUT-4–containing endosomes into the cell membrane. The GLUT-4 transporters then mediate glucose transport into the cell.

In the tissues in which insulin increases the number of GLUTs in cell membranes, the rate of phosphorylation of the glucose, once it has entered the cells, is regulated by other hormones. Growth hormone and cortisol both inhibit phosphorylation in certain tissues. Transport is normally so rapid that it is not a rate-limiting step in glucose metabolism. However, it is rate-limiting in B cells.

Insulin also increases the entry of glucose into liver cells, but it does not exert this effect by increasing the number of GLUT-4 transporters in the cell membranes. Instead, it induces glucokinase, and this increases the phosphorylation of glucose, so that the intracellular free glucose concentration stays low, facilitating the entry of glucose into the cell.

Insulin-sensitive tissues also contain a population of GLUT-4 vesicles that move into the cell membrane in response to exercise, a process that occurs independent of the action of insulin. This is why exercise lowers blood sugar. A 5′-adenosine monophosphate (AMP)–activated kinase may trigger the insertion of these vesicles into the cell membrane.

INSULIN PREPARATIONS

The maximal decline in plasma glucose occurs 30 min after intravenous injection of insulin. After subcutaneous administration, the maximal fall occurs in 2–3 h. A wide variety of insulin preparations are now available commercially. These include insulins that have been complexed with protamine and other polypeptides to delay absorption and degradation, and synthetic insulins in which there have been changes in amino acid residues. In general, they fall into three categories: rapid, intermediate-acting, and long-acting (24–36 h).

Relation To Potassium

Insulin causes K^+ to enter cells, with a resultant lowering of the extracellular K^+ concentration. Infusions of insulin and glucose significantly lower the plasma K^+ level in normal individuals and are very effective for the temporary relief of hyperkalemia in patients with renal failure. **Hypokalemia** often develops when patients with diabetic acidosis are treated with insulin. The reason for the intracellular migration of K^+ is still uncertain. However, insulin increases the activity of Na, K ATPase in cell membranes, so that more K^+ is pumped into cells.

OTHER ACTIONS

The hypoglycemic and other effects of insulin are summarized in temporal terms in Table 24–1, and the net effects on various tissues are summarized in Table 24–2. The action on glycogen synthase fosters glycogen storage, and the actions on glycolytic enzymes favor glucose metabolism to two carbon fragments (see Chapter 1), with resulting promotion of lipogenesis. Stimulation of protein synthesis from amino acids entering the cells and inhibition of protein degradation foster growth.

The anabolic effect of insulin is aided by the protein-sparing action of adequate intracellular glucose supplies. Failure to grow is a symptom of diabetes in children, and insulin stimulates the growth of immature hypophysectomized rats to almost the same degree as growth hormone.

MECHANISM OF ACTION

INSULIN RECEPTORS

Insulin receptors are found on many different cells in the body, including cells in which insulin does not increase glucose uptake.

The insulin receptor, which has a molecular weight of approximately 340,000, is a tetramer made up of two α and two β glycoprotein subunits **(Figure 24–5)**. All these are synthesized on a single mRNA and then proteolytically separated and bound to each other by disulfide bonds. The gene for the insulin receptor has 22 exons and in humans is located on chromosome 19. The α subunits bind insulin and are extracellular, whereas the β subunits span the membrane. The intracellular portions of the β subunits have tyrosine kinase activity. The α and β subunits are both glycosylated, with sugar residues extending into the interstitial fluid.

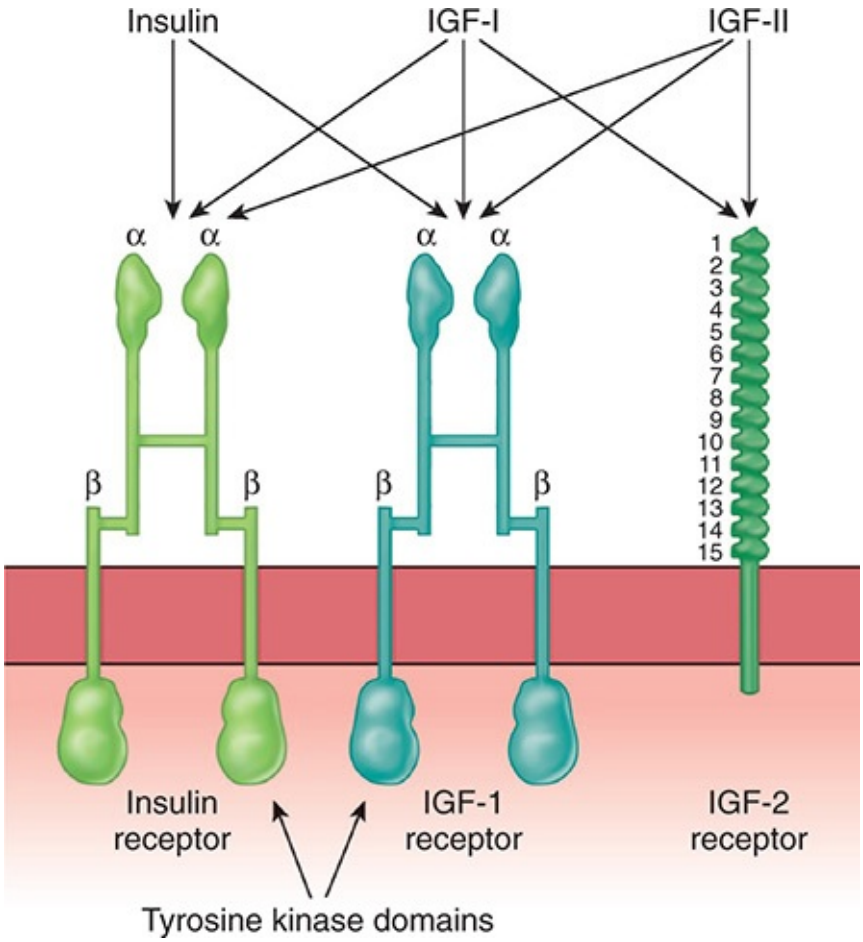


FIGURE 24–5 Insulin, IGF-I, and IGF-II receptors. Each hormone binds primarily to its own receptor, but insulin also binds to the IGF-I receptor, and IGF-I and IGF-II bind to all three. The purple boxes are intracellular tyrosine

kinase domains. Note the marked similarity between the insulin receptor and the IGF-I receptor; also note the 15 repeat sequences in the extracellular portion of the IGF-II receptor.

Binding of insulin triggers the tyrosine kinase activity of the β subunits, producing autophosphorylation of the β subunits on tyrosine residues. The autophosphorylation, which is necessary for insulin to exert its biologic effects, triggers phosphorylation of some cytoplasmic proteins and dephosphorylation of others, mostly on serine and threonine residues. Insulin receptor substrate (IRS-1) mediates some of the effects in humans but there are other effector systems as well **(Figure 24–6)**. For example, mice in which the insulin receptor gene is knocked out show marked growth retardation in utero, have abnormalities of the central nervous system (CNS) and skin, and die at birth of respiratory failure, whereas IRS-1 knockouts show only moderate growth retardation in utero, survive, and are insulin-resistant but otherwise nearly normal.

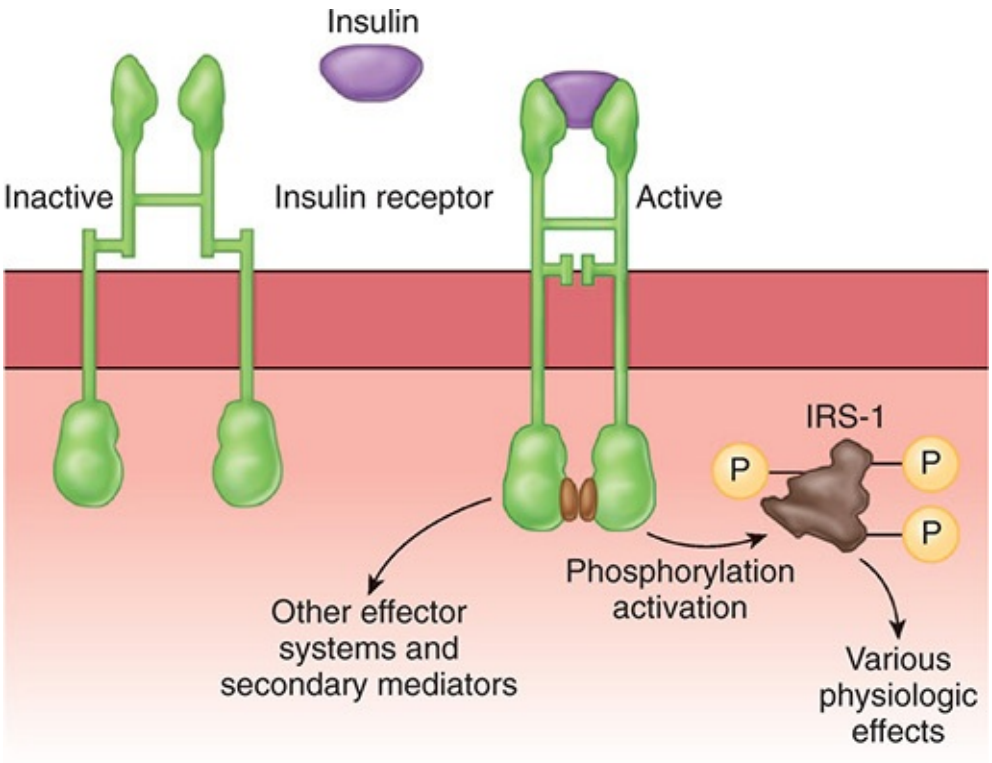


FIGURE 24–6 Intracellular responses triggered by insulin binding to the insulin receptor. Circles labeled P represent phosphate groups. IRS-1, insulin receptor substrate-1.

The growth-promoting protein anabolic effects of insulin are mediated via

phosphatidylinositol 3-kinase (PI3K), and evidence indicates that in invertebrates, this pathway is involved in the growth of nerve cells and axon guidance in the visual system.

It is interesting to compare the insulin receptor with other related receptors. The insulin receptor is very similar to the receptor for IGF-I but different from the receptor for IGF-II (Figure 24–5). Other receptors for growth factors and receptors for various oncogenes also are tyrosine kinases. However, the amino acid composition of these receptors is quite different.

When insulin binds to its receptors, they aggregate in patches and are taken up into the cell by receptor-mediated endocytosis (see Chapter 2). Eventually, the insulin–receptor complexes enter lysosomes, where the receptors are broken down or recycled. The half-life of the insulin receptor is about 7 h.

CONSEQUENCES OF INSULIN DEFICIENCY

The far-reaching physiologic effects of insulin are highlighted by a consideration of the extensive and serious consequences of insulin deficiency **(Clinical Box 24–1)**.

In humans, insulin deficiency is a common pathologic condition. In animals, it can be produced by pancreatectomy; by administration of alloxan, streptozocin, or other toxins that in appropriate doses cause selective destruction of the B cells of the pancreatic islets; by administration of drugs that inhibit insulin secretion; and by administration of anti-insulin antibodies. Strains of mice, rats, hamsters, guinea pigs, miniature swine, and monkeys that have a high incidence of spontaneous diabetes mellitus have also been described.

CLINICAL BOX 24–1

Diabetes Mellitus

The constellation of abnormalities caused by insulin deficiency is called **diabetes mellitus.** Greek and Roman physicians used the term “diabetes” to refer to conditions in which the cardinal finding was a large urine volume, and two types were distinguished: “diabetes mellitus,” in which the urine tasted sweet; and “diabetes insipidus,” in which the urine had little taste. Today, the term “diabetes insipidus” is reserved for conditions in which there is a deficiency of the production or action of vasopressin (see Chapter 38), and the unmodified word “diabetes” is generally used as a synonym for

diabetes mellitus.

The cause of clinical diabetes is always a deficiency of the effects of insulin at the tissue level. **Type 1 diabetes,** or **insulin-dependent diabetes mellitus (IDDM),** is due to insulin deficiency caused by autoimmune destruction of the B cells in the pancreatic islets, and it accounts for 3–5% of cases and usually presents in children. **Type 2 diabetes, or non–insulin-dependent diabetes mellitus (NIDDM),** is characterized by the dysregulation of insulin release from the B cells, along with insulin resistance in peripheral tissues such as skeletal muscle, brain, and liver. Type 2 diabetes historically presented in overweight or obese adults, although it is increasingly being diagnosed in children as childhood obesity increases.

Diabetes is characterized by polyuria (passage of large volumes of urine), polydipsia (excessive drinking), weight loss in spite of polyphagia (increased appetite), hyperglycemia, glycosuria, ketosis, acidosis, and coma. Widespread biochemical abnormalities are present, but the fundamental defects to which most of the abnormalities can be traced are (1) reduced entry of glucose into various "peripheral" tissues and (2) increased liberation of glucose into the circulation from the liver. Therefore, there is an extracellular glucose excess and, in many cells, an intracellular glucose deficiency—a situation that has been called "starvation in the midst of plenty." Also, the entry of amino acids into muscle is decreased and lipolysis is increased.

THERAPEUTIC HIGHLIGHTS

In type 1 diabetes, the mainstay of therapy is provision of exogenous insulin, carefully titrated to dietary intake of glucose. In type 2 diabetes, lifestyle changes such as alterations in the diet or increased exercise can often delay symptoms in early disease, but these are difficult to secure. Insulin-sensitizing drugs represent second-line agents (see Chapter 16).

GLUCOSE TOLERANCE

In diabetes, glucose piles up in the bloodstream, especially after meals. If a glucose load is given to a diabetic, the plasma glucose rises higher and returns to the baseline more slowly than it does in normal individuals. The response to a standard oral test dose of glucose, the **oral glucose tolerance test,** is used in the clinical diagnosis of diabetes **(Figure 24–7)**.

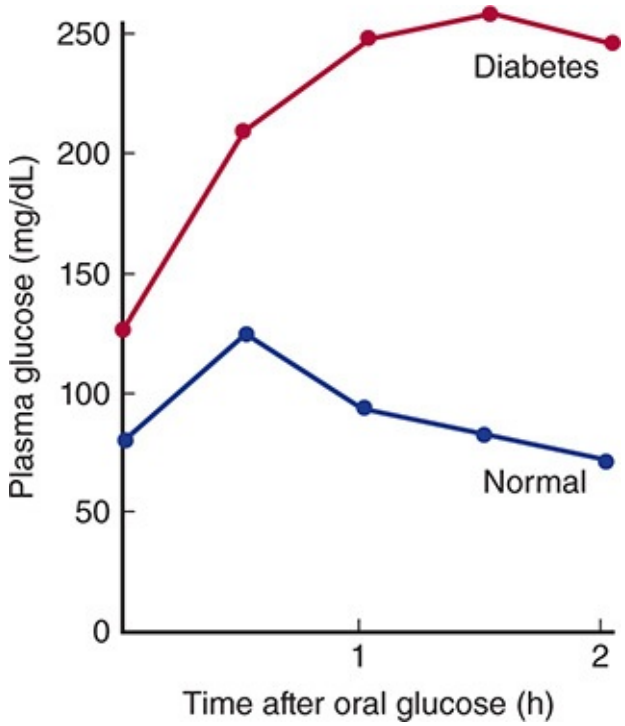


FIGURE 24–7 Oral glucose tolerance test. Adults are given 75 g of glucose in 300 mL of water. In normal individuals, the fasting venous plasma glucose is less than 115 mg/dL, the 2-hour value is less than 140 mg/dL, and no value is greater than 200 mg/dL. Diabetes mellitus is present if the 2-hour value and one other value are greater than 200 mg/dL. Impaired glucose tolerance is diagnosed when the values are above the upper limits of normal but below the values diagnostic of diabetes.

Impaired glucose tolerance in diabetes is due in part to reduced entry of glucose into cells **(decreased peripheral utilization).** In the absence of insulin, the entry of glucose into skeletal, cardiac, and smooth muscle and other tissues is decreased **(Figure 24–8)**. Glucose uptake by the liver is also reduced, but the effect is indirect. Intestinal absorption of glucose is unaffected, as is its reabsorption from the urine by the cells of the proximal tubules of the kidneys. Glucose uptake by most of the brain and the red blood cells is also normal.

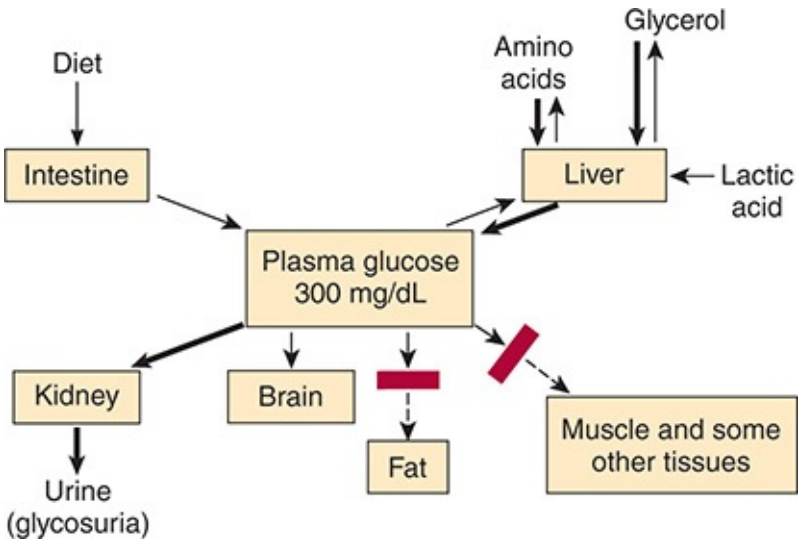


FIGURE 24–8 Disordered plasma glucose homeostasis in insulin deficiency. The heavy arrows indicate reactions that are accentuated. The rectangles across arrows indicate reactions that are blocked.

The second and the major cause of hyperglycemia in diabetes is derangement of the glucostatic function of the liver (see Chapter 28). The liver takes up glucose from the bloodstream and stores it as glycogen, but because the liver contains glucose-6-phosphatase it also discharges glucose into the bloodstream. Insulin facilitates glycogen synthesis and inhibits hepatic glucose output. When the plasma glucose is high, insulin secretion is normally increased and hepatic glucogenesis is decreased. This response does not occur in type 1 diabetes mellitus (as insulin is absent) or in type 2 diabetes mellitus (as tissues are insulin-resistant). Glucagon can contribute to hyperglycemia as it stimulates gluconeogenesis. Glucose output by the liver can be stimulated by catecholamines, cortisol, and growth hormone (ie, during a stress response).

EFFECTS OF HYPERGLYCEMIA

Hyperglycemia by itself can cause symptoms resulting from the hyperosmolality of the blood. In addition, there is glycosuria because the renal capacity for glucose reabsorption is exceeded. Excretion of the osmotically active glucose molecules entails the loss of large amounts of water (osmotic diuresis; see Chapter 38). The resultant dehydration activates the mechanisms regulating water intake, leading to polydipsia. There is an appreciable urinary loss of Na^+ and K^+ as well. For every gram of glucose excreted, 4.1 kcal is lost from the

body. Increasing the oral caloric intake to cover this loss simply raises the plasma glucose further and increases the glycosuria, so mobilization of endogenous protein and fat stores and weight loss are not prevented.

When plasma glucose is episodically elevated over time, small amounts of hemoglobin A are nonenzymatically glycated to form **HbA1c** (see Chapter 31). Careful control of the diabetes with insulin reduces the amount formed and consequently HbA1c concentration is measured clinically as an integrated index of diabetic control for the 4- to 6-week period before the measurement.

The role of chronic hyperglycemia in production of the long-term complications of diabetes is discussed below.

EFFECTS OF INTRACELLULAR GLUCOSE DEFICIENCY

The abundance of glucose outside the cells in diabetes contrasts with the intracellular deficit. Glucose catabolism is normally a major source of energy for cellular processes, and in diabetes energy requirements can be met only by drawing on protein and fat reserves. Mechanisms are activated that greatly increase the catabolism of protein and fat, and one of the consequences of increased fat catabolism is ketosis.

Deficient glucose utilization and deficient hormone sensing (insulin, leptin, CCK) in the cells of the hypothalamus that regulate satiety are the probable causes of hyperphagia in diabetes. The feeding area of the hypothalamus is not inhibited and thus satiety is not sensed so food intake is increased.

Glycogen depletion is a common consequence of intracellular glucose deficit, and the glycogen content of liver and skeletal muscle in diabetic animals is usually reduced.

CHANGES IN PROTEIN METABOLISM

In diabetes, the rate at which amino acids are catabolized to CO_2 and H_2O is increased. In addition, more amino acids are converted to glucose in the liver. The increased gluconeogenesis has many causes. Glucagon stimulates gluconeogenesis, and hyperglucagonemia is generally present in diabetes. Adrenal glucocorticoids also contribute to increased gluconeogenesis when they are elevated in severely ill diabetics. The supply of amino acids is increased for gluconeogenesis because, in the absence of insulin, less protein synthesis occurs

in muscle and hence blood amino acid levels rise. Alanine is particularly easily converted to glucose. In addition, the activity of the enzymes that catalyze the conversion of pyruvate and other two-carbon metabolic fragments to glucose is increased. These include phosphoenolpyruvate carboxykinase, which facilitates the conversion of oxaloacetate to phosphoenolpyruvate (see Chapter 1). They also include fructose 1,6-diphosphatase, which catalyzes the conversion of fructose diphosphate to fructose 6-phosphate, and glucose 6-phosphatase, which controls the entry of glucose into the circulation from the liver. Increased acetyl-CoA increases pyruvate carboxylase activity, and insulin deficiency increases the supply of acetyl-CoA because lipogenesis is decreased. Pyruvate carboxylase catalyzes the conversion of pyruvate to oxaloacetate (see Figure 1–22).

In diabetes, the net effect of accelerated protein conversion to CO_2, H_2O, and glucose, plus diminished protein synthesis, is protein depletion and wasting. Protein depletion from any cause is associated with poor "resistance" to infections.

FAT METABOLISM IN DIABETES

The principal abnormalities of fat metabolism in diabetes are accelerated lipid catabolism, with increased formation of ketone bodies, and decreased synthesis of fatty acids and triglycerides. The manifestations of the disordered lipid metabolism are so prominent that diabetes has been called "more a disease of lipid than of carbohydrate metabolism."

Fifty percent of an ingested glucose load is normally burned to CO_2 and H_2O; 5% is converted to glycogen; and 30–40% is converted to fat in the fat depots. In diabetes, less than 5% of ingested glucose is converted to fat, despite a decrease in the amount burned to CO_2 and H_2O, and no change in the amount converted to glycogen. Therefore, glucose accumulates in the bloodstream and spills over into the urine.

The role of lipoprotein lipase and hormone-sensitive lipase in the regulation of the metabolism of fat depots is discussed in Chapter 1. In diabetes, conversion of glucose to fatty acids in the depots is decreased because of the intracellular glucose deficiency. Insulin inhibits the hormone-sensitive lipase in adipose tissue, and, in the absence of this hormone, the plasma level of **free fatty acids** (NEFA, UFA, FFA) is more than doubled. The increased glucagon also contributes to the mobilization of FFA. Thus, the FFA level parallels the plasma glucose level in diabetes and in some ways is a better indicator of the severity of

the diabetic state. In the liver and other tissues, the fatty acids are catabolized to acetyl-CoA. Some of the acetyl-CoA is burned along with amino acid residues to yield CO_2 and H_2O in the citric acid cycle. However, the supply exceeds the capacity of the tissues to catabolize the acetyl-CoA.

In addition to the previously mentioned increase in gluconeogenesis and marked outpouring of glucose into the circulation, the conversion of acetyl-CoA to malonyl-CoA and thence to fatty acids is markedly impaired. This is due to a deficiency of acetyl-CoA carboxylase, the enzyme that catalyzes the conversion. The excess acetyl-CoA is converted to ketone bodies **(Clinical Box 24–2)**.

In uncontrolled diabetes, the plasma concentration of triglycerides and chylomicrons as well as FFA is increased, and the plasma is often lipemic. The rise in these constituents is mainly due to decreased removal of triglycerides into the fat depots. The decreased activity of lipoprotein lipase contributes to this decreased removal.

CLINICAL BOX 24–2

Ketosis

When excess acetyl-CoA is present in the body, some of it is converted to acetoacetyl-CoA and then, in the liver, to acetoacetate. Acetoacetate and its derivatives, acetone and β-hydroxybutyrate, enter the circulation in large quantities (see Chapter 1).

These circulating ketone bodies are an important source of energy in fasting. Half of the metabolic rate in fasted normal dogs is said to be due to metabolism of ketones. The rate of ketone utilization in diabetics is also appreciable. It has been calculated that the maximal rate at which fat can be catabolized without significant ketosis is 2.5 g/kg body weight/d in diabetic humans. In untreated diabetes, production is much greater than this, and ketone bodies pile up in the bloodstream.

ACIDOSIS

As noted in Chapter 1, acetoacetate and β-hydroxybutyrate are anions of the fairly strong acids acetoacetic acid and β-hydroxybutyric acids. The hydrogen ions from these acids are buffered, but the buffering capacity is soon exceeded if

production is increased. The resulting acidosis stimulates respiration, producing the rapid, deep respiration described by Kussmaul as “air hunger” and named (for him) **Kussmaul breathing.** The urine becomes acidic. However, when the ability of the kidneys to replace the plasma cations accompanying the organic anions with H^+ and NH_4^+ is exceeded, Na^+ and K^+ are lost in the urine. The electrolyte and water losses lead to dehydration, hypovolemia, and hypotension. Finally, the acidosis and dehydration depress consciousness to the point of coma. Diabetic acidosis is a medical emergency. Now that the infections that used to complicate the disease can be controlled with antibiotics, acidosis is the most common cause of early death in persons with clinical diabetes.

In severe acidosis, total body Na^+ is markedly depleted, and when Na^+ loss exceeds water loss, plasma Na^+ may also be low. Total body K^+ is also low, but the plasma K^+ is usually normal, partly because extracellular fluid (ECF) volume is reduced and partly because K^+ moves from cells to ECF when the ECF H^+ concentration is high. Another factor tending to maintain the plasma K^+ is the lack of insulin-induced entry of K^+ into cells.

COMA

Coma in diabetes can be due to acidosis and dehydration. However, the plasma glucose can be elevated to such a degree that independent of plasma pH, the hyperosmolarity of the plasma causes unconsciousness **(hyperosmolar coma).** Accumulation of lactate in the blood **(lactic acidosis)** may also complicate diabetic ketoacidosis if the tissues become hypoxic, and lactic acidosis may itself cause coma. Brain edema occurs in about 1% of children with ketoacidosis, and it can cause coma. Its cause is unsettled, but it is a serious complication, with a mortality rate of about 25%.

CHOLESTEROL METABOLISM

In diabetes, the plasma cholesterol level is usually elevated and this plays a role in the accelerated development of the atherosclerotic vascular disease that is a major long-term complication of diabetes in humans. The rise in plasma cholesterol level is due to an increase in the plasma concentration of very low-density lipoprotein (VLDL) and low-density lipoprotein (LDL) (see Chapter 1). These in turn may be due to increased hepatic production of VLDL or decreased removal of VLDL and LDL from the circulation.

SUMMARY

Because of the complexities of the metabolic abnormalities in diabetes, a summary is in order. One of the key features of insulin deficiency **(Figure 24–9)** is decreased entry of glucose into many tissues (decreased peripheral utilization). Also, the net release of glucose from the liver is increased (increased production), due in part to glucagon excess. The resultant hyperglycemia leads to glycosuria and a dehydrating osmotic diuresis. Dehydration leads to polydipsia. In the face of intracellular glucose deficiency, appetite is stimulated, glucose is formed from protein (gluconeogenesis), and energy supplies are maintained by metabolism of proteins and fats. Weight loss, debilitating protein deficiency, and inanition are the result.

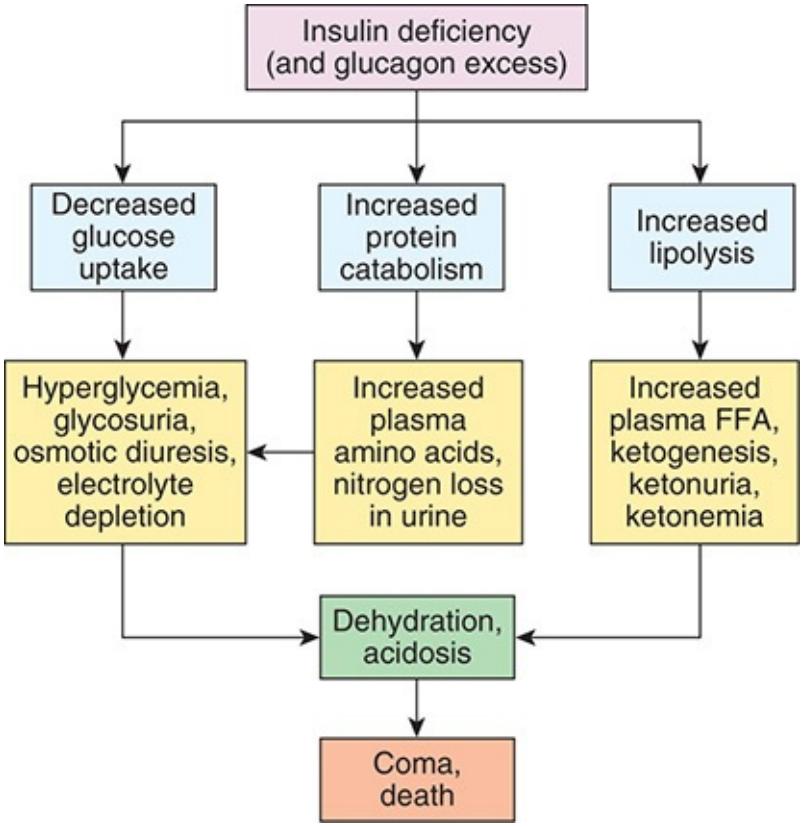


FIGURE 24–9 Effects of insulin deficiency. FFA, free fatty acids. (Used with permission of RJ Havel.)

Fat catabolism is increased and the system is flooded with triglycerides and FFA. Fat synthesis is inhibited and the overloaded catabolic pathways cannot handle the excess acetyl-CoA that is formed. In the liver, the acetyl-CoA is

converted to ketone bodies. Two of these are organic acids, and metabolic acidosis develops as ketones accumulate. Na^+ and K^+ depletion is added to the acidosis because these plasma cations are excreted with the organic anions not covered by the H^+ and NH_4^+ secreted by the kidneys. Finally, the acidotic, hypovolemic, hypotensive, depleted animal or patient becomes comatose because of the toxic effects of acidosis, dehydration, and hyperosmolarity on the nervous system and dies if treatment is not instituted.

All of these abnormalities are corrected by administration of insulin. Although emergency treatment of acidosis also includes administration of alkali to combat the acidosis as well as parenteral water, Na^+, and K^+ to replenish body stores, only insulin repairs the fundamental defects in a way that permits a return to normal.

INSULIN EXCESS

SYMPTOMS

All the known consequences of insulin excess are manifestations, directly or indirectly, of the effects of hypoglycemia on the nervous system. Except in individuals who have been fasting for some time, glucose is the only fuel used in appreciable quantities by the brain. The carbohydrate reserves in neural tissue are very limited and normal function depends on a continuous glucose supply. As the plasma glucose level falls, the first symptoms are palpitations, sweating, and nervousness due to autonomic discharge. These appear at plasma glucose values slightly lower than the value at which autonomic activation first begins, because the threshold for symptoms is slightly above the threshold for initial activation. At lower plasma glucose levels, so-called **neuroglycopenic symptoms** begin to appear. These include hunger as well as confusion and the other cognitive abnormalities. At even lower plasma glucose levels, lethargy, coma, convulsions, and eventually death occur. Obviously, the onset of hypoglycemic symptoms calls for prompt treatment with glucose or glucose-containing drinks such as orange juice. Although a dramatic disappearance of symptoms is the usual response, abnormalities ranging from intellectual dulling to coma may persist if the hypoglycemia was severe or prolonged.

COMPENSATORY MECHANISMS

One important compensation for hypoglycemia is cessation of the secretion of endogenous insulin. Inhibition of insulin secretion is complete at a plasma glucose level of about 80 mg/dL **(Figure 24–10)**. In addition, hypoglycemia triggers increased secretion of at least four counterregulatory hormones: glucagon, epinephrine, growth hormone, and cortisol. The epinephrine response is reduced during sleep. Glucagon and epinephrine increase the hepatic output of glucose by increasing glycogenolysis. Growth hormone decreases the utilization of glucose in various peripheral tissues, and cortisol has a similar action. The keys to counterregulation appear to be epinephrine and glucagon: if the plasma concentration of either increases, the decline in the plasma glucose level is reversed; but if both fail to increase, there is little if any compensatory rise in the plasma glucose level. The actions of the other hormones are supplementary.

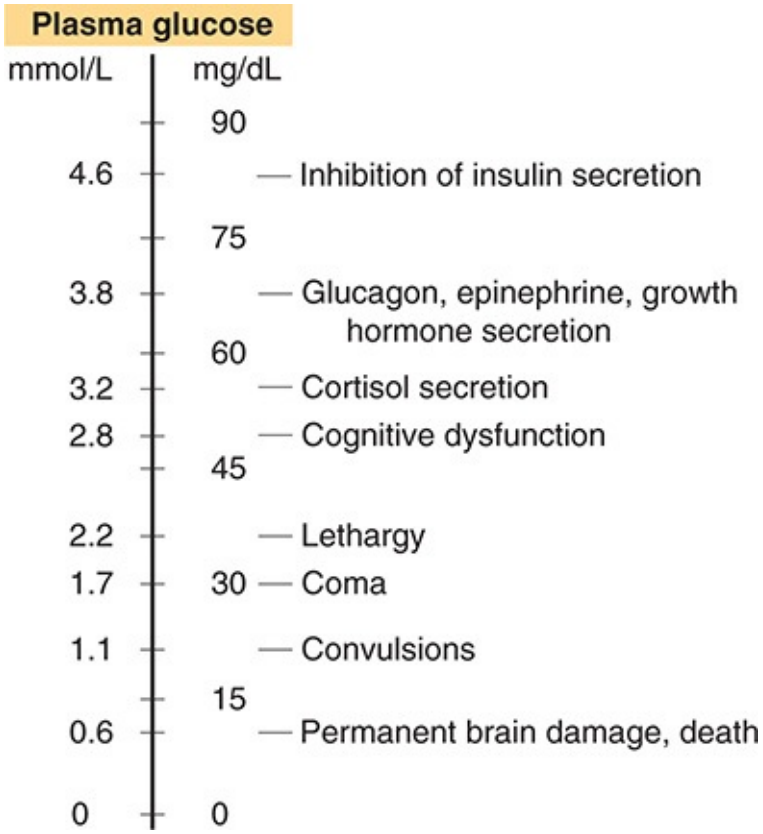


FIGURE 24–10 Plasma glucose levels at which various effects of hypoglycemia appear.

Note that the autonomic discharge and release of counterregulatory hormones normally occur at a higher plasma glucose level than the cognitive deficits and other more serious CNS changes (Figure 24–10). For diabetics treated with

insulin, the symptoms caused by the autonomic discharge serve as a warning to seek glucose replacement. However, particularly in long-term diabetics who have been tightly regulated, the autonomic symptoms may not occur, and the resulting **hypoglycemia unawareness** can be a clinical problem of some magnitude.

REGULATION OF INSULIN SECRETION

The normal concentration of insulin measured by radioimmunoassay in the peripheral venous plasma of fasting normal humans is 0–70 μU/mL (0–502 pmol/L). The amount of insulin secreted in the basal state is about 1 unit/h, with a 5-fold to 10-fold increase following ingestion of food. Therefore, the average amount secreted per day in a normal human is about 40 units (287 nmol).

Factors that stimulate and inhibit insulin secretion are summarized in **Table 24–4**.

TABLE 24–4 Factors affecting insulin secretion.

Stimulators	Inhibitors
Glucose	Somatostatin
Mannose	2-Deoxyglucose
Amino acids (leucine, arginine, others)	Mannoheptulose
Intestinal hormones (GIP, GLP-1 [7–36], gastrin, secretin, CCK; others?)	α-Adrenergic stimulators (norepinephrine, epinephrine)
β-Keto acids	β-Adrenergic blockers (propranolol)
Acetylcholine	
Glucagon	Galanin
Cyclic AMP and various cAMP-generating substances	Diazoxide Thiazide diuretics
β-Adrenergic stimulators	K^+ depletion
Theophylline	Phenytoin
Sulfonylureas	Alloxan Microtubule inhibitors Insulin

EFFECTS OF THE PLASMA GLUCOSE LEVEL

It has been known for many years that glucose acts directly on pancreatic B cells to increase insulin secretion. The response to glucose is biphasic; there is a rapid but short-lived increase in secretion followed by a more slowly developing prolonged increase.

Glucose enters the B cells via GLUT-2 transporters and is phosphorylated by glucokinase then metabolized to pyruvate in the cytoplasm **(Figure 24–11)**. The pyruvate enters the mitochondria and is metabolized to CO_2 and H_2O via the citric acid cycle with the formation of ATP by oxidative phosphorylation. The ATP enters the cytoplasm, where it inhibits ATP-sensitive K^+ channels, reducing K^+ efflux. This depolarizes the B cell, and Ca^{2+} enters the cell via voltage-gated

Ca^{2+} channels. The Ca^{2+} influx causes exocytosis of a readily releasable pool of insulin-containing secretory granules, producing the initial spike of insulin secretion.

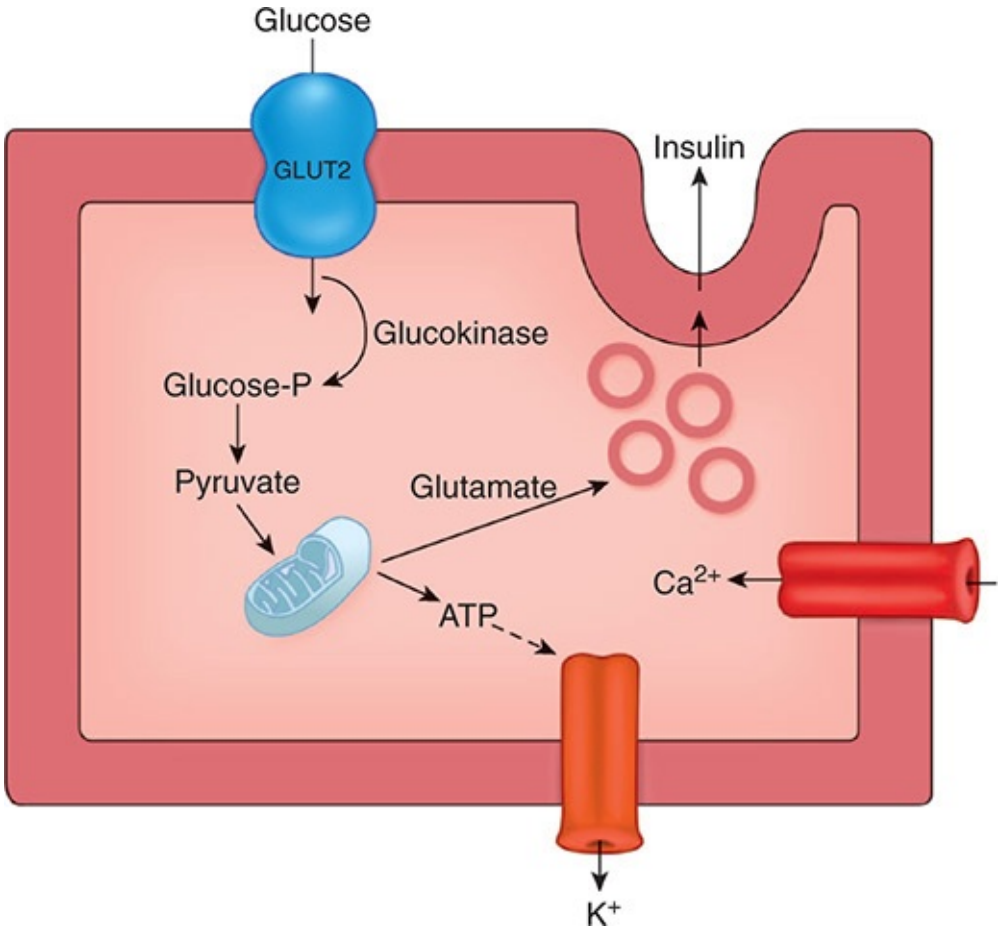


FIGURE 24–11 Insulin secretion. Glucose enters B cells by GLUT-2 transporters. It is phosphorylated and metabolized to pyruvate (Pyr) in the cytoplasm. The Pyr enters the mitochondria and is metabolized via the citric acid cycle. The ATP formed by oxidative phosphorylation inhibits ATP-sensitive K^+ channels, reducing K^+ efflux. This depolarizes the B cell, and Ca^{2+} influx is increased. The Ca^{2+} stimulates release of insulin by exocytosis. Glutamate (Glu) is also formed, and this primes secretory granules, preparing them for exocytosis.

Metabolism of pyruvate via the citric acid cycle also causes an increase in intracellular glutamate. The glutamate appears to act on a second pool of secretory granules, committing them to the releasable form. The action of glutamate may be to decrease the pH in the secretory granules, a necessary step

in their maturation. The release of these granules then produces the prolonged second phase of the insulin response to glucose. Thus, glutamate appears to act as an intracellular second messenger that primes secretory granules for secretion.

The feedback control of plasma glucose on insulin secretion normally operates with great precision so that plasma glucose and insulin levels parallel each other with remarkable consistency.

PROTEIN & FAT DERIVATIVES

Insulin stimulates the incorporation of amino acids into proteins and combats the fat catabolism that produces the β-keto acids. Therefore, it is not surprising that arginine, leucine, and certain other amino acids stimulate insulin secretion, as do β-keto acids such as acetoacetate. Like glucose, these compounds generate ATP when metabolized, and this closes ATP-sensitive K^+ channels in the B cells. In addition, L-arginine is the precursor of NO, and NO stimulates insulin secretion.

ORAL HYPOGLYCEMIC AGENTS

Tolbutamide and other sulfonylurea derivatives such as acetohexamide, tolazamide, glipizide, and glyburide are orally active hypoglycemic agents that lower blood glucose by increasing the secretion of insulin. They only work in patients with some remaining B cells and are ineffective after pancreatectomy or in type 1 diabetes. They bind to the ATP-inhibited K^+ channels in the B cell membranes and inhibit channel activity, depolarizing the B cell membrane and increasing Ca^{2+} influx and hence insulin release, independent of increases in plasma glucose.

Persistent hyperinsulinemic hypoglycemia of infancy is a condition in which plasma insulin is elevated despite the hypoglycemia. The condition is caused by mutations in the genes for various enzymes in B cells that decrease K^+ efflux via the ATP-sensitive K^+ channels. Treatment consists of administration of diazoxide, a drug that increases the activity of the K^+ channels or, in more severe cases, subtotal pancreatectomy.

The biguanide metformin is an oral hypoglycemic agent that acts in the absence of insulin. Metformin acts primarily by reducing gluconeogenesis and therefore decreasing hepatic glucose output. It is sometimes combined with a sulfonylurea in the treatment of type 2 diabetes. Metformin can cause lactic acidosis, but the incidence is usually low.

Troglitazone (Rezulin) and related **thiazolidinediones** are also used in the treatment of diabetes because they increase insulin-mediated peripheral glucose disposal, thus reducing insulin resistance. They bind to and activate peroxisome proliferator-activated receptor γ (PPARγ) in the nucleus of cells. Activation of this receptor, which is a member of the superfamily of hormone-sensitive nuclear transcription factors, has a unique ability to normalize a variety of metabolic functions.

CYCLIC AMP & INSULIN SECRETION

Stimuli that increase cAMP levels in B cells increase insulin secretion, including β-adrenergic agonists, glucagon, and phosphodiesterase inhibitors such as theophylline.

Catecholamines have a dual effect on insulin secretion; they inhibit insulin secretion via α_2-adrenergic receptors and stimulate insulin secretion via β-adrenergic receptors. The net effect of epinephrine and norepinephrine is usually inhibition. However, if catecholamines are infused after administration of α-adrenergic blocking drugs, the inhibition is converted to stimulation.

EFFECT OF AUTONOMIC NERVES

Branches of the right vagus nerve innervate the pancreatic islets, and stimulation of this parasympathetic pathway causes increased insulin secretion via M_3 receptors (see Table 7–2). Atropine blocks the response and acetylcholine stimulates insulin secretion. The effect of acetylcholine, like that of glucose, is due to increased cytoplasmic Ca^{2+}, but acetylcholine activates phospholipase C, with the released IP_3 releasing the Ca^{2+} from the endoplasmic reticulum.

Stimulation of the sympathetic nerves to the pancreas inhibits insulin secretion. The inhibition is produced by released norepinephrine acting on α_2-adrenergic receptors. However, if α-adrenergic receptors are blocked, stimulation of the sympathetic nerves causes increased insulin secretion mediated by β_2-adrenergic receptors. The polypeptide galanin is found in some of the autonomic nerves innervating the islets, and galanin inhibits insulin secretion by activating the K^+ channels that are inhibited by ATP. Thus, although the denervated pancreas responds to glucose, the autonomic innervation of the pancreas is involved in the overall regulation of insulin secretion **(Clinical Box 24–3)**.

INTESTINAL HORMONES

Orally administered glucose exerts a greater insulin-stimulating effect than intravenously administered glucose, and orally administered amino acids also produce a greater insulin response than intravenous amino acids. These observations led to exploration of the possibility that a substance secreted by the gastrointestinal mucosa stimulated insulin secretion. Glucagon, glucagon derivatives, secretin, cholecystokinin (CCK), gastrin, and gastric inhibitory peptide (GIP) all have such an action (see Chapter 25), and CCK potentiates the insulin-stimulating effects of amino acids. However, GIP is the only one of these peptides that produces stimulation when administered in doses that reflect blood GIP levels produced by an oral glucose load.

CLINICAL BOX 24-3

Effects of K^+ Depletion

K^+ depletion decreases insulin secretion, and K^+-depleted patients, for example, patients with primary hyperaldoster-onism (see Chapter 20), develop diabetic glucose tolerance curves. These curves are restored to normal by K^+ repletion.

THERAPEUTIC HIGHLIGHTS

The thiazide diuretics, which cause loss of K^+ as well as Na^+ in the urine (see Chapter 37), decrease glucose tolerance and make diabetes worse. They apparently exert this effect primarily because of their K^+-depleting effects, although some of them also cause pancreatic islet cell damage. Potassium-sparing diuretics, such as amiloride, should be substituted in the diabetic patient who needs such treatment.

Recently, attention has focused on glucagon-like polypeptide 1 (7–36) (GLP-1 [7–36]) as an additional gut factor that stimulates insulin secretion. This polypeptide is a product of preproglucagon. B cells have GLP-1 (7–36) receptors as well as GIP receptors, and GLP-1 (7–36) is a more potent insulinotropic hormone than GIP. GIP and GLP-1 (7–36) both appear to act by increasing Ca^{2+} influx through voltage-gated Ca^{2+} channels.

The possible roles of pancreatic somatostatin and glucagon in the regulation of insulin secretion are discussed below.

LONG-TERM CHANGES IN B CELL RESPONSES

The magnitude of the insulin response to a given stimulus is determined in part by the secretory history of the B cells. Individuals fed a high-carbohydrate diet for several weeks not only have higher fasting plasma insulin levels but also show a greater secretory response to a glucose load than individuals fed an isocaloric low-carbohydrate diet.

Although B cells respond to stimulation with hypertrophy like other endocrine cells, they become exhausted and stop secreting **(B cell exhaustion)** when the stimulation is marked or prolonged. The pancreatic reserve is large and it is difficult to produce B cell exhaustion in normal animals, but if the pancreatic reserve is reduced by partial pancreatectomy, exhaustion of the remaining B cells can be initiated by any procedure that chronically raises the plasma glucose level. For example, diabetes can be produced in animals with limited pancreatic reserves by anterior pituitary extracts, growth hormone, thyroid hormones, or the prolonged continuous infusion of glucose alone. The diabetes precipitated by hormones in animals is at first reversible, but with prolonged treatment it becomes permanent. The transient diabetes is usually named for the agent producing it, for example, “hypophysial diabetes” or “thyroid diabetes.” Permanent diabetes persisting after treatment has been discontinued is indicated by the prefix meta-, for example, **“metahypophysial diabetes”** or **“metathyroid diabetes.”** When insulin is administered along with the diabetogenic hormones, the B cells are protected, probably because the plasma glucose is lowered, and diabetes does not develop.

It is interesting in this regard that genetic factors may be involved in the control of B cell reserve. In mice in which the gene for IRS-1 has been knocked out (see above), a robust compensatory B cell response occurs. However, in IRS-2 knockouts, the compensation is reduced and a more severe diabetic phenotype is produced.

GLUCAGON

CHEMISTRY

Human glucagon, a linear polypeptide with a molecular weight of 3485, is produced by the A cells of the pancreatic islets and the upper gastrointestinal tract. It contains 29 amino acid residues. All mammalian glucagons appear to have the same structure. Human preproglucagon **(Figure 24–12)** is a 179-amino-acid protein that is found in pancreatic A cells, in L cells in the lower gastrointestinal tract, and in the brain. It is the product of a single mRNA, but it is processed differently in different tissues. In A cells, it is processed primarily to glucagon and **major proglucagon fragment (MPGF).** In L cells, it is processed primarily to **glicentin,** a polypeptide that consists of glucagon extended by additional amino acid residues at either end, plus **glucagon-like polypeptides 1 and 2 (GLP-1** and **GLP-2).** Some **oxyntomodulin** is also formed, and in both A and L cells, residual **glicentin-related polypeptide (GRPP)** is left. Glicentin has some glucagon activity. GLP-1 and GLP-2 have no definite biologic activity by themselves. However, GLP-1 is processed further by removal of its amino-terminal amino acid residues and the product, **GLP-1 (7–36),** is a potent stimulator of insulin secretion that also increases glucose utilization (see above). GLP-1 and GLP-2 are also produced in the brain. The function of GLP-1 in this location is uncertain, but GLP-2 appears to be the mediator in a pathway from the nucleus tractus solitarius (NTS) to the dorsomedial nuclei of the hypothalamus, and injection of GLP-2 lowers food intake. Oxyntomodulin inhibits gastric acid secretion, though its physiologic role is unsettled, and GRPP does not have any established physiologic effects.

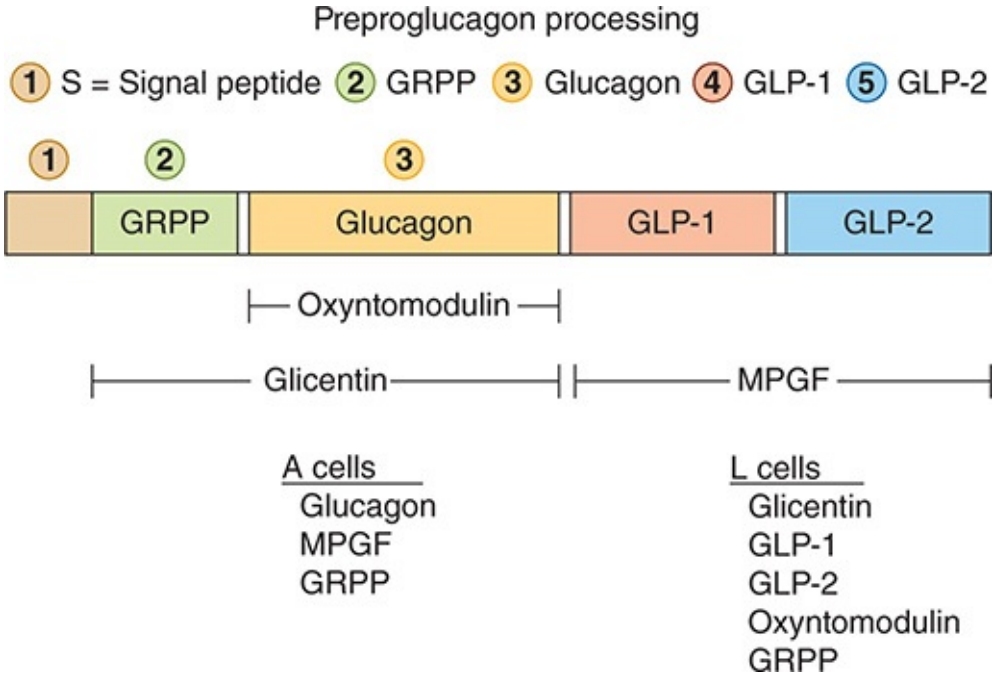


FIGURE 24–12 Posttranslational processing of preproglucagon in A and L cells. S, signal peptide; GRPP, glicentin-related polypeptide; GLP, glucagon-like polypeptide; Oxy, oxyntomodulin; MPGF, major proglucagon fragment. (Modified with permission from Drucker DJ: Glucagon and glucagon-like peptides. Pancreas 1990; July; 5(4):484–488.)

ACTION

Glucagon is glycogenolytic, gluconeogenic, lipolytic, and ketogenic. It acts on G-protein-coupled receptors with a molecular weight of about 190,000. In the liver, it acts via G_s to activate adenylyl cyclase and increase intracellular cAMP. This leads via protein kinase A to activation of phosphorylase and therefore to increased breakdown of glycogen and an increase in plasma glucose. However, glucagon acts on different glucagon receptors located on the same hepatic cells to activate phospholipase C, and the resulting increase in cytoplasmic Ca^{2+} also stimulates glycogenolysis. Protein kinase A also decreases the metabolism of glucose-6-phosphate **(Figure 24–13)** by inhibiting the conversion of phosphoenolpyruvate to pyruvate. It also decreases the concentration of fructose 2,6-diphosphate and this in turn inhibits the conversion of fructose 6-phosphate to fructose 1,6-diphosphate. The resultant buildup of glucose-6-phosphate leads to increased glucose synthesis and release.

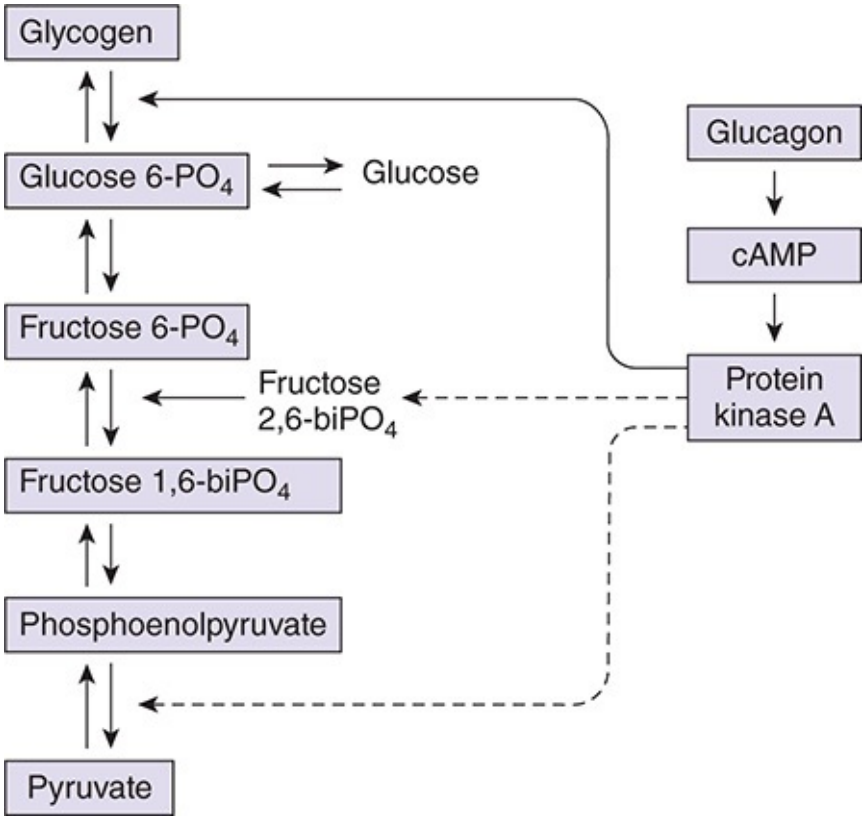


FIGURE 24–13 Mechanisms by which glucagon increases glucose output from the liver. Solid arrows indicate facilitation; dashed arrows indicate inhibition.

Glucagon does not cause glycogenolysis in muscle. It increases gluconeogenesis from available amino acids in the liver and elevates the metabolic rate. It increases ketone body formation by decreasing malonyl-CoA levels in the liver. Its lipolytic activity, which leads in turn to increased ketogenesis, is discussed in Chapter 1. The calorigenic action of glucagon is not due to the hyperglycemia per se but probably to the increased hepatic deamination of amino acids.

Large doses of exogenous glucagon exert a positive inotropic effect on the heart (see Chapter 30) without producing increased myocardial excitability, presumably because they increase myocardial cAMP. Use of this hormone in the treatment of heart disease has been advocated, but there is no evidence for a physiologic role of glucagon in the regulation of cardiac function. Glucagon also stimulates the secretion of growth hormone, insulin, and pancreatic somatostatin.

METABOLISM

Glucagon has a half-life in the circulation of 5–10 min. It is degraded by many tissues but particularly by the liver. Because glucagon is secreted into the portal vein and reaches the liver before it reaches the peripheral circulation, peripheral blood levels are relatively low. The rise in peripheral blood glucagon levels produced by excitatory stimuli is exaggerated in patients with cirrhosis, presumably because of decreased hepatic degradation of the hormone.

REGULATION OF SECRETION

The principal factors known to affect glucagon secretion are summarized in **Table 24–5**. Secretion is increased by hypoglycemia and decreased by a rise in plasma glucose. Pancreatic B cells contain GABA, and evidence suggests that coincident with the increased insulin secretion produced by hyperglycemia, GABA is released and acts on the A cells to inhibit glucagon secretion by activating $GABA_A$ receptors. The $GABA_A$ receptors are Cl^- channels, and the resulting Cl^- influx hyperpolarizes the A cells.

TABLE 24–5 Factors affecting glucagon secretion.

Stimulators	Inhibitors
Amino acids (particularly the glucogenic amino acids: alanine, serine, glycine, cysteine, and threonine)	Glucose
CCK, gastrin	Somatostatin
Cortisol	Secretin
Exercise	FFA
Infections	Ketones
Other stresses	Insulin
β-Adrenergic stimulators	Phenytoin
Theophylline	α-Adrenergic stimulators
Acetylcholine	GABA

Secretion is also increased by stimulation of the sympathetic nerves to the

pancreas, and this sympathetic effect is mediated via β-adrenergic receptors and cAMP. It appears that the A cells are like the B cells in that stimulation of β-adrenergic receptors increases secretion and stimulation of α-adrenergic receptors inhibits secretion. However, the pancreatic response to sympathetic stimulation in the absence of blocking drugs is increased secretion of glucagon, so the effect of β-receptors predominates in the glucagon-secreting cells. The stimulatory effects of various stresses and possibly of exercise and infection are mediated at least in part via the sympathetic nervous system. Vagal stimulation also increases glucagon secretion.

A protein meal and infusion of various amino acids increase glucagon secretion. It seems appropriate that the glucogenic amino acids are particularly potent in this regard, since these are the amino acids that are converted to glucose in the liver under the influence of glucagon. The increase in glucagon secretion following a protein meal is also valuable, since the amino acids stimulate insulin secretion and the secreted glucagon prevents the development of hypoglycemia while the insulin promotes storage of the absorbed carbohydrates and lipids. Glucagon secretion increases during starvation. It reaches a peak on the third day of a fast, at the time of maximal gluconeogenesis. Thereafter, the plasma glucagon level declines as fatty acids and ketones become the major sources of energy.

During exercise, there is an increase in glucose utilization that is balanced by an increase in glucose production caused by an increase in circulating glucagon levels.

The glucagon response to oral administration of amino acids is greater than the response to intravenous infusion of amino acids, suggesting that a glucagon-stimulating factor is secreted from the gastrointestinal mucosa. CCK and gastrin increase glucagon secretion, whereas secretin inhibits it. Because CCK and gastrin secretion are both increased by a protein meal, either hormone could be the gastrointestinal mediator of the glucagon response. The inhibition produced by somatostatin is discussed below.

Glucagon secretion is also inhibited by FFA and ketones. However, this inhibition can be overridden, since plasma glucagon levels are high in diabetic ketoacidosis.

INSULIN–GLUCAGON MOLAR RATIOS

As noted previously, insulin is glycogenic, antigluconeogenetic, antilipolytic, and antiketotic in its actions. It thus favors storage of absorbed nutrients and is a

“hormone of energy storage.” Glucagon, on the other hand, is glycogenolytic, gluconeogenetic, lipolytic, and ketogenic. It mobilizes energy stores and is a “hormone of energy release.” Because of their opposite effects, the blood levels of both hormones must be considered in any given situation. It is convenient to think in terms of the molar ratios of these hormones.

The insulin–glucagon molar ratios fluctuate markedly because the secretion of glucagon and insulin are both modified by the conditions that preceded the application of any given stimulus **(Table 24–6)**. Thus, for example, the insulin–glucagon molar ratio on a balanced diet is approximately 2.3. An infusion of arginine increases the secretion of both hormones and raises the ratio to 3.0. After 3 days of starvation, the ratio falls to 0.4, and an infusion of arginine in this state lowers the ratio to 0.3. Conversely, the ratio is 25 in individuals receiving a constant infusion of glucose and rises to 170 on ingestion of a protein meal during the infusion (Table 24–6). The rise occurs because insulin secretion rises sharply, while the usual glucagon response to a protein meal is abolished. Thus, when energy is needed during starvation, the insulin–glucagon molar ratio is low, favoring glycogen breakdown and gluconeogenesis; conversely, when the need for energy mobilization is low, the ratio is high, favoring the deposition of glycogen, protein, and fat **(Clinical Box 24–4)**.

TABLE 24–6 Insulin–glucagon molar ratios (I/G) in blood in various conditions.

Condition	Hepatic Glucose Storage (S) or Production (P)[a]	I/G
Glucose availability		
Large carbohydrate meal	4+ (S)	70
Intravenous glucose	2+ (S)	25
Small meal	1+ (S)	7
Glucose need		
Overnight fast	1+ (P)	2.3
Low-carbohydrate diet	2+ (P)	1.8
Starvation	4+ (P)	0.4

[a]1+ to 4+ indicate relative magnitude.

Courtesy of RH Unger.

OTHER ISLET CELL HORMONES

In addition to insulin and glucagon, the pancreatic islets secrete somatostatin and pancreatic polypeptide into the bloodstream. In addition, somatostatin may be involved in regulatory processes within the islets that adjust the pattern of hormones secreted in response to various stimuli.

SOMATOSTATIN

Somatostatin and its receptors are discussed in Chapter 7. Somatostatin 14 (SS 14) and its amino terminal-extended form somatostatin 28 (SS 28) are found in the D cells of pancreatic islets. Both forms inhibit the secretion of insulin, glucagon, and pancreatic polypeptide and act locally within the pancreatic islets in a paracrine fashion. SS 28 is more active than SS 14 in inhibiting insulin secretion, and it apparently acts via the SSTR5 receptor (see Chapter 7). Patients with somatostatin-secreting pancreatic tumors **(somatostatinomas)** develop hyperglycemia and other manifestations of diabetes that disappear when the tumor is removed. Dyspepsia also develops as a result of slow gastric emptying and decreased gastric acid secretion, and gallstones, which are precipitated by decreased gallbladder contraction due to inhibition of CCK secretion. The secretion of pancreatic somatostatin is increased by several of the same stimuli that increase insulin secretion, that is, glucose and amino acids, particularly arginine and leucine. It is also increased by CCK. Somatostatin is released from the pancreas and the gastrointestinal tract into the peripheral blood.

CLINICAL BOX 24–4

Macrosomia & GLUT-1 Deficiency

Infants born to diabetic mothers often have high birth weights and large organs (**macrosomia**). This condition is caused by excess circulating insulin in the fetus, which in turn is caused in part by stimulation of the fetal pancreas by high blood glucose and amino acids from the diabetic mother.

Free insulin in maternal blood is destroyed by proteases in the placenta, but antibody-bound insulin is protected, so it reaches the fetus.

Infants with **GLUT-1 deficiency** have defective transport of glucose across the blood-brain barrier. They have low cerebrospinal fluid glucose in the presence of normal plasma glucose, seizures, and developmental delay.

PANCREATIC POLYPEPTIDE

Human pancreatic polypeptide is a linear polypeptide that contains 36 amino acid residues and is produced by F cells in the islets. It is closely related to two other 36-amino acid polypeptides, **polypeptide YY,** a gastrointestinal peptide (see Chapter 25), and **neuropeptide Y,** which is found in the brain and the autonomic nervous system (see Chapter 7). All end in tyrosine and are amidated at their carboxyl terminal. At least in part, pancreatic polypeptide secretion is under cholinergic control; plasma levels fall after administration of atropine. Its secretion is increased by a meal containing protein and by fasting, exercise, and acute hypoglycemia. Secretion is decreased by somatostatin and intravenous glucose. Infusions of leucine, arginine, and alanine do not affect it, so the stimulatory effect of a protein meal may be mediated indirectly. Pancreatic polypeptide slows the absorption of food in humans, and it may smooth out the peaks and valleys of absorption. However, its exact physiologic function is still uncertain.

ORGANIZATION OF THE PANCREATIC ISLETS

The presence in the pancreatic islets of hormones that affect the secretion of other islet hormones suggests that the islets function as secretory units in the regulation of nutrient homeostasis. Somatostatin inhibits the secretion of insulin, glucagon, and pancreatic polypeptide **(Figure 24–14)**; insulin inhibits the secretion of glucagon; and glucagon stimulates the secretion of insulin and somatostatin. As noted above, A and D cells and pancreatic polypeptide-secreting cells are generally located around the periphery of the islets, with the B cells in the center. There are clearly two types of islets, glucagon-rich islets and pancreatic polypeptide-rich islets, but the functional significance of this separation is not known. The islet cell hormones released into the ECF probably diffuse to other islet cells and influence their function (paracrine communication; see Chapter 25). It has been demonstrated that gap junctions are present between A, B, and D cells and that these permit the passage of ions and other small molecules from one cell to another, which could coordinate their secretory functions.

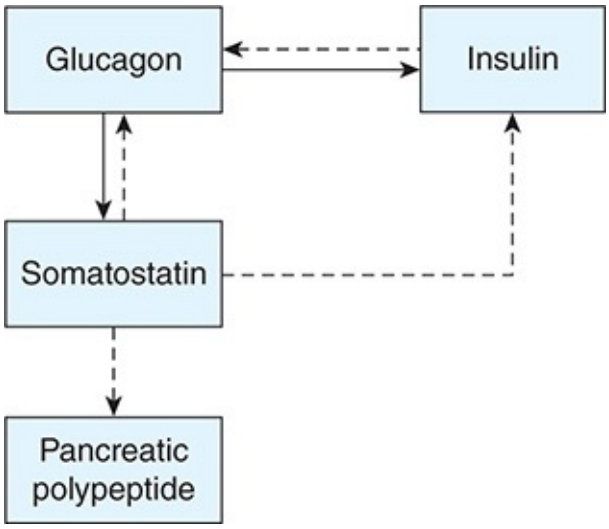


FIGURE 24–14 Effects of islet cell hormones on the secretion of other islet cell hormones. Solid arrows indicate stimulation; dashed arrows indicate inhibition.

EFFECTS OF OTHER HORMONES & EXERCISE ON CARBOHYDRATE METABOLISM

Exercise has direct effects on carbohydrate metabolism. Many hormones in addition to insulin, IGF-I, IGF-II, glucagon, and somatostatin also have important roles in the regulation of carbohydrate metabolism. They include epinephrine, thyroid hormones, glucocorticoids, and growth hormone. The other functions of these hormones are considered elsewhere, but it seems wise to summarize their effects on carbohydrate metabolism in the context of the present chapter.

EXERCISE

The entry of glucose into skeletal muscle is increased during exercise in the absence of insulin by causing an insulin- independent increase in the number of GLUT-4 transporters in muscle cell membranes (see above). This increase in glucose entry persists for several hours after exercise, and regular exercise training can also produce prolonged increases in insulin sensitivity. Exercise can precipitate hypoglycemia in diabetics not only because of the increase in muscle uptake of glucose but also because absorption of injected insulin is more rapid during exercise. Patients with diabetes should take in extra calories or reduce

their insulin dosage when they exercise.

CATECHOLAMINES

The activation of phosphorylase in liver by catecholamines is discussed in Chapter 1. Activation occurs via β-adrenergic receptors, which increase intracellular cAMP, and α-adrenergic receptors, which increase intracellular Ca^{2+}. Hepatic glucose output is increased, producing hyperglycemia. In muscle, the phosphorylase is also activated via cAMP and presumably via Ca^{2+}, but the glucose-6-phosphate formed can be catabolized only to pyruvate because of the absence of glucose-6- phosphatase. For reasons that are not entirely clear, large amounts of pyruvate are converted to lactate, which diffuses from the muscle into the circulation **(Figure 24–15)**. The lactate is oxidized in the liver to pyruvate and converted to glycogen. Therefore, the response to an injection of epinephrine is an initial glycogenolysis followed by a rise in hepatic glycogen content. Lactate oxidation may be responsible for the calorigenic effect of epinephrine (see Chapter 19). Epinephrine and norepinephrine also liberate FFA into the circulation, and epinephrine decreases peripheral utilization of glucose.

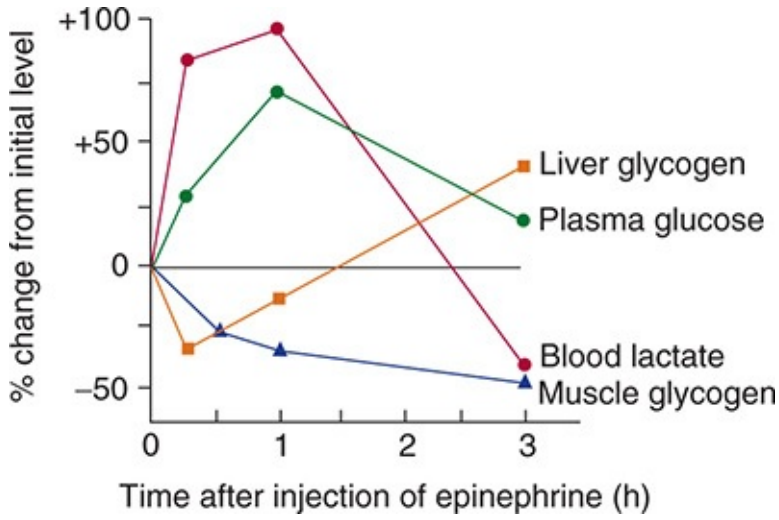


FIGURE 24–15 Effect of epinephrine on tissue glycogen, plasma glucose, and blood lactate levels in fed rats. (Reproduced with permission from Ruch TC, Patton HD [editors]: *Physiology and Biophysics*, 20th ed, St. Louis, MO: Saunders; 1973.)

THYROID HORMONES

Thyroid hormones make experimental diabetes worse; thyrotoxicosis aggravates clinical diabetes; and metathyroid diabetes can be produced in animals with decreased pancreatic reserve. The principal diabetogenic effect of thyroid hormones is to increase absorption of glucose from the intestine, but the hormones also cause (probably by potentiating the effects of catecholamines) some degree of hepatic glycogen depletion. Glycogen-depleted liver cells are easily damaged. When the liver is damaged, the glucose tolerance curve is diabetic because the liver takes up less of the absorbed glucose. Thyroid hormones may also accelerate the degradation of insulin. All these actions have a hyperglycemic effect and, if the pancreatic reserve is low, may lead to B cell exhaustion.

ADRENAL GLUCOCORTICOIDS

Glucocorticoids from the adrenal cortex (see Chapter 19) elevate blood glucose and produce a diabetic type of glucose tolerance curve. In humans, this effect may occur only in individuals with a genetic predisposition to diabetes. Glucose tolerance is reduced in 80% of patients with Cushing syndrome (see Chapter 19), and 20% of these patients have frank diabetes. The glucocorticoids are necessary for glucagon to exert its gluconeogenic action during fasting. They are gluconeogenic themselves, but their role is mainly permissive. In adrenal insufficiency, the blood glucose is normal as long as food intake is maintained, but fasting precipitates hypoglycemia and collapse. The plasma-glucose-lowering effect of insulin is greatly enhanced in patients with adrenal insufficiency. In animals with experimental diabetes, adrenalectomy markedly ameliorates the diabetes. The major diabetogenic effects are an increase in protein catabolism with increased gluconeogenesis in the liver; increased hepatic glycogenesis and ketogenesis; and a decrease in peripheral glucose utilization relative to the blood insulin level that may be due to inhibition of glucose phosphorylation.

GROWTH HORMONE

Human growth hormone makes clinical diabetes worse, and 25% of patients with growth hormone–secreting tumors of the anterior pituitary have diabetes. Hypophysectomy ameliorates diabetes and decreases insulin resistance even more than adrenalectomy, whereas growth hormone treatment increases insulin resistance.

The effects of growth hormone are partly direct and partly mediated via IGF-I (see Chapter 18). Growth hormone mobilizes FFA from adipose tissue, thus favoring ketogenesis. It decreases glucose uptake into some tissues ("anti-insulin action"), increases hepatic glucose output, and may decrease tissue binding of insulin. Indeed, it has been suggested that the ketosis and decreased glucose tolerance produced by starvation are due to hypersecretion of growth hormone. Growth hormone does not stimulate insulin secretion directly, but the hyperglycemia it produces secondarily stimulates the pancreas and may eventually exhaust the B cells.

HYPOGLYCEMIA & DIABETES MELLITUS IN HUMANS

HYPOGLYCEMIA

"Insulin reactions" are common in type 1 diabetics and occasional hypoglycemic episodes are the price of good diabetic control in most diabetics. Glucose uptake by skeletal muscle and absorption of injected insulin both increase during exercise (see above).

Symptomatic hypoglycemia also occurs in nondiabetics, and a review of some of the more important causes serves to emphasize the variables affecting plasma glucose homeostasis. Chronic mild hypoglycemia can cause incoordination and slurred speech, and the condition can be mistaken for drunkenness. Mental aberrations and convulsions in the absence of frank coma also occur. When the level of insulin secretion is chronically elevated by an **insulinoma,** a rare, insulin-secreting tumor of the pancreas, symptoms are most common in the morning. This is because a night of fasting has depleted hepatic glycogen reserves. However, symptoms can develop at any time, and in such patients, the diagnosis may be missed. Some cases of insulinoma have been erroneously diagnosed as epilepsy or psychosis. Hypoglycemia also occurs in some patients with large malignant tumors that do not involve the pancreatic islets, and the hypoglycemia in these cases is apparently due to excess secretion of IGF-II.

As noted above, the autonomic discharge caused by lowered blood glucose that produces shakiness, sweating, anxiety, and hunger normally occurs at plasma glucose levels that are higher than the glucose levels that cause cognitive dysfunction, thereby serving as a warning to ingest sugar. However, in some individuals, these warning symptoms fail to occur before the cognitive

symptoms, due to cerebral dysfunction (desensitization), and this **hypoglycemia unawareness** is potentially dangerous. The condition is prone to develop in patients with insulinomas and in diabetics receiving intensive insulin therapy, so it appears that repeated bouts of hypoglycemia cause the eventual development of hypoglycemia unawareness. If blood sugar rises again for some time, the warning symptoms again appear at a higher plasma glucose level than cognitive abnormalities and coma. The reason why prolonged hypoglycemia causes loss of the warning symptoms is unsettled.

In liver disease, the glucose tolerance curve is diabetic but the fasting plasma glucose level is low **(Figure 24–16)**. In **functional hypoglycemia,** the plasma glucose rise is normal after a test dose of glucose, but the subsequent fall overshoots to hypoglycemic levels, producing symptoms 3–4 h after meals. This pattern is sometimes seen in individuals in whom diabetes develops later. Patients with this syndrome should be distinguished from the more numerous patients with similar symptoms due to psychological or other problems who do not have hypoglycemia when blood is drawn during the symptomatic episode. It has been postulated that the overshoot of the plasma glucose is due to insulin secretion stimulated by impulses in the right vagus, but cholinergic blocking agents do not routinely correct the abnormality. In some thyrotoxic patients and in patients who have had gastrectomies or other operations that speed the passage of food into the intestine, glucose absorption is abnormally rapid. The plasma glucose rises to a high, early peak, but it then falls rapidly to hypoglycemic levels because the wave of hyperglycemia evokes a greater than normal rise in insulin secretion. Symptoms characteristically occur about 2 h after meals.

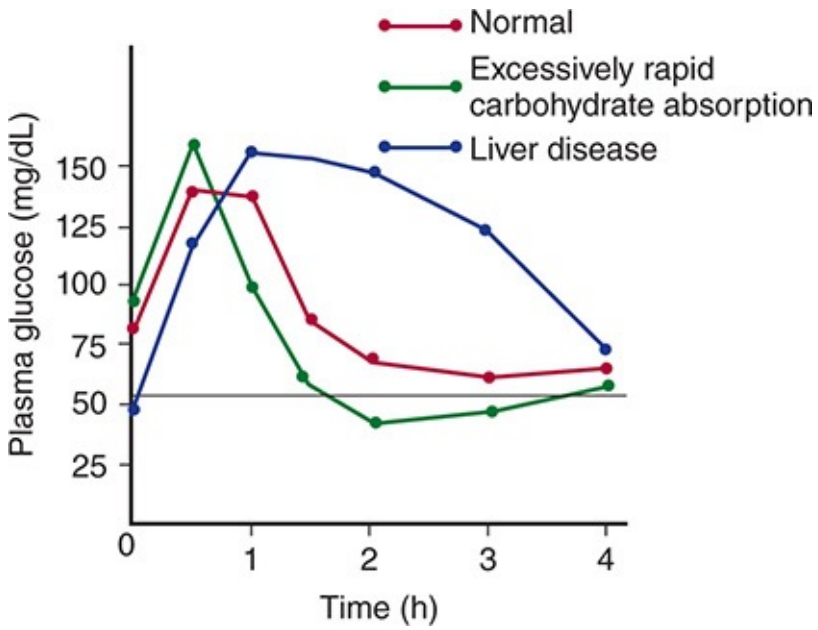


FIGURE 24–16 Typical glucose tolerance curves after an oral glucose load in liver disease and in conditions causing excessively rapid absorption of glucose from the intestine. The horizontal line is the approximate plasma glucose level at which hypoglycemic symptoms may appear.

DIABETES MELLITUS

The incidence of diabetes mellitus in the human population has reached epidemic proportions worldwide and it is increasing at a rapid rate. In 2010 an estimated 285 million people worldwide had diabetes, according to the International Diabetes Federation. The federation predicts as many as 438 million will have diabetes by 2030. Ninety percent of the present cases are type 2 diabetes, and most of the increase will be in type 2, paralleling the increase in the incidence of obesity.

Diabetes is sometimes complicated by acidosis and coma, and in long-standing diabetes, additional complications occur. These include microvascular, macrovascular, and neuropathic disease. The microvascular abnormalities are proliferative scarring of the retina **(diabetic retinopathy)** leading to blindness and renal disease **(diabetic nephropathy)** leading to chronic kidney disease. The macrovascular abnormalities are due to accelerated atherosclerosis, which is secondary to increased plasma LDL. The result is an increased incidence of stroke and myocardial infarction. The neuropathic abnormalities **(diabetic neuropathy)** involve the autonomic nervous system and peripheral nerves. The

neuropathy plus the atherosclerotic circulatory insufficiency in the extremities and reduced resistance to infection can lead to chronic ulceration and gangrene, particularly in the feet.

The ultimate cause of the microvascular and neuropathic complications is chronic hyperglycemia, and tight control of the diabetes reduces their incidence. Intracellular hyperglycemia activates the enzyme aldose reductase. This increases the formation of sorbitol in cells, which in turn reduces cellular Na, K ATPase. In addition, intracellular glucose can be converted to so-called Amadori products, and these in turn can form **advanced glycosylation end products (AGEs),** which cross-link matrix proteins. This damages blood vessels. The AGEs also interfere with leukocyte responses to infection.

TYPES OF DIABETES

The cause of clinical diabetes is always a deficiency of the effects of insulin at the tissue level, but the deficiency may be relative. One of the common forms, **type 1,** or **insulin-dependent diabetes mellitus (IDDM),** is due to insulin deficiency caused by autoimmune destruction of the B cells in the pancreatic islets; the A, D, and F cells remain intact. The second common form, **type 2,** or **noninsulin-dependent diabetes mellitus (NIDDM),** is characterized by insulin resistance.

In addition, some cases of diabetes are due to other diseases or conditions such as chronic pancreatitis, total pancreatectomy, Cushing syndrome (see Chapter 19), and acromegaly (see Chapter 18). These make up 5% of the total cases and are sometimes classified as **secondary diabetes.**

Type 1 diabetes usually develops before the age of 40 and hence is called **juvenile diabetes.** Patients with this disease are not obese and they have a high incidence of ketosis and acidosis. Various anti-B cell antibodies are present in plasma, but the current thinking is that type 1 diabetes is primarily a T lymphocyte–mediated disease. Definite genetic susceptibility is present as well; if the disease develops in one identical twin, the chances are 1 in 3 that it will also develop in the other twin. In other words, the **concordance rate** is about 33%. The main genetic abnormality is in the major histocompatibility complex on chromosome 6, making individuals with certain types of histocompatibility antigens (see Chapter 3) much more prone to the disease. Other genes are also involved.

Immunosuppression with drugs such as cyclosporine ameliorate type 1 diabetes if given early in the disease before all islet B cells are lost. Attempts

have been made to treat type 1 diabetes by transplanting pancreatic tissue or isolated islet cells, but results to date have been poor, largely because B cells are easily damaged and it is difficult to transplant enough of them to normalize glucose responses.

As mentioned above, type 2 is the most common type of diabetes and is usually associated with obesity. It usually develops after age 40 and is not associated with total loss of the ability to secrete insulin. It has an insidious onset, is rarely associated with ketosis, and is usually associated with normal B cell morphology and insulin content if the B cells have not become exhausted. The genetic component in type 2 diabetes is actually stronger than the genetic component in type 1 diabetes; in identical twins, the concordance rate is higher, ranging in some studies to nearly 100%.

In some patients, type 2 diabetes is due to defects in identified genes. Over 60 of these defects have been described. They include defects in glucokinase (about 1% of the cases), the insulin molecule itself (about 0.5% of the cases), the insulin receptor (about 1% of the cases), GLUT-4 (about 1% of the cases), or IRS-1 (about 15% of the cases). In maturity-onset diabetes occurring in young individuals (MODY), which accounts for about 1% of the cases of type 2 diabetes, loss-of-function mutations have been described in six different genes. Five of the genes code for transcription factors affecting the production of enzymes involved in glucose metabolism. The sixth is the gene for glucokinase (Figure 24–11), the enzyme that controls the rate of glucose phosphorylation and hence its metabolism in the B cells. However, the vast majority of cases of type 2 diabetes are almost certainly polygenic in origin, and the actual genes involved are still unknown.

OBESITY, THE METABOLIC SYNDROME, & TYPE 2 DIABETES

Obesity is increasing in incidence and relates to the regulation of food intake and energy balance and overall nutrition. It deserves additional consideration in this chapter because of its special relation to disordered carbohydrate metabolism and diabetes. As body weight increases, insulin resistance increases, that is, there is a decreased ability of insulin to move glucose into fat and muscle and to shut off glucose release from the liver. Weight reduction decreases insulin resistance. Associated with obesity there is hyperinsulinemia, dyslipidemia (characterized by high circulating triglycerides and low high-density lipoprotein [HDL]), and

accelerated development of atherosclerosis. This combination of findings is commonly called the **metabolic syndrome,** or **syndrome X.** Some of the patients with the syndrome are prediabetic, whereas others have frank type 2 diabetes. It has not been proved but it is logical to assume that the hyperinsulinemia is a compensatory response to the increased insulin resistance and that frank diabetes develops in individuals with reduced B cell reserves.

These observations and other data strongly suggest that fat produces a chemical signal or signals that act on muscles and the liver to increase insulin resistance. Evidence for this includes the recent observation that when GLUTs are selectively knocked out in adipose tissue, there is an associated decrease in glucose transport in muscle in vivo, but when the muscles of those animals are tested in vitro their transport is normal.

One possible signal is the circulating level of FFAs, which is elevated in many insulin-resistant states. Other possibilities are peptides and proteins secreted by fat cells. It is now clear that white fat depots are not inert lumps but are actually endocrine tissues that secrete not only leptin but also other hormones that affect fat metabolism. These adipose-derived hormones are commonly termed **adipokines** as they are *cytokines* secreted by *adipose tissue*. Known adipokines are leptin, adiponectin, and resistin.

Some adipokines decrease, rather than increase, insulin resistance. Leptin and adiponectin, for example, decrease insulin resistance, whereas resistin increases insulin resistance. Further complicating the situation, marked insulin resistance is present in the rare metabolic disease **congenital lipodystrophy,** in which fat depots fail to develop. This resistance is reduced by leptin and adiponectin. Finally, a variety of knockouts of intracellular second messengers have been reported to increase insulin resistance. It is unclear how, or indeed if, these findings fit together to provide an explanation of the relation of obesity to insulin tolerance, but the topic is obviously an important one and it is under intensive investigation.

CHAPTER SUMMARY

- Four polypeptides with hormonal activity are secreted by the pancreas: insulin, glucagon, somatostatin, and pancreatic polypeptide.
- Insulin increases the entry of glucose into cells. In skeletal muscle cell it increases the number of GLUT-4 transporters in the cell membranes. In liver it induces glucokinase, which increases the phosphorylation of glucose, facilitating the entry of glucose into the cell.

- Insulin causes K^+ to enter cells, with a resultant lowering of the extracellular K^+ concentration. Insulin increases the activity of Na, K ATPase in cell membranes, so that more K^+ is pumped into cells. Hypokalemia often develops when patients with diabetic acidosis are treated with insulin.
- Insulin receptors are found on many different cells in the body and have two subunits, α and β. Binding of insulin to its receptor triggers a signaling pathway that involves autophosphorylation of the β subunits on tyrosine residues. This triggers phosphorylation of some cytoplasmic proteins and dephosphorylation of others, mostly on serine and threonine residues.
- The constellation of abnormalities caused by insulin deficiency is called diabetes mellitus. Type 1 diabetes is due to insulin deficiency caused by autoimmune destruction of the B cells in the pancreatic islets. Type 2 diabetes is characterized by the dysregulation of insulin release from the B cells, along with insulin resistance in peripheral tissues such as skeletal muscle, brain, and liver.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. Which of the following are *incorrectly* paired?

   A. B cells: insulin

   B. D cells: somatostatin

   C. A cells: glucagons

   D. Pancreatic exocrine cells: chymotrypsinogen

   E. F cells: gastrin

2. Which of the following are *incorrectly* paired?

   A. Epinephrine: increased glycogenolysis in skeletal muscle

   B. Insulin: increased protein synthesis

   C. Glucagon: increased gluconeogenesis

   D. Progesterone: increased plasma glucose level

   E. Growth hormone: increased plasma glucose level

3. Which of the following would be *least* likely to be seen 14 days after a rat is injected with a drug that kills all of its pancreatic B cells?

   A. A rise in the plasma H^+ concentration

B. A rise in the plasma glucagon concentration

C. A fall in the plasma HCO_3^- concentration

D. A fall in the plasma amino acid concentration

E. A rise in plasma osmolality

4. When the plasma glucose concentration falls to low levels, a number of different hormones help combat the hypoglycemia. After intravenous administration of a large dose of insulin, the return of a low blood sugar level to normal is delayed in

A. adrenal medullary insufficiency.

B. glucagon deficiency.

C. combined adrenal medullary insufficiency and glucagon deficiency.

D. thyrotoxicosis.

E. acromegaly.

5. Insulin increases the entry of glucose into

A. all tissues.

B. renal tubular cells.

C. the mucosa of the small intestine.

D. most neurons in the cerebral cortex.

E. skeletal muscle.

6. Glucagon increases glycogenolysis in liver cells but ACTH does not because

A. cortisol increases the plasma glucose level.

B. liver cells have an adenylyl cyclase different from that in adrenocortical cells.

C. ACTH cannot enter the nucleus of liver cells.

D. the membranes of liver cells contain receptors different from those in adrenocortical cells.

E. liver cells contain a protein that inhibits the action of ACTH.

7. A meal rich in proteins containing the amino acids that stimulate insulin secretion but low in carbohydrates does not cause hypoglycemia because

A. the meal causes a compensatory increase in T_4 secretion.

B. cortisol in the circulation prevents glucose from entering muscle.

C. glucagon secretion is also stimulated by the meal.

D. the amino acids in the meal are promptly converted to glucose.

E. insulin does not bind to insulin receptors if the plasma concentration of amino acids is elevated.

SECTION IV

Gastrointestinal Physiology

For unicellular organisms that exist in a sea of nutrients, it is possible to satisfy nutritional requirements simply with the activity of membrane transport proteins that permit the uptake of specific molecules into the cytosol. However, for multicellular organisms, including humans, the challenges of delivering nutrients to appropriate sites in the body are significantly greater, particularly if the organisms are terrestrial. Further, most of the food we eat is in the form of macromolecules, and even when these are digested to their component monomers, most of the end products are water-soluble and do not readily cross cell membranes (a notable exception are the constituents of dietary lipids). Thus, the gastrointestinal system has evolved to permit nutrient acquisition and assimilation into the body, while prohibiting the uptake of undesirable substances (toxins and microbial products, as well as microbes themselves). The latter situation is complicated by the fact that the intestine maintains a lifelong relationship with a rich microbial ecosystem residing in its lumen, a relationship that is largely mutually beneficial if the microbes are excluded from the systemic compartment.

The intestine is a continuous tube that extends from mouth to anus and is formally contiguous with the external environment. A single cell layer of columnar epithelial cells comprises the semipermeable barrier across which controlled uptake of nutrients takes place. Various glandular structures empty into the intestinal lumen at points along its length, providing for digestion of food components, signaling to distal segments, and regulation of the microbiota. There are also important motility functions that move the intestinal contents and resulting waste products along the length of the gut, and a rich innervation that regulates motility, secretion, and nutrient uptake, in many cases in a manner that is independent of the central nervous system (CNS). There are also a large number of endocrine cells that release hormones that work together with

neurotransmitters to coordinate overall regulation of the gastrointestinal system. In general, there is a considerable redundancy of control systems as well as excess capacity for nutrient digestion and uptake. This served humans well in ancient times when food sources were scarce but may now contribute to the modern epidemic of obesity.

The liver, while playing important roles in whole body metabolism, is usually considered a part of the gastrointestinal system for two main reasons. First, it provides for excretion from the body of lipid-soluble waste products that cannot enter the urine. These are secreted into the bile and thence into the intestine to be excreted with the feces. Second, the blood flow draining the intestine is arranged such that substances that are absorbed pass first through the liver, allowing for the removal and metabolism of any toxins that have inadvertently been taken up, as well as clearance of particulates, such as small numbers of enteric bacteria.

In this section, the function of the gastrointestinal system and liver will be considered, and the ways in which the various segments communicate to provide an integrated response to a mixed meal (proteins, carbohydrates, and lipids). The relevance of gastrointestinal physiology for the development of digestive diseases will also be considered. While many are rarely life-threatening (with some notable exceptions, such as specific cancers) digestive diseases represent a substantial burden in terms of morbidity and lost productivity. A 2009 report of the US National Institutes of Diabetes, Digestive and Kidney Diseases found that on an annual basis, for every 100 US residents, there were 35 ambulatory care visits and nearly 5 overnight hospital stays that involved a gastrointestinal diagnosis. Digestive diseases also appear to be increasing in this population (although mortality, principally from cancers, is thankfully in decline). On the other hand, digestive diseases, and in particular infectious diarrhea, remain important causes of mortality in developing countries where clean sources of food and water cannot be assured. In any event, the burden of digestive diseases provides an important impetus for gaining a full understanding of gastrointestinal physiology, since it is a failure of such physiology that most often leads to disease. Conversely, an understanding of specific digestive conditions can often illuminate physiologic principles, as will be stressed in this section.

CHAPTER 25

Overview of Gastrointestinal Function & Regulation

OBJECTIVES

After studying this chapter, you should be able to:

- Understand the functional significance of the gastrointestinal system, and in particular, its roles in nutrient assimilation, excretion, and immunity.
- Describe the structure of the gastrointestinal tract, the glands that drain into it, and its subdivision into functional segments.
- List the major gastrointestinal secretions, their components, and the stimuli that regulate their production.
- Identify the nature and sources of the major hormones, other peptides, and key neurotransmitters of the gastrointestinal system, and their targets and mechanisms of action.
- Describe the special features of the enteric nervous system, the mucosal immune system and the splanchnic circulation.
- Describe water balance in the gastrointestinal tract and explain how the level of luminal fluidity is adjusted to allow for digestion and absorption.

INTRODUCTION

The primary function of the gastrointestinal tract is to serve as a portal whereby nutrients and water can be absorbed into the body. In fulfilling this function, the meal is mixed with a variety of secretions that arise from both the gastrointestinal tract itself and organs that drain into it, such as the pancreas, gallbladder, and salivary glands. Likewise, the intestine displays a variety of motility patterns that serve to mix the meal with digestive secretions and move it along the length of the gastrointestinal tract. Ultimately, residues of the meal that cannot be absorbed, along with cellular debris, are expelled from the body. All of these functions are tightly regulated in concert with the ingestion of meals. Thus, the gastrointestinal system has evolved a large number of regulatory mechanisms that act both locally and over long distances to coordinate the function of the gut and the organs that drain into it. In keeping with its functional continuity with the external environment and its lifelong interaction with a complex microbiota, moreover, the gastrointestinal tract also harbors well-developed innate and adaptive immune systems. Nevertheless, it remains an important portal for infection.

STRUCTURAL CONSIDERATIONS

The parts of the gastrointestinal tract that are encountered by the meal or its residues include, in order, the mouth, esophagus, stomach, duodenum, jejunum, ileum, cecum, colon, rectum, and anus **(Figure 25–1)**. Throughout the length of the intestine, glandular structures deliver secretions into the lumen, particularly in the stomach and mouth. Also important in the process of digestion are secretions from the pancreas and the biliary system of the liver. The intestine itself also has a very substantial surface area, which is important for its absorptive function. The intestinal tract is functionally divided into segments by means of muscle rings known as **sphincters,** which restrict the flow of intestinal contents to optimize digestion and absorption. These sphincters include the upper and lower esophageal sphincters, the pylorus that retards emptying of the stomach, the ileocecal valve that retains colonic contents (including large numbers of bacteria) in the large intestine, and the inner and outer anal sphincters. After toilet training, the latter permits delaying the elimination of wastes until a time when it is socially convenient.

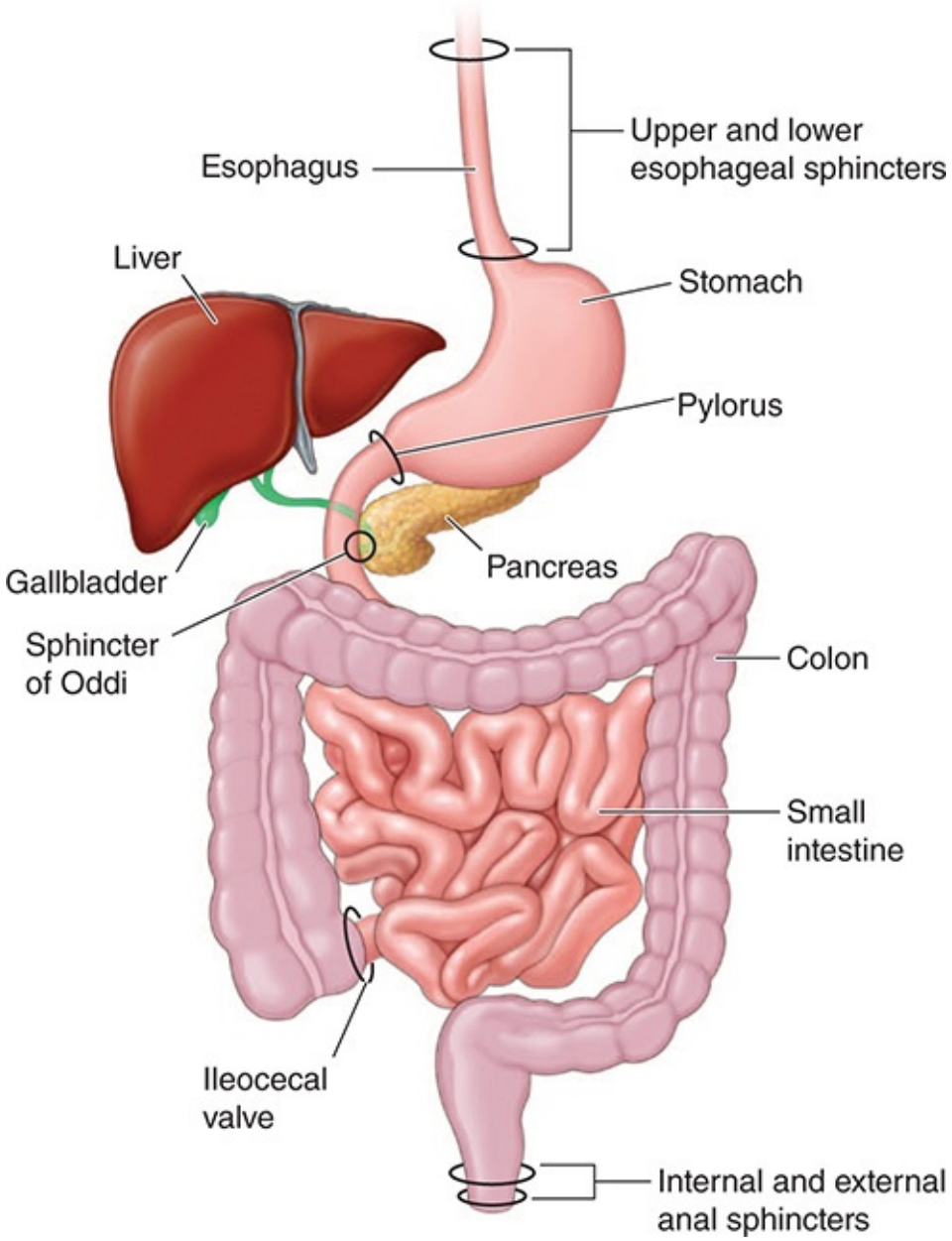


FIGURE 25–1 Overall anatomy of the gastrointestinal system The figure also shows division of the gastrointestinal tract into functional segments by sphincters and valves.

The intestine is composed of functional layers **(Figure 25–2)**. Immediately adjacent to nutrients in the lumen is a single layer of columnar epithelial cells. This represents the barrier that nutrients must traverse to enter the body. Below the epithelium is a layer of loose connective tissue known as the lamina propria, which contains many immune and inflammatory cells even in health and in turn is surrounded by concentric layers of smooth muscle, oriented circumferentially

and then longitudinally to the axis of the gut (the circular and longitudinal muscle layers, respectively). Sandwiched between the two muscle layers, and within the submucosa, are the myenteric and submucosal nerve plexuses of the enteric nervous system, respectively. Sensory nerves project toward the epithelium, and secretomotor nerves innervate both the epithelium and the muscle layers. The intestine is also amply supplied with blood vessels and lymphatics.

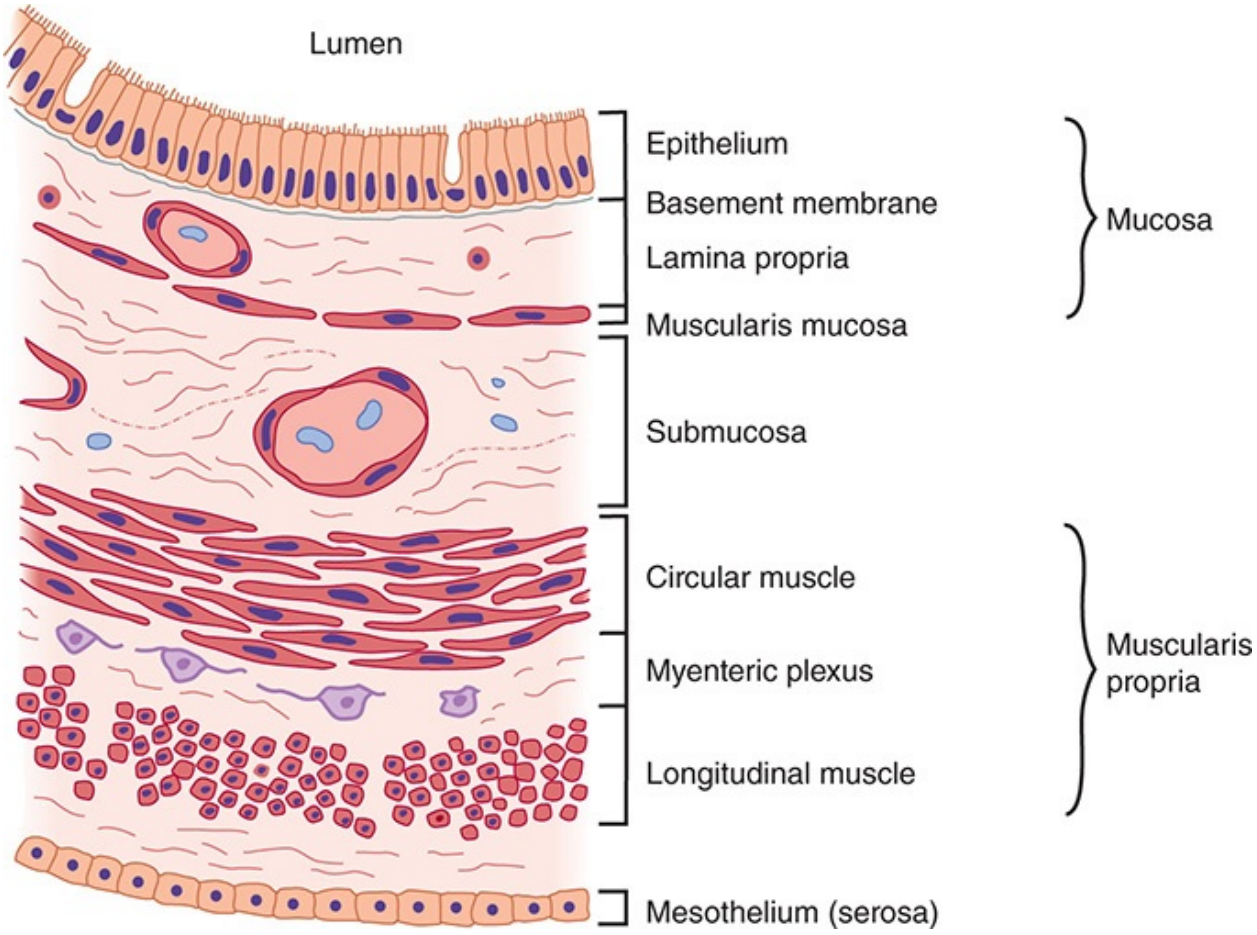


FIGURE 25–2 Organization of the wall of the intestine into functional layers. (Adapted with permission from Yamada T: *Textbook of Gastroenterology*, 4th ed. New York, NY: Lippincott Williams & Wilkins; 2003.)

The epithelium of the intestine is also further specialized in a way that maximizes the surface area available for nutrient absorption. Throughout the small intestine, it is folded up into fingerlike projections called villi **(Figure 25–3)**. Between the villi are infoldings known as crypts. Stem cells that give rise to both crypt and villus epithelial cells reside toward the base of the crypts and are

responsible for completely renewing the epithelium every few days or so. Indeed, the gastrointestinal epithelium is one of the most rapidly dividing tissues in the body. Daughter cells undergo several rounds of cell division in the crypts then migrate out onto the villi, where they are eventually shed and lost in the stool. The villus epithelial cells are also notable for the extensive microvilli that characterize their apical membranes. These microvilli are endowed with a dense glycocalyx (the brush border) that probably protects the cells to some extent from the effects of digestive enzymes. Some digestive enzymes are also actually part of the brush border, being membrane-bound proteins. These so-called “brush border hydrolases” perform the final steps of digestion for specific nutrients.

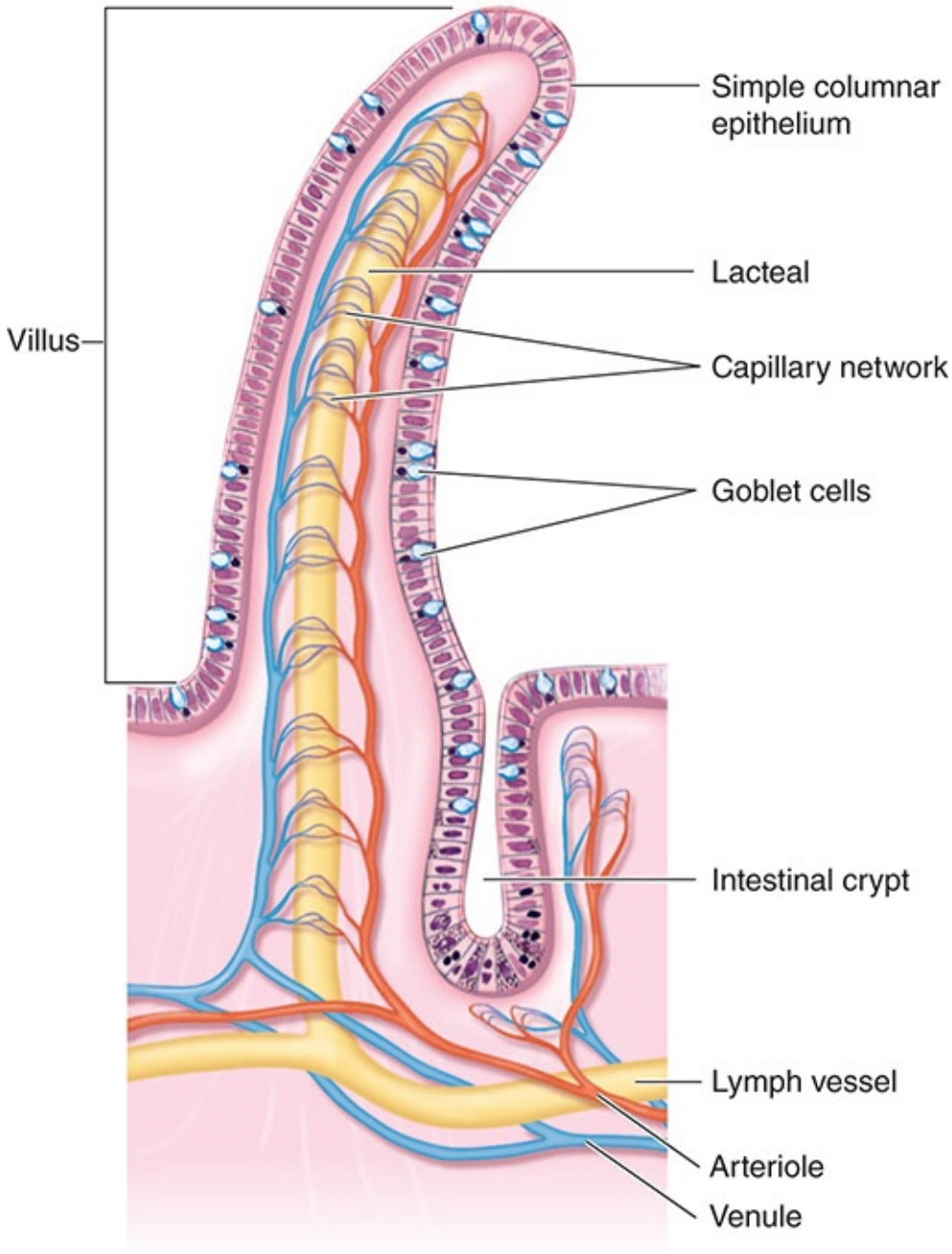


FIGURE 25–3 The structure of intestinal villi and crypts. The epithelial layer also contains scattered endocrine cells and intraepithelial lymphocytes. The crypt base contains Paneth cells, which secrete antimicrobial peptides, as well as the stem cells that provide for continual turnover of the crypt and villus epithelium. The epithelium turns over every 3–5 days in healthy adult humans. (Reproduced with permission from Fox SI: *Human Physiology*, 10th ed. New York, NY: McGraw-Hill; 2008.)

GASTROINTESTINAL SECRETIONS

SALIVARY SECRETION

The ability of the gastrointestinal tract to digest nutrients rests on the sequential exposure of the meal to a series of specific secretions. The first secretion encountered when food is ingested is saliva. Saliva is produced by three pairs of salivary glands (the **parotid, submandibular,** and **sublingual glands**) that drain into the oral cavity. It has a number of organic constituents that serve to initiate digestion (particularly of starch, mediated by amylase) and which also protect the oral cavity from bacteria (such as immunoglobulin A and lysozyme). Saliva also serves to lubricate the food bolus (aided by mucins). Saliva is hypotonic compared with plasma and alkaline; the latter feature is important to neutralize any gastric secretions that reflux into the esophagus.

The salivary glands consist of blind end pieces (acini) that produce the primary secretion containing the organic constituents dissolved in a fluid that is essentially identical in its composition to plasma. The salivary glands are extremely active when maximally stimulated, secreting their own weight in saliva every minute. To accomplish this, they are richly endowed with surrounding blood vessels that dilate when salivary secretion is initiated. The composition of the saliva is then modified as it flows from the acini out into ducts that eventually coalesce and deliver the saliva into the mouth. Na^+ and Cl^- are extracted and K^+ and bicarbonate are added. Because the ducts are relatively impermeable to water, the loss of NaCl renders the saliva hypotonic, particularly at low secretion rates. As the rate of secretion increases, there is less time for NaCl to be extracted and the tonicity of the saliva rises, but it always stays somewhat hypotonic with respect to plasma. Overall, the three pairs of salivary glands that drain into the mouth supply 1000–1500 mL of saliva per day.

Salivary secretion is almost entirely controlled by neural influences, with the parasympathetic branch of the autonomic nervous system playing the most prominent role **(Figure 25–4)**. Sympathetic input slightly modifies the composition of saliva (by increasing proteinaceous content), but has little influence on volume. Secretion is triggered by reflexes that are stimulated by the physical act of chewing, but is actually initiated even before the meal is taken into the mouth as a result of central triggers that are prompted by thinking about, seeing, or smelling food. Indeed, salivary secretion can readily be conditioned, as in the classic experiments of Pavlov where dogs were conditioned to salivate in response to a ringing bell by associating this stimulus with a meal. Salivary

secretion is also prompted by nausea but inhibited by fear or during sleep.

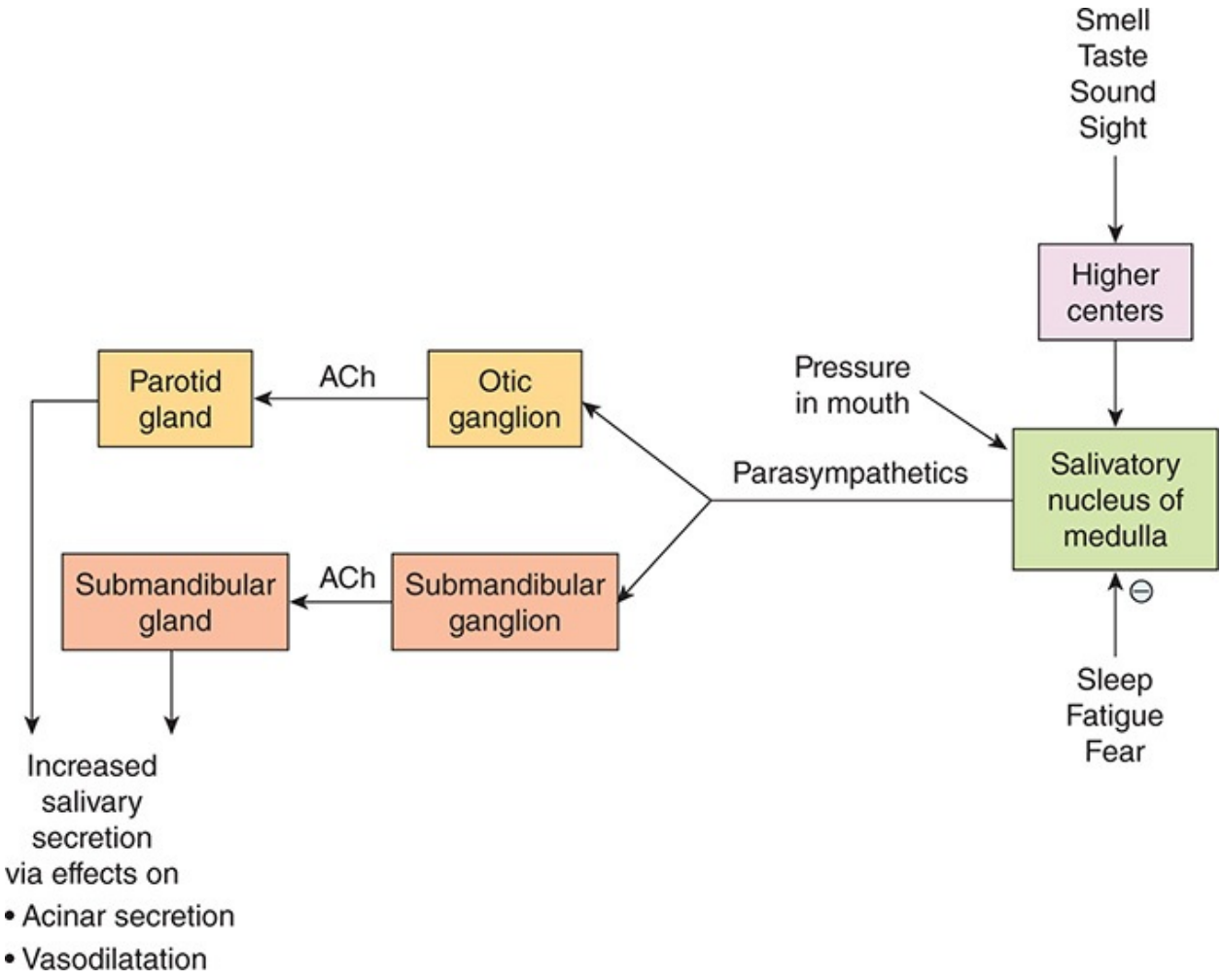


FIGURE 25–4 Regulation of salivary secretion by the parasympathetic nervous system. ACh, acetylcholine. Saliva is also produced by the sublingual glands (not depicted), but these are the minor contributors to both resting and stimulated salivary flows. (Adapted with permission from Barrett KE: *Gastrointestinal Physiology*. New York, NY: McGraw-Hill; 2006.)

Saliva performs a number of important functions: it facilitates swallowing, keeps the mouth moist, serves as a solvent for the molecules that stimulate the taste buds, aids speech by facilitating movements of the lips and tongue, and keeps the mouth and teeth clean. The saliva also has some antibacterial action, and patients with deficient salivation **(xerostomia)** have a higher than normal incidence of dental caries. The buffers in saliva help maintain the oral pH at about 7.0.

GASTRIC SECRETION

Food is stored in the stomach; mixed with acid, mucus, and pepsin; and released at a controlled, steady rate into the duodenum.

ANATOMIC CONSIDERATIONS

The gross anatomy of the stomach is shown in **Figure 25–5**. The gastric mucosa contains many deep glands. In the cardia and the pyloric region, the glands secrete mucus. In the body of the stomach, including the fundus, the glands also contain **parietal cells** and **chief cells** **(Figure 25–6)**. These secretions mix with mucus secreted by the cells in the necks of the glands. Several of the glands open onto a common chamber **(gastric pit)** that opens in turn onto the surface of the mucosa. Mucus is also secreted along with HCO_3^- by mucus cells on the surface of the epithelium between glands.

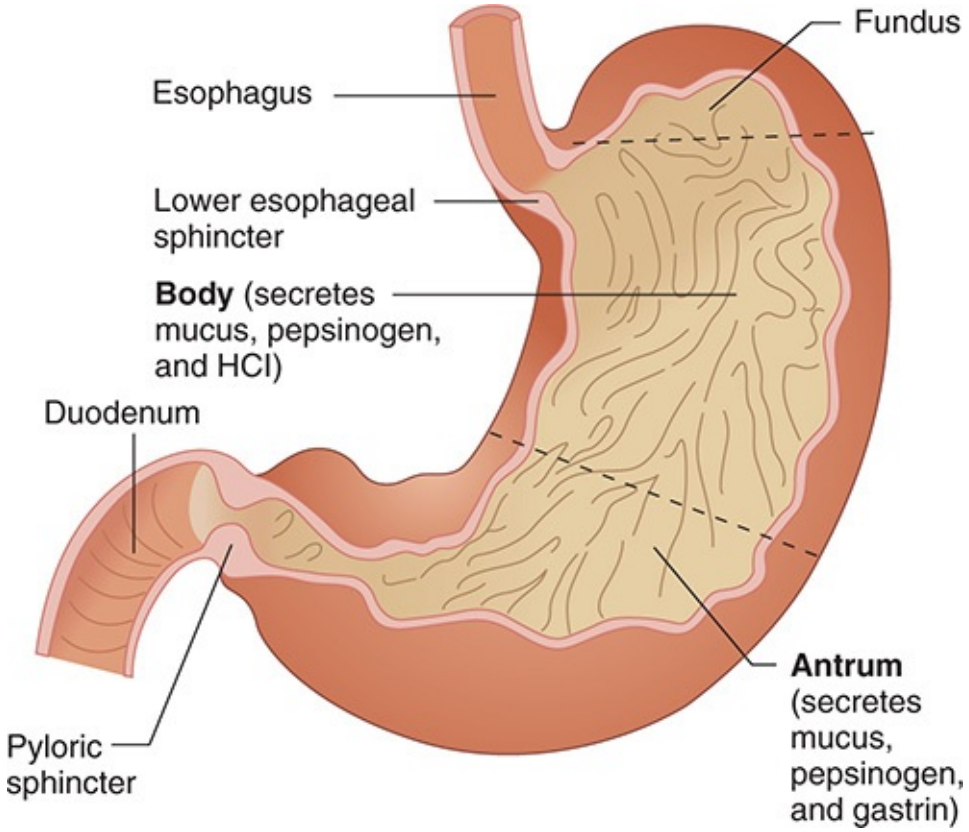


FIGURE 25–5 Anatomy of the stomach. The principal secretions of the body and antrum are listed in parentheses. (Reproduced with permission from

Widmaier EP, Raff H, Strang KT: *Vander's Human Physiology: The Mechanisms of Body Function*, 11th ed. New York, NY: McGraw-Hill; 2008.)

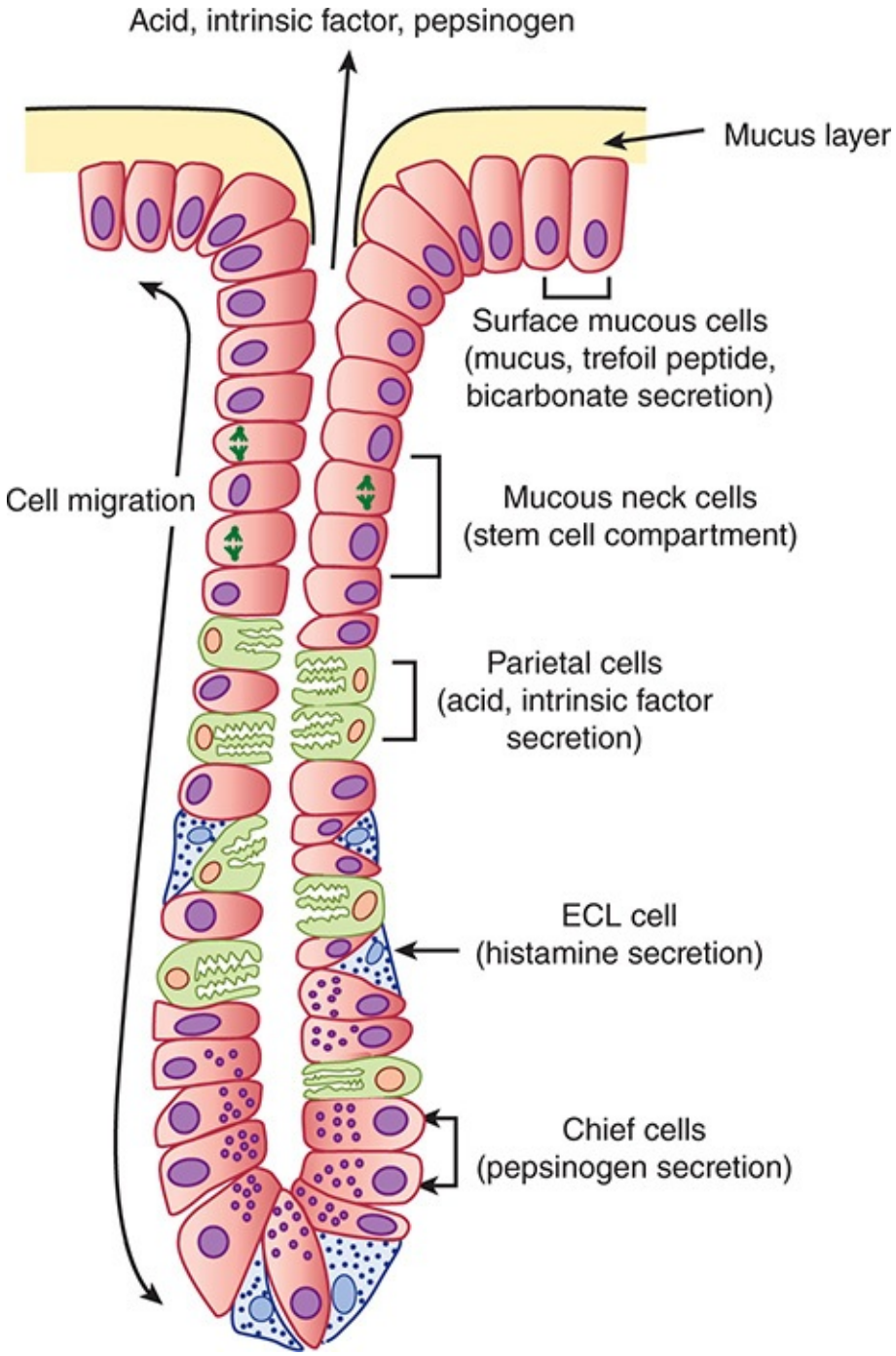


FIGURE 25–6 Structure of a gastric gland from the fundus or body of the stomach. These acid- and pepsinogen-producing glands are referred to as "oxyntic" glands in some sources. Similarly, some sources refer to parietal cells as oxyntic cells. ECL, enterochromaffin-like. (Adapted with permission from

Barrett KE: *Gastrointestinal Physiology*. New York, NY: McGraw-Hill; 2006.)

The stomach has a very rich blood and lymphatic supply. Its parasympathetic nerve supply comes from the vagi and its sympathetic supply from the celiac plexus.

ORIGIN & REGULATION OF GASTRIC SECRETION

The stomach adds a significant volume of digestive juices to the meal. Like salivary secretion, the stomach readies itself to receive the meal before it is actually taken in, during the so-called cephalic phase that can be influenced by food preferences. Subsequently, there is a gastric phase of secretion that is quantitatively the most significant, and finally an intestinal phase once the meal has left the stomach. Each phase is closely regulated by both local and distant triggers.

The gastric secretions **(Table 25–1)** arise from glands in the wall of the stomach that drain into its lumen, and also from the surface cells that secrete primarily mucus and bicarbonate to protect the stomach from digesting itself, as well as substances known as trefoil peptides that stabilize the mucus-bicarbonate layer. The glandular secretions of the stomach differ in different regions of the organ. The most characteristic secretions derive from the glands in the fundus or body of the stomach. These contain the distinctive parietal cells, which secrete hydrochloric acid and intrinsic factor; and chief cells, which produce pepsinogens and gastric lipase (Figure 25–5). The acid secreted by parietal cells serves to sterilize the meal and also to begin the hydrolysis particularly of dietary protein. Intrinsic factor is important for the later absorption of vitamin B_{12}, or cobalamin. Pepsinogen is the precursor of pepsin, which initiates protein digestion. Lipase similarly begins the digestion of dietary fats.

TABLE 25–1 Contents of normal gastric juice (fasting state).

Cations: Na^+, K^+, Mg^{2+}, H^+ (pH approximately 3.0)
Anions: Cl^-, HPO_4^{2-}, SO_4^{2-}
Pepsins
Lipase
Mucus

Intrinsic factor

There are three primary stimuli of gastric secretion, each with a specific role to play in matching the rate of secretion to functional requirements **(Figure 25–7)**. Gastrin is a hormone that is released by G cells in the antrum of the stomach both in response to a specific neurotransmitter released from enteric nerve endings, known as gastrin-releasing peptide (GRP) and also in response to the presence of oligopeptides in the gastric lumen. Gastrin is then carried through the bloodstream to the fundic glands, where it binds to receptors not only on parietal (and likely, chief) cells to activate secretion, but also on so-called enterochromaffin-like cells (ECL cells) that are located in the gland, and release histamine. Histamine is also a trigger of parietal cell secretion, via binding to H_2-receptors. Finally, parietal and chief cells can also be stimulated by acetylcholine, released from enteric nerve endings in the fundus.

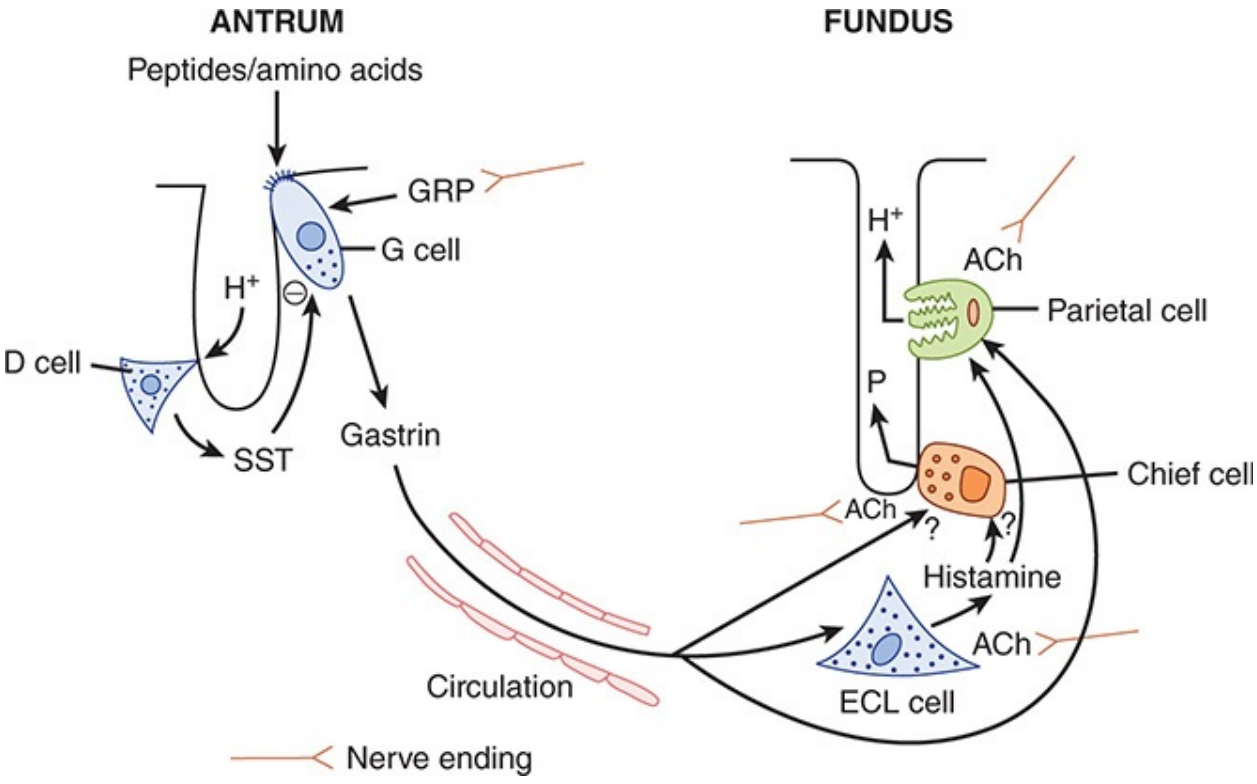


FIGURE 25–7 Regulation of gastric acid and pepsin secretion by soluble mediators and neural input. Gastrin is released from G cells in the antrum in response to gastrin-releasing peptide (GRP) and travels through the circulation to influence the activity of enterochromaffin-like (ECL) cells and parietal cells. ECL cells release histamine, which also acts on parietal cells. Acetylcholine

(ACh), released from nerves, is an agonist for ECL cells, chief cells, and parietal cells. Other specific agonists of the chief cell are not well understood. Gastrin release is negatively regulated by luminal acidity via the release of somatostatin from antral D cells. P, pepsinogen. (Adapted with permission from Barrett KE: *Gastrointestinal Physiology*. New York, NY: McGraw-Hill; 2006.)

Gastric secretion that occurs during the cephalic phase is activated predominantly by vagal input that originates from the brain region known as the dorsal vagal complex, which coordinates input from higher centers. Vagal outflow to the stomach then releases GRP and acetylcholine, thereby initiating secretory function. However, before the meal enters the stomach, there are few additional triggers and thus the amount of secretion is limited. Once the meal is swallowed, on the other hand, meal constituents trigger substantial release of gastrin and the physical presence of the meal also distends the stomach and activates stretch receptors, which provoke a "vago-vagal" as well as local reflexes that further amplify secretion during the gastric phase. The presence of the meal also buffers gastric acidity. Acidity otherwise would serve as a feedback inhibitory signal to shut off secretion, secondary to the release of somatostatin, which inhibits both G and ECL cells as well as secretion by parietal cells themselves (Figure 25–7). This probably represents a key mechanism whereby gastric secretion is terminated after the meal moves from the stomach into the small intestine.

Gastric parietal cells are highly specialized for their unusual task of secreting concentrated acid **(Figure 25–8)**. The cells are packed with mitochondria that supply energy to drive the apical H^+, K^+-ATPase, or proton pump, that moves H^+ ions out of the parietal cell against a concentration gradient of more than a million-fold. At rest, the proton pumps are sequestered within the parietal cell in a series of membrane compartments known as tubulovesicles. When the parietal cell begins to secrete, on the other hand, these vesicles fuse with invaginations of the apical membrane known as canaliculi, thereby substantially amplifying the apical membrane area and positioning the proton pumps to begin acid secretion **(Figure 25–9)**. The apical membrane also contains potassium channels, which supply the K^+ ions to be exchanged for H^+, and Cl^- channels that supply the counterion for HCl secretion **(Figure 25–10)**. The secretion of protons is also accompanied by the release of equivalent numbers of bicarbonate ions into the bloodstream, which are later used to neutralize gastric acidity once its function is complete (Figure 25–10).

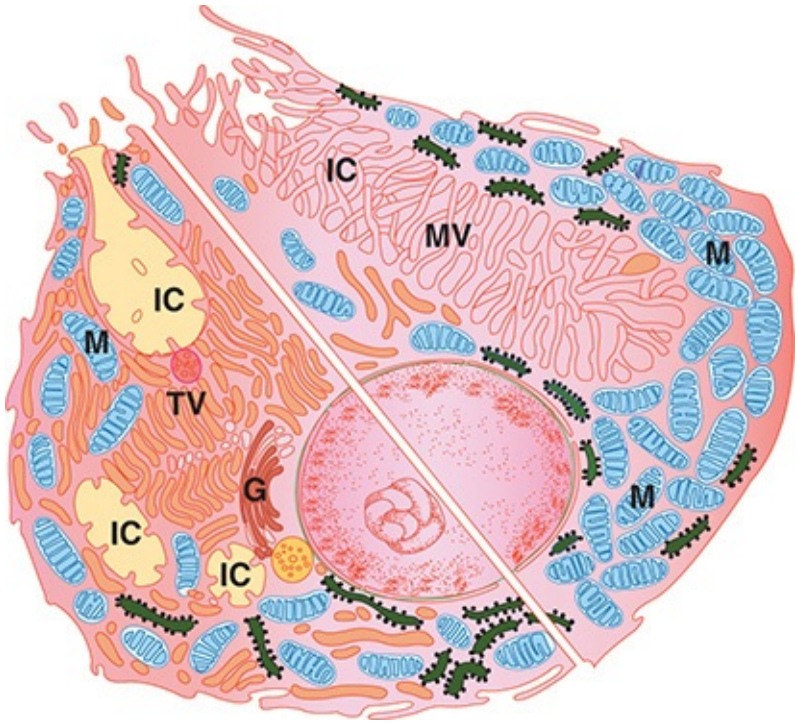


FIGURE 25–8 Composite diagram of a parietal cell, showing the resting state (lower left) and the active state (upper right). The resting cell has intracellular canaliculi (IC), which open on the apical membrane of the cell, and many tubulovesicular structures (TV) in the cytoplasm. When the cell is activated, the TVs fuse with the cell membrane and microvilli (MV) project into the canaliculi, so the area of cell membrane in contact with gastric lumen is greatly increased. M, mitochondrion; G, Golgi apparatus. (Reproduced with pemission of Ito S, Schofield GC: Studies on the depletion and accumulation of microvilli and changes in the tubulovesicular compartment of mouse parietal cells in relation to gastric acid secretion. J Cell Biol 1974; Nov; 63(2 Pt 1):364–382.)

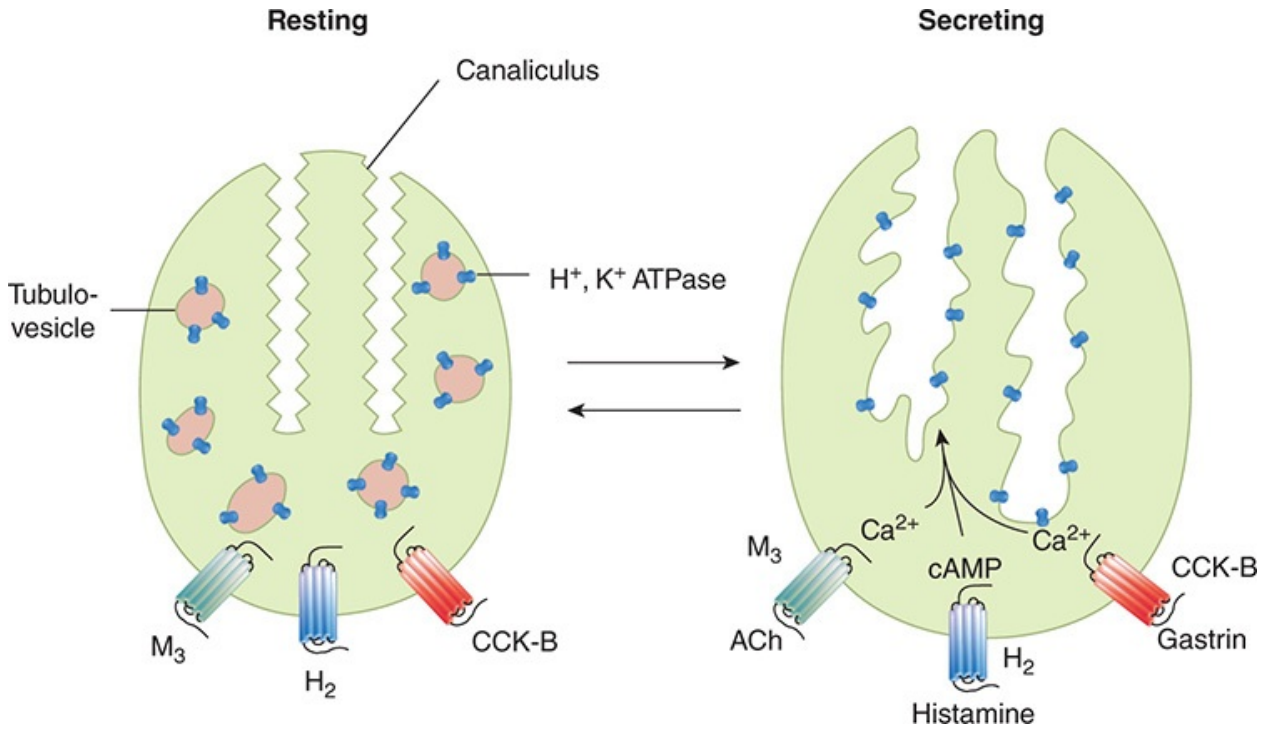


FIGURE 25–9 Parietal cell receptors and schematic representation of the morphologic changes depicted in Figure 25–8. Amplification of the apical surface area is accompanied by an increased density of H^+, K^+–ATPase molecules at this site. Note that acetylcholine (ACh) and gastrin signal via calcium, whereas histamine signals via cAMP. (Adapted with permission from Barrett KE: *Gastrointestinal Physiology*. New York, NY: McGraw-Hill; 2006.)

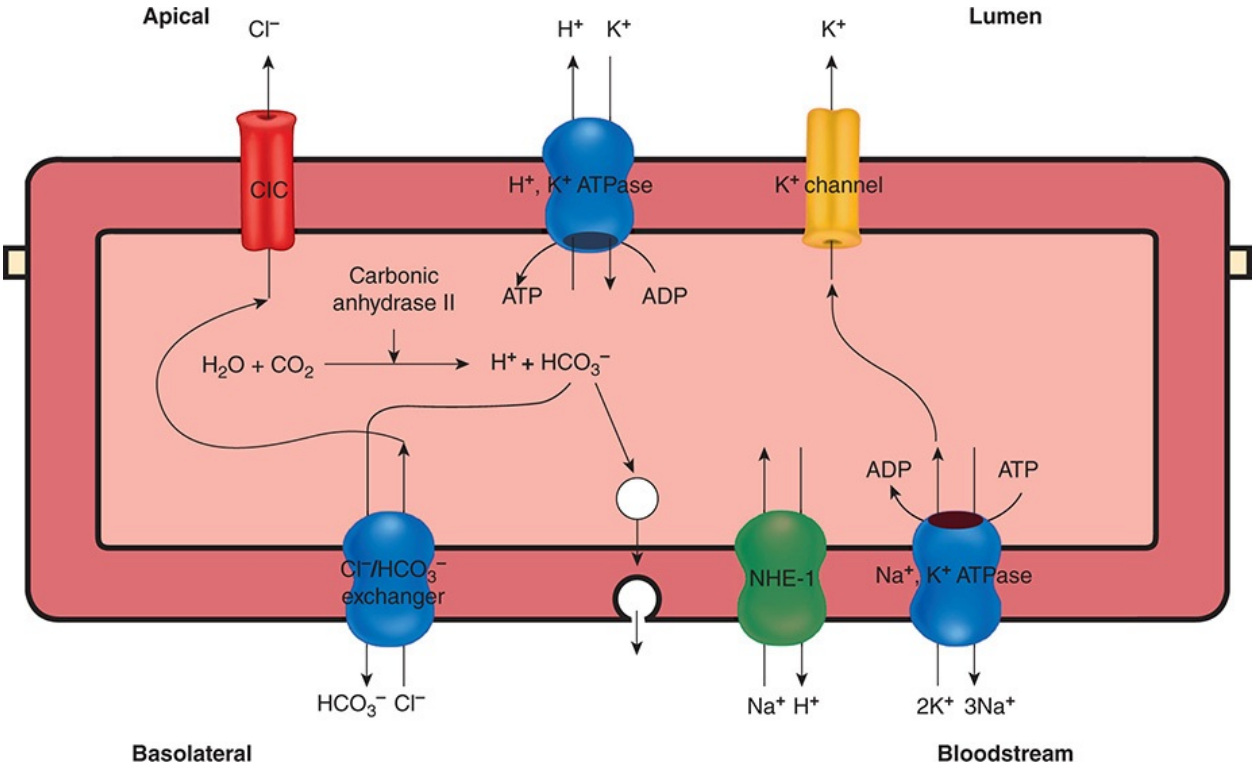


FIGURE 25–10 Ion transport proteins of parietal cells. Protons are generated in the cytoplasm via the action of carbonic anhydrase II. Bicarbonate ions are exported from the basolateral pole of the cell either by vesicular fusion or via a chloride/bicarbonate exchanger. The sodium/hydrogen exchanger, NHE1, on the basolateral membrane is considered a "housekeeping" transporter that maintains intracellular pH in the face of cellular metabolism during the unstimulated state.

The three agonists of the parietal cell—gastrin, histamine, and acetylcholine—each bind to distinct receptors on the basolateral membrane (Figure 25–9). Gastrin and acetylcholine promote secretion by elevating cytosolic free calcium concentrations, whereas histamine increases intracellular cyclic adenosine 3′,5′-monophosphate (cAMP). The net effects of these second messengers are the transport and morphologic changes described above. However, it is important to be aware that the two distinct pathways for activation are synergistic, with a greater than additive effect on secretion rates when histamine plus gastrin or acetylcholine, or all three, are present simultaneously. The physiologic significance of this synergism is that high rates of secretion can be stimulated with relatively small changes in availability of each of the stimuli. Synergism is also therapeutically significant because secretion can be markedly inhibited by blocking the action of only one of the triggers (most commonly that of histamine, via H_2-antagonists that are widely used therapies for adverse effects

of excessive gastric secretion) **(Clinical Box 25–1)**.

Gastric secretion adds about 2.5 L/day to the intestinal contents. However, despite their substantial volume and fine control, gastric secretions are dispensable for the full digestion and absorption of a meal, with the exception of cobalamin absorption. This illustrates an important facet of gastrointestinal physiology, namely that digestive and absorptive capacities are markedly in excess of normal requirements. On the other hand, if gastric secretion is chronically reduced, individuals may display increased susceptibility to infections acquired via the oral route.

PANCREATIC SECRETION

The pancreatic juice contains enzymes that are of major importance in digestion **(see Table 25–2)**. Its secretion is controlled in part by a reflex mechanism and in part by the gastrointestinal hormones secretin and cholecystokinin (CCK).

TABLE 25–2 Principal digestive enzymes.[a]

Source	Enzyme	Activator	Substrate	Catalytic Function or Products
Salivary glands	Salivary α-amylase	Cl^-	Starch	Hydrolyzes 1:4α linkages, producing α-limit dextrins, maltotriose, and maltose
Stomach	Pepsins (pepsinogens)	HCl	Proteins and polypeptides	Cleave peptide bonds adjacent to aromatic amino acids
	Gastric lipase		Triglycerides	Fatty acids and glycerol
Exocrine pancreas	Trypsin (trypsinogen)	Enteropeptidase	Proteins and polypeptides	Cleave peptide bonds on carboxyl side of basic amino acids (arginine or lysine)
	Chymotrypsins (chymotrypsinogens)	Trypsin	Proteins and polypeptides	Cleave peptide bonds on carboxyl side of aromatic amino acids
	Elastase (proelastase)	Trypsin	Elastin, some other proteins	Cleaves bonds on carboxyl side of aliphatic amino acids
	Carboxypeptidase A (procarboxypeptidase A)	Trypsin	Proteins and polypeptides	Cleave carboxyl terminal amino acids that have aromatic or branched aliphatic side chains
	Carboxypeptidase B (procarboxypeptidase B)	Trypsin	Proteins and polypeptides	Cleave carboxyl terminal amino acids that have basic side chains
	Colipase (procolipase)	Trypsin	Fat droplets	Binds pancreatic lipase to oil droplet in the presence of bile acids
	Pancreatic lipase	…	Triglycerides	Monoglycerides and fatty acids
	Cholesteryl ester hydrolase	…	Cholesteryl esters	Cholesterol
	Pancreatic α-amylase	Cl^-	Starch	Same as salivary α-amylase
	Ribonuclease	…	RNA	Nucleotides
	Deoxyribonuclease	…	DNA	Nucleotides
	Phospholipase A_2 (pro-phospholipase A_2)	Trypsin	Phospholipids	Fatty acids, lysophospholipids
Intestinal mucosa	Enteropeptidase	…	Trypsinogen	Trypsin
	Aminopeptidases	…	Polypeptides	Cleave amino terminal amino acid from peptide
	Carboxypeptidases	…	Polypeptides	Cleave carboxyl terminal amino acid from peptide
	Endopeptidases	…	Polypeptides	Cleave between residues in midportion of peptide
	Dipeptidases	…	Dipeptides	Two amino acids
	Maltase	…	Maltose, maltotriose	Glucose
	Lactase	…	Lactose	Galactose and glucose
	Sucrase[b]	…	Sucrose; also maltotriose and maltose	Fructose and glucose
	Isomaltase[b]	…	α-Limit dextrins, maltose Maltotriose	Glucose
	Nuclease and related enzymes	…	Nucleic acids	Pentoses and purine and pyrimidine bases
Cytoplasm of mucosal cells	Various peptidases	…	Di-, tri-, and tetrapeptides	Amino acids

[a]Corresponding proenzymes, where relevant, are shown in parentheses.

[b]Sucrase and isomaltase are separate subunits of a single protein.

ANATOMIC CONSIDERATIONS

The portion of the pancreas that secretes pancreatic juice is a compound alveolar gland resembling the salivary glands. This **exocrine pancreas** should be

distinguished from the **endocrine pancreas** that produces insulin and other hormones from the islets of Langerhans (see Chapter 24), although both are contained within the same organ. Granules containing the digestive enzymes **(zymogen granules)** are formed in the cell and discharged by exocytosis (see Chapter 2) from the apexes of the cells into the lumens of the pancreatic ducts **(Figure 25–11)**. The small duct radicles coalesce into a single duct (pancreatic duct of Wirsung), which joins the bile duct to form the ampulla of the bile duct (also known as the ampulla of Vater) **(Figure 25–12)**. The ampulla opens through the duodenal papilla, and its orifice is encircled by the sphincter of Oddi.

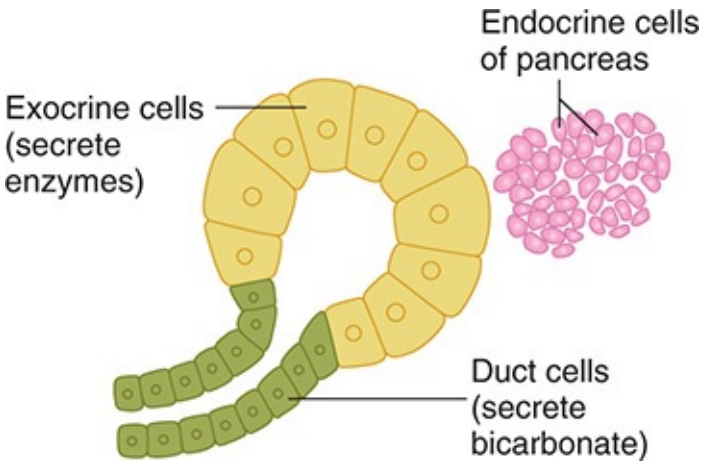


FIGURE 25–11 Structure of the pancreas. (Reproduced with permission from Widmaier EP, Raff H, Strang KT: *Vander's Human Physiology: The Mechanisms of Body Function*, 11th ed. New York, NY: McGraw-Hill; 2008.)

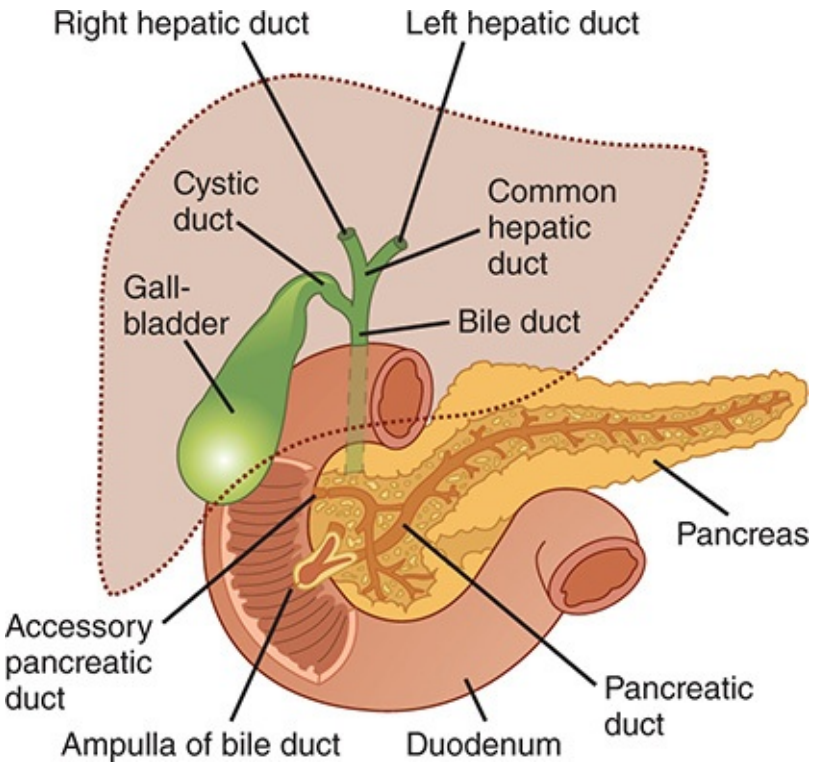


FIGURE 25–12 Connections of the ducts of the gallbladder, liver, and pancreas. (Adapted with permission from Bell GH, Emslie-Smith D, Paterson CR: *Textbook of Physiology and Biochemistry*, 9th ed. Churchill Livingstone, 1976.)

CLINICAL BOX 25–1

Peptic Ulcer Disease

Gastric and duodenal ulceration in humans is related primarily to a breakdown of the barrier that normally prevents irritation and autodigestion of the mucosa by the gastric secretions. Infection with the bacterium *Helicobacter pylori* disrupts this barrier, as do aspirin and other nonsteroidal anti-inflammatory drugs (NSAIDs), which inhibit the production of prostaglandins and consequently decrease mucus and HCO_3^- secretion. NSAIDs are widely used to combat pain and treat arthritis. An additional cause of ulceration is prolonged excess secretion of acid. An example of this is the ulcers that occur in the **Zollinger–Ellison syndrome**. This syndrome is seen in patients with gastrinomas. These tumors can occur in the stomach and duodenum, but most of them are found in the pancreas. The gastrin causes

prolonged hypersecretion of acid, and severe ulcers are produced.

THERAPEUTIC HIGHLIGHTS

Gastric and duodenal ulcers can be given a chance to heal by inhibition of acid secretion with drugs such as omeprazole and related drugs that inhibit H^+–K^+ ATPase ("proton pump inhibitors") or with histamine H_2-receptor antagonists. If present, *H. pylori* can be eradicated with antibiotics, and NSAID-induced ulcers can be treated by stopping the NSAID or, when this is not advisable, by treatment with the prostaglandin agonist misoprostol. Gastrinomas can sometimes be removed surgically.

COMPOSITION OF PANCREATIC JUICE

The pancreatic juice is alkaline **(Table 25–3)** and has a high HCO_3^- content (approximately 113 mEq/L vs 24 mEq/L in plasma). About 1500 mL of pancreatic juice is secreted per day. Bile and intestinal juices are also neutral or alkaline, and these three secretions neutralize the gastric acid, raising the pH of the duodenal contents to 6.0–7.0. By the time the chyme reaches the jejunum, its pH is nearly neutral.

TABLE 25–3 Composition of normal human pancreatic juice.

Cations: Na^+, K^+, Ca^{2+}, Mg^{2+} (pH approximately 8.0)
Anions: HCO_3^-, Cl^-, SO_4^{2-}, HPO_4^{2-}
Digestive enzymes (see Table 25–2; 95% of protein in juice)
Other proteins

The pancreatic juice also contains a range of digestive enzymes, but most of these are released in inactive forms and only activated when they reach the intestinal lumen (see Chapter 26). The enzymes are activated following proteolytic cleavage by trypsin, itself a pancreatic protease that is released as an inactive precursor (trypsinogen). The potential danger of the release within the pancreas of a small amount of active trypsin is apparent; the resulting chain reaction would produce active enzymes that could digest the organ. It is therefore not surprising that the pancreas also normally secretes a trypsin

inhibitor.

Another enzyme activated by trypsin is phospholipase A_2. This enzyme splits a fatty acid off phosphatidylcholine (PC), forming lyso-PC. Lyso-PC damages cell membranes. It has been hypothesized that in **acute pancreatitis,** a severe and sometimes fatal disease, phospholipase A_2 is activated prematurely in the pancreatic ducts, with the formation of lyso-PC from the PC that is a normal constituent of bile. This causes disruption of pancreatic tissue and necrosis of surrounding fat.

Small amounts of pancreatic digestive enzymes normally leak into the circulation, but in acute pancreatitis, the circulating levels of the digestive enzymes rise markedly. Measurement of the plasma amylase or lipase concentration is therefore of value in diagnosing the disease.

REGULATION OF THE SECRETION OF PANCREATIC JUICE

Secretion of pancreatic juice is primarily under hormonal control. Secretin acts on the pancreatic ducts to cause copious secretion of a very alkaline pancreatic juice that is rich in HCO_3^- and poor in enzymes. The effect on duct cells is due to an increase in intracellular cAMP. Secretin also stimulates bile secretion. CCK acts on the acinar cells to cause the release of zymogen granules and production of pancreatic juice rich in enzymes but low in volume. Its effect is mediated by phospholipase C (see Chapter 2). Thus, by acting together, CCK and secretin add enzymes to the pancreatic juice and ensure that they are washed into the intestine.

The response to intravenous secretin is shown in **Figure 25–13**. Note that as the volume of pancreatic secretion increases, its Cl^- concentration falls and its HCO_3^- concentration increases. Although HCO_3^- is secreted in the small ducts, it is reabsorbed in the large ducts in exchange for Cl^- **(Figure 25–14)**. The magnitude of the exchange is inversely proportional to the rate of flow.

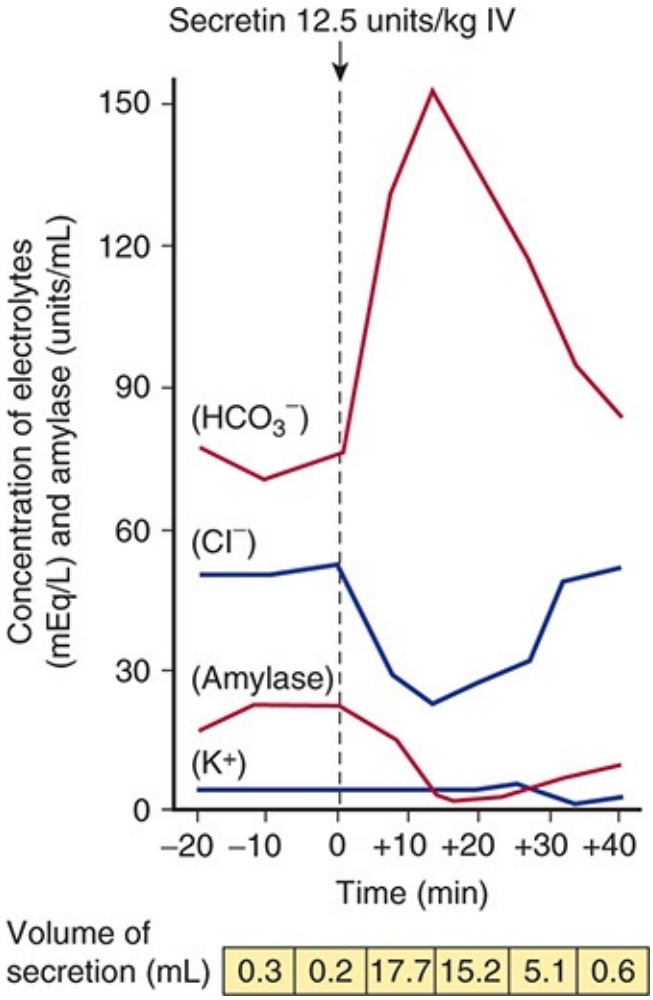


FIGURE 25–13 Effect of a single dose of secretin on the composition and volume of the pancreatic juice in humans. Note the reciprocal changes in the concentrations of chloride and bicarbonate after secretin is infused. The fall in amylase concentration reflects dilution as the volume of pancreatic juice increases.

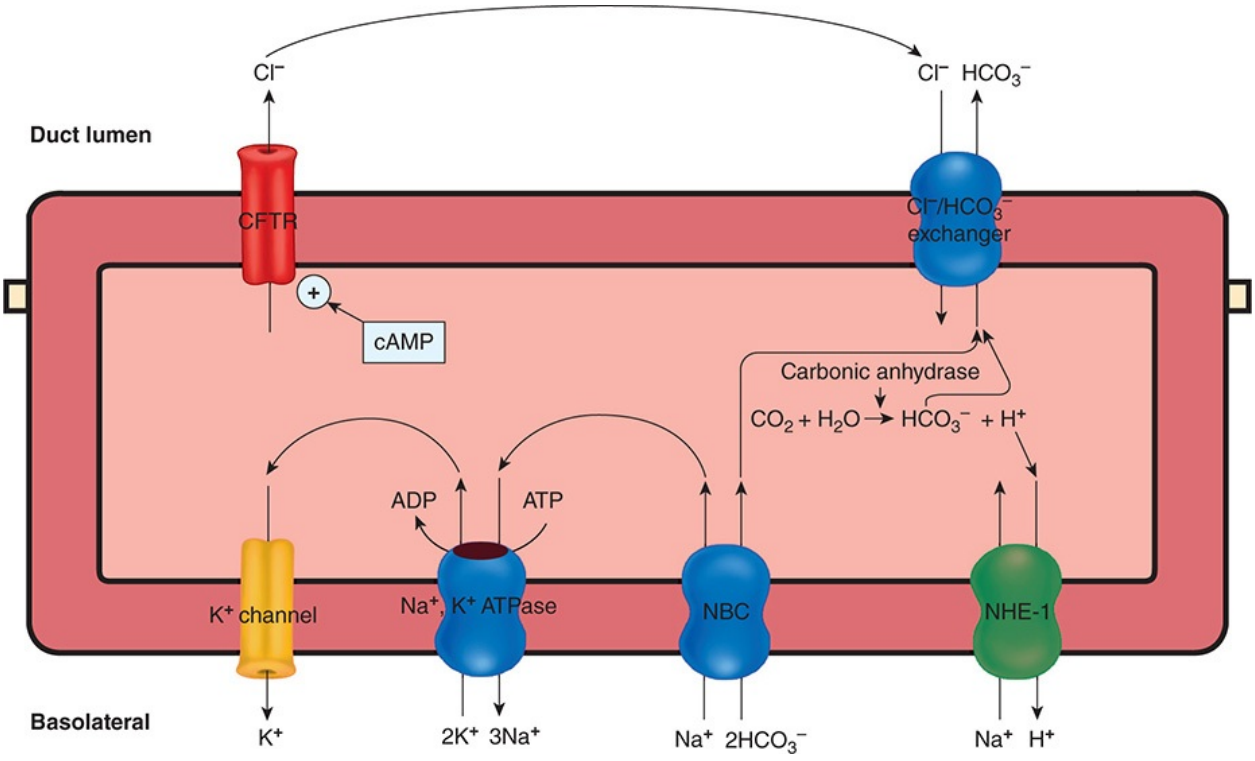


FIGURE 25–14 Ion transport pathways present in pancreatic duct cells. CFTR, cystic fibrosis transmembrane conductance regulator; NHE-1, sodium/hydrogen exchanger-1; NBC, sodium-bicarbonate cotransporter.

Like CCK, acetylcholine acts on acinar cells via phospholipase C to cause discharge of zymogen granules, and stimulation of the vagi causes secretion of a small amount of pancreatic juice rich in enzymes. There is evidence for vagally mediated, conditioned reflex secretion of pancreatic juice in response to the sight or smell of food.

BILIARY SECRETION

An additional secretion important for gastrointestinal function, bile, arises from the liver. The bile acids contained therein are important in the digestion and absorption of fats. In addition, bile serves as a critical excretory fluid by which the body disposes of lipid soluble end products of metabolism as well as lipid soluble xenobiotics. Bile is also the only route by which the body can dispose of cholesterol—either in its native form, or following conversion to bile acids. In this chapter and the next, the role of bile as a digestive fluid will be emphasized. In Chapter 28, a more general consideration of the transport and metabolic

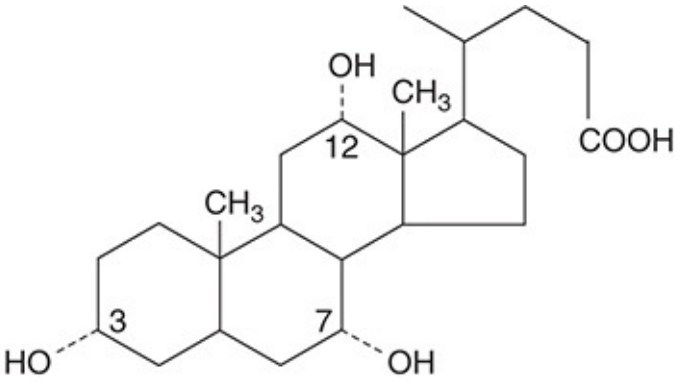
functions of the liver will be presented.

Bile

Bile is made up of the bile acids, bile pigments, and other substances dissolved in an alkaline electrolyte solution that resembles pancreatic juice. About 500 mL is secreted per day. Some of the components of the bile are reabsorbed in the intestine and then excreted again by the liver **(enterohepatic circulation)**.

The glucuronides of the **bile pigments,** bilirubin and biliverdin, are responsible for the golden yellow color of bile. The formation of these breakdown products of hemoglobin is discussed in detail in Chapter 28.

When considering bile as a digestive secretion, it is the **bile acids** that represent the most important components. They are synthesized from cholesterol and secreted into the bile conjugated to glycine or taurine. The four major bile acids found in humans are listed in **Figure 25–15**. In common with vitamin D, cholesterol, and a variety of steroid hormones, bile acids contain the steroid nucleus (see Chapter 19). The two principal (primary) bile acids formed in the liver are cholic acid and chenodeoxycholic acid. In the colon, bacteria convert cholic acid to deoxycholic acid and chenodeoxycholic acid to lithocholic acid. In addition, small quantities of ursodeoxycholic acid are formed from chenodeoxycholic acid. Because they are formed by bacterial action, deoxycholic, lithocholic, and ursodeoxycholic acids are called secondary bile acids.



Cholic acid

	Group at position			Percent in human bile
	3	7	12	
Cholic acid	OH	OH	OH	50
Chenodeoxycholic acid	OH	OH	H	30
Deoxycholic acid	OH	H	OH	15
Lithocholic acid	OH	H	H	5

FIGURE 25–15 Human bile acids. The numbers in the formula for cholic acid refer to the positions in the steroid ring.

The bile acids have a number of important actions: they reduce surface tension and, in conjunction with phospholipids and monoglycerides, are responsible for the emulsification of fat preparatory to its digestion and absorption in the small intestine (see Chapter 26). They are **amphipathic,** that is, they have both hydrophilic and hydrophobic domains; one surface of the molecule is hydrophilic because the polar peptide bond and the carboxyl and hydroxyl groups are on that surface, whereas the other surface is hydrophobic. Therefore, the bile acids tend to form cylindrical disks called **micelles (Figure 25–16)**. Their hydrophilic portions face out and their hydrophobic portions face in. Above a certain concentration, called the **critical micellar concentration,** all bile salts added to a solution form micelles. Ninety to 95% of the bile acids are absorbed from the small intestine. Once they are deconjugated, they can be absorbed by nonionic diffusion, but most are absorbed in their conjugated forms from the terminal ileum **(Figure 25–17)** by an extremely efficient Na^+–bile salt cotransport system (ABST) whose activity is secondarily driven by the low intracellular sodium concentration established by the basolateral Na^+, K^+ ATPase. The remaining 5–10% of the bile salts enter the colon and are converted to the salts of deoxycholic acid and lithocholic acid. Lithocholate is relatively

insoluble and is mostly excreted in the stools; only 1% is absorbed. However, deoxycholate is absorbed.

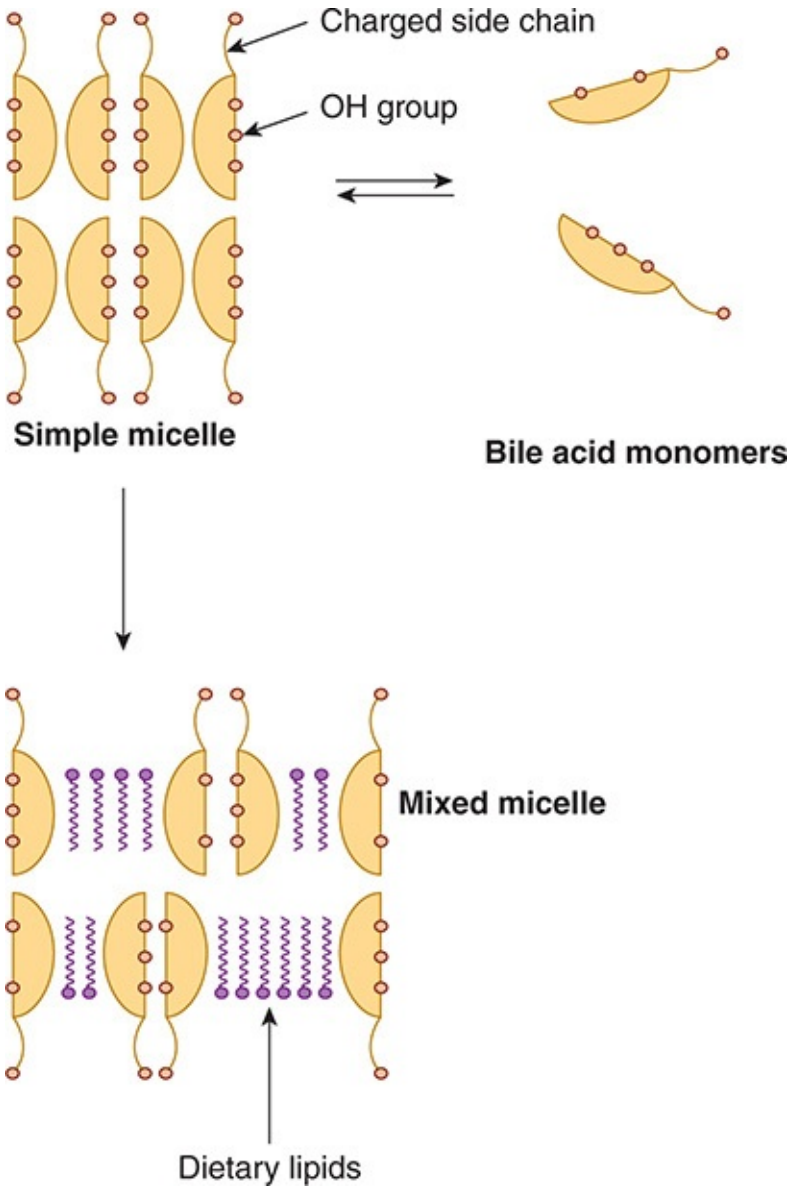


FIGURE 25–16 Physical forms adopted by bile acids in solution. Micelles are shown in cross-section and are actually thought to be cylindrical in shape. Mixed micelles of bile acids present in intestinal contents also incorporate dietary lipids. (Adapted with permission from Barrett KE: *Gastrointestinal Physiology*. New York, NY: McGraw-Hill; 2006.)

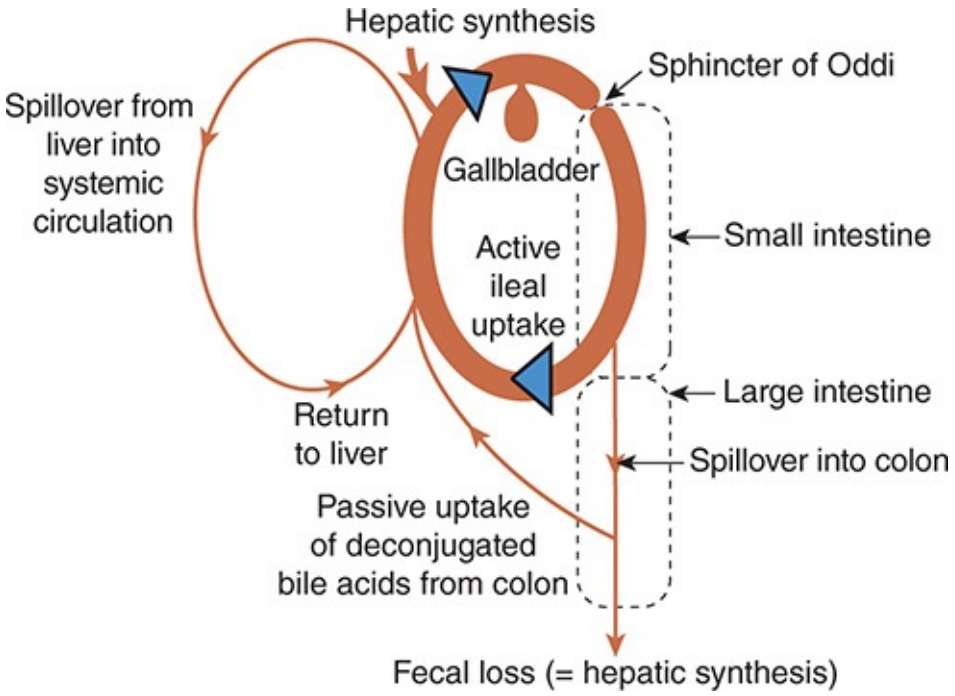


FIGURE 25–17 Quantitative aspects of the circulation of bile acids. The majority of the bile acid pool circulates between the small intestine and liver. A minority of the bile acid pool is in the systemic circulation (due to incomplete hepatocyte uptake from the portal blood) or spills over into the colon and is lost to the stool. Fecal loss must be equivalent to hepatic synthesis of bile acids at steady state. (Adapted with permission from Barrett KE: *Gastrointestinal Physiology*. New York, NY: McGraw-Hill; 2006.)

The absorbed bile acids are transported back to the liver in the portal vein and reexcreted in the bile (enterohepatic circulation) (Figure 25–17). Those lost in the stool are replaced by synthesis in the liver; the normal rate of bile acid synthesis is 0.2–0.4 g/day. The total bile acid pool of approximately 3.5 g recycles repeatedly via the enterohepatic circulation; it has been calculated that the entire pool recycles twice per meal and 6–8 times per day.

GASTROINTESTINAL REGULATION

The various functions of the gastrointestinal tract, including secretion, digestion, and absorption (Chapter 26), and motility (Chapter 27), must be regulated in an integrated way to ensure efficient assimilation of nutrients after a meal. There are three main modalities for gastrointestinal regulation that operate in a complementary fashion to ensure that function is appropriate. First, **endocrine**

regulation is mediated by the release of hormones by triggers associated with the meal. These hormones travel through the bloodstream to change the activity of a distant segment of the gastrointestinal tract, an organ draining into it (eg, the pancreas), or both. Second, some similar mediators are not sufficiently stable to persist in the bloodstream, but instead alter the function of cells in the local area where they are released, in a **paracrine** fashion. Finally, the intestinal system is endowed with extensive neural connections. These include connections to the CNS **(extrinsic innervation),** but also the activity of the largely autonomous **enteric nervous system** that comprises both sensory and secretomotor neurons. The enteric nervous system integrates central input to the gut but can also regulate gut function independently in response to changes in the luminal environment. In some cases, the same substance can mediate regulation by endocrine, paracrine, and neurocrine pathways (eg, CCK, see below).

HORMONES/PARACRINES

Biologically active polypeptides that are secreted by nerve cells and gland cells in the mucosa act in a paracrine fashion, but they also enter the circulation. Measurement of their concentrations in blood after a meal has shed light on the roles these **gastrointestinal hormones** play in the regulation of gastrointestinal secretion and motility.

When large doses of the hormones are given, their actions overlap. However, their physiologic effects appear to be relatively discrete. On the basis of structural similarity and, to a degree, similarity of function, the key hormones fall into one of two families: the gastrin family, the primary members of which are gastrin and CCK; and the secretin family, the primary members of which are secretin, glucagon, vasoactive intestinal peptide (VIP; actually a neurotransmitter, or neurocrine), and gastric inhibitory polypeptide (also known as glucose-dependent insulinotropic peptide, or GIP). There are also other regulatory peptides that do not fall into either of these families.

ENTEROENDOCRINE CELLS

More than 15 types of hormone-secreting **enteroendocrine cells** have been identified in the mucosa of the stomach, small intestine, and colon. Many of these secrete only one hormone and are identified by letters (G cells, S cells, etc). Others manufacture serotonin or histamine and are called **enterochromaffin** or **ECL cells,** respectively.

GASTRIN

Gastrin is produced by cells called G cells in the antral portion of the gastric mucosa **(Figure 25–18)**. G cells are flask-shaped, with a broad base containing many gastrin granules and a narrow apex that reaches the mucosal surface. Microvilli project from the apical end into the lumen. Receptors mediating gastrin responses to changes in gastric contents are present on the microvilli. Other cells in the gastrointestinal tract that secrete hormones have a similar morphology.

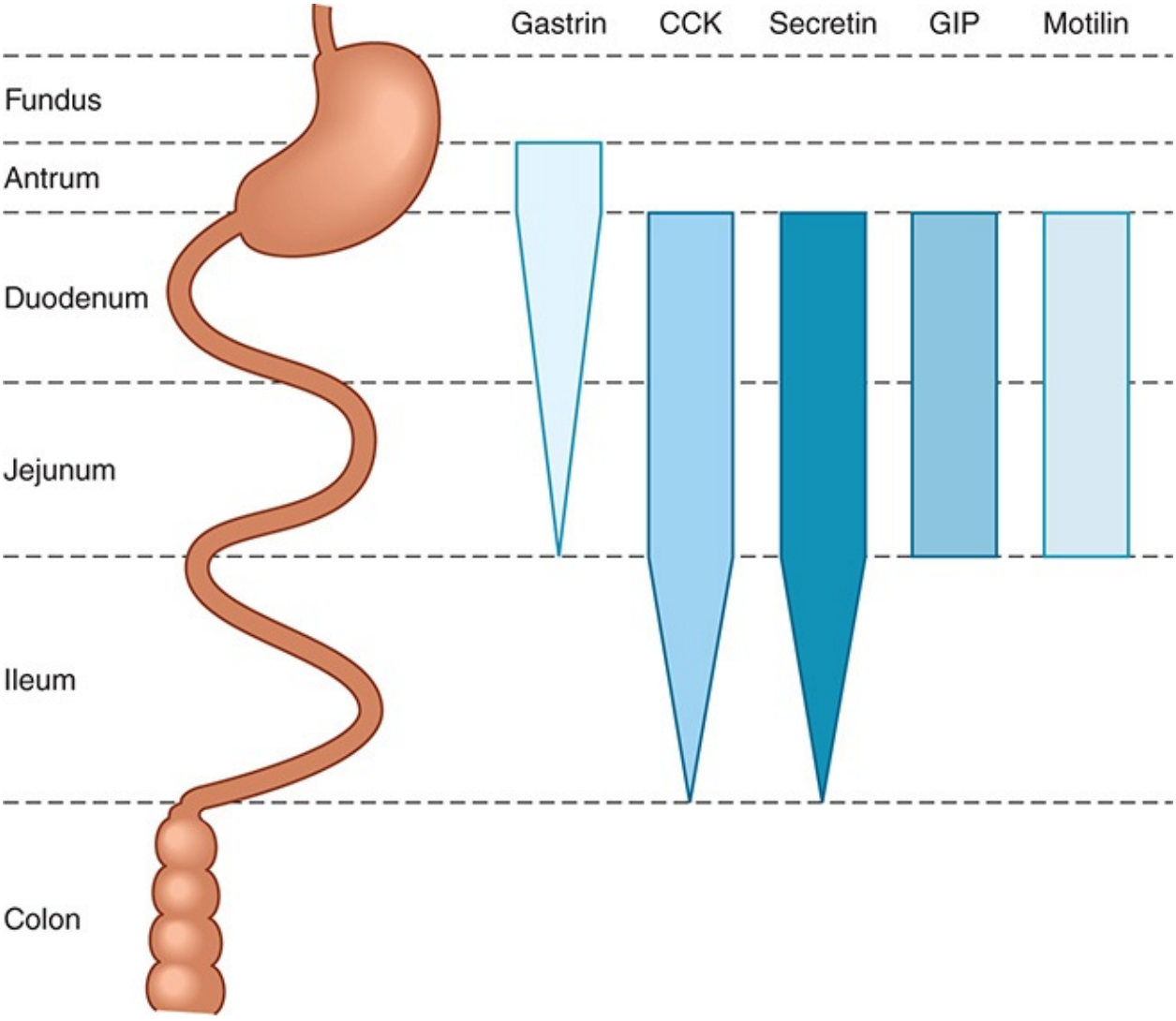


FIGURE 25–18 Sites of production of the five gastrointestinal hormones along the length of the gastrointestinal tract. The width of the bars reflects the relative abundance at each location.

The precursor for gastrin, preprogastrin, is processed into fragments of various sizes. Three main fragments contain 34, 17, and 14 amino acid residues. All have the same carboxyl terminal. These forms are also known as G 34, G 17, and G 14 gastrins, respectively. Gastrins may also be derivatized with sulfation of the tyrosine that is the sixth amino acid residue from the carboxyl terminal. Approximately equal amounts of nonsulfated and sulfated forms are present in blood and tissues, and they are equally active. The carboxyl terminal phenylalanine is also amidated, which likely enhances the peptide's stability in the plasma by rendering it resistant to carboxypeptidases. G 14 and G 17 have half-lives of 2–3 min in the circulation, whereas G 34 has a half-life of 15 min. Gastrins are inactivated primarily in the kidney and small intestine.

The principal physiologic actions of gastrin are stimulation of gastric acid and pepsin secretion and stimulation of the growth of the mucosa of the stomach and small and large intestines **(trophic action)**. Gastrin secretion is affected by the contents of the stomach, the rate of discharge of the vagus nerves, and bloodborne factors **(Table 25–4)**. Gastrin secretion is also increased by the presence of the products of protein digestion in the stomach, particularly amino acids, which act directly on the G cells. Phenylalanine and tryptophan are particularly effective. Gastrin acts via a receptor (CCK-B) that is related to the primary receptor (CCK-A) for cholecystokinin (see below). This likely reflects the structural similarity of the two hormones, and may result in some overlapping actions if excessive quantities of either hormone are present (eg, in the case of a gastrin-secreting tumor, or gastrinoma).

TABLE 25–4 Stimuli that affect gastrin secretion.

Stimuli that increase gastrin secretion
Luminal
Peptides and amino acids
Distention
Neural
Increased vagal discharge via GRP
Bloodborne
Calcium
Epinephrine
Stimuli that inhibit gastrin secretion
Luminal
Acid
Somatostatin
Bloodborne
Secretin, GIP, VIP, glucagon, calcitonin

Acid in the antrum inhibits gastrin secretion, partly by a direct action on G cells and partly by release of somatostatin, a relatively potent inhibitor of gastrin secretion. The effect of acid is the basis of a negative feedback loop regulating gastrin secretion. Increased secretion of the hormone increases acid secretion, but the acid then feeds back to inhibit further gastrin secretion. In conditions such as pernicious anemia in which the acid-secreting cells of the stomach are damaged, gastrin secretion is chronically elevated.

CHOLECYSTOKININ

CCK is secreted by endocrine cells known as I cells in the mucosa of the upper small intestine. It has a plethora of actions in the gastrointestinal system, but the most important appear to be the stimulation of pancreatic enzyme secretion; the contraction of the gallbladder (the action for which it was named); and relaxation of the sphincter of Oddi, which allows both bile and pancreatic juice to flow into the intestinal lumen.

Like gastrin, CCK is produced from a larger precursor. Prepro-CCK is also processed into many fragments, all of which have the same five amino acids at the carboxyl terminal as gastrin. The carboxyl terminal is amidated, and the tyrosine that is the seventh amino acid residue from the carboxyl terminal is

sulfated. Unlike gastrin, the nonsulfated form of CCK has not been found in tissues. The half-life of circulating CCK is about 5 min, but little is known about its metabolism.

In addition to its secretion by I cells, CCK is found in nerves in the distal ileum and colon. It is also found in neurons in the brain, especially the cerebral cortex, and in nerves in many parts of the body (see Chapter 7). In the brain, it may be involved in the regulation of food intake.

In addition to its primary actions, CCK augments the action of secretin in producing secretion of an alkaline pancreatic juice. It also inhibits gastric emptying, exerts a trophic effect on the pancreas, increases the synthesis of enterokinase (see chapter 26), and may enhance the motility of the small intestine and colon. There is some evidence that, along with secretin, it augments the contraction of the pyloric sphincter, thus preventing the reflux of duodenal contents into the stomach. Two CCK receptors have been identified. CCK-A receptors are primarily located in the periphery, whereas both CCK-A and CCK-B (gastrin) receptors are found in the brain. Both activate PLC, and signal via calcium (see Chapter 2).

The secretion of CCK is increased by contact of the intestinal mucosa with the products of digestion, particularly peptides and amino acids, and also by the presence in the duodenum of fatty acids containing more than 10 carbon atoms. There are also two protein-releasing factors that activate CCK secretion, known as CCK-releasing peptide and monitor peptide, which derive from the intestinal mucosa and pancreas, respectively. Because the bile and pancreatic juice that enter the duodenum in response to CCK enhance the digestion of protein and fat, and the products of this digestion stimulate further CCK secretion, a sort of positive feedback operates in the control of CCK secretion. However, the positive feedback is terminated when the products of digestion move on to the lower portions of the gastrointestinal tract, and also because CCK-releasing peptide and monitor peptide are degraded by proteolytic enzymes once these are no longer occupied in digesting dietary proteins.

SECRETIN

Secretin occupies a unique position in the history of physiology. In 1902, Bayliss and Starling first demonstrated that the excitatory effect of duodenal stimulation on pancreatic secretion was due to a bloodborne factor. Their research led to the identification of the first hormone, secretin. Secretin is secreted by S cells that are located deep in the glands of the mucosa of the upper portion of the small

intestine. The structure of secretin is different from that of CCK and gastrin, but very similar to that of GIP, glucagon, and VIP. Only one active form of secretin has been isolated. Its half-life is about 5 min, but little is known about its metabolism.

Secretin increases the secretion of bicarbonate by the duct cells of the pancreas and biliary tract. It thus causes the secretion of a watery, alkaline pancreatic juice. Its action on pancreatic duct cells is mediated via cAMP. It also augments the action of CCK in producing pancreatic secretion of digestive enzymes. It decreases gastric acid secretion and may cause contraction of the pyloric sphincter.

The secretion of secretin is increased by the products of protein digestion and by acid bathing the mucosa of the upper small intestine. The release of secretin by acid is another example of feedback control: Secretin causes alkaline pancreatic juice to flood into the duodenum, neutralizing the acid from the stomach and thus inhibiting further secretion of the hormone.

GIP

GIP contains 42 amino acid residues and is produced by K cells in the mucosa of the duodenum and jejunum. Its secretion is stimulated by glucose and fat in the duodenum, and because in large doses it inhibits gastric secretion and motility, it was named gastric inhibitory peptide. However, it now appears that it does not have significant gastric inhibiting activity when administered in amounts comparable to those seen after a meal. In the meantime, it was found that GIP stimulates insulin secretion at physiological levels. For this reason, it is often called **glucose-dependent insulinotropic peptide**. The glucagon derivative GLP-1 (7–36) (see Chapter 24) also stimulates insulin secretion and may also be a physiologic B cell–stimulating hormone of the gastrointestinal tract.

The integrated action of gastrin, CCK, secretin, and GIP in facilitating digestion and utilization of absorbed nutrients is summarized in **Figure 25–19**.

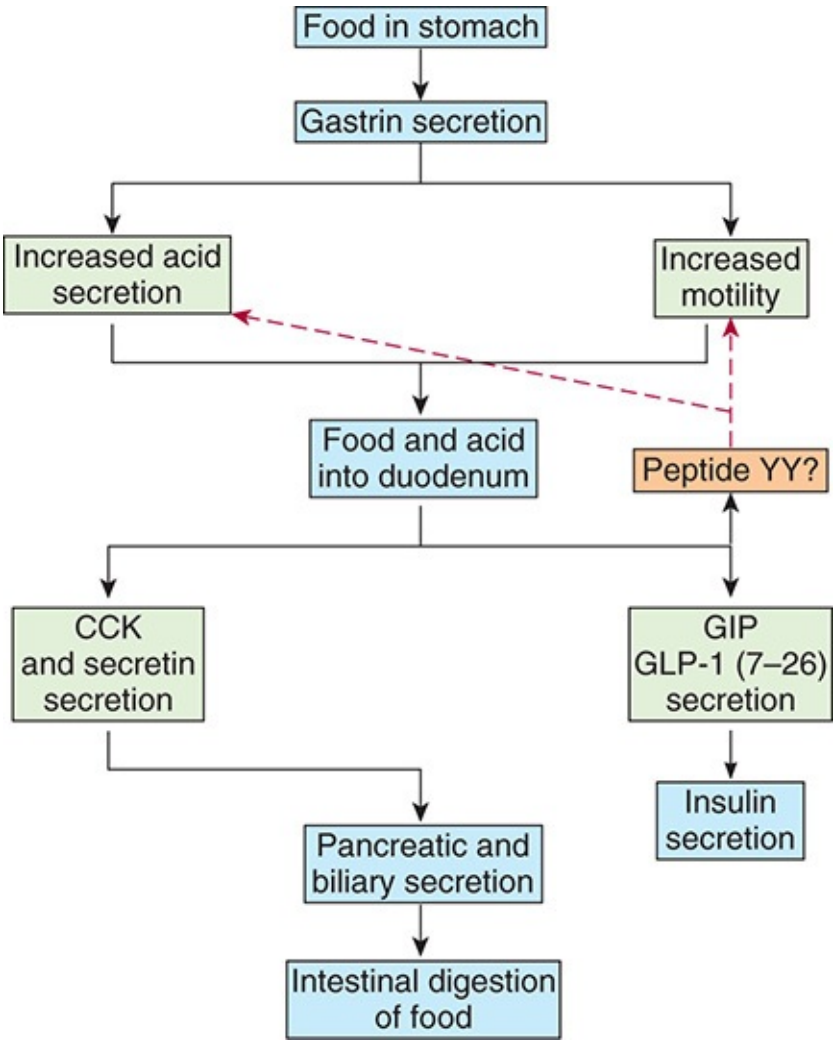


FIGURE 25–19 Integrated action of gastrointestinal hormones in regulating digestion and utilization of absorbed nutrients. The dashed arrows indicate inhibition. The exact identity of the hormonal factor or factors from the intestine that inhibit(s) gastric acid secretion and motility is unsettled, but it may be peptide YY.

VIP

VIP contains 28 amino acid residues. It is found in nerves in the gastrointestinal tract and thus is not itself a hormone, despite its similarities to secretin. VIP is, however, found in blood, in which it has a half-life of about 2 min. In the intestine, it markedly stimulates intestinal secretion of electrolytes and hence of water. Its other actions include relaxation of intestinal smooth muscle, including sphincters; dilation of peripheral blood vessels; and inhibition of gastric acid

secretion. It is also found in the brain and many autonomic nerves (see Chapter 7), where it often occurs in the same neurons as acetylcholine. It potentiates the action of acetylcholine in salivary glands. However, VIP and acetylcholine do not coexist in neurons that innervate other parts of the gastrointestinal tract. VIP-secreting tumors (VIPomas) have been described in patients with severe diarrhea.

MOTILIN

Motilin is a polypeptide containing 22 amino acid residues that is secreted by enterochromaffin cells and Mo cells in the stomach, small intestine, and colon. It acts on G-protein-coupled receptors on enteric neurons in the duodenum and colon and produces contraction of smooth muscle in the stomach and intestines in the period between meals (see Chapter 27).

SOMATOSTATIN

Somatostatin, the growth hormone–inhibiting hormone originally isolated from the hypothalamus, is secreted as a paracrine by D cells in the pancreatic islets (see Chapter 24) and by similar D cells in the gastrointestinal mucosa. It exists in tissues in two forms, somatostatin 14 and somatostatin 28, and both are secreted. Somatostatin inhibits the secretion of gastrin, VIP, GIP, secretin, and motilin. Its secretion is stimulated by acid in the lumen, and it acts in a paracrine fashion to mediate the inhibition of gastrin secretion produced by acid. It also inhibits pancreatic exocrine secretion; gastric acid secretion and motility; gallbladder contraction; and the absorption of glucose, amino acids, and triglycerides.

OTHER GASTROINTESTINAL PEPTIDES

Peptide YY

The structure of peptide YY is discussed in Chapter 24. It inhibits gastric acid secretion and motility and is a good candidate to be the gastric inhibitory peptide (Figure 25–19). Its release from the jejunum is stimulated by fat.

Others

Ghrelin is secreted primarily by the stomach and appears to play an important role in the central control of feeding (see Chapter 26). It also stimulates growth hormone secretion by acting directly on receptors in the pituitary (see Chapter 18). Its levels increase before a meal and exogenous administration of this hormone markedly enhances appetite and food intake. Secretion is markedly reduced in patients who have undergone gastric bypass surgery for the treatment of severe obesity, which may contribute to the efficacy of the treatment.

Substance P is found in endocrine and nerve cells in the gastrointestinal tract and may enter the circulation. It increases the motility of the small intestine. The neurotransmitter **GRP** is present in the vagal nerve endings that terminate on G cells and is the neurotransmitter producing vagally mediated increases in gastrin secretion. **Glucagon** from the gastrointestinal tract may be responsible (at least in part) for the hyperglycemia seen after pancreatectomy.

Guanylin is a gastrointestinal polypeptide that binds to guanylyl cyclase. It is made up of 15 amino acid residues and is secreted by cells of the intestinal mucosa. Stimulation of guanylyl cyclase increases the concentration of intracellular cyclic 3′,5′-guanosine monophosphate (cGMP), and this in turn causes increased secretion of Cl^- into the intestinal lumen. Guanylin appears to act predominantly in a paracrine fashion, and it is produced in cells from the pylorus to the rectum. In an interesting example of molecular mimicry, the heat-stable enterotoxin of certain diarrhea-producing strains of *Escherichia coli* has a structure very similar to guanylin and activates guanylin receptors in the intestine. Guanylin receptors are also found in the kidneys, the liver, and the female reproductive tract, and guanylin may act in an endocrine fashion to regulate fluid movement in these tissues as well, and particularly to integrate the actions of the intestine and kidneys.

THE ENTERIC NERVOUS SYSTEM

Two major networks of nerve fibers are intrinsic to the gastrointestinal tract: the **myenteric plexus** between the outer longitudinal and middle circular muscle layers, and the **submucous plexus** between the middle circular layer and the mucosa (Figure 25–2). Collectively, these neurons constitute the **enteric nervous system.** The system contains about 100 million sensory neurons, interneurons, and motor neurons in humans—as many as are found in the whole spinal cord—and the system is probably best viewed as a displaced part of the CNS that is concerned with the regulation of gastrointestinal function. It is sometimes referred to as the “little brain” for this reason. It is connected to the

CNS by parasympathetic and sympathetic fibers but can function autonomously without these connections (see below). The myenteric plexus innervates the longitudinal and circular smooth muscle layers and is concerned primarily with motor control, whereas the submucous plexus innervates the glandular epithelium, intestinal endocrine cells, and submucosal blood vessels and is primarily involved in the control of intestinal secretion. The neurotransmitters in the system include acetylcholine, the amines norepinephrine and serotonin, the amino acid γ-aminobutyrate (GABA), the purine adenosine triphosphate (ATP), the gases NO and CO, and many different peptides and polypeptides. Some of these peptides also act in a paracrine fashion, and some enter the bloodstream, becoming hormones. Not surprisingly, most of them are also found in the brain.

EXTRINSIC INNERVATION

The intestine receives a dual extrinsic innervation from the autonomic nervous system, with parasympathetic cholinergic activity generally increasing the activity of intestinal smooth muscle and sympathetic noradrenergic activity generally decreasing it while causing sphincters to contract. The preganglionic parasympathetic fibers consist of about 2000 vagal efferents and other efferents in the sacral nerves. They generally end on cholinergic nerve cells of the myenteric and submucous plexuses. The sympathetic fibers are postganglionic, but many of them end on postganglionic cholinergic neurons, where the norepinephrine they secrete inhibits acetylcholine secretion by activating α_2 presynaptic receptors. Other sympathetic fibers appear to end directly on intestinal smooth muscle cells. The electrical properties of intestinal smooth muscle are discussed in Chapter 5. Still other fibers innervate blood vessels, where they produce vasoconstriction. It appears that the intestinal blood vessels have a dual innervation: they have an extrinsic noradrenergic innervation and an intrinsic innervation by fibers of the enteric nervous system. VIP and NO are among the mediators in the intrinsic innervation, which seems, among other things, to be responsible for the increase in local blood flow **(hyperemia)** that accompanies digestion of food. It is unsettled whether the blood vessels have an additional cholinergic innervation.

GASTROINTESTINAL (MUCOSAL) IMMUNE SYSTEM

The mucosal immune system was mentioned in Chapter 3, but it bears repeating here that the continuity of the intestinal lumen with the outside world also makes the gastrointestinal system an important portal for infection. Similarly, the intestine benefits from interactions with a complex community of commensal (ie, nonpathogenic) bacteria that provide beneficial metabolic functions as well as likely increasing resistance to pathogens. In the face of this constant microbial stimulation, it is not surprising that the intestine of mammals has developed a sophisticated set of both innate and adaptive immune mechanisms to distinguish friend from foe. Indeed, the intestinal mucosa contains more lymphocytes than are found in the circulation, as well as large numbers of inflammatory cells that are placed to rapidly defend the mucosa if epithelial defenses are breached. It is likely that immune cells, and their products, also impact the physiologic function of the epithelium, endocrine cells, nerves and smooth muscle, particularly at times of infection and if inappropriate immune responses are perpetuated, such as in inflammatory bowel diseases (see Chapter 3).

A massive amount of information is emerging about the nature and complexity of the microbes that make up the intestinal microbiota, and the potential roles of these microbes in health and disease both within and beyond the gastrointestinal system. The number of microbes increases dramatically as one moves aborally, and is highest in the colon where the microbiota is dominated by strict anaerobes. The microbes supply a large number of metabolites that cannot be produced by mammalian cells, including vitamins, and salvage nutrients that cannot otherwise be digested by pancreatic enzymes. Thus, dietary fiber is digested to yield short chain fatty acids that can subsequently be absorbed across the colonic epithelium. Bacteria also deconjugate bile acids, allowing their unconjugated forms (which are more hydrophobic) to be passively absorbed back into the portal circulation. It is also likely that the collective properties of the microbiota increase resistance to colonization by pathogenic microorganisms, educate the mucosal immune system during early post-natal life, and send signals to the brain that modify behavior. When the balance of the gut microbial community is deranged (known as **dysbiosis**) as a result of disease or the use of broad spectrum antibiotics, there may be an impact on intestinal physiology and/or an opportunity for pathogens to colonize the gut. For example, *Clostridium difficile* is a pathogen that often overgrows in the gut of hospitalized patients who have received antibiotics. It causes significant gastrointestinal symptoms and can be very difficult to eradicate. Recent clinical studies have shown that patients suffering from recurrent *C. difficile* infections can often benefit dramatically from the transplant (via enema) of the fecal microbes from a healthy donor.

GASTROINTESTINAL (SPLANCHNIC) CIRCULATION

A final general point that should be made about the gastrointestinal tract relates to its unusual circulatory features. The blood flow to the stomach, intestines, pancreas, and liver is arranged in a series of parallel circuits, with all the blood from the intestines and pancreas draining via the portal vein to the liver **(Figure 25–20)**. The blood from the intestines, pancreas, and spleen drains via the hepatic portal vein to the liver and from the liver via the hepatic veins to the inferior vena cava. The viscera and the liver receive about 30% of the cardiac output via the celiac, superior mesenteric, and inferior mesenteric arteries. The liver receives about 1300 mL/min from the portal vein and 500 mL/min from the hepatic artery during fasting, and the portal supply increases still further after meals.

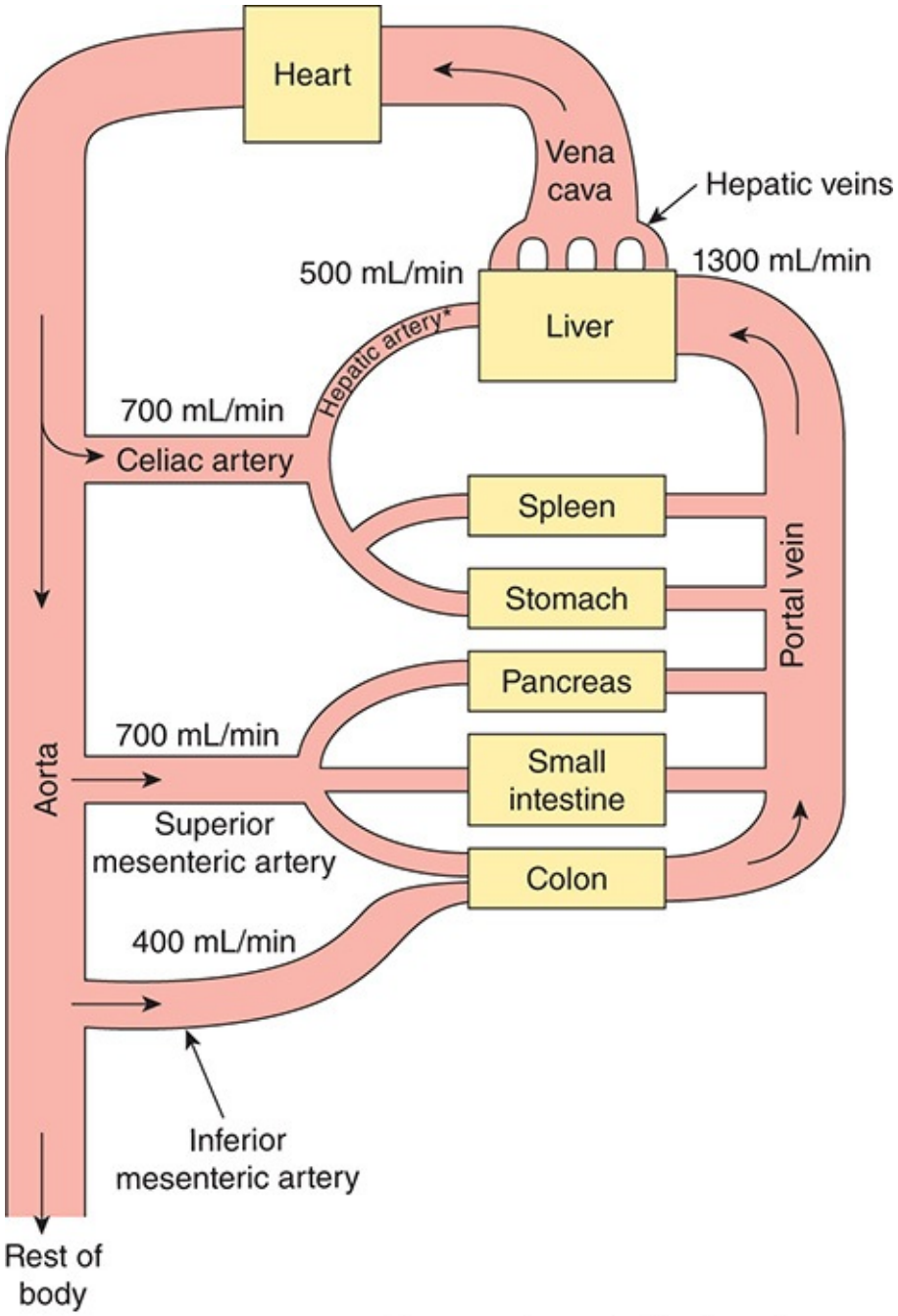


FIGURE 25–20 Schematic of the splanchnic circulation under fasting conditions. Note that even during fasting, the liver receives the majority of its blood supply via the portal vein.

INTESTINAL FLUID & ELECTROLYTE

TRANSPORT

The intestine supplies a fluid environment in which the processes of digestion and absorption can occur. Then, when the meal has been assimilated, fluid used during digestion and absorption is reclaimed by transport back across the epithelium to avoid dehydration. Water moves passively into and out of the gastrointestinal lumen, driven by electrochemical gradients established by the active transport of ions and other solutes. In the period after a meal, much of the fluid reuptake is driven by the coupled transport of nutrients, such as glucose, with sodium ions. In the period between meals, absorptive mechanisms center exclusively around electrolytes. In both cases, secretory fluxes of fluid are largely driven by the active transport of chloride ions into the lumen, although absorption still predominates overall.

Overall water balance in the gastrointestinal tract is summarized in **Table 25–5**. The intestines are presented each day with about 2000 mL of ingested fluid plus 7000 mL of secretions from the mucosa of the gastrointestinal tract and associated glands. Ninety-eight percent of this fluid is reabsorbed, with a daily fluid loss of only 200 mL in the stools.

TABLE 25–5 Daily water turnover (mL) in the gastrointestinal tract.

Ingested		2000
Endogenous secretions		7000
Salivary glands	1500	
Stomach	2500	
Bile	500	
Pancreas	1500	
Intestine	+1000	
	7000	
Total input		9000
Reabsorbed		8800
Jejunum	5500	
Ileum	2000	
Colon	+1300	
	8800	
Balance in stool		200

Data from Moore EW: *Physiology of Intestinal Water and Electrolyte Absorption.* American Gastroenterological Association, 1976.

In the small intestine, secondary active transport of Na^+ is important in bringing about absorption of glucose, some amino acids, and other substances such as bile acids (see above). Conversely, the presence of glucose in the intestinal lumen facilitates the reabsorption of Na^+. In the period between meals, when nutrients are not present, sodium and chloride are absorbed together from the lumen by the coupled activity of a sodium/hydrogen exchanger (NHE) and chloride/bicarbonate exchanger in the apical membrane, in a so-called electroneutral mechanism **(Figure 25–21)**. Water then follows to maintain an osmotic balance. In the colon, moreover, an additional electrogenic mechanism for sodium absorption is expressed, particularly in the distal colon. In this mechanism, sodium enters across the apical membrane via an ENaC (epithelial sodium) channel that is identical to that expressed in the distal tubule of the kidney **(Figure 25–22)**. This underpins the ability of the colon to desiccate the stool and ensure that only a small portion of the fluid load used daily in the digestion and absorption of meals is lost from the body. Following a low-salt diet, increased expression of ENaC in response to aldosterone increases the ability to reclaim sodium from the stool.

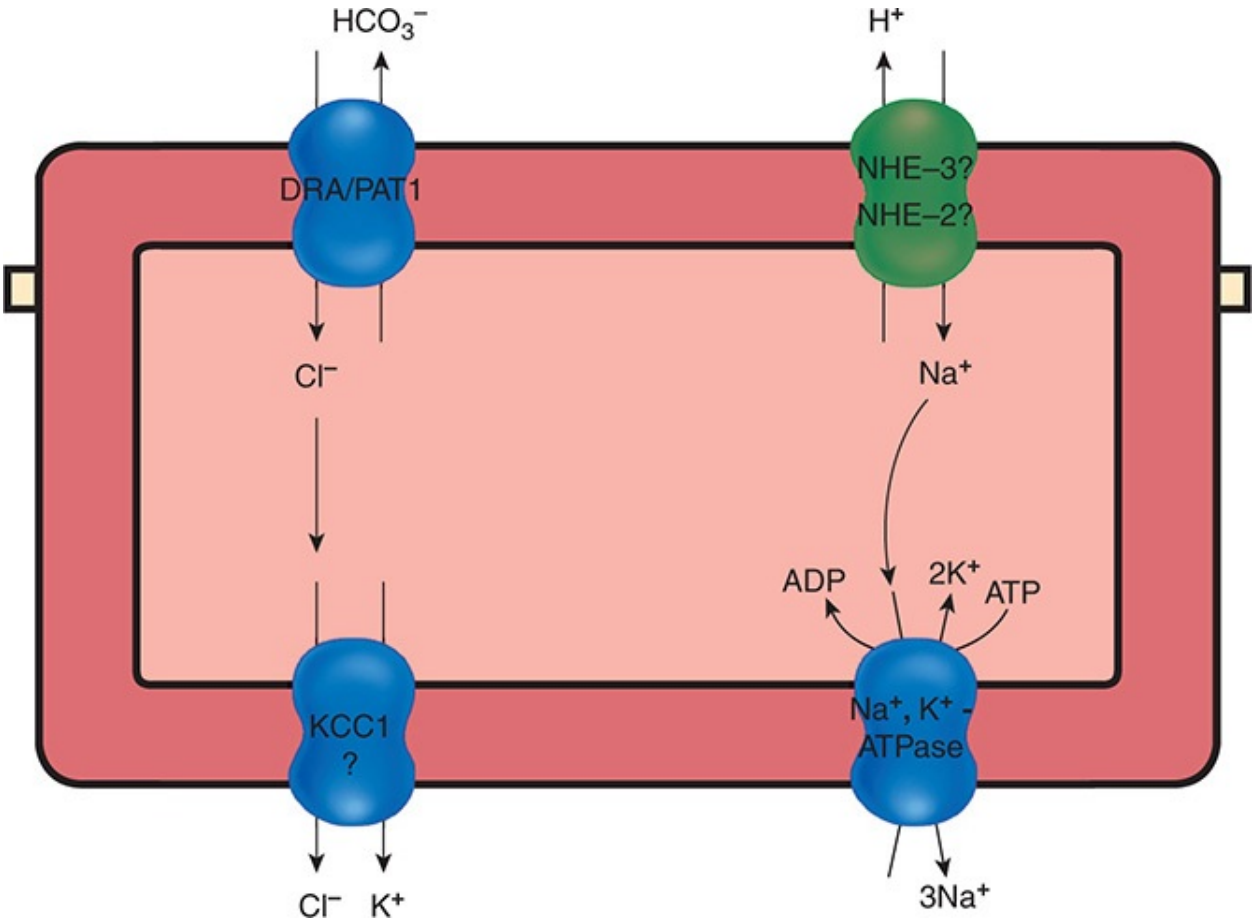


FIGURE 25–21 Electroneutral NaCl absorption in the small intestine and colon. NaCl enters across the apical membrane via the coupled activity of a sodium/hydrogen exchanger (NHE) and a chloride/bicarbonate exchanger (down-regulated in adenoma (DRA), or putative anion transporter-1 (PAT1)). A putative potassium/chloride cotransporter (KCC1) in the basolateral membrane provides for chloride exit, whereas sodium is extruded by the Na^+, K^+ ATPase.

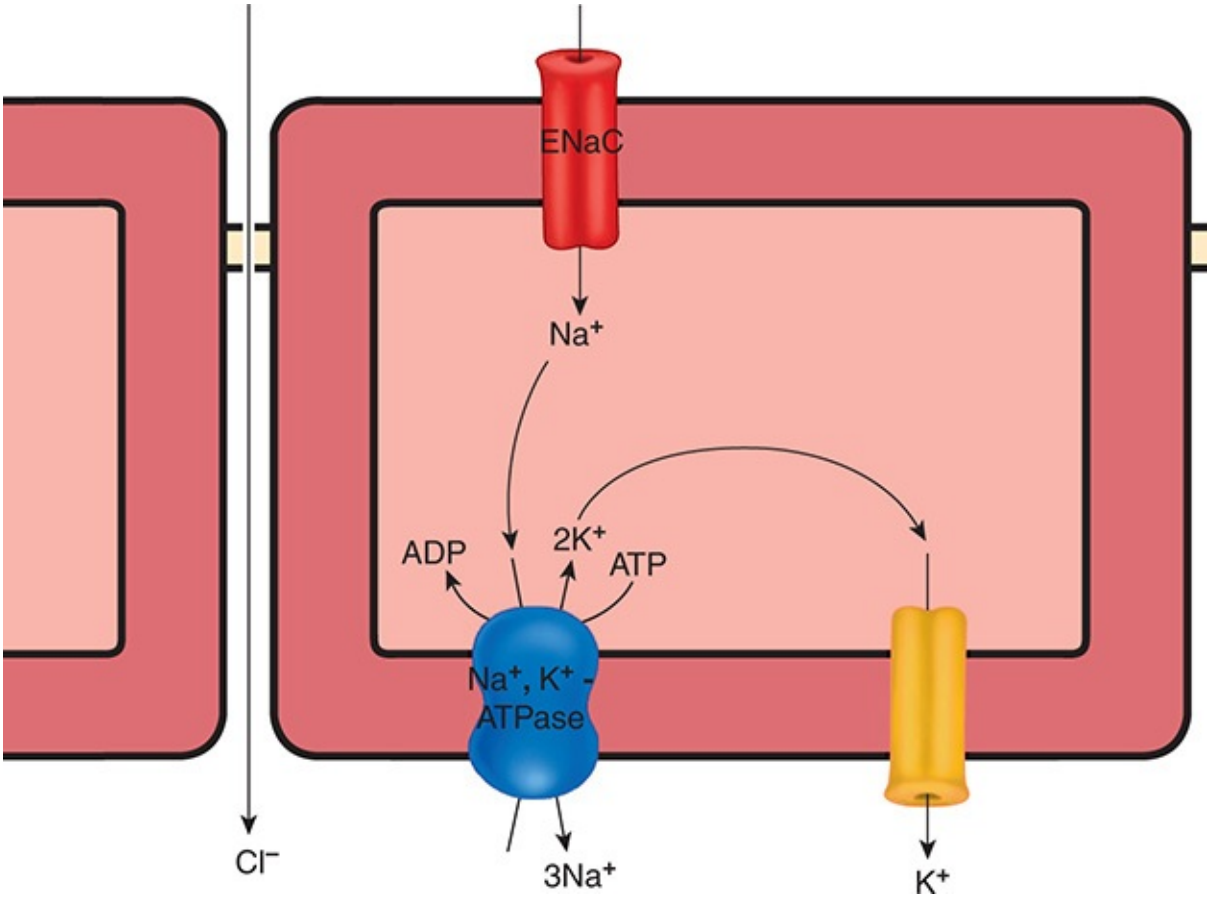


FIGURE 25–22 Electrogenic sodium absorption in the colon. Sodium enters the epithelial cell via apical epithelial sodium channels (ENaC), and exits via the Na^+, K^+ ATPase.

Despite the predominance of absorptive mechanisms, secretion also takes place continuously throughout the small intestine and colon to adjust the local fluidity of the intestinal contents as needed for mixing, diffusion, and movement of the meal and its residues along the length of the gastrointestinal tract. Cl^- normally enters enterocytes from the interstitial fluid via Na^+–K^+–$2Cl^-$ cotransporters in their basolateral membranes **(Figure 25–23)**, and the Cl^- is then secreted into the intestinal lumen via channels that are regulated by various protein kinases. The cystic fibrosis transmembrane conductance regulator (CFTR) channel that is defective in the disease of cystic fibrosis is quantitatively most important, and is activated by protein kinase A and hence by cAMP **(Clinical Box 25–2)**.

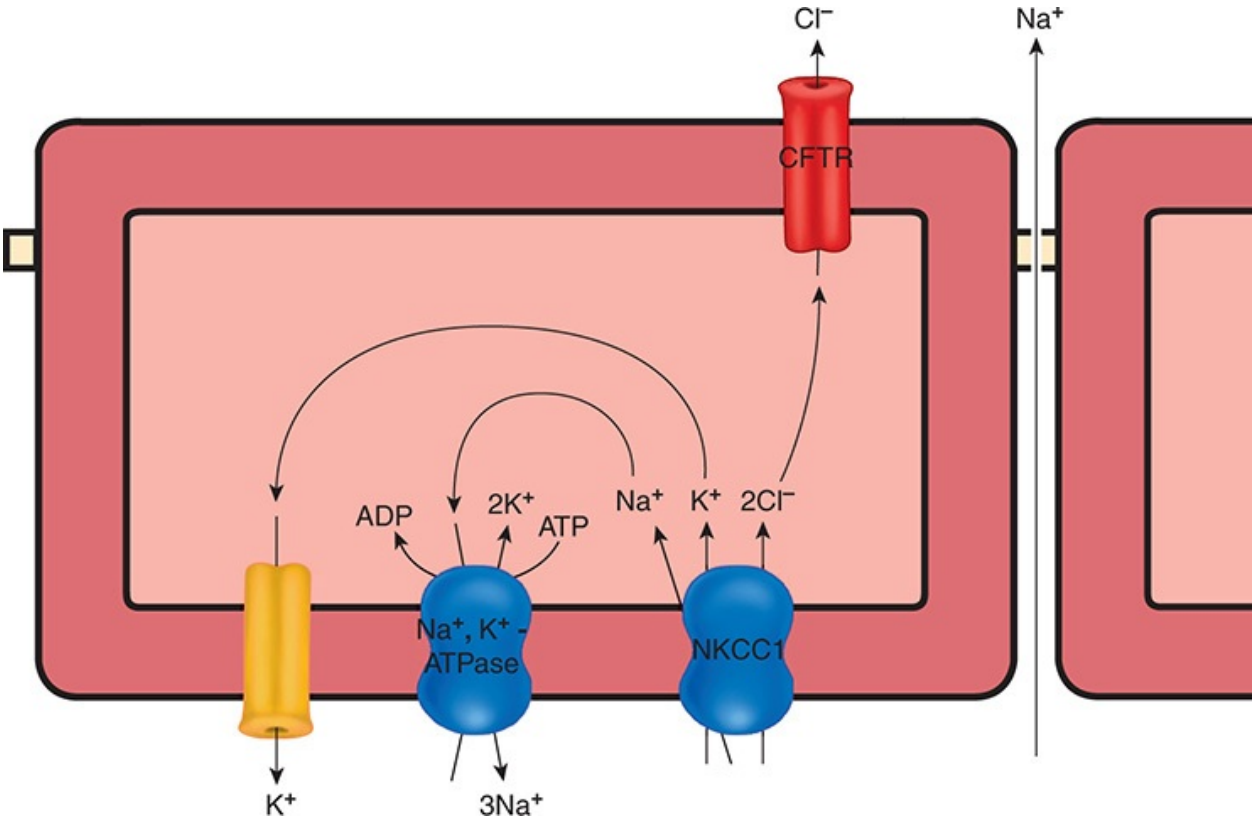


FIGURE 25–23 Chloride secretion in the small intestine and colon. Chloride uptake occurs via the sodium/potassium/2 chloride cotransporter, NKCC1. Chloride exit is via the cystic fibrosis transmembrane conductance regulator (CFTR) as well as perhaps via other chloride channels, not shown.

Water moves into or out of the intestine until the osmotic pressure of the intestinal contents equals that of the plasma. The osmolality of the duodenal contents may be hypertonic or hypotonic, depending on the meal ingested, but by the time the meal enters the jejunum, its osmolality is close to that of plasma. This osmolality is maintained throughout the rest of the small intestine; the osmotically active particles produced by digestion are removed by absorption, and water moves passively out of the gut along the osmotic gradient thus generated. In the colon, Na^+ is absorbed and water moves passively with it, again along the osmotic gradient. **Saline cathartics** such as magnesium sulfate are poorly absorbed salts that retain their osmotic equivalent of water in the intestine, thus increasing intestinal volume and consequently exerting a laxative effect. The amount of fluid in the lumen will also depend on the rate of intestinal motility. When motility is slow, there is more time for absorption. Several antidiarrheal drugs exert their effects by desynchronizing muscle contraction in the gut and slowing propulsive motility (see also Chapter 27).

CLINICAL BOX 25–2

Cholera

Cholera is a severe secretory diarrheal disease that often occurs in epidemics associated with natural disasters where normal sanitary practices break down. Along with other secretory diarrheal illnesses produced by bacteria and viruses, cholera causes a significant amount of morbidity and mortality, particularly among the young and in developing countries. The cAMP concentration in intestinal epithelial cells is increased in cholera. The cholera bacillus stays in the intestinal lumen, but it produces a toxin that binds to receptors on the apical membrane of intestinal epithelial cells, and this permits the A subunit of the toxin to enter the cell. The toxin binds adenosine diphosphate ribose to the α subunit of G_s, inhibiting its GTPase activity (see Chapter 2). Thereafter, the constitutively activated G-protein produces prolonged stimulation of adenylyl cyclase and a marked increase in the intracellular cAMP concentration. In addition to increased Cl^- secretion, the function of NHE3 is reduced, thus reducing NaCl absorption. The resultant increase in electrolyte and water content of the intestinal contents causes the diarrhea. However, Na^+, K^+ ATPase, and the Na^+/glucose cotransporter are unaffected, so coupled reabsorption of glucose and Na^+ bypasses the defect.

THERAPEUTIC HIGHLIGHTS

Treatment for cholera is mostly supportive, since the infection will eventually clear, although antibiotics are sometimes used. The most important therapeutic approach is to ensure that the large volumes of fluid, along with electrolytes, lost to the stool are replaced to avoid dehydration. Stool volumes can approach 20 L per day. When sterile supplies are available, fluids and electrolytes can most conveniently be replaced intravenously. However, this is often not possible in the setting of an epidemic. Instead, the persistent activity of the Na^+/glucose cotransporter provides a physiologic basis for the treatment of Na^+ and water loss by oral administration of solutions containing NaCl and glucose. Oral rehydration solution, a prepackaged mixture of sugar and salt to be dissolved in water, is a simple remedy that has dramatically reduced mortality in epidemics of cholera and other diarrheal diseases in developing countries.

Some K^+ is secreted into the intestinal lumen, especially as a component of mucus. K^+ channels are present in the luminal as well as the basolateral membrane of the enterocytes of the colon, so K^+ is secreted into the colon. In addition, K^+ moves passively down its electrochemical gradient. The accumulation of K^+ in the colon is partially offset by H^+–K^+ ATPase in the luminal membrane of cells in the distal colon, with resulting active transport of K^+ into the cells. Nevertheless, loss of ileal or colonic fluids in chronic diarrhea can lead to severe hypokalemia. When the dietary intake of K^+ is high for a prolonged period, aldosterone secretion is increased and more K^+ enters the colonic lumen. This is due in part to the appearance of more Na^+, K^+ ATPase pumps in the basolateral membranes of the cells, with a consequent increase in intracellular K^+ and K^+ diffusion across the luminal membranes of the cells.

CHAPTER SUMMARY

- The gastrointestinal system evolved as a portal to permit controlled nutrient uptake in multicellular organisms and to excrete the residues of meals, as well as endogenous lipid-soluble metabolic waste products. It is functionally continuous with the outside environment and has a well-developed immune system.
- The gastrointestinal tract is lined by a continuous layer of columnar epithelial cells that are continuously replaced, and folded into structures known as crypts and villi. The lamina propria and submucosa lie below the epithelium and contain immune cells, blood vessels, and lymphatics. Circular and longitudinal muscle layers provide for motility functions.
- Digestive secretions serve to chemically alter the components of meals (particularly macromolecules) such that their constituents can be absorbed across the epithelium. Meal components are acted on sequentially by saliva, gastric juice, pancreatic juice, and bile, which contain enzymes, ions, water, and other specialized components.
- Gastrointestinal functions are regulated in an integrated fashion by endocrine, paracrine, and neurocrine mechanisms. Hormones and paracrine factors are released from enteroendocrine cells in response to signals coincident with the intake of meals.
- The enteric nervous system conveys information from the central nervous system to the gastrointestinal tract, but also often can activate programmed

responses of secretion and motility in an autonomous fashion.

- The intestine harbors an extensive mucosal immune system that regulates responses to the complex microbiota normally resident in the lumen, as well as defending the body against invasion by pathogens.
- The intestine has an unusual circulation, in that the majority of its venous outflow does not return directly to the heart, but rather is directed initially to the liver via the portal vein.
- The intestine and the organs that drain into it secrete about 8 L of fluid per day, which is added to water consumed in food and beverages. Most of this fluid is reabsorbed, leaving only approximately 200 mL to be lost to the stool. Fluid secretion and absorption are both dependent on the active epithelial transport of ions, nutrients, or both.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. A researcher conducts a study of the regulation of salivary secretion in a group of volunteer medical students under various conditions. Which of the following conditions would be expected to be associated with the lowest rates of secretion?

   A. Chewing gum

   B. Undergoing a mock dental exam

   C. Sleep

   D. Exposure to a nauseating odor

   E. Resting control conditions

2. A patient suffering from anemia comes to his physician complaining of frequent bouts of gastroenteritis. A blood test reveals antibodies directed against gastric parietal cells. The anemia in this patient is attributable to hyposecretion of which gastric product?

   A. Histamine

   B. Gastrin

   C. Pepsinogen

   D. Intrinsic factor

   E. Hydrochloric acid

3. A 50-year-old man comes to see his clinician complaining of severe epigastric

pain, frequent heartburn, and unexplained weight loss of 20 pounds over a 6-month period. He claims to have obtained no relief from over-the-counter H_2 antihistamine drugs. He is referred to a gastroenterologist, and upper endoscopy reveals erosions and ulcerations in the proximal duodenum and an increased output of gastric acid in the fasting state. The patient is most likely to have a tumor secreting which of the following hormones?

A. Secretin

B. Somatostatin

C. Motilin

D. Gastrin

E. Cholecystokinin

4. Which of the following has the highest pH?

A. Gastric juice

B. Colonic luminal contents

C. Pancreatic juice

D. Saliva

E. Contents of the intestinal crypts

5. A 60-year-old woman undergoes total pancreatectomy because of the presence of a tumor. Which of the following outcomes would *not* be expected after she recovers from the operation?

A. Steatorrhea

B. Hyperglycemia

C. Metabolic acidosis

D. Weight gain

E. Decreased absorption of amino acids

6. In a germ-free animal, which of the following will not be present?

A. Cholic acid

B. Lithocholic acid

C. Gastrin

D. Cholecystokinin

E. Trypsin

7. A woman suffering from Crohn's disease undergoes a surgical resection of her diseased terminal ileum. Following her recovery from surgery and reintroduction of a normal diet, which of the following will not be increased?

A. Hepatic synthesis of bile acids

B. Levels of conjugated bile acids in the portal vein

C. Stool fat

D. Stool volume

E. Hepatic uptake of unconjugated bile acids

8. Water is absorbed in the jejunum, ileum, and colon and excreted in the feces. Arrange these in order of the amount of water absorbed or excreted from the greatest to the smallest.

A. Colon, jejunum, ileum, feces

B. Feces, colon, ileum, jejunum

C. Jejunum, ileum, colon, feces

D. Colon, ileum, jejunum, feces

E. Feces, jejunum, ileum, colon

9. Following a natural disaster in Haiti, there is an outbreak of cholera among displaced persons living in a tent encampment. The affected individuals display severe diarrheal symptoms because of which of the following changes in intestinal transport?

A. Increased Na^+–K^+ cotransport in the small intestine

B. Increased K^+ secretion into the colon

C. Reduced K^+ absorption in the crypts of Lieberkühn

D. Increased Na^+ absorption in the small intestine

E. Increased Cl^- secretion into the intestinal lumen

CHAPTER 26

Digestion & Absorption of Nutrients

OBJECTIVES

After studying this chapter, you should be able to:

- Understand the basic principles of nutrition, how nutrients are delivered to the body, and the chemical processes needed to convert them to a form suitable for absorption.
- List the major dietary carbohydrates and define the luminal and brush border processes that produce absorbable monosaccharides as well as the transport mechanisms that provide for the uptake of these hydrophilic molecules.
- Understand the process of protein assimilation, and the ways in which it is comparable to, or converges from, that used for carbohydrates.
- Define the stepwise processes of lipid digestion and absorption and the role of bile acids in solubilizing the products of lipolysis.
- Understand the basis of the anatomic reserve for nutrient absorption, the adaptive processes that maintain absorption even when a portion of the bowel is lost, and the basis of the malabsorption syndrome and its impact.
- Identify the source and functions of short-chain fatty acids in the colon.
- Delineate the requirements for, and mechanisms of uptake for vitamins and minerals.

- Explain the processes that regulate overall food intake, as well as the mechanisms and consequences of obesity.

INTRODUCTION

The gastrointestinal system is the portal through which nutritive substances, vitamins, minerals, and fluids enter the body. Proteins, fats, and complex carbohydrates are broken down into absorbable units **(digested),** principally, although not exclusively, in the small intestine. The products of digestion and the vitamins, minerals, and water cross the mucosa and enter the lymph or the blood **(absorption).** The digestive and absorptive processes are the subject of this chapter.

Digestion of the major foodstuffs is an orderly process involving the action of a large number of **digestive enzymes** discussed in the previous chapter. Enzymes from the salivary glands attack carbohydrates (and fats in some species); enzymes from the stomach attack proteins and fats; and enzymes from the exocrine portion of the pancreas attack carbohydrates, proteins, lipids, DNA, and RNA. Other enzymes that complete the digestive process are found in the luminal membranes and the cytoplasm of the cells that line the small intestine. The action of the enzymes is aided by the hydrochloric acid secreted by the stomach and the bile secreted by the liver.

Most substances pass from the intestinal lumen into the enterocytes and then out of the enterocytes to the interstitial fluid. The processes responsible for movement across the luminal cell membrane are often quite different from those responsible for movement across the basal and lateral cell membranes to the interstitial fluid.

NUTRITIONAL PRINCIPLES

Certain substances are essential constituents of any human diet. An optimal diet includes, in addition to sufficient water (see Chapter 37), adequate calories, protein, fat, minerals, and vitamins.

CALORIC INTAKE & DISTRIBUTION

The caloric value of the dietary intake must be approximately equal to the energy

expended if body weight is to be maintained. In addition to the 2000 kcal/d necessary to meet basal needs, 500–2500 kcal/d (or more) are required to meet the energy demands of daily activities.

The distribution of the calories among carbohydrate, protein, and fat is determined partly by physiologic factors and partly by taste and economic considerations. A daily protein intake of 1 g/kg body weight to supply the eight nutritionally essential amino acids **(Table 26–1)** and other amino acids is desirable. The **essential amino acids** are those that cannot be synthesized by the human body, and thus must be obtained from dietary protein. The source of the protein is also important. **Grade I proteins,** the animal proteins of meat, fish, dairy products, and eggs, contain all amino acids, including the essential amino acids, in approximately the proportions required for protein synthesis and other uses. On the other hand, most plant proteins are **grade II** because they supply different proportions of amino acids and some lack one or more of the essential amino acids. The protein needs of vegetarians can be met by ingesting strategic mixtures of grade II proteins, but the intake must often be large because of amino acid wastage.

TABLE 26–1 Amino acids that are essential for human nutrition

Amino Acid Class	Amino Acids
Basic	Lysine, histidine
Aliphatic	Valine, leucine, isoleucine
Aromatic	Phenylalanine, tryptophan
Hydroxyl	Threonine
Sulfur	Methionine

Fat is the most compact form of food, since it supplies 9.3 kcal/g. However, often it is also the most expensive. Indeed, internationally there is a reasonably good positive correlation between fat intake and standard of living. Typically, Western diets have contained large amounts (100 g/d or more). In Central and South American Indian communities where corn (carbohydrate) is the dietary staple, adults live without ill effects for years on a very low fat intake. Therefore, provided that the needs for essential fatty acids (FA) are met, a low-fat intake does not seem to be harmful, and a diet low in saturated fats is desirable.

Carbohydrate is the cheapest source of calories and provides 50% or more of

the calories in most diets. In the average middle-class American diet, approximately 50% of the calories come from carbohydrate, 15% from protein, and 35% from fat. When calculating dietary needs, it is usual to meet the protein requirement first and then split the remaining calories between fat and carbohydrate, depending on taste, income, and other factors. For example, a 65-kg man who is moderately active needs about 2800 kcal/d to maintain his weight. He should eat at least 65 g of protein daily, supplying 267 (65 × 4.1) kcal. A reasonable figure for fat intake is 50–60 g. The rest of the caloric requirement can be met by supplying carbohydrate.

DIGESTION & ABSORPTION: CARBOHYDRATES

DIGESTION

The principal dietary carbohydrates are polysaccharides, disaccharides, and monosaccharides. Starches (glucose polymers) and their derivatives are the only polysaccharides that are digested to any degree in the human gastrointestinal tract by human enzymes. Amylopectin, which typically constitutes around 75% of dietary starch, is a branched molecule, whereas amylose is a straight chain with only α1:4 linkages **(Figure 26–1)**. The disaccharides **lactose** (milk sugar) and **sucrose** (table sugar) are also ingested, along with the monosaccharides fructose and glucose.

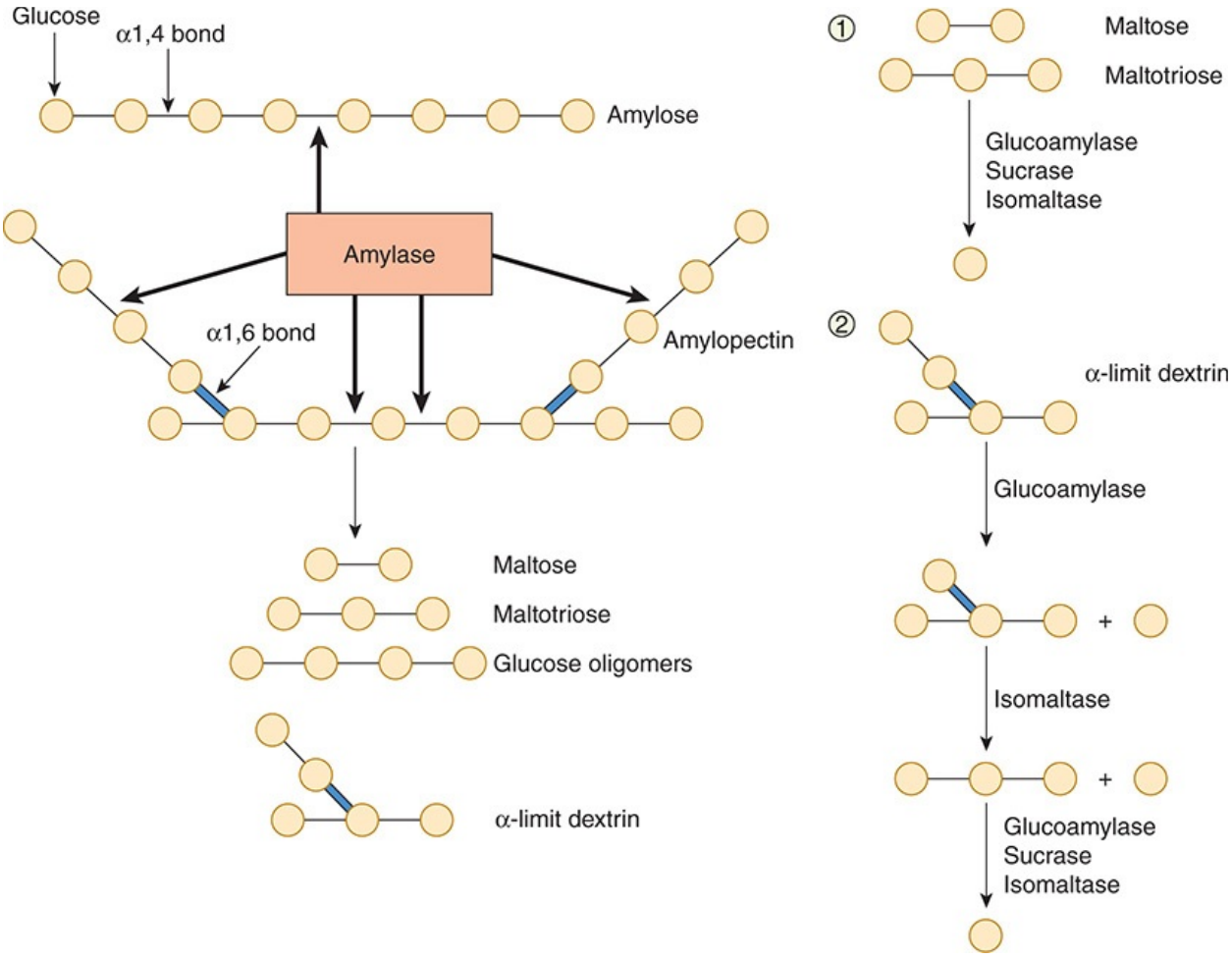


FIGURE 26–1 Left: Structure of amylose and amylopectin, which are polymers of glucose (indicated by circles). These molecules are partially digested by the enzyme amylase, yielding the products shown at the bottom of the figure. **Right:** Brush border hydrolases responsible for the sequential digestion of the products of luminal starch digestion (1, linear oligomers; 2, α-limit dextrins).

In the mouth, starch is attacked by salivary α-amylase. The optimal pH for this enzyme is 6.7. However, it remains partially active even once it moves into the stomach, despite the acidic gastric juice, because the active site is protected in the presence of substrate to some degree. In the small intestine, both the salivary and the pancreatic α-amylase also act on the ingested polysaccharides. Both enzymes hydrolyze internal α1:4 linkages but spare α1:6 linkages and terminal α1:4 linkages. Consequently, the end products of α-amylase digestion are oligosaccharides: the disaccharide **maltose;** the trisaccharide **maltotriose;** and **α-limit dextrins,** branched polymers of glucose containing an average of

about eight glucose molecules with α1:6 linkages (Figure 26–1).

The oligosaccharidases responsible for the further digestion of the starch derivatives are located in the brush border of small intestinal epithelial cells (Figure 26–1). Some of these enzymes have more than one substrate. **Isomaltase** is mainly responsible for hydrolysis of α1:6 linkages. Along with **maltase** and **sucrase,** it also breaks down maltotriose and maltose. Sucrase and isomaltase are initially synthesized as a single glycoprotein chain that is inserted into the brush border membrane. It is then hydrolyzed by pancreatic proteases into sucrase and isomaltase subunits.

Sucrase hydrolyzes sucrose into a molecule of glucose and a molecule of fructose. In addition, **lactase** hydrolyzes lactose to glucose and galactose. Overall, brush border digestion of carbohydrates generates monosaccharides at high concentrations at precisely the location where they will be absorbed. This may serve to sequester them from bacteria and avoid bacterial overgrowth in the small intestine.

Deficiency of one or more of the brush border oligosaccharidases may cause diarrhea, bloating, and flatulence after ingestion of sugar **(Clinical Box 26–1)**. The diarrhea is due to the increased number of osmotically active oligosaccharide molecules that remain in the intestinal lumen, causing the volume of the intestinal contents to increase. In the colon, bacteria break down some of the oligosaccharides, further increasing the number of osmotically active particles. The bloating and flatulence are due to the production of gas (CO_2 and H_2) from disaccharide residues in the lower small intestine and colon.

ABSORPTION

Hexoses are rapidly absorbed across the wall of the small intestine **(Table 26–2)**. Essentially all the hexoses are removed before the remains of a meal reach the terminal part of the ileum. The sugar molecules pass from the mucosal cells to the blood in the capillaries draining into the portal vein.

TABLE 26–2 Normal transport of substances by the intestine and location of maximum absorption or secretion.[a]

Absorption of	Small Intestine			Colon
	Upper[b]	Mid	Lower	
Sugars (glucose, galactose, etc)	++	+++	++	0
Amino acids	++	++	++	0
Water-soluble and fat-soluble vitamins except vitamin B_{12}	+++	++	0	0
Betaine, dimethylglycine, sarcosine	+	++	++	?
Antibodies in newborns	+	++	+++	?
Pyrimidines (thymine and uracil)	+	+	?	?
Long-chain fatty acid absorption and conversion to triglyceride	+++	++	+	0
Bile acids	+	+	+++	+
Vitamin B_{12}	0	+	+++	0
Na^+	+++	++	+++	+++
K^+	+	+	+	Sec
Ca^{2+}	+++	++	+	?
Fe^{2+}	+++	+	+	?
Cl^-	+++	++	+	+
SO_4^{2-}	++	+	0	?

[a]Amount of absorption is graded + to +++. Sec, secreted when luminal K^+ is low.

[b]Upper small intestine refers primarily to jejunum, although the duodenum is similar in most cases studied (with the notable exception that the duodenum secretes HCO_3^- and shows little net absorption or secretion of NaCl).

The transport of glucose and galactose depends on Na^+ in the intestinal lumen; a high concentration of Na^+ on the mucosal surface of the cells facilitates sugar influx into the epithelial cells while a low concentration inhibits sugar influx into the epithelial cells. This is because these sugars and Na^+ share the same **cotransporter,** or **symport,** the **sodium-dependent glucose transporter** (SGLT, Na^+ glucose cotransporter) **(Figure 26–2)**. There are two members of this family of transporters, SGLT-1 and SGLT-2. SGLT-1 is responsible for uptake of dietary glucose from the gut, whereas SGLT-2 is responsible for glucose transport out of the renal tubules (see Chapter 37).

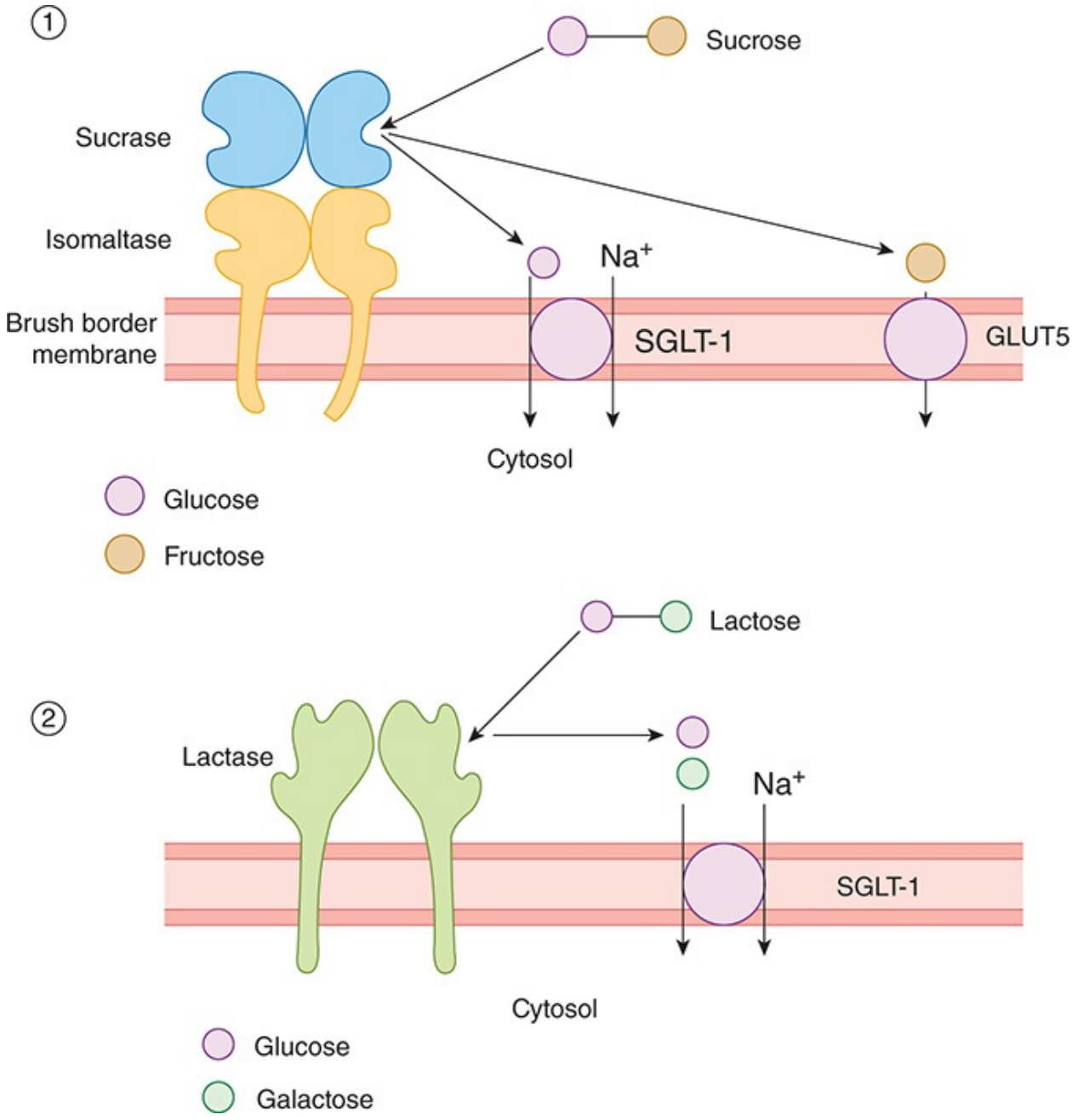


FIGURE 26–2 Brush border digestion and assimilation of the disaccharides sucrose (panel 1) and lactose (panel 2). Uptake of glucose and galactose is driven secondarily by the low intracellular sodium concentration established by the basolateral Na^+, K^+ ATPase (not shown). SGLT-1, sodium-glucose cotransporter-1.

Because the intracellular Na^+ concentration is low in intestinal epithelial cells (as it is in other cells), Na^+ moves into the cell along its concentration gradient.

Glucose moves with the Na^+ and is released in the cell (Figure 26–2). The Na^+ is then exported basolaterally by the Na^+, K^+ ATPase, and any glucose that is not needed for cellular requirements leaves the cell by facilitated diffusion via GLUT2 into the interstitium and thence to the capillaries. Thus, glucose transport is an example of secondary active transport (see Chapter 2); the energy for glucose transport is provided indirectly by the active transport of Na^+ out of the cell. This maintains the concentration gradient across the luminal border of the cell, so that more Na^+ and consequently more glucose enter. When the Na^+/glucose cotransporter is congenitally defective, the resulting **glucose/galactose malabsorption** causes severe diarrhea that is often fatal if glucose and galactose are not promptly removed from the diet. Glucose and its polymers can also be used to retain Na^+ in diarrheal disease, as was discussed in Chapter 25.

CLINICAL BOX 26–1

Lactose Intolerance

In most mammals and in many races of humans, intestinal lactase activity is high at birth, then declines to low levels during childhood and adulthood. The low lactase levels are associated with intolerance to milk **(lactose intolerance).** Most Europeans and their American descendants retain sufficient intestinal lactase activity in adulthood; the incidence of lactase deficiency in northern and western Europeans is only about 15%. However, the incidence in blacks, American Indians, Asians, and Mediterranean populations is 70–100%. When such individuals ingest dairy products, they are unable to digest lactose sufficiently, and so symptoms such as bloating, pain, gas, and diarrhea are produced by the unabsorbed osmoles that are subsequently digested by colonic bacteria.

THERAPEUTIC HIGHLIGHTS

The simplest treatment for lactose intolerance is to avoid dairy products in the diet, but this can sometimes be challenging (or undesirable for the individual who loves ice cream). Symptoms can be ameliorated by administration of commercial lactase preparations, but this is expensive.

As indicated, SGLT-1 also transports galactose, but fructose utilizes a different mechanism. Its absorption is independent of Na^+ or the transport of glucose and galactose; it is transported instead by facilitated diffusion from the intestinal lumen into the enterocytes by GLUT5 and out of the enterocytes into the interstitium by GLUT2. Some fructose is also converted to glucose in mucosal cells.

Insulin has little effect on intestinal transport of sugars. In this respect, intestinal absorption resembles glucose reabsorption in the proximal convoluted tubules of the kidneys (see Chapter 37); neither process requires phosphorylation, and both are essentially normal in diabetes but are depressed by the drug phlorizin. The maximal rate of glucose absorption from the intestine is about 120 g/h.

PROTEINS & NUCLEIC ACIDS

PROTEIN DIGESTION

Protein digestion begins in the stomach, where pepsin cleaves some of the peptide linkages. Like many of the other enzymes concerned with protein digestion, pepsin is secreted in the form of an inactive precursor **(proenzyme)** and activated in the gastrointestinal tract. The pepsin precursor is called pepsinogen and is activated by gastric acid.

Pepsin hydrolyzes the bonds between aromatic amino acids such as phenylalanine or tyrosine and a second amino acid, so the products of peptic digestion are polypeptides of very diverse sizes. Because pepsin has a pH optimum of 1.6–3.2, its action is terminated when the gastric contents are mixed with the alkaline pancreatic juice in the duodenum and jejunum. The pH of the intestinal contents in the duodenal bulb is 3.0–4.0, but rapidly rises; in the rest of the duodenum it is about 6.5.

In the small intestine, the polypeptides formed by digestion in the stomach are further digested by the powerful proteolytic enzymes of the pancreas and intestinal mucosa. Trypsin, chymotrypsin, and elastase act at interior peptide bonds in the peptide molecules and are called **endopeptidases.** These powerful protein-splitting enzymes of the pancreatic juice are secreted as inactive proenzymes. The formation of the active endopeptidases occurs only when they have reached their site of action, secondary to the action of the brush border hydrolase, **enterokinase** (**Figure 26–3**). Thus, trypsinogen is converted to the active enzyme trypsin by **enterokinase** when the pancreatic juice enters the

duodenum. Enterokinase contains 41% polysaccharide, and this high polysaccharide content apparently prevents it from being digested itself before it can exert its effect. Trypsin converts chymotrypsinogen into chymotrypsin and other proenzymes into active enzymes (Figure 26–3). Trypsin can also activate trypsinogen; therefore, once some trypsin is formed, there is an auto-catalytic chain reaction. Enterokinase deficiency occurs as a congenital abnormality and leads to protein malnutrition.

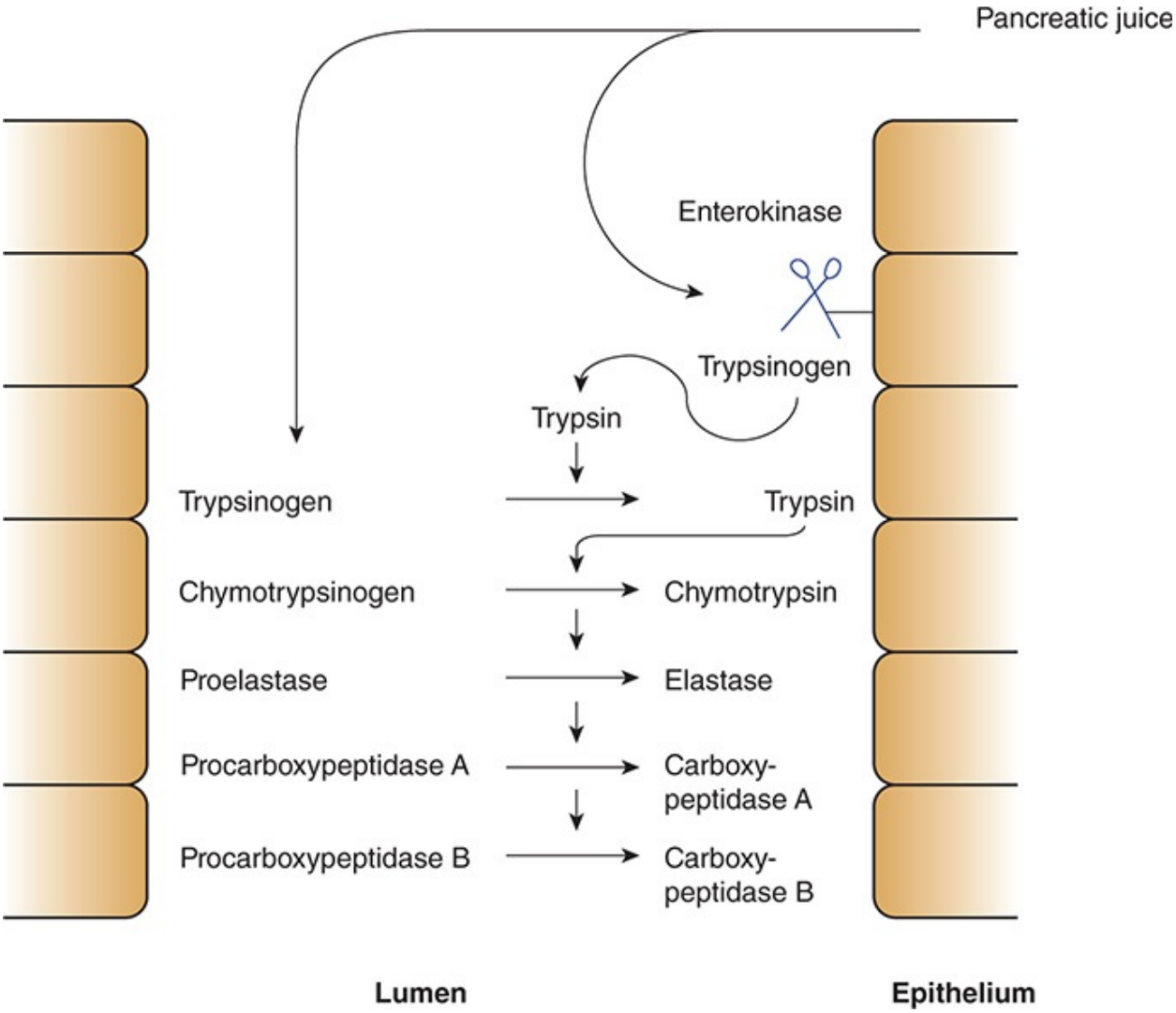


FIGURE 26–3 Mechanism to avoid activation of pancreatic proteases until they are in the duodenal lumen. Pancreatic juice contains proteolytic enzymes in their inactive, precursor forms. When the juice enters the duodenal lumen, trypsinogen contacts enterokinase expressed on the apical surface of enterocytes. Trypsinogen is thereby cleaved to trypsin, which in turn can activate additional trypsin molecules as well as the remaining proteolytic enzymes.

The carboxypeptidases of the pancreas are **exopeptidases** that hydrolyze the amino acids at the carboxyl ends of the polypeptides **(Figure 26–4)**. Some free amino acids are liberated in the intestinal lumen by this mechanism, but others are liberated at the cell surface by aminopeptidases, carboxypeptidases, endopeptidases, and dipeptidases present in the brush border of the mucosal cells. Some dipeptides and tripeptides are actively transported into the intestinal cells and hydrolyzed by intracellular peptidases, with the amino acids entering the bloodstream. Thus, the final digestion to amino acids occurs in three locations: the intestinal lumen, the brush border, and the cytoplasm of the mucosal cells.

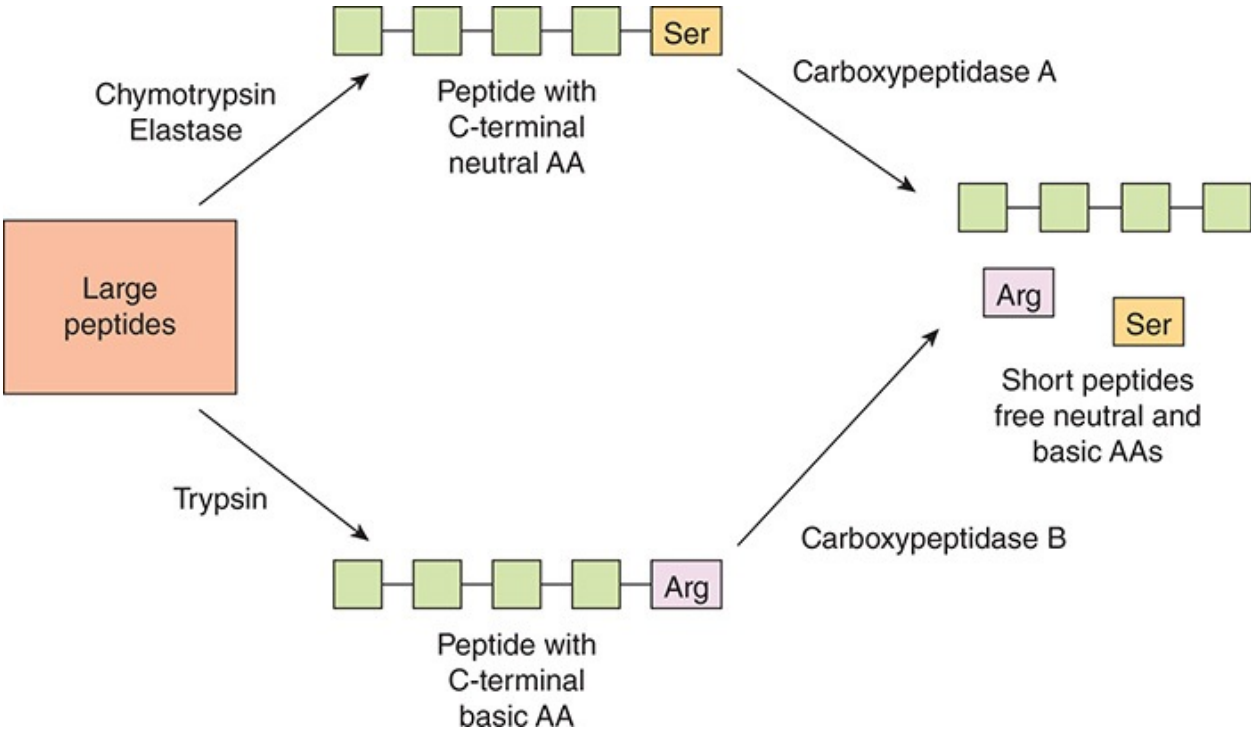


FIGURE 26–4 Luminal digestion of peptides by pancreatic endopeptidases and exopeptidases. Individual amino acids (AAs) are shown as squares.

ABSORPTION

At least seven different transport systems transport amino acids into enterocytes. Five of these require Na^+ and cotransport amino acids and Na^+ in a fashion similar to the cotransport of Na^+ and glucose (Figure 26–2). Two of these five also require Cl^-. In two systems, transport is independent of Na^+.

Dipeptides and tripeptides are transported into enterocytes by a system known

as PepT1 (or peptide transporter 1) that requires H^+ instead of Na^+ **(Figure 26–5)**. There is very little absorption of larger food-derived peptides after the neonatal period. In the enterocytes, amino acids released from the peptides by intracellular hydrolysis plus the amino acids absorbed from the intestinal lumen and brush border are transported out of the enterocytes along their basolateral borders by at least five transport systems. From there, they enter the hepatic portal blood.

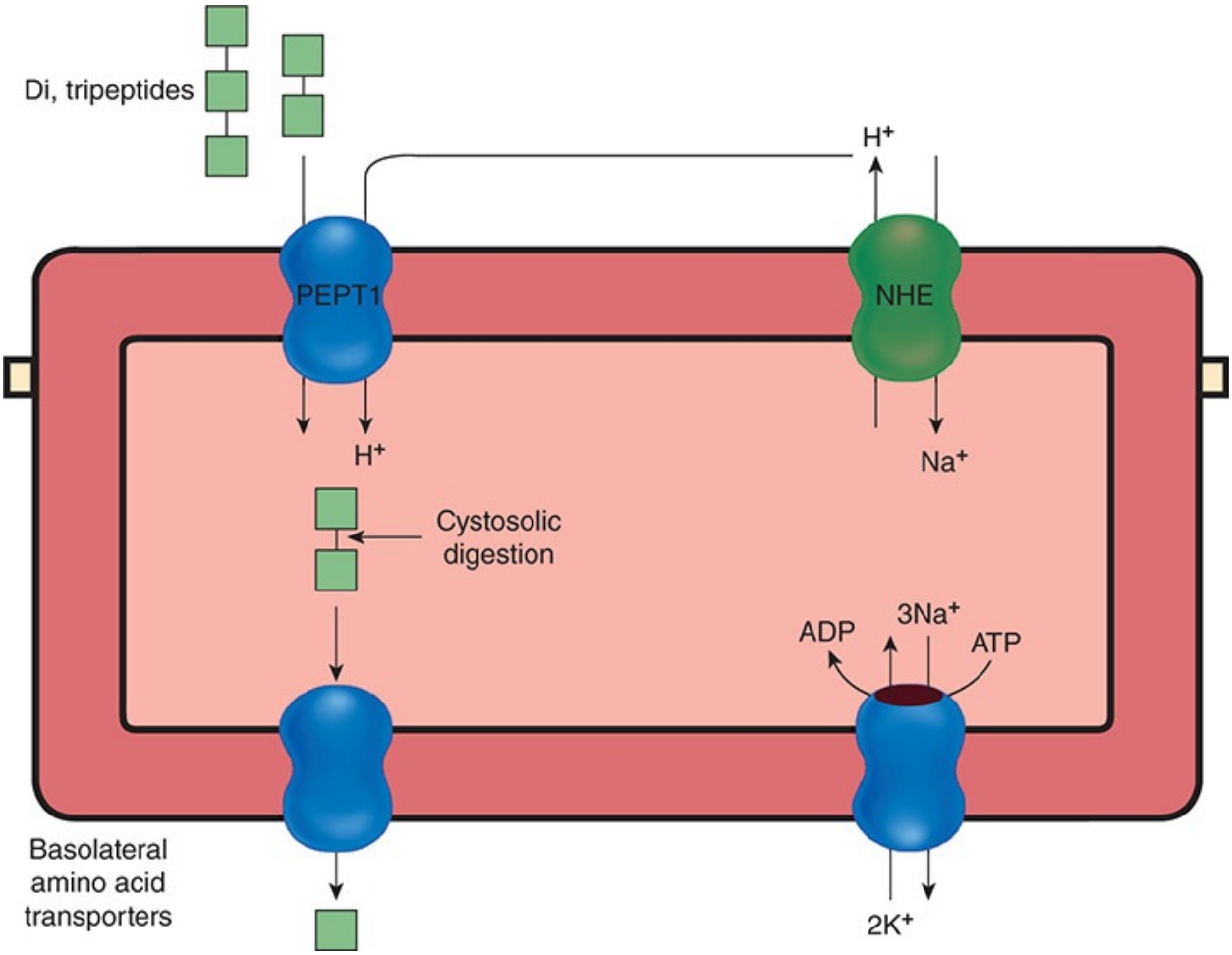


FIGURE 26–5 Disposition of short peptides in intestinal epithelial cells. Peptides are absorbed together with a proton supplied by an apical sodium/hydrogen exchanger (NHE) by the peptide transporter 1 (PepT1). Absorbed peptides are digested by cytosolic proteases, and any amino acids that are surplus to the needs of the epithelial cell are transported into the bloodstream by a series of basolateral transport proteins.

Absorption of amino acids is rapid in the duodenum and jejunum. There is a

little absorption in the ileum in health, because the majority of the free amino acids have already been assimilated at that point. Approximately 50% of the digested protein comes from ingested food, 25% from proteins in digestive juices, and 25% from desquamated mucosal cells. Only 2–5% of the protein in the small intestine escapes digestion and absorption. Some of this is eventually digested by bacterial action in the colon. Almost all of the protein in the stools is not of dietary origin but comes from bacteria and cellular debris. Evidence suggests that the peptidase activities of the brush border and the mucosal cell cytoplasm are increased by resection of part of the ileum and that they are independently altered in starvation. Thus, these enzymes appear to be subject to homeostatic regulation. In humans, a congenital defect in the mechanism that transports neutral amino acids in the intestine and renal tubules causes **Hartnup disease.** A congenital defect in the transport of basic amino acids causes **cystinuria.** However, most patients do not experience nutritional deficiencies of these amino acids because peptide transport compensates.

In infants, moderate amounts of undigested proteins are also absorbed. The protein antibodies in maternal colostrum are largely secretory immunoglobulins (IgAs), the production of which is increased in the breast in late pregnancy. They cross the mammary epithelium by transcytosis and enter the circulation of the infant from the intestine, providing passive immunity against infections. Absorption is by endocytosis and subsequent exocytosis.

Absorption of intact proteins declines sharply after weaning, but adults still absorb small quantities. Foreign proteins that enter the circulation provoke the formation of antibodies, and the antigen–antibody reaction occurring on subsequent entry of more of the same protein may cause allergic symptoms. Thus, absorption of proteins from the intestine may explain the occurrence of allergic symptoms after eating certain foods. The incidence of food allergy in children is said to be as high as 8%. However, in most individuals food allergies do not occur, and there is an evidence for a genetic component in susceptibility.

NUCLEIC ACIDS

Nucleic acids are split into nucleotides in the intestine by the pancreatic nucleases, and the nucleotides are split into the nucleosides and phosphoric acid by enzymes that appear to be located on the luminal surfaces of the mucosal cells. The nucleosides are then split into their constituent sugars and purine and pyrimidine bases. The bases are absorbed by active transport. Families of equilibrative (ie, passive) and concentrative (ie, secondary active) nucleoside

transporters have recently been identified and are expressed on the apical membrane of enterocytes.

LIPIDS

FAT DIGESTION

A lingual lipase is secreted by Ebner glands on the dorsal surface of the tongue in some species, and the stomach also secretes a lipase (Table 26–1). They are of little quantitative significance for lipid digestion other than in the setting of pancreatic insufficiency, but they may generate free fatty acids (FFA) that signal to more distal parts of the gastrointestinal tract (eg, causing the release of CCK; see Chapter 25).

Most fat digestion therefore begins in the duodenum, with pancreatic lipase being one of the most important enzymes involved. This enzyme hydrolyzes the 1- and 3-bonds of the triglycerides (triacylglycerols) with relative ease but acts on the 2-bonds at a very low rate, so the principal products of its action are FFA and 2-monoglycerides (2-monoacylglycerols). It acts on fats that have been emulsified (see below). **Colipase,** an accessory factor, is also secreted in the pancreatic juice, and helps to stabilize pancreatic lipase in an active conformation. Colipase is secreted in an inactive proform (Table 26–1) and is activated in the intestinal lumen by trypsin. Colipase is also critical for the action of lipase because it allows lipase to remain associated with droplets of dietary lipid even in the presence of bile acids.

Another pancreatic lipase that is activated by bile acids has been characterized. This **cholesterol esterase** represents about 4% of the total protein in pancreatic juice. In adults, pancreatic lipase is 10–60 times more active, but unlike pancreatic lipase, cholesterol esterase catalyzes the hydrolysis of cholesterol esters, esters of fat-soluble vitamins, and phospholipids, as well as triglycerides. A very similar enzyme is found in human milk.

Fats are relatively insoluble, which limits their ability to cross the unstirred layer and reach the surface of the mucosal cells. However, they are finely emulsified in the small intestine by the detergent action of bile acids, phosphatidylcholine, and monoglycerides. When the concentration of bile acids in the intestine is high after ingestion of a meal and contraction of the gallbladder, lipids and bile acids interact spontaneously to form **micelles** (**Figure 26–6**). These cylindrical aggregates take up lipids, and although their lipid concentration varies, they generally contain FA, monoglycerides, and cholesterol

in their hydrophobic centers. Micellar formation further solubilizes the lipids and provides a mechanism for their transport to the enterocytes. Thus, the micelles move down their concentration gradient through the unstirred layer to the brush border of the mucosal cells. The lipids diffuse out of the micelles, and a saturated aqueous solution of the lipids is maintained in contact with the brush border of the mucosal cells (Figure 26–6).

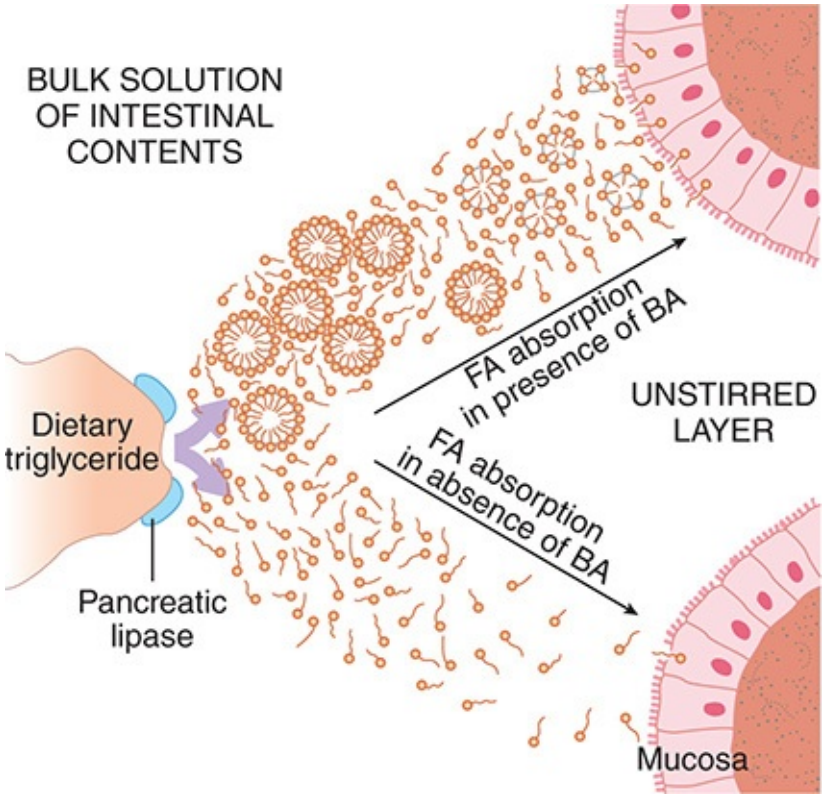


FIGURE 26–6 Lipid digestion and passage to intestinal mucosa. Fatty acids (FA) are liberated by the action of pancreatic lipase on dietary triglycerides and, in the presence of bile acids (BA), form micelles (the circular structures), which diffuse through the unstirred layer to the mucosal surface. Not shown, colipase binds to bile acids on the surface of the triglyceride droplet to anchor lipase to the surface and allow for its lipolytic activity. (Modified with permission from Westergaard H, Dietschy JM: Normal mechanisms of fat absorption and derangements induced by various gastrointestinal diseases. Med Clin North Am Nov; 58(6):1413–1427.)

STEATORRHEA

Pancreatectomized animals and patients with diseases that destroy the exocrine portion of the pancreas have fatty, bulky, clay-colored stools **(steatorrhea)** because of the impaired digestion and absorption of fat. This form of steatorrhea is mostly due to lipase deficiency. However, acid inhibits the lipase, and the lack of alkaline secretion from the pancreas also contributes by lowering the pH of the intestine contents. In some cases, hypersecretion of gastric acid can cause steatorrhea. Another cause of steatorrhea is defective reabsorption of bile acids in the distal ileum (see Chapter 28).

When bile is completely excluded from the intestine, up to 50% of ingested fat may appear in the feces. A severe malabsorption of fat-soluble vitamins also results. When bile acid reabsorption is prevented by resection of the terminal ileum or by disease in this portion of the small intestine, the amount of fat in the stools may also be increased because when the enterohepatic circulation is interrupted, the liver may not be able to increase the rate of bile acid production to a sufficient degree to compensate for the loss. Nevertheless, in some individuals, steatorrhea may not result even in the absence of micelles because the gut has a considerable reserve of absorptive surface area across which lipids may be absorbed in their molecular form, albeit more slowly than in the case of lipids in micellar form. This is referred to as the **anatomic reserve**, and applies also to the assimilation of the digested products of protein and carbohydrate. In individuals who have lost a significant proportion of their bowel and/or absorptive surface area to disease or surgery, however, malabsorption of one or more classes of nutrients may result **(Clinical Box 26–2)**.

FAT ABSORPTION

Traditionally, lipids were thought to enter the enterocytes by passive diffusion, but some evidence now suggests that carriers are involved. Inside the cells, the lipids are rapidly esterified, maintaining a favorable concentration gradient from the lumen into the cells **(Figure 26–7)**. There are also carriers that export certain lipids back into the lumen, thereby limiting their oral availability. This is the case for plant sterols as well as cholesterol.

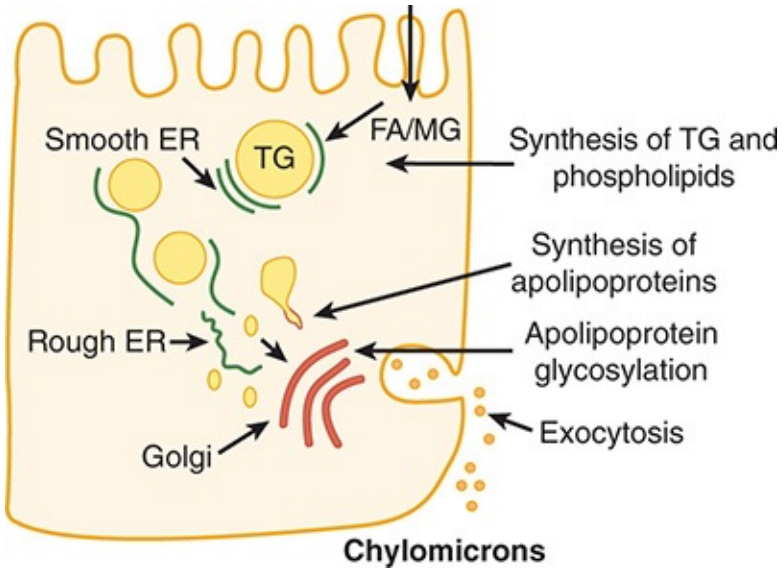


FIGURE 26–7 Intracellular handling of the products of lipid digestion. Absorbed fatty acids (FA) and monoglycerides (MG) are reesterified to form triglyceride (TG) in the smooth endoplasmic reticulum (ER). Apoproteins synthesized in the rough ER are coated around lipid cores, and the resulting chylomicrons are secreted from the basolateral pole of epithelial cells by exocytosis.

The fate of the FA in enterocytes depends on their size. FA containing less than 10–12 carbon atoms are sufficiently water-soluble that they pass through the enterocyte unmodified and are actively transported into the portal blood. They then circulate as free (unesterified) FA. FA containing more than 10–12 carbon atoms are too insoluble for this. They are therefore reesterified to triglycerides in the enterocytes. In addition, some of the absorbed cholesterol is esterified. The triglycerides and cholesterol esters are then coated with a layer of protein, cholesterol, and phospholipid to form chylomicrons. These leave the cell by exocytosis and enter the lymphatics, because they are too large to pass through the junctions between capillary endothelial cells (Figure 26–7).

CLINICAL BOX 26–2

The Malabsorption Syndrome

The digestive and absorptive functions of the small intestine are essential for life. However, the digestive and absorptive capacity of the intestine is larger than needed for normal function (the anatomic reserve). Removal of short

segments of the jejunum or ileum generally does not cause severe symptoms, and compensatory hypertrophy and hyperplasia of the remaining mucosa occur. However, when more than 50% of the small intestine is resected or bypassed **(short gut syndrome),** the absorption of nutrients and vitamins is so compromised that it is very difficult to prevent malnutrition and wasting **(malabsorption).** Resection of the terminal ileum also prevents the absorption of bile acids, and this leads in turn to deficient fat absorption. It also causes diarrhea because the unabsorbed bile acids enter the colon, where they activate chloride secretion (see Chapter 25). Other complications of intestinal resection or bypass include hypocalcemia, arthritis, and possibly fatty infiltration of the liver, followed by cirrhosis. Various disease processes can also impair absorption without a loss of intestinal length. The pattern of deficiencies that results is sometimes called the **malabsorption syndrome.** This pattern varies somewhat with the cause, but it can include deficient absorption of amino acids, with marked body wasting and, eventually, hypoproteinemia and edema. Carbohydrate and fat absorption are also depressed. Because of defective fat absorption, the fat-soluble vitamins (vitamins A, D, E, and K) are not absorbed in adequate amounts. One of the most interesting conditions causing the malabsorption syndrome is the autoimmune disease **celiac disease.** This disease occurs in genetically predisposed individuals in whom gluten and closely related proteins cause intestinal T cells to mount an inappropriate immune response that damages the intestinal epithelial cells and results in a loss of villi and a flattening of the mucosa. The proteins are found in wheat, rye, barley, and to a lesser extent in oats—but not in rice or corn. When grains containing gluten are omitted from the diet, bowel function is generally restored to normal.

THERAPEUTIC HIGHLIGHTS

Treatment of malabsorption depends on the underlying cause. In celiac disease, the mucosa returns to normal if foods containing gluten are strictly excluded from the diet, although this may be difficult to achieve. The diarrhea that accompanies bile acid malabsorption can be treated with a resin (cholestyramine) that binds the bile acids in the lumen and prevents their secretory action on colonocytes. Patients who become deficient in fat-soluble vitamins may be given these compounds as water-soluble derivatives. For serious cases of short bowel syndrome, it may be necessary to supply nutrients parenterally. There is hope that small bowel transplantation may eventually become routine, but of course transplantation carries its own long-term

disadvantages and also requires a reliable supply of donor tissues.

Absorption of long-chain FA is greatest in the upper parts of the small intestine, but appreciable amounts are also absorbed in the ileum. On a moderate fat intake, 95% or more of the ingested fat is absorbed. The processes involved in fat absorption are not fully mature at birth, and infants fail to absorb 10–15% of ingested fat. Thus, they are more susceptible to the ill effects of disease processes that reduce fat absorption.

SHORT-CHAIN FATTY ACIDS IN THE COLON

Short-chain fatty acids (SCFAs) are produced in the colon and absorbed from it. SCFAs are 2–5-carbon weak acids that have an average normal concentration of about 80 mmol/L in the lumen. About 60% of this total is acetate, 25% propionate, and 15% butyrate. They are formed by the action of colonic bacteria (**fermentation**) on complex carbohydrates, resistant starches, and other components of the dietary fiber, that is, the material that escapes digestion in the upper gastrointestinal tract and enters the colon.

Absorbed SCFAs are metabolized and make a significant contribution to the total caloric intake. In addition, they exert a trophic effect on the colonic epithelial cells; combat inflammation; and are absorbed in part by exchange for H^+, helping maintain acid–base equilibrium. SCFAs are absorbed by specific transporters present in colonic epithelial cells. SCFAs also promote the absorption of Na^+.

ABSORPTION OF MINERALS & VITAMINS

MINERALS

A number of minerals must be ingested daily for the maintenance of health. Besides those for which recommended daily dietary allowances have been set, a variety of different trace elements should be included. Trace elements are defined as elements found in tissues in minute amounts. In many cases, the mechanisms that provide for their uptake from the diet are poorly understood. Those believed to be essential for life, at least in experimental animals, are listed in **Table 26–3**. In humans, iron deficiency causes anemia. Cobalt is part of the

vitamin B_{12} molecule, and vitamin B_{12} deficiency leads to megaloblastic anemia (see Chapter 31). Iodine deficiency causes thyroid disorders (see Chapter 19). Zinc deficiency causes skin ulcers, depressed immune responses, and hypogonadal dwarfism. Copper deficiency causes anemia and changes in ossification. Chromium deficiency causes insulin resistance. Fluorine deficiency increases the incidence of dental caries. Sodium and potassium are also essential minerals, but listing them is academic, because it is very difficult to prepare a sodium-free or potassium-free diet. A **low-salt** diet is, however, well tolerated for prolonged periods because of the compensatory mechanisms that conserve Na^+. The mechanisms for sodium and potassium transport from the gut were discussed in Chapter 25.

TABLE 26–3 Trace elements believed essential for life.

Arsenic	Manganese
Chromium	Molybdenum
Cobalt	Nickel
Copper	Selenium
Fluorine	Silicon
Iodine	Vanadium
Iron	Zinc

Conversely, some minerals can be toxic when present in the body in excess. For example, severe iron overload with toxic effects is seen in hemochromatosis (see below). Similarly, copper excess causes brain damage (Wilson disease).

The remainder of this section addresses the intestinal transport mechanisms for two important minerals, calcium and iron.

CALCIUM

A total of 30–80% of ingested calcium is absorbed. The absorptive process and its relation to 1,25-dihydroxycholecalciferol are discussed in Chapter 21. Through this vitamin D derivative, Ca^{2+} absorption is adjusted to body needs; absorption is increased in the presence of Ca^{2+} deficiency and decreased in the presence of Ca^{2+} excess. Ca^{2+} absorption is also facilitated by protein. It is

inhibited by phosphates and oxalates because these anions form insoluble salts with Ca^{2+} in the intestine.

IRON

In adults, the amount of iron lost from the body is relatively small. The losses are generally unregulated, and total body stores of iron are regulated by changes in the rate at which it is absorbed from the intestine. Men lose about 0.6 mg/d, largely in the stools. Premenopausal women have a variable, larger loss averaging about twice this value because of the additional iron lost during menstruation. The average daily iron intake in the United States and Europe is about 20 mg, but the amount absorbed is equal only to the losses. Thus, the amount of iron absorbed is normally about 3–6% of the amount ingested. Various dietary factors affect the availability of iron for absorption; for example, the phytic acid found in cereals reacts with iron to form insoluble compounds in the intestine, as do phosphates and oxalates.

Most of the iron in the diet is in the ferric (Fe^{3+}) form, whereas it is the ferrous (Fe^{2+}) form that is absorbed. Fe^{3+} reductase activity is associated with the iron transporter in the brush borders of the enterocytes **(Figure 26–8)**. Gastric secretions dissolve the iron and permit it to form soluble complexes with ascorbic acid and other substances that aid its reduction to the Fe^{2+} form. The importance of this function in humans is indicated by the fact that iron deficiency anemia is a troublesome and relatively frequent complication of partial gastrectomy.

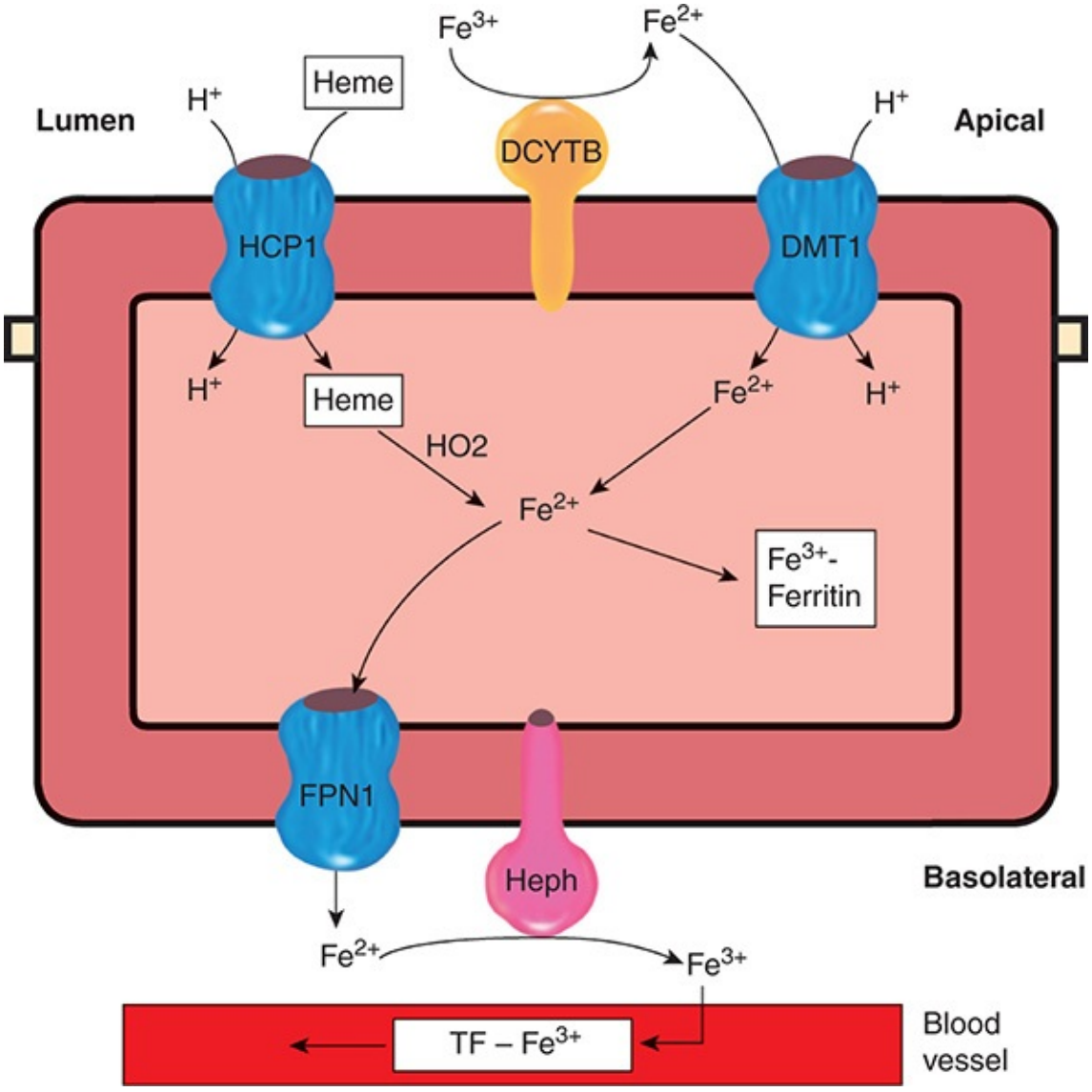


FIGURE 26–8 Intestinal absorption of iron. Fe^{3+} is converted to Fe^{2+} by the ferric reductase DCYTB, and Fe^{2+} is transported into the enterocyte by the apical membrane iron transporter DMT1. Heme is transported into the enterocyte by a separate heme transporter (most likely heme carrier protein 1, HCP1), and heme oxygenase-2 (HO2) releases Fe^{2+} from the heme. Some of the intracellular Fe^{2+} is converted to Fe^{3+} and bound to ferritin. The rest binds to the basolateral Fe^{2+} transporter ferroportin-1 (FPN1) and is transported to the interstitial fluid. The transport is aided by hephaestin (Heph) which converts Fe^{2+} to Fe^{3+}. In plasma, Fe^{3+} is transported bound to the iron transport protein transferrin (TF).

Almost all iron absorption occurs in the duodenum. Transport of Fe^{2+} into the

enterocytes occurs via divalent metal transporter 1 **(DMT1)** (Figure 26–8). Some is stored in ferritin, and the remainder is transported out of the enterocytes by a basolateral transporter named **ferroportin 1.** A protein called **hephaestin (Hp)** is associated with ferroportin 1. It is not a transporter itself, but it facilitates basolateral transport. In the plasma, Fe^{2+} is converted to Fe^{3+} and bound to the iron transport protein **transferrin.** This protein has two iron-binding sites. Normally, transferrin is about 35% saturated with iron, and the normal plasma iron level is about 130 µg/dL (23 µmol/L) in men and 110 µg/dL (19 µmol/L) in women.

Heme (see Chapter 31) binds to an apical transport protein in enterocytes and is carried into the cytoplasm. In the cytoplasm, HO-2, a subtype of heme oxygenase, removes Fe^{2+} from the porphyrin and adds it to the intracellular Fe^{2+} pool.

Seventy percent of the iron in the body is in hemoglobin, 3% in myoglobin, and the rest in ferritin, which is present not only in enterocytes, but also in many other cells. Apoferritin is a globular protein made up of 24 subunits. Ferritin molecules in lysosomal membranes may aggregate in deposits that contain as much as 50% iron. These deposits are called **hemosiderin.**

CLINICAL BOX 26–3

Disorders of Iron Uptake

Iron deficiency causes anemia. Conversely, iron overload causes hemosiderin to accumulate in the tissues, producing **hemosiderosis.** Large amounts of **hemosiderin** can damage tissues, such as is seen in the common genetic disorder of hemochromatosis. This syndrome is characterized by pigmentation of the skin, pancreatic damage with diabetes (“bronze diabetes”), cirrhosis of the liver, a high incidence of hepatic carcinoma, and gonadal atrophy. Hemochromatosis may be hereditary or acquired. The most common cause of the hereditary forms is a mutated *HFE* gene that is common in the white population. It is located on the short arm of chromosome 6 and is closely linked to the HLA-A locus. It is still unknown precisely how mutations in *HFE* cause hemochromatosis, but individuals who are homogenous for *HFE* mutations absorb excess amounts of iron because *HFE* normally inhibits expression of the duodenal transporters that participate in iron uptake. Acquired hemochromatosis occurs when the iron-regulating system is overwhelmed by excess iron loads due to chronic destruction of red

blood cells, liver disease, or repeated transfusions in diseases such as intractable anemia.

THERAPEUTIC HIGHLIGHTS

If hereditary hemochromatosis is diagnosed before excessive amounts of iron accumulate in the tissues, life expectancy can be prolonged substantially by repeated withdrawal of blood.

Intestinal absorption of iron is regulated by three factors: recent dietary intake of iron, the state of the iron stores in the body, and the state of erythropoiesis in the bone marrow. The normal operation of the factors that maintain iron balance is essential for health **(Clinical Box 26–3)**.

VITAMINS

Vitamins were discovered when it was observed that certain diets otherwise adequate in calories, essential amino acids, fats, and minerals failed to maintain health (for example, in sailors engaged in long voyages without access to fresh fruits and vegetables). The term **vitamin** has now come to refer to any organic dietary constituent necessary for life, health, and growth that does not function by supplying energy, and which cannot be synthesized endogenously (at least in adequate amounts).

Most vitamins are absorbed in the upper small intestine, but vitamin B_{12} is absorbed in the ileum. This vitamin binds to intrinsic factor, a protein secreted by the parietal cells of the stomach, and the complex is the form that is absorbed across the ileal mucosa.

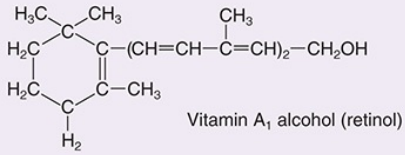
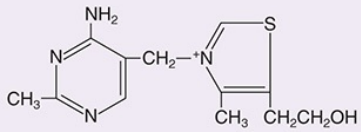
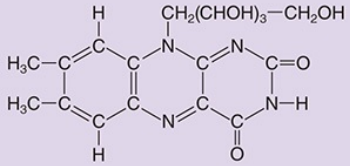
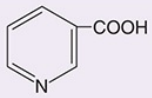
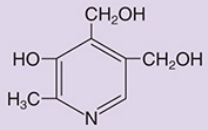
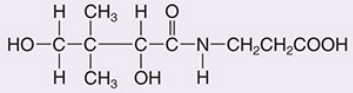
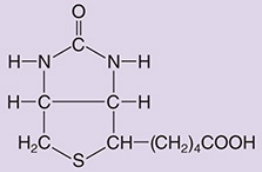
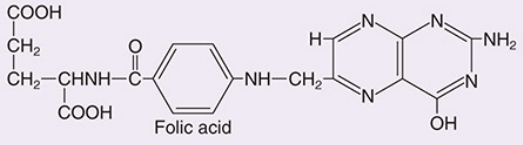
Vitamin B_{12} absorption and folate absorption are Na^+-independent, but all seven of the remaining water-soluble vitamins—thiamin, riboflavin, niacin, pyridoxine, pantothenate, biotin, and ascorbic acid—are absorbed by carriers that are Na^+ cotransporters.

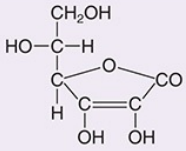
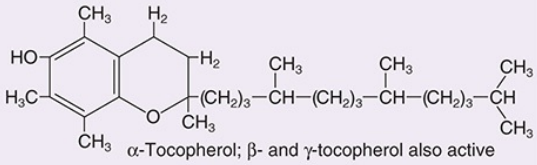
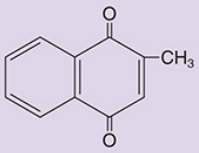
The water-soluble vitamins are easily absorbed, but the fat-soluble vitamins (vitamins A, D, E, and K) are poorly absorbed in the absence of bile and/or pancreatic juice because their absorption depends almost entirely on micellar solubilization. Some dietary fat intake is also necessary for their absorption, and in obstructive jaundice or disease of the exocrine pancreas, deficiencies of the

fat-soluble vitamins can develop even if their intake is adequate. Vitamin A and vitamin D are bound to transfer proteins in the circulation. The α-tocopherol form of vitamin E is normally bound to chylomicrons. In the liver, it is transferred to very low-density lipoprotein (VLDL) and distributed to tissues by an α-tocopherol transfer protein. When this protein is abnormal due to mutation of its gene in humans, there is cellular deficiency of vitamin E and the development of a condition resembling Friedreich ataxia.

Because there are minor differences in metabolism between mammalian species, some substances are vitamins in one species and not in another. The sources and functions of the major vitamins in humans are listed in **Table 26–4**. Most vitamins have important functions in intermediary metabolism or the special metabolism of the various organ systems. The diseases caused by deficiency of each of the vitamins are also listed in Table 26–4.

TABLE 26–4 Vitamins essential or probably essential to human nutrition.[a]

Vitamin	Action	Deficiency Symptoms	Sources	Chemistry
A (A_1, A_2)	Constituents of visual pigments (see Chapter 9: Vision); necessary for fetal development and for cell development throughout life	Night blindness, dry skin	Yellow vegetables and fruit	 Vitamin A_1 alcohol (retinol)
B complex				
Thiamin (vitamin B_1)	Cofactor in decarboxylations	Beriberi, neuritis	Liver, unrefined cereal grains	
Riboflavin (vitamin B_2)	Constituent of flavoproteins	Glossitis, cheilosis	Liver, milk	
Niacin	Constituent of NAD^+ and $NADP^+$	Pellagra	Yeast, lean meat, liver	 Can be synthesized in body from tryptophan
Pyridoxine (vitamin B_6)	Forms prosthetic group of certain decarboxylases and transaminases. Converted in body into pyridoxal phosphate and pyridoxamine phosphate	Convulsions, hyperirritability	Yeast, wheat, corn, liver	
Pantothenic acid	Constituent of CoA	Dermatitis, enteritis, alopecia, adrenal insufficiency	Eggs, liver, yeast	
Biotin	Catalyzes CO_2 "fixation" (in fatty acid synthesis, etc)	Dermatitis, enteritis	Egg yolk, liver, tomatoes	
Folates (folic acid) and related compounds	Coenzymes for "1-carbon" transfer; involved in methylating reactions	Sprue, anemia. Neural tube defects in children born to folate-deficient women	Leafy green vegetables	 Folic acid

Cyanocobalamin (vitamin B_{12})	Coenzyme in amino acid metabolism. Stimulates erythropoiesis	Pernicious anemia (see Chapter 25: Overview of Gastrointestinal Function & Regulation)	Liver, meat, eggs, milk	Complex of four substituted pyrrole rings around a cobalt atom (see Chapter 25: Overview of Gastrointestinal Function & Regulation)
C	Maintains prosthetic metal ions in their reduced form; scavenges free radicals	Scurvy	Citrus fruits, leafy green vegetables	 Ascorbic acid (synthesized in most mammals except guinea pigs and primates, including humans)
D group	Increase intestinal absorption of calcium and phosphate (see Chapter 21: Hormonal Control of Calcium & Phosphate Metabolism & the Physiology of Bone)	Rickets	Fish liver	Family of sterols (see Chapter 21: Hormonal Control of Calcium & Phosphate Metabolism & the Physiology of Bone)
E group	Antioxidants; cofactors in electron transport in cytochrome chain?	Ataxia and other symptoms and signs of spinocerebellar dysfunction	Milk, eggs, meat, leafy vegetables	 α-Tocopherol; β- and γ-tocopherol also active
K group	Catalyze γ carboxylation of glutamic acid residues on various proteins concerned with blood clotting	Hemorrhagic phenomena	Leafy green vegetables	 Vitamin K_3; a large number of similar compounds have biological activity

[a]Choline is synthesized in the body in small amounts, but it has recently been added to the list of essential nutrients.

It is worth remembering, however, that very large doses of the fat-soluble vitamins are definitely toxic. **Hypervitaminosis A** is characterized by anorexia, headache, hepatosplenomegaly, irritability, scaly dermatitis, patchy loss of hair, bone pain, and hyperostosis. Acute vitamin A intoxication was first described by Arctic explorers, who developed headache, diarrhea, and dizziness after eating polar bear liver, which is particularly rich in vitamin A. **Hypervitaminosis D** is associated with weight loss, calcification of many soft tissues, and acute kidney injury. **Hypervitaminosis K** is characterized by gastrointestinal disturbances and anemia. Large doses of water-soluble vitamins have been thought to be less likely to cause problems because they can be rapidly cleared from the body. However, it has been demonstrated that ingestion of megadoses of pyridoxine (vitamin B_6) can produce peripheral neuropathy.

CONTROL OF FOOD INTAKE

The intake of nutrients is under complex control involving signals from both the periphery and the central nervous system. Complicating the picture, higher functions also modulate the response to both central and peripheral cues that either trigger or inhibit food intake. Thus, food preferences, emotions, environment, lifestyle, and circadian rhythms may all have profound effects on whether food is sought, and the type of food that is ingested.

Many of the hormones and other factors that are released coincident with a meal, and may play other important roles in digestion and absorption (see Chapter 25) are also involved in the regulation of feeding behavior **(Figure 26–9)**. For example, CCK either produced by I cells in the intestine, or released by nerve endings in the brain, inhibits further food intake and thus is defined as a **satiety factor** or **anorexin.** CCK and other similar factors have attracted great interest from the pharmaceutical industry in the hopes that derivatives might be useful as aids to dieting, an objective that is lent greater urgency given the current epidemic of obesity in Western countries **(Clinical Box 26–4)**.

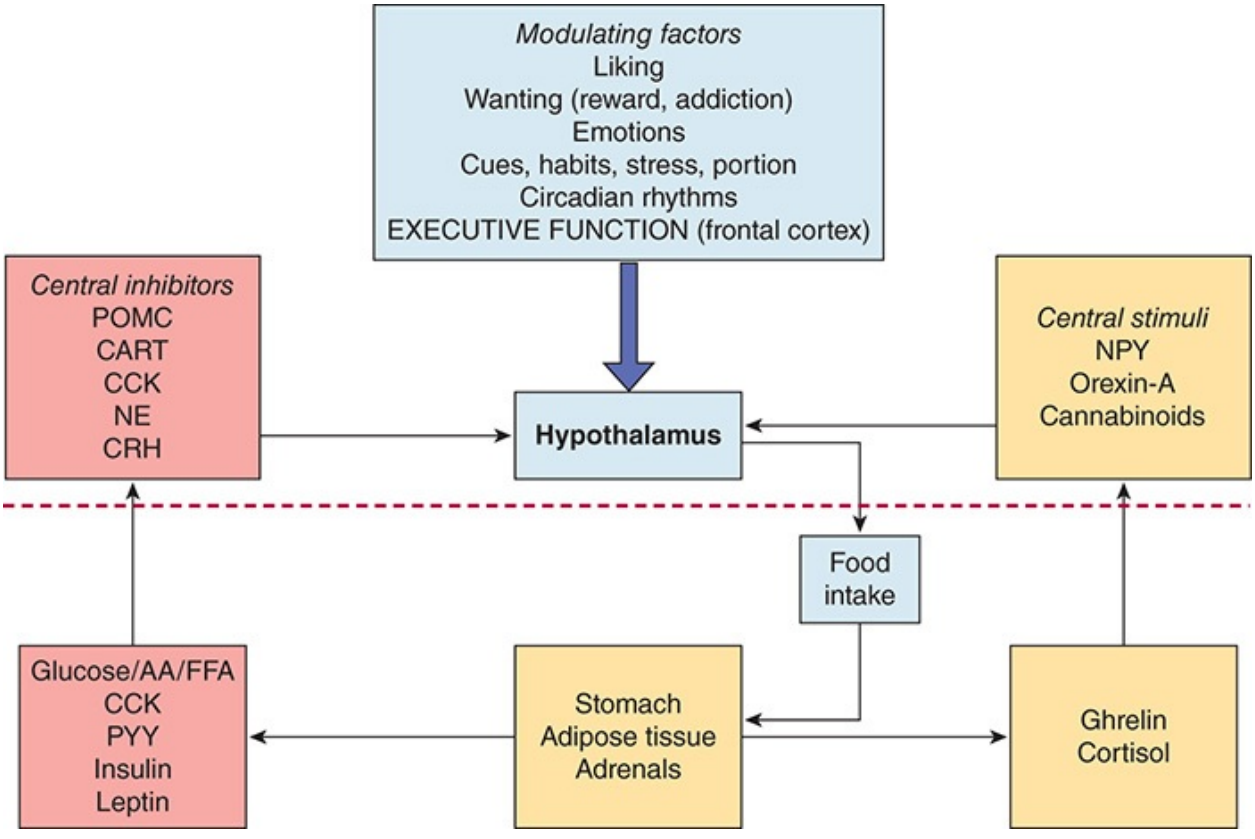


FIGURE 26–9 Summary of mechanisms controlling food intake. Peripheral

stimuli and inhibitors, release in anticipation of or in response to food intake, cross the blood-brain barrier (indicated by the broken red line) and activate the release and/or synthesis of central factors in the hypothalamus that either increase or decrease subsequent food intake. Food intake can also be modulated by signals from higher centers, as shown. Not shown, peripheral orexins can reduce production of central inhibitors, and vice versa. AA, amino acid; CART, cocaine- and amphetamine-regulated transcript; CCK, cholecystokinin; CRH, corticotropin-releasing hormone; FFA, free fatty acids; NE, norepinephrine; NPY, neuropeptide Y; POMC, pro-opiomelanocortin; PYY, Peptide YY. (Used with permission of Dr Samuel Klein, Washington University.)

Leptin and ghrelin are peripheral factors that act reciprocally on food intake, and have emerged as critical regulators in this regard. Both activate their receptors in the hypothalamus that initate signaling cascades leading to changes in food intake. Leptin is produced by adipose tissue, and signals the status of the fat stores therein. As adipocytes increase in size, they release greater quantities of leptin and this tends to decrease food intake, in part by increasing the expression of other anorexigenic factors in the hypothalamus such as pro-opiomelanocortin (POMC), cocaine- and amphetamine-regulated transcript (CART), neurotensin, and corticotropin-releasing hormone (CRH). Leptin also stimulates the metabolic rate. Animal studies have shown that it is possible to become resistant to the effects of leptin, however, and in this setting, food intake persists despite adequate (or even growing) adipose stores—obesity therefore results.

CLINICAL BOX 26–4

Obesity

Obesity is the most common and most expensive nutritional problem in the United States. A convenient and reliable indicator of body fat is the **body mass index (BMI),** which is body weight (in kilograms) divided by the square of height (in meters). Values above 25 are abnormal. Individuals with values of 25–30 are considered overweight, and those with values >30 are obese. In the United States, 34% of the population is overweight and 34% is obese. The incidence of obesity is also increasing in other countries. Indeed, the Worldwatch Institute has estimated that although starvation continues to be a problem in many parts of the world, the number of overweight people in

the world is now as great as the number of underfed. Obesity is a problem because of its complications. It is associated with accelerated atherosclerosis and an increased incidence of gallbladder and other diseases. Its association with type 2 diabetes is especially striking. As weight increases, insulin resistance increases and frank diabetes appears. At least in some cases, glucose tolerance is restored when weight is lost. In addition, the mortality rates from many kinds of cancer are increased in obese individuals. The causes of the high incidence of obesity in the general population are probably multiple. Studies of twins raised apart show a definite genetic component. It has been pointed out that through much of human evolution, famines were common, and mechanisms that permitted increased energy storage as fat had survival value. Now, however, food is plentiful in many countries, and the ability to gain and retain fat has become a liability. As noted above, the fundamental cause of obesity is still an excess of energy intake in food over energy expenditure. If human volunteers are fed a fixed high-calorie diet, some gain weight more rapidly than others, but those with slower weight gain can be shown to exhibit increased energy expenditure in the form of small, fidgety movements **(nonexercise activity thermogenesis; NEAT).** Body weight generally increases at a slow but steady rate throughout adult life. Decreased physical activity is undoubtedly a factor in this increase, but decreased sensitivity to leptin may also play a role.

THERAPEUTIC HIGHLIGHTS

Obesity is such a vexing medical and public health problem because its effective treatment depends so dramatically on lifestyle changes. Long-term weight loss can only be achieved with decreased food intake, increased energy expenditure, or, ideally, some combination of both. Exercise alone is rarely sufficient because it typically induces the patient to ingest more calories. For those who are seriously obese and who have developed serious health complications as a result, a variety of surgical approaches have been developed that reduce the size of the stomach reservoir and/or bypass it altogether. These surgical maneuvers are intended to reduce the size of meals that can be tolerated, but also have dramatic metabolic effects even before significant weight loss occurs, perhaps as a result of reduced production of peripheral orexins, such as ghrelin, by the gut. Pharmaceutical companies are also actively exploring the science of orexins and anorexins to develop drugs that might act centrally to modify food intake (Figure 26–9).

Ghrelin, on the other hand, is a predominantly fast-acting **orexin** that stimulates food intake. It is produced mainly by the stomach, as well as other tissues such as the pancreas and adrenal glands in responses to changes in nutritional status—circulating ghrelin levels increase preprandially, then decrease after a meal. It is believed to be involved primarily in meal initiation, unlike the longer-term effects of leptin. Like leptin, however, the effects of ghrelin are produced mostly via actions in the hypothalamus. It increases synthesis and/or release of central orexins, including neuropeptide Y and cannabinoids, and suppresses the ability of leptin to stimulate the anorexigenic factors discussed above. Loss of the activity of ghrelin may account in part for the effectiveness of gastric bypass procedures for obesity. Its secretion may also be inhibited by leptin, underscoring the reciprocity of these hormones. There is some evidence to suggest, however, that the ability of leptin to reduce ghrelin secretion is lost in the setting of obesity.

CHAPTER SUMMARY

- A balanced diet is important for health, and certain substances obtained from the diet are essential to life. The caloric value of dietary intake must be approximately equal to energy expenditure for homeostasis.
- A typical mixed meal consists of carbohydrates, proteins, and lipids (the latter largely in the form of triglycerides). Each must be digested to allow its uptake into the body. Specific transporters carry the products of digestion into the body.
- In the process of carbohydrate assimilation, the epithelium can only transport monomers, whereas for proteins, short peptides can be absorbed in addition to amino acids.
- The protein assimilation machinery, which rests heavily on the proteases in pancreatic juice, is arranged such that these enzymes are not activated until they reach their substrates in the small intestinal lumen. This is accomplished by the restricted localization of an activating enzyme, enterokinase.
- Lipids face special challenges to assimilation given their hydrophobicity. Bile acids solubilize the products of lipolysis in micelles and accelerate their ability to diffuse to the epithelial surface. The assimilation of triglycerides is enhanced by this mechanism, whereas that of cholesterol and fat-soluble vitamins absolutely requires it.

- Food intake is regulated by a complex network of signals both from the periphery and in the central nervous system. Ghrelin released by the stomach is a critical peripheral orexin that stimulates centers in the hypothalamus to trigger food intake. Leptin is released by adipocytes and signals the status of fat stores. Unless leptin resistance has developed, it stimulates the release of anorexins in the hypothalamus to terminate feeding behavior.
- When energy intake exceeds expenditures, the increasingly prevalent modern disease of obesity results.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. A newborn baby is brought to the pediatrician suffering from severe diarrhea that worsens with meals. The symptoms diminish when nutrients are delivered intravenously. The child most likely has a mutation in which of the following intestinal transporters?

   A. Na^+, K^+ ATPase

   B. NHE3

   C. SGLT-1

   D. H^+, K^+ ATPase

   E. NKCC1

2. An infant that was previously healthy but is displaying symptoms of acute diarrhea and dehydration after a presumed enteric infection is given Pedialyte, a solution of glucose and electrolytes. What membrane protein accounts for the ability of this solution to provide more rapid hydration than plain water?

   A. Sucrase-isomaltase

   B. SGLT-1

   C. CFTR

   D. Chloride-bicarbonate exchanger

   E. Lactose-phlorizin hydrolase

3. In the situation described in question 2, the child's mother does not have any Pedialyte available, so gives the infant some boiled water in which she has dissolved some salt and table sugar. Activity of which enzyme will be needed to make this solution effective?

A. Salivary amylase
B. Pancreatic amylase
C. Glucoamylase
D. Lactose-phlorizin hydrolase
E. Sucrase-isomaltase

4. A decrease in which of the following would be expected in a child exhibiting a congenital absence of enterokinase?

A. Incidence of pancreatitis
B. Glucose absorption
C. Bile acid reabsorption
D. Gastric pH
E. Protein assimilation

5. In Hartnup disease (a defect in the transport of neutral amino acids), patients do not become deficient in these amino acids due to the activity of

A. PepT1.
B. brush border peptidases.
C. Na^+, K^+ ATPase.
D. cystic fibrosis transmembrane conductance regulator (CFTR).
E. trypsin.

6. A patient with hypercholesterolemia is treated with cholestyramine, a resin that binds bile acids in the intestinal lumen. Overall levels of absorption of which of the following is likely to be abnormal in this patient?

A. Long-chain triglyceride
B. Medium-chain triglyceride
C. Starch
D. Vitamin D
E. Vitamin B_6

7. Maximum absorption of short-chain fatty acids produced by bacteria occurs in the

A. stomach.
B. duodenum.
C. jejunum.
D. ileum.

E. colon.

8. A premenopausal woman who is physically active seeks advice from her primary care clinician regarding measures she can take to ensure adequate availability of dietary calcium to ensure bone health later in life. Which of the following dietary components should enhance calcium uptake?

A. Protein

B. Oxalates

C. Iron

D. Vitamin D

E. Sodium

9. A patient with obstructive jaundice who is scheduled for gallbladder surgery is found to have an elevated prothrombin time. This laboratory finding is most likely due to malabsorption of which of the following vitamins?

A. A

B. C

C. B_{12}

D. K

E. E

10. A 40-year-old man undergoes bariatric surgery for the treatment of morbid obesity. Levels of which of the following substances would be expected to be reduced in his brain following recovery?

A. Glucose

B. Ghrelin

C. Leptin

D. Pro-opiomelanocortin

E. Neurotensin

CHAPTER 27

Gastrointestinal Motility

OBJECTIVES

After studying this chapter, you should be able to:

- List the major forms of motility in the gastrointestinal tract and their roles in digestion and excretion; distinguish between peristalsis and segmentation.
- Explain the electrical basis of gastrointestinal contractions and the role of basic electrical activity in governing motility patterns.
- Describe how gastrointestinal motility changes during fasting.
- Understand how food is swallowed and transferred to the stomach.
- Define the factors that govern gastric emptying and the abnormal response of vomiting.
- Define how the motility patterns of the colon subserve its function to desiccate and evacuate the stool.

INTRODUCTION

The digestive and absorptive functions of the gastrointestinal system outlined in the previous chapter depend on a variety of mechanisms that soften the food, propel it through the length of the gastrointestinal tract **(Table 27–1)**, and mix it

with bile from the gallbladder and digestive enzymes secreted by the salivary glands and pancreas. Some of these mechanisms depend on intrinsic properties of the intestinal smooth muscle. Others involve the operation of reflexes involving the neurons intrinsic to the gut, reflexes involving the central nervous system (CNS), paracrine effects of chemical messengers, and gastrointestinal hormones.

TABLE 27–1 Mean lengths of various segments of the gastrointestinal tract as measured by intubation in living humans.

Segment	Length (cm)
Pharynx, esophagus, and stomach	65
Duodenum	25
Jejunum and ileum	260
Colon	110

Data from Hirsch JE, Ahrens EH Jr, Blankenhorn DH: Measurement of human intestinal length in vivo and some causes of variation. Gastroenterology 1956;31:274.

GENERAL PATTERNS OF MOTILITY

PERISTALSIS

Peristalsis is a reflex response that is initiated when the gut wall is stretched by the contents of the lumen, and it occurs in all parts of the gastrointestinal tract from the esophagus to the rectum. The stretch initiates a circular contraction behind the stimulus and an area of relaxation in front of it **(Figure 27–1)**. The wave of contraction then moves in an oral-to-caudal direction, propelling the contents of the lumen forward at rates that vary from 2 to 25 cm/s. Peristaltic activity can be increased or decreased by the autonomic input to the gut, but its occurrence is independent of extrinsic innervation. Indeed, progression of the contents is not blocked by removal and resuture of a segment of intestine in its original position and is blocked only if the segment is reversed before it is sewn back into place. Peristalsis is an excellent example of the integrated activity of the enteric nervous system. It appears that local stretch releases serotonin, which activates sensory neurons that activate the myenteric plexus. Cholinergic neurons passing in a retrograde direction in this plexus activate neurons that release substance P and acetylcholine, causing smooth muscle contraction behind

the bolus. At the same time, cholinergic neurons passing in an anterograde direction activate neurons that secrete NO and vasoactive intestinal polypeptide (VIP), producing the relaxation ahead of the stimulus.

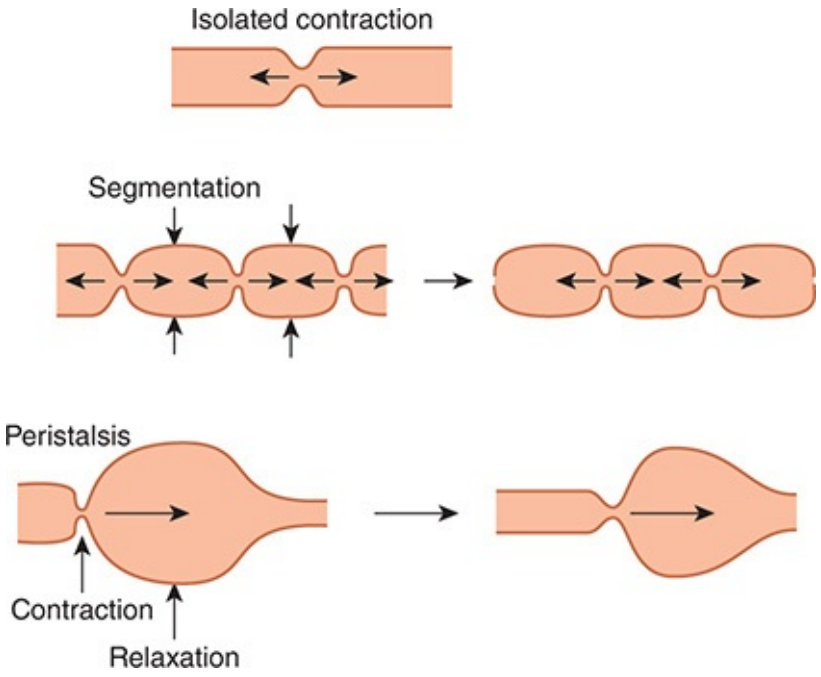


FIGURE 27–1 Patterns of gastrointestinal motility and propulsion. An isolated contraction moves contents orally and aborally. Segmentation mixes contents over a short stretch of intestine, as indicated by the time sequence from left to right. In the diagram on the left, the vertical arrows indicate the sites of subsequent contraction. Peristalsis involves both contraction and relaxation, and moves contents aborally.

SEGMENTATION & MIXING

When the meal is present, the enteric nervous system promotes a motility pattern that is related to peristalsis but is designed to retard the movement of the intestinal contents along the length of the intestinal tract to provide time for digestion and absorption (Figure 27–1). This motility pattern is known as segmentation, and it provides for ample mixing of the intestinal contents (known as **chyme**) with the digestive juices. A segment of bowel contracts at both ends, and then a second contraction occurs in the center of the segment to force the chyme both backward and forward. Unlike peristalsis, therefore, retrograde movement of the chyme occurs routinely in the setting of segmentation. This

mixing pattern persists for as long as nutrients remain in the lumen to be absorbed. It presumably reflects programmed activity of the bowel dictated by the enteric nervous system, and can occur independent of central input, although the latter can modulate it.

BASIC ELECTRICAL ACTIVITY & REGULATION OF MOTILITY

Except in the esophagus and the proximal portion of the stomach, the smooth muscle of the gastrointestinal tract has spontaneous rhythmic fluctuations in membrane potential between about −65 and −45 mV. This **basic electrical rhythm (BER)** is initiated by the **interstitial cells of Cajal,** which are stellate mesenchymal pacemaker cells with smooth muscle-like features that send long multiply branched processes into the intestinal smooth muscle. In the stomach and the small intestine, these cells are located in the outer circular muscle layer near the myenteric plexus; in the colon, they are at the submucosal border of the circular muscle layer. In the stomach and small intestine, there is a descending gradient in pacemaker frequency, and as in the heart, the pacemaker with the highest frequency usually dominates.

The BER itself rarely causes muscle contraction, but **spike potentials** superimposed on the most depolarizing portions of the BER waves do increase muscle tension **(Figure 27–2)**. The depolarizing portion of each spike is due to Ca^{2+} influx, and the repolarizing portion is due to K^{+} efflux. Many polypeptides and neurotransmitters affect the BER. For example, acetylcholine increases the number of spikes and the tension of the smooth muscle, whereas epinephrine decreases the number of spikes and the tension. The rate of the BER is about 4/min in the stomach. It is about 12/min in the duodenum and falls to about 8/min in the distal ileum. In the colon, the BER rate rises from about 2/min at the cecum to about 6/min at the sigmoid. The function of the BER is to coordinate peristaltic and other motor activity, such as setting the rhythm of segmentation; contractions can occur only during the depolarizing part of the waves. After vagotomy or transection of the stomach wall, for example, peristalsis in the stomach becomes irregular and chaotic.

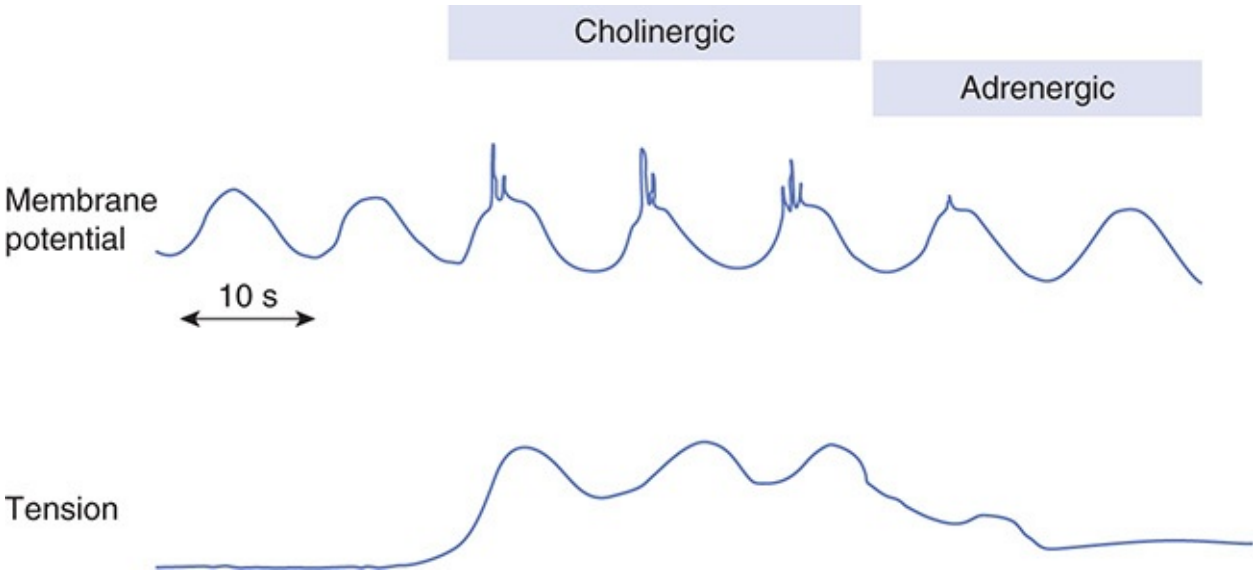


FIGURE 27–2 Basic electrical rhythm (BER) of gastrointestinal smooth muscle. Top: Membrane potential showing spike potentials under conditions of maximal cholinergic tone and inhibition under adrenergic tone. **Bottom:** Associated changes in muscle tension.

MIGRATING MOTOR COMPLEX

During fasting between periods of digestion, the pattern of electrical and motor activity in gastrointestinal smooth muscle becomes modified so that cycles of motor activity migrate from the stomach to the distal ileum. Each cycle, or **migrating motor complex (MMC),** starts with a quiescent period (phase I), continues with a period of irregular electrical and mechanical activity (phase II), and ends with a burst of regular activity (phase III) **(Figure 27–3)**. The MMCs are initiated by motilin. The circulating level of this hormone increases at intervals of approximately 100 min in the interdigestive state, coordinated with the contractile phases of the MMC. The contractions migrate aborally at a rate of about 5 cm/min, and also occur at intervals of approximately 100 min. Gastric secretion, bile flow, and pancreatic secretion increase during each MMC. They likely serve to clear the stomach and small intestine of luminal contents in preparation for the next meal.

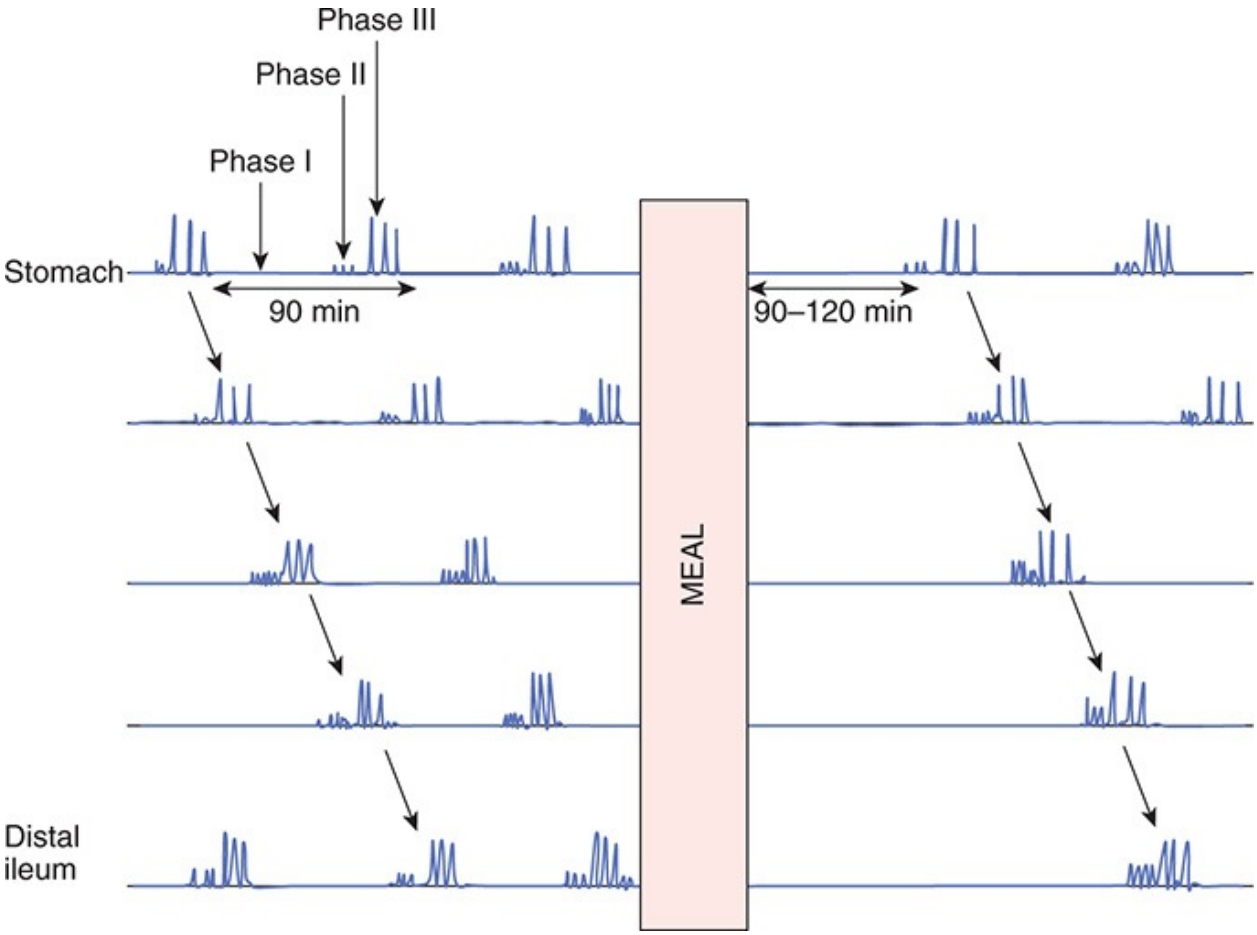


FIGURE 27–3 Migrating motor complexes (MMCs). The three phases include a quiescent phase (phase I); a phase consisting of small, irregular contractions that do not propagate (phase II); and a phase of regular activity lasting about 5 minutes (phase III), which sweeps along the length of the intestine. The entire cycle repeats every 90–100 minutes under fasting conditions. Note that the complexes are completely inhibited by a meal and resume 90–120 min later.

Conversely, when a meal is ingested, secretion of motilin is suppressed (ingestion of food suppresses motilin release via mechanisms that have not yet been elucidated), and the MMC is abolished, until digestion and absorption are complete.

SEGMENT-SPECIFIC PATTERNS OF MOTILITY

MOUTH & ESOPHAGUS

In the mouth, food is mixed with saliva and propelled into the esophagus. Peristaltic waves in the esophagus move the food into the stomach.

MASTICATION

Chewing **(mastication)** breaks up large food particles and mixes the food with the secretions of the salivary glands. This wetting and homogenizing action aids swallowing and subsequent digestion. Large food particles can be digested, but they cause strong and often painful contractions of the esophageal musculature. Particles that are small tend to disperse in the absence of saliva and also make swallowing difficult because they do not form a bolus. The number of chews that is optimal depends on the food, but usually ranges from 20 to 25.

Edentulous patients are generally restricted to a soft diet and have considerable difficulty eating dry food.

SWALLOWING

Swallowing (deglutition) is a reflex response that is triggered by afferent impulses in the trigeminal, glossopharyngeal, and vagus nerves **(Figure 27–4)**. These impulses are integrated in the nucleus of the tractus solitarius and the nucleus ambiguus. The efferent fibers pass to the pharyngeal musculature and the tongue via the trigeminal, facial, and hypoglossal nerves. Swallowing is initiated by the voluntary action of collecting the oral contents on the tongue and propelling them backward into the pharynx. This starts a wave of involuntary contraction in the pharyngeal muscles that pushes the material into the esophagus. Inhibition of respiration and glottic closure are part of the reflex response. A peristaltic ring contraction of the esophageal muscle forms behind the material, which is then swept down the esophagus at a speed of approximately 4 cm/s. When humans are in an upright position, liquids and semisolid foods generally fall by gravity to the lower esophagus ahead of the peristaltic wave. However, if any food remains in the esophagus, it is cleared by a second wave of peristalsis. It is therefore possible to swallow food while standing on one’s head.

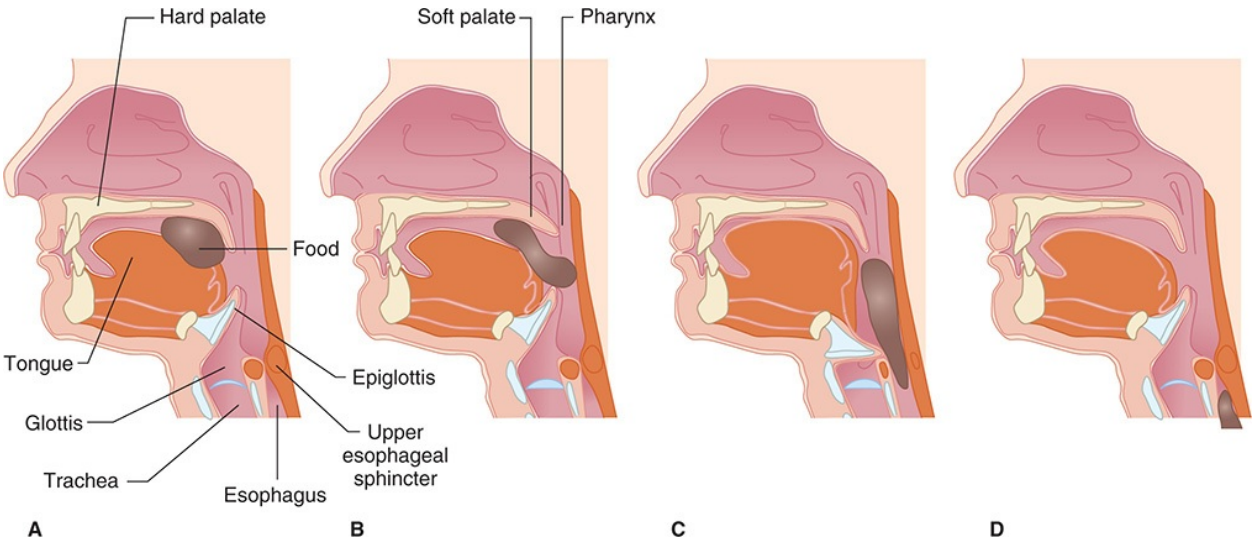


FIGURE 27–4 Movement of food through the pharynx and upper esophagus during swallowing. A) The tongue pushes the food bolus to the back of the mouth. **B)** The soft palate elevates to prevent food from entering the nasal passages. **C)** The epiglottis covers the glottis to prevent food from entering the trachea and the upper esophageal sphincter relaxes. **D)** Food descends into the esophagus.

LOWER ESOPHAGEAL SPHINCTER

Unlike the rest of the esophagus, the musculature of the gastroesophageal junction **(lower esophageal sphincter; LES)** is tonically active but relaxes on swallowing. The tonic activity of the LES between meals prevents reflux of gastric contents into the esophagus. The LES is made up of three components **(Figure 27–5)**. The esophageal smooth muscle is more prominent at the junction with the stomach (intrinsic sphincter). Fibers of the crural portion of the diaphragm, a skeletal muscle, surround the esophagus at this point (extrinsic sphincter) and exert a pinchcock-like action on the esophagus. In addition, the oblique or sling fibers of the stomach wall create a flap valve that helps close off the esophagogastric junction and prevent regurgitation when intragastric pressure rises.

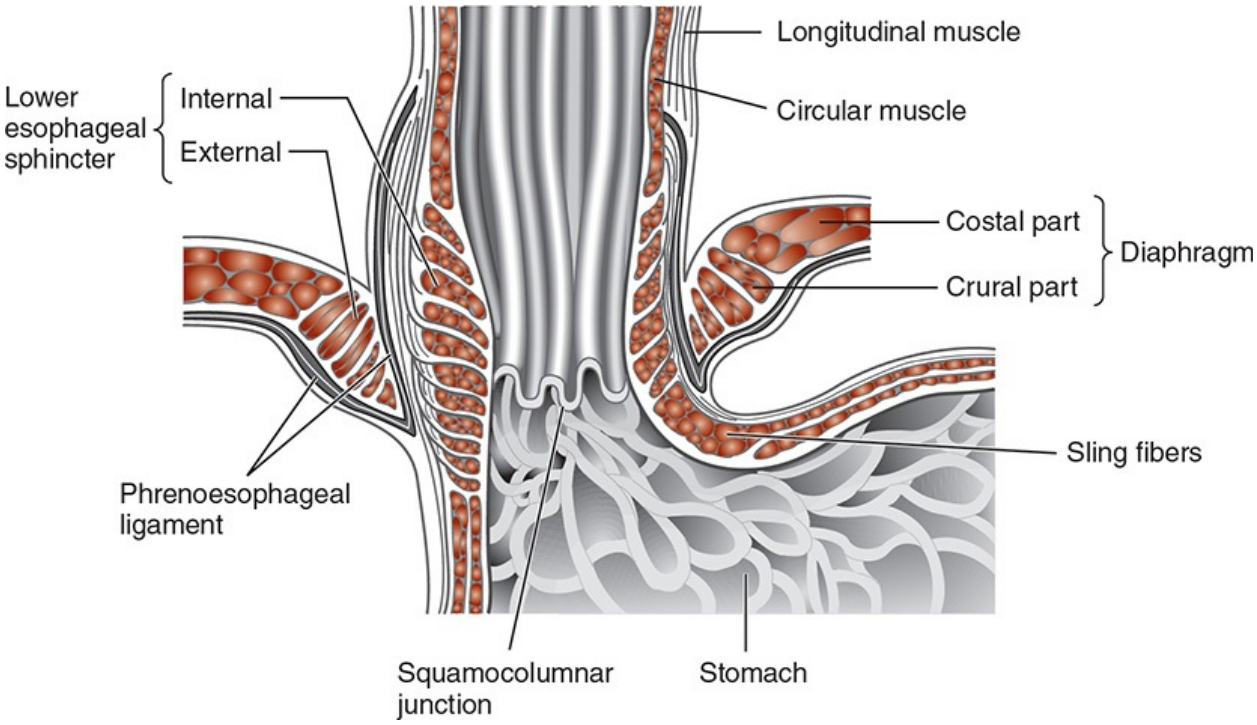


FIGURE 27–5 Esophagogastric junction. Note that the lower esophageal sphincter (intrinsic sphincter) is supplemented by the crural portion of the diaphragm (extrinsic sphincter), and that the two are anchored to each other by the phrenoesophageal ligament. (Reproduced with permission from Mittal RK, Balaban DH: The esophagogastric junction. N Engl J Med 1997; Mar 27; 336(13):924–932.)

The tone of the LES is under neural control. Release of acetylcholine from vagal endings causes the intrinsic sphincter to contract, and release of NO and VIP from interneurons innervated by other vagal fibers causes it to relax. Contraction of the crural portion of the diaphragm, which is innervated by the phrenic nerves, is coordinated with respiration and contractions of chest and abdominal muscles. Thus, the intrinsic and extrinsic sphincters operate together to permit orderly flow of food into the stomach and to prevent reflux of gastric contents into the esophagus **(Clinical Box 27–1)**.

AEROPHAGIA & INTESTINAL GAS

Some air is unavoidably swallowed in the process of eating and drinking **(aerophagia).** Some of the swallowed air is regurgitated (belching), and some of the gases it contains are absorbed, but much of it passes on to the colon. Here,

some of the oxygen is absorbed, and hydrogen, hydrogen sulfide, carbon dioxide, and methane formed by the colonic bacteria from carbohydrates and other substances are added to it. It is then expelled as **flatus.** The smell is largely due to sulfides. The volume of gas normally found in the human gastrointestinal tract is about 200 mL, and the daily production is 500–1500 mL. In some individuals, gas in the intestines causes cramps, **borborygmi** (rumbling noises), and abdominal discomfort.

STOMACH

Food is stored in the stomach; mixed with acid, mucus, and pepsin; and released at a controlled, steady rate into the duodenum.

GASTRIC MOTILITY & EMPTYING

When food enters the stomach, the fundus and upper portion of the body relax and accommodate the food with little if any increase in pressure **(receptive relaxation).** Peristalsis then begins in the lower portion of the body, mixing and grinding the food and permitting small, semiliquid portions of it to pass through the pylorus and enter the duodenum.

Receptive relaxation is, in part, vagally mediated and triggered by movement of the pharynx and esophagus. Intrinsic reflexes also lead to relaxation as the stomach wall is stretched. Peristaltic waves controlled by the gastric BER begin soon thereafter and sweep toward the pylorus. The contraction of the distal stomach caused by each wave is sometimes called **antral systole** and can last up to 10 s. Waves occur 3–4 times per minute.

In the regulation of gastric emptying, the antrum, pylorus, and upper duodenum apparently function as a unit. Contraction of the antrum is followed by sequential contraction of the pyloric region and the duodenum. In the antrum, partial contraction ahead of the advancing gastric contents prevents solid masses from entering the duodenum, and they are mixed and crushed instead. The more liquid gastric contents are squirted a bit at a time into the small intestine. Normally, regurgitation from the duodenum does not occur, because the contraction of the pyloric segment tends to persist slightly longer than that of the duodenum. The prevention of regurgitation may also be due to the stimulating action of cholecystokinin (CCK) and secretin on the pyloric sphincter.

CLINICAL BOX 27–1

Motor Disorders of the Esophagus

Achalasia (literally, failure to relax) is a condition in which food accumulates in the esophagus and the organ can become massively dilated. It is due to increased resting LES tone and incomplete relaxation on swallowing. The myenteric plexus of the esophagus is deficient at the LES and the release of NO and VIP is defective. The opposite condition is LES incompetence, which permits reflux of acid gastric contents into the esophagus **(gastroesophageal reflux disease).** This common condition is the most frequent digestive disorder causing patients to seek care from a clinician. It causes heartburn and esophagitis and can lead to ulceration and stricture of the esophagus due to scarring. In severe cases, the intrinsic sphincter or the extrinsic sphincter, and sometimes both, are weak, but less severe cases are caused by intermittent periods of poorly understood decreases in the neural drive to both sphincters.

THERAPEUTIC HIGHLIGHTS

Achalasia can be treated by pneumatic dilation of the sphincter or incision of the esophageal muscle (myotomy). Inhibition of acetylcholine release by injection of botulinum toxin into the LES is also effective and produces relief that lasts for several months. Gastroesophageal reflux disease can be treated by inhibition of acid secretion with H_2-receptor blockers or proton pump inhibitors (see Chapter 25). Surgical treatment in which a portion of the fundus of the stomach is wrapped around the lower esophagus so that the LES is inside a short tunnel of stomach **(fundoplication)** can also be tried, although in many patients who undergo this procedure the symptoms eventually return.

REGULATION OF GASTRIC MOTILITY & EMPTYING

The rate at which the stomach empties into the duodenum depends on the type of food ingested. Food rich in carbohydrate leaves the stomach in a few hours. Protein-rich food leaves more slowly, and emptying is slowest after a meal containing fat **(Figure 27–6)**. The rate of emptying also depends on the osmotic

pressure of the material entering the duodenum. Hyperosmolality of the duodenal contents is sensed by “duodenal osmoreceptors” that initiate a decrease in gastric emptying, which is probably neural in origin.

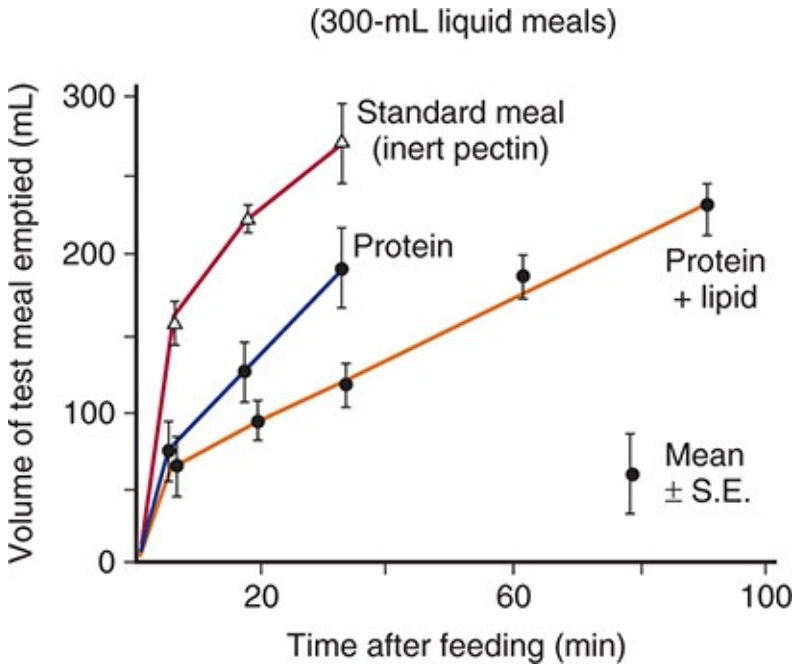


FIGURE 27–6 Effect of protein and fat on the rate of emptying of the human stomach. Persons were fed 300-mL liquid meals. (Reproduced with permission from Brooks FP: Integrative lecture. Response of the GI tract to a meal. *Undergraduate Teaching Project*. American Gastroenterological Association; 1974.)

Fats, carbohydrates, and acid in the duodenum inhibit gastric acid and pepsin secretion and gastric motility via neural and hormonal mechanisms. The messenger involved is probably peptide YY. CCK has also been implicated as an inhibitor of gastric emptying **(Clinical Box 27–2)**.

VOMITING

The protective response of vomiting is an example of central regulation of gut motility functions. Vomiting starts with salivation and the sensation of nausea. Reverse peristalsis empties material from the upper part of the small intestine into the stomach. The glottis closes, preventing aspiration of vomitus into the trachea. The breath is held in mid inspiration. The muscles of the abdominal wall contract, and because the chest is held in a fixed position, the contraction

increases intra-abdominal pressure. The LES and the esophagus relax, and the gastric contents are ejected. The “vomiting center” in the reticular formation of the medulla **(Figure 27–7)** consists of various scattered groups of neurons in this region that control the different components of the vomiting act.

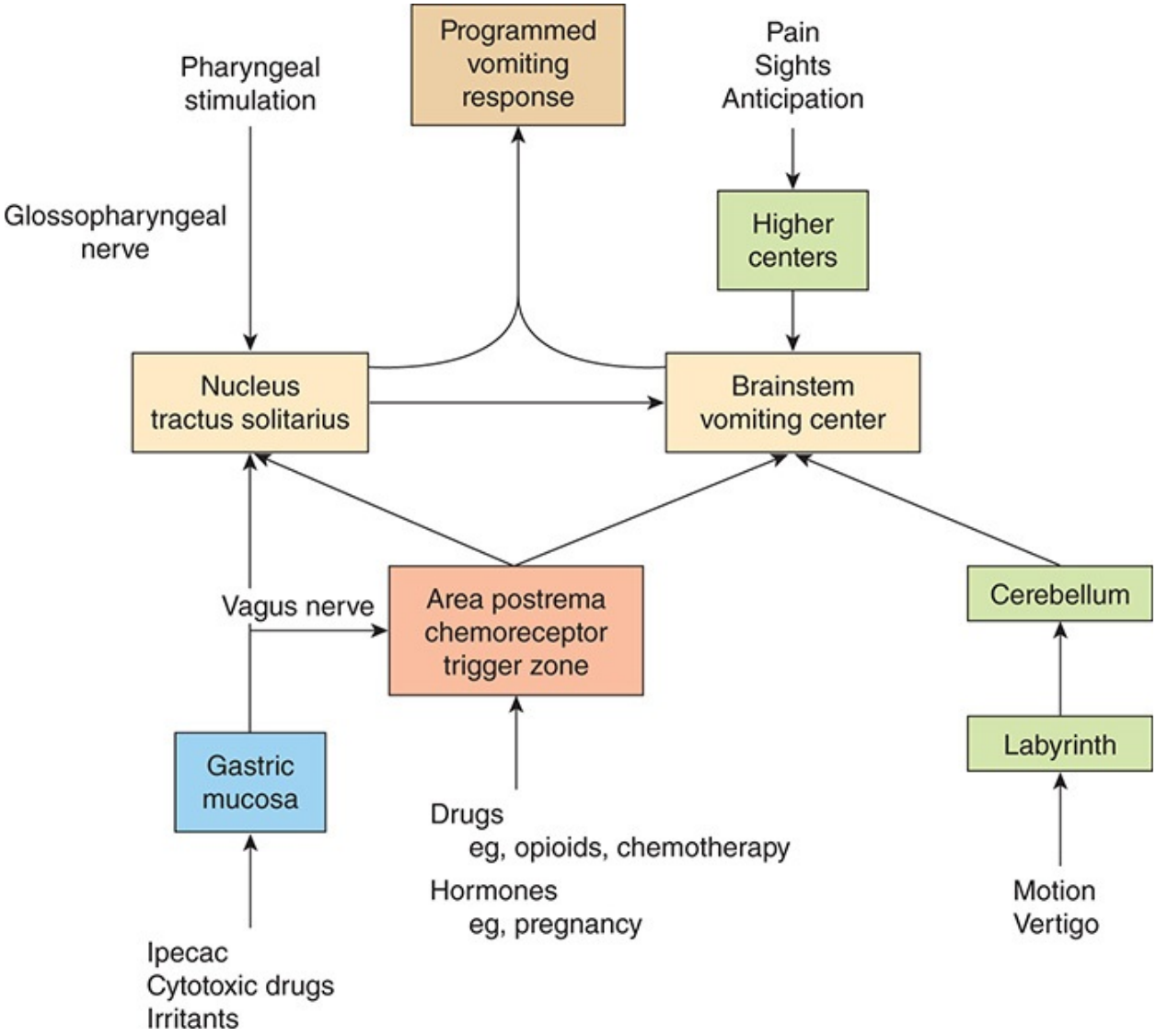


FIGURE 27–7 Neural pathways leading to the initiation of vomiting in response to various stimuli.

CLINICAL BOX 27–2

Consequences of Gastric Bypass Surgery

Patients who are morbidly obese can undergo a surgical procedure in which the stomach is stapled or otherwise reduced so that most of it is bypassed, and thus the reservoir function of the stomach is lost, along with gastric signals, such as ghrelin, that trigger food ingestion. As a result of the loss of the gastric reservoir, patients can eat only small meals. If larger meals are eaten, because of rapid absorption of glucose from the intestine and the resultant hyperglycemia and abrupt rise in insulin secretion, hypoglycemic symptoms sometimes develop about 2 h after meals. This weakness, dizziness, and sweating after meals, due in part to hypoglycemia, are part of the picture of the **"dumping syndrome,"** a distressing syndrome that can also develop in patients in whom portions of the stomach have been removed or the jejunum has been anastomosed to the stomach. Another cause of the symptoms is rapid entry of hypertonic meals into the intestine; this provokes the movement of so much water into the gut that significant hypovolemia and hypotension are produced.

THERAPEUTIC HIGHLIGHTS

There are no treatments, per se, for the dumping syndrome, other than avoiding large meals, and particularly those with high concentrations of simple sugars. Indeed, its occurrence may contribute to the overall success of bypass surgery in reducing food intake, and thus obesity, in many patients who undergo this surgery.

Irritation of the mucosa of the upper gastrointestinal tract is one trigger for vomiting. Impulses are relayed from the mucosa to the medulla over visceral afferent pathways in the sympathetic nerves and vagi. Other causes of vomiting can arise centrally. For example, afferents from the vestibular nuclei mediate the nausea and vomiting of motion sickness. Other afferents presumably reach the vomiting control areas from the diencephalon and limbic system, because emetic responses to emotionally charged stimuli also occur.

Chemoreceptor cells in the medulla can also initiate vomiting when they are stimulated by certain circulating chemical agents. The **chemoreceptor trigger zone** in which these cells are located (Figure 27–7) is in the **area postrema,** a V-shaped band of tissue on the lateral walls of the fourth ventricle near the obex. This structure is one of the circumventricular organs (see Chapter 33) and is not protected by the blood-brain barrier. Lesions of the area postrema have little effect on the vomiting response to gastrointestinal irritation or motion sickness,

but abolish the vomiting that follows injection of apomorphine and a number of other emetic drugs. Such lesions also decrease vomiting in uremia and radiation sickness, both of which may be associated with endogenous production of circulating emetic substances.

Serotonin (5-HT) released from enterochromaffin cells in the small intestine appears to initiate impulses via 5-HT_3 receptors that trigger vomiting. In addition, there are dopamine D_2 receptors and 5-HT_3 receptors in the area postrema and adjacent nucleus of the solitary tract. 5-HT_3 antagonists such as ondansetron and D_2 antagonists such as chlorpromazine and haloperidol are effective antiemetic agents. Corticosteroids, cannabinoids, and benzodiazepines, alone or in combination with 5-HT_3 and D_2 antagonists, are also useful in treatment of the vomiting produced by chemotherapy. The mechanisms of action of corticosteroids and cannabinoids are unknown, whereas the benzodiazepines probably reduce the anxiety associated with chemotherapy.

SMALL INTESTINE

In the small intestine, the intestinal contents are mixed with the secretions of the mucosal cells and with pancreatic juice and bile. The chyme is also retained in the small intestine long enough for nutrient absorption to take place.

INTESTINAL MOTILITY

The MMCs that pass along the intestine at regular intervals in the fasting state, and their replacement by peristaltic and other contractions controlled by the BER, are described above. In the small intestine, there are an average of 12 BER cycles/min in the proximal jejunum, declining to 8/min in the distal ileum. There are three types of smooth muscle contractions: peristaltic waves, segmentation contractions, and tonic contractions. **Peristalsis** is described above. It propels the intestinal contents toward the large intestines. **Segmentation contractions** (Figure 27–1), also described above, move the chyme to and fro and increase its exposure to the mucosal surface. These contractions are initiated by focal increases in Ca^{2+} influx with waves of increased Ca^{2+} concentration spreading from each focus. **Tonic contractions** are relatively prolonged contractions that in effect isolate one segment of the intestine from another. Note that these last two types of contractions slow transit in the small intestine to the point that the transit time is longer in the fed than in the fasted state. This permits longer

contact of the chyme with the enterocytes and fosters absorption **(Clinical Box 27–3)**.

CLINICAL BOX 27–3

Ileus

When the intestines are traumatized, there is a direct inhibition of smooth muscle, which causes a decrease in intestinal motility. It is due in part to activation of opioid receptors. When the peritoneum is irritated, reflex inhibition occurs due to increased discharge of noradrenergic fibers in the splanchnic nerves. Both types of inhibition operate to cause **paralytic (adynamic) ileus** after abdominal surgeries. Because of the diffuse decrease in peristaltic activity in the small intestine, its contents are not propelled into the colon, and it becomes irregularly distended by pockets of gas and fluid. Intestinal peristalsis returns in 6–8 h, followed by gastric peristalsis, but colonic activity takes 2–3 days to return.

THERAPEUTIC HIGHLIGHTS

Adynamic ileus can be relieved by passing a tube through the nose down to the small intestine and aspirating the fluid and gas for a few days until peristalsis returns. The occurrence of ileus has been reduced by more widespread use of minimally invasive (eg, laparoscopic) surgery. Postsurgical regimens also encourage early ambulation, where possible, which tends to enhance intestinal motility. There are also ongoing trials of specific opioid antagonists in this condition.

COLON

The colon serves as a reservoir for the residues of meals that cannot be digested or absorbed **(Figure 27–8)**. Motility in this segment is likewise slowed to allow the colon to absorb water, Na^+, and other minerals. By removing about 90% of the fluid, it converts the 1000–2000 mL of isotonic chyme that enters it each day from the ileum to about 200 mL of semisolid feces.

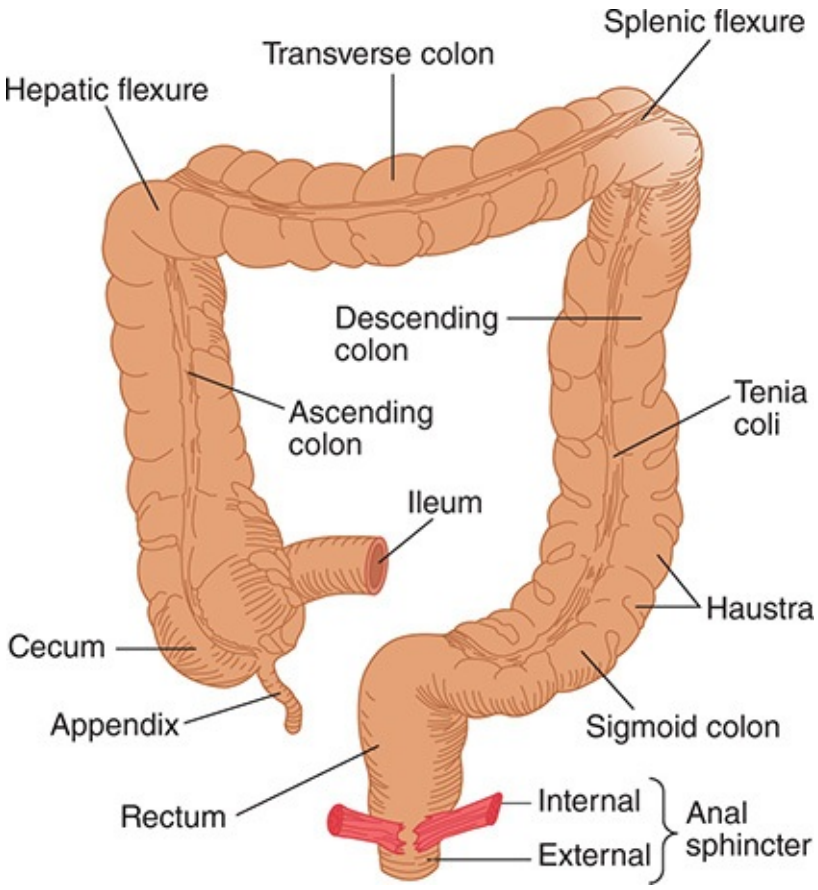


FIGURE 27–8 The human colon.

MOTILITY OF THE COLON

The ileum is linked to the colon by a structure known as the ileocecal valve, which restricts reflux of colonic contents, and particularly the larger numbers of commensal bacteria, into the relatively sterile ileum. The portion of the ileum containing the ileocecal valve projects slightly into the cecum, so that increases in colonic pressure squeeze it shut, whereas increases in ileal pressure open it. It is normally closed. Each time a peristaltic wave reaches it, it opens briefly, permitting some of the ileal chyme to squirt into the cecum. When food leaves the stomach, the cecum relaxes and the passage of chyme through the ileocecal valve increases **(gastroileal reflex).** This is presumably a vagovagal reflex.

The movements of the colon include segmentation contractions and peristaltic waves like those occurring in the small intestine. Segmentation contractions mix the contents of the colon and, by exposing more of the contents to the mucosa, facilitate absorption. Peristaltic waves propel the contents toward the rectum,

although weak antiperistalsis is sometimes seen. A third type of contraction that occurs only in the colon is the **mass action contraction,** occurring about 10 times per day, in which there is simultaneous contraction of the smooth muscle over large confluent areas. These contractions move material from one portion of the colon to another **(Clinical Box 27–4)**. They also move material into the rectum, and rectal distension initiates the defecation reflex (see below).

The movements of the colon are coordinated by the BER of the colon. The frequency of this wave, unlike the wave in the small intestine, increases along the colon, from about 2/min at the ileocecal valve to 6/min at the sigmoid.

TRANSIT TIME IN THE SMALL INTESTINE & COLON

The first part of a test meal reaches the cecum in about 4 h in most individuals, and all the undigested portions have entered the colon in 8 or 9 h. On average, the first remnants of the meal traverse the first third of the colon in 6 h, the second third in 9 h, and reach the terminal part of the colon (the sigmoid colon) in 12 h. From the sigmoid colon to the anus, transport is much slower **(Clinical Box 27–5)**. When small colored beads are fed with a meal, an average of 70% of them are recovered in the stool in 72 h, but total recovery requires more than a week. Transit time, pressure fluctuations, and changes in pH in the gastrointestinal tract can also be observed by monitoring the progress of a small pill that contains sensors and a miniature radio transmitter.

CLINICAL BOX 27–4

Hirschsprung Disease

Some children have a genetically determined condition of abnormal colonic motility known as Hirschsprung disease or **aganglionic megacolon,** which is characterized by abdominal distension, anorexia, and lassitude. The disease is typically diagnosed in infancy, and affects as many as 1 in 5000 live births. It is due to a congenital absence of the ganglion cells in both the myenteric and submucous plexuses of a segment of the distal colon, as a result of failure of the normal cranial-to-caudal migration of neural crest cells during development. The action of endothelins on the endothelin B receptor (see Chapter 7) are necessary for normal migration of certain neural crest cells;

megacolon developed in knockout mice lacking endothelin B receptors. In addition, one cause of congenital aganglionic megacolon in humans appears to be a mutation in the endothelin B receptor gene. The absence of peristalsis in patients with this disorder causes feces to pass the aganglionic region with difficulty, and children with the disease may defecate as infrequently as once every 3 weeks.

THERAPEUTIC HIGHLIGHTS

The symptoms of Hirschsprung disease can be relieved completely if the aganglionic portion of the colon is resected and the portion of the colon above it anastomosed to the rectum. However, this is not possible if an extensive segment is involved. In this case, patients may require a colectomy.

DEFECATION

Distension of the rectum with feces initiates reflex contractions of its musculature and the desire to defecate. In humans, the sympathetic nerve supply to the internal (involuntary) anal sphincter is excitatory, whereas the parasympathetic supply is inhibitory. This sphincter relaxes when the rectum is distended. The nerve supply to the external anal sphincter, a skeletal muscle, comes from the pudendal nerve. The sphincter is maintained in a state of tonic contraction, and moderate distension of the rectum increases the force of its contraction **(Figure 27–9)**. The urge to defecate first occurs when rectal pressure increases to about 18 mm Hg. When this pressure reaches 55 mm Hg, the external as well as the internal sphincter relaxes and there is reflex expulsion of the contents of the rectum. This is why reflex evacuation of the rectum can occur even in the setting of spinal injury.

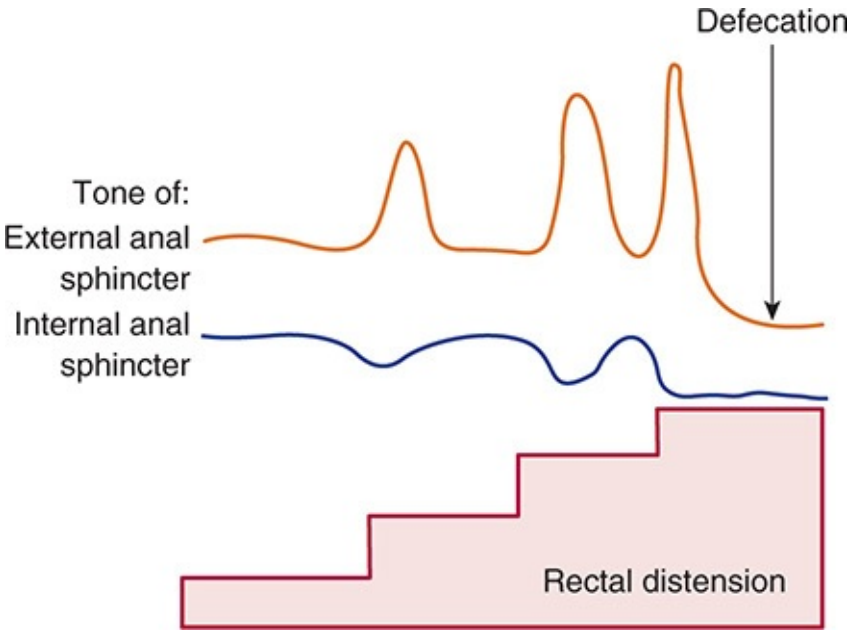


FIGURE 27–9 Motor responses of the anal sphincters to distension of the rectum. Distension produces passive tension due to stretching of the wall of the rectum, and additional active tension when the smooth muscle in the wall contracts. The internal and external sphincters respectively relax and contract, then accommodate, with each stepwise increase in distension until the pressure threshold for defecation is reached.

CLINICAL BOX 27–5

Constipation

Constipation refers to a pathologic decrease in bowel movements. It was previously considered to reflect changes in motility, but the recent success of a drug designed to enhance chloride secretion for the treatment of chronic constipation suggests alterations in the balance between secretion and absorption in the colon could also contribute to symptom generation. Patients with persistent constipation, and particularly those with a recent change in bowel habits, should be examined carefully to rule out underlying organic disease. However, many normal humans defecate only once every 2–3 days, even though others defecate once a day and some as often as three times a day. Furthermore, the only symptoms caused by constipation are slight anorexia and mild abdominal discomfort and distension. These symptoms are not due to absorption of “toxic substances,” because they are promptly

relieved by evacuating the rectum and can be reproduced by distending the rectum with inert material. In western societies, the amount of misinformation and undue apprehension about constipation probably exceeds that about any other health topic. Symptoms other than those described above that are attributed by the lay public to constipation are likely due to anxiety or other causes.

THERAPEUTIC HIGHLIGHTS

Most cases of constipation are relieved by a change in the diet to include more fiber, or the use of laxatives that retain fluid in the colon, thereby increasing the bulk of the stool and promoting reflexes that lead to evacuation. As noted above, lubiprostone has recently joined the armamentarium for the treatment of constipation, and is assumed to act by enhancing chloride, and thus water, secretion into the colon thereby increasing the fluidity of the colonic contents.

Before the pressure that relaxes the external anal sphincter is reached, voluntary defecation can be initiated by straining. Normally, the angle between the anus and the rectum is approximately 90–100° **(Figure 27–10)**, and this, added to contraction of the puborectalis muscle, inhibits defecation. With straining, the abdominal muscles contract, the pelvic floor is lowered 1–3 cm, and the puborectalis muscle relaxes. The anorectal angle is straightened, and when this is combined with relaxation of the external anal sphincter, defecation occurs. Defecation is therefore a spinal reflex that can be voluntarily inhibited by keeping the external sphincter contracted or facilitated by relaxing the sphincter and contracting the abdominal muscles.

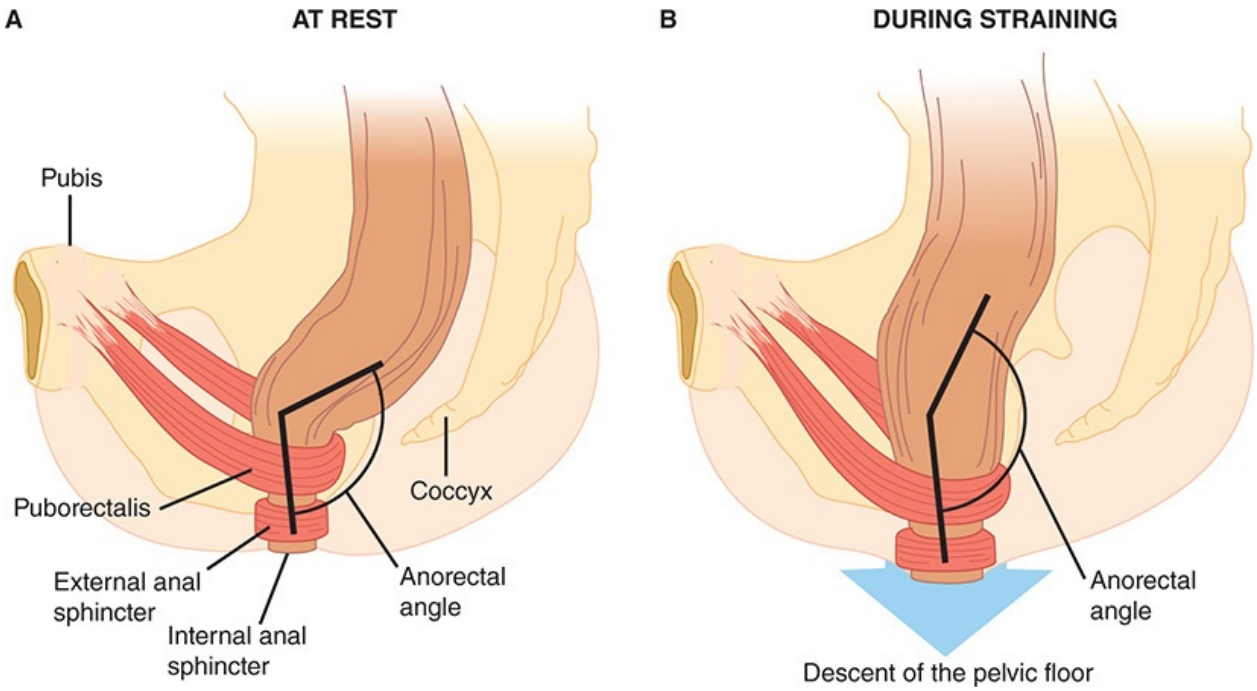


FIGURE 27–10 Sagittal view of the anorectal area at rest (A) and during straining (B). Note the straightening of the anorectal angle and lowering of the pelvic floor during straining. (Modified with permission from Lembo A, Camilleri M: Chronic constipation. N Engl J Med 2003; Oct 2; 349(14):1360–1368.)

Distension of the stomach by food initiates contractions of the rectum and, frequently, a desire to defecate. The response is called the **gastrocolic reflex,** and may be amplified by an action of gastrin on the colon. Because of the response, defecation after meals is frequent in children. In adults, habit and cultural factors play a large role in determining when defecation occurs.

CHAPTER SUMMARY

- The regulatory factors that govern gastrointestinal secretion also regulate its motility to soften the food, mix it with secretions, and propel it along the length of the tract.
- Two major patterns of motility are peristalsis and segmentation, which serve to propel or retard/mix the luminal contents, respectively. Peristalsis involves coordinated contractions and relaxations above and below the food bolus.
- The membrane potential of the majority of gastrointestinal smooth muscle

undergoes rhythmic fluctuations that sweep along the length of the gut. The rhythm varies in different gut segments and is established by pacemaker cells known as interstitial cells of Cajal. This basic electrical rhythm (BER) provides for sites of muscle contraction when stimuli superimpose spike potentials on the depolarizing portion of the BER waves.

- In the period between meals, the intestine is relatively quiescent, but every 90 min or so it is swept through by a large peristaltic wave triggered by the hormone motilin. This migrating motor complex presumably serves a "housekeeping" function.
- Swallowing is triggered centrally and is coordinated with a peristaltic wave along the length of the esophagus that drives the food bolus to the stomach, even against gravity. Relaxation of the lower esophageal sphincter is timed to just precede the arrival of the bolus, thereby limiting reflux of the gastric contents. Nevertheless, gastroesophageal reflux disease is one of the most common gastrointestinal complaints.
- The stomach accommodates the meal by a process of receptive relaxation. This permits an increase in volume without a significant increase in pressure. The stomach then serves to mix and grind the meal and to control its delivery to downstream segments.
- Motility responses in the small intestine mix the meal with pancreatic juice and bile, and propel the meal along the length of the intestine so that its digested constituents are exposed to the epithelial surface, allowing absorption.
- Luminal contents move slowly through the colon, which enhances water recovery. Distension of the rectum causes reflex contraction of the internal anal sphincter and the desire to defecate. After toilet training, defecation can be delayed until a convenient time via voluntary contraction of the external anal sphincter.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. In governing peristalsis, which of the following directions of nerve impulses and neurotransmitter is correctly matched?

   A. Retrograde; VIP

   B. Retrograde; nitric oxide

C. Anterograde; glutamate

D. Anterograde; nitric oxide

E. Anterograde; acetylcholine

2. Conversion of the basic electrical rhythm to an increase in intestinal muscle tension is produced by which of the following?

A. Activation of interstitial cells of Cajal

B. Adrenergic discharge

C. Release of acetylcholine

D. Release of VIP

E. Inhibition of nitric oxide synthase

3. The migrating motor complex is triggered by which of the following?

A. Motilin

B. NO

C. CCK

D. Somatostatin

E. Secretin

4. A patient is referred to a gastroenterologist because of persistent difficulties with swallowing. Endoscopic examination reveals that the lower esophageal sphincter fails to fully open as the bolus reaches it, and a diagnosis of achalasia is made. During the examination, or in biopsies taken from the sphincter region, a decrease would be expected in which of the following?

A. Esophageal peristalsis

B. Expression of neuronal NO synthase

C. Acetylcholine receptors

D. Substance P release

E. Contraction of the crural diaphragm

5. Gastric pressures seldom rise above the levels that breach the lower esophageal sphincter, even when the stomach is filled with a meal, due to which of the following processes?

A. Peristalsis

B. Gastroileal reflex

C. Segmentation

D. Stimulation of the vomiting center

E. Receptive relaxation

6. The physiological gastric pressure response to feeding described in Question 5 could be partly inhibited experimentally by all of the following treatments except

A. cholinergic antagonist.

B. nitric oxide synthase inhibitor.

C. cholecystokinin antagonist.

D. histamine antagonist.

E. VIP antagonist.

7. A patient who has undergone anastomosis of his jejunum to his stomach reports to his primary care physician that he has experienced several episodes of nausea, cramping, dizziness, sweating, and a rapid heart rate an hour or two after ingesting large volumes of sugary beverages. His symptoms are caused in part by

A. increased blood pressure.

B. increased secretion of glucagon.

C. increased secretion of CCK.

D. hypoglycemia.

E. hyperglycemia.

8. For the patient in Question 7, what is the most likely diagnosis?

A. Constipation-predominant irritable bowel syndrome (IBS-C)

B. Hirschsprung disease

C. Achalasia

D. Peptic ulcer disease

E. Dumping syndrome

9. In infants, defecation often follows a meal. The cause of colonic contractions in this situation is

A. histamine.

B. increased circulating levels of CCK.

C. the gastrocolic reflex.

D. increased circulating levels of somatostatin.

E. the enterogastric reflex.

10. Following a forceps delivery of her third child, a woman returns to her physician complaining of minor fecal incontinence when lifting her older children, without any urinary incontinence. Her symptoms are most likely

attributable to injury to which of the following?

A. Anal sensory nerves

B. Internal anal sphincter

C. External anal sphincter

D. Pudendal nerves

E. Puborectalis muscle

CHAPTER 28

Transport & Metabolic Functions of the Liver

OBJECTIVES

After studying this chapter, you should be able to:

- Understand the functional anatomy of the liver and the relative arrangements of hepatocytes, cholangiocytes, endothelial cells, and Kupffer cells.
- Define the characteristics of the hepatic circulation and its role in subserving the liver's functions.
- Describe the major functions of the liver with respect to metabolism, detoxification, and excretion of hydrophobic substances.
- Identify the plasma proteins that are synthesized by the liver.
- Outline the mechanisms by which the liver contributes to whole body ammonia homeostasis and the consequences of the failure of these mechanisms, particularly for brain function.
- Describe the constituents of bile, its formation, and its role in the excretion of cholesterol and bilirubin.
- Identify the mechanisms that permit normal functioning of the gallbladder and the basis of gallstone disease.

INTRODUCTION

The liver is the largest gland in the body. It is essential for life because it conducts a vast array of biochemical and metabolic functions, including ridding the body of substances that would otherwise be injurious if allowed to accumulate, and excreting drug metabolites. It is also the first port of call for most nutrients absorbed across the gut wall, supplies most of the plasma proteins, and synthesizes the bile that optimizes the absorption of fats as well as serving as an excretory fluid. The liver and associated biliary system have therefore evolved an array of structural and physiologic features that underpin this broad range of critical functions.

THE LIVER

FUNCTIONAL ANATOMY

An important function of the liver is to serve as a filter between the blood coming from the gastrointestinal tract and the blood in the rest of the body. Blood from the intestines and other viscera reach the liver via the portal vein. This blood percolates in sinusoids between plates of hepatic cells and eventually drains into the hepatic veins, which in turn empty into the inferior vena cava. Hepatic artery blood also enters the sinusoids. During its passage through the hepatic plates, the blood is extensively modified chemically. Bile is formed on the other side of each plate **(Figure 28–1)**.

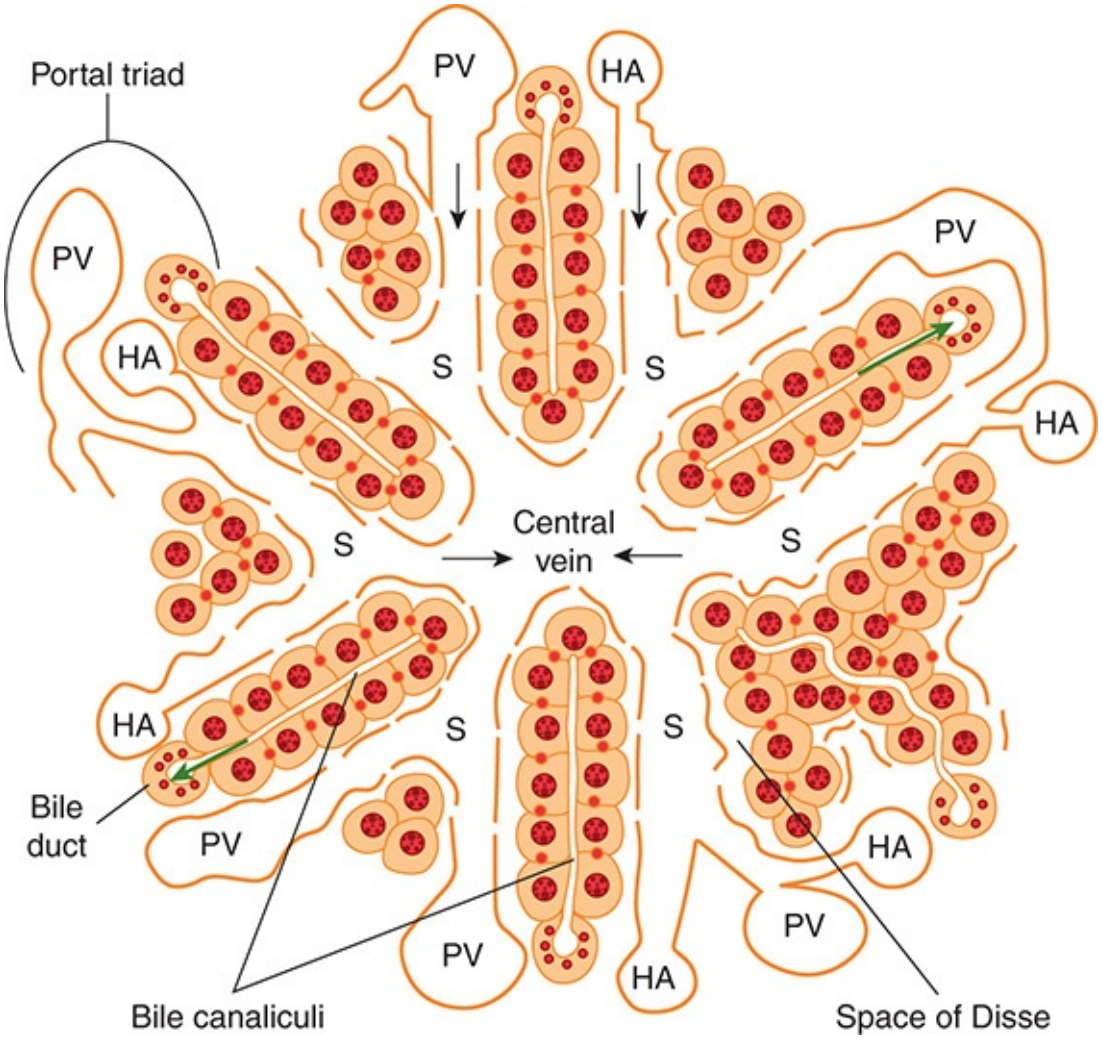


FIGURE 28–1 Schematic anatomy of the liver. Hepatocytes are arranged radially in plates surrounding a central vein. Blood is supplied to the liver by branches of the portal vein (PV) and hepatic artery (HA), which empty into sinusoids (S) surrounding the hepatocytes. The direction of blood flow is indicated with black arrows. The endothelial cells that line the sinusoids are fenestrated and thus provide little hindrance to the transfer of substances from the sinusoids to the space of Disse, which abuts the basolateral membrane of the hepatocytes. The apical membranes of adjacent hepatocytes form bile canaliculi, which transfer bile to the bile ducts lined by cholangiocytes. Bile flows in the opposite direction to blood (green arrows). The bile duct, portal vein, and hepatic artery comprise the "portal triad." (Adapted with permission from Paulsen DF: *Histology and Cell Biology: Examination and Board Review*. 5th edition. New York, NY: McGraw-Hill; 2010.)

In each hepatic lobule, the plates of hepatic cells are usually only one cell

thick. Plasma is in intimate contact with these cells **(Figure 28–2)**. The average transit time for blood across the liver lobule from the portal venule to the central hepatic vein is about 8.4 s. Additional details of the features of the hepatic microcirculation and macrocirculation, which are critical to organ function, are provided below. Numerous macrophages **(Kupffer cells)** are anchored to the endothelium of the sinusoids and project into the lumen. The functions of these phagocytic cells are discussed in Chapter 3.

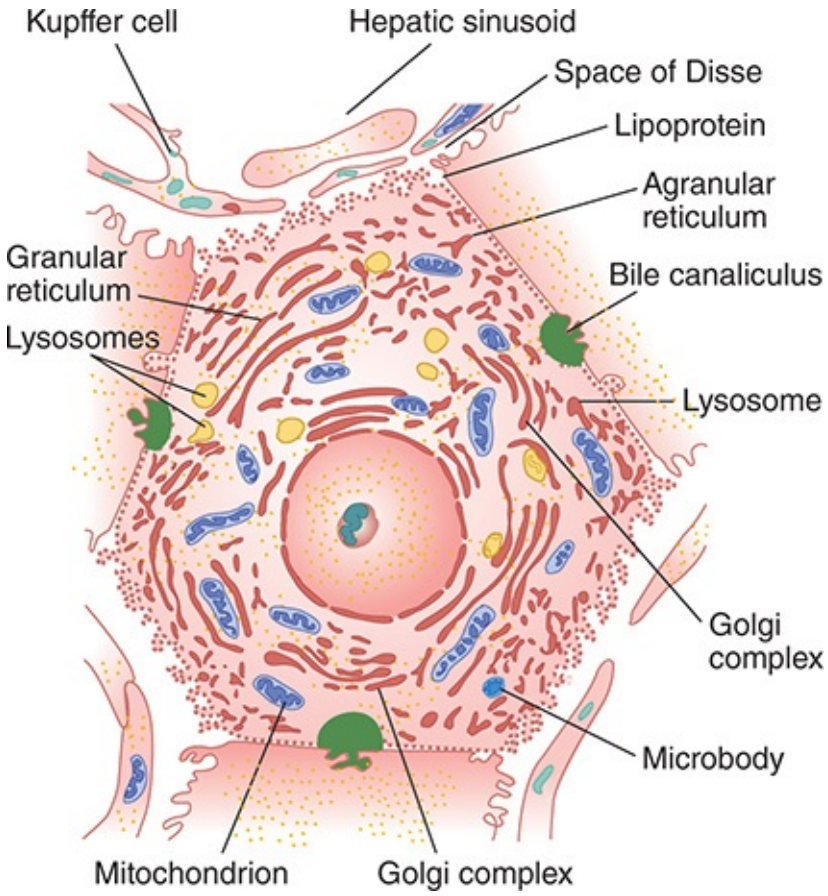


FIGURE 28–2 Hepatocyte. Note the relation of the cell to bile canaliculi and sinusoids. Note also the wide openings (fenestrations) between the endothelial cells next to the hepatocyte. (Used with permission of Sylvia Colard Keene.)

Each liver cell is opposed to several **bile canaliculi** (Figure 28–2). The canaliculi drain into intralobular bile ducts, and these coalesce via interlobular bile ducts to form the right and left hepatic ducts. These ducts join outside the liver to form the common hepatic duct. The cystic duct drains the gallbladder. The hepatic duct unites with the cystic duct to form the common bile duct (Figure 28–1). The common bile duct enters the duodenum at the duodenal

papilla. Its orifice is surrounded by the **sphincter of Oddi,** and it usually unites with the main pancreatic duct just before entering the duodenum. The sphincter is usually closed, but when the gastric contents enter the duodenum, cholecystokinin (CCK) is released and the gastrointestinal hormone relaxes the sphincter and makes the gallbladder contract.

The walls of the extrahepatic biliary ducts and the gallbladder contain fibrous tissue and smooth muscle. They are lined by a layer of columnar cells known as **cholangiocytes** with scattered mucous glands. In the gallbladder, the surface is extensively folded; this increases its surface area and gives the interior of the gallbladder a honeycombed appearance. The cystic duct is also folded to form the so-called spiral valves. This arrangement is believed to increase the turbulence of bile as it flows out of the gallbladder, thereby reducing the risk that it will precipitate and form gallstones.

HEPATIC CIRCULATION

Large gaps occur between endothelial cells in the walls of hepatic sinusoids, and the sinusoids are highly permeable. The way the intrahepatic branches of the hepatic artery and portal vein converge on the sinusoids and drain into the central lobular veins of the liver is shown in Figure 28–1. The functional unit of the liver is the acinus. Each acinus is at the end of a vascular stalk containing terminal branches of portal veins, hepatic arteries, and bile ducts. Blood flows from the center of this functional unit to the terminal branches of the hepatic veins at the periphery **(Figure 28–3)**. This is why the central portion of the acinus, sometimes called zone 1, is well oxygenated, the intermediate zone (zone 2) is moderately well oxygenated, and the peripheral zone (zone 3) is least well oxygenated and most susceptible to anoxic injury. The acini have been likened to grapes or berries, each on a vascular stem. The human liver contains about 100,000 acini.

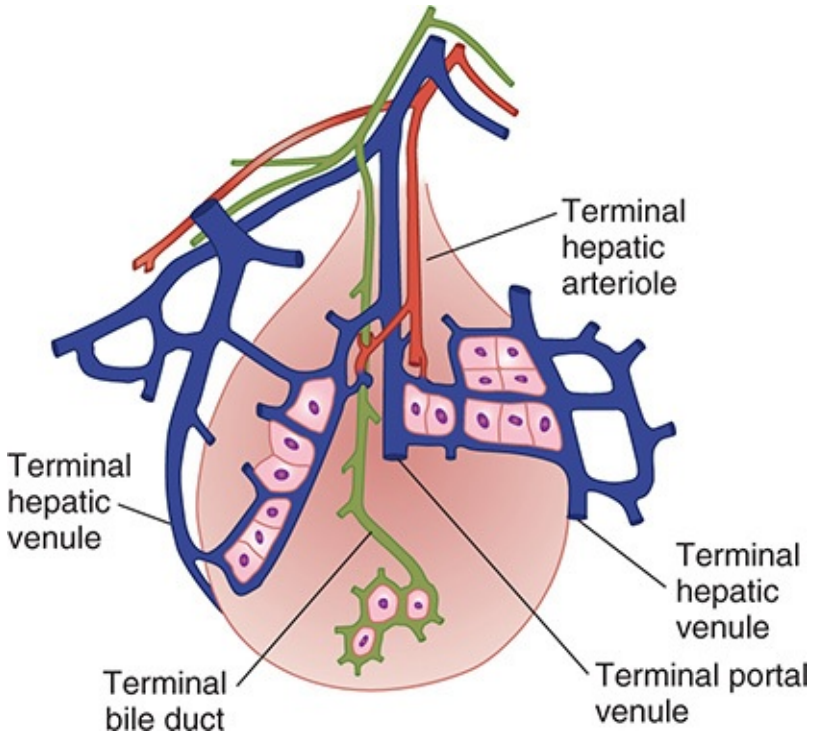


FIGURE 28–3 Concept of the acinus as the functional unit of the liver. In each acinus, blood in the portal venule and hepatic arteriole enters the center of the acinus and flows outward to the hepatic venule. (Adapted with permission from Rappaport AM: The microcirculatory hepatic unit. Microvasc Res 1973 Sep; 6(2):212–228.)

Portal venous pressure is normally about 10 mm Hg in humans, and hepatic venous pressure is approximately 5 mm Hg. The mean pressure in the hepatic artery branches that converge on the sinusoids is about 90 mm Hg, but the pressure in the sinusoids is lower than the portal venous pressure, so a marked pressure drop occurs along the hepatic arterioles. This pressure drop is adjusted so that there is an inverse relationship between hepatic arterial and portal venous blood flow. This inverse relationship may be maintained in part by the rate at which adenosine is removed from the region around the arterioles. According to this hypothesis, adenosine is produced by metabolism at a constant rate. When portal flow is reduced, it is washed away more slowly, and the local accumulation of adenosine dilates the terminal arterioles. In the period between meals, moreover, many of the sinusoids are collapsed. Following a meal, on the other hand, when portal flow to the liver from the intestine increases considerably, these “reserve” sinusoids are recruited. This arrangement means that portal pressures do not increase linearly with portal flow until all sinusoids

have been recruited. This may be important to prevent fluid loss from the highly permeable liver under normal conditions. Indeed, if hepatic pressures are increased in disease states (such as the hardening of the liver that is seen in cirrhosis), many liters of fluid can accumulate in the peritoneal cavity as **ascites.**

The intrahepatic portal vein radicles have smooth muscle in their walls that is innervated by noradrenergic vasoconstrictor nerve fibers reaching the liver via the 3rd to 11th thoracic ventral roots and the splanchnic nerves. The vasoconstrictor innervation of the hepatic artery comes from the hepatic sympathetic plexus. When systemic venous pressure rises, the portal vein radicles are dilated passively and the amount of blood in the liver increases. In heart failure, this hepatic venous congestion may be extreme. Conversely, when diffuse noradrenergic discharge occurs in response to a drop in systemic blood pressure, the intrahepatic portal radicles constrict, portal pressure rises, and blood flow through the liver is brisk, bypassing most of the organ. Most of the blood in the liver enters the systemic circulation. Constriction of the hepatic arterioles diverts blood from the liver, and constriction of the mesenteric arterioles reduces portal inflow. In severe shock, hepatic blood flow may be reduced to such a degree that patchy necrosis of the liver takes place.

FUNCTIONS OF THE LIVER

The liver has many complex functions that are summarized in **Table 28–1**. Several will be touched upon briefly here.

TABLE 28–1 Principal functions of the liver.

Formation and secretion of bile
Nutrient and vitamin metabolism
Glucose and other sugars
Amino acids
Lipids
Fatty acids
Cholesterol
Lipoproteins
Fat-soluble vitamins
Water-soluble vitamins
Inactivation of various substances
Toxins
Steroids
Other hormones
Synthesis of plasma proteins
Acute-phase proteins
Albumin
Clotting factors
Steroid-binding and other hormone-binding proteins
Immunity
Kupffer cells

METABOLISM & DETOXIFICATION

It is beyond the scope of this volume to touch upon all of the metabolic functions of the liver. Instead, this chapter will focus on those aspects most closely aligned to gastrointestinal physiology. First, the liver plays key roles in carbohydrate metabolism, including glycogen storage, conversion of galactose and fructose to glucose, and gluconeogenesis, as well as many of the reactions covered in Chapter 1. The substrates for these reactions derive from the products of carbohydrate digestion and absorption that are transported from the intestine to the liver in the portal blood. The liver also plays a major role in maintaining the stability of blood glucose levels in the postprandial period, removing excess glucose from the blood and returning it as needed—the so-called **glucose buffer**

function of the liver. In liver failure, hypoglycemia is commonly seen. Similarly, the liver contributes to fat metabolism. It supports a high rate of fatty acid oxidation for energy supply to the liver itself and other organs. Amino acids and two carbon fragments derived from carbohydrates are also converted in the liver to fats for storage. The liver also synthesizes most of the lipoproteins required by the body and preserves cholesterol homeostasis by synthesizing this molecule and also converting excess cholesterol to bile acids.

The liver also detoxifies the blood of substances originating from the gut or elsewhere in the body. Part of this function is physical in nature—bacteria and other particulates are trapped in and broken down by the strategically located Kupffer cells. The remaining reactions are biochemical, and mediated in their first stages by the large number of cytochrome P450 enzymes expressed in hepatocytes. These convert xenobiotics and other toxins to inactive, less lipophilic metabolites. Detoxification reactions are divided into phase I (oxidation, hydroxylation, and other reactions mediated by cytochrome P450s) and phase II (esterification). Ultimately, metabolites are secreted into the bile for elimination via the gastrointestinal tract. In this regard, in addition to disposing of drugs, the liver is responsible for metabolism of essentially all steroid hormones. Liver disease can therefore result in the apparent overactivity of the relevant hormone systems.

SYNTHESIS OF PLASMA PROTEINS

The principal proteins synthesized by the liver are listed in Table 28–1. Albumin is quantitatively the most significant, and accounts for the majority of plasma oncotic pressure. Many of the products are **acute-phase proteins,** proteins synthesized and secreted into the plasma on exposure to stressful stimuli (see Chapter 3). Others are proteins that transport steroids and other hormones in the plasma, and still others are clotting factors. Following blood loss, the liver replaces the plasma proteins in days to weeks. The only major class of plasma proteins not synthesized by the liver is the immunoglobulins.

AMMONIA METABOLISM & EXCRETION

The liver is critical for ammonia handling in the body **(Clinical Box 28–1)**. Ammonia levels must be carefully controlled because it is toxic to the central nervous system (CNS), and freely permeable across the blood-brain barrier. The

liver is the only organ in which the complete urea cycle (also known as the Krebs-Henseleit cycle) is expressed (Figure 1–20). This converts circulating ammonia to urea, which can then be excreted in the urine **(Figure 28–4)**.

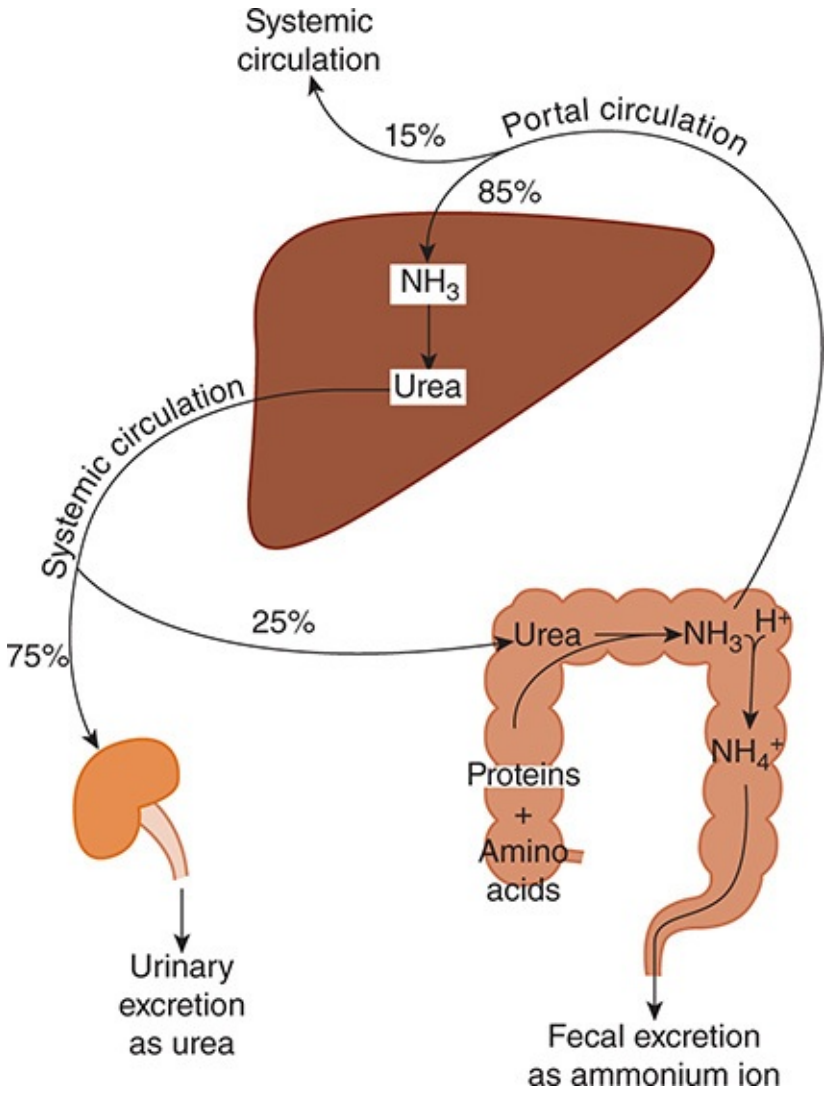


FIGURE 28–4 Whole body ammonia homeostasis in health. The majority of ammonia produced by the body is excreted by the kidneys in the form of urea.

Ammonia in the circulation comes primarily from the colon and kidneys with lesser amounts deriving from the breakdown of red blood cells and from metabolism in the muscles. As it passes through the liver, almost all of the ammonia in the circulation is cleared into the hepatocytes. There, it is converted in the mitochondria to carbamoyl phosphate, which in turn reacts with ornithine to generate citrulline. A series of subsequent cytoplasmic reactions eventually produce arginine, and this can be dehydrated to urea and ornithine. The latter

returns to the mitochondria to begin another cycle, and urea, as a small molecule, diffuses readily back out into the sinusoidal blood. It is then filtered in the kidneys and lost from the body in the urine.

CLINICAL BOX 28–1

Hepatic Encephalopathy

The clinical importance of hepatic ammonia metabolism is seen in liver failure, when increased levels of circulating ammonia cause the condition of hepatic encephalopathy. Initially, patients may seem merely confused, but if untreated, the condition can progress to coma and irreversible changes in cognition. The disease results not only from the loss of functional hepatocytes, but also shunting of portal blood around the hardened liver, meaning that less ammonia is removed from the blood by the remaining hepatic mass. Additional substances that are normally detoxified by the liver likely also contribute to the mental status changes.

THERAPEUTIC HIGHLIGHTS

The cognitive symptoms of advanced liver disease can be minimized by reducing the load of ammonia coming to the liver from the colon (eg, by feeding the nonabsorbable carbohydrate, lactulose, which is converted into short-chain fatty acids in the colonic lumen and thereby traps luminal ammonia in its ionized form). However, in severe disease, the only truly effective treatment is to perform a liver transplant, although the paucity of available organs means that there is great interest in artificial liver assist devices that could clean the blood.

BILE

Bile is made up of the bile acids, bile pigments, and other substances dissolved in an alkaline electrolyte solution that resembles pancreatic juice **(Table 28–2)**. About 500 mL is secreted per day. Some of the components of the bile are reabsorbed in the intestine and then excreted again by the liver **(enterohepatic circulation).** In addition to its role in digestion and absorption of fats (Chapter 26), bile (and subsequently the feces) is the major excretory route for lipid-

soluble waste products.

TABLE 28–2 Composition of human hepatic duct bile.

Water	97.0%
Bile acids	0.7%
Bile pigments	0.2%
Cholesterol	0.06%
Inorganic salts	0.7%
Fatty acids	0.15%
Phosphatidylcholine	0.2%
Fat	0.1%
Alkaline phosphatase	…

The glucuronides of the **bile pigments,** bilirubin and biliverdin, are responsible for the golden yellow color of bile. The formation of these breakdown products of hemoglobin is discussed in detail in Chapter 31, and their excretion is discussed in the following text.

BILIRUBIN METABOLISM & EXCRETION

Most of the bilirubin in the body is formed in the tissues by the breakdown of hemoglobin (see Chapter 31 and **Figure 28–5**). The bilirubin is bound to albumin in the circulation. Most of it is tightly bound, but some of it can dissociate in the liver, and free bilirubin enters liver cells via a member of the organic anion transporting polypeptide (OATP) family, and then becomes bound to cytoplasmic proteins **(Figure 28–6)**. It is next conjugated to glucuronic acid in a reaction catalyzed by the enzyme **glucuronyl transferase** (UDP-glucuronosyltransferase). This enzyme is located primarily in the smooth endoplasmic reticulum. Each bilirubin molecule reacts with two uridine diphosphoglucuronic acid (UDPGA) molecules to form bilirubin diglucuronide. This glucuronide, which is more water-soluble than the free bilirubin, is then transported against a concentration gradient most likely by an active transporter known as multidrug resistance protein-2 (MRP-2) into the bile canaliculi. A small amount of the bilirubin glucuronide escapes into the blood, where it is

bound less tightly to albumin than is free bilirubin, and is excreted in the urine. Thus, the total plasma bilirubin normally includes free bilirubin plus a small amount of conjugated bilirubin. Most of the bilirubin glucuronide passes via the bile ducts to the intestine.

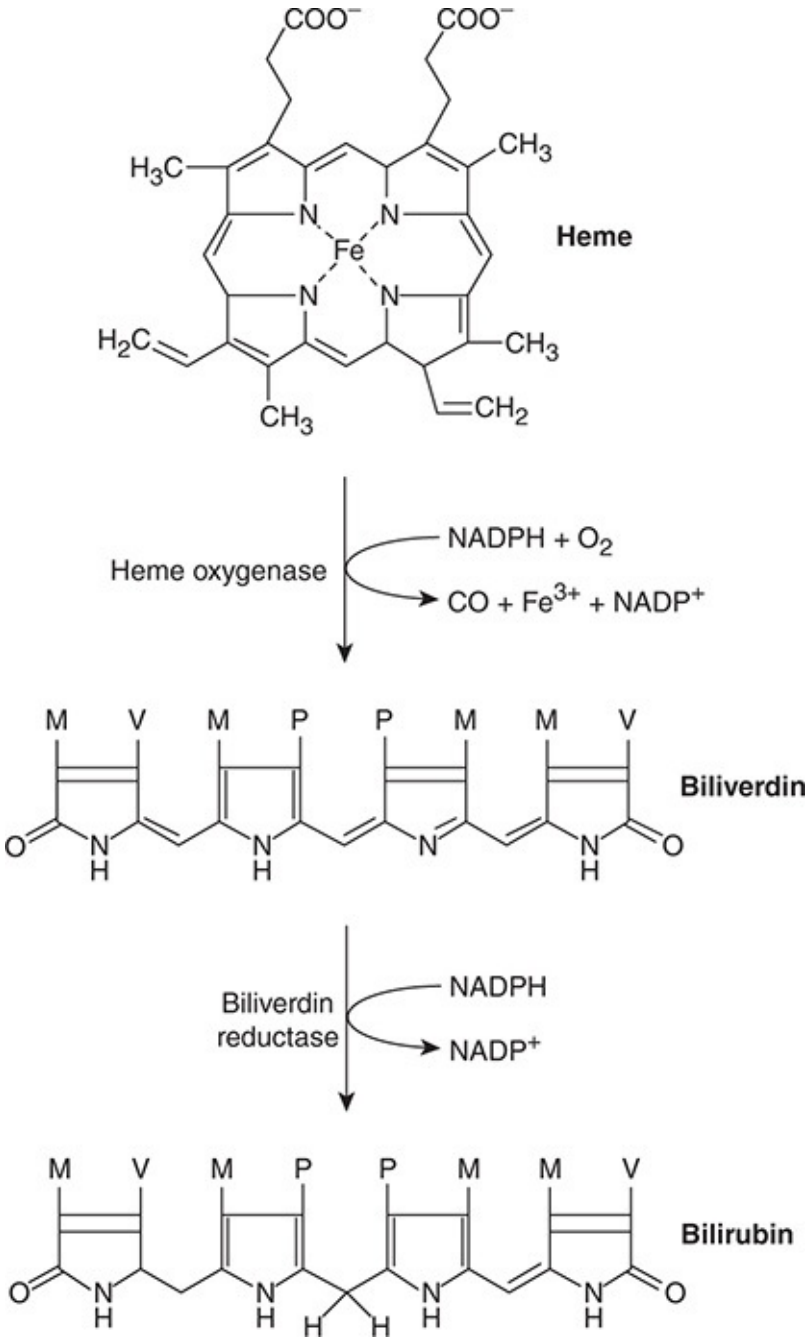


FIGURE 28–5 Conversion of heme to bilirubin is a two-step reaction catalyzed by heme oxygenase and biliverdin reductase. M, methyl; P, propionate; V, vinyl.

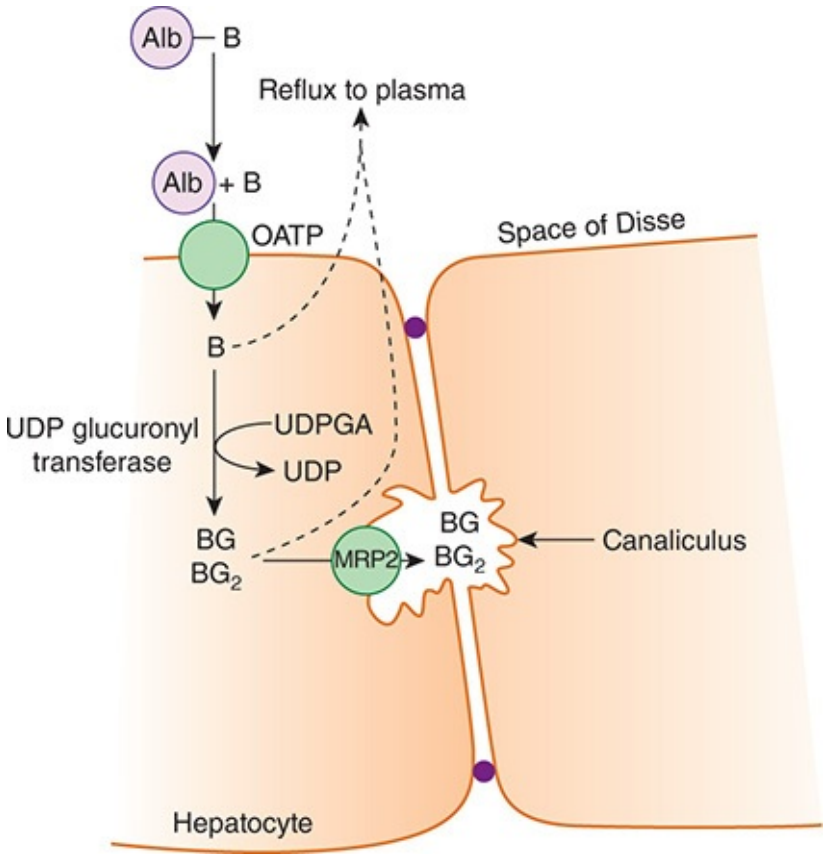


FIGURE 28-6 Handling of bilirubin by hepatocytes. Albumin (Alb)-bound bilirubin (B) enters the space of Disse adjacent to the basolateral membrane of hepatocytes, and bilirubin is selectively transported into the hepatocyte. Here, it is conjugated with glucuronic acid (G). The conjugates are secreted into bile via the multidrug resistance protein 2 (MRP-2). Some unconjugated and conjugated bilirubin also refluxes into the plasma. The purple circles linking the two adjacent cells represent the tight junctions. BG, bilirubin monoglucuronide; BG_2, bilirubin diglucuronide; OATP, organic anion transporting polypeptide.

The intestinal mucosa is relatively impermeable to conjugated bilirubin but is permeable to unconjugated bilirubin and to urobilinogens, a series of colorless derivatives of bilirubin formed by the action of bacteria in the intestine. Consequently, some of the bile pigments and urobilinogens are reabsorbed in the portal circulation. Some of the reabsorbed substances are again excreted by the liver (enterohepatic circulation), but small amounts of urobilinogens enter the general circulation and are excreted in the urine.

JAUNDICE

When free or conjugated bilirubin accumulates in the blood, the skin, scleras, and mucous membranes turn yellow. This yellowness is known as **jaundice** (icterus) and is usually detectable when the total plasma bilirubin is greater than 2 mg/dL (34 µmol/L). Hyperbilirubinemia may be due to (1) excess production of bilirubin (hemolytic anemia, etc; see Chapter 31), (2) decreased uptake of bilirubin into hepatic cells, (3) disturbed intracellular protein binding or conjugation, (4) disturbed secretion of conjugated bilirubin into the bile canaliculi, or (5) intrahepatic or extrahepatic bile duct obstruction. When it is due to one of the first three processes, the free bilirubin rises. When it is due to disturbed secretion of conjugated bilirubin or bile duct obstruction, bilirubin glucuronide regurgitates into the blood, and it is predominantly the conjugated bilirubin in the plasma that is elevated.

OTHER SUBSTANCES CONJUGATED BY GLUCURONYL TRANSFERASE

The glucuronyl transferase system in the smooth endoplasmic reticulum catalyzes the formation of the glucuronides of a variety of substances in addition to bilirubin. As discussed above, the list includes steroids (see Chapter 20) and various drugs. These other compounds can compete with bilirubin for the enzyme system when they are present in appreciable amounts. In addition, several barbiturates, antihistamines, anticonvulsants, and other compounds cause marked proliferation of the smooth endoplasmic reticulum in the hepatic cells, with a concurrent increase in hepatic glucuronyl transferase activity. Phenobarbital has been used successfully for the treatment of a congenital disease in which there is a relative deficiency of glucuronyl transferase (type 2 UDP-glucuronosyltransferase deficiency).

OTHER SUBSTANCES EXCRETED IN THE BILE

Cholesterol and alkaline phosphatase are excreted in the bile. In patients with jaundice due to intrahepatic or extrahepatic obstruction of the bile duct, the blood levels of these two substances usually rise. A much smaller rise is generally seen when the jaundice is due to nonobstructive hepatocellular disease. Adrenocortical and other steroid hormones and a number of drugs are excreted in the bile and subsequently reabsorbed (enterohepatic circulation).

THE BILIARY SYSTEM

BILE FORMATION

Each of its major constituents derived from the hepatocytes enters the bile by means of a specific canalicular transporter. It is the active secretion of bile acids, however, that is believed to be the primary driving force for the initial formation of canalicular bile. Because they are osmotically active, the canalicular bile is transiently hypertonic. However, the tight junctions that join adjacent hepatocytes are relatively permeable and thus a number of additional substances passively enter the bile from the plasma by diffusion. These substances include water, glucose, calcium, glutathione, amino acids, and urea.

Phosphatidylcholine that enters the bile forms mixed micelles with the bile acids and cholesterol. The ratio of bile acids:phosphatidylcholine:cholesterol in canalicular bile is approximately 10:3:1. Deviations from this ratio may cause cholesterol to precipitate, leading to one type of gallstones **(Figure 28–7)**.

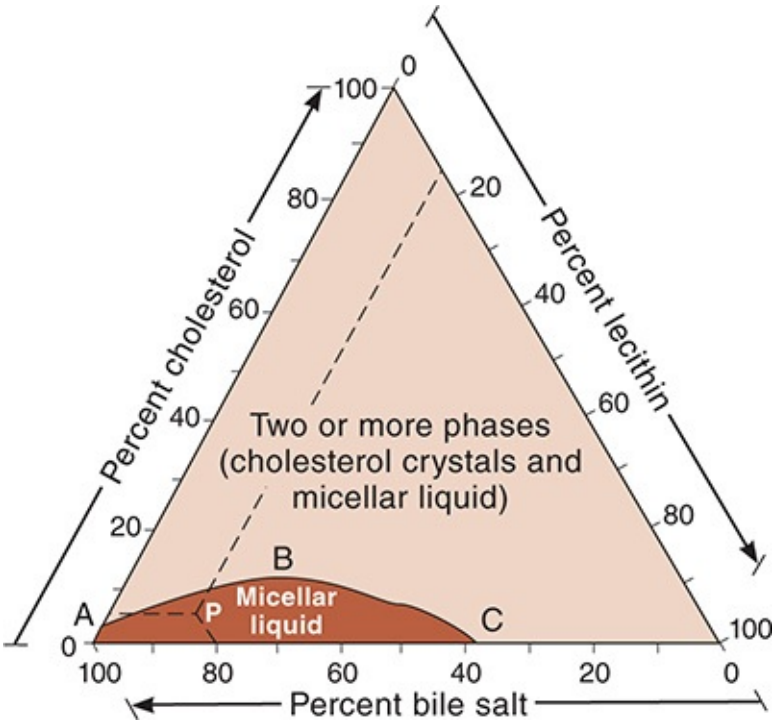


FIGURE 28–7 Cholesterol solubility in bile as a function of the proportions of lecithin (phosphatidylcholine), bile salts (bile acids), and cholesterol. In bile that has a composition described by any point below line ABC (eg, point P), cholesterol is solely in micellar solution; points above line ABC describe bile in which there are cholesterol crystals as well. (Reproduced with permission from

Small DM: Gallstones. N Engl J Med 1968; Sep 12; 279(11):588–593.)

The bile is transferred to progressively larger bile ductules and ducts, where it undergoes modification of its composition. The bile ductules are lined by cholangiocytes, specialized columnar epithelial cells. Their tight junctions are less permeable than those of the hepatocytes, although they remain freely permeable to water and thus bile remains isotonic. The ductules scavenge plasma constituents, such as glucose and amino acids, and return them to the circulation by active transport. Glutathione is also hydrolyzed to its constituent amino acids by an enzyme, gamma glutamyltranspeptidase (GGT), expressed on the apical membrane of the cholangiocytes. Removal of glucose and amino acids is likely important to prevent bacterial overgrowth in the bile, particularly during gallbladder storage (see below). The ductules also secrete bicarbonate in response to secretin in the postprandial period, as well as IgA and mucus for protection.

FUNCTIONS OF THE GALLBLADDER

In normal individuals, bile flows into the gallbladder when the sphincter of Oddi is closed (ie, the period in between meals). In the gallbladder, the bile is concentrated by absorption of water. The degree of this concentration is shown by the increase in the concentration of solids **(Table 28–3)**; hepatic bile is 97% water, whereas the average water content of gallbladder bile is 89%. However, because the bile acids are a micellar solution, the micelles simply become larger, and since osmolarity is a colligative property, bile remains isotonic. However, bile becomes less alkaline as sodium ions are exchanged for protons (although the overall concentration of sodium ions rises with a concomitant loss of chloride and bicarbonate as the bile is concentrated).

TABLE 28–3 Comparison of human hepatic duct bile and gallbladder bile.

	Hepatic Duct Bile	Gallbladder Bile
Percentage of solids	2–4	10–12
Bile acids (mmol/L)	10–20	50–200
Ph	7.8–8.6	7.0–7.4

REGULATION OF BILIARY SECRETION

When food enters the mouth, the resistance of the sphincter of Oddi decreases under both neural and hormonal influences **(Figure 28–8)**. Fatty acids and amino acids in the duodenum release CCK, which causes gallbladder contraction.

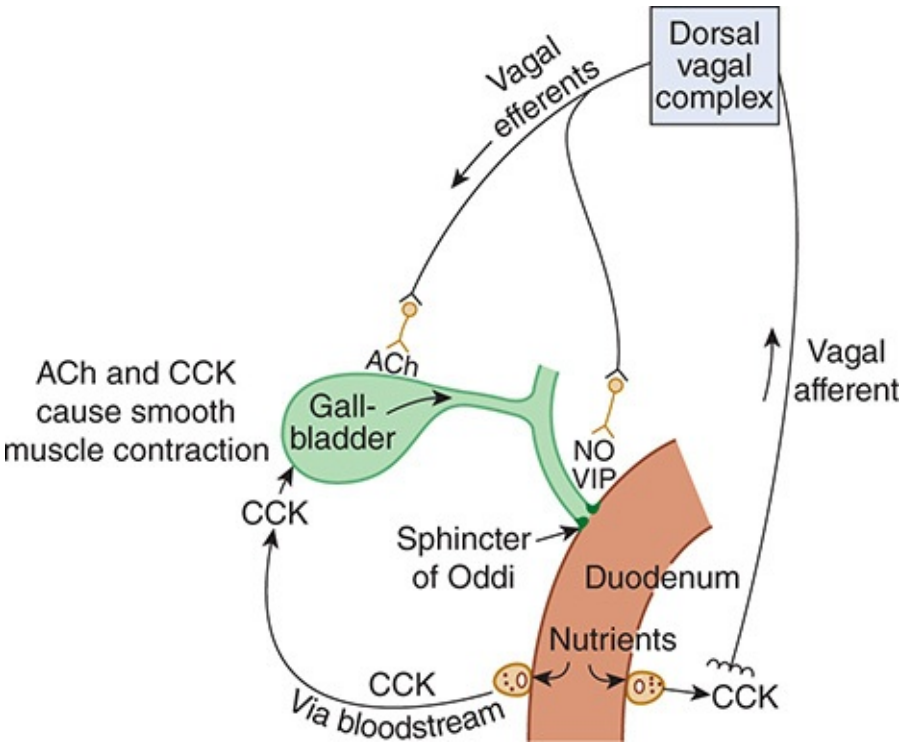


FIGURE 28–8 Neurohumoral control of gallbladder contraction and biliary secretion. Endocrine release of cholecystokinin (CCK) in response to nutrients causes gallbladder contraction. CCK also activates vagal afferents to trigger a vagovagal reflex that reinforces gallbladder contraction (via acetylcholine [ACh]) and relaxation of the sphincter of Oddi to permit bile outflow (via NO and vasoactive intestinal polypeptide [VIP]).

The production of bile is increased by stimulation of the vagus nerves and by the hormone secretin, which increases the water and HCO_3^- content of bile. Substances that increase the secretion of bile are known as **choleretics.** Bile acids themselves are among the most important physiologic choleretics.

EFFECTS OF CHOLECYSTECTOMY

The periodic discharge of bile from the gallbladder aids digestion but is not

essential for it. Cholecystectomized patients maintain good health and nutrition with a constant slow discharge of bile into the duodenum, although eventually the bile duct becomes somewhat dilated, and more bile tends to enter the duodenum after meals than at other times.

VISUALIZING THE GALLBLADDER

Exploration of the right upper quadrant with an ultrasonic beam **(ultrasonography)** and computed tomography (CT) have become the most widely used methods for visualizing the gallbladder and detecting gallstones. A third method of diagnosing gallbladder disease is **nuclear cholescintigraphy.** When administered intravenously, technetium-99m-labeled derivatives of iminodiacetic acid are excreted in the bile and provide excellent gamma camera images of the gallbladder and bile ducts. The response of the gallbladder to CCK can then be observed following intravenous administration of the hormone. The biliary tree can also be visualized by injecting contrast media from an endoscope channel maneuvered into the sphincter of Oddi, in a procedure known as endoscopic retrograde cholangiopancreatography (ERCP). It is even possible to insert small instruments with which to remove gallstone fragments that may be obstructing the flow of bile, the flow of pancreatic juice, or both **(Clinical Box 28–2)**.

CLINICAL BOX 28–2

Gallstones

Cholelithiasis, that is, the presence of gallstones, is a common condition. Its incidence increases with age, so that in the United States, for example, 20% of the women and 5% of the men between the ages of 50 and 65 have gallstones. The stones are of two types: calcium bilirubinate stones and cholesterol stones. In the United States and Europe, 85% of the stones are cholesterol stones. Three factors appear to be involved in the formation of cholesterol stones. One is bile stasis; stones form in the bile that is sequestrated in the gallbladder rather than the bile that is flowing in the bile ducts. A second is supersaturation of the bile with cholesterol. Cholesterol is very insoluble in bile, and it is maintained in solution in micelles only at certain concentrations of bile salts and lecithin. At concentrations above line ABC in Figure 28–7, the bile is supersaturated and contains small crystals of

cholesterol in addition to micelles. However, many healthy individuals in whom gallstones do not develop also have supersaturated bile. The third factor is a mix of nucleation factors that favors formation of stones from the supersaturated bile. Outside the body, bile from patients with cholelithiasis forms stones in 2–3 days, whereas it takes more than 2 weeks for stones to form in bile from normal individuals. The exact nature of the nucleation factors is unsettled, although glycoproteins in gallbladder mucus have been implicated. In addition, it is unsettled whether stones form as a result of excess production of components that favor nucleation or decreased production of antinucleation components that prevent stones from forming in normal individuals.

Gallstones that obstruct bile outflow from the liver can result in **obstructive jaundice.** If the flow of bile out of the liver is completely blocked, substances normally excreted in the bile, such as cholesterol, accumulate in the bloodstream. The interruption of the enterohepatic circulation of bile acids also induces the liver to synthesize bile acids at a greater rate. Some of these bile acids can be excreted by the kidney, and thus represent a mechanism for indirect excretion of at least a portion of cholesterol. However, retained biliary constituents may also cause liver toxicity.

THERAPEUTIC HIGHLIGHTS

The treatment of gallstones depends on their nature and the severity of any symptoms. Many, particularly if small and retained in the gallbladder, may be asymptomatic. Larger stones that cause obstruction may need to be removed surgically, or via ERCP. Oral dissolution agents may dissolve small stones composed of cholesterol, but the effect is slow and stones often return once therapy is stopped. A definitive cure for patients who suffer from recurrent attacks of symptomatic cholelithiasis is to have the gallbladder removed, which is usually now performed laparoscopically.

CHAPTER SUMMARY

- The liver conducts a huge number of metabolic reactions and serves to detoxify and dispose of many exogenous substances, as well as metabolites endogenous to the body that would be harmful if allowed to accumulate.

- The structure of the liver is such that it can filter large volumes of blood and remove even hydrophobic substances that are protein-bound. This function is provided for by a fenestrated endothelium. The liver also receives essentially all venous blood from the intestine prior to its delivery to the remainder of the body.
- The liver serves to buffer blood glucose, synthesize the majority of plasma proteins, contribute to lipid metabolism, and preserve cholesterol homeostasis.
- Bilirubin is an end product of heme metabolism that is glucuronidated by the hepatocyte to permit its excretion in bile. Bilirubin and its metabolites impart color to the bile and stools.
- The liver removes ammonia from the blood and converts it to urea for excretion by the kidneys. An accumulation of ammonia as well as other toxins cause hepatic encephalopathy in the setting of liver failure.
- Bile contains substances actively secreted across the canalicular membrane by hepatocytes, and notably bile acids, phosphatidylcholine, and cholesterol. The composition of bile is modified as it passes through the bile ducts and is stored in the gallbladder. Gallbladder contraction is regulated to coordinate bile availability with the timing of meals.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. Which of the following cell types protects against sepsis secondary to translocation of intestinal bacteria?

   A. Hepatic stellate cell

   B. Cholangiocyte

   C. Kupffer cell

   D. Hepatocyte

   E. Gallbladder epithelial cell

2. A 60-year-old man comes to his physician complaining of a progressive increase in his girth despite attempts to diet. He is also jaundiced and complains of nausea and malaise. When a large needle is inserted into his abdomen, several liters of tan fluid drain out. An increase in which of the following is not involved in this fluid accumulation?

A. Portal pressure
B. Hepatic collagen
C. Plasma albumin
D. Stellate cell activity
E. Plasma transudation

3. P450s (CYPs) are highly expressed in hepatocytes. In which of the following do they *not* play an important role?

A. Bile acid formation
B. Carcinogenesis
C. Steroid hormone formation
D. Detoxification of drugs
E. Glycogen synthesis

4. A surgeon is studying new methods of liver transplantation. She performs a complete hepatectomy in an experimental animal. Before the donor liver is grafted, a rise in the blood level of which of the following would be expected?

A. Glucose
B. Fibrinogen
C. 25-Hydroxycholecalciferol
D. Conjugated bilirubin
E. Estrogens

5. A scientist studying the function of the different cell types in the liver creates a mouse in which the activity of one specific cell type is transiently ablated. The mice exhibit behavioral defects consistent with encephalopathy. The function of which of the following cell types was most likely targeted?

A. Hepatic stellate cell
B. Cholangiocyte
C. Kupffer cell
D. Hepatocyte
E. Gallbladder epithelial cell

6. A patient suffering from severe ulcerative colitis undergoes a total colectomy with formation of a stoma. After a full recovery from surgery, and compared to his condition prior to surgery, which of the following would be expected to be decreased?

A. Ability to absorb lipids

B. Ability to clot the blood

C. Circulating levels of conjugated bile acids

D. Urinary urea

E. Urinary urobilinogen

7. Compared to hepatic bile, gallbladder bile contains a reduced concentration of which of the following?

A. Bile acids

B. Chloride ions

C. Protons

D. Glucose

E. Calcium ions

8. A newborn delivered vaginally is noted to be mildly jaundiced, but no bilirubin is found in the urine. The child's symptoms are most likely attributable to a developmental delay in the expression or establishment of which of the following:

A. Colonic bacterial colonization

B. MDR2

C. UDP glucuronyl transferase

D. Heme oxygenase

E. Biliverdin reductase

9. A 40-year-old woman comes to her primary care clinician complaining of severe, episodic abdominal pain that is particularly intense after she ingests a fatty meal. An imaging procedure reveals that her gallbladder is acutely dilated, and a diagnosis of cholelithiasis is made. A gallstone lodged in which location will also increase her risk of pancreatitis?

A. Left hepatic duct

B. Right hepatic duct

C. Cystic duct

D. Common bile duct

E. Sphincter of Oddi

10. A 45-year-old woman is brought to the emergency room with a 3-day history of colicky epigastric pain that suddenly increased in severity after a large meal. A gallstone is found to be occluding her spincter of Oddi. Which of the following substances would be found at reduced levels in her circulation?

A. Unconjugated bile acids
B. Conjugated bile acids
C. Cholesterol
D. Phosphatidylcholine
E. Amylase

SECTION V

Cardiovascular Physiology

Cells exist within a body fluid compartment known as the interstitial fluid, and the cardiovascular system has evolved to ensure that the composition of the interstitial fluid is maintained within a narrow range. Homeostasis is accomplished by pumping a separate fluid compartment—plasma—around the body, where it can be “conditioned” as it passes through specific organs that add nutrients, oxygen, hormones and needed metabolites, and/or remove waste products. The plasma then delivers needed substances to other organs and tissues. Efficient transfer of substances between the cells and the plasma is accomplished by dense networks of capillaries, which offer little resistance to the transfer of substances across their walls, and provide for short diffusion distances between the capillaries and the sites at which products will be utilized. The pumping function in this system is provided by the heart, a four-chambered organ that drives blood around two circuits in series, one that perfuses the lungs and one that serves the remainder of the body.

In principle, this sounds like a simple system. However, in practice, exquisite minute-to-minute regulation is needed to ensure that organs receive the substances that they need when they need them, particularly in the face of ever-changing demands. For example, when an individual begins to exercise, there is a prompt demand for additional oxygen and glucose in the contracting muscles to sustain muscle activity. In the brain, there is no capacity to store glucose and blood flow must be maintained to ensure consciousness, even in the face of hydrostatic challenges (eg, moving from a recumbent to a standing position). The cardiovascular system must therefore be able to adjust the rate at which the plasma is circulated around the body as a whole and to redirect plasma flow to the places where it is most needed. Further, the body is an “open” system, meaning that some bodily constituents (eg, water) are constantly being lost to the environment. The circulation, and the organs that condition it, must respond

promptly to these threats to homeostasis to ensure the proper functioning of vital body systems, which typically operate within a narrow range of osmolarity, pH, oxygen saturation, etc.

In this section, the components of the cardiovascular system that permit it to serve the body's needs for substance transfer will be considered. First, the electrical activity that allows the chambers of the heart to contract in an ordered fashion, to move the circulation unidirectionally, will be discussed. Then, the properties of blood and its components that suit them to transport dissolved solutes to and from the interstitial fluid will be considered. The properties of the circulatory "plumbing," or blood vessels, will be addressed, along with the mechanisms that regulate them. Finally, the specialized properties of the circulation in areas of the body with unique needs will be discussed.

Obviously, a functional cardiovascular system is vital for life, and irreversible damage quickly ensues in a number of organs if the heart stops beating. Less drastic disorders of the cardiovascular system also represent a considerable burden of disease. In fact, cardiovascular diseases, collectively, are the leading cause of death and responsible for significant disability worldwide. In the United States, heart disease and stroke are the first and third most common causes of death, and an estimated one in three American adults has some form of cardiovascular disorder. Cardiovascular diseases are also leading causes of hospitalization, and account for the highest economic burden of any disease category. Finally, while there have been impressive gains in treatment and prevention of some cardiovascular diseases, the growing epidemic of obesity, and the rise in the proportion of the population with at least one cardiovascular risk factor, has been cause for considerable alarm among public health officials. All of these facts underscore the relevance of a thorough understanding of cardiovascular physiology for the aspiring health professional.

CHAPTER 29

Origin of the Heartbeat & the Electrical Activity of the Heart

OBJECTIVES

After studying this chapter, you should be able to:

- Describe the structure and function of the conduction system of the heart and compare the action potentials in each part.
- Describe the way the electrocardiogram (ECG) is recorded, the waves of the ECG, and the relationship of the ECG to the electrical axis of the heart.
- Name the common cardiac arrhythmias and describe the processes that produce them.
- List the principal early and late ECG manifestations of myocardial infarction and explain the early changes in terms of the underlying ionic events that produce them.
- Describe the ECG changes and the changes in cardiac function produced by alterations in the ionic composition of the body fluids.

INTRODUCTION

The parts of the heart normally beat in orderly sequence: Contraction of the atria

(atrial systole) is followed by contraction of the ventricles **(ventricular systole),** and during **diastole** all four chambers are relaxed (also see Chapter 30). The cardiac electric activity that triggers heartbeat originates in a specialized **cardiac conduction system** and spreads via this system to all parts of the myocardium. The structures that make up the conduction system are the **sinoatrial node (SA node),** the **internodal atrial pathways,** the **atrioventricular node (AV node),** the **bundle of His** and its branches, and the **Purkinje system.** The various parts of the conduction system and, under abnormal conditions, parts of the myocardium, are capable of spontaneous discharge. However, the SA node normally discharges most rapidly, with depolarization spreading from it to the other regions before they discharge spontaneously. The SA node is therefore the normal **cardiac pacemaker,** with its rate of discharge determining the rate at which the heart beats. Impulses generated in the SA node pass through the atrial pathways to the AV node, through this node to the bundle of His, and through the branches of the bundle of His via the Purkinje system to the ventricular muscle. Each of the cell types in the heart contains a unique electrical discharge pattern; the sum of these electrical discharges can be recorded as the electrocardiogram (ECG or EKG).

ORIGIN & SPREAD OF CARDIAC EXCITATION

ANATOMIC CONSIDERATIONS

In the human heart, the SA node is located at the junction of the superior vena cava with the right atrium. The AV node is located in the right posterior portion of the interatrial septum **(Figure 29–1)**. There are three bundles of atrial fibers that contain Purkinje-type fibers and connect the SA node to the AV node: the anterior, middle (tract of Wenckebach), and posterior (tract of Thorel) tracts. Bachmann bundle is sometimes used to identify a branch of the anterior intermodal tract that connects the right and left atria. Conduction also occurs through atrial myocytes, but it is more rapid in these bundles. The AV node is continuous with the bundle of His, which gives off a left bundle branch at the top of the interventricular septum and continues as the right bundle branch. The left bundle branch divides into an anterior fascicle and a posterior fascicle. The branches and fascicles run subendocardially down either side of the septum and come into contact with the Purkinje system, whose fibers spread to all parts of the ventricular myocardium.

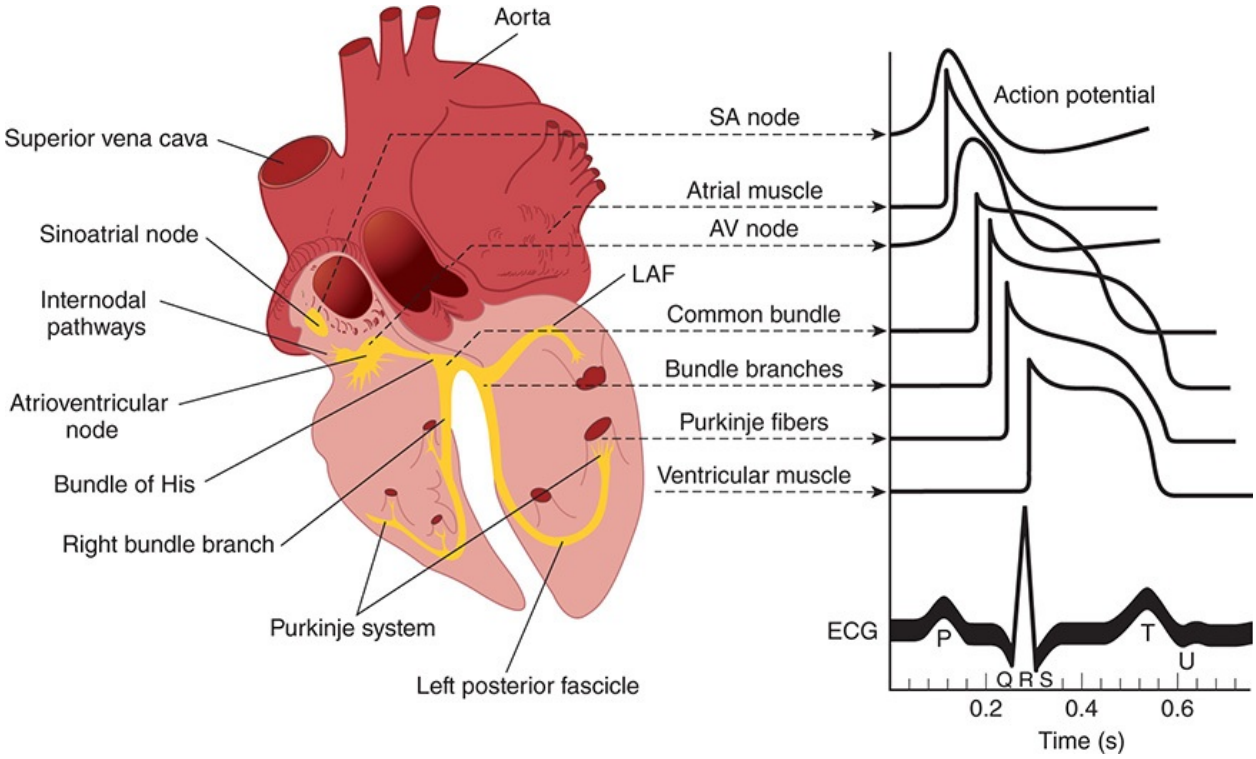


FIGURE 29–1 Conducting system of the heart. Left: Anatomic depiction of the human heart with additional focus on areas of the conduction system. **Right:** Typical transmembrane action potentials for the SA and AV nodes, other parts of the conduction system, and the atrial and ventricular muscles are shown along with the correlation to the extracellularly recorded electrical activity, that is, the electrocardiogram (ECG). The action potentials and ECG are plotted on the same time axis but with different zero points on the vertical scale for comparison. AV, atrioventricular; LAF, left anterior fascicle; SA, sinoatrial. (Data from Donahue JG, Choo PW, Manson JE, et al. The incidence of herpes zoster. Arch Intern Med 1995;155:1605–1609; Choo PW, Galil K, Donahue JG, et al. Risk factors for postherpetic neuralgia. Arch Intern Med 1997;157:1217–1224.)

The histology of a typical cardiac muscle cell (eg, a ventricular myocyte) is described in Chapter 5. For the most part, the conduction system is composed of modified cardiac muscle that has fewer striations and indistinct boundaries. Individual cells within regions of the heart have unique histologic features. Purkinje fibers, specialized conducting cells, are large with fewer mitochondria and striations and distinctly different from a myocyte specialized for contraction. Compared with Purkinje fibers, cells within the SA node and, to a lesser extent, the AV nodes are smaller and sparsely striated and are less conductive due to

their higher internal resistance. The atrial muscle fibers are separated from those of the ventricles by a fibrous tissue ring, and normally the only conducting tissue between the atria and ventricles is the bundle of His.

The SA node develops from structures on the right side of the embryo and the AV node from structures on the left. This is why in the adult the right vagus is distributed mainly to the SA node and the left vagus mainly to the AV node. Similarly, the sympathetic innervation on the right side is distributed primarily to the SA node and the sympathetic innervation on the left side primarily to the AV node. On each side, most sympathetic fibers come from the stellate ganglion. Noradrenergic fibers are epicardial, whereas the vagal fibers are endocardial. However, connections exist for reciprocal inhibitory effects of the sympathetic and parasympathetic innervation of the heart on each other. Thus, acetylcholine acts presynaptically to reduce norepinephrine release from the sympathetic nerves, and conversely, neuropeptide Y released from noradrenergic endings may inhibit the release of acetylcholine.

PROPERTIES OF CARDIAC MUSCLE

The electrical responses of cardiac muscle and nodal tissue and the ionic fluxes that underlie them are discussed in detail in Chapter 5 and are briefly reviewed here for comparison with the pacemaker cells below. Myocardial fibers have a resting membrane potential of approximately −90 mV **(Figure 29–2A)**. The individual fibers are separated by membranes, but membrane depolarization spreads radially through them as if they were a syncytium because of the presence of gap junctions. The transmembrane action potential of single cardiac muscle cells is characterized by rapid depolarization (phase 0), an initial rapid repolarization (phase 1), a plateau (phase 2), and a slow repolarization process (phase 3) that allows return to the resting membrane potential (phase 4). The initial depolarization is due to Na^+ influx through rapidly opening voltage-gated Na^+ channels (the Na^+ current, I_{Na}). The inactivation of voltage-gated Na^+ channels contributes to the rapid repolarization phase. Ca^{2+} influx through more slowly opening voltage-gated Ca^{2+} channels (the Ca^{2+} current, I_{Ca}) produces the plateau phase, and repolarization is due to net K^+ efflux through multiple types of K^+ channels. Recorded extracellularly, the summed electrical activity of all the cardiac muscle fibers is the ECG (discussed below). The timing of the discharge of the individual units relative to the ECG is shown in Figure 29–1 (right panel). Note that the ECG is a combined electrical record and thus the

overall shape reflects electrical activity from cells from different regions of the heart.

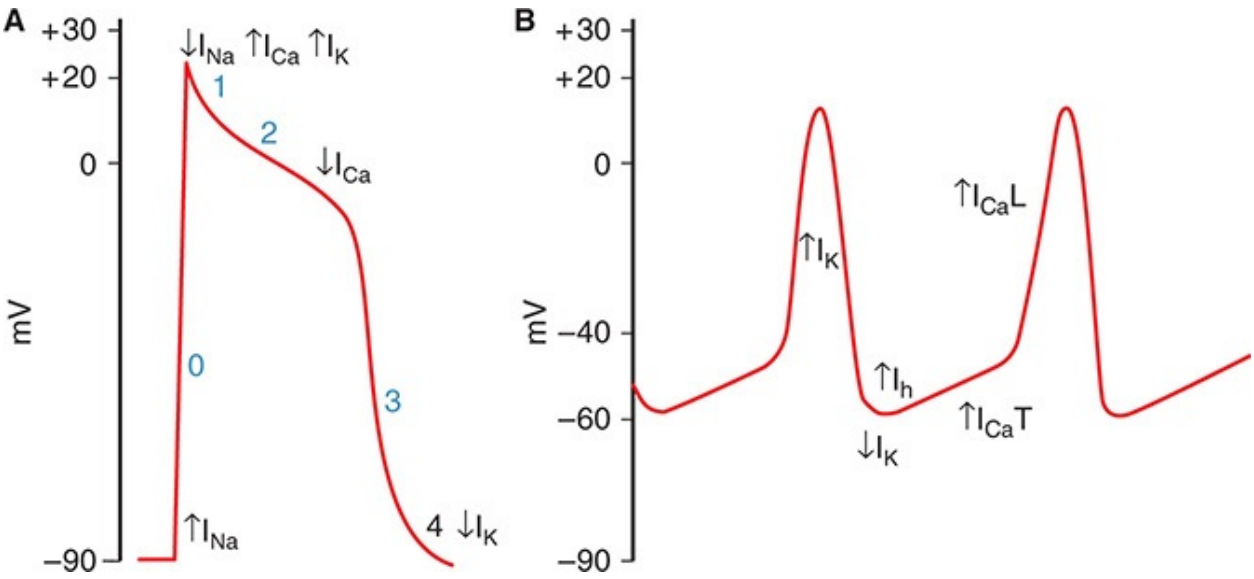


FIGURE 29–2 Comparison of action potentials in ventricular muscle and diagram of the membrane potential of pacemaker tissue. A) Phases of action potential in ventricular myocyte (0–4, see text for details) are superimposed with principal changes in current that contribute to changes in membrane potential. **B)** The principal current responsible for each part of the potential of pacemaker tissue is shown under or beside the component. L, long-lasting; T, transient. Other ion channels contribute to the electrical response. Note that the resting membrane potential of pacemaker tissue is somewhat lower than that of atrial and ventricular muscle.

PACEMAKER POTENTIALS

Rhythmically discharging cells have a membrane potential that, after each impulse, declines to the firing level. Thus, this **prepotential** or **pacemaker potential** (Figure 29–2B) triggers the next impulse. At the peak of each impulse, I_K begins and brings about repolarization. I_K then declines, and a channel permeable to both Na^+ and K^+ is activated. Because this channel is activated following hyperpolarization, it is referred to as an "h" channel and the current through the h channels is called "h" current (I_h); however, because of its unusual (funny) activation it has also been dubbed an "f" channel and the current produced as "funny current." As I_h increases, the membrane begins to

depolarize, forming the first part of the prepotential. When prepotential reaches the activation threshold of voltage-gated Ca^{2+} channels, Ca^{2+} channels then open. There are two types of these voltage-gated Ca^{2+} channels in the heart, the **T** (for transient) **channels** and the **L** (for long-lasting) **channels.** The Ca^{2+} current (I_{Ca}) due to opening of T channels completes the prepotential, and I_{Ca} due to opening of L channels produces the impulse. Other ion channels are also involved, and there is evidence that local Ca^{2+} release from the sarcoplasmic reticulum **(Ca^{2+} sparks)** occurs during the prepotential.

The action potentials in the SA and AV nodes are largely due to Ca^{2+} invlus, with no contribution by Na^+ influx. Consequently, there is no sharp, rapid depolarizing spike before the plateau, as there is in other parts of the conduction system and in the atrial and ventricular fibers. In addition, prepotentials are normally prominent only in the SA and AV nodes. However, “latent pacemakers” are present in other portions of the conduction system that can take over when the SA and AV nodes are depressed or conduction from them is blocked. Atrial and ventricular muscle fibers do not have prepotentials, and they discharge spontaneously only when injured or abnormal.

When the cholinergic vagal fibers to nodal tissue are stimulated, the membrane becomes hyperpolarized and the slope of the prepotentials is decreased **(Figure 29–3)** because the acetylcholine released at the nerve endings increases the K^+ conductance of nodal tissue. This action is mediated by M_2 muscarinic receptors, which, via the βγ subunit of a G protein, open a special set of K^+ channels. The resulting K^+ currents (I_{KAch}) through these G protein-gated K^+ channels slows the depolarizing effect of I_h. In addition, activation of the M_2 receptors decreases cyclic adenosine 3′,5′-monophosphate (cAMP) in the cells, and this slows the opening of Ca^{2+} channels. The result is a decrease in firing rate. Strong vagal stimulation may abolish spontaneous discharge for some time.

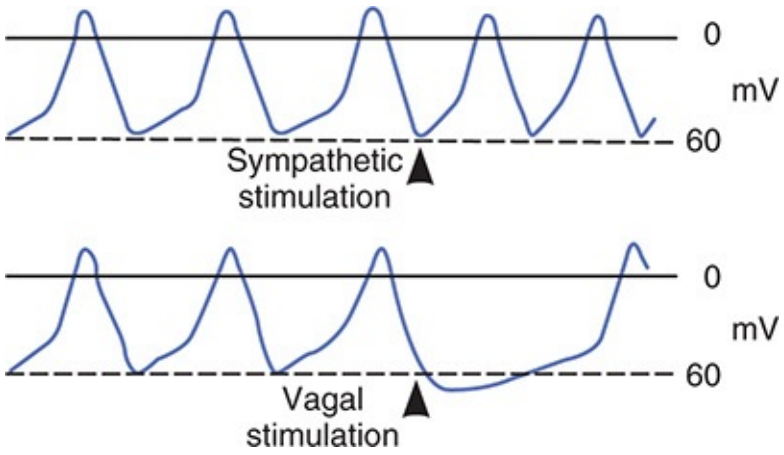


FIGURE 29–3 Effect of sympathetic (noradrenergic) and vagal (cholinergic) stimulation on the membrane potential of the SA node. Note the reduced slope of the prepotential after vagal stimulation and the increased spontaneous discharge after sympathetic stimulation. SA, sinoatrial.

CLINICAL BOX 29–1

Use of Digitalis

Digitalis, or its clinically useful preparations (digoxin and digitoxin), has been described in medical literature for over 200 years. It was originally derived from the foxglove plant. (*Digitalis purpurea* is the name of the common foxglove.) Correct administration can strengthen cardiac contractions through digitalis inhibitory effects on the Na, K ATPase, resulting in greater amounts of Ca^{2+} release and subsequent changes in contraction forces. Digitalis can also have an electrical effect in decreasing AV nodal conduction velocity and thus altering AV transmission to the ventricles.

THERAPEUTIC HIGHLIGHTS

Digitalis has been used for treatment of systolic heart failure. It augments contractility, thereby improving cardiac output, improving left ventricle emptying, and decreasing ventricular filling pressures. Digitalis has also been used to treat atrial fibrillation and atrial flutter. In this scenario, digitalis reduces the number of impulses transmitted through the AV node and thus, provides effective rate control.

In both these instances alternative treatments developed over the past 20 years and the need to tightly regulate dose due to significant potential for side effects have reduced the use of digitalis. However, with better understanding of mechanism and toxicity, digitalis and its clinically prepared derivatives remain important drugs in modern medicine.

Conversely, stimulation of the sympathetic cardiac nerves speeds the depolarizing effect of I_h, and the rate of spontaneous discharge increases (Figure 29–3). Norepinephrine secreted by the sympathetic endings binds to β_1 receptors, and the resulting increase in intracellular cAMP facilitates the opening of L channels, increasing I_{Ca} and the rapidity of the depolarization phase of the impulse.

The rate of discharge of the SA node and other nodal tissue is influenced by temperature and by drugs. The discharge frequency is increased when the temperature rises, and this may contribute to the tachycardia associated with fever. Digitalis depresses nodal tissue and exerts an effect like that of vagal stimulation, particularly on the AV node **(Clinical Box 29–1;** also see Clinical Box 5–6).

SPREAD OF CARDIAC EXCITATION

Depolarization initiated in the SA node spreads radially through the atria, then converges on the AV node. Atrial depolarization is complete in about 0.1 s. Because conduction in the AV node is slow **(Table 29–1)**, a delay of about 0.1 s **(AV nodal delay)** occurs before excitation spreads to the ventricles. It is interesting to note here that when there is a lack of contribution of I_{Na} in the depolarization (phase 0) of the action potential, a marked loss of conduction is observed. This delay is shortened by stimulation of the sympathetic nerves to the heart and lengthened by stimulation of the vagi. From the top of the septum, the wave of depolarization spreads in the rapidly conducting Purkinje fibers to all parts of the ventricles in 0.08–0.1 s. In humans, depolarization of the ventricular muscle starts at the left side of the interventricular septum and moves first to the right across the mid portion of the septum. The wave of depolarization then spreads down the septum to the apex of the heart. It returns along the ventricular walls to the AV groove, proceeding from the endocardial to the epicardial surface **(Figure 29–4)**. The last parts of the heart to be depolarized are the

posterobasal portion of the left ventricle, the pulmonary conus, and the uppermost portion of the septum.

TABLE 29–1 Conduction speeds in cardiac tissue.

Tissue	Conduction Rate (m/s)
SA node	0.05
Atrial pathways	1
AV node	0.05
Bundle of His	1
Purkinje system	4
Ventricular muscle	1

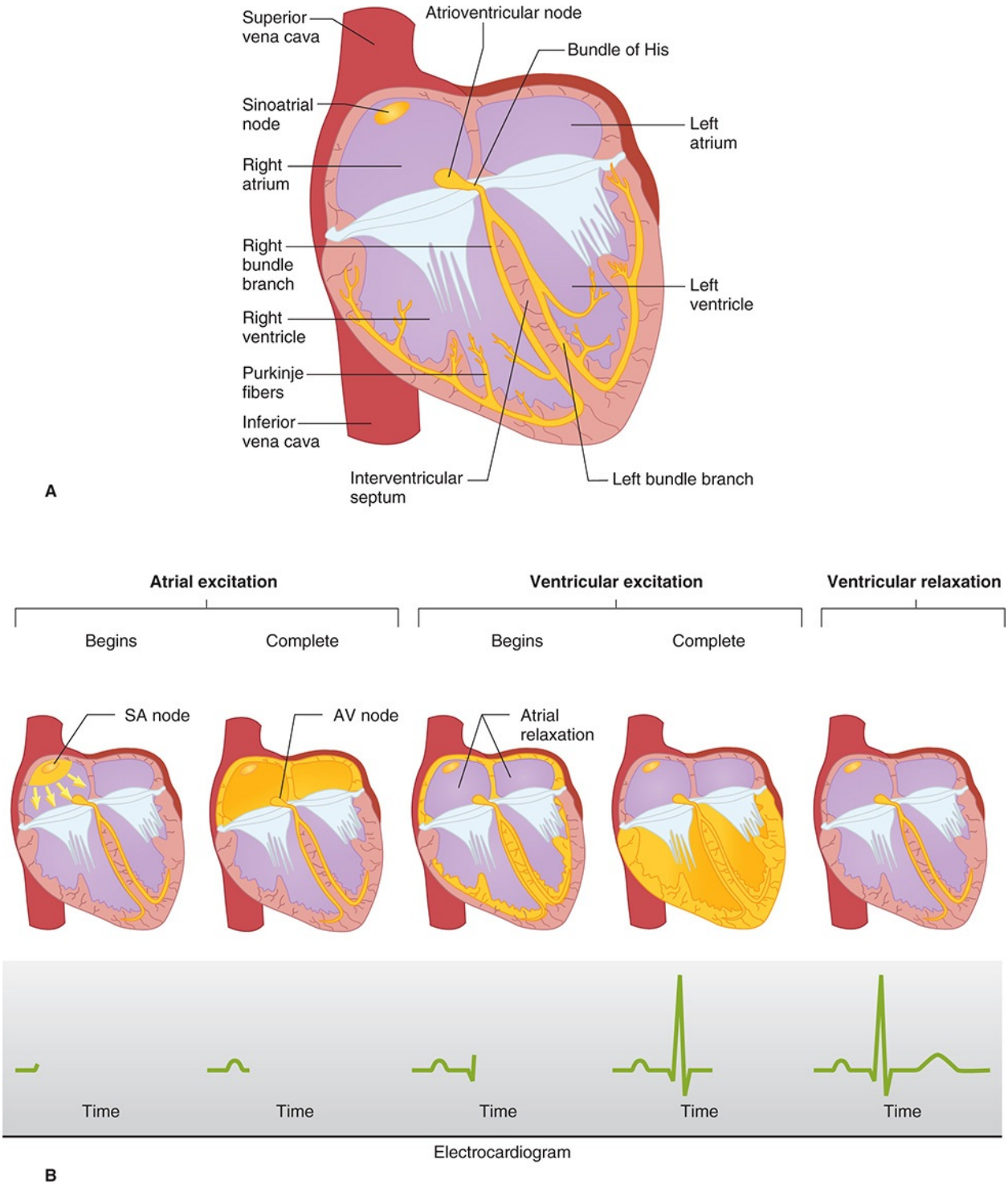


FIGURE 29–4 Normal spread of electrical activity in the heart. A) Conducting system of the heart. **B)** Sequence of cardiac excitation. **Top:** Anatomic position of electrical activity. **Bottom:** Corresponding electrocardiogram. The yellow color denotes areas that are depolarized. AV, atrioventricular; SA, sinoatrial. (Reproduced with permission from Goldman MJ:

Principles of Clinical Electrocardiography, 12th ed. Originally published by Appleton & Lange. Copyright © 1986 by McGraw-Hill.)

THE ELECTROCARDIOGRAM

Because the body fluids are good conductors (ie, because the body is a **volume conductor**), fluctuations in potential, representing the algebraic sum of the action potentials of myocardial fibers, can be recorded extracellularly. The record of these fluctuations in potential during the cardiac cycle is the **ECG or EKG.**

The ECG may be recorded by using an **active** or **exploring electrode** connected to an indifferent electrode at zero potential **(unipolar recording)** or by using two active electrodes **(bipolar recording).** In a volume conductor, the sum of the potentials at the points of an equilateral triangle with a current source in the center is zero at all times. A triangle with the heart at its center **(Einthoven triangle,** see below) can be approximated by placing electrodes on both arms and on the left leg. These are the three **standard limb leads** used in electrocardiography. If these electrodes are connected to a common terminal, an indifferent electrode that stays near zero potential is obtained. Depolarization moving toward an active electrode in a volume conductor produces a positive deflection, whereas depolarization moving in the opposite direction produces a negative deflection.

The names of the various waves and segments of the ECG in humans are shown in **Figure 29–5**. By convention, an upward deflection is written when the active electrode becomes positive relative to the indifferent electrode, and a downward deflection is written when the active electrode becomes negative. As can be seen in Figure 29–1, the P wave is primarily produced by atrial depolarization, the QRS complex is dominated by ventricular depolarization, and the T wave by ventricular repolarization. The U wave is an inconstant finding that may be due to ventricular myocytes with long action potentials. However, the contributions to this segment are still undetermined. The intervals between the various waves of the ECG and the events in the heart that occur during these intervals are shown in **Table 29–2**.

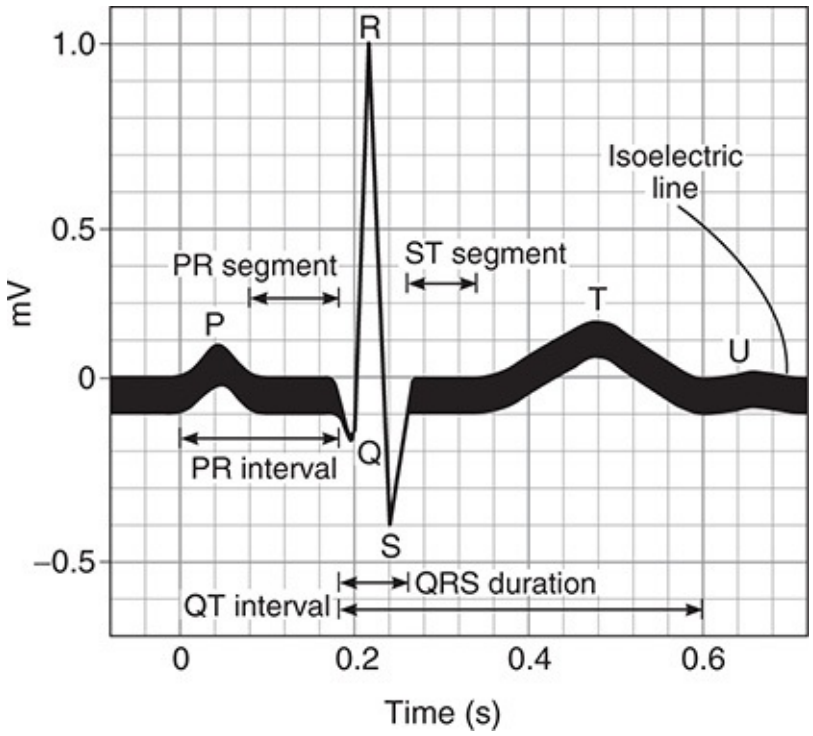


FIGURE 29–5 Waves of the ECG. Standard names for individual waves and segments that make up the ECG are shown. Electrical activity that contributes the observed deflections are discussed in the text and in Table 29–2. ECG, electrocardiogram.

TABLE 29–2 ECG intervals.

Intervals	Normal Durations (second, s)		Events in the Heart during Interval
	Average	Range	
PR interval[a]	0.18[b]	0.12–0.20	Atrioventricular conduction
QRS duration	0.08	to 0.10	Ventricular depolarization
QT interval	0.40[c]	to 0.43	Ventricular action potential
ST interval (QT minus QRS)	0.32	…	Plateau portion of the ventricular action potential

[a]Measured from the beginning of the P wave to the beginning of the QRS complex.
[b]Shortens as heart rate increases from average of 0.18 s at a rate of 70 beats/min to 0.14 s at a rate of 130 beats/min.
[c]Can be lower (0.35 s) depending on the heart rate.

BIPOLAR LEADS

Bipolar leads were used before unipolar leads were developed. The **standard limb leads** each record the differences in potential between two limbs. Because current flows only in the body fluids, the records obtained are those that would be obtained if the electrodes were at the points of attachment of the limbs, no

matter where on the limbs the electrodes are placed. In lead I, the electrodes are connected so that an upward deflection is inscribed when the left arm becomes positive relative to the right (left arm positive). In lead II, the electrodes are on the right arm and left leg, with the leg positive; and in lead III, the electrodes are on the left arm and left leg, with the leg positive.

UNIPOLAR (V) LEADS

An additional nine unipolar leads, that is, leads that record the potential difference between an **exploring electrode** and an **indifferent electrode,** are commonly used in clinical electrocardiography. There are six unipolar chest leads (precordial leads) designated V_1–V_6 **(Figure 29–6)** and three unipolar limb leads: VR (right arm), VL (left arm), and VF (left foot). The indifferent electrode is constructed by connecting electrodes placed on the two arms and the left leg to a central terminal. This "V" lead effectively records a "zero" potential because they are situated such that the electrical activity should be cancelled out. **Augmented limb leads,** designated by the letter a (aVR, aVL, aVF), are generally used. The augmented limb leads do not use the "V" electrode as the zero, rather, they are recordings between the one, augmented limb and the other two limbs. This increases the size of the potentials by 50% without any change in configuration from the nonaugmented record.

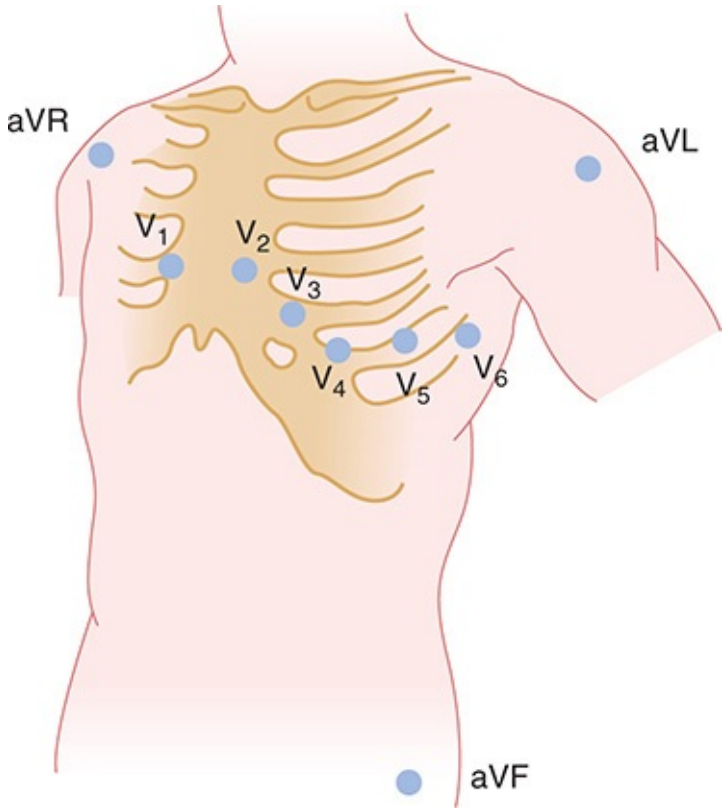


FIGURE 29–6 Unipolar electrocardiographic leads. Positional for standard unipolar leads are shown. The augmented extremity leads (aVR, aVL, and aVF) are shown on the right arm, left arm, and left leg, respectively. The six chest leads (V_1–V_6) are shown in their proper placement.

Unipolar leads can also be placed at the tips of catheters and inserted into the esophagus or heart. Although sensitivity can be increased, this is obviously more invasive and thus, not a first step in obtaining electrical readings.

NORMAL ECG

The ECG tracings of a normal individual are shown in Figure 29–4b and **Figure 29–7**. The sequence in which the parts of the heart are depolarized (Figure 29–4) and the position of the heart relative to the electrodes (Figure 29–7) are the important considerations in interpreting the configurations of the waves in each lead. The atria are located posteriorly in the chest. The ventricles form the base and anterior surface of the heart, and the right ventricle is anterolateral to the left. Thus, aVR "looks at" the cavities of the ventricles. Atrial depolarization, ventricular depolarization, and ventricular repolarization move away from the

exploring electrode, and the P wave, QRS complex, and T wave are therefore all negative (downward) deflections; aVL and aVF look at the ventricles, and the deflections are therefore predominantly positive or biphasic. There is no Q wave in V_1 and V_2, and the initial portion of the QRS complex is a small upward deflection because ventricular depolarization first moves across the midportion of the septum from left to right toward the exploring electrode. The wave of excitation then moves down the septum and into the left ventricle away from the electrode, producing a large S wave. Finally, it moves back along the ventricular wall toward the electrode, producing the return to the isoelectric line. Conversely, in the left ventricular leads (V_4–V_6) there may be an initial small Q wave (left to right septal depolarization), and there is a large R wave (septal and left ventricular depolarization) followed in V_4 and V_5 by a moderate S wave (late depolarization of the ventricular walls moving back toward the AV junction). It should be noted that there is considerable variation in the position of the normal heart, and the position affects the configuration of the electrocardiographic complexes in the various leads.

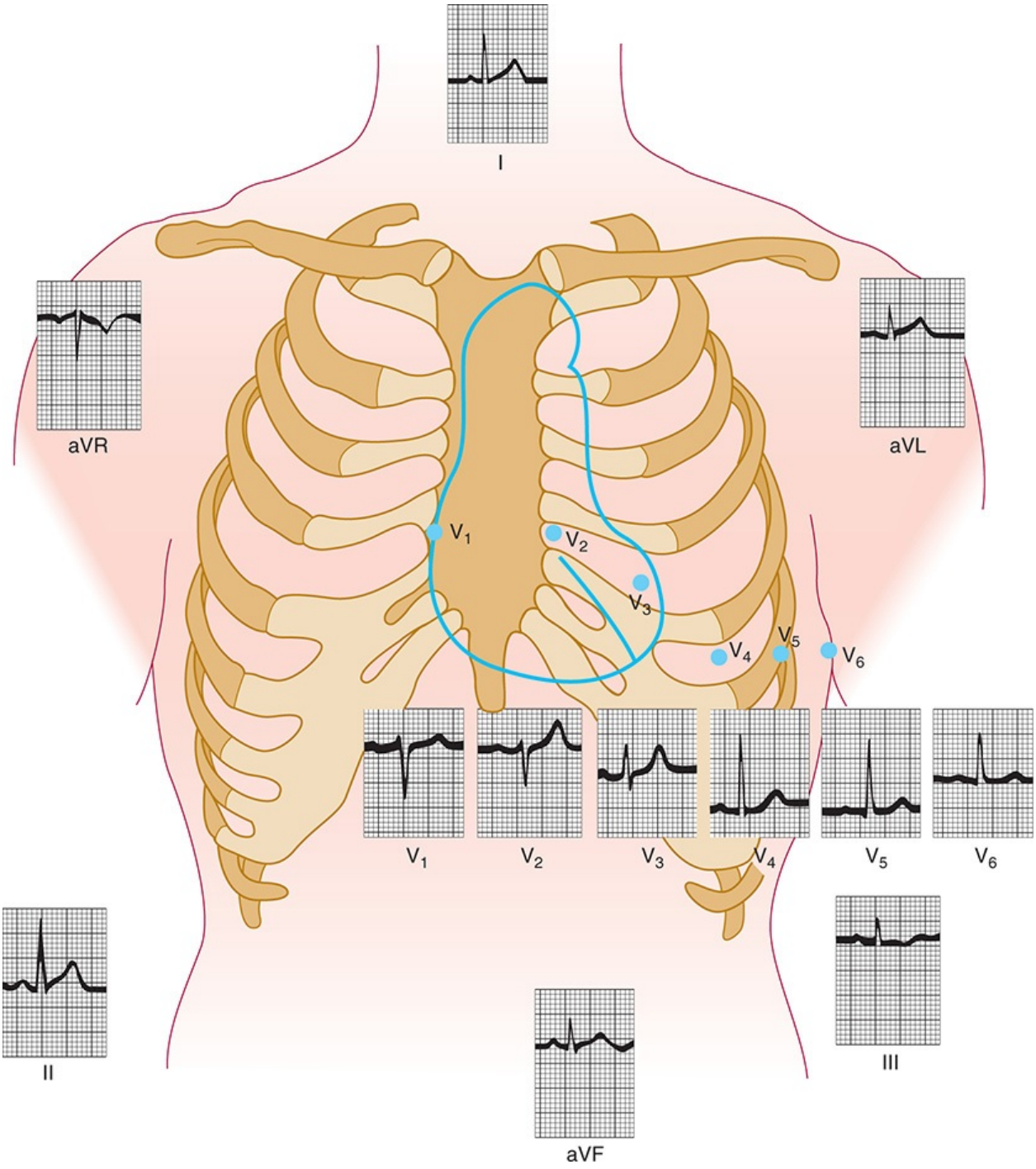


FIGURE 29–7 Normal ECG. Tracings from individual electrodes (positions marked in figure) are shown for a normal ECG. See text for additional details. ECG, electrocardiogram. (Reproduced with permission from Goldman MJ: *Principles of Clinical Electrocardiography*, 12th ed. Originally published by Appleton & Lange. Copyright © 1986 by McGraw-Hill.)

BIPOLAR LIMB LEADS & THE CARDIAC VECTOR

Because the standard limb leads are records of the potential differences between two points, the deflection in each lead at any instant indicates the magnitude and direction of the electromotive force generated in the heart in the axis of the lead **(cardiac vector** or **axis).** The vector at any given moment in the two dimensions of the frontal plane can be calculated from any two standard limb leads **(Figure 29–8)** if it is assumed that the three electrode locations form the points of an equilateral triangle (Einthoven triangle) and that the heart lies in the center of the triangle. These assumptions are not completely warranted, but calculated vectors are useful approximations. An approximate **mean QRS vector** (“electrical axis of the heart”) is often plotted by using the average QRS deflection in each lead, as shown in Figure 29–8. This is a **mean** vector as opposed to an **instantaneous** vector, and the average QRS deflections should be measured by integrating the QRS complexes. However, they can be approximated by measuring the net differences between the positive and negative peaks of the QRS. The normal direction of the mean QRS vector is generally said to be from −30 to +110° on the coordinate system shown in Figure 29–8. **Left** or **right axis deviation** is said to be present if the calculated axis falls to the left of −30° or to the right of +110°, respectively. Right axis deviation suggests right ventricular hypertrophy. Left axis deviation may be due to left ventricular hypertrophy, but there are better and more reliable electrocardiographic criteria for this condition.

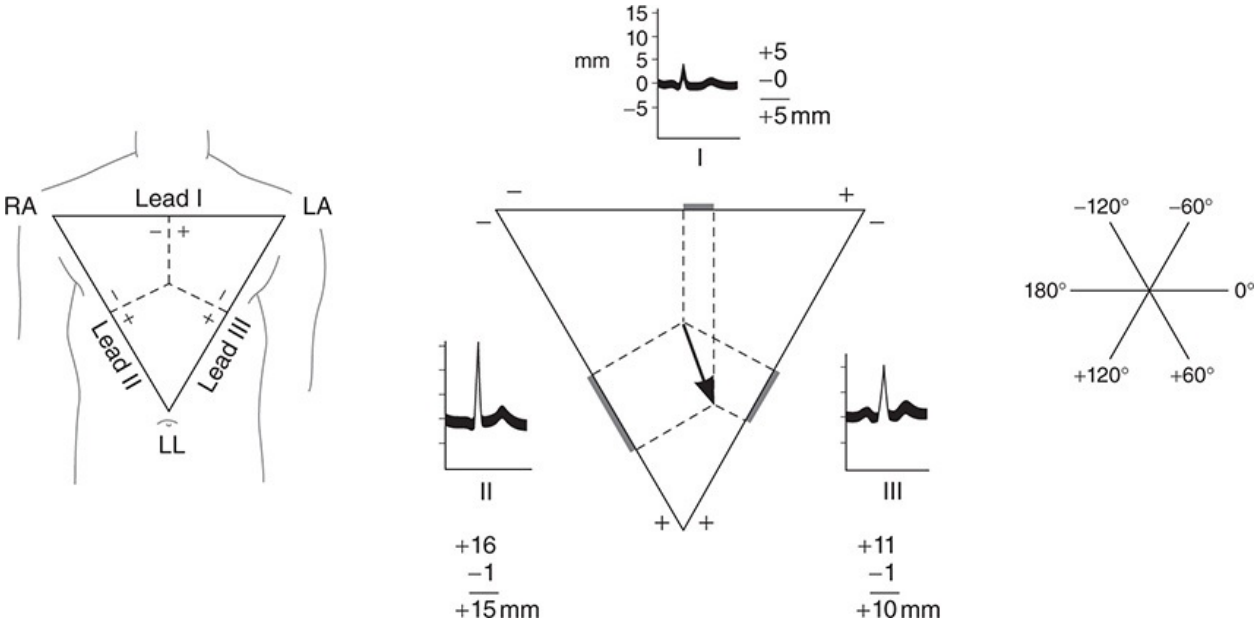


FIGURE 29–8 Cardiac vector. Left: Einthoven triangle. Perpendiculars dropped from the midpoints of the sides of the equilateral triangle intersect at the center of electrical activity. RA, right arm; LA, left arm; LL, left leg. **Center:** Calculation of mean QRS vector. In each lead, distances equal to the height of the R wave minus the height of the largest negative deflection in the QRS complex are measured off from the midpoint of the side of the triangle representing that lead. An arrow drawn from the center of electrical activity to the point of intersection of perpendiculars extended from the distances measured off on the sides represents the magnitude and direction of the mean QRS vector. **Right:** Reference axes for determining the direction of the vector.

HIS BUNDLE ELECTROGRAM

In patients with heart block, the electrical events in the AV node, bundle of His, and Purkinje system are frequently studied with a catheter containing an electrode at its tip that is passed through a vein to the right side of the heart and manipulated into a position close to the tricuspid valve. Three or more standard electrocardiographic leads are recorded simultaneously. The record of the electrical activity obtained with the catheter **(Figure 29–9)** is the **His bundle electrogram (HBE).** It normally shows an A deflection when the AV node is activated, an H spike during transmission through the His bundle, and a V deflection during ventricular depolarization. With the HBE and the standard electrocardiographic leads, it is possible to time three intervals accurately: (1) the PA interval, the time from the first appearance of atrial depolarization to the A wave in the HBE, which represents conduction time from the SA node to the AV node; (2) the AH interval, from the A wave to the start of the H spike, which represents the AV nodal conduction time; and (3) the HV interval, the time from the start of the H spike to the start of the QRS deflection in the ECG, which represents conduction in the bundle of His and the bundle branches. The approximate normal values for these intervals in adults are PA, 27 ms; AH, 92 ms; and HV, 43 ms. These values illustrate the relative slowness of conduction in the AV node.

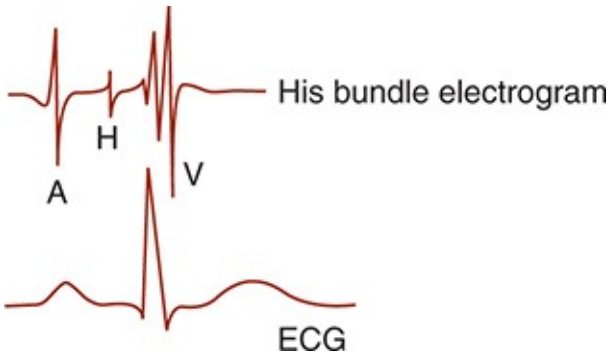


FIGURE 29–9 Normal His bundle electrogram (HBE) with simultaneously recorded ECG. An HBE recorded with an invasive electrode is superimposed on a standard ECG reading. Timing of depolarizations of the HBE are described in the text. ECG, electrocardiogram.

MONITORING

The ECG has long been used in patient care. In the past, it was often recorded continuously in hospital coronary care units, with alarms arranged to sound at the onset of life-threatening arrhythmias. Using a small portable tape recorder **(Holter monitor),** it is also possible to record the ECG in ambulatory individuals as they go about their normal activities. The recording is later played back at high speed and analyzed. Recordings obtained with monitors have proved valuable in the diagnosis of arrhythmias and in planning the treatment of patients recovering from myocardial infarctions. Currently, modern systems can be hooked up to individuals and obtain and store heart rhythm data over days to better evaluate long-term electrical activity.

CLINICAL APPLICATIONS: CARDIAC ARRHYTHMIAS

NORMAL CARDIAC RATE

In the normal human heart, each beat originates in the SA node **(normal sinus rhythm, NSR).** The heart beats about 70 times a minute at rest. The rate is slowed **(bradycardia)** during sleep and accelerated **(tachycardia)** by emotion, exercise, fever, and many other stimuli. In healthy young individuals breathing at a normal rate, the heart rate varies with the phases of respiration: It accelerates during inspiration and decelerates during expiration, especially if the depth of

breathing is increased. This **sinus arrhythmia** **(Figure 29–10)** is a normal phenomenon and is primarily due to fluctuations in parasympathetic output to the heart. During inspiration, impulses in the vagi from the stretch receptors in the lungs inhibit the cardio-inhibitory area in the medulla oblongata. The tonic vagal discharge that keeps the heart rate slow decreases, and the heart rate rises. Disease processes affecting the sinus node lead to marked bradycardia accompanied by dizziness and syncope **(Clinical Box 29–2)**.

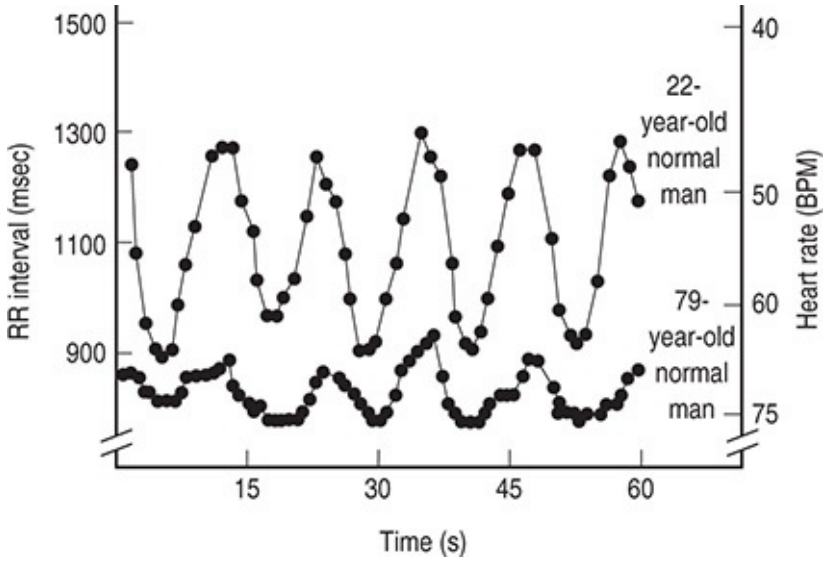


FIGURE 29–10 Sinus arrhythmia in a young man and an old man. Each subject breathed five times per minute. With each inspiration the RR interval (the interval between R waves) declined, indicating an increase in heart rate. Note the marked reduction in the magnitude of the arrhythmia in the older man. These records were obtained after β-adrenergic blockade but would have been generally similar in its absence. (Reproduced with permission from Pfeifer MA, Weinberg CR, Cook D, Best JD, Reenan A, Halter JB: Differential changes of autonomic nervous system function with age in man. Am J Med 1983;75:249.)

ABNORMAL PACEMAKERS

The AV node and other portions of the conduction system can, in abnormal situations, become the cardiac pacemaker. In addition, diseased atrial and ventricular muscle fibers can have their membrane potentials reduced and discharge repetitively.

As noted above, the discharge rate of the SA node is more rapid than that of the other parts of the conduction system, and this is why the SA node normally

controls the heart rate. When conduction from the atria to the ventricles is completely interrupted, **complete (third-degree) heart block** results, and the ventricles beat at a low rate **(idioventricular rhythm)** independently of the atria **(Figure 29–11)**. The block may be due to disease in the AV node **(AV nodal block)** or in the conducting system below the node **(infranodal block).** In patients with AV nodal block, the remaining nodal tissue becomes the pacemaker and the rate of the idioventricular rhythm is approximately 45 beats/min. In patients with infranodal block due to disease in the bundle of His, the ventricular pacemaker is located more peripherally in the conduction system and the ventricular rate is lower; it averages 35 beats/min, but in individual cases it can be as low as 15 beats/min. In such individuals, there may also be periods of asystole lasting a minute or more. The resultant cerebral ischemia causes dizziness and fainting **(Stokes–Adams syndrome).** Causes of third-degree heart block include septal myocardial infarction and damage to the bundle of His during surgical correction of congenital interventricular septal defects.

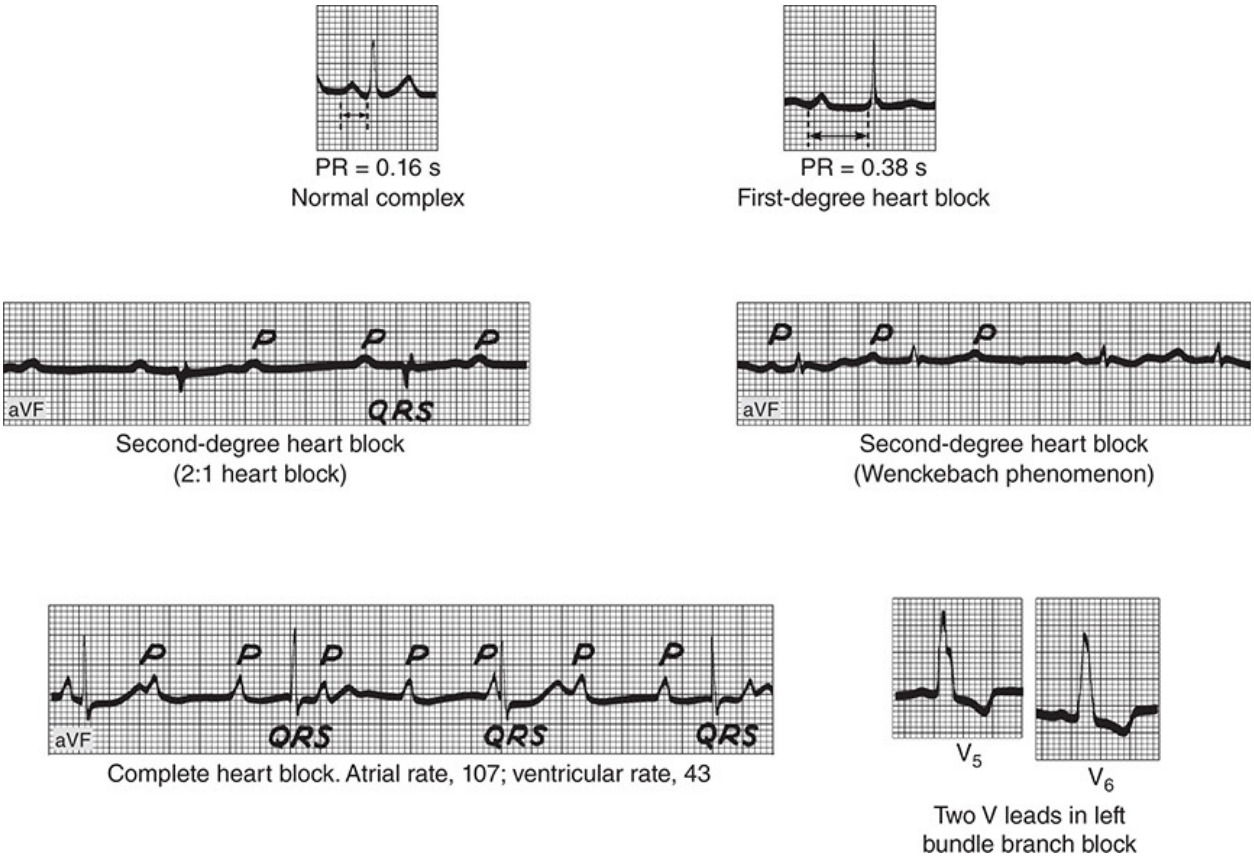


FIGURE 29–11 ECG with heart block. Individual traces that depict various forms of heart block are shown. When appropriate, unipolar leads are noted. See text for further details. ECG, electrocardiogram.

When conduction between the atria and ventricles is slowed but not completely interrupted, **incomplete heart block** is present. In the form called **first-degree heart block,** all the atrial impulses reach the ventricles but the PR interval is abnormally long. In the form called **second-degree heart block,** not all atrial impulses are conducted to the ventricles. For example, a ventricular beat may follow every second or every third atrial beat (2:1 block, 3:1 block, etc). In another form of incomplete heart block, there are repeated sequences of beats in which the PR interval lengthens progressively until a ventricular beat is dropped **(Wenckebach phenomenon)**. The PR interval of the cardiac cycle that follows each dropped beat is usually normal or only slightly prolonged (Figure 29–11).

CLINICAL BOX 29–2

Sick Sinus Syndrome

Sick sinus syndrome (bradycardia-tachycardia syndrome; sinus node dysfunction) is a collection of heart rhythm disorders that include **sinus bradycardia** (slow heart rates from the natural pacemaker of the heart), **tachycardias** (fast heart rates), and **bradycardia-tachycardia** (alternating slow and fast heart rhythms). Sick sinus syndrome is relatively uncommon and is usually found in people older than 50, in whom the cause is often a nonspecific, scar-like degeneration of the heart's conduction system. When found in younger people, especially in children, a common cause of sick sinus syndrome is heart surgery, especially on the upper chambers. Holter monitoring is an effective tool for diagnosing sick sinus syndrome because of the episodic nature of the disorder. Extremely slow heart rate and prolonged pauses may be seen during Holter monitoring, along with episodes of atrial tachycardias.

THERAPEUTIC HIGHLIGHTS

Treatment depends on the severity and type of disease. Tachycardias are frequently treated with medication. When there is marked bradycardia in patients with sick sinus syndrome or third-degree heart block, an electronic pacemaker is frequently implanted. These devices, which have become sophisticated and reliable, are useful in patients with sinus node dysfunction, AV block, and bifascicular or trifascicular block. They are useful also in patients with severe neurogenic syncope, in whom carotid sinus stimulation

produces pauses of more than 3 s between heartbeats.

Sometimes one branch of the bundle of His is interrupted, causing **right** or **left bundle branch block.** In bundle branch block, excitation passes normally down the bundle on the intact side and then sweeps back through the muscle to activate the ventricle on the blocked side. The ventricular rate is therefore normal, but the QRS complexes are prolonged and deformed (Figure 29–11). Block can also occur in the anterior or posterior fascicle of the left bundle branch, producing the condition called **hemiblock** or **fascicular block.** Left anterior hemiblock produces abnormal left axis deviation in the ECG, whereas left posterior hemiblock produces abnormal right axis deviation. It is not uncommon to find combinations of fascicular and branch blocks **(bifascicular** or **trifascicular block).** The HBE permits detailed analysis of the site of block when there is a defect in the conduction system.

ECTOPIC FOCI OF EXCITATION

Normally, myocardial cells do not discharge spontaneously, and the possibility of spontaneous discharge of the His bundle and Purkinje system is low because the normal pacemaker discharge of the SA node is more rapid than their rate of spontaneous discharge. However, in abnormal conditions, the His–Purkinje fibers or the myocardial fibers may discharge spontaneously. In these conditions, **increased automaticity** of the heart is said to be present. If an irritable **ectopic focus** discharges once, the result is a beat that occurs before the expected next normal beat and transiently interrupts the cardiac rhythm (atrial, nodal, or ventricular **extrasystole** or **premature beat**). If the focus discharges repetitively at a rate higher than that of the SA node, it produces rapid, regular tachycardia (atrial, ventricular, or nodal **paroxysmal tachycardia** or **atrial flutter**).

REENTRY

A more common cause of paroxysmal arrhythmias is a defect in conduction that permits a wave of excitation to propagate continuously within a closed circuit **(circus movement).** For example, if a transient block is present on one side of a portion of the conducting system, the impulse can go down the other side. If the block then wears off, the impulse may conduct in a retrograde direction in the